3.0 DESIGN OF STRUCTURES, COMPONENTS, EQUIPMENT, AND SYSTEMS

This chapter identifies, describes, and discusses the principal architectural and engineering design features of those structures, components, equipment, and systems that are necessary to assure:

A. The integrity of the reactor coolant pressure boundary.

B. The capability to shut down the reactor and maintain it in a safe shutdown condition.

C. The capability to prevent or mitigate the consequences of accidents which could result in potential offsite exposures comparable to the guideline values of 10 CFR 100.

3.1 CONFORMANCE WITH NUCLEAR REGULATORY COMMISSION (NRC) GENERAL DESIGN CRITERIA (GDC)

This section briefly discusses the extent to which the design criteria for VEGP structures, systems, and components important to safety comply with Title 10, Code of Federal Regulations, Part 50 (10 CFR 50), Appendix A, General Design Criteria for Nuclear Power Plants. As presented in this section, each criterion is first quoted and then discussed in enough detail to demonstrate compliance of the VEGP with each criterion. For some criteria, additional information may be required for a complete discussion. In such cases, detailed evaluations of compliance with the various general design criteria are incorporated in more appropriate sections, but are located by reference.

3.1.1 OVERALL REQUIREMENTS

CRITERION 1 - QUALITY STANDARDS AND RECORDS

"Structures, systems, and components important to safety shall be designed, fabricated, erected, and tested to quality standards commensurate with the importance of the safety function to be performed. Where generally recognized codes and standards are used, they shall be identified and evaluated to determine their applicability, adequacy, and sufficiency and shall be supplemented or modified, as necessary, to assure a quality product, in keeping with the required safety function.

A quality assurance program shall be established and implemented in order to provide adequate assurance that these structures, systems, and components will satisfactorily perform their safety functions. Appropriate records of the design, fabrication, erection, and testing of structures, systems, and components important to safety shall be maintained by or under the control of the nuclear power unit licensee throughout the life of the unit."

DISCUSSION

The quality assurance program of the VEGP and Southern Nuclear Operating Company (SNC), together with the quality assurance, quality engineering, and quality control programs of the major contractors and their vendors, ensure that structures, systems, and components important to safety are designed, fabricated, erected, and tested to quality standards commensurate with the safety functions to be performed. This is accomplished through the use

of recognized codes, standards, and design criteria. As necessary, additional supplemental standards, design criteria, and requirements are developed by the VEGP and the major contractors' engineering organizations. Appropriate records associated with the engineering and design, fabrication, erection, and testing which document the compliance with recognized codes, standards, and design criteria are maintained throughout the life of the units either by or under the control of SNC. Quality assurance is described in chapter 17.

The principal design criteria, design bases, codes, and standards applied to the facility are described in section 3.2. Additional detail may be found in the pertinent section of the document dealing with structures, systems, and components important to safety; e.g., the containment as described in subsection 3.8.1.

CRITERION 2 - DESIGN BASES FOR PROTECTION AGAINST NATURAL PHENOMENA

"Structures, systems, and components important to safety shall be designed to withstand the effects of natural phenomena such as earthquakes, tornadoes, hurricanes, floods, tsunami, and seiches without the loss of the capability to perform their safety functions. The design bases for these structures, systems, and components shall reflect: (1) appropriate consideration of the most severe of the natural phenomena that have been historically reported for the site and surrounding area, with sufficient margin for the limited accuracy, quantity, and period of time in which the historical data have been accumulated, (2) appropriate combinations of the effects of normal and accident conditions with the effects of the natural phenomena, and (3) the importance of the safety functions to be performed."

DISCUSSION

The structures, systems, and components important to safety are designed either to withstand the effects of natural phenomena without loss of the capability to perform their safety functions, or are designed such that their response or failure will be in a safe condition. Those structures, systems, and components vital to the shutdown capability of the reactor are designed to withstand the maximum probable natural phenomena at the site, determined from recorded data for the site vicinity, with appropriate margin to account for uncertainties in historical data. Appropriate combinations of structural loadings from normal, accident, and natural phenomena are considered in the plant design. The nature and magnitude of the natural phenomena considered in the design of this plant are discussed in chapter 2. Chapter 3 discusses the design of the plant in relationship to natural events. Seismic and quality group classifications, as well as other pertinent standards and information, are given in the sections discussing individual structures and components and in table 3.2.2-1.

CRITERION 3 - FIRE PROTECTION

"Structures, systems, and components important to safety shall be designed and located to minimize, consistent with other safety requirements, the probability and effect of fires and explosions. Noncombustible and heat-resistant materials shall be used wherever practical throughout the unit, particularly in locations such as the containment and control room. Fire detection and fighting systems of appropriate capacity and capability shall be provided and designed to minimize the adverse effects of fires on structures, systems, and components important to safety. Firefighting systems shall be designed to assure that their rupture or inadvertent operation does not significantly impair the safety capability of these structures, systems, and components."

DISCUSSION

The plant is designed to minimize the probability and effect of fires and explosions. Noncombustible and fire-resistant materials are used in the containment, control room, components of safety features systems, and throughout the unit wherever fire is a potential risk to safety-related systems. For example, electrical cables have a fire-retardant jacketing, and fire barriers and fire stops are utilized as described in subsection 9.5.1. Equipment and facilities for fire protection, including detection, alarm, and extinguishment, are provided to protect both plant equipment and personnel from fire, explosion, and the resultant release of toxic vapors.

Fire protection is provided by deluge systems (water spray), sprinklers, Halon 1301, and portable extinguishers.

Firefighting systems are designed to assure that their rupture or inadvertent operation will not prevent systems important to safety from performing their design functions.

The following codes, guides, and standards are used as guidelines in the design of the fire protection system and equipment. Where required by law, the system and equipment substantially conform to the applicable portions of the following standards:

A. National Fire Protection Association (NFPA) National Fire Codes.

B. Nuclear Mutual Limited (NML) Property Loss Prevention Standards for Nuclear Generating Stations.

C. BTP-APCSB 9.5-1, Appendix A, Guidelines for Fire Protection for Nuclear Power Plants, May 1, 1976.

D. BTP-CMEB 9.5-1, Guidelines for Fire Protection for Nuclear Power Plants, July 1981.

CRITERION 4 - ENVIRONMENTAL AND MISSILE DESIGN BASES

"Structures, systems, and components important to safety shall be designed to accommodate the effects of and to be compatible with the environmental conditions associated with normal operation, maintenance, testing, and postulated accidents, including loss-of-coolant accidents (LOCAs). These structures, systems, and components shall be appropriately protected against dynamic effects, including the effects of missiles, pipe whipping, and discharging fluids, that may result from equipment failures and from events and conditions outside the nuclear power unit."

DISCUSSION

Structures, systems, and components important to safety are designed to accommodate the effects of and to be compatible with the environmental conditions associated with normal operation, maintenance, testing, and postulated accidents, including LOCAs. Criteria are presented in chapter 3, and the environmental conditions are described in section 3.11.

These structures, systems, and components are appropriately protected against dynamic effects, including the effects of missiles, pipe whipping, and discharging fluids, that may result from equipment failures and from events and conditions outside the nuclear power unit. Details of the design, environmental testing, and construction of these systems, structures, and components are included in chapters 3, 5, 6, 7, 9, and 10. Evaluation of the performance of the safety features is contained in chapter 15.

CRITERION 5 - SHARING OF STRUCTURES, SYSTEMS, AND COMPONENTS

"Structures, systems, and components important to safety shall not be shared among nuclear power units unless it can be shown that such sharing will not significantly impair their ability to perform their safety functions, including, in the event of an accident in one unit, an orderly shutdown and cooldown of the remaining unit."

DISCUSSION

The VEGP is a two-unit plant with the following common safety-related structures:

A. Control building.

B. Auxiliary building.

C. Fuel handling building.

Within these buildings are shared spaces, such as the control room, which contain physically separated safety-related equipment. A detailed description of plant structures is provided in section 3.8.

Safety-related systems are not shared, with the exception of the fuel handling building post-accident exhaust system. (See subsection 9.4.2.) Common heating, ventilation, and air-conditioning system (HVAC) ducting headers are used in some instances for redundant HVAC units. Systems or portions of systems and spaces that are shared by units 1 and 2 are listed in paragraphs 1.2.2.1 and 1.2.2.2. Where common structures, systems, and components are utilized, such sharing has been evaluated to ensure that there are no adverse impacts on safety functions.

3.1.2 PROTECTION BY MULTIPLE FISSION PRODUCT BARRIERS

CRITERION 10 - REACTOR DESIGN

"The reactor core and associated coolant, control, and protection systems shall be designed with appropriate margin to assure that specified acceptable fuel design limits are not exceeded during any condition of normal operation, including the effects of anticipated operational occurrences."

DISCUSSION

The reactor core and associated coolant, control, and protection systems are designed to the following criteria:

A. No fuel damage will occur during normal core operation and operational transients (Condition 1) or any transient conditions arising from occurrences of moderate frequency (Condition 2) beyond a small fraction of clad defects for which various aspects of the plant are designed. Fuel damage, as used here, is defined as penetration of the fission product barrier; i.e., the fuel rod clad. Conditions 1 and 2, as used here, are defined by American National Standards Institute (ANSI) N18.2-1973. The small number of clad defects that may occur are within the capability of the plant cleanup system and are consistent with the plant design bases.

B. The reactor can be returned to a safe shutdown state following a Condition 3 event with only a small fraction of the fuel rods damaged, although sufficient fuel damage might occur to preclude the immediate resumption of operation. Condition 3, as used here, is defined by ANSI N18.2-1973.

C. The core will remain intact with acceptable heat transfer geometry following transients arising from occurrences of limiting faults (Condition 4). Condition 4, as used here, is defined by ANSI N18.2-1973.

The reactor trip system is designed to actuate a reactor trip whenever necessary to ensure that the fuel design limits are not exceeded. The core design, together with the process and decay heat removal systems, provide for this capability under all expected conditions of normal operation with appropriate margins for uncertainties and anticipated transient situations, including the effects of the loss of reactor coolant flow, trip of the turbine generator, loss of normal feedwater, and loss of both normal and preferred power sources.

Chapter 4 discusses the design bases and design evaluation of core components. Details of the control and protection systems' instrumentation design and logic are discussed in chapter 7. This information supports the accident analyses of chapter 15 which show that the acceptable fuel design limits are not exceeded for Condition 1 and 2 occurrences.

CRITERION 11 - REACTOR INHERENT PROTECTION

"The reactor core and associated coolant systems shall be designed so that in the power-operating range the net effect of the prompt inherent nuclear feedback characteristics tends to compensate for a rapid increase in reactivity."

DISCUSSION

Whenever the reactor is critical, prompt compensatory reactivity feedback effects are assured by the negative fuel temperature effect (Doppler effect) and during the initial cycle, by the nonpositive operational limit on the moderator temperature coefficient of reactivity. (The moderator temperature coefficient may be slightly positive for parts of reload cycles as discussed in chapter 15.) The negative Doppler coefficient of reactivity is assured by the inherent design, using low enrichment fuel. The nonpositive moderator temperature coefficient of reactivity at full power (100 percent) is assured by administratively controlling the dissolved absorber concentration or by using burnable poison.

Reactivity coefficients and their effects are discussed in chapter 4.

CRITERION 12 - SUPPRESSION OF REACTOR POWER OSCILLATIONS

"The reactor core and associated coolant, control, and protection systems shall be designed to assure that power oscillations which can result in conditions exceeding specified acceptable fuel design limits are not possible or can be reliably and readily detected and suppressed."

DISCUSSION

Power oscillations of the fundamental mode are inherently eliminated by negative Doppler and nonpositive moderator temperature coefficient of reactivity. During parts of reload cycles the moderator temperature coefficient may be slightly positive as discussed in chapter 15.

Oscillations, due to xenon spatial effects, in the radial, diametral, and azimuthal overtone modes are heavily damped due to the inherent design and due to the negative Doppler and nonpositive moderator temperature coefficients of reactivity.

Oscillations, due to xenon spatial effects, may occur in the axial first overtone mode. Assurance that fuel design limits are not exceeded by xenon axial oscillations is provided by reactor trip functions, using the measured axial power imbalance as an input.

If necessary to maintain axial imbalance within the limits of the Core Operating Limits Report; i.e., imbalances which are alarmed to the operator and are within the imbalance trip setpoints; the operator can suppress xenon axial oscillations by control rod motions and/or temporary power reductions.

Oscillations, due to xenon spatial effects, in axial modes higher than the first overtone are heavily damped due to the inherent design and due to the negative Doppler coefficient of reactivity.

The stability of the core against xenon-induced power oscillations and the functional requirements of instrumentation for monitoring and measuring core power distribution are discussed in chapter 4. Details of the instrumentation design and logic are discussed in chapter 7.

CRITERION 13 - INSTRUMENTATION AND CONTROL

"Instrumentation shall be provided to monitor variables and systems over their anticipated ranges for normal operation, for anticipated operational occurrences, and for accident conditions as appropriate to assure adequate safety, including those variables and systems that can affect the fission process, the integrity of the reactor core, the reactor coolant pressure boundary, and the containment and its associated systems. Appropriate controls shall be provided to maintain these variables and systems within prescribed operating ranges."

DISCUSSION

Instrumentation and controls are provided to monitor and control neutron flux, control rod position, fluid temperatures, pressures, flows, and levels, as necessary, to assure that adequate plant safety can be maintained. Instrumentation is provided in the reactor coolant system, steam and power conversion system, containment, engineered safety features systems, radioactive waste management systems, and other auxiliary systems. Parameters that must be provided for operator use under normal operating and accident conditions are indicated in the control room in proximity to the controls for maintaining the indicated parameters in their proper ranges.

The quantity and types of process instrumentation provided ensure safe and orderly operation of all systems over the full design range of the plant. These systems are described in chapters 6, 7, 8, 9, 10, 11, and 12.

CRITERION 14 - REACTOR COOLANT PRESSURE BOUNDARY

"The reactor coolant pressure boundary shall be designed, fabricated, erected, and tested so as to have an extremely low probability of abnormal leakage, of rapidly propagating failure, and of gross rupture."

DISCUSSION

The reactor coolant pressure boundary (RCPB) is designed to accommodate the system pressures and temperatures attained under the expected modes of plant operation, including anticipated transients, with stresses within applicable limits. Consideration is given to loadings under normal operating conditions and to abnormal loadings, such as pipe rupture and seismic loadings, as discussed in chapter 3. The piping is protected from overpressure by means of

pressure-relieving devices, as required by American Society of Mechanical Engineers (ASME), Section III.

Reactor coolant pressure boundary materials and fabrication techniques are such that there is a low probability of gross rupture or significant leakage. (Refer to Criterion 31 for further discussion of reactor coolant pressure boundary.)

Coolant chemistry is controlled to protect from corrosion the materials of construction of the RCPB.

The RCPB welds are accessible for inservice inspections to assess the structural and leaktight integrity. The details are given in chapter 5. For the reactor vessel, a material surveillance program conforming to applicable codes is provided. Chapter 5 has additional details.

Instrumentation is provided to detect significant leakage from the RCPB with indication in the control room, as discussed in chapter 5.

CRITERION 15 - REACTOR COOLANT SYSTEM DESIGN

"The reactor coolant system and associated auxiliary, control, and protection systems shall be designed with sufficient margin to assure that the design conditions of the reactor coolant pressure boundary are not exceeded during normal operation, including anticipated operational occurrences."

DISCUSSION

Steady-state and transient analyses are performed to ensure that reactor coolant system (RCS) design conditions are not exceeded during normal operation. Protection and control setpoints are based on these analyses.

Additionally, RCPB components have a large margin of safety through application of proven materials and design codes, use of proven fabrication techniques, nondestructive shop testing, and integrated hydrostatic testing of assembled components.

The effect of radiation embrittlement is considered in reactor vessel design, and surveillance samples monitor adherence to expected conditions throughout the plant life.

Multiple safety and relief valves are provided for the RCS. These valves and their setpoints meet the ASME criteria for overpressure protection. The ASME criteria are satisfactory, based on a long history of industrial use. Chapter 5 discusses the RCS design.

CRITERION 16 - CONTAINMENT DESIGN

"The reactor containment and associated systems shall be provided to establish an essentially leak-tight barrier against the uncontrolled release of radioactivity to the environment and to assure that the containment design conditions important to safety are not exceeded for as long as postulated accident conditions require."

DISCUSSION

A steel-lined, prestressed, post-tensioned concrete containment structure encloses the entire RCS. It is designed to sustain, without loss of required integrity, the effects of LOCAs up to and including the double-ended rupture of the largest pipe in the RCS or double-ended rupture of a steam or feedwater pipe. Engineered safety features comprising the emergency core cooling system, containment spray system, and the containment air coolers serve to cool the reactor core and return the containment to near atmospheric pressure. The containment structure and

engineered safety features systems are designed to ensure the required functional capability of containing any uncontrolled release of radioactivity. The concrete radiological shielding and the liner within the containment limit the uncontrolled release of radioactivity to the environment.

Refer to chapters 3, 6, and 15.

CRITERION 17 - ELECTRICAL POWER SYSTEMS

"An onsite electric power system and an offsite electric power system shall be provided to permit the functioning of structures, systems, and components important to safety. The safety function for each system (assuming that the other system is not functioning) shall be to provide sufficient capacity and capability to assure that (1) specified acceptable fuel design limits and design conditions of the reactor coolant pressure boundary are not exceeded as a result of anticipated operational occurrences and (2) the core is cooled and containment integrity and other vital functions are maintained in the event of postulated accidents."

"The onsite electric power supplies, including the batteries, and the onsite electric distribution system shall have sufficient independence, redundancy, and testability to perform their safety functions, assuming a single failure."

"Electric power from the transmission network to the onsite electric distribution system shall be supplied by two physically independent circuits (not necessarily on separate rights-of-way) designed and located so as to minimize, to the extent practical, the likelihood of their simultaneous failure under operating and postulated accident and environmental conditions. A switchyard common to both circuits is acceptable. Each of these circuits shall be designed to be available in sufficient time, following the loss of all onsite alternating current power sources and the offsite electric power circuit, to assure that specified acceptable fuel design limits and design conditions of the reactor coolant pressure boundary are not exceeded. One of these circuits shall be designed to be available within a few seconds following a LOCA to assure that core cooling, containment integrity, and other vital safety functions are maintained."

"Provisions shall be included to minimize the probability of losing electric power from any of the remaining supplies as a result of, or coincident with, the loss of power generated by the nuclear power unit, the loss of power from the transmission network, or the loss of power from the onsite electric power supplies."

DISCUSSION

An onsite electric power system and an offsite electric power system are provided to permit the functioning of structures, systems, and components important to safety. As discussed in chapter 8, each Class 1E electric power system is designed with adequate independence, capacity, redundancy, and testability to ensure the functioning of engineered safety features (ESF). Independence is provided by physical separation and electrical isolation of components and cables.

The onsite ac power system includes a Class 1E system and a non-Class 1E system. Onsite ac power is supplied from either the 230-kV switchyard through reserve auxiliary transformers (which feed the non-Class 1E and Class 1E buses), or from a 13.8-kV underground circuit emanating from Georgia Power Company Plant Wilson switchyard (which connects through the standby auxiliary transformer (SAT) located in the Vogtle low voltage switchyard) to a Class 1E 4.16-kV bus and some non-Class 1E 4.16-kV loads. The Class 1E ac power system is the power source used in (or associated with) shutting down the reactor and preventing or limiting the release of radioactive material following a design basis event. The system is divided into

two independent ac power trains, train A and train B, each fed from an independent Class 1E bus.

Each Class 1E bus is provided with two (normal and alternate) offsite preferred power sources and a standby onsite power source. With at least two offsite sources available per Unit, each Class 1E bus is supplied from a separate reserve auxiliary transformer. However, one Class 1E bus may be directly connected to the SAT instead of a RAT for its offsite preferred power source feed. This direct connection is performed under administrative controls and with the use of key interlocked disconnect switches.

The Class 1E ac system distributes power to all safety-related loads. Also, the Class 1E ac system supplies power to certain selected loads which are not safety related but are important to the plant operation; however, these loads are tripped when a safety injection signal is received.

The non-Class 1E ac system supplies preferred (offsite) power to the Class 1E ac system through the reserve auxiliary transformer 4160-V windings. Each reserve auxiliary transformer has the capacity to supply all connected non-Class 1E running loads and to start and run the loads of one Class 1E train. The SAT may also supply preferred (offsite) power to either one of the Class 1E 4160 volt buses. The SAT has the capacity to start and run the loads of one Class 1E train, and to supply some additional non Class 1E loads. Non Class 1E loading of the SAT is administratively controlled.

A failure of a single component will not prevent the safety-related systems from performing their function. Each of the connected preferred offsite power circuits is designed to be available in sufficient time, following a loss of all onsite power sources and the other offsite electric power circuit, to assure that specified acceptable fuel design limits and design conditions of the reactor coolant pressure boundary are not exceeded.

Emergency onsite ac power is furnished by two diesel generators per unit. Each diesel generator is connected to a Class 1E bus. The ESF loads are divided between the Class 1E buses in balanced, redundant load groupings. Each diesel generator is capable of supplying sufficient power in sufficient time for the operation of the ESF required for the unit during a postulated LOCA. During a postulated LOCA, both diesel generators start automatically. If preferred power is available to the Class 1E bus following a LOCA, the ESF loads will be started sequentially. However, in the event that preferred power is lost, the load sequencing system will shed all loads, connect each diesel generator to its associated Class 1E bus, and sequentially start the ESF equipment. The diesel generators are arranged so that a failure of a single component will not prevent the safe shutdown of the reactor. The onsite Class 1E dc power supply consists of four independent battery systems. Failure of a single component in the dc power supply will not impair function of the ESF required to maintain the reactor in a safe condition.

CRITERION 18 - INSPECTION AND TESTING OF ELECTRIC POWER SYSTEMS

"Electric power systems important to safety shall be designed to permit appropriate periodic inspection and testing of important areas and features, such as wiring, insulation, connections, and switchboards, to assess the continuity of the systems and the condition of their components. The systems shall be designed with a capability to test periodically (1) the operability and functional performance of the components of the systems, such as onsite power sources, relays, switches, and buses, and (2) the operability of the systems as a whole and, under conditions as close to design as practical, the full operation sequence that brings the systems into operation, including operation of applicable portions of the protection system, and

the transfer of power among the nuclear power unit, the offsite power system, and the onsite power system."

DISCUSSION

Class 1E electric power systems are designed as described below in order that the following aspects of the system can be periodically tested:

A. The operability and functional performance of the components of Class 1E electric power systems (diesel generators, ESF buses, dc system).

B. The operability of these electric power systems as a whole and under conditions as close to design as practical, including the full operational sequence that actuates these systems.

The non-Class 1E switchyard circuit breakers will be inspected, maintained, and tested on a routine basis without affecting the rest of the system. Protective relaying will be periodically tested, and transmission lines will be periodically inspected.

Any one of the non-Class 1E reserve auxiliary transformers or the standby auxiliary transformer, and its circuit to the Class 1E buses can be taken out of service and tested periodically. Each transformer includes the capacity to supply power to one train of Class 1E loads. The 4160-V and 480-V circuit breakers and the associated equipment will be tested one at a time only while redundant equipment is operational.

The dc system is provided with detectors to indicate and alarm when there is a ground existing on any part of the system. During plant operation, normal maintenance may be performed.

Provisions for the testing of Class 1E ac electric power systems, Class 1E dc power systems, and the standby power supplies (diesel generators) are described in chapter 8.

CRITERION 19 - CONTROL ROOM

"A control room shall be provided from which actions can be taken to operate the nuclear power unit safely under normal conditions and to maintain it in a safe condition under accident conditions, including LOCAs. Adequate radiation protection shall be provided to permit access and occupancy of the control room under accident conditions without personnel receiving radiation exposures in excess of 5 rem whole body, or its equivalent, to any part of the body, for the duration of the accident.

"Equipment at appropriate locations outside the control room shall be provided (1) with a design capability for prompt hot shutdown of the reactor, including necessary instrumentation and controls to maintain the unit in a safe condition during hot shutdown and (2) with a potential capability for subsequent cold shutdown of the reactor through the use of suitable procedures."

DISCUSSION

A shared control room is provided for the control of the VEGP units from which actions can be taken to operate each nuclear power unit safely under normal conditions and to maintain it in a safe manner under accident conditions, including LOCAs. Operator action outside of the control room to mitigate the consequences of an accident is permitted. The control room and its post-accident ventilation systems are designed to satisfy Seismic Category 1 requirements, as discussed in chapter 3. Adequate concrete shielding and radiation protection are provided against direct gamma radiation and inhalation doses resulting from a postulated release of fission products inside the containment structure based on the assumptions contained in

Regulatory Guide 1.4. The shielding and the control room standby air-conditioning system allow access to and occupancy of the control rooms under accident conditions without personnel receiving radiation exposures in excess of 5 rem whole body or its equivalent to any part of the body for the duration of the accident. (Refer to chapter 15.) Fission product removal is provided in the control room recirculation equipment to remove iodine and particulate matter, thereby minimizing the thyroid dose which could result from the accident. The control room habitability features are described in chapter 6.

In the event that the operators are forced to abandon the control room, panel-mounted instrumentation and controls are provided on the train-related shutdown panels to achieve and maintain the plant in the safe shutdown condition. (See section 7.4.)

3.1.3 PROTECTION AND REACTIVITY CONTROL SYSTEMS

CRITERION 20 - PROTECTION SYSTEM FUNCTIONS

"The protection system shall be designed (1) to initiate automatically the operation of appropriate systems, including the reactivity control systems, to assure that specified acceptable fuel design limits are not exceeded as a result of anticipated operational occurrences and (2) to sense accident conditions and to initiate the operation of systems and components important to safety."

DISCUSSION

A fully automatic protection system with appropriate redundant channels is provided to cope with transient events where insufficient time is available for manual corrective action. The design basis for all protection systems is in accordance with the guidelines of Institute of Electrical and Electronic Engineers (IEEE) Standards 279-1971 and 379-1972. The reactor protection system automatically initiates a reactor trip when any variable monitored by the system or combination of monitored variables exceeds the normal operating range. Setpoints are designed to provide an envelope of safe operating conditions with adequate margin for uncertainties to ensure that the fuel design limits are not exceeded.

Reactor trip is initiated by removing power to the rod drive mechanisms of all the rod cluster control assemblies. This causes the rods to insert by gravity, thus rapidly reducing the reactor power. The response and adequacy of the protection system have been verified by analysis of anticipated transients.

The ESF actuation system automatically initiates emergency core cooling and other safety functions by sensing accident conditions, using redundant analog channels measuring diverse variables. Manual actuation of safety features may be performed where ample time is available for operator action. The ESF actuation system automatically trips the reactor on a manual or automatic safety injection signal.

CRITERION 21 - PROTECTION SYSTEM RELIABILITY AND TESTABILITY

"The protection system shall be designed for high functional reliability and inservice testability commensurate with the safety functions to be performed. Redundancy and independence designed into the protection system shall be sufficient to assure that (1) no single failure results in the loss of the protection function and (2) removal from service of any component or channel does not result in the loss of the required minimum redundancy unless the acceptable reliability of operation of the protection system can be otherwise demonstrated. The protection system

shall be designed to permit periodic testing of its functioning when the reactor is in operation, including a capability to test channels independently to determine failures and losses of redundancy that may have occurred."

DISCUSSION

The protection system is designed for functional reliability and inservice testability. The design employs redundant logic trains and measurement and equipment diversity.

The protection system, including the ESF test cabinet, is designed to meet Regulatory Guide 1.22 and conform to the requirements of IEEE Standards 279-1971 and 379-1972. Functions that cannot be tested with the reactor at power are tested during shutdown, as allowed by the regulatory guide and the above standards.

In cases where actuated equipment cannot be tested at power, the channels and logic associated with this equipment, up to the final actuation device, have the capability for testing at power. Such testing discloses failures of reduction in redundancy which may have occurred.

Removal from service of any single channel or component does not result in the loss of minimum required redundancy. For example, a two-of-three function is placed in the one-of-two mode when one channel is removed. (Note that distinction is made between channels and trains in this discussion. A train may be removed from service only during testing.) Bypassed and inoperable status indication for safety-related systems is provided in accordance with Regulatory Guide 1.47.

Semiautomatic testers are built into each of the two logic trains of the protection system. These testers have the capability of testing the system logic very rapidly while the reactor is at power. A self-testing provision is designed into each tester. (For a detailed description of reliability and testability of the protection system, refer to section 7.2.)

CRITERION 22 - PROTECTION SYSTEM INDEPENDENCE

"The protection system shall be designed to assure that the effects of natural phenomena, and of normal operating, maintenance, testing, and postulated accident conditions on redundant channels do not result in the loss of the protection function or shall be demonstrated to be acceptable on some other defined basis. Design techniques, such as functional diversity or diversity in component design and principles of operation, shall be used to the extent practical to prevent loss of the protection function."

DISCUSSION

Design of the protection systems includes consideration of natural phenomena, normal maintenance, testing, and accident conditions so that the protection functions are always available.

Protection system components are designed, arranged, and qualified for operation in the environment accompanying any emergency situation in which the components are required to function.

Functional diversity has been designed into the system. The extent of this functional diversity has been evaluated for a variety of postulated accidents. Diverse protection functions will automatically terminate an accident before intolerable consequences can occur.

Sufficient redundancy and independence are designed into the protection systems to assure that no single failure or removal from service of any component or channel of a system would result in loss of the protection function. Functional diversity and consequential location diversity

are designed into the system. Automatic reactor trips are based upon neutron flux measurements, reactor coolant loop temperature measurements, pressurizer pressure and level measurements, and reactor coolant pump power supply underfrequency, undervoltage measurements, and other parameters. Trips may also be initiated manually or by a safety injection signal. See chapter 7 for details.

High quality components, conservative design and applicable quality control, inspection, calibration, and tests are utilized to guard against common-mode failure. Qualification testing and analysis is performed on the various safety systems to demonstrate functional operation at normal and post-accident conditions of temperature, humidity, pressure, and radiation for specified periods, if required. Typical protection system equipment is subjected to type tests under simulated seismic conditions, using conservatively large accelerations and applicable frequencies. The test results indicate no loss of the protection function. (Refer to sections 3.10.B, 3.10.N, 3.11.B, and 3.11.N for further details.)

CRITERION 23 - PROTECTION SYSTEM FAILURE MODES

"The protection system shall be designed to fail into a safe state or into a state demonstrated to be acceptable on some other defined basis if conditions such as disconnection of the system, loss of energy (e.g., electric power, instrument air) or postulated adverse environments (e.g., extreme heat or cold, fire, pressure, steam, water, and radiation) are experienced."

DISCUSSION

The protection system is designed with consideration of the most probable failure modes of the components under various perturbations of the environment and energy sources. Each reactor trip channel, with the exception of the reactor coolant pump (RCP) underfrequency (UF) and undervoltage (UV) trip input to the reactor trip system, is designed on the deenergize-to-trip principle so loss of power, disconnection, open channel faults, and the majority of the internal channel short circuit faults cause the channel to go into its tripped mode. The RCP UF and UV relay logic input to the reactor trip system is energize-to-trip. This meets the requirements of Criterion 23 due to the redundancy and train separation of the RCP UF and UV trip channels. Failure of one channel will not prevent a reactor trip in the event of an actual UF or UV event.

Similarly, that portion of the ESF actuation system provided for actuation of auxiliary feedwater system and containment ventilation isolation is designed to fail into a safe state, except for the final output relays. The relays are energized to actuate, as are the pumps and motor-operated valves of the actuated equipment.

For a more detailed description of the protection system, refer to chapter 7.

CRITERION 24 - SEPARATION OF PROTECTION AND CONTROL SYSTEMS

"The protection system shall be separated from the control systems to the extent that failure of any single control system component or channel, or failure or removal from service of any single protection system component or channel which is common to the control and protection systems, leaves intact a system satisfying all reliability, redundancy, and independence requirements of the protection system. Interconnection of the protection and control systems shall be limited so as to assure that safety is not significantly impaired."

DISCUSSION

The protection system is separate and distinct from the control systems, as described in chapter 7. Control systems are, in some cases, dependent on the protection system in that

control signals are derived from protection system measurements, where applicable. These signals are transferred to the control system by isolation devices which are classified as protection components. The adequacy of the system isolation has been verified by testing under conditions of postulated credible faults. The failure of any single control system component or channel, or the failure or removal from service of any single protection system component or channel which is common to the control and protection system, leaves intact a system which satisfies the requirements of the protection system. The removal of a train from service is allowed only during testing of the train. Distinction between channel and train is made in the discussions.

CRITERION 25 - PROTECTION SYSTEM REQUIREMENTS FOR REACTIVITY CONTROL MALFUNCTIONS

"The protection system shall be designed to assure that specified acceptable fuel design limits are not exceeded for any single malfunction of the reactivity control systems, such as accidental withdrawal (not ejection or dropout) of the control rods."

DISCUSSION

The protection system is designed to limit reactivity transients so that the fuel design limits are not exceeded. Reactor shutdown by control rod insertion is completely independent of the normal control function since the trip breakers interrupt power to the rod mechanisms regardless of existing control signals. Thus, in the postulated accidental withdrawal of a control rod or control rod bank (assumed to be initiated by a control malfunction) neutron flux, temperature, pressure, level, and flow signals would be generated independently. Any of these signals (trip demands) would operate the breakers to trip the reactor.

Analyses of the effects of possible malfunctions are discussed in chapter 15. These analyses show that for postulated boron dilution during refueling, startup, or manual or automatic operation at power, the operator has ample time to determine the cause of dilution, terminate the source of dilution, and initiate reboration before the shutdown margin is lost. The analyses show that acceptable fuel damage limits are not exceeded even in the event of a single malfunction of either system.

CRITERION 26 - REACTIVITY CONTROL SYSTEM REDUNDANCY AND CAPABILITY

"Two independent reactivity control systems of different design principles shall be provided. One of the systems shall use control rods, preferably including a positive means for inserting the rods, and shall be capable of reliably controlling reactivity changes to assure that under conditions of normal operation, including anticipated operational occurrences, and with appropriate margin for malfunctions such as stuck rods, specified acceptable fuel design limits are not exceeded. The second reactivity control system shall be capable of reliably controlling the rate of reactivity changes resulting from planned, normal power changes (including xenon burnout) to assure that the acceptable fuel design limits are not exceeded. One of the systems shall be capable of holding the reactor core subcritical under cold conditions."

DISCUSSION

Two reactivity control systems are provided. These are rod cluster control assemblies (RCCAs) and chemical shim (boric acid). The RCCAs are inserted into the core by the force of gravity.

During operation, the shutdown rod banks are fully withdrawn. The control rod system automatically maintains a programmed average reactor temperature compensating for reactivity

effects associated with scheduled and transient load changes. The shutdown rod banks, along with the control banks, are designed to shut down the reactor with adequate margin under conditions of normal operation and anticipated operational occurrences, thereby ensuring that specified fuel design limits are not exceeded. The most restrictive period in the core life is assumed in all analyses, and the most reactive rod cluster is assumed to be in the fully withdrawn position.

The boron system will maintain the reactor in the cold shutdown state independent of the position of the control rods and can compensate for xenon burnout transients.

Details of the construction of the RCCAs are presented in chapter 4, and the operation is discussed in chapter 7. The means of controlling the boric acid concentration is described in chapter 9. Performance analyses under accident conditions are included in chapter 15.

CRITERION 27 - COMBINED REACTIVITY CONTROL SYSTEMS CAPABILITY

"The reactivity control systems shall be designed to have a combined capability, in conjunction with poison addition by the emergency core cooling system, of reliably controlling reactivity changes to assure that under postulated accident conditions and with appropriate margin for stuck rods the capability to cool the core is maintained."

DISCUSSION

The facility is provided with means of making and holding the core subcritical under any anticipated conditions and with appropriate margin for contingencies. These means are discussed in detail in chapters 4 and 9. Combined use of the rod cluster control system and the chemical shim control system permits the necessary shutdown margin to be maintained during long-term xenon decay and plant cooldown. The single highest worth RCCA is assumed to be stuck full out upon trip for this determination.

CRITERION 28 - REACTIVITY LIMITS

"The reactivity control systems shall be designed with appropriate limits on the potential amount and rate of reactivity increase to assure that the effects of postulated reactivity accidents can neither (1) result in damage to the reactor coolant pressure boundary greater than limited local yielding nor (2) sufficiently disturb the core, its support structures, or other reactor pressure vessel internals to impair significantly the capability to cool the core. These postulated reactivity accidents shall include consideration of rod ejection (unless prevented by positive means), rod dropout, steam line rupture, changes in reactor coolant temperature and pressure, and cold water addition."

DISCUSSION

The maximum reactivity worth of the control rods and the maximum rates of reactivity insertion employing control rods and boron removal are limited to values that prevent any reactivity increase from rupturing the RCS boundary or disrupting the core or vessel internals to a degree that could impair the effectiveness of emergency core cooling.

The appropriate reactivity insertion rate for the withdrawal of RCCAs and the dilution of the boric acid in the reactor coolant systems are limited by design. The reactor protection system and engineered safety features actuation system provide protection for such events as a rod ejection accident and steamline break. Protection system setpoints are contained in the Technical Specifications. Reactivity insertion rates, dilution, and withdrawal limits are also discussed in chapter 4. The capability of the chemical and volume control system to avoid an

inadvertent excessive rate of boron dilution is discussed in chapter 9. The relationship of the reactivity insertion rates to plant safety is discussed in chapter 15.

Core cooling capability following accidents, such as rod ejection, steam line break, etc., is assured by keeping the reactor coolant pressure boundary stresses within faulted condition limits, as specified by applicable ASME codes. Structural deformations are also checked and limited to values that do not jeopardize the operation of needed safety features.

CRITERION 29 - PROTECTION AGAINST ANTICIPATED OPERATIONAL OCCURRENCES

"The protection and reactivity control systems shall be designed to assure an extremely high probability of accomplishing their safety functions in the event of anticipated operational occurrences."

DISCUSSION

The protection and reactivity control systems have an extremely high probability of performing their required safety functions in any anticipated operational occurrences. Diversity and redundancy, coupled with a quality assurance program and analyses, support this probability as does operating experience in plants using the same basic design. Failure modes of system components are designed to be safe modes. Loss of power to the protection system results in a reactor trip. Details of system design are covered in chapters 4 and 7.

3.1.4 FLUID SYSTEMS

CRITERION 30 - QUALITY OF REACTOR COOLANT PRESSURE BOUNDARY

"Components which are part of the reactor coolant pressure boundary shall be designed, fabricated, erected, and tested to the highest quality standards practical. Means shall be provided for detecting and, to the extent practical, identifying the location of the source of reactor coolant leakage."

DISCUSSION

All RCS components are designed, fabricated, inspected, and tested in conformance with the ASME Boiler and Pressure Vessel Code, Section III.

All balance of plant components are classified according to Regulatory Guide 1.26, and all nuclear steam supply system (NSSS) components are classified according to ANSI N18.2-1973 and ANSI N18.2A-1975 (which is an acceptable alternative to Regulatory Guide 1.26) and are accorded all the quality measures appropriate to these classifications. The design bases and evaluations of the RCS are discussed in chapter 5.

A number of methods are available for detecting reactor coolant leakage. The reactor vessel closure joint is provided with a temperature monitored leakoff between double gaskets. Leakage inside the reactor containment is drained to the containment building and reactor cavity sumps, where the level is monitored. Leakage is also detected by measuring the airborne activity and humidity of the containment. Monitoring the inventory of reactor coolant in the system at the pressurizer, volume control tank, and reactor coolant drain tank provides an accurate indication of integrated leakage. Refer to chapter 5 for complete description of the RCPB leakage detection system.

CRITERION 31 - FRACTURE PREVENTION OF REACTOR COOLANT PRESSURE BOUNDARY

"The reactor coolant pressure boundary shall be designed with sufficient margin to assure that when stressed under operating, maintenance, testing, and postulated accident conditions (1) the boundary behaves in a nonbrittle manner and (2) the probability of rapidly propagating fracture is minimized. The design shall reflect consideration of service temperatures and other conditions of the boundary material under operating, maintenance, testing, and postulated accident conditions and the uncertainties in determining (1) material properties, (2) the effects of irradiation on material properties, (3) residual, steady state, and transient stresses, and (4) size of flaws."

DISCUSSION

Close control is maintained over material selection and fabrication for the RCS to assure that the boundary behaves in a nonbrittle manner. The RCS materials which are exposed to the coolant are corrosion-resistant stainless steel or Inconel. The nil ductility transition reference temperature (RT_{NDT}) of the reactor vessel structural steel is established by Charpy V-notch and drop weight tests in accordance with 10 CFR 50, Appendix G, Fracture Toughness Requirements.

The reactor vessel specification imposes the following requirements which are not specified by the ASME code.

A. The performance of a 100-percent volumetric ultrasonic shear wave test of reactor vessel plate and a post hydro-test ultrasonic map of all welds in the pressure vessel are required. Cladding bond ultrasonic inspection to more restrictive requirements than those specified in the code is also required to preclude interpretation problems during inservice inspection.

B. In the surveillance programs, the evaluation of the radiation damage is based on pre-irradiation testing of Charpy V-notch and tensile specimens and postirradiation testing of Charpy V-notch, tensile, and 1/2 T compact tension specimens. These programs are directed toward evaluation of the effect of radiation on the fracture toughness of reactor vessel steels based on the reference transition temperature approach and the fracture mechanics approach, and are in accordance with American Society of Testing Materials (ASTM) E-185-82, Standard Recommended Practice for Surveillance Tests for Nuclear Reactor Vessels, and the requirements of 10 CFR 50, Appendix H, Reactor Vessel Material Surveillance Program Requirements.

C. Reactor vessel core region material chemistry (copper, phosphorous, and vanadium) is controlled to reduce sensitivity to embrittlement due to irradiation over the life of the plant.

The fabrication and quality control techniques used in the fabrication of the RCS are equivalent to those used for the reactor vessel. The inspections of reactor vessel, pressurizer, piping, pumps, and steam generators are governed by ASME code requirements. (Refer to chapter 5 for details.)

Allowable pressure-temperature relationships for plant heatup and cooldown rates are calculated, using methods derived from the ASME Code, Section III, Appendix G, Protection Against NonDuctile Failure. The approach specifies that the allowable stress intensity factors for all vessel operating conditions do not exceed the reference stress intensity factor (KIR) for

the metal temperature at any time. Operating specifications include conservative margins for predicted changes in the material reference temperatures (RT_{NDT}) due to irradiation.

CRITERION 32 - INSPECTION OF REACTOR COOLANT PRESSURE BOUNDARY

"Components which are part of the reactor coolant pressure boundary shall be designed to permit (1) periodic inspection and testing of important areas and features to assess their structural and leak-tight integrity and (2) an appropriate material surveillance program for the reactor pressure vessel."

DISCUSSION

The design of the RCPB provides accessibility to the entire internal surfaces of the reactor vessel and most external zones of the vessel, including the nozzle to reactor coolant piping welds, the top and bottom heads, and external surfaces of the reactor coolant piping, except for the area of pipe within the primary shielding concrete. The inspection capability complements the leakage detection systems in assessing the pressure boundary components' integrity. The RCPB will be periodically inspected under the provisions of the ASME Code, Section XI.

Monitoring of changes in the fracture toughness properties of the reactor vessel core region plates, forgings, weldments, and associated heat-treated zones is performed in accordance with 10 CFR 50, Appendix H. Samples of reactor vessel plate materials are retained and catalogued in case future engineering development shows the need for further testing.

The material properties surveillance program includes not only the conventional tensile and impact tests, but also fracture mechanics specimens. The observed shifts in RT_{NDT} of the core region materials with irradiation will be used to confirm the allowable limits calculated for all operational transients.

The design of the RCPB piping provides for accessibility of all welds requiring inservice inspection under the provisions of the ASME Code, Section XI. Removable insulation is provided at all welds requiring inservice inspection. The inservice inspection program is discussed in detail in chapter 6.

CRITERION 33 - REACTOR COOLANT MAKEUP

"A system to supply reactor coolant makeup for protection against small breaks in the reactor coolant pressure boundary shall be provided. The system safety function shall be to assure that specified acceptable fuel design limits are not exceeded as a result of reactor coolant loss due to leakage from the reactor coolant pressure boundary and rupture of small piping or other small components which are part of the boundary. The system shall be designed to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished using the piping, pumps, and valves used to maintain coolant inventory during normal reactor operation."

DISCUSSION

The chemical and volume control system provides a means of reactor coolant makeup and adjustment of the boric acid concentration. Makeup is added automatically if the level in the volume control tank falls below a preset level. The normal charging pump or centrifugal charging pump is used as the normal means of reactor coolant makeup. This pump is powered from the offsite power system.

The centrifugal charging pumps are a backup method of providing reactor coolant makeup. The centrifugal charging pumps are capable of supplying the required makeup and reactor coolant seal injection flow when power is available from either onsite or offsite electric power systems. Functional reliability is assured by provision of standby components assuring a safe response to probable modes of failure. Details of system design, including descriptions of the effects of small piping and component ruptures, are provided in sections 6.3 and 9.3 and in chapter 15; details of the electric power system are included in chapter 8.

CRITERION 34 - RESIDUAL HEAT REMOVAL

"A system to remove residual heat shall be provided. The system safety function shall be to transfer fission product decay heat and other residual heat from the reactor core at a rate such that specified acceptable fuel design limits and the design conditions of the reactor coolant pressure boundary are not exceeded.

"Suitable redundancy in components and features and suitable interconnections, leak detection, and isolation capabilities shall be provided to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure."

DISCUSSION

The residual heat removal (RHR) system, in conjunction with the steam and power conversion system, is designed to transfer the fission product decay heat and other residual heat from the reactor core at a rate which keeps the fuel within acceptable limits. The RHR system functions when temperature and pressure are below approximately 350°F and 400 psig, respectively.

Redundancy of the RHR system is provided by two residual heat removal pumps (located in separate flood-proof compartments, with means available for draining and monitoring leakage), two heat exchangers, and associated piping, cabling, and electric power sources. (For a more detailed description of RHR system redundancy, refer to subsection 5.4.7.) The RHR system is able to operate on either the onsite or offsite electrical power system.

Redundancy of heat removal at temperatures above approximately 350°F is provided by the four steam generators, four atmospheric relief valves, and the auxiliary feedwater system.

Details of the system design are provided in subsection 5.4.7, chapter 9, and chapter 10.

CRITERION 35 - EMERGENCY CORE COOLING

"A system to provide abundant emergency core cooling shall be provided. The system safety function shall be to transfer heat from the reactor core following any loss of reactor coolant at a rate such that (1) fuel and clad damage that could interfere with continued effective core cooling is prevented and (2) clad metal-water reaction is limited to negligible amounts.

"Suitable redundancy in components and features and suitable interconnections, leak detection, isolation, and containment capabilities shall be provided to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure."

DISCUSSION

The emergency core cooling system (ECCS) has the capability to mitigate the effects of any LOCA within the design bases. Cooling water is provided in an emergency to transfer heat from the core at a rate sufficient to maintain the core in a coolable geometry and to assure that clad metal-water reaction is limited to less than 1 percent. Design provisions assure performance of the required safety functions even with a postulated single failure.

Emergency core cooling is provided even if there should be a failure of any component in the system. A passive system of four accumulators which do not require any external signals or source of power to operate provide the short-term cooling requirements for large reactor coolant pipe system breaks. Three independent and redundant pumping systems are provided: the charging system, safety injection system, and residual heat removal system. The charging system is a high-pressure, low-flow system capable of providing the required emergency cooling for small breaks. The safety injection system is an intermediate-pressure, intermediate-flow system capable of providing the required emergency cooling for medium-sized breaks. The charging system can be operated to complement the safety injection system. The RHR system is a low-pressure, high-flow system capable of providing the required emergency cooling for large breaks. The charging system and safety injection system can be operated to complement the RHR system. These systems are arranged so that the single failure of any active component does not interfere with meeting the short-term cooling requirements.

The primary function of the ECCS is to deliver borated cooling water to the reactor core in the event of a LOCA. This limits the fuel-clad temperature; ensures that the core will remain intact and in place, with its essential heat transfer geometry preserved; and prevents a return to criticality. This protection is afforded for:

A. All pipe breaks sizes up to and including the hypothetical circumferential rupture of the largest pipe of a reactor coolant loop.

B. A loss-of-coolant associated with a rod ejection accident.

The ECCS is described in chapter 6. The LOCA, including an evaluation of consequences, is discussed in chapter 15.

CRITERION 36 - INSPECTION OF EMERGENCY CORE COOLING SYSTEM

"The emergency core cooling system shall be designed to permit appropriate periodic inspection of important components, such as spray rings in the reactor pressure vessel, water injection nozzles, and piping, to assure the integrity and capability of the system."

DISCUSSION

The ECCS is accessible for visual inspection and for nondestructive inservice inspection, to satisfy the ASME Code, Section XI.

Components outside the containment are accessible for leaktightness inspection during operation of the reactor.

Details of the inspection program for the ECCS are discussed in section 6.3, the Inservice Inspection Program, and the Technical Specifications.

CRITERION 37 - TESTING OF EMERGENCY CORE COOLING SYSTEM

"The emergency core cooling system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leak-tight integrity of its components, (2) the operability and performance of the active components of the system, and (3) the operability of the system as a whole and under conditions as close to design as practical, the performance of the full operational sequence that brings the system into operation, including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of the associated cooling water system."

DISCUSSION

The design of the ECCS permits periodic testing of both active and passive components of the ECCS.

Preoperational performance tests of the ECCS components are performed by the manufacturer. Initial system hydrostatic and functional flow tests demonstrate structural and leaktight integrity of components and proper functioning of the system. Thereafter, periodic tests demonstrate that components are functioning properly.

Each active component of the ECCS may be individually operated on the normal power source or transferred to standby power sources at any time during normal plant operation to demonstrate operability. The centrifugal charging pumps are available for operation as necessary during plant operation. The test of the safety injection pumps employs the minimum flow recirculation test line which connects back to the refueling water storage tank. Remote-operated valves are exercised and actuation circuits tested. The automatic actuation circuitry, valves, and pump breakers may be checked during integrated system tests performed during a planned cooldown of the RCS.

Design provisions include special instrumentation, testing, and sampling lines to perform the tests during plant shutdown to demonstrate proper automatic operation of the ECCS. (Refer to section 1.9 for a discussion of Regulatory Guide 1.22.) A test signal is applied to initiate automatic action, and verification is made that the safety injection pumps attain required discharge heads. The test demonstrates the operation of the valves, pump circuit breakers, and automatic circuitry. In addition, the periodic recirculation to the refueling water storage tank can verify the ECCS delivery capability. This recirculation test includes all but the last valve, which connects to the reactor coolant piping.

The design provides for capability to test initially, to the extent practical, the full operational sequence up to the design conditions, including transfer to alternate power sources for the ECCS to demonstrate the state of readiness and capability of the system. This functional test is performed with the water level below the safety injection signal setpoint in the pressurizer and with the RCS initially cold and depressurized. The ECCS valving is set to initially simulate the system alignment for plant power operation.

Details of the ECCS are found in chapter 6. Performance under accident conditions is evaluated in chapter 15. Surveillance requirements are identified in the Technical Specifications.

CRITERION 38 - CONTAINMENT HEAT REMOVAL SYSTEM

"A system to remove heat from the reactor containment shall be provided. The system safety function shall be to reduce rapidly, consistent with the functioning of other associated systems, the containment pressure and temperature following any LOCA and maintain them at acceptably low levels.

"Suitable redundancy in components and features and suitable interconnections, leak detection, isolation, and containment capabilities shall be provided to assure that for onsite electrical power system operation (assuming offsite power is not available) and for offsite electrical power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure."

DISCUSSION

The containment spray and containment fan cooler systems, in conjunction with the ECCS, are capable of removing sufficient energy and subsequent decay energy from the containment following the hypothesized LOCA to maintain the containment pressure below the containment design pressure. During the post-accident injection phase, water for the containment spray system and ECCS is drawn from the refueling water storage tank. During the later recirculation phase, spray water and ECCS water are pumped from the containment sump.

Each of the containment spray and containment fan cooler systems consists of two independent subsystems supplied from separate Class 1E power buses. No single failure, including loss of onsite or offsite electrical power, can cause loss of more than half of the installed 200-percent cooling capacity. The containment spray system and containment fan coolers are discussed in chapter 6. Electrical facilities are described in chapter 8. A containment pressure and temperature analysis following a LOCA is given in chapter 6, with additional results found in chapter 15.

CRITERION 39 - INSPECTION OF CONTAINMENT HEAT REMOVAL SYSTEM

"The containment heat removal system shall be designed to permit appropriate periodic inspection of important components, such as the torus, sumps, spray nozzles and piping, to assure the integrity and capability of the system."

DISCUSSION

The essential equipment of the containment spray system (CSS) is outside the containment, except for risers, distribution header piping, spray nozzles, and the containment sumps. The containment sumps, spray piping, and nozzles can be inspected during shutdown. Portions of the containment spray suction piping and the RHR suction piping from the containment recirculation sumps are embedded in concrete and are not accessible for inspection. Associated equipment outside the containment can be visually inspected.

The containment air coolers and associated cooling water system piping inside the containment can be inspected during shutdowns.

These periodic inspections assure that the capability of these heat removal systems as specified in the Technical Specifications is met.

(For details on the containment air coolers and containment spray system, see chapter 6.)

CRITERION 40 - TESTING OF CONTAINMENT HEAT REMOVAL SYSTEM

"The containment heat removal system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leak-tight integrity of its components, (2) the operability and performance of the active components of the system, and (3) the operability of the system as a whole, and, under conditions as close to the design as practical, the performance of the full operational sequence that brings the system into operation,

including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of the associated cooling water system."

DISCUSSION

The containment spray system and the containment fan cooling system are designed to permit periodic testing to assure the structural and leaktight integrity of their components and to assure the operability and performance of the active components of the systems. All active components of the CSS and delivery piping up to the last powered valve before the spray nozzle have the capability to be tested during reactor power operation. In addition, when the unit is shut down, smoke or air can be blown through the test connections for visual verification of the flow path. All safety-related active components of the containment fan cooling system can be tested to verify operability during reactor power operation. In addition, since the containment fan cooling system is a normally operating system, the performance and operability of portions of the system are continuously verified during normal reactor power operation. The facility design allows, under conditions as close to the design as practicable, the performance of a full operational sequence that brings these systems into operation. More complete discussions of the testing of these systems are in chapters 6 and 8 and the Technical Specifications.

CRITERION 41 - CONTAINMENT ATMOSPHERE CLEANUP

"Systems to control fission products, hydrogen, oxygen, and other substances which may be released into the reactor containment shall be provided, as necessary, to reduce, consistent with the functioning of other associated systems, the concentration and quantity of fission products released to the environment following postulated accidents and to control the concentration of hydrogen or oxygen and other substances in the containment atmosphere following postulated accidents to assure that containment integrity is maintained.

"Each system shall have suitable redundancy in components and features and suitable interconnections, leak detection, isolation, and containment capabilities to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) its safety function can be accomplished, assuming a single failure."

DISCUSSION

The CSS serves to remove radioiodine and other airborne particulate fission products from the containment atmosphere following a LOCA. The system consists of two independent systems, each supplied from separate electrical power buses, as described in chapter 8. Either subsystem alone can provide the fission product removal capacity for which credit is taken in chapter 15, in conformance with Regulatory Guide 1.4.

The generation of hydrogen in the containment under post-accident conditions has been evaluated, using the assumptions of Regulatory Guide 1.7. (See chapter 6.) A post-accident hydrogen recombiner system is provided with redundancy of vital components so that a single failure does not prevent timely operation of the system. This system is described in subsection 6.2.5. The post-LOCA purge exhaust system is provided as a backup. No single failure causes both subsystems to fail to operate.

CRITERION 42 - INSPECTION OF CONTAINMENT ATMOSPHERE CLEANUP SYSTEM

"The containment atmosphere cleanup systems shall be designed to permit appropriate periodic inspection of important components, such as filter frames, ducts, and piping, to assure the integrity and capability of the systems."

DISCUSSION

The containment atmosphere cleanup systems are designed and located so that they can be inspected periodically, as required. The essential equipment of the CSS is outside the containment, except for risers, distribution header piping, and spray nozzles in the containment. The hydrogen recombiners are located inside the containment. The post-LOCA purge exhaust filter unit and the hydrogen monitors are located outside the containment. The equipment outside the containment may be inspected during normal power operation. Components of the CSS, the post-LOCA purge exhaust system, and the hydrogen recombiner and monitoring system located inside the containment, can be inspected during refueling shutdowns. (See chapter 6 for details on these systems.)

CRITERION 43 - TESTING OF CONTAINMENT ATMOSPHERE CLEANUP SYSTEMS

"The containment atmosphere cleanup systems shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leak-tight integrity of its components, (2) the operability and performance of the active components of the systems such as fans, filters, dampers, pumps, and valves, and (3) the operability of the systems as a whole and, under conditions as close to design as practical, the performance of the full operational sequence that brings the systems into operation, including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of associated systems."

DISCUSSION

The CSS which serves as the containment atmosphere cleanup system can be tested. The operation of the spray pumps can be tested by recirculation to the refueling water storage tank through a test line. The system valves can be operated through their full travel. The system is checked for leaktightness during testing. (See subsection 6.2.2.2 for details and chapter 8 for electrical power details.) The spray headers and nozzles can be smoke or air tested, as described in the response to Criterion 40.

CRITERION 44 - COOLING WATER

"A system to transfer heat from structures, systems, and components important to safety to an ultimate heat sink shall be provided. The system safety function shall be to transfer the combined heat load of these structures, systems, and components under normal operating and accident conditions.

"Suitable redundancy in components and features and suitable interconnections, leak detection, and isolation capabilities shall be provided to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished assuming a single failure."

DISCUSSION

The component cooling water (CCW) and nuclear service cooling water (NSCW) systems are provided to transfer heat from plant safety-related components to the ultimate heat sink. These systems are designed to transfer their respective heat loads under all anticipated normal and accident conditions. Suitable redundancy, leak detection, systems interconnection, and isolation capabilities are incorporated in the design of these systems to assure the required safety function, assuming a single failure, with either onsite or offsite power.

Complete descriptions of the NSCW system and the CCW system are given in chapter 9.

CRITERION 45 - INSPECTION OF COOLING WATER SYSTEM

"The cooling water system shall be designed to permit appropriate periodic inspection of important components, such as heat exchangers and piping, to assure the integrity and capability of the system."

DISCUSSION

The CCW system and portions of the NSCW system are capable of being monitored during normal operation. The important components are located in accessible areas with the exception of any underground piping for the NSCW system. These components have suitable manholes, handholes, inspection ports, or other appropriate design and layout features to allow periodic inspection. The integrity of any underground piping will be demonstrated by pressure and functional tests. Piping to and from the containment air coolers is accessible for inspection during reactor shutdown and refueling periods. These systems are discussed in chapter 9.

CRITERION 46 - TESTING OF COOLING WATER SYSTEM

"The cooling water system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leak-tight integrity of its components, (2) the operability and the performance of the active components of the system, and (3) the operability of the system as a whole and, under conditions as close to design as practical, the performance of the full operational sequence that brings the system into operation for reactor shutdown and for LOCA, including operation of applicable portions of the protection system and the transfer between normal and emergency power sources."

DISCUSSION

The CCW and NSCW systems operate continuously during normal plant operation and shutdown, under flow and pressure conditions that approximate the accident conditions. These operations demonstrate the operability, performance, and structural and leaktight integrity of all cooling water system components.

These cooling water systems are designed to include the capability for testing through the full operational sequence that brings the system into operation for reactor shutdown and for LOCAs, including operation of applicable portions of the protection system and the transfer between normal and emergency power sources. The CCW system and NSCW system are capable of being tested during normal operation by alternating operation of the systems between the redundant trains.

For a detailed description of the cooling water systems, refer to section 9.2.

3.1.5 REACTOR CONTAINMENT

CRITERION 50 - CONTAINMENT DESIGN BASIS

"The reactor containment structure, including access opening, penetrations, and the containment heat removal system, shall be designed so that the containment structure and its internal compartments can accommodate, without exceeding the design leakage rate and with sufficient margin, the calculated pressure and temperature conditions resulting from any LOCA. This margin shall reflect consideration of (1) the effects of potential energy sources which have not been included in the determination of the peak conditions, such as energy in steam generators and energy from metal-water and other chemical reactions that may result from degraded emergency core cooling functioning, (2) the limited experience and experimental data available for defining accident phenomena and containment responses, and (3) the conservatism of the calculational model and input parameters."

DISCUSSION

The design of the containment structure is based on the containment design basis accidents, which include the rupture of a reactor coolant pipe in the RCS or the rupture of a main steam line. In either case, the pipe rupture is assumed to be coupled with partial loss of the redundant safety feature systems (minimum safety features). The maximum pressure and temperature reached for a containment design basis accident are presented in chapter 6. The containment design, as discussed in subsection 3.8.1, provides ample margin to the design basis limits.

See chapters 3 and 6 for details.

CRITERION 51 - FRACTURE PREVENTION OF CONTAINMENT PRESSURE BOUNDARY

"The reactor containment boundary shall be designed with sufficient margin to assure that under operating, maintenance, testing, and postulated accident conditions (1) its ferritic materials behave in a nonbrittle manner and (2) the probability of rapidly propagating fracture is minimized. The design shall reflect consideration of service temperatures and other conditions of the containment boundary material during operation, maintenance, testing, and postulated accident conditions, and the uncertainties in determining (1) material properties, (2) residual, steady-state, and transient stresses, and (3) size of flaws."

DISCUSSION

Principal load-carrying components of ferritic materials exposed to the external environment are selected (as discussed in subsection 3.8.1) so that their temperatures under normal operating and testing conditions are not less than 30°F above nil ductility transition temperature.

Refer to subsection 3.8.1 for details.

CRITERION 52 - CAPABILITY FOR CONTAINMENT LEAKAGE RATE TESTING

"The reactor containment and other equipment which may be subjected to containment test conditions shall be designed so that periodic integrated leakage rate testing can be conducted at containment design pressure."

DISCUSSION

The containment system is designed and constructed and the necessary equipment is provided to permit periodic integrated leakage rate tests during plant lifetime, in accordance with the requirements of Appendix J of 10 CFR 50. Details concerning the conduct of periodic integrated leakage rate tests are included in chapter 6.

CRITERION 53 - PROVISIONS FOR CONTAINMENT TESTING AND INSPECTION

"The reactor containment shall be designed to permit (1) appropriate periodic inspection of all important areas, such as penetrations, (2) an appropriate surveillance program, and (3) periodic testing at containment design pressure of the leak-tightness of penetrations which have resilient seals and expansion bellows."

DISCUSSION

Provisions exist for conducting individual leakage rate tests on containment penetrations. Penetrations are visually inspected and pressure tested for leaktightness at periodic intervals. Other inspections are performed as required by Appendix J of 10 CFR 50. (Refer to chapter 6.)

CRITERION 54 - PIPING SYSTEMS PENETRATING CONTAINMENT

"Piping systems penetrating the primary reactor containment shall be provided with leak detection, isolation and containment capabilities having redundancy, reliability, and performance capabilities which reflect the importance to safety of isolating these piping systems. Such piping systems shall be designed with a capability to test periodically the operability of the isolation valves and associated apparatus and to determine if valve leakage is within acceptable limits."

DISCUSSION

Piping systems penetrating the primary reactor containment are provided with containment isolation valves. Penetrations which must be closed for containment isolation have redundant valving and associated apparatus. Automatic isolation valves with air or motor operators, which do not restrict normal plant operation, are periodically tested to assure operability. Secondary system piping inside the containment is considered an extension of the containment boundary, as described in subsection 6.2.4. The isolation valve arrangements are discussed in chapter 6.

Piping that penetrates the containment has been equipped with test connections and test vents or has other provisions to allow periodic leak rate testing to ensure that leakage is within the acceptable limit as defined by the Technical Specifications and Appendix J to 10 CFR 50, as described in chapter 6.

The fuel transfer tube is not classified as a fluid system penetration. The blind flange and the portion of the transfer tube inside the containment are an extension of the containment boundary. The blind flange isolates the transfer tube at all times, except when the reactor is shutdown for refueling. This assembly is a penetration in the same sense as are equipment hatches and personnel locks.

CRITERION 55 - REACTOR COOLANT PRESSURE BOUNDARY PENETRATING CONTAINMENT

"Each line that is part of the reactor coolant pressure boundary and that penetrates the primary reactor containment shall be provided with containment isolation valves as follows, unless it can

be demonstrated that the containment isolation provisions for a specific class of lines, such as instrument lines, are acceptable on some other defined basis:

1. One locked closed isolation valve inside and one locked closed isolation valve outside the containment; or
2. One automatic isolation valve inside and one locked closed isolation valve outside the containment; or
3. One locked closed isolation valve inside and one automatic isolation valve outside the containment. A simple check valve may not be used as the automatic isolation valve outside containment; or
4. One automatic isolation valve inside and one automatic isolation valve outside the containment. A simple check valve may not be used as the automatic isolation valve outside the containment.

"Isolation valves outside the containment shall be located as close to the containment as practical and, upon loss of actuating power, automatic isolation valves shall be designed to take the position that provides greater safety.

"Other appropriate requirements to minimize the probability or consequences of an accidental rupture of these lines or of lines connected to them shall be provided, as necessary, to assure adequate safety. Determination of the appropriateness of these requirements, such as higher quality in design, fabrication and testing additional provisions for inservice inspection, protection against more severe natural phenomena, and additional isolation valves and containment, shall include consideration of the population density and use characteristics and physical characteristics of the site environs."

DISCUSSION

Each line that is a part of the reactor coolant pressure boundary and penetrates the containment is provided with isolation valves meeting the intent of this criterion, except that the reactor shutdown cooling lines (RHR system) which are part of the RCPB and which penetrate the containment are provided with two isolation valves in series, both inside the containment. This system is a closed system outside the containment and is constructed to ASME Code, Section III, Class 2, specifications and is considered the second passive barrier to fission product release, as described in chapter 6. The arrangement and type of valves utilized are discussed in chapter 6. Containment penetrations are Seismic Category 1 and are protected against possible environmental effects, including missiles.

CRITERION 56 - PRIMARY CONTAINMENT ISOLATION

"Each line that connects directly to the containment atmosphere and penetrates the primary reactor containment shall be provided with containment isolation valves as follows, unless it can be demonstrated that the containment isolation provisions for a specific class of lines, such as instrument lines, are acceptable on some other defined basis:

1. One locked closed isolation valve inside and one locked closed isolation valve outside the containment; or
2. One automatic isolation valve inside and one locked closed isolation valve outside the containment; or
3. One locked closed isolation valve inside and one automatic isolation valve outside containment. A simple check valve may not be used as the automatic isolation valve outside containment; or

4. One automatic isolation valve inside and one automatic isolation valve outside the containment. A simple check valve may not be used as the automatic isolation valve outside the containment.

"Isolation valves outside the containment shall be located as close to the containment as practical and, upon loss of actuating power automatic isolation valves shall be designed to take the position that provides greater safety."

DISCUSSION

Lines which communicate directly with the containment atmosphere and which penetrate the reactor containment are normally provided with two isolation valves in series, one inside and one outside the containment, in accordance with one of the above acceptable arrangements. Several penetrations use alternative arrangements which satisfy containment isolation on some other defined bases. Special cases are described in chapter 6.

Valving arrangements are combinations of locked-shut isolation valves and automatic isolation valves or remote-manual isolation valves. No simple check valves are utilized as automatic isolation valves outside the containment. Where necessary, provision for leak detection is provided for lines outside the containment.

Instrument lines satisfy other acceptable criteria, as described in chapter 6.

CRITERION 57 - CLOSED SYSTEM ISOLATION VALVES

"Each line that penetrates the primary reactor containment and is neither part of the reactor coolant pressure boundary nor connected directly to the containment atmosphere shall have at least one containment isolation valve which shall be either automatic, locked closed, or capable of remote manual operation. This valve shall be outside the containment and located as close to the containment as practical. A simple check valve may not be used as the automatic isolation valve."

DISCUSSION

Lines which penetrate the containment and are neither part of the RCPB nor connected directly to the containment atmosphere are considered closed systems within the containment and are equipped with at least one containment isolation valve of one of the following types:

A. An automatic isolation valve (a simple check valve is not used as this automatic valve).

B. A locked-closed valve.

C. A valve capable of remote manual operation.

This valve is located outside the containment and as close to the containment wall as practical. Valve locations are discussed in detail in subsection 6.2.4.

3.1.6 FUEL AND REACTIVITY CONTROL

CRITERION 60 - CONTROL OF RELEASES OF RADIOACTIVE MATERIALS TO THE ENVIRONMENT

"The nuclear power unit design shall include means to control suitably the release of radioactive materials in gaseous and liquid effluents and to handle radioactive solid wastes produced during normal reactor operation, including anticipated operational occurrences. Sufficient holdup capacity shall be provided for the retention of gaseous and liquid effluents containing radioactive materials, particularly where unfavorable site environmental conditions can be expected to impose unusual operational limitations upon the release of such effluents to the environment."

DISCUSSION

Means are provided to control the release of radioactive materials in gaseous and liquid effluents and to handle radioactive solid wastes produced during normal reactor operation, including anticipated operational occurrences. The radioactive waste management systems are designed to minimize the potential for an inadvertent release of radioactivity from the facility and to assure that the discharge of radioactive wastes is maintained as low as practicable below regulatory limits of 10 CFR 20 during normal operation. The radioactive waste management systems, the design criteria, and the amounts of estimated releases of radioactive effluents to the environment are described in chapter 11.

CRITERION 61 - FUEL STORAGE AND HANDLING AND RADIOACTIVITY CONTROL

"The fuel storage and handling, radioactive waste, and other systems which may contain radioactivity shall be designed to assure adequate safety under normal and postulated accident conditions. These systems shall be designed (1) with a capability to permit appropriate periodic inspection and testing of components important to safety, (2) with suitable shielding for radiation protection, (3) with appropriate containment, confinement, and filtering systems, (4) with a residual heat removal capability having reliability and testability that reflects the importance to safety of decay heat and other residual heat removal, and (5) to prevent significant reduction in fuel storage coolant inventory under accident conditions."

DISCUSSION

The spent fuel pool and associated cooling system, fuel handling system, independent spent fuel storage installation (ISFSI), and radioactive waste processing system are designed to assure adequate safety under normal and postulated accident conditions.

The spent fuel pool cooling system provides cooling to remove residual heat from the fuel stored in the spent fuel pool. The system is designed with redundancy and testability to assure continued heat removal. The spent fuel pool cooling system is described in subsection 9.1.3.

The spent fuel pool is designed so that no postulated accident could cause excessive loss-of-coolant inventory. Accidents are discussed in chapter 15.

Spent fuel in the ISFSI is stored in casks designed to remove residual heat from the fuel using passive cooling. Each cask is designed to withstand credible and postulated accidents without breaching the confinement barrier resulting in loss of coolant inventory and radiological release to the public.

Structures, components, and systems are designed and located so that appropriate periodic inspection and testing may be performed.

Adequate shielding is provided as described in chapter 12. Radiation monitoring is provided as discussed in chapters 11 and 12.

Individual components that contain significant radioactivity are in confined areas adequately ventilated through appropriate filtering systems.

CRITERION 62 - PREVENTION OF CRITICALITY IN FUEL STORAGE AND HANDLING

"Criticality in the fuel storage and handling system shall be prevented by physical systems or processes, preferably by use of geometrically safe configurations."

DISCUSSION

The restraints and interlocks provided for the safe handling and storage of new and spent fuel are discussed and illustrated in chapter 9.

The reactivity of the spent fuel rack is analyzed such that K_{eff} remains less than 1.0 under No Soluble Boron 95/95 K_{eff} conditions as described in paragraph 4.3.2.6.1. To provide safety margin in the criticality analysis of the spent fuel racks, credit is taken for the soluble boron in the spent fuel pool water as described in FSAR paragraph 4.3.2.6.1. New fuel in the new fuel storage racks is stored with enough center-to-center distance to ensure a k_{eff} <0.98, under conditions of optimum moderation.

The design of the spent fuel storage rack assembly is such that it is configurationally impossible to insert the spent fuel assemblies in other than prescribed locations, without physically modifying the rack, thereby preventing any possibility of accidental criticality.

Layout of the fuel handling area is such that the spent fuel cask cannot traverse the spent fuel storage pool.

CRITERION 63 - MONITORING FUEL AND WASTE STORAGE

"Appropriate systems shall be provided in the fuel storage and radioactive waste systems and associated handling areas (1) to detect conditions that may result in the loss of residual heat removal capability and excessive radiation levels and (2) to initiate appropriate safety actions."

DISCUSSION

Instrumentation is provided to detect and alarm, in the control room, excessive temperature or low water level in the spent fuel storage pool. Area radiation monitors are provided in the fuel storage area for personnel protection and general surveillance. These area monitors alarm locally and in the control room. Normally, the fuel building ventilation system removes radioactivity from the atmosphere above the spent fuel storage pool and discharges it by way of the plant vent. The ventilation system is continuously monitored by gaseous, particulate, and radioiodine radiation monitors.

If radiation levels reach a predetermined point, an alarm is sounded in the control room and the ventilation discharge path is automatically transferred through filter adsorber units which provide adequate filtration before discharge from the plant vent. (See chapters 7, 9, and 12 for details.)

CRITERION 64 - MONITORING RADIOACTIVITY RELEASES

"Means shall be provided for monitoring the reactor containment atmosphere, spaces containing components for recirculation of LOCA fluids, effluent discharge paths, and the plant environs for radioactivity that may be released from normal operations, including anticipated operational occurrences, and from postulated accidents."

DISCUSSION

The containment atmosphere is continually monitored during normal and transient station operations, using the containment particulate, gaseous, and radioiodine radiation monitors. Under accident conditions, samples of the containment atmosphere provide data on existing airborne radioactive concentrations within the containment. Portable radiation detection instruments are provided to periodically monitor radiation levels in the auxiliary building spaces which contain components for recirculation of LOCA fluids and components for processing radioactive wastes. Radioactivity levels contained in the facility effluent and discharge paths and in the plant environs are continually monitored during normal and accident conditions by the plant radiation monitoring systems. In addition to the installed detectors, periodic plant environmental surveillance is established. Measurement capability and reporting of effluents are based on the guidelines of Regulatory Guides 1.4 and 1.21. Radiation monitoring systems are discussed in section 11.5 and subsection 12.3.4.

3.2 CLASSIFICATION OF STRUCTURES, COMPONENTS, EQUIPMENT, AND SYSTEMS

This section provides a guide to the classification method of structures, components, and systems.

3.2.1 SEISMIC CLASSIFICATION

General Design Criterion 2 of Appendix A to 10 CFR 50, General Design Criteria for Nuclear Power Plants, requires that nuclear power plant structures, systems, and components important to safety be designed to withstand the effects of natural phenomena such as earthquakes without loss of capability to perform necessary safety functions. Appendix A to 10 CFR 100, Seismic and Geologic Siting Criteria for Nuclear Power Plants, sets forth the principal seismic and geologic considerations which are used in the evaluation of the suitability of plant design bases established in consideration of the site seismic and geologic characteristics.

3.2.1.1 Definitions

Seismic Category 1 structures, components, and systems are classified in accordance with Regulatory Guide 1.29. Safety-related, Seismic Category 1 structures, components, and systems are those necessary to ensure the following:

A. The integrity of the reactor coolant pressure boundary.

B. The capability to shut down the reactor and maintain it in a safe-shutdown condition.

C. The capability to prevent or mitigate the consequences of accidents that could result in potential offsite exposures comparable to the guideline exposures of 10 CFR 100.

Seismic Category 1 structures, components, and systems are designed to withstand the appropriate seismic loads, as discussed in section 3.7, and other applicable loads without loss of function. Seismic Category 1 structures are sufficiently isolated from non-Category 1 structures, or they are analyzed to ensure that their structural integrity is maintained during the postulated safe shutdown earthquake (SSE). Non-Seismic Category 1 systems, equipment, and components installed in Seismic Category 1 structures whose failure could result in loss of required safety function of Seismic Category 1 structures, equipment, systems, or components are either separated by distance or barrier from the affected structure, system, equipment, or component or designed together with their anchorages to maintain their structural integrity during the SSE.

Structures, equipment, and systems not classified as Seismic Category 1 are classified as Seismic Category 2.

Safety-related structures, systems, and components that are classified Seismic Category 1 are in compliance with the quality assurance requirements of 10 CFR 50, Appendix B.

The criteria used for the design of Seismic Category 1 structures, equipment, systems, and components; Seismic Category 2 items; and Seismic Category 2 items whose failure could result in the loss of required safety function of Seismic Category 1 items are discussed in section 3.7. Also discussed are the additional seismic criteria for which the radwaste buildings are designed.

3.2.1.2 Classifications

Table 3.2.2-1 provides a listing of structures, components, and systems and identifies those that are Seismic Category 1.

Where only portions of systems are identified as Seismic Category 1 in table 3.2.2-1, the boundaries of the Seismic Category 1 portions of the system are shown on the piping and instrumentation diagrams in appropriate sections of the FSAR. Conformance of the above seismic classifications with Regulatory Guide 1.29 is discussed in section 1.9.

3.2.2 VEGP CLASSIFICATION SYSTEM

Equipment, components, and structures in the VEGP are categorized according to nuclear safety, seismic category, and codes and standards by the VEGP project classification system. This system conforms to 10 CFR 50 and Regulatory Guides 1.26 and 1.29. A project classification is assigned to each item in the plant. The three element project classification indicates, in sequence, the nuclear safety class, the seismic category, and the applicable codes and standards. The project classification system provides an easily recognizable means of identifying the extent to which components, equipment, and structures are related to nuclear safety and seismic qualification requirements. In addition, the project classification system provides the means whereby the codes and/or standards that govern the design of a component or structure can be located. Table 3.2.2-1 provides a listing of the principal VEGP structures, systems, components, and the associated project classifications. Table 3.2.2-3 summarizes the construction codes and standards for VEGP components.

3.2.2.1 Nuclear Safety Classifications

The first element of a project classification identifies the nuclear safety class. The nuclear safety class designators used on VEGP are as follows:

Designator	Definition
0	Nuclear Safety Class 0 is assigned to safety-related mechanical components not within the purview of Regulatory Guide 1.26. Nuclear Safety Class 0 is also assigned to safety-related structures and structural components.
1	Nuclear Safety Class 1 parallels Group A as defined in 10 CFR 50.55a and 10 CFR 50.2(v) and is assigned to components of the reactor coolant pressure boundary. Nuclear Safety Class 1 is also assigned to safety-related instruments, controls, and electrical components.
2	Nuclear Safety Class 2 parallels Group B as defined in Regulatory Guide 1.26 and is assigned to emergency and auxiliary systems serving the reactor coolant system.
3	Nuclear Safety Class 3 parallels Group C as defined in Regulatory Guide 1.26 and is assigned to other safety-related auxiliary systems.
4	Nuclear Safety Class 4 parallels Group D as defined in

Designator	Definition
	Regulatory Guide 1.26 and is assigned to nonsafety-related systems.
6	Nuclear Safety Class 6 is assigned to nonsafety-related components not within the purview of Regulatory Guide 1.26. This designation is also assigned to nonsafety-related structures and structural components, instrumentation, controls, and electrical components.

3.2.2.2 Seismic Classification

The second element of a project classification is either 1 or 2, which designates the appropriate seismic category. Seismic classification is discussed in subsection 3.2.1.

3.2.2.3 Codes and Standards

The third element of a project classification indicates the primary codes and/or standards applicable to plant equipment, components, and structures. The codes and standards designators used on VEGP are as follows:

Designator	Codes and Standards
1	American Society of Mechanical Engineers Boiler and Pressure Vessel (ASME B&PV) Code, Section III, Class 1.
2	ASME B&PV Code, Section III, Class 2.
3	ASME B&PV Code, Section III, Class 3.
4	Regulatory Guide 1.26 - Table 1 – Quality Group D.
5	Special designator associated with Nuclear Safety Classes 0 or 4. (Refer to paragraph 3.2.2.3.1.)
6	Special designator associated with Nuclear Safety Class 6. (Refer to paragraph 3.2.2.3.2.)
7	Regulatory Guide 1.143 - Table 1. (Refer to paragraph 3.2.2.3.3.)
8	ASME B&PV Code, Section I.
9	Applicable National Fire Protection Association (NFPA) codes per the guidelines of Branch Technical Position CMEB 9.5-1, section C4. (Refer to paragraph 3.2.2.3.4.)
C	Structures or structural components designed to codes and standards as defined in the design bases.(d)
E	Electrical equipment designed to codes and standards as defined in the design bases.(a)
J	Instrumentation and control equipment designed to codes

Designator	Codes and Standards
	and standards as defined in the design bases.[d]

[a] For fire protection-related features, BTP CMEB 9.5-1, section C4 applies as described in section 9.5.1.1.4.

A more complete listing of applicable codes and standards is provided in table 3.2.2-2.

3.2.2.3.1 Codes and Standards Designator 5

Codes and standards designator 5 is associated only with Nuclear Safety Class 0 or 4.

When associated with Nuclear Safety Class 0, codes and standards designator 5 requires standards for construction which are of sufficient quality to ensure acceptable performance for the intended safety-related function of the component, as defined in the design bases for the particular system.

When associated with Nuclear Safety Class 4, codes and standards designator 5 specifies that materials, components, parts, appurtenances, and piping subassemblies shall be procured in accordance with the ASME B&PV Code, Section III, Class 3; however, the system shall be designed and installed in accordance with American National Standards Institute (ANSI) B31.1. Conformance with these aspects of the ASME code is required only for initial procurement. Subsequent conformance for maintenance and replacement will be subject to the owner's discretion.

3.2.2.3.2 Codes and Standards Designator 6

Codes and standards designator 6 is associated only with Nuclear Safety Class 6. Codes and standards designator 6 requires standards for construction which are of sufficient quality to ensure acceptable performance for the intended nonsafety-related function of the component, as defined in the design bases for the particular system.

3.2.2.3.3 Codes and Standards Designator 7

The codes and standards used for the construction of radioactive waste management and steam generator blowdown systems are provided in Regulatory Guide 1.143. Quality assurance requirements are to be applied to radioactive waste management systems as described in Regulatory Guide 1.143.

3.2.2.3.4 Codes and Standards Designator 9

The design, fabrication, construction, and testing of fire protection systems are performed in accordance with the applicable portions of the NFPA codes, which invoke ANSI B31.1, American Water Works Association (AWWA), American Petroleum Institute (API), and other codes, depending upon service. Quality assurance program requirements are implemented to ensure that the requirements for design, procurement, installation, testing, and administrative controls for the fire protection program are satisfied. The quality assurance requirements that apply to the fire protection program are described in section 9.5.1.1.4. These quality assurance requirements also apply to fire protection-related features under codes and standards designators C, E, J, 4, and 6.

3.2.2.4 Clarification to VEGP Classification System

Instrumentation and control project classes 11J, 61J, and 62J are normally assigned to instrumentation. Piping, tubing, fittings, control valves, instrument valves, and other mechanical components associated with instrumentation are assigned mechanical project classifications consistent with the project class assigned to process equipment (tanks, piping, vessels, etc.) to which the instrumentation is connected.

Equipment which is classified 61J and 61E is seismically designed not to fail in a manner that would compromise the functioning of safety-related equipment during or after a safe shutdown earthquake. However, the 61J and 61E equipment will not necessarily remain functional.

TABLE 3.2.2-1 CONTENTS

CLASSIFICATION OF STRUCTURES, COMPONENTS, AND SYSTEMS

Contents	Sheet No.
NUCLEAR STEAM SYSTEM SUPPLIER (NSSS) AND NUCLEAR AUXILIARY SYSTEMS	
Reactor Coolant System	1
Reactor Head Vent System	3
Safety Injection System	3
Residual Heat Removal System	5
Containment Spray System	6
Chemical and Volume Control System	7
Boron Recycle System	12
Containment Isolation System	13
WATER SYSTEMS	
Nuclear Service Cooling Water (NSCW) System	14
Component Cooling Water System	15
Auxiliary Component Cooling Water System	16
Spent Fuel Cooling and Purification System	16
Reactor Makeup Water System	17
RADWASTE SYSTEMS	
Waste Processing System - Liquid	18
Waste Processing System - Gaseous	23
Steam Generator Blowdown System	24
Backflushable Filter System	25
SECONDARY CYCLE SYSTEMS	
Main Steam System	25
Auxiliary Feedwater System	27
Condensate and Feedwater System	28
Power Conversion System	29
Condenser Air Ejector System	30
Circulating Water System	30
River Intake Structure Systems	31
Condensate Chemical Injection System	32
Condensate Filter Demineralizer System	33
Plant Makeup Water Well System	34
Plant Makeup Water Treatment System	35
Plant Makeup Water Treatment System Waste Neutralization System	36

TABLE 3.2.2-1 CONTENTS

Contents	Sheet No.
SERVICE SYSTEMS	
Demineralized Water System	36
Diesel Generator Systems	36
Fire Protection Systems	38
Fire Protection Systems - Seismic Category 1	39
Auxiliary Gas Systems (N_2, H_2, and 0_2)	39
NSSS Liquid Sampling System	39
NSSS Gas Sampling System	40
Post-Accident Sampling System	40
Turbine Plant Sampling System	40
Control Building Drain System	41
Auxiliary Building Flood Retaining Rooms, Alarms, and Drains	41
Containment, Auxiliary Building, and Miscellaneous Drain Systems	42
Turbine Building Drain System	44
Turbine Lube Oil Storage and Filtration System	44
Auxiliary Steam System	45
Turbine Generator H_2 and H_2 Seal Oil System	45
Turbine Generator CO_2 System	46
Turbine Plant Closed Cooling Water System	46
Turbine Plant Cooling Water System	46
Waste Evaporator Steam Supply System	47
Waste Water Effluent System	48
Instrument and Service Air System	50
Potable Water System	52
Reactor Coolant System Leak Detection System	52
Miscellaneous Leak Detection System	53
Fuel Handling Building Drain System	53
Utility Water System	53
HEATING, VENTILATING, AND AIR CONDITIONING (HVAC) SYSTEMS	
Containment Air Cooling System	53
Containment Lower Level Air Circulating System	53
Containment Preaccess Filter System	54
Containment Normal Preaccess Purge Exhaust System	54
Containment Normal Preaccess Purge Supply System	55
Containment Post-Loss-of-Coolant Accident (LOCA) Purge Exhaust System	55
Containment Control Rod Drive Mechanism (CRDM) Cooling System	56
Containment Cavity Cooling System	56
Containment Reactor Support Cooling System	56
H_2 Recombiner and Monitoring System	56
Containment Auxiliary Air Cooling System	57
Containment Post-LOCA Cavity Purge System	57

TABLE 3.2.2-1 CONTENTS

Contents	Sheet No.
Equipment Building HVAC System	57
Control Room HVAC System (Essential Portion)	58
Control Room HVAC System (Normal Portion)	59
Safety Feature Electrical Equipment Room HVAC System	60
Control Building Levels A, B, 1, and 2 Normal System	60
Control Building Lab Hood Vent System	62
Control Building Locker and Toilet Exhaust System	62
Control Building Cable Spreading Room HVAC System	62
Electrical Tunnel Ventilation System	64
Fuel Handling Building HVAC System	65
Fuel Handling Building Post-Accident Exhaust System	67
Auxiliary Building Outside Air Supply, Normal HVAC, Radioactive Filter Exhaust, and Continuous Exhaust Systems	67
Auxiliary Building Engineered Safety Feature (ESF) Room Coolers	68
Piping Penetration and MSIV Area Ventilation System	72
Piping Penetration Filter Exhaust System	72
Diesel Generator Building HVAC System	73
Turbine Building Condenser Vacuum Exhaust Filtration System	74
Turbine Building HVAC System	74
Technical Support Center HVAC System	75
Fire Pump and Well Pumphouse HVAC System	76
Normal Chilled Water System	76
Essential Chilled Water System	77
Special Chilled Water Systems	77
Auxiliary Feedwater Pumphouse HVAC System	78
Miscellaneous HVAC Systems	78
ELECTRICAL AND CONTROL SYSTEMS	
Main Control Board	78
Nuclear Instrumentation System	79
Process Control System	79
Protection System Nuclear Steam System (NSS)	79
Rod Control Power System	79
Full Length Rod Control System	79
Rod Position Indication System	79
Radiation Monitoring System	80
ESF Actuation System	80
Reactor Instrumentation	80
Reactor Control System	80
Post-Accident Monitoring System	80
Plant Auxiliary Control Boards	80
Safety-Related Systems Bypass/Inoperable Status and Trip/Monitoring Indicating Lights	81

TABLE 3.2.2-1 CONTENTS

Contents	Sheet No.
Incore Instrumentation	81
Turbine Protection System	81
Turbine Supervisory Instrumentation	81
Electrohydraulic Control System	81
Anticipated Transient Without Scram Mitigation System Actuation Circuitry (AMSAC)	81
Computer System	81
Annunciator System	81
Telephone Page System	81
PABX System	82
Sound-Powered System	82
Seismic Monitoring Equipment	82
Plant Security System	82
Offsite Power System	82
ac System - 480 V	82
ac System - 4160 V	83
dc System - Class 1E	83
120-V ac Power System-Class 1E	83
Lighting System	83
Cable System	84
Fire Detection System	84
Heat Tracing Systems	84
Electrical Penetration System	84
Standby Power System	85
ac System - 25 kV	85
dc System - Non-Class 1E	85
120-V ac Power System - Non-Class 1E	85
Switchyard Interfaces	85
Multisystem Panels and Boards	85
Iso-Phase Bus System	85
ac System - 13.8 kV	85
High Voltage Switchyard	86
STRUCTURES, BUILDINGS, AND CRANES	
Structures and Buildings	86
Containment Building Polar Bridge Crane	89
Fuel Handling System	89
Raceway Systems	90

TABLE 3.2.2-1 (SHEET 1 OF 98)

CLASSIFICATION OF STRUCTURES, COMPONENTS, AND SYSTEMS

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
REACTOR COOLANT SYSTEM													
1.	Reactor vessel and head	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
2.	Vessel internals	C	C	W	B	2	1	2	III-CS	Y	Y	1-A	Note af
3.	Fuel assemblies and appurtenances	C	C	W	NA	1	1	1	mfg	Y	Y	1-A	
4.	CRDM housing	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
5.	CRDM head adapter plug	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
6.	Steam generator	C	C	W	A	1,2	1	1,2	III-1,2	Y	Y	1-A	
7.	Pressurizer	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
8.	Pressurizer surge line	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
9.	Pressurizer relief lines (upstream of relief valves)	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
10.	Pressurizer relief lines (downstream) of relief valve	C	C	B	D	4	1	5	III-3	N	N		Note r
11.	Pressurizer safety and relief valves	C	C	W	A	1	1	1	III-1	Y	y	1-A	
12.	Pressurizer relief tank	C-171′	C-171′	W	D	4	2	5	III-3	N	N		Note r
13.	Normal pressurizer heaters	C	C	W	NA	6	2	E	mfg	N	N		
14.	Backup pressurizer heaters	C	C	W	NA	6	1	E	mfg	N	N		Note m
15.	Reactor coolant pump casing	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
16.	RCP seal standpipe	C	C	W	D	4	2	4	VIII	N	N		
17.						(This line has been intentionally left blank.)							
18.	Seal 1 housing	C	C	W	A	1	1	1	III-1	Y	Y	1-A	

TABLE 3.2.2-1 (SHEET 2 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
19.	Thermal barrier	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
20.	Seal 2 housing	C	C	W	B	2	1	2	III-2	Y	Y	1-A	
21.	Pressure retaining bolts	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
22.	Main flange	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
23.	RCP motor	C	C	W	NA	6	2	E	NEMA MG1	N	N		
24.	Shaft coupling	C	C	W	NA	2	1	NA	NA	Y	Y	1-A	
25.	Spool piece	C	C	W	B	2	1	2	Mfg	Y	Y	1-A	
26.	Armature	C	C	W	B	2	1	2	NEMA MG1	Y	Y	1-A	
27.	Flywheel	C	C	W	NA	2	1	NA	mfg	Y	Y	1-A	
28.	Motor bolting	C	C	W	NA	2	1	NA	NEMA MG1	Y	Y	1-A	
29.	Upper oil cooler (tube side - ACCW)	C	C	W	C	3	1	3	III-3	Y	Y		
30.	Upper oil cooler (shell side - oil)	C	C	W	C	3	1	3	III-3	Y	Y	1-A	
31.	Lower oil cooling coil	C	C	W	C	3	1	3	III-3	Y	Y		
32.						(This line has been intentionally left blank.)							
33.	RTD thermowells	C	C	W	A	1	1	1	III-1	Y	Y	1-A	
34.	RCS loop piping	C	C	W	A	1	1	1	III-1	Y	Y		
35.	Valves >O_2 in.	C	C	W	A	1	1	1	III-1	Y	Y		
36.	Other piping and valves ≤2 in.	C	C	B	A	1	1	1	III-1	Y	Y		
37.	Safety-related valve operators	C	C	W	NA	1	1	E	NEMA MG1	Y	Y		
38.	Reactor vessel supports	C	C	W	NA	0	1	C	III-NF	Y	Y		
39.	Steam generator supports	C	C	W	NA	0	1	C	III-NF	Y	Y		
40.	Pressurizer supports	C	C	W	NA	0	1	C	III-NF	Y	Y		

TABLE 3.2.2-1 (SHEET 3 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	((g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
41.	Reactor coolant pump supports	C	C	W	NA	0	1	C	III-NF	Y	Y		
42.	Other safety-related piping supports and hangers	C	C	W,B	NA	See Note 4		III-NF	III-NF	Y	Y		
43.	Lube oil drain tanks	C-171′	C-171′	B	D	4	1	4	API-650	N	N		Note ac
44.	Bottom-mounted instrument tubing	C	C	W	A	1	1	1	III-1	Y	Y		
45.	Control rods	C	C	W	NA	0	1	5	mfg	Y	Y		
46.	Reactor coolant pump articulated arm	C	C	W	NA	6	2	6	mfg	N	N		
REACTOR HEAD VENT SYSTEM													
1.	Piping and valves through second isolation valve			W	A	1	1	1	III-1	Y	Y		
2.	All other piping and valves			W,B	B	2	1	2	III-2	Y	Y		
3.	Instrumentation			W,B	NA	1	1	J	mfg	Y	Y		Note s
SAFETY INJECTION SYSTEM													
1.	Accumulators	C-171′	C-171′	W	B	2	1	2	III-2	Y	Y	VIII	
2.	Boron injection tank	AB-B11	Deleted	W	B	2	1	2	III-2	Y	Y	VIII	Note ak
3.						(This line has been intentionally left blank.)							
4.						(This line has been intentionally left blank.)							
5.						(This line has been intentionally left blank.)							
6.						(This line has been intentionally left blank.)							
7.	Safety injection pumps	AB-B15 AB-B19	AB-B117 AB-B119	W	B	2	1	2	III-2	Y	Y	VIII	
8.	Safety injection pump motors	AB-B15 AB-B19	AB-B117 AB-B119	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 4 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9. Cold leg injection piping and valves downstream of 8 check valves, 143-150		W,B	A	1	1	1	III-1	Y	Y		
10. Hot leg injection piping and valves downstream of 6 check valves, 120-123, 128, 129		W,B	A	1	1	1	III-1	Y	Y		
11. Accumulator discharge piping and valves downstream of MOVs 8808A-D		W,B	A	1	1	1	III-1	Y	Y		
12. Piping and valves from RWST and containment sumps to 8 check valves, 139-142 and 120-123		W,B	B	2	1	2	III-2	Y	Y		
13. Piping from accumulators to MOVs 8808A-D		B	B	2	1	2	III-2	Y	Y		
14. Boron injection piping and valves downstream of check valve 013		W,B	A	1	1	1	III-1	Y	Y		
15. Boron injection piping and valves from charging pumps to check valve 013		W,B	B	2	1	2	III-2	Y	Y		
16. Safety-related instrumentation		W	NA	1	1	J	mfg	Y	Y		Note s
17. Safety-related valve operators		W	NA	1	1	E	NEMA MG1	Y	Y		

TABLE 3.2.2-1 (SHEET 5 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
18.	Sludge mixing pump	O	O	B	D	4	2	4	mfg	N	N		
19.	Electric circulation heater	O	O	B	D	4	2	4	mfg	N	N		
20.	Mixing eductors	O	O	B	D	4	2	4	mfg	N	N		
21.	Safety injection pump lube oil coolers	AB-B15 AB-B19	AB-B117 AB-B119	W	NA	0	1	5	III-3	Y	Y	VIII	
22.	Miniflow orifices			W	B	2	1	2	III-2	Y	Y		
RESIDUAL HEAT REMOVAL SYSTEM													
1.	RHR pumps	AB-D48 AB-D49	AB-D21 AB-D22	W	B	2	1	2	III-2	Y	Y	VIII	
2.	RHR pump Motors	AB-D48 AB-D49	AB-D21 AB-D22	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	
3.	RHR HXs:												
	Tube side, RHR	AB-C90 & C91	AB-C25 & C26	W	B	2	1	2	III-2, TEMA-R	Y	Y	VIII	Note t
	Shell side, CCW	AB-C90 AB-C91	AB-C25 AB-C26	W	C	3	1	3	III-3, TEMA-R	Y	Y	VIII	
4.	Piping and valves from hot legs to pump suction MOVs 8701A and 8702A			W,B	A	1	1	1	III-1	Y	Y		
5.	Piping and valves from pump suction MOVs 8701A and 8702A to cold leg injection check valves 147-150 and to hot leg injection check valves 128 and 129			W,B	B	2	1	2	III-2	Y	Y		
6.	Safety-related instrumentation			W	NA	1	1	J	mfg	Y	Y		Note s

TABLE 3.2.2-1 (SHEET 6 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
7. Safety-related valve operators			W	NA	1	1	E	NEMA MG1	Y	Y		
8. RHR pump seal coolers	AB-D48 AB-D49	AB-D21 AB-D22	B	B	2	1	2	III-2	Y	Y	VIII	
9. Emergency sump screen	C	C	SNC	NA	0	1	C	MFG	Y	Y		
10. Encapsulation vessels	FB-C07 AB-C105	FB-C01 AB-C04	B	NA	0	1	C	III-MC	Y	Y	VII	
CONTAINMENT SPRAY SYSTEM												
1. Containment spray pumps	AB-D76 AB-D77	AB-D04 AB-D05	W	B	2	1	2	III-2	Y	Y	VIII	
2. Containment spray pump motors	AB-D76 AB-D77	AB-D04 AB-D05	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	
3. Spray nozzles	C	C	W	B	2	1	2	III-2	Y	Y	VIII	
4. Spray additive tank	AB-D74	AB-D04	W	C	3	1	3	III-3	Y	Y	VIII	Note aq
5. Spray eductors	AB-D76 AB-D77	AB-D04 AB-D05	W	B	2	1	2	III-2	Y	Y	VIII	Note aq
6. Process valves and piping from RWST, containment sumps, and educator inlet check valves to spray nozzles			W,B	B	2	1	2	III-2	Y	Y		Note aq
7. Process valves and piping downstream of spray additive tank to eductor inlet check valves			W,B	C	3	1	3	III-3	Y	Y		
8. Safety-related instrumentation			W	NA	1	1	J	mfg	Y	Y		Note s
9. Safety-related valve operators			W	NA	1	1	E	NEMA MG1	Y	Y		
10. Encapsulation vessels	FB-C08 AB-C105	FB-C02 AB-C04	B	NA	0	1	C	III-MC	Y	Y	VIII	
11. Trisodium phosphate baskets	C	C	S	NA	6	1	C	AISC	N	N		

TABLE 3.2.2-1 (SHEET 7 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CHEMICAL AND VOLUME CONTROL SYSTEM												
1. Volume control tank	AB-A48	AB-A80	W	B	2	1	2	III-2	Y	Y	VIII	
2. Boric acid storage tank	AB-D69	AB-D09	B	C	3	1	3	III-3	Y	Y	VIII	
3. Boric acid batching tank	AB-C113	shared	W	D	4	2	4	API-650	N	N		
4. Boric acid batching tank agitator	AB-C113	shared	W	D	4	2	4	mfg	N	N		
5. Boric acid transfer pumps	AB-D72 & D119	AB-D106 & D123	W	C	3	1	3	III-3	Y	Y	VIII	
6. Boric acid transfer pump motors	AB-D72 & D119	AB-D106 AB-D123	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	Note m
7. Centrifugal charging pumps	AB-C115 AB-C118	AB-C16 AB-C17	W	B	2	1	2	III-2	Y	Y	VIII	
8. Centrifugal charging pump motors	AB-C115 & C118	AB-C16 & C117	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	
9. Deleted												
10. Deleted												
11. Regenerative HX	C-171′	C-171′	W	B	2	1	2	III-2 TEMA-R	Y	Y	1-A	
12. Letdown HX:			W									
Tube side, CVCS	AB-A07	AB-A100		B	2	1	2	III-2, TEMA-R	Y	Y	VIII	
Shell side, ACCW	AB-A07	AB-A100		D	4	1	5	III-3, TEMA-R	N	N		Note t
13. Excess letdown HX:			W									
Tube side, CVCS	C-171′	C-171′		B	2	1	2	III-2, TEMA-R	Y	Y	1-A	
Shell side, ACCW	C-171′	C-171′		D	4	1	5	III-2, TEMA-R	N	N		Note t

TABLE 3.2.2-1 (SHEET 8 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
14. Seal water HX:			W									
Tube side, CVCS	AB-B20	AB-B116		B	2	1	2	III-2, TEMA-R	Y	Y	VIII	
Shell side, ACCW	AB-B20	AB-B116		D	4	1	5	III-3, TEMA-R	N	N		Note t
15. Letdown reheat HX:			W									
Tube side, CVCS	AB-A07	AB-A100		B	2	1	2	III-2, TEMA-R	Y	Y	VIII	
Shell side, CVCS	AB-A07	AB-A100		C	3	1	3	III-3, TEMA-R	Y	Y		
16. Moderating HX	AB-106	AB-154	W	C	3	1	3	III-3, TEMA-R	Y	Y	VIII	
17. Letdown chiller HX:			W									
Tube side	AB-135	AB-160		C	3	1	3	III-3, TEMA-R	Y	Y	VIII	
Shell side	AB-135	AB-160		D	4	2	4	VIII, TEMA-C	N	N		
18. Reactor coolant filter housing	AB-B	AB-B	B	B	2	1	2	III-2	Y	Y	VIII	
19. Seal water return backflushable filter housing	AB-B	AB-B	B	B	2	1	2	III-2	Y	Y	VIII	
20. Boric acid filter	AB-D90	AB-D81	W	C	3	1	3	III-3	Y	Y	VIII	
21. Seal injection filter housings	AB-B	AB-B	B	B	2	1	2	III-2	Y	Y	VIII	
22. Letdown orifices			W	C	2	1	2	III-3	Y	Y		
23. Boric acid transfer pump orifices			B	C	3	1	3	III-3	Y	Y		
24. RCP seal bypass orifice			W	B	2	1	2	III-2	Y	Y		
25. Centrifugal charging pump miniflow orifice			W	B	2	1	2	III-2	Y	Y		

TABLE 3.2.2-1 (SHEET 9 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
26. CVCS chillers	AB-124		G	D	4	2	4	VIII	N	N		
27. CVCS chiller surge tank	AB-124		W	D	4	2	4	API-650	N	N		
28. CVCS chiller pump	AB-124		W	D	4	2	4	mfg	N	N		
29. CVCS chiller pump motor	AB-124		W	NA	6	2	E	NEMA MG1	N	N		
30. Deleted												
31. Thermal regenerative demineralizers	AB-A	AB-A	W	C	3	1	3	III-3	Y	Y	VIII	
32. CVCS cation bed demineralizer	AB-A	AB-A	W	C	3	1	3	III-3	Y	Y	VIII	
33. Mixed bed demineralizers	AB-A	AB-A	W	C	3	1	3	III-3	Y	Y	VIII	
34. Boron meter	AB-C80	AB-C33	W	D	4	2	4	B31.1	N	N		
35. This line has been intentionally left blank.												
36. This line has been intentionally left blank.												
37. Chemical mixing tank	AB-A39	AB-A69	W	D	4	2	4	VIII	N	N		
38. Safety-related CVCS instrumentation			W,B	NA	1	1	J	mfg	Y	Y		Note s
39. Piping from valve 273 to boric acid storage tank			B	C	3	1	3	III-3	Y	Y		
40. Piping and valves from boric acid storage tank to MOV HV-8104, boric acid mixing tee valve 188 and valve 505			W,B,S	C	3	1	3	III-3	Y	Y		

TABLE 3.2.2-1 (SHEET 10 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
41. Chemical mixing tank inlet and discharge piping from valve 176 to valve 181		B	D	4	2	4	B31.1	N	N		
42. Piping and valves from reactor makeup water supply valve 177 to valves 183, 176, and FV-0110B		B	C	3	1	3	III-3	Y	Y		
43. RCP seal injection piping and valves from check valves 006, 359, 360, and 361 to RCPs		B	A	1	1	1	III-1	Y	Y		
44. Letdown piping from valve LV-0459 to volume control tank		W,B	B	2	1	2	III-2	Y	Y		
45. Charging piping from volume control tank to valves 035 and 037		W,B	B	2	1	2	III-2	Y	Y		
46. Mixed bed demineralizer piping and valves from valve TV-0129 to check valve 083		W,B	C	3	1	3	III-3	Y	Y		
47. Mixed bed demineralizer piping from check valve 083 to reactor coolant filter		W,B	B	2	1	2	III-2	Y	Y		

TABLE 3.2.2-1 (SHEET 11 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
48. Moderating HX, letdown chiller HX (tube side) and letdown reheat HX(shell side) piping and valves			W,B	C	3	1	3	III-3	Y	Y		
49. Thermal regenerative demineralizer piping and valves			W,B	C	3	1	3	III-3	Y	Y		
50. Letdown reheat HX (tube side) piping and valves			W,B	B	2	1	2	III-2	Y	Y		
51. Letdown chiller HX (shell side) piping and valves			W,B	D	4	2	4	B31.1	N	N		
52. Piping and valves from boric acid batching tank to valve 304			W,B	D	4	2	4	B31.1	N	N		
53. Safety-related valve operators			W	NA	1	1	E	NEMA MG1	Y	Y		
54. Seal return piping and valves from RCP to seal water HX			W,B	B	2	1	2	III-2	Y	Y		
55. Piping and valves between seal water HX and volume control tank discharge piping			W,B	B	2	1	2	III-2	Y	Y		
56. Centrifugal charging pump lube oil coolers	AB-C115 & C118	AB-C16 & C17	W	NA	0	1	5	III-3	Y	Y	VIII	
57. Normal charging pump	AB-C111	AB-C12	S	B	2	1	2	III-2	Y	Y	VIII	
58. Normal charging pump motor	AB-C111	AB-C12	S	NA	6	2	E	NEMA MG1	N	N		
59. Normal charging pump minimum flow orifice	AB-C112	AB-C09	S	B	2	1	2	III-2	Y	Y		

TABLE 3.2.2-1 (SHEET 12 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
BORON RECYCLE SYSTEM												
1. Recycle evaporator feed pumps	AB-D34 & D35	shared	W	D	4	1	4	mfg	N	N		au
2. Recycle evaporator feed pump motors	AB-D34 & D35	shared	W	NA	6	2	E	NEMA MG1	N	N		au
3. Boron recycle holdup tanks	AB-D57 & D33	shared	B	C	3	1	3	III-3	Y	Y	VIII	au
4. Recycle evaporator feed backflushable filter housings	AB-B	shared	B	D	4	1	4	VIII	N	N		au
6. Recycle evaporator feed demineralizers	AB-C141	shared	W	D	4	1	4	VIII	N	N		au
7. Recycle evaporator condensate demineralizer	AB-C140	shared	W	D	4	2	4	VIII	N	N		au
10. Recycle evaporator condensate filter	AB-D96	shared	W	D	4	2	5	III-3	N	N		au
11. Piping and valves between recycle holdup tanks and recycle evaporator package			W,B	D	4	1	4	B31.1	N	N		au

TABLE 3.2.2-1 (SHEET 13 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
12. Condensate piping and valves between recycle evaporator package and reactor makeup water storage tank inlet check valve 002		W,B	D	4	2	4	B31.1	N	N		
13. Piping and valves between recycle evaporator feed demineralizers and recycle holdup tanks		W,B	D	4	1	4	B31.1	N	N		au
14. Letdown piping and valves from valve LV-0112A to recycle evaporator feed demineralizers		W,B	D	4	1	4	B31.1	N	N		
15. Piping and valves from recycle evaporator to boric acid storage tank inlet valve 273		B	D	4	2	4	B31.1	N	N		au
16. Recycle holdup tank vent eductor		W	D	4	1	7	B31.1	N	N		Note ag
17. Valve operators		W	NA	6	1	E	NEMA MG1	N	N		
18. Instrumentation		W	NA	6	2	J	mfg	N	N		Note s
19. Electronic metering pump		G	NA	4	2	4	mfg	N	N	VIII	
20. Zinc addition batch tank		G	NA	4	2	4	mfg	N	N	VIII	
CONTAINMENT ISOLATION SYSTEM											
1. Valves and piping		W,B	B	2	1	2	III-2	Y	Y		
2. Valve operators		W,B	NA	1	1	E	NEMA MG1	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 14 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3. Instrumentation and controls			W,B	NA	1	1	J	mfg	Y	Y		Note s
NUCLEAR SERVICE COOLING WATER SYSTEM												
1. NSCW pumps	NSW	NSW	B	C	3	1	3	III-3	Y	Y	III	
2. NSCW pump motors	NSW	NSW	B	NA	1	1	E	NEMA MG1	Y	Y	III	
3. NSCW transfer pumps	NSW	NSW	B	C	3	1	3	III-3	Y	Y	III	
4. NSCW transfer pump motors	NSW	NSW	B	NA	1	1	E	NEMA MG1	Y	Y	III	Note m
5.					(This line has been intentionally left blank.)							
6.					(This line has been intentionally left blank.)							
7.					(This line has been intentionally left blank.)							
8. Tower fans	NSW	NSW	B	NA	0	1	5	AMCA	Y	Y	III	
9. Fan motors	NSW	NSW	B	NA	1	1	E	NEMA MG1	Y	Y	III	
10. Valves and piping (outside containment)			B	C	3	1	3	III-3	Y	Y		
11. Valves and piping inside containment)			B	B	2	1	2	III-2	Y	Y		
12. Safety-related valve operators			B	NA	1	1	E	NEMA MG1	Y	Y		
13. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
14. Containment auxiliary air coolers (tube side)	C-261′	C-261′	B	B	2	1	2	III-2	Y	Y	1-B	
15. Containment cavity coolers (tube side)	C-206′	C-206′	B	B	2	1	2	III-2	Y	Y	1-A	
16. Piping penetration area coolers (tube side)	AB-209′	AB-206′	B	C	3	1	3	III-3	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 15 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
17. Containment spray pump motor coolers	AB-D76 & D77	AB-D04 & D05	W	C	3	1	3	III-3	Y	Y	VIII	
18. Safety injection pump motor coolers	AB-215 & B19	AB-B117 & B119	W	C	3	1	3	III-3	Y	Y	VIII	
19. RHR pump motor coolers	AB-D48 & D49	AB-D21 & D22	W	C	3	1	3	III-3	Y	Y	VIII	
20. CCW pump motor coolers	AB-A03 & A05	AB-A96 & A98	B	C	3	1	3	III-3	Y	Y	VIII,B4	
21. Centrifugal charging pump motor coolers	AB-C115 & C118	AB-C16 & C17	W	C	3	1	3	III-3	Y	Y	VIII	
22. Containment air coolers (tube side)	C-238′	C-238′	B	B	2	1	2	III-2	Y	Y	1-B	
23. NSCW pump motor coolers	NSW	NSW	B	NA	0	1	5	mfg	Y	Y	VI	
24. NSCW tower basin transfer line (buried pipe)			B	C	3	1	3	III-3	Y	Y		
COMPONENT COOLING WATER SYSTEM												
1. CCW surge tanks	AB-213 & 214	AB-201 &202	B	C	3	1	3	III-3	Y	Y	VIII	
2. CCW pumps	AB-A03 & A05	AB-A96 & A98	B	C	3	1	3	III-3	Y	Y	VIII,B4	
3. CCW HXs	AB-213 & 214	AB-201 & 202	B	C	3	1	3	III-3, TEMA-R	Y	Y	VIII	
4. CCW pump motors	AB-A03 & A05	AB-A96 & A98	B	NA	1	1	E	NEMA MG1	Y	Y	VIII,B4	
5. CCW chemical addition tanks	AB-A03 & A05	AB-A96 & A98	B	D	4	2	4	VIII	N	N		
6. Chemical addition tank valves and piping			B	D	4	2	4	B31.1	N	N		
7. Other process valves and piping			B	C	3	1	3	III-3	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 16 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
8. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
AUXILIARY COMPONENT COOLING WATER SYSTEM												
1. ACCW surge tank	AB-132	AB-105	B	D	4	1	5	III-3	N	N		Note z
2. ACCW pumps	AB-B23 & B24	AB-B112 & B113	B	D	4	1	5	III-3	N	N		for 415 components
3. ACCW pump motors	AB-B23 & B24	AB-B112 & B113	B	NA	1	1	E	NEMA MG1	Y	Y	VIII,B4	Note m
4. ACCW HXs	AB-132 & 134	AB-103 & 105	B	C	3	1	3	III-3, TEMA-R	Y	Y	VIII	
5. Containment penetration piping and valves			B	B	2	1	2	III-2	Y	Y		
6. Chemical addition feeder tank	AB-B23	AB-B112	B	D	4	2	4	VIII	N	N		
7. Other safety-related valves and piping			B	C	3	1	3	III-3	Y	Y		
8. All other valves and piping			B	D	4	1	5	III-3	N	N		
9. Safety-related valve operators			B	NA	1	1	E	NEMA MG1	Y	Y		
10. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
11. All other instrumentation			B	NA	6	2	J	mfg	N	N		
SPENT FUEL COOLING AND PURIFICATION SYSTEM												
1. SFP HXs	AB-A53 FB-A07	AB-A91 FB-A04	W	C	3	1	3	III-3, TEMA-R	Y	Y	VIII,VII	
2. SFP pumps	AB-A53 FB-A07	AB-A91 FB-A04	W	C	3	1	3	III-3	Y	Y	VIII	
3. SFP pump motors	AB-A53 FB-A07	AB-A91 FB-A04	W	NA	1	1	E	NEMA MG1	Y	Y	VIII	Note m

TABLE 3.2.2-1 (SHEET 17 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
4.	Refueling water purification pump	AB-A40	AB-A89	B	D	4	2	4	mfg	N	N		
5.	Refueling water purification pump motor	AB-A40	AB-A89	B	NA	6	2	E	NEMA MG1	N	N		
6.	SFP demineralizer	AB-A	AB-A	W	D	4	2	4	III-3	N	N		Note r
7.	SFP strainers			W	D	4	2	4	mfg	N	N	VIII	
8.	SFP filter housing	AB-B	AB-B	B	D	4	2	4	VIII	N	N		Note r
9.	SFP skimmer pump	AB-A53	AB-A91	W	D	4	2	4	mfg	N	N		
10.	SFP skimmer pump motor	AB-A53	AB-A91	W	NA	6	2	E	NEMA MG1	N	N		
11.	SFP skimmer filter	AB-D94	AB-D83	W	D	4	2	4	VIII	N	N		
12.	SFP skimmer strainers	FB	FB	W	D	4	2	4	VIII	N	N		
13.	Purification and skimmer-related valves and piping			B	D	4	2	4	B31.1	N	N		
14.	Cooling-related valves and piping W2 in.			W,B	C	3	1	3	III-3	Y	Y		
15.	All other cooling-related valves and piping K2 in.			B	C	3	1	3	III-3	Y	Y		
16.	Safety-related instrumentation			W,B	NA	1	1	J	mfg	Y	Y		Note s
17.	Safety-related valve operators			W	NA	1	1	E	NEMA MG1	Y	Y		
REACTOR MAKEUP WATER SYSTEM													
1.	Makeup pumps	AB-B22	AB-B114	B	D	4	1	5	III-3	N	N		Note r

TABLE 3.2.2-1 (SHEET 18 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
2.	Makeup pump motors	AB-B22	AB-B114	B	NA	6	2	E	NEMA MG1	N	N		Note m
3.	Degasifier vacuum pumps	O	O	B	D	4	2	4	mfg	N	N		
4.	Degasifier feed pump	O	O	B	D	4	2	4	mfg	N	N		
5.	Degasifier feed/transfer pump	O	O	B	D	4	2	4	mfg	N	N		
6.	Degasifier transfer pump	O	O	B	D	4	2	4	mfg	N	N		
7.	Vacuum degasifier	O	O	B	D	4	2	4	mfg	N	N		
8.	Makeup water process piping and valves to SFP			B	D	4	1	5	III-3	N	N		Note r
9.	Degasifier piping and valves			B	D	4	2	4	B31.1	N	N		
10.	Instrumentation			B	NA	6	2	J	mfg	N	N		
11.	Degasifier pump motors	O	O	B	NA	6	2	E	NEMA MG1	N	N		
12.	All other piping			B	D	4	2	4	B31.1	N	N		
WASTE PROCESSING SYSTEM - LIQUID													Note w Note r for all 417 and 427 components designated as III-3.
1.	Waste holdup tank	AB-D63	AB-D13	W	D	4	1	7	III-3	N	N		
2.	Waste evaporator feed pump	AB-D62	AB-D14	W	D	4	1	7	III-3	N	N		
3.	Waste evaporator-feed pump motor	AB-D62	AB-D14	W	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 19 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
4. Waste evaporator feed backflushable filter housing	AB-B	AB-B	B	D	4	1	7	III-3	N	N		Note an
5. Deleted												
6. Deleted												
7. Deleted												
8. Deleted												
9. Waste evaporator reagent tank	AB-C63	AB-C40	W	D	4	2	4	VIII	N	N		av
10. Waste evaporator condensate demineralizer	AB-C146	AB-C149	W	D	4	2	7	III-3	N	N		
11. Waste evaporator condensate filter	AB-D91	AB-D86	W	D	4	2	7	III-3	N	N		
12. Waste evaporator condensate pump	AB-D64	AB-D12	W	D	4	2	7	III-3	N	N		
13. Waste evaporator condensate pump motor	AB-D64	AB-D12	W	NA	6	2	E	NEMA MG1	N	N		
14. Waste evaporator condensate tank	AB-C80	AB-C33	W	D	4	2	7	III-3	N	N		
15. Chemical drain tank	AB-D45	shared	W	D	4	2	7	VIII	N	N		
16. Chemical drain tank pump	AB-D47	shared	W	D	4	2	7	VIII	N	N		
17. Chemical drain tank pump motor	AB-D47	shared	W	NA	6	2	E	NEMA MG1	N	N		
18. Spent resin storage tank	AB-D36	AB-D37	W	D	4	1	7	III-3	N	N		

TABLE 3.2.2-1 (SHEET 20 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
19.	Spent resin sluice pump	AB-D39	AB-D40	W	D	4	1	7	mfg	N	N		
20.	Spent resin sluice pump motor	AB-D39	AB-D40	W	NA	6	2	E	NEMA MG1	N	N		
21.	Sample vessels	AB-C63	AB-C40	B	D	4	2	7	VIII	N	N		
22.	Solidification strainer			W	D	4	2	7	B 31.1	N	N		
23.	Spent resin sluice back-flushable filter housing	AB-B	AB-B	B	D	4	1	7	III-3	N	N		
24.	Floor drain tank	AB-D46	AB-D24	W	D	4	2	7	III-3	N	N		
25.	Floor drain tank pump		AB-D29	S	D	4	2	7	mfg	N	N		
		AB-D50		S	D	4	2	7	mfg	N	N		
26.	Floor drain tank pump motor	AB-D50	AB-D29	W	NA	6	2	E	NEMA MG1	N	N		
27.	Floor drain tank strainer	AB-D	AB-D	W	D	4	2	7	mfg	N	N		
28.	Floor drain tank backflushable filter housing	AB-B	AB-B	B	D	4	2	7	III-3	N	N		
29.	Waste monitor tank	AB-C81 & C82	AB-C34 & C35	W	D	4	2	7	III-3	N	N		
30.	Waste monitor tank pump	AB-D58 & D59	AB-D17 & D18	W	D	4	2	7	III-3	N	N		
31.	Waste monitor tank pump motor	AB-D58 & D59	AB-D17 & D18	W	NA	6	2	E	NEMA MG1	N	N		
32.	Waste monitor tank backflushable filter housing	AB-B	AB-B	B	D	4	2	7	III-3	N	N		
33.	Waste monitor tank demineralizer	AB-C143	AB-C139	W	D	4	2	7	III-3	N	N		
33a.	Auxiliary waste monitor tank	N/A	AB-D08	B	D	4	2	7	API-650	N	N		
33b.	Auxiliary waste monitor tank pump	N/A	AB-D109	B	D	4	2	7	mfg	N	N		

TABLE 3.2.2-1 (SHEET 21 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
33c.	Auxiliary waste monitor tank pump motor	N/A	AB-D109	B	N/A	6	2	E	mfg	N	N		
34.	Laundry and hot shower tank	AB-D25	shared	W	D	4	2	7	VIII	N	N		
35.	Laundry and hot shower tank pump	AB-D26	shared	W	D	4	2	7	mfg	N	N		
36.	Laundry and hot shower tank pump motor	AB-D26	shared	W	NA	6	2	E	NEMA MG1	N	N		
37.	Laundry and hot shower tank strainer	AB-D26	shared	W	D	4	2	7	B31.1	N	N		
38.	Laundry and hot shower tank filter	AB-D88	shared	W	D	4	2	7	B31.1	N	N		Note ao
39.	Reactor coolant drain tank	C-171′	C-171′	W	D	4	2	7	III-3	N	N		
40.	Reactor coolant drain tank pump	C-171′	C-171′	W	D	4	2	7	III-3	N	N		
41.	Reactor coolant drain tank pump motor	C-171′	C-171′	W	NA	6	2	E	NEMA MG1	N	N		
42.	Reactor coolant drain tank HX:			W									
	Tube side	C-171′	C-171′		D	4	2	7	III-3, TEMA-R	N	N		
	Shell side, ACCW	C-171′	C-171′		D	4	1	5	III-2, TEMA-R	N	N		Note t
43.	Piping and valves from waste evaporator and evaporator condensate demineralizer through evaporator condensate pump discharge valve119			B	D	4	2	7	B31.1	N	N		av

TABLE 3.2.2-1 (SHEET 22 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
44. Piping and valves from waste holdup tank to waste evaporator		B	D	4	1	7	III-3	N	N		
45. Piping and valves from spent resin storage tank to valves HV-7325 and 143 and from valves HV-7305 and 058 to spent resin storage tank		W,B	D	4	1	7	III-3	N	N		
46. Process piping and valves downstream of spent resin tank valves HV-7325 and 143		W,B	D	4	2	7	B31.1	N	N		
47. Piping and valves from chemical drain tank to RPF		B	D	4	2	7	B31.1	N	N		
48. Process piping and valves from laundry and hot shower tank to waste monitor tank, from floor drain tank to waste monitor tank and demineralizer, from demineralizer to waste monitor tank, and discharge from waste monitor tanks		W,B	D	4	2	7	B31.1	N	N		

TABLE 3.2.2-1 (SHEET 23 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
49. Reactor coolant drain tank inlet and discharge piping and valves to valve LV-1003			W,B	D	4	2	7	B31.1	N	N		
50. RPF Demineralizers	RPF		S	D	4	2	7	B31.1	N	N		
51. Instrumentation			W,B	NA	6	2	J	mfg	N	N		Note s
52. RPF Demineralizers Filters			S	D	4	2	7	B31.1	N	N		
WASTE PROCESSING SYSTEM - GASEOUS												
1. Gas decay tanks	AB-B33 B36,B38 B42,B43 B40,B41	AB-B78 B82,B83 B88,B90 B84,B85	W	C	3	1	3	III-3	Y	Y	VIII	
2. Gas decay tank drain pump	AB-B35	AB-B89	W	D	4	2	4	mfg	N	N		
3. Gas decay tank drain pump motor	AB-B35	AB-B89	W	NA	6	2	E	NEMA MG1	N	N		
4. Waste gas drain filter	AB-D95	AB-D82	W	D	4	2	4	VIII	N	N		
5. Sample vessel	AB-B45	AB-B79	B	D	4	2	4	VIII	N	N		
6. Waste gas compressor package	AB-B64 & B68	AB-B100 & B101	W	D	4	1	7	B31.1	N	N	VIII	Note ag
7. Catalytic H_2 recombiner and gas analyzer package	AB-B58 & B59	AB-B74 & B76	W	D	4	1	7	B31.1	N	N	VIII	Note ag
8. Waste gas decay shutdown tank	AB-B46 & B47	shared	W	C	3	1	3	III-3	Y	Y	VIII	
9. Process piping and valves			W,B	C	3	1	3	III-3	Y	Y		Note ay
10. Drain piping and valves			B	D	4	2	4	B31.1	N	N		
11. Safety-related valve operators			W	NA	1	1	E	NEMA MG1	Y	Y		
12. Safety-related instrumentation			W	NA	1	1	J	mfg	Y	Y		Note s
13. Gas traps			S,W	D	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 24 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
STEAM GENERATOR BLOWDOWN SYSTEM												
1. Blowdown HXs	AB-C108	AB-C02	W	D	4	2	7	VIII, TEMA-C	N	N		
2. Steam generator drain pump	AB-C108	AB-C02	W	D	4	2	7	mfg	N	N		
3. Blowdown backflushable filters	AB-B	AB-B	B	D	4	2	7	VIII	N	N		
4. Demineralizers	AB-A	AB-A	W	D	4	2	7	VIII	N	N		
5. Spent resin storage tank	AB-C75	AB-C54	W	D	4	2	7	VIII	N	N		
6. Spent resin sluice pump	AB-C74	AB-C55	W	D	4	2	7	mfg	N	N		
7. Spent resin sluice pump motor	AB-C74	AB-C55	W	NA	6	2	E	NEMA MG1	N	N		
8. Spent resin sluice filter	AB-A58	AB-A58	W	D	4	2	7	VIII	N	N		
9. Process valves and piping b2 in. outside containment			W,B	D	4	2	7	B31.1	N	N		
10. Process valves and piping K2 in.			B	D	4	2	7	B31.1	N	N		
11. Process valves and piping inside containment through outer isolation valves			W,B	B	2	1	2	III-2	Y	Y	VIII	
12. Blowdown trim HX	AB-B03	AB-B124	W	D	4	2	7	VIII, TEMA-C	N	N		
13. Steam generator drain pump motor	AB-C108	AB-C02	W	NA	6	2	E	NEMA MG1	N	N		
14. Blowdown outlet filters	AB-A58	AB-A58	B	D	4	2	7	VIII	N	N		
15. Instrumentation			W,B	NA	6	2	J	mfg	N	N		
16. Steam generator blowdown cartridge filter	AB-A49		S	D	4	2	7	VIII	N	N		

TABLE 3.2.2-1 (SHEET 25 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
BACKFLUSHABLE FILTER SYSTEM												
1. Crud tank	AB-C	AB-C	B	D	4	2	7	VIII	N	N		
2. Crud tank pumps	AB-C93 & C94	AB-C28 & C29	B	D	4	2	7	mfg	N	N		
3. Crud tank pump motors	AB-C93 & C94	AB-C28 & C29	B	NA	6	2	E	NEMA MG1	N	N		
4. N_2 accumulator	AB-B63	AB-B103	B	D	4	2	4	VIII	N	N		
5. N_2 system valves and piping			B	D	4	2	4	B31.1	N	N		
6. Flush valves and piping			B	D	4	2	4	B31.1	N	N		
7. Instrumentation			B	NA	6	2	J	mfg	N	N		
MAIN STEAM SYSTEM												
1. Steam generator (shell side)	C	C	W	B	2	1	2	III-2	Y	Y	1-A	
2. Steam diffuser	C	C	W	B	2	1	2	III-2	Y	Y	1-A	
3. Piping from SG to weld 1 after 5-way restraint			B	B	2	1	2	III-2	Y	Y	1-A	
4. Piping downstream of 5-way restraint to turbines			B	D	4	2	4	B31.1	N	N		
5. Safety valves	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	B	2	1	2	III-2	Y	Y	II	
6. Atmospheric power-operated relief valves	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	B	2	1	2	III-2	Y	Y	II	
7. Atmospheric power-operated relief valves operators	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	NA	1	1	E	NEMA MG1	Y	Y	II	
8. Main steam isolation valves	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	B	2	1	2	III-2	Y	Y	II	

TABLE 3.2.2-1 (SHEET 26 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9. Main steam isolation valve actuators	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	NA	1	1	E	mfg	Y	Y	II	
10. Steam packing exhauster condenser	TB-245′	TB-245′	S	D	4	2	4	mfg TEMA-C	N	N		
11. Safety-related valve operators			B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
12. Drain and test valves and piping upstream of forged section			B	B	2	1	2	III-2	Y	Y		
13. Wet layup pumps	C-180′	C-180′	B	D	4	2	4	mfg	N	N		
14. Wet layup pump motors	C-180′	C-180′	B	NA	6	2	E	NEMA MG1	N	N		
15. Wet layup piping and valves			B	D	4	2	4	B31.1	N	N		
16. Mechanical pressure, flow, and level instruments			W	B	2	1	2	III-2	Y	Y		
17. Safety-related instrumentation			W	NA	1	1	J	mfg	Y	Y		Note s
20. Steam flow limiters			W	B	2	1	2	III-2	Y	Y		
21. Steam packing exhauster blower	TB-245′	TB-245′	S	D	4	2	4	mfg	N	N		
22. Steam packing exhauster blower motor	TB-245′	TB-245′	S	NA	6	2	E	NEMA MG1	N	N		
23. Main steam isolation valve bypass valves	AB-108 EG-123 & 122	AB-159 EB-123 & 122	B	B	2	1	2	III-2	Y	Y	II	
24. Main steam isolation valve bypass actuators	AB-108 EB-123 & 122	AB-159 EB-123 & 122	B	NA	1	1	E	mfg	Y	Y	II	

TABLE 3.2.2-1 (SHEET 27 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
AUXILIARY FEEDWATER SYSTEM													
1.	Auxiliary feed pump turbine	AFP	AFP	B	NA	0	1	5	NEMA SM23	Y	Y	VI	
2.	Auxiliary feed pumps	AFP	AFP	B	C	3	1	3	III-3	Y	Y	VI	
3.	Auxiliary feed pump motors	AFP	AFP	B	NA	1	1	E	NEMA MG1	Y	Y	VI	
4.	Auxiliary feed turbine steam valve (auto)	AFP	AFP	B	C	3	1	3	III-3	Y	Y	VI	
5.	Auxiliary feed turbine steam supply valve motor	AFP	AFP	B	NA	1	1	E	NEMA MG1	Y	Y	VI	
6.	AFW piping up to AFW flow control MOVs			B	C	3	1	3	III-3	Y	Y		
7.	AFW piping and valves including AFW flow control MOVs			B	B	2	1	2	III-2	Y	Y		
8.	AFW flow control MOV motors			B	NA	1	1	E	NEMA MG1	Y	Y		
9.	AFW pump suction valves and piping			B	C	3	1	3	III-3	Y	Y		
10.	AFW pump suction MOV motors			B	NA	1	1	E	NEMA MG1	Y	Y		
11.	Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
12.	Flow limiting orifices			B	B	2	1	2	III-2	Y	Y		
13.	Degasifier feed pump	O	O	B	D	4	2	4	mfg	N	N		
14.	Degasifier feed/transfer pump	O	O	B	D	4	2	4	mfg	N	N		
15.	Degasifier transfer pump	O	O	B	D	4	2	4	mfg	N	N		
16.	Vacuum degasifier	O	O	B	D	4	2	4	mfg	N	N		
17.	Degasifier silencer/separator	O	O	B	D	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 28 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
18. Degasifier ejector	O	O	B	D	4	2	4	VIII	N	N		
19. Degasifier piping and valves			B	D	4	2	4	B31.1	N	N		
20. Degasifier pump motors	O	O	B	NA	6	2	E	NEMA MG1	N	N		
21. AFW turbine lube oil cooler:												
Oil side	AFP	AFP	B	NA	0	1	5	III-3	Y	Y	II	
Water side			B	C	3	1	3	III-3	Y	Y	II	
CONDENSATE AND FEEDWATER SYSTEM												
1. Main feed line isolation valves			B	B	2	1	2	III-2	Y	Y		
2. Main feed line isolation valve actuators			B	NA	1	1	E	NEMA MG1	Y	Y		
3. Main and auxiliary feed inlet check valves			B	B	2	1	2	III-2	Y	Y		
4. Piping from forged section to SG			B	B	2	1	2	III-2	Y	Y		
5. Piping upstream of forged section			B	D	4	2	4	B31.1	N	N		
6. Main feed regulating valves			W	C	3	1	3	III-3	Y	Y		
7. Main feed regulating bypass valves			B	C	3	1	3	III-3	Y	Y		
8. Condensate pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
9. Condensate pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
10. Feed pumps	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
11. Feed pump turbines	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
12. Speed control valves			S	D	4	2	4	B31.1	N	N		
13. Other piping and valves			S	D	4	2	4	B31.1	N	N		

TABLE 3.2.2-1 (SHEET 29 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
14. Feed isolation instrumentation and controls			B	NA	1	1	J	mfg	Y	Y		Note s
15. Other instrumentation and controls			S	NA	6	2	J	mfg	N	N		
16.	(This line has been intentionally left blank.)											
17. Feedwater isolation bypass valves			B	B	2	1	2	III-2	Y	Y	II	
18. Feedwater isolation bypass valves actuator			B	NA	1	1	E	mfg	Y	Y	II	
POWER CONVERSION SYSTEM												
1. Main turbines	TB-270′	TB-270′	S	D	4	2	4	mfg	N	N		
2. Generator	TB-270′	TB-270′	S	D	4	2	4	mfg	N	N		
3. Main condensers	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
4. Feedwater heaters	TB-220′ 245′ & 290′	TB-220′ 245′ & 290′	S	D	4	2	4	VIII	N	N		
5. Moisture separator reheaters	TB-270′	TB-270′	S	D	4	2	4	VIII	N	N		
6. Moisture separator drain tanks	TB-245′	TB-245′	S	D	4	2	4	VIII	N	N		
7. Heater drain tanks	TB-220′	TB-220′		D	4	2	4	VIII	N	N		
8. Reheater drain tanks	TB-245′	TB-245′	S	D	4	2	4	VIII	N	N		
9. Heater drain pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
10. Heater drain pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
11. Extraction steam valves and piping			S	D	4	2	4	B31.1	N	N		
12. Other valves and piping			S	D	4	2	4	B31.1	N	N		
13. Instrumentation			S	NA	6	2	J	mfg	N	N		
14. Generator H_2 cooler	TB-270′	TB-270′	S	D	4	2	4	mfg	N	N		
15. Generator stator coolant pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 30 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
16. Generator stator coolant pump motor	TB-220′	TB-220′	S	NA	6	2	E	mfg	N	N		
17. Generator stator cooler	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
CONDENSER AIR EJECTOR SYSTEM												
1. Air ejectors	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
2. Air ejector condensers	TB-220′	TB-220′	S	D	4	2	4	mfg TEMA-C	N	N		
3. Vacuum pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
4. Vacuum pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
5. Vacuum pump exhaust silencers	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
6. Vacuum pump seal water HXs	TB-195′	TB-195′	S	D	4	2	4	VIII, TEMA-C	N	N		
7. Seal water pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
8. Valves and piping			S	D	4	2	4	B31.1	N	N		
9. Seal water pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
10. Instrumentation			S	NA	6	2	J	mfg	N	N		
CIRCULATING WATER SYSTEM												
1. Circulating water pumps	O	O	S	D	4	2	4	mfg	N	N		
2. Circulating water pump motors	O	O	S	NA	6	2	E	NEMA MG1	N	N		
3. Condenser water box drain pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
4. Condenser water box drain pump motor	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
5. Piping			S	NA	6	2	6	B31.1	N	N		
6. Valves			S	NA	6	2	6	B31.1	N	N		
7. Instrumentation			S	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 31 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
RIVER INTAKE STRUCTURE SYSTEMS												
1. Mixing chamber	VB	shared	S	NA	6	2	C	mfg	N	N		
2. River makeup and dilution pumps	O	shared	S	NA	6	2	6	mfg	N	N		
3. Mixing chamber sample unit	VB	shared	S	NA	6	2	6	mfg	N	N		
4. Mixing chamber sample pump	VB	shared	S	NA	6	2	6	mfg	N	N		
5. Traveling and stationary screens	O	shared	S	NA	6	2	6	mfg	N	N		
6. Process piping and valves			S	NA	6	2	6	B31.1	N	N		
7. Pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
8. Instrumentation			S	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 32 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONDENSATE CHEMICAL INJECTION SYSTEM												
1. Methoxypropylamine (MPA) storage tank	O	shared	S	D	4	2	4	VIII	N	N		
2. MPA transfer pumps	O	O	S	D	4	2	4	mfg	N	N		
3. Hydrazine day tank	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
4. MPA day tank	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
5. Hydrazine storage tank	O	shared	S	D	4	2	4	mfg	N	N		
6. Hydrazine transfer pump	O	O	S	D	4	2	4	mfg	N	N		
7. Hydrazine mixing pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
8. Deleted												
9. Hydrazine dispensing pump (low volume)	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
10. Hydrazine dispensing pump (high volume)	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
11. Continuous condensate feed pumps (Hydrazine and MPA)	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
12. Batch tank	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
13. Batch mixing pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
14. Steam generator layup pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
15. Feedwater layup pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
16. Pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
17. Piping and valves			S	D	4	2	4	B31.1	N	N		
18. Instrumentation			S	NA	6	2	J	mfg	N	N		
19. Ammonium Chloride Pump	TB-195′	O	S	NA	4	2	4	mfg	N	N		
20. Ammonium Chloride Tank	TB-195′	O	S	NA	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 33 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONDENSATE FILTER DEMINERALIZER SYSTEM												
1. Spent resin transfer pump	RTB	RTB	B	D	4	2	4	mfg	N	N		
2. Spent resin transfer pump motor	RTB	RTB	B	NA	6	2	E	NEMA MG1	N	N		
3. Backwash recovery pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
4. Backwash recovery pump motor	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
5. Backwash pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
6. Backwash pump motor	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
7. Decant transfer pump	RTB	RTB	B	D	4	2	4	mfg	N	N		
8. Decant transfer pump motor	RTB	RTB	B	NA	6	2	E	NEMA MG1	N	N		
9. Spent resin pump	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
10. Spent resin pump motor	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
11. Precoat tank	TB-220′	TB-220′	S	D	4	2	4	API 650	N	N		
12. Precoat tank agitator	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
13. Overlay tank	TB-220′	TB-220′	S	D	4	2	4	API 650	N	N		
14. Overlay tank agitator	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
15. Holding pumps	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
16. Holding pump motors	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
17. Resin traps	TB-220′	TB-220′	S	D	4	2	4	VIII	N	N		
18. Precoat pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 34 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
19. Precoat pump motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
20. Overlay pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
21. Overlay pump motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
22. Decant collection tank	RTB	RTB	B	D	4	2	4	API 650	N	N		
23. Backwash recovery tank	TB-195′	TB-195′	S	D	4	2	4	API 650	N	N		
24. Backwash recovery tank scraper	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
25. Phase separator tank	RTB	RTB	B	D	4	2	4	API 650	N	N		
26. Powdex vessels	TB-220′	TB-220′	S	D	4	2	4	VIII	N	N		
27. Air receiver	TB-195′	TB-195′	S	D	4	2	4	VIII	N	N		
28. Valves and piping			S,B	D	4	2	4	B31.1	N	N		
29. Instrumentation			S	NA	6	2	J	mfg	N	N		
30. Dirty spent resin holding tank	TB-195′	shared	G	D	4	2	4	API 650	N	N		
31. Clean spent resin holding tank	TB-195′	shared	G	D	4	2	4	API 650	N	N		
32. Recirculation pump	TB-195′	shared	G	D	4	2	4	mfg	N	N		
33. Recirculation pump motor	TB-195′	shared	G	N/A	6	2	E	NEMA MG1	N	N		
34. Dewatering pump	TB-195′	shared	G	D	4	2	4	mfg	N	N		
35. Dewatering pump motor	TB-195′	shared	G	N/A	6	2	E	NEMA MG1	N	N		
36. Dewatering filters	TB-195′	shared	G	D	4	2	4	mfg	N	N		
37. pressure filter skid	TB-195′	shared	G	D	4	2	4	mfg	N	N		
PLANT MAKEUP WATER WELL SYSTEM												
1. Demineralizer booster pumps	VB	shared	S	NA	6	2	6	mfg	N	N		
2. Demineralizer backwash pump	VB	shared	S	NA	6	2	6	mfg	N	N		ba
3. Plant makeup well pumps	O	shared	S	NA	6	2	6	mfg	N	N		

TABLE 3.2.2-1 (SHEET 35 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
4. Makeup well water storage tank	O	shared	S	NA	6	2	6	API 650	N	N		
5. Pump motors			S	NA	6	2	E	NEMA MG1	N	N		
6. Piping and valves			S	NA	6	2	6	B31.1	N	N		
7. Instrumentation			S	NA	6	2	J	mfg	N	N		
8. Flushing water storage tank Pump (Note 1)	O	shared	G	NA	6	2	6	mfg	N	N		
9. Flushing water storage tank (Note 1)	O	shared	G	NA	6	2	6	API 650	N	N		
PLANT MAKEUP WATER TREATMENT SYSTEM												
1. Vendor supplied water treatment system	VB	shared	S	NA	6	2	6	mfg	N	N		

Notes:

1. The flushing water storage tank and pump have been retired in place.

TABLE 3.2.2-1 (SHEET 36 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
2. All other pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
3. Piping and valves	VB	shared	S	NA	6	2	6	B31.1	N	N		
4. Instrumentation	VB	shared	S	NA	6	2	J	mfg	N	N		
PLANT MAKEUP WATER TREATMENT SYSTEM WASTE NEUTRALIZATION SYSTEM												
1. Neutralizer transfer pumps	O	shared	S	NA	6	2	6	mfg	N	N		
2. Pump motors	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
3. Valves and piping			S	NA	6	2	6	B31.1	N	N		
4. Instrumentation			S	NA	6	2	J	mfg	N	N		
DEMINERALIZED WATER SYSTEM												
1. Demineralized water storage tank	O	shared	S	NA	6	2	4	API 650	N	N		
2. Transfer pumps	VB	shared	S	NA	6	2	4	mfg	N	N		
3. Transfer pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
4. Valves and piping			S,B	NA	6	2	4	B31.1	N	N		
5. Transfer booster pumps	AB-A17	shared	B	NA	6	2	4	mfg	N	N		
6. Transfer booster pump motors	AB-A17	shared	B	NA	6	2	E	NEMA MG1	N	N		
7. Instrumentation			S,B	NA	6	2	J	mfg	N	N		
DIESEL GENERATOR SYSTEMS												
1. Day tanks	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
2. Fuel transfer pumps	VB	VB	B	NA	0	1	3	III-3	Y	Y	IV	

TABLE 3.2.2-1 (SHEET 37 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	((g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3.	Fuel transfer pump motors	VB	VB	B	NA	1	1	E	NEMA MG1	Y	Y	IV	
4.	Duplex fuel filters	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
5.	Duplex fuel strainers	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
6.	Engine driven fuel pump	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
7.	Engine driven lube oil pump	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
8.	Lube oil HX	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
9.	Lube oil heater	DB	DB	B	NA	0	1	5	mfg	Y	Y		Casing is ASME VIII
10.	Lube oil keep warm pump	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
11.	Lube oil keep warm pump motor	DB	DB	B	NA	6	2	E	mfg	N	N		
12.	Lube oil strainers	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
13.	Lube oil filters	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
14.	Cooling jacket water heater	DB	DB	B	NA	0	1	5	mfg	Y	Y		Casing is ASME VIII
15.	Engine driven jacket water pump	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
16.	Jacket water keep warm pump	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
17.	Jacket water keep warm pump motor	DB	DB	B	NA	6	2	E	mfg	N	N		
18.	Jacket water HX	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
19.	Jacket water standpipe	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
20.	Air compressors	DB	DB	B	NA	6	2	6	mfg	N	N		
21.	Air receivers	DB	DB	B	NA	0	1	3	III-3	Y	Y	IV	
22.	Air dryers	DB	DB	B	NA	6	2	6	mfg	N	N		
23.	Air compressor after coolers	DB	DB	B	NA	6	2	6	mfg	N	N		
24.	Intake and exhaust silencers	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
25.	Intake air filter	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	

TABLE 3.2.2-1 (SHEET 38 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
26. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y	IV	Note s
27. Diesel generators	DB	DB	B	NA	0	1	5	DEMA, NEMA	Y	Y	IV	
28. Fuel oil pressure regulating valve	DB	DB	B	NA	0	1	5	mfg	Y	Y	IV	
29. Engine boundary piping and valves			B	NA	0	1	5	mfg	Y	Y		
30. Engine auxiliaries piping and valves			B	NA	0	1	3	III-3	Y	Y		
31. Flame arresters	VB	VB	B	NA	6	2	6	mfg	N	N		
32. Lube oil sump	DB	DB	B	NA	0	1	3	III-3	Y	Y		
33. Fuel oil storage tanks	VB	VB	B	NA	0	1	3	III-3	Y	Y		
34. Exhaust piping			B	NA	0	1	5	mfg	Y	Y		
35. Category 1 buried pipe			B	NA	0	1	3	III-3	Y	Y		
FIRE PROTECTION SYSTEMS												Note v
1. Diesel fire pumps	FPH	FPH	B	NA	6	2	9	NFPA	N	N		
2. Diesel engines	FPH	FPH	B	NA	6	2	9	mfg	N	N		
3. Motor driven fire pump	FPH	FPH	B	NA	6	2	9	NFPA	N	N		
4. Pump motors	FPH	FPH	B	NA	6	2	E	NEMA MG1	N	N		
5. Jockey pumps	FPH	FPH	B	NA	6	2	9	NFPA	N	N		
6. Diesel fuel oil tanks	O	shared	B	NA	6	2	9	NFPA	N	N		
7. Water storage tanks	O	shared	B	NA	6	2	9	NFPA	N	N		
8. Water system piping and valves			B,S	NA	6	2	9	NFPA	N	N		
9. Halon system piping, valves, and components			B	NA	6	2	9	NFPA	N	N		
10. Instrumentation			B	NA	6	2	J	mfg	N	N		
11. Hydrant, hose-houses	O	shared	S	NA	6	2	9	NFPA	N	N		

TABLE 3.2.2-1 (SHEET 39 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
12.	Hose cabinet and fire extinguishers	VB	VB	S	NA	6	2	9	NFPA	N	N		
13.	Local suppression indication panel	VB	VB	B	NA	6	2	9	mfg	N	N		
FIRE PROTECTION SYSTEMS - SEISMIC CATEGORY 1													
1.	Valves and piping			B	C	3	1	3	III-3	N	N		Note x
AUXILIARY GAS SYSTEMS (N_2 , H_2 , AND 0_2)													
1.	N_2 low pressure vaporizer	O	shared	S	NA	6	2	6	mfg	N	N		
2.	N_2 high pressure vaporizer	O	shared	S	NA	6	2	6	mfg	N	N		
3.	N_2 cryogenic pump	O	shared	S	NA	6	2	6	mfg	N	N		
4.	N_2 cryogenic pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
5.	Liquid N_2 storage tank	O	shared	S	NA	6	2	6	VIII	N	N		
6.	Gaseous N_2 active storage tanks	O	shared	S	NA	6	2	6	VIII	N	N		
7.						(This line has been intentionally left blank.)							
8.	O_2 active storage tanks	O	shared	S	NA	6	2	6	VIII	N	N		
9.	O_2 reserve storage tanks	O	shared	S	NA	6	2	6	VIII	N	N		
10.	H_2 active (liquid storage tanks)			REMOVED									Note ap
11.						(This line has been intentionally left blank.)							
12.	Piping and valves	O	shared	S	NA	6	2	6	B31.1	N	N		
13.	Instrumentation	O	shared	S	NA	6	2	J	mfg	N	N		
NSSS LIQUID SAMPLING SYSTEM													
1.	Process sample valves and piping			B	D	4	2	4	B31.1	N	N		See note u

TABLE 3.2.2-1 (SHEET 40 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
2. RCS sample valves and piping through outside containment isolation valve			B	B	2	1	2	III-2	Y	Y		
3. Sample vessels	CB-144	CB-144	B	D	4	2	4	VIII	N	N		
4. Sample coolers	FB-A10	FB-A01	B	D	4	2	4	VIII	N	N		
5. Deleted												
6. Instrumentation			B	NA	6	2	J	mfg	N	N		
NSSS GAS SAMPLING SYSTEM												
1. Sample vessels	CB-144	CB-144	B	D	4	2	4	VIII	N	N		
2. Valves and piping inside vent hood			B	D	4	2	4	B31.1	N	N		
3. Instrumentation			B	NA	6	2	J	mfg	N	N		
POST-ACCIDENT SAMPLING SYSTEM												
1. Deleted												See paragraph 9.3.2.2.5
2. Deleted												
3. Deleted												
4. Sample piping and valves			B	D	4	1	4	B31.1	N	N		
5. Containment penetration piping and valves			B	B	2	1	2	III-2	Y	Y		
6. Other piping and valves			B	D	4	2	4	B31.1	N	N		
7. Deleted												
TURBINE PLANT SAMPLING SYSTEM												
1. Hotwell sample pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
2. Hotwell sample pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 41 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3. Sample coolers	TB-220′	TB-220′	S	D	4	2	4	mfg, TEMA C	N	N		
4. Process valves and piping			S	D	4	2	4	B31.1	N	N		
5. Chilled water circulating pump	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
6. Chilled water circulating pump motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
7. Condenser circulating water sample pump	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
8. Condenser circulating water sample pump motor	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
9. Instrumentation			S	NA	6	2	J	mfg	N	N		
CONTROL BUILDING DRAIN SYSTEM												
1. Sump pumps	CB-C	CB-C	B	D	4	2	4	mfg	N	N		
2. Sump pump motors	CB-C	CB-C	B	NA	6	2	E	NEMA MG1	N	N		
3. Piping, valves, and floor drain boxes			B	D	4	1	4	B31.1, API 650	N	N		
4. Instrumentation			B	NA	6	2	J	mfg	N	N		
AUXILIARY BUILDING FLOOD RETAINING ROOMS, ALARMS, AND DRAINS												
1. ESF pump room: Alarm units			B	NA	1	1	J	mfg	Y	Y		Notes as, at
Piping to room isolation valve			B	D	4	1	4	B31.1	N	N		
Isolation valves			B	C	3	1	3	III-3	Y	Y		
2. Penetration rooms:												
Alarm units			B	NA	6	2	J	mfg	N	N		
Valves and piping			B	D	4	1	4	B31.1	N	N		
3. Instrumentation			B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 42 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONTAINMENT, AUXILIARY BUILDING, AND MISCELLANEOUS DRAIN SYSTEMS												
1. Radioactive drain sump pumps	AB-D55	AB-D20	B	D	4	2	4	mfg	N	N		
2. Reactor cavity sump pumps	C-171′	C-171′	B	D	4	1	4	mfg	N	N		
3. Containment sump pumps	C-171′	C-171′	B	D	4	1	4	mfg	N	N		
4. Penetration room sump pumps	AB-D73	AB-D107	B	D	4	1	4	mfg	N	N		
5. Auxiliary building sump pumps	AB-D51	AB-D28	B	D	4	2	4	mfg	N	N		
6. Component cooling water drain tank	AB-D75	AB-D06	B	D	4	2	4	API 650	N	N		
7. Component cooling water drain tank pump	AB-D75	AB-D06	B	D	4	2	4	mfg	N	N		
8. Clean water sump pumps	AB-D36	AB-D32	B	D	4	2	4	mfg	N	N		
9. NSCW pumphouse sump pumps	NSW	NSW	B	D	4	2	4	mfg	N	N		
10. Diesel electrical tunnel sump pumps	VB	VB	B	D	4	2	4	mfg	N	N		
11. Main steam and feedwater tunnel sump pumps	VB	VB	B	D	4	2	4	mfg	N	N		
12. Diesel building oily waste sump pumps	DB	DB	B	D	4	2	4	mfg	N	N		
13. Deleted												
14. Deleted												
15. Auxiliary feedwater pumphouse sump pumps	AFP	AFP	B	D	4	2	4	mfg	N	N		
16. Electric boiler building sump pumps	VB	Shared	B	D	4	2	4	mfg	N	N		

TABLE 3.2.2-1 (SHEET 43 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
17. Containment, reactor cavity, and penetration room sump pump motors			B	NA	6	1	E	mfg	N	N		
18. All other pump motors			B	NA	6	2	E	NEMA MG1	N	N		
19. Instrumentation			B	NA	6	2	J	mfg	N	N		
20. Drain isolation valves for ESF pump, HX, and valve rooms			B	C	3	1	3	III-3	Y	Y		Notes as, at
21. Drain piping valves inside containment			B	D	4	2	4	B31.1	N	N		
22. All other valves and all piping			B	D	4	1	4	B31.1	N	N		
23. Tendon Gallery Sump Pumps	C-148′	C-148′	G	D	4	2	4	mfg	N	N		
24. Tendon Gallery Sump Pump Motors	C-148′	C-148′	G	NA	6	2	E	mfg	N	N		
25. Tendon Gallery Sump Pump discharge valves and piping			G	D	4	2	4	B31.1	N	N		

TABLE 3.2.2-1 (SHEET 44 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
TURBINE BUILDING DRAIN SYSTEM												
1. Turbine building drain tanks	TB-195′	TB-195′	S	D	4	2	4	API-650	N	N		
2. Turbine building drain transfer pumps	TB-195′	TB-195′	S	D	4	2	4	mfg	N	N		
3. Turbine building drain transfer pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
4. Turbine building sump pumps	TB-195′	TB-195′	S	D	4	2	4	API-610	N	N		
5. Turbine building sump pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
6. Turbine building drain discharge filter	AB-D93	AB-D84	W	D	4	2	4	VIII	N	N		
7. Turbine building mixed bed demineralizers	AB-C144 & C145	AB-C147 & C148	B	D	4	2	4	VIII	N	N		
8. Turbine building drain feed filter	AB-D92	AB-D85	B	D	4	2	4	VIII	N	N		
9. Turbine building drain oil separator	AB-D89	AB-D87	W	D	4	2	4	VIII	N	N		
10. Valves and piping			S,B	D	4	2	4	B31.1	N	N		
11. Instrumentation			S,B	NA	6	2	J	mfg	N	N		
TURBINE LUBE OIL STORAGE AND FILTRATION SYSTEM												
1. Main lube oil reservoir package	TB-245′	TB-245′	S	NA	6	2	6	mfg	N	N		
2. FW pump turbine lube oil reservoir packages	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
3. Clean lube oil storage tank	O	shared	S	NA	6	2	6	API 650	N	N		
4. Dirty lube oil storage tank	O	shared	S	NA	6	2	6	API 650	N	N		
5. Lube oil storage tank transfer pump	O	shared	S	NA	6	2	6	mfg	N	N		

TABLE 3.2.2-1 (SHEET 45 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
6. Vapor extractors	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
7. Lube oil filter pumps	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
8. Pump motors			S	NA	6	2	E	NEMA MG1	N	N		
9. Main lube oil conditioner package	TB-220′	TB-220′	S	NA	6	2	6	B31.1	N	N		
10. FW pump lube oil conditioner package	TB-195′	TB-195′	S	NA	6	2	6	B31.1	N	N		
11. Oil coolers	TB-270′	TB-270′	S	NA	6	2	6	mfg	N	N		
12. Main lube oil conditioner backflow tank			S	NA	6	2	6	mfg	N	N		
13. Lube oil drain pump	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
14. Valves and piping			S	NA	6	2	6	B31.1	N	N		
15. Instrumentation			S	NA	6	2	J	mfg	N	N		
AUXILIARY STEAM SYSTEM												
1. Deleted												
2. Deleted												
3. Other mechanical equipment	VB	shared	S	NA	6	2	6	mfg	N	N		
4. Electrical equipment			S	NA	6	2	E	mfg	N	N		
5. Instrumentation			S	NA	6	2	J	mfg	N	N		
6. Valves and piping			S	NA	6	2	6	B31.1	N	N		
TURBINE GENERATOR H_2 AND H_2 SEAL OIL SYSTEM												
1. Main seal oil pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
2. Recirculating seal oil pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
3. Seal oil vacuum pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
4. Emergency seal oil pump	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
5. Pump motors	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 46 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
6. Valves and piping			S	D	4	2	4	B31.1	N	N		
7. Instrumentation			S	NA	6	2	J	mfg	N	N		
TURBINE GENERATOR CO_2 SYSTEM												
1. All mechanical components	TB-220′	TB-220′	S	D	4	2	4	mfg	N	N		
TURBINE PLANT CLOSED COOLING WATER SYSTEM												
1. Closed cooling HXs	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		
2. Closed cooling circulating pumps	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
3. Closed cooling water makeup surge tank	TB-270′	TB-270′	S	NA	6	2	6	VIII	N	N		Note ac
4. Closed cooling water drain tank	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		Note ac
5. Closed cooling water booster pump	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
6. Pump motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
7. Piping and valves			S	NA	6	2	6	B31.1	N	N		
8. Chemical addition pot	TB	TB	S	NA	6	2	6	mfg	N	N		
9. Instrumentation			S	NA	6	2	J	mfg	N	N		
TURBINE PLANT COOLING WATER SYSTEM												
1. Turbine plant cooling water pumps	VB	VB	S	D	4	2	4	mfg	N	N		
2. Turbine plant cooling water pump motors	VB	VB	S	NA	6	2	E	NEMA MG1	N	N		
3. Piping and valves			S,B	NA	6	2	6	mfg	N	N		
4. Instrumentation			S	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 47 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
WASTE EVAPORATOR STEAM SUPPLY SYSTEM												
1. Electric boiler feed-water pumps	AB-D52	shared	B	D	4	2	4	mfg	N	N		av
2. Electric boiler condensate receiver tank	AB-D52	shared	B	D	4	2	4	API 650	N	N		av

TABLE 3.2.2-1 (SHEET 48 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3. Steam isolation valves		shared										av
4. Steam isolation piping		shared										av
5. Manual valve upstream of steam isolation valves		shared										av
WASTE WATER EFFLUENT SYSTEM												
1. Waste water retention basin transfer pumps	O	shared	S	NA	6	2	6	mfg	N	N		
2. Waste water retention basin transfer pump motors	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
3. Low voltage transformer area sump pump	O	shared	S	NA	6	2	6	mfg	N	N		
4. Low voltage transformer area sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
5. Firewater pumphouse oily waste separator pumps	O	shared	S	NA	6	2	6	mfg	N	N		
6. Firewater pumphouse oily waste separator pump motors	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
7. Auxiliary boiler room sump pumps	VB	shared	S	NA	6	2	6	mfg	N	N		

TABLE 3.2.2-1 (SHEET 49 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
8. Auxiliary boiler room sump pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
9. Waste water sample booster pump	VB	shared	S	NA	6	2	6	mfg	N	N		
10. Waste water sample booster pump motor	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
11. Water treatment building floor drain sump pump	O	shared	S	NA	6	2	6	mfg	N	N		
12. Water treatment building floor drain sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
13. Lube oil storage area sump pump	O	shared	S	NA	4	2	4	mfg	N	N		
14. Lube oil storage area sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
15. Station service transformer area sump pumps	O	O	S	NA	6	2	6	mfg	N	N		
16. Station service transformer area sump pumps motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
17. Outdoor pipe trench sump pump	O	shared	S	NA	6	2	6	mfg	N	N		
18. Outdoor pipe trench sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
19. Automatic transformer area sump pump	O	shared	S	NA	6	2	6	mfg	N	N		
20. Automatic transformer area sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 50 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
21. Switch house station service transformer area sump pump	O	shared	S	NA	6	2	6	mfg	N	N		
22. Switch house station service transformer area sump pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
23. Lined startup pond discharge pump	O	shared	S	NA	6	2	6	mfg	N	N		
24. Lined startup pond discharge pump motor	O	shared	S	NA	6	2	E	NEMA MG1	N	N		
25. Waste water dechlorination package	VB	shared	S	NA	6	2	6	mfg	N	N		
26. Piping and valves			S	NA	6	2	6	mfg	N	N		
27. Instrumentation			S	NA	6	2	J	mfg	N	N		
INSTRUMENT AND SERVICE AIR SYSTEM												
1. Reciprocating air compressors	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
2. Reciprocating air compressor motors	TB-195′	TB-195′	S	NA	6	2	E	NEMA MG1	N	N		
3. Rotary air compressors	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
4. Rotary air compressor motors	TB-195′	TB-195	S	NA	6	2	E	NEMA MG1	N	N		
5. Air receivers	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		
6. Intake filter silencers	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
7. Aftercoolers	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		
8. Moisture separators	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		

TABLE 3.2.2-1 (SHEET 51 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9. Coolant heat exchangers	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		
10. Air coolant receivers	TB-195′	TB-195′	S	NA	6	2	6	VIII	N	N		
11. Deleted												
12. Deleted												
13. Service air prefilters	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
14. Service air afterfilters	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
15. Instrument air prefilters	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
16. Instrument air afterfilters	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
17. Service air dryers	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
18. Instrument air dryers	TB-195′	TB-195′	S	NA	6	2	6	mfg	N	N		
19. Safety-related piping and valves (other than containment isolation)			B	NA	0	1	3	III-3	Y	Y		Note ad
20. Containment penetration piping and valves			B	B	2	1	2	III-2	Y	Y		
21. All other piping and valves			S	NA	6	2	6	B31.1	N	N		
22. Instrumentation			S	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 52 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
POTABLE WATER SYSTEM												
1. Hot water recirculating pumps	CB	shared	B	NA	6	2	6	mfg	N	N		
2. Hot water recirculating pump motors	CB	shared	B	NA	6	2	E	NEMA MG1	N	N		
3. Water heaters	CB	shared	B	NA	6	2	E	UL	N	N		
4. Piping and valves			B	NA	6	2	6	B31.1	N	N		
5. Instrumentation			B	NA	6	2	J	mfg	N	N		
6. Potable water booster pumps	VB	shared	S	NA	6	2	6	mfg	N	N		
7. Potable water jockey pump	VB	shared	S	NA	6	2	6	mfg	N	N		
8. Chlorine injection pump	VB	shared	S	NA	6	2	6	mfg	N	N		
9. Chlorine injector	VB	shared	S	NA	6	2	6	mfg	N	N		
10. Chlorine dispenser	VB	shared	S	NA	6	2	6	mfg	N	N		
11. Potable water storage tank	O	shared	S	NA	6	2	6	mfg	N	N		
12. Pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
REACTOR COOLANT SYSTEM LEAK DETECTION SYSTEM												
1. Radiation monitors	C	C	W	NA	6	1	J	mfg	N	N		
2. Condensate measuring instruments	C	C	B	NA	6	2	J	mfg	N	N		
3. Condensate measuring piping and valves	C	C	B	D	4	2	4	B31.1	N	N		
4. Humidity instruments	C	C	B	NA	6	2	J	mfg	N	N		
5. Tank and sump level instruments	C	C	B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 53 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
MISCELLANEOUS LEAK DETECTION SYSTEM												
1. Piping and valves			B	D	4	2	4	B31.1	N	N		
FUEL HANDLING BUILDING DRAIN SYSTEM												
1. Sump pumps	FB-C09	FB-C03	B	D	4	2	4	mfg	N	N		
2. Sump pump motors	FB-C09	FB-C03	B	NA	6	2	E	NEMA MG1	N	N		
3. Piping and valves			B	D	4	1	4	B31.1	N	N		
4. Instrumentation			B	NA	6	2	J	mfg	N	N		
UTILITY WATER SYSTEM												
1. Utility water booster pumps	VB	shared	S	NA	6	2	6	mfg	N	N		
2. Utility water booster pump motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
3. Utility water loop header	O	shared	S	NA	6	2	C	mfg	N	N		
4. Valves and all other piping			S,B	NA	6	2	6	mfg	N	N		
5. Instrumentation			S	NA	6	2	J	mfg	N	N		
CONTAINMENT AIR COOLING SYSTEM												Note ah
1. Cooling fans	C-238′	C-238′	B	NA	0	1	5	AMCA	Y	Y	1-B	
2. Fan motors	C-238′	C-238′	B	NA	1	1	E	NEMA MG1	Y	Y	1-B	
3. Cooling coils	C-238′	C-238′	B	B	2	1	2	III-2	Y	Y	1-B	
4. Ductwork	C	C	B	NA	0	1	5	ANSI N509	Y	Y		
5. Dampers	C	C	B	NA	0	1	5	ANSI N509	Y	Y	1-B	
6. Damper motors	C	C	B	NA	1	1	E	NEMA MG1	Y	Y		
7. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
CONTAINMENT LOWER LEVEL AIR CIRCULATING SYSTEM												Note ah
1. Fans	C-176′	C-176′	B	NA	6	2	6	AMCA	N	N		
2. Fan motors	C-176′	C-176′	B	NA	6	2	E	NEMA MG1	N	N		
3. Ductwork			B	NA	6	2	6	SMACNA	N	N		
4. Instrumentation			B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 54 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONTAINMENT PREACCESS FILTER SYSTEM												Note ah
1. Fans	C-268′	C-268′	B	NA	6	2	6	AMCA	N	N		
2. Fan motors	C-268′	C-268′	B	NA	6	2	E	NEMA MG1	N	N		
3. HEPA filters	C-261′	C-261′	B	NA	6	2	6	ANSI N509	N	N		
4. Charcoal filters	C-261′	C-261′	B	NA	6	2	6	ANSI N509	N	N		
5. Heaters	C-261′	C-261′	B	NA	6	2	E	UL	N	N		
6. Moisture eliminators	C-261′	C-261′	B	NA	6	2	6	ANSI N509	N	N		
7. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT NORMAL PREACCESS PURGE EXHAUST SYSTEM												Note ah
1. Preaccess purge fan	EB-227′	EB-227′	B	NA	6	2	6	AMCA	N	N		
2. Preaccess purge fan motor	EB-227′	EB-227′	B	NA	6	2	E	NEMA MG1	N	N		
3. Minipurge fan	EB-227′	EB-227′	B	NA	6	2	6	AMCA	N	N		
4. Minipurge fan motor	EB-227′	EB-227′	B	NA	6	2	E	NEMA MG1	N	N		
5. Moisture eliminators	EB-220′	EB-220′	B	NA	6	2	6	ANSI N509	N	N		
6. Heaters	EB-220′	EB-220′	B	NA	6	2	E	UL	N	N		
7. HEPA filters	EB-220′	EB-220′	B	NA	6	2	6	ANSI N509	N	N		
8. Charcoal filters	EB-220′	EB-220′	B	NA	6	2	6	ANSI N509	N	N		
9. Preaccess purge exhaust unit housing	EB-220′	EB-220′	B	NA	6	1	6	ANSI N509	N	N		
10. Containment penetration ducting			B	B	2	1	2	III-2	Y	Y		
11. All other ductwork			B	NA	6	2	6	SMACNA	N	N		
12. Isolation dampers			B	B	2	1	2	III-2	Y	Y		
13. Isolation damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
14. Discharge dampers			B	NA	6	2	6	ANSI N509	N	N		
15. Instrumentation for containment isolation			B	NA	1	1	J	mfg	Y	Y		Note s

TABLE 3.2.2-1 (SHEET 55 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
16. All other instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT NORMAL PREACCESS PURGE SUPPLY SYSTEM												Note ah
1. Preaccess purge fan	EB-223′	EB-223′	B	NA	6	2	6	AMCA	N	N		
2. Preaccess purge fan motor	EB-223′	EB-223′	B	NA	6	2	E	NEMA MG1	N	N		
3. Minipurge fan	EB-223′	EB-223′	B	NA	6	2	6	AMCA	N	N		
4. Minipurge fan motor	EB-223′	EB-223′	B	NA	6	2	E	NEMA MG1	N	N		
5. Prefilters	EB-220′	EB-220′	B	NA	6	2	6	ASHRAE	N	N		
6. Heaters	EB-220′	EB-220′	B	NA	6	2	E	UL	N	N		
7. Containment penetration ducting			B	B	2	1	2	III-2	Y	Y		
8. All other ductwork			B	NA	6	2	6	SMACNA	N	N		
9. Isolation dampers			B	B	2	1	2	III-2	Y	Y		
10. Isolation damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
11. Inlet damper			B	NA	6	2	6	ANSI N509	N	N		
12. Instrumentation for containment isolation			B	NA	1	1	J	mfg	Y	Y		Note s
13. All other instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT POST-LOCA PURGE EXHAUST SYSTEM												Note ah
1. Containment penetration ducting			B	B	2	1	2	III-2	Y	Y		
2. Welded ductwork			B	NA	6	1	6	ANSI N509	N	N		
3. All other ductwork			B	NA	6	2	6	SMACNA	N	N		
4. Isolation dampers			B	B	2	1	2	III-2	Y	Y		
5. Isolation damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
6. Heaters	EB-222′	EB-222′	B	NA	6	2	E	UL	N	N		
7. Moisture eliminators	EB-222′	EB-222′	B	NA	6	2	6	ANSI N509	N	N		
8. HEPA filters	EB-222′	EB-222′	B	NA	6	2	6	ANSI N509	N	N		

TABLE 3.2.2-1 (SHEET 56 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9. Charcoal filters	EB-222′	EB-222′	B	NA	6	2	6	ANSI N509	N	N		
10. Post-LOCA purge exhaust unit housing	EB-222′	EB-222′	B	NA	6	1	6	ANSI N509	N	N		
11. Inlet globe valve and discharge check valve			B	NA	6	1	6	B31.1	N	N		
12. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT CRDM COOLING SYSTEM												Note ah
1. Fans	C-240′	C-240′	W	NA	6	2	6	AMCA	N	N		
2. Fan motors	C-240′	C-240′	W	NA	6	2	E	NEMA MG1	N	N		Note m
3. Ductwork	C	C	W	NA	6	2	6	SMACNA	N	N		
4. Instrumentation			W	NA	6	2	J	mfg	N	N		
CONTAINMENT CAVITY COOLING SYSTEM												Note ah
1. Fans	C-206′	C-206′	B	NA	6	1	6	AMCA	N	N		
2. Fan motors	C-206′	C-206′	B	NA	6	1	E	NEMA MG1	N	N		Note m
3. Ductwork	C	C	B	NA	6	2	6	ANSI N509	N	N		
4. Dampers	C	C	B	NA	6	2	6	ANSI N509	N	N		
5. Cooling coils	C-206′	C-206′	B	B	2	1	2	III-2	Y	Y	1-B	
6. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT REACTOR SUPPORT COOLING SYSTEM												Note ah
1. Fans	C-171′	C-171′	B	NA	6	1	6	AMCA	N	N		
2. Fan motors	C-171′	C-171′	B	NA	6	1	E	NEMA MG1	N	N		Note m
3. Ductwork			B	NA	6	1	6	ANSI N509	N	N		Note ae
4. Dampers			B	NA	6	1	6	ANSI N509	N	N		
5. Damper motors			B	NA	6	1	E	NEMA MG1	N	N		
6. Instrumentation			B	NA	6	2	J	mfg	N	N		
H_2 RECOMBINER AND MONITORING SYSTEM												
1. Penetration piping and valves			B	B	2	1	2	III-2	Y	Y		
2. Other piping and valves			B	B	2	1	2	III-2	Y	Y		
3. H_2 monitors			B	NA	1	1	J	mfg	Y	Y		

TABLE 3.2.2-1 (SHEET 57 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
4. H_2 recombiners	C-261′	C-261′	W	NA	0	1	5	III-2	Y	Y		Note m
5. Safety-related recombiner instrumentation			W,B	NA	1	1	J	mfg		Y	Y	Note s
6. Safety-related valve operators			B	NA	1	1	E	NEMA MG1	Y	Y		
7. All other instrumentation			W,G	NA	6	1	J	mfg	N	N		
CONTAINMENT AUXILIARY AIR COOLING SYSTEM												Note ah
1. Fans	C-261′	C-261′	B	NA	6	1	6	AMCA	N	N		
2. Fan motors	C-261′	C-261′	B	NA	6	1	E	NEMA MG1	N	N		
3. Cooling coils	C-261′	C-261′	B	B	2	1	2	III-2	Y	Y	1-A	
4. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTAINMENT POST-LOCA CAVITY PURGE SYSTEM												Note ah
1. Fans	C-199′	C-199′	B	NA	0	1	5	AMCA	Y	Y	1-A	
2. Fan motors	C-199′	C-199′	B	NA	1	1	E	NEMA MG1	Y	Y	1-A	
3. Ductwork and backdraft damper			B	NA	0	1	5	ANSI N509	Y	Y		
4. Embedded ductwork			B	NA	0	1	5	B31.1	Y	Y		
5. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
EQUIPMENT BUILDING HVAC SYSTEM												Note ah
1. Equipment building ventilation fans	EB-236′	EB-236′	B	NA	6	2	6	AMCA	N	N		
2. Equipment building ventilation fan motors	EB-236′	EB-236′	B	NA	6	2	E	NEMA MG1	N	N		
3. Tendon gallery ventilation fans	EB-177′	EB-177′	B	NA	6	2	6	AMCA	N	N		
4. Tendon gallery ventilation fan motors	EB-177′	EB-177′	B	NA	6	2	E	NEMA MG1	N	N		
5. Ductwork			B	NA	6	2	6	ANSI N509	N	N		
6. Dampers			B	NA	6	2	6	ANSI N509	N	N		
7. Damper motors			B	NA	6	2	E	NEMA MG1	N	N		
8. Instrumentation			B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 58 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONTROL ROOM HVAC SYSTEM (ESSENTIAL PORTION)													Note ah
1.	Filter unit fans	CB-312 & 321	CB-305 & 311	B	NA	0	1	5	AMCA	Y	Y	IX	
2.	Filter unit fans motors	CB-312 & 321	CB-305 & 311	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
3.	Return air fans	CB-312 & 321	CB-305 & 311	B	NA	0	1	5	AMCA	Y	Y	IX	Note ar
4.	Return air fans motor	CB-312 & 321	CB-305 & 311	B	NA	1	1	E	NEMA MG1	Y	Y	IX	Note ar
5.	Moisture eliminators	CB-312 & 321	CB-305 & 311	B	NA	0	1	5	ANSI N509	Y	Y	IX	
6.	Heaters	CB-312 & 321	CB-305 & 311	B	NA	1	1	E	UL, ANSI N509	Y	Y	IX	
7.	HEPA filters	CB-312 & 321	CB-305 & 311	B	NA	0	1	5	ANSI N509	Y	Y	IX	
8.	Charcoal filters	CB-312 & 321	CB-305 & 321	B	NA	0	1	5	ANSI N509	Y	Y	IX	
9.	Cooling coils	CB-312 & 321	CB-305 & 311	B	C	3	1	3	III-3	Y	Y	IX	
10.	Dampers			B	NA	0	1	5	ANSI N509	Y	Y		
11.	Damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
12.	Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
13.	Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
14.	ESF chiller room exhaust fans	CB-313 & 320	CB-308 & 310	B	NA	0	1	5	AMCA	Y	Y	IX	
15.	ESF chiller room exhaust fan motors	CB-313 & 320	CB-308 & 310	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
16.	ESF chiller room electric heaters	CB-313 & 320	CB-308 & 310	B	NA	6	2	E	UL	N	N		
17.	ESF chiller room electric heater motors	CB-313 & 320	CB-308 & 310	B	NA	6	2	E	NEMA MG1	N	N		
18.	Deleted												
19.	Duct silencers	CB-313 & 320	CB-305, 308, 310 & 123	B	NA	0	1	5	ASTM E-477	Y	Y		

TABLE 3.2.2-1 (SHEET 59 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
CONTROL ROOM HVAC SYSTEM (NORMAL PORTION)													Note ah
1.	Return and exhaust fans	CB-403	shared	B	NA	6	2	6	AMCA	N	N		
2.	Return and exhaust fan motors	CB-403	shared	B	NA	6	2	E	NEMA MG1	N	N		
3.	Kitchen, toilet, and conference room fan	CB-325	shared	B	NA	6	2	6	AMCA	N	N		
4.	Kitchen, toilet, and conference room fan motor	CB-325	shared	B	NA	6	2	E	NEMA MG1	N	N		
5.	A/C fans	CB-403	shared	B	NA	6	2	6	AMCA	N	N		
6.	A/C fan motors	CB-403	shared	B	NA	6	2	E	NEMA MG1	N	N		
7.	Cooling coils	CB-403	shared	B	NA	6	2	6	ARI	N	N		
8.	High efficiency and prefilters	CB-403	shared	B	NA	6	2	6	ANSI N509, ASHRAE	N	N		
9.	Suction and discharge duct-work including isolation dampers			B	NA	0	1	5	ANSI N509	Y	Y		
10.	All other ductwork and dampers			B	NA	6	1	6	ANSI N509	N	N		Note ae
11.	Electric duct heaters	CB-220′	shared	B	NA	6	2	E	UL	N	N		
12.	Instrumentation			B	NA	6	2	J	mfg	N	N		
13.	Filter room electric heaters	CB-260′	CB-260′	B	NA	6	2	E	UL	N	N		
14.	Filter room electric heater motors	CB-260′	CB-260′	B	NA	6	2	E	NEMA MG1	N	N		
15.	Normal A/C equipment room exhaust fan	CB-260′	shared	B	NA	6	2	6	AMCA	N	N		
16.	Normal A/C equipment room exhaust fan motor	CB-260′	shared	B	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 60 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
17. A/C equipment room electric heater	CB-260′ & 280′	CB-260′ & 280′	B	NA	6	2	E	UL	N	N		
18. A/C equipment room electric heater motor	CB-260′ & 280′	CB-260′ & 280′	B	NA	6	2	E	NEMA MG1	N	N		
19. Normal chiller room exhaust fan	CB-320′	CB-308′	B	NA	6	2	6	AMCA	N	N		
20. Normal chiller exhaust fan motor	CB-320	CB-308	B	NA	6	2	E	NEMA MG1	N	N		
21. Normal chiller room electric heater	CB-320	CB-308	B	NA	6	2	E	UL	N	N		
22. Normal chiller room electric heater motor	CB-320	CB-308	B	NA	6	2	E	NEMA MG1	N	N		
SAFETY FEATURE ELECTRICAL EQUIPMENT ROOM HVAC SYSTEM												Note ah
1. Battery room exhaust fans	CB-180′	CB-180′	B	NA	0	1	5	AMCA	Y	Y	IX	
2. Battery room exhaust fan motors	CB-180′	CB-180′	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
3. A/C unit fans	CB-B60 & B62	CB-B16 & B17	B	NA	0	1	5	AMCA	Y	Y	IX	
4. A/C unit fan motors	CB-B60 & B62	CB-B16 & B17	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
5. Cooling coils	CB-B60 & B62	CB-B16 & B17	B	C	3	1	3	III-3	Y	Y	IX	
6. Prefilters	CB-B60 & B62	CB-B16 & B17	B	NA	0	1	5	ASHRAE	Y	Y	IX	
7. Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
8. Dampers			B	NA	0	1	5	ANSI N509	Y	Y		
9. Damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
10. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
CONTROL BUILDING LEVELS A, B, 1, AND 2 NORMAL SYSTEM												Note ah
1. Wing area A/C fan	CB-B05	CB-B75	B	NA	6	2	6	AMCA	N	N		
2. Wing area A/C fan motor	CB-B05	CB-B75	B	NA	6	2	E	NEMA MG1	N	N		

TABLE 3.2.2-1 (SHEET 61 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3.	Switchgear A/C fan	CB-B40	CB-B38	B	NA	6	2	6	AMCA	N	N		
4.	Switchgear A/C fan motor	CB-B40	CB-B38	B	NA	6	2	E	NEMA MG1	N	N		
5.	Prefilters	CB-B05 & B40	CB-B75 & B38	B	NA	6	2	6	ASHRAE	N	N		
6.	Cooling coils	CB-B05 & B40	CB-B75 & B38	B	NA	6	2	6	ARI	N	N		
7.	Ductwork and dampers			B	NA	6	2	6	SMACNA/ ANSI N509	N	N		
8.	Instrumentation			B	NA	6	2	J	mfg	N	N		
9.	Wing area exhaust fan	CB-180′	CB-180′	B	NA	6	2	6	AMCA	N	N		
10.	Wing area exhaust fan motor	CB-180′	CB-180′	B	NA	6	2	E	NEMA MG1	N	N		
11.	Service area normal return and exhaust fan	CB-240′	CB-240′	B	NA	6	2	6	AMCA	N	N		
12.	Service area normal return and exhaust fan motor	CB-240′	CB-240′	B	NA	6	2	E	NEMA MG1	N	N		
13.	Service area normal A/C fan	CB-249′	shared	B	NA	6	2	6	AMCA	N	N		
14.	Service area normal A/C fan motor	CB-249′	shared	B	NA	6	2	E	NEMA MG1	N	N		
15.	Cooling coils	CB-249′	shared	B	NA	6	2	6	ARI	N	N		
16.	Prefilter	CB-249′	shared	B	NA	6	2	6	ASHRAE	N	N		
17.	Service area electric wall heater	CB-220′ CB-240′	CB-220′ CB-240′	B	NA	6	2	E	UL	N	N		
18.	Service area electric duct heater	CB-240′	CB-240′	B	NA	6	2	E	UL	N	N		
19.	Central alarm station standby fan	CB-240′	shared	B	NA	6	2	6	AMCA	N	N		
20.	Central alarm station standby fan motor	CB-240′	shared	B	NA	6	2	E	NEMA MG1	N	N		
21.	Cooling coils	CB-240′	shared	B	NA	6	2	6	ARI	N	N		
22.	Prefilter	CB-240′	shared	B	NA	6	2	6	ASHRAE	N	N		

TABLE 3.2.2-1 (SHEET 62 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
23. Smoke exhaust fan	CB-280′	shared	B	NA	6	2	6	AMCA	N	N		
24. Smoke exhaust fan motor	CB-280′	shared	B	NA	6	2	E	NEMA MG1	N	N		
CONTROL BUILDING LAB HOOD VENT SYSTEM												Note ah
1. Fumehoods supply fan	CB-240′	shared	B	NA	6	2	6	AMCA	N	N		
2. Fumehoods supply fan motor	CB-240′	shared	B	NA	6	2	E	NEMA MG1	N	N		
3. Fumehood filter unit fan	CB-248′	shared	B	NA	6	2	6	AMCA	N	N		
4. Fumehood filter unit fan motor	CB-248	shared	B	NA	6	2	E	NEMA MG1	N	N		
5. Moisture eliminator	CB-248	shared	B	NA	6	2	6	ANSI N509	N	N		
6. Heater	CB-248	shared	B	NA	6	2	E	UL	N	N		
7. HEPA filters	CB-248	shared	B	NA	6	2	6	ANSI N509	N	N		
8. Charcoal filter	CB-248	shared	B	NA	6	2	6	ANSI N509	N	N		
9. Ductwork and dampers			B	NA	6	2	6	SMACNA/ ANSI N509	N	N		
10. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTROL BUILDING LOCKER AND TOILET EXHAUST SYSTEM												Note ah
1. Exhaust fan	CB-240′	shared	B	NA	6	2	6	AMCA	N	N		
2. Exhaust fan motor	CB-240′	shared	B	NA	6	2	E	NEMA MG1	N	N		
3. Ductwork			B	NA	6	2	6	SMACNA	N	N		
4. Instrumentation			B	NA	6	2	J	mfg	N	N		
CONTROL BUILDING CABLE SPREADING ROOM HVAC SYSTEM												Note ah
1. Cable spreading room A/C fans	CB-A32 CB-325	shared	B	NA	6	2	6	AMCA	N	N		
2. Cable spreading room A/C fan motors	CB-A32 CB-325	shared	B	NA	6	2	E	NEMA MG1	N	N		
3. Auxiliary relay room ESF A/C fans	CB-226	CB-223	B	NA	0	1	5	AMCA	Y	Y	IX	

TABLE 3.2.2-1 (SHEET 63 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
4.	Auxiliary relay room ESF A/C fan motors	CB-226	CB-223	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
5.	Prefilters	CB-A32 CB-325′	shared	B	NA	6	2	6	ASHRAE	N	N		
6.	Auxiliary relay room ESF A/C	CB-200′ CB-240′	CB-200′ CB-240′										
	(cooling coils)			B	C	3	1	3	III-3	Y	Y	IX	
	(ductwork) fan coolers			B	NA	0	1	5	mfg	Y	Y		
7.	All other coolers	CB-A32 CB-325′	shared	B	NA	6	2	6	ARI	N	N		
8.	Electric duct heaters			B	NA	6	2	E	UL	N	N		
9.	All other ductwork			B	NA	6	2	6	SMACNA	N	N		
10.	Auxiliary relay room ductwork and dampers			B	NA	0	1	5	ANSI N509	Y	Y		
11.	Smoke exhaust fan	CB-325	shared	B	NA	6	2	6	AMCA	N	N		
12.	Smoke exhaust fan motor	CB-325	shared	B	NA	6	2	E	NEMA MG1	N	N		
13.	Computer room A/C fans	CB-A38	CB-A30	G	NA	6	2	6	AMCA	N	N		
14.	Computer room A/C fan motors	CB-A38	CB-A30	G	NA	6	2	E	NEMA MG1	N	N		
15.	Computer room A/C unit humidifier	CB-A38	CB-A30	G	NA	6	2	E	ANSI N509	N	N		
16.	Computer room A/C unit cooling coils	CB-A38	CB-A30	G	NA	6	2	6	ARI	N	N		
17.	Auxiliary relay room ESF A/C instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
18.	All other instrumentation			B	NA	6	2	J	mfg	N	N		
19.	Normal AC room ESF AC vent	CB-325	CB-325	B	NA	0	1	5		Y	Y		
	Normal chilled water cooling coil	CB-325	-	B	NA	6	2	6	ARI	N	N		
	ESF chilled water cooling coils	CB-325	CB-325	B	C	3	1	3	III-3	Y	Y		

TABLE 3.2.2-1 (SHEET 64 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
20. Electric equipment room ESF AC unit	CB-322	-	B	NA	0	1	5		Y	Y		
Normal chilled water cooling coil	CB-322	-	B	NA	6	2	6	ARI	N	N		
ESF chilled water cooling coil	CB-322	-	B	C	3	1	3	III-3	Y	Y		
ELECTRICAL TUNNEL VENTILATION SYSTEM												Note ah
1. Electrical tunnel supply fan (normal)	VB	VB	B	NA	6	2	6	AMCA	N	N		
2. Electrical tunnel supply fan motor	VB	VB	B	NA	6	2	E	NEMA MG1	N	N		
3. Turbine building chase to control building tunnel fan	CB-180′	CB-180′	B	NA	6	2	6	AMCA	N	N		
4. Turbine building chase to control building tunnel fan motor	CB-180′	CB-180′	B	NA	6	2	E	NEMA MG1	N	N		
5. Turbine and auxiliary building tunnel supply fan	AB-260′	AB-260′	B	NA	0	1	5	AMCA	Y	Y	VIII	
6. Turbine and auxiliary building tunnel supply fan motor	AB-260′	AB-260′	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
7. Diesel power cable tunnel exhaust fans	VB	VB	B	NA	0	1	5	AMCA	Y	Y	III	
8. Diesel power cable tunnel exhaust fan motors	VB	VB	B	NA	1	1	E	NEMA MG1	Y	Y	III	

TABLE 3.2.2-1 (SHEET 65 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9.	NSCW tower cable tunnel fans	VB	VB	B	NA	0	1	5	AMCA	Y	Y	III	
10.	NSCW tower cable tunnel fan motors	VB	VB	B	NA	1	1	E	NEMA MG1	Y	Y	III	
11.	Prefilters	VB	VB	B	NA	6	2	6	ASHRAE	N	N		
12.	Turbine building chase and electrical tunnel ductwork			B	NA	6	2	6	SMACNA	N	N		
13.	All other ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
14.	Turbine building chase and electrical tunnel instrumentation			B	NA	6	2	J	mfg	N	N		
15.	All other instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
FUEL HANDLING BUILDING HVAC SYSTEM													Note ah
1.	Fuel pool recirculating air handling unit fans	FB-220′	FB-220′	B	NA	6	2	6	AMCA	N	N		
2.	Fuel pool recirculating fan motors	FB-220′	FB-220′	B	NA	6	2	E	NEMA MG1	N	N		
3.	Recirculating fan coolers	FB-220′	FB-220′	B	NA	6	2	6	ARI	N	N		
4.	Fuel pool electric duct fan heaters	FB-220′	FB-220′	B	NA	6	2	E	UL	N	N		
5.	Railroad corridor recirculating fan	AB-127′	shared	B	NA	6	2	6	AMCA	N	N		
6.	Railroad corridor recirculating fan motor	AB-127	shared	B	NA	6	2	E	NEMA MG1	N	N		
7.	Railroad corridor recirculating fan cooler	AB-127	shared	B	NA	6	2	6	ARI	N	N		

TABLE 3.2.2-1 (SHEET 66 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
8. Railroad corridor infrared heaters	AB-127	shared	B	NA	6	2	E	UL	N	N		
9. FHB normal A/C fan	CB-403	shared	B	NA	6	2	6	AMCA	N	N		
10. FHB normal A/C fan motor	CB-403	shared	B	NA	6	2	E	NEMA MG1	N	N		
11. FHB normal A/C prefilters	CB-403	shared	B	NA	6	2	6	ASHRAE	N	N		
12. FHB normal A/C heaters	CB-403	shared	B	NA	6	2	E	UL	N	N		
13. FHB normal A/C coolers	CB-403	shared	B	NA	6	2	6	ARI	N	N		
14. FHB exhaust A/C fan	FB-301	shared	B	NA	6	2	6	AMCA	N	N		
15. FHB exhaust A/C fan motor	FB-301	shared	B	NA	6	2	E	NEMA MG1	N	N		
16. FHB exhaust A/C HEPA filters	FB-301	shared	B	NA	6	2	6	ANSI N509	N	N		
17. FHB exhaust A/C heaters	FB-301	shared	B	NA	6	2	E	UL	N	N		
18. FHB exhaust A/C moisture eliminators	FB-301	shared	B	NA	6	2	6	ANSI N509	N	N		
19. FHB exhaust A/C charcoal filters	FB-301	shared	B	NA	6	2	6	ANSI N509	N	N		
20. HEPA filters	FB-301	shared	B	NA	6	2	6	ANSI N509	N	N		
21. Process duct-work and dampers			B	NA	6	2	6	SMACNA/ ANSI N509	N	N		
22. Damper motors			B	NA	6	2	E	NEMA MG1	N	N		
23. Negative pressure boundary penetration ductwork and dampers			B	NA	0	1	5	ANSI N509	Y	Y		
24. Instrumentation			B	NA	6	2	J	mfg	N	N		
25. FHB normal equipment reheat coil	FB-301	shared	B	NA	6	2	E	UL	N	N		

TABLE 3.2.2-1 (SHEET 67 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
26. FHB accident equipment reheat coil	FB-260′	shared	B	NA	6	2	E	UL	N	N		
FUEL HANDLING BUILDING POST-ACCIDENT EXHAUST SYSTEM												Note ah
1. Fans	FB-303, 304	shared	B	NA	0	1	5	AMCA	Y	Y	VII	
2. Fan motors	FB-303, 304	shared	B	NA	1	1	E	NEMA MG1	Y	Y	VII	
3. Moisture eliminators	FB-303, 304	shared	B	NA	0	1	5	ANSI N509	Y	Y	VII	
4. Heaters	FB-303, 304	shared	B	NA	1	1	E	UL, ANSI N509	Y	Y	VII	
5. HEPA filters	FB-303, 304	shared	B	NA	0	1	5	ANSI N509	Y	Y	VII	
6. Charcoal filters	FB-303, 304	shared	B	NA	0	1	5	ANSI N509	Y	Y	VII	
7. Ductwork			B	NA	0	1	5	ANSI N509	Y	Y	VII	
8. Dampers			B	NA	0	1	5	ANSI N509	Y	Y	VII	
9. Dampers motors			B	NA	1	1	E	NEMA MG1	Y	Y	VII	
10. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y	VII	Note s
AUXILIARY BUILDING OUTSIDE AIR SUPPLY, NORMAL HVAC, RADIOACTIVE FILTER EXHAUST, AND CONTINUOUS EXHAUST SYSTEMS												Note ah
1. Exhaust unit fans	AB-212	AB-221	B	NA	6	2	6	AMCA	N	N		
2. Exhaust unit fan motors	AB-212	AB-221	B	NA	6	2	E	NEMA MG1	N	N		
3. A/C fans	AB-212	AB-221	B	NA	6	2	6	AMCA	N	N		
4. A/C fan motors	AB-212	AB-221	B	NA	6	2	E	NEMA MG1	N	N		
5. HEPA filters	AB-212	AB-221	B	NA	6	2	6	ANSI N509	N	N		
6. Charcoal filters	AB-212	AB-221	B	NA	6	2	6	ANSI N509	N	N		
7. Moisture eliminators	AB-212	AB-221	B	NA	6	2	6	ANSI N509	N	N		
8. Electric heaters	AB-212	AB-221	B	NA	6	2	E	UL	N	N		
9. Cooling coils	AB-212	AB-221	B	NA	6	2	6	ARI	N	N		
10. Ductwork			B	NA	6	2	6	SMACNA	N	N		
11. Dampers			B	NA	6	2	6	ANSI N509	N	N		
12. Damper motors			B	NA	6	2	E	NEMA MG1	N	N		
13. Instrumentation			B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 68 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
AUXILIARY BUILDING ESF ROOM COOLERS												Note ah
1. Electrical, switchgear, and MCC room cooler A, level D:												
Fan	AB-D79	AB-D02	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-D79	AB-D02	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-D79	AB-D02	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-D79	AB-D02	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
2. Electrical, switchgear, and MCC room cooler B, level 2:												
Fan	AB-212	AB-221	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-212	AB-221	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-212	AB-221	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-212	AB-221	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
3. Electrical, switchgear, and MCC room cooler A, level C:												
Fan	AB-B13	AB-B123	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-B13	AB-B123	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-B13	AB-B123	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-B13	AB-B123	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
4. Electrical, switchgear, and MCC room cooler B, level B:												
Fan	AB-B16	AB-B122	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-B16	AB-B122	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 69 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
Normal chilled water cooling coils	AB-B16	AB-B122	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-B16	AB-B122	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
5. Electrical, switchgear, and MCC room cooler A, level 1:												
Fan	AB-118	AB-149	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-118	AB-149	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-118	AB-149	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-118	AB-149	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
6. Electrical, switchgear, and MCC room cooler B, level 1:												
Fan	AB-116	AB-147	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-116	AB-147	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-116	AB-147	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-116	AB-147	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
7. RHR pump room cooler A:												
Fan	AB-D122	AB-D112	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-D122	AB-D112	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-D122	AB-D112	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-D122	AB-D112	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		

TABLE 3.2.2-1 (SHEET 70 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
8. RHR pump room cooler B:												
Fan	AB-D122	AB-D112	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-D122	AB-D112	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-D122	AB-D112	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-D122	AB-D112	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
9. Containment spray pump room cooler A:												
Fan	AB-D79	AB-D02	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-D79	AB-D02	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-D79	AB-D02	B	C	3	1	3	III-3	Y	Y	VIII	
Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
10. Containment spray pump room cooler B:												
Fan	AB-D77	AB-D04	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-D77	AB-D04	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-D77	AB-D04	B	C	3	1	3	III-3	Y	Y	VIII	
11. CCW pump room cooler A:												
Fan	AB-A05	AB-A98	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-A05	AB-A98	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-A05	AB-A98	B	C	3	1	3	III-3	Y	Y	VIII	
12. CCW pump room cooler B:												
Fan	AB-A03	AB-A96	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-A03	AB-A96	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-A03	AB-A96	B	C	3	1	3	III-3	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 71 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
13. Charging pump room cooler A:												
Fan	AB-C115	AB-C16	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-C115	AB-C16	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-C115	AB-C16	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-C115	AB-C16	B	C	3	1	3	III-3	Y	Y	VIII	
14. Charging pump room cooler B:												
Fan	AB-C118	AB-C17	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-C118	AB-C17	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-C118	AB-C17	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-C118	AB-C17	B	C	3	1	3	III-3	Y	Y	VIII	
15. SI pump room cooler A:												
Fan	AB-B15	AB-B119	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-B15	AB-B119	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-B15	AB-B119	B	C	3	1	3	III-3	Y	Y	VIII	
16. SI pump room cooler B:												
Fan	AB-B19	AB-B117	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-B19	AB-B117	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
ESF cooling coils	AB-B19	AB-B117	B	C	3	1	3	III-3	Y	Y	VIII	
17. SFP pump and HX room cooler A:												
Fan	AB-A53	AB-A91	B	NA	0	1	5	AMCA	Y	Y	VIII	
Fan motor	AB-A53	AB-A91	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
Normal chilled water cooling coils	AB-A53	AB-A91	B	C	3	1	3	III-3	Y	Y	VIII	
ESF chilled water cooling coils	AB-A53	AB-A91	B	C	3	1	3	III-3	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 72 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
18.	SFP pump and HX room cooler B:												
	Fan	FB-A07	FB-A04	B	NA	0	1	5	AMCA	Y	Y	VIII	
	Fan motor	FB-A07	FB-A04	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
	Normal chilled water cooling coils	FB-A07	FB-A04	B	C	3	1	3	III-3	Y	Y	VIII	
	ESF chilled water cooling coils	FB-A07	FB-A04	B	C	3	1	3	III-3	Y	Y	VIII	
19.	Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
PIPING PENETRATION AND MSIV VENTILATION SYSTEM													Note ah
1.	Restraint cooling fans	VB	VB	B	NA	6	2	6	AMCA	N	N		
2.	Restraint cooling fan motors	VB	VB	B	NA	6	2	E	NEMA MG1	N	N		Note m
3.	Restraint cooling ductwork			B	NA	6	2	6	SMACNA	N	N		
4.	Restraint cooling backdraft dampers			B	NA	6	2	6	ANSI N509	N	N		
5.	Restraint cooling instrumentation			B	NA	6	2	J	mfg	N	N		
6.	MSIV AHU fans	VB	VB	S	NA	6	2	6	AMCA	N	N		
7.	MSIV AHU fan motors	VB	VB	S	NA	6	2	E	NEMA MG1	N	N		
8.	MSIV ductwork			S	NA	6	2	6	SMACNA	N	N		
9.	MSIV instrumentation		S	NA	6	2	J	mfg	N	N			
10.	MSIV ventilation instrumentation			S	NA	6	2	J	mfg	N	N		
PIPING PENETRATION FILTER EXHAUST SYSTEM													Note ah
1.	Fans	AB-209 & 220	AB-219 & 220	B	NA	0	1	5	AMCA	Y	Y	VIII	
2.	Fan motors	AB-209 & 220	AB-219 & 220	B	NA	1	1	E	NEMA MG1	Y	Y	VIII	
3.	Moisture eliminators	AB-209 & 220	AB-219 & 220	B	NA	0	1	5	ANSI N509	Y	Y	VIII	
4.	Electrical heaters	AB-209 & 220	AB-219 & 220	B	NA	1	1	E	UL, ANSI N509	Y	Y	VIII	

TABLE 3.2.2-1 (SHEET 73 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
5. Infrared heaters	AB-209 & 220	AB-219 & 220	B	NA	6	2	E	UL	N	N		
6. HEPA filters	AB-209 & 220	AB-219 & 220	B	NA	0	1	5	ANSI N509	Y	Y	VIII	
7. Charcoal filters	AB-209 & 220	AB-219 & 220	B	NA	0	1	5	ANSI N509	Y	Y	VIII	
8. Dampers			B	NA	0	1	5	ANSI N509	Y	Y		
9. Damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
10. Ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
11. Area coolers	AB-209 & 220	AB-219 & 220	B	C	3	1	3	III-3	Y	Y	VIII	
12. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
DIESEL GENERATOR BUILDING HVAC SYSTEM												Note ah
1. ESF supply fans	DB-270′	DB-270′	B	NA	0	1	5	AMCA	Y	Y	IV	
2. ESF supply fan motors	DB-270′	DB-270′	B	NA	1	1	E	NEMA MG1	Y	Y	IV	
3. Non-ESF exhaust fans	DB-255′	DB-255′	B	NA	6	2	6	AMCA	N	N		
4. Non-ESF exhaust fan motors	DB-255′	DB-255′	B	NA	6	2	E	NEMA MG1	N	N		
5. Building unit heaters	DB-234′	DB-234′	B	NA	6	2	E	UL	N	N		

TABLE 3.2.2-1 (SHEET 74 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
6. ESF ductwork			B	NA	0	1	5	ANSI N509	Y	Y		
7. ESF dampers			B	NA	0	1	5	ANSI N509	Y	Y		
8. Non-ESF ductwork			B	NA	6	2	6	SMACNA	N	N		
9. ESF damper motors			B	NA	1	1	E	NEMA MG1	Y	Y		
10. ESF instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
11. Non-ESF instrumentation			B	NA	6	2	J	mfg	N	N		
TURBINE BUILDING CONDENSER VACUUM EXHAUST FILTRATION SYSTEM												Note ah
1. HEPA filters	TB-245′	TB-245′	S	NA	6	2	6	ANSI N509	N	N		
2. Charcoal filter	TB-245′	TB-245′	S	NA	6	2	6	ANSI N509	N	N		
3. Demister	TB-245′	TB-245′	S	NA	6	2	6	ANSI N509	N	N		
4. Heater	TB-245′	TB-245′	S	NA	6	2	E	UL	N	N		
5. Piping			S	D	4	2	4	B31.1	N	N		
6. Dampers			S	NA	6	2	6	ANSI N509	N	N		
7. Instrumentation			S	NA	6	2	J	mfg	N	N		
TURBINE BUILDING HVAC SYSTEM												Note ah
1. Turbine building supply fans	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
2. Turbine building supply fan motors	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
3. Toilet exhaust fans	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
4. Toilet exhaust fan motors	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
5. Water analysis A/C fan	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
6. Water analysis A/C fan motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
7. Battery room A/C fan	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
8. Battery room A/C fan motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
9. Switchgear room level 1 A/C fan	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
10. Switchgear room level 1 A/C fan motor	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
11. Switchgear room level 2 A/C fan	TB-245′	TB-245′	S	NA	6	2	6	AMCA	N	N		

TABLE 3.2.2-1 (SHEET 75 OF 98)

Principal System and Components		(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
12.	Switchgear room level 2 A/C fan motor	TB-245′	TB-245′	S	NA	6	2	E	NEMA MG1	N	N		
13.	Stairwell exhaust fans	TB-270′	TB-270′	S	NA	6	2	6	AMCA	N	N		
14.	Stairwell exhaust fan motors	TB-270′	TB-270′	S	NA	6	2	E	NEMA MG1	N	N		
15.	Turbine building exhaust fans	TB-270′	TB-270′	S	NA	6	2	6	AMCA	N	N		
16.	Turbine building exhaust fan motors	TB-270′	TB-270′	S	NA	6	2	E	NEMA MG1	N	N		
17.	Prefilters	TB	TB	S	NA	6	2	6	ASHRAE	N	N		
18.	Cooling coils	TB	TB	S	NA	6	2	6	ARI	N	N		
19.	Heaters	TB	TB	S	NA	6	2	E	UL	N	N		
20.	Heater fans	TB	TB	S	NA	6	2	6	AMCA	N	N		
21.	Ductwork and dampers			S	NA	6	2	6	SMACNA	N	N		
22.	Damper motors			S	NA	6	2	E	NEMA MG1	N	N		
23.	Instrumentation			S	NA	6	2	J	mfg	N	N		
24.	Battery room exhaust fans	TB-220′	TB-220′	S	NA	6	2	6	AMCA	N	N		
25.	Battery room exhaust fan motors	TB-220′	TB-220′	S	NA	6	2	E	NEMA MG1	N	N		
TECHNICAL SUPPORT CENTER HVAC SYSTEM													Note ah
1.	TSC filter fan	TSC	shared	B	NA	6	2	6	AMCA	N	N		
2.	TSC filter fan motor	TSC	shared	B	NA	6	2	E	NEMA MG1	N	N		
3.	HEPA filters	TSC	shared	B	NA	6	2	6	ANSI N509	N	N		
4.	Charcoal filter	TSC	shared	B	NA	6	2	6	ANSI N509	N	N		
5.	Prefilter	TSC	shared	B	NA	6	2	6	ASHRAE	N	N		
6.	Filter unit heater	TSC	shared	B	NA	6	2	E	UL	N	N		
7.	TSC A/C fan	TSC	shared	B	NA	6	2	6	AMCA	N	N		
8.	TSC A/C fan motor	TSC	shared	B	NA	6	2	E	NEMA MG1	N	N		
9.	Moisture eliminator	TSC	shared	B	NA	6	2	6	ANSI N509	N	N		
10.	TSC toilet exhaust fan	TSC	shared	B	NA	6	2	6	AMCA	N	N		

TABLE 3.2.2-1 (SHEET 76 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
11. TSC toilet exhaust fan motor	TSC	shared	B	NA	6	2	E	NEMA MG1	N	N		
12. TSC battery room exhaust fan	TSC	shared	B	NA	6	2	6	AMCA	N	N		
13. TSC battery room exhaust fan motor	TSC	shared	B	NA	6	2	E	NEMA MG1	N	N		
14. Electric duct heaters	TSC	shared	B	NA	6	2	E	UL	N	N		
15. Ductwork	TSC	shared	B	NA	6	2	6	SMACNA	N	N		
16. Dampers	TSC	shared	B	NA	6	2	6	ANSI N509	N	N		
17. Damper operators	TSC	shared	B	NA	6	2	E	mfg	N	N		
18. Instrumentation	TSC	shared	B	NA	6	2	J	mfg	N	N		
FIRE PUMP AND WELL PUMPHOUSE HVAC SYSTEM												Note ah
1. Exhaust fans	VB	shared	S	NA	6	2	6	AMCA	N	N		
2. Exhaust fan motors	VB	shared	S	NA	6	2	E	NEMA MG1	N	N		
3. Heaters	VB	shared	S	NA	6	2	E	UL	N	N		
4. Instrumentation	VB	shared	S	NA	6	2	J	mfg	N	N		
NORMAL CHILLED WATER SYSTEM												
1. Expansion tank	CB-410	shared	B	D	4	2	4	VIII	N	N		
2. Chemical feed pot	CB-410	shared	B	D	4	2	4	VIII	N	N		
3. Air separator/ strainer	CB-410	shared	B	D	4	2	4	VIII	N	N		
4. Chilled water pumps	CB-410	shared	B	D	4	2	4	mfg	N	N		
5. Chilled water pump motors	CB-410	shared	B	NA	6	2	E	NEMA MG1	N	N		
6. Chillers	CB-410	shared	B	NA	6	2	6	VIII	N	N		
7. Piping and valves in safety-related equipment room			B	D	4	2	4	B31.1	N	N		
8. Other piping and valves			B	D	4	2	4	B31.1	N	N		
9. Instrumentation			B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 77 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
ESSENTIAL CHILLED WATER SYSTEM												
1. Expansion tanks	CB-320 & 313	CB-310 & 308	B	C	3	1	3	III-3	Y	Y	IX	
2. Chemical feed pots	CB-320 & 313	CB-320 & 313	B	NA	4	2	4	VIII	N	N		
3. ESF chilled water pumps	CB-320 & 313	CB-310 & 308	B	C	3	1	3	III-3	Y	Y	IX	
4. ESF chilled water pump motors	CB-320 & 313	CB-310 & 308	B	NA	1	1	E	NEMA MG1	Y	Y	IX	
5. Valves, piping, and cooling coils			B	C	3	1	3	III-3	Y	Y		
6. Chillers	CB-320	CB-310									IX	
(tube side)	& 313	&308		C	3	1	3	III-3	Y	Y		
(shell side)				NA	0	1	5	VIII	Y	Y		
7. Safety-related instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
SPECIAL CHILLED WATER SYSTEMS												
1. TSC chilled water pump			B	NA	6	2	6	mfg	N	N		
2. TSC chilled water pump motor			B	NA	6	2	E	NEMA MG1	N	N		
3. TSC chilled water expansion tank			B	D	4	2	4	VIII	N	N		
4. TSC chilled water air separator/strainer			B	D	4	2	4	B31.1	N	N		
5. TSC chilled water chemical feed pot			B	D	4	2	4	VIII	N	N		
6. TSC chillers			B	NA	6	2	6	VIII	N	N		
7. CAS chilled water pump			B	NA	6	2	6	mfg	N	N		
8. CAS chilled water pump motor			B	NA	6	2	E	NEMA MG1	N	N		
9. CAS chilled water expansion tank			B	D	4	2	4	VIII	N	N		
10. CAS chilled water air separator/strainer			B	D	4	2	4	B31.1	N	N		

TABLE 3.2.2-1 (SHEET 78 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
11. CAS chilled water chemical feed pot			B	D	4	2	4	VIII	N	N		
12. CAS chillers			B	NA	6	2	6	VIII	N	N		
13. Piping and valves			B	D	4	2	4	B31.1	N	N		
14. Instrumentation			B	NA	6	2	J	mfg	N	N		
AUXILIARY FEEDWATER PUMPHOUSE HVAC SYSTEM												Note ah
1. ESF supply fan	AFP	AFP	B	NA	0	1	5	AMCA	Y	Y	VI	
2. ESF supply fan motors	AFP	AFP	B	NA	1	1	E	NEMA MG1	Y	Y	VI	
3. Non-ESF supply fan	AFP	AFP	B	NA	6	2	6	AMCA	N	N		
4. Non-ESF supply fan motor	AFP	AFP	B	NA	6	2	E	NEMA MG1	N	N		
5. Building unit heaters	AFP	AFP	B	NA	6	2	E	UL	N	N		
6. ESF Dampers			B	NA	0	1	5	ANSI N509	Y	Y		
7. ESF instrumentation			B	NA	1	1	J	mfg	Y	Y		Note s
8. Non-ESF instrumentation			B	NA	6	2	J	mfg	N	N		
MISCELLANEOUS HVAC SYSTEMS												Note ah
1. Mechanical components	VB	VB	B	NA	6	2	6	mfg	N	N		
2. Electrical equipment	VB	VB	B	NA	6	2	E	NEMA MG1	N	N		
3. Instrumentation			B	NA	6	2	J	mfg	N	N		
MAIN CONTROL BOARD												Notes l, p and s
1. PAMS instrumentation			W	NA	1	1	J	mfg	Y	Y		
2. Hand switches and controls for safety-related equipment			W	NA	1	1	J	mfg	Y	Y		
3. All other instruments and controls			W	NA	6	1	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 79 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
NUCLEAR INSTRUMENTATION SYSTEM											
1. All instruments inputting to reactor protection system		W	NA	1	1	J	mfg	Y	Y		
PROCESS CONTROL SYSTEM											
1. NSSS BOP safety-related instrumentation and controls		W,B	NA	1	1	J	mfg	Y	Y		
2. NSSS BOP nonsafety-related instrumentation and controls		W,B	NA	6	1	J	mfg	N	N		
PROTECTION SYSTEM NSS											
1. Protection instrumentation and controls		W	NA	1	1	J	mfg	Y	Y		Note s
ROD CONTROL POWER SYSTEM											
1. Reactor trip switchgear		W	NA	1	1	E	mfg	Y	Y		
2. Other switchgear		W	NA	6	2	E	mfg	N	N		
FULL LENGTH ROD CONTROL SYSTEM											
1. Rod control equipment		W	NA	6	1	J	mfg	N	N		
ROD POSITION INDICATION SYSTEM											
1. Rod position instrumentation		W	NA	6	1	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 80 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
RADIATION MONITORING SYSTEM											
1. Safety-related portions		W	NA	1	1	J	mfg	Y	Y		
2. Nonsafety-related, seismic Category 1 portions		W	NA	6	1	J	mfg	N	N		
3. Other portions		W	NA	6	2	J	mfg	N	N		
ESF ACTUATION SYSTEM											
1. All portions		W	NA	1	1	J	mfg	Y	Y		
REACTOR INSTRUMENTATION											
1. All portions inputting to reactor protection		W	NA	1	1	J	mfg	Y	Y		
2. Other portions		W	NA	6	2	J	mfg	N	N		
REACTOR CONTROL SYSTEM											
1. Protection-related portions		W	NA	1	1	J	mfg	Y	Y		
2. Other portions		W	NA	6	2	J	mfg	N	N		
POST-ACCIDENT MONITORING SYSTEM											Note p
1. Safety-related portions		W,B	NA	1	1	J	mfg	Y	Y		
2. Nonsafety-related, seismic Category 1 portions		W,B	NA	6	1	J	mfg	N	N		
3. Other portions		W,B	NA	6	2	J	mfg	N	N		
PLANT AUXILIARY CONTROL BOARDS											
1. Safety-related portions		W,B	NA	1	1	J	mfg	Y	Y		Note s
2. Nonsafety-related portions		W,B	NA	6	2	J	mfg	N	N		

TABLE 3.2.2-1 (SHEET 81 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
SAFETY-RELATED SYSTEMS BYPASS/INOPERABLE STATUS AND TRIP/MONITORING INDICATING LIGHTS												
1. Trip monitoring lights			W	NA	6	1	J	mfg	N	N		
2. All other portions			W,B	NA	1	1	J	mfg	Y	Y		
INCORE INSTRUMENTATION												
1. All portions			W	NA	6	1	J	mfg	N	N		
TURBINE PROTECTION SYSTEM												
1. All portions			S	NA	6	2	J	mfg	N	N		
TURBINE SUPERVISORY INSTRUMENTATION												
1. All portions			S	NA	6	2	J	mfg	N	N		
ELECTROHYDRAULIC CONTROL SYSTEM												
1. EHC panel			S	NA	6	2	J	mfg	N	N		Unit 2 only
2. DEHC Mark Vle			SNC	NA	6	2	J	mfg	N	N		Unit 1 only
3. All other I & C portions			S	NA	6	2	J	mfg	N	N		
4. Hydraulic fluid power units	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
5. Control coolers	TB-220′	TB-220′	S	NA	6	2	6	mfg	N	N		
6. Piping & valves			S	NA	6	2	6	B31.1	N	N		
AMSAC												
1. Nonsafety-related portion			W	NA	6	1	J	mfg	N	N		
2. Safety-related portion			W	NA	1	1	J	mfg	Y	Y		
COMPUTER SYSTEM												
1. All portions			S	NA	6	2	J	mfg	N	N		
ANNUNCIATOR SYSTEM												
1. All portions			W	NA	6	1	J	mfg	N	N		
TELEPHONE PAGE SYSTEM												

TABLE 3.2.2-1 (SHEET 82 OF 98)

1.	All portions	B	NA	6	2	E	mfg	N	N

TABLE 3.2.2-1 (SHEET 83 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
PABX SYSTEM											
1. All portions		G	NA	6	2	E	mfg	N	N		
SOUND-POWERED SYSTEM											
1. Maintenance and refueling		B	NA	6	2	E	mfg	N	N		
2. Shutdown		B	NA	6	2	E	mfg	N	N		Note v
SEISMIC MONITORING EQUIPMENT											
1. “Free Field” accelerograph near river intake structure		B	NA	6	2	J	mfg	N	N		Note al
2. All other portions		B	NA	6	1	J	mfg	N	N		Note ax
PLANT SECURITY SYSTEM											
1. Fencing		B	NA	6	2	6	mfg	N	N		
2. Instrumentation		B	NA	6	2	J	mfg	N	N		
3. Electrical equipment		B	NA	6	2	E	mfg	N	N		
OFFSITE POWER SYSTEM											Note n
1. All portions		S	NA	6	2	E	mfg	N	N		
ac SYSTEM - 480 V (Class 1E portion)											
1. 4160/480 V transformers		B	NA	1	1	E	mfg	Y	Y		
2. Load centers		B	NA	1	1	E	mfg	Y	Y		
3. Motor control centers		B	NA	1	1	E	mfg	Y	Y		
4. Instrumentation and control		B	NA	1	1	E	mfg	Y	Y		
ac SYSTEM, 480 V (Non-class 1E portions)											
1. 4160/480V transformers		B	NA	6	2	E	mfg	N	N		
2. Load centers		B	NA	6	2	E	mfg	N	N		
3. Motor control centers		B	NA	6	2	E	mfg	N	N		
4. Instrumentation and control		B	NA	6	2	E	mfg	N	N		

TABLE 3.2.2-1 (SHEET 84 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
ac SYSTEM - 4160 V (Class 1E portions)											
1. 4.16 kV buses and switchgear		B	NA	1	1	E	mfg	Y	Y		
2. Instrumentation and controls		B	NA	1	1	E	mfg	Y	Y		
ac SYSTEM - 4160 V (Non-class 1E portion)											
1. 4.16 kV buses and switchgear		B	NA	6	2	E	mfg	N	N		
2. Instrumentation and control		B	NA	6	2	E	mfg	N	N		
dc SYSTEM - CLASS 1E											
1. Batteries		B	NA	1	1	E	mfg	Y	Y		
2. Chargers		B	NA	1	1	E	mfg	Y	Y		
3. Breakers, buswork, and switchgear		B	NA	1	1	E	mfg	Y	Y		
4. Instrumentation and controls		B	NA	1	1	E	mfg	Y	Y		
5. Motor control center		B	NA	1	1	E	mfg	Y	Y		
6. Distribution panels		B	NA	1	1	E	mfg	Y	Y		
120-V ac POWER SYSTEM - CLASS 1E											
1. Transformers		B	NA	1	1	E	mfg	Y	Y		
2.		(This line has been intentionally left blank.)									
3. dc-ac inverters		B,W	NA	1	1	E	mfg	Y	Y		
4. Instrumentation and control		B	NA	1	1	E	mfg	Y	Y		Note s
5. Distribution panels		B	NA	1	1	E	mfg	Y	Y		
LIGHTING SYSTEM											
1. Emergency lighting (control room, shutdown, diesel, and auxiliary feedwater panels including access and egress route to these areas)		B	NA	6	1	E	Mfg	N	N		Note o, v

TABLE 3.2.2-1 (SHEET 85 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
2. Other emergency lighting		B	NA	6	2	E	mfg	N	N		Note v
3. Essential lighting		B	NA	6	2	E	mfg	N	N		
4. Other lighting		B	NA	6	2	E	mfg	N	N		
5. Lighting isolation transformers		B	NA	1	1	E	mfg	Y	Y		
CABLE SYSTEM											
1. Safety-related power, control, and instrument cables		B	NA	1	2	E	mfg	Y	Y		
2. Nonsafety-related portions		B	NA	6	2	E	mfg	N	N		
FIRE DETECTION SYSTEM											Note v
1. Detector and alarm panels		B	NA	6	2	E	mfg	N	N		
2. Signaling systems		B	NA	6	2	E	mfg	N	N		
3. Local zone indicating panel		B	NA	6	2	E	mfg	N	N		
4. Local display cabinets		B	NA	6	2	E	mfg	N	N		
HEAT TRACING SYSTEMS											
1. Boric acid injection heat tracing, sensors and controls		B	NA	6	2	E	mfg	N	N		
2. Other heat tracing, sensors, and controls		B,S	NA	6	2	E	mfg	N	N		
ELECTRICAL PENETRATION SYSTEM											
1. Penetration assemblies		B	NA	1	1	E	IEEE-317, III-MC	Y	Y		

TABLE 3.2.2-1 (SHEET 86 OF 98)

Principal System and Components	(a) Location Unit 1 Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
STANDBY POWER SYSTEM											
1. Diesel generator package		B	NA	1	1	E	DEMA	Y	Y		
2. Instrumentation and controls		B	NA	1	1	E	mfg	Y	Y		Note s
3. Diesel generator jacket water chemical addition system		S	NA	6	2	6	mfg	N	N		
ac SYSTEM - 25 kV											
1. All portions		B	NA	6	2	E	mfg	N	N		
dc SYSTEM - NON-CLASS 1E											
1. All portions		B	NA	6	2	E	mfg	N	N		
120-V ac POWER SYSTEM - NON-CLASS 1E											
1. All portions		B,S	NA	6	2	E	mfg	N	N		
SWITCHYARD INTERFACES											
1. All portions		B,S	NA	6	2	E	mfg	N	N		
MULTISYSTEM PANELS AND BOARDS											
1. Safety-related portions		B	NA	1	1	E	mfg	Y	Y		
2. Other portions		B,S	NA	6	1,2	E	mfg	N	N		
ISO-PHASE BUS SYSTEM											
1. All portions		S	NA	6	2	E	mfg	N	N		
ac SYSTEM - 13.8 kV											
1. 13.8-kV buses and switchgear		B,S	NA	6	2	E	mfg	N	N		
2. 13.8-kV RCP 1E breakers		B	NA	1	1	E	mfg	Y	Y		

TABLE 3.2.2-1 (SHEET 87 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
HIGH VOLTAGE SWITCHYARD												
1. All portions			S	NA	6	2	E	mfg	N	N		
STRUCTURES AND BUILDINGS												
1. Containment building			B	NA	0	1	C	ASME III	Y	Y		Code identified is for design only. See section 3.8.1 for details. Material, fabrication, and erection only.
2. Equipment hatch and personnel locks	C	C	B	NA	0	1	C	Div. 2, CC-3000 III-MC	Y	Y		
3. Liner plate system	C	C	B	NA	0	1	C	ASME III	Y	Y		
4. Penetration assemblies	C	C	B	NA	0	1	C	ASME III	Y	Y		
5. Fuel transfer tube housing expansion bellows and supports	C,FB	C,FB	B	NA	0	1	C	Div. 1, III-MC, NF	Y	Y		
6. Equipment building			B	NA	6	2	C	AISC-69, ACI 318-71	N	N		
7. NSCW cooling towers			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
8. Diesel generator building			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
9. Auxiliary building			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
10. Fuel handling building			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
11. Control building			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
12. Refueling water storage tank and dike	O	O	B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
13. Condensate storage tank and dike	O	O	B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
14. Diesel fuel oil storage tank pumphouse			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
15. Category 1 tunnels			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		

TABLE 3.2.2-1 (SHEET 88 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
16. Auxiliary feedwater pumphouse			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
17. Spent fuel pool and refueling canal liner plate	FB	FB	B	NA	0	1	C	AISC-69	Y	Y		
18. Fuel pool gate	FB	FB	B	NA	0	1	C	AISC-69	Y	Y		
19. Turbine building			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
20. River intake structure			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
21. Plant water makeup wells			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
22. Reactor makeup water storage tank and dike			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
23. Circulating water intake structure			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
24. Natural draft cooling tower			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
25. Circulating water canals			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
26. Circulating water piping			B	NA	6	2	C	AWWA C-301	N	N		
27. Minor plant structures and pads			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
28. Containment internal structures			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		
29. Turbine generator pedestal			S	NA	6	2	C	AISC-69, ACI 318-71, UBC-76	N	N		
30. Storm drain system			S	NA	6	2	C	mfg	N	N		Note Y
31. River makeup water piping			S	NA	6	2	C	AWWA C-200	N	N		
33. NSCW tower valve house			B	NA	0	1	C	AISC-69, ACI 318-71	Y	Y		

TABLE 3.2.2-1 (SHEET 89 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
34. Radwaste transfer building			B	NA	6	2	C	AISC-69 ACI 318-71, UBC-76	N	N		
35. Category 1 electrical cable tray and conduit supports			B	NA	0	1	C	AISC-69, AISI-68	Y	Y		Note 2
36. Category 1 HVAC duct supports			B	NA	0	1	C	AISC-69	Y	Y		Note 2
37. Category 1 pipe supports			B		See Note 4			AISC-69 III-NF	Y	Y		Note 2
38. Pipe whip restraints			W,B	NA	0	1	C	AISC-69	Y	Y		
39. Water tight doors and seals			B	NA	0	1	C	mfg	Y	Y		
40. Waterproofing and water stops			B	NA	6	2	C	mfg	N	N		
41. Category 1 backfill			B	NA	0	1	C	See DC-1000-C	Y	Y		
42. Category 1 tank liner plate	O	O	B	NA	0	1	C	AISC-69	Y	Y		
43. Underground Category 1 conduits			B	NA	0	1	C	AISC-69,	Y	Y		
44. Alternate radwaste building, control room and dress out area	shared		B	NA	6	2	C	UBC-76 AISC-69 ACI-318-71	N	N		
45. Fire dampers	VB	VB	B	NA	6	1,2	6	mfg	N	N		Note v
46. Fire doors	VB	VB	B	NA	6	2	C	mfg	N	N		Note v
47. Fire-rated penetration seals	VB	VB	B	NA	6	2	C	mfg	N	N		Note v
48. Structural steel fire proofing	VB	VB	B	NA	6	2	C	mfg	N	N		Note v
49. Radiant energy shields	C	C	B	NA	6	2	9	mfg	N	N		Note v
50. Electrical raceway fireproofing	VB	VB	B	NA	6	2	9	mfg	N	N		Note v
51. 3-hour plaster walls	VB	VB	B	NA	6	2	C	mfg	N	N		Note v

TABLE 3.2.2-1 (SHEET 90 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
52. Seismic gap fire seals	VB	VB	B	NA	6	2	C	mfg	N	N		Note v
53. Alternate radwaste bldg. dressout area and control room HVAC	Shared		S	NA	6	2	6	mfg	N	N		
54. Alternate radwaste bldg bridge crane	Shared		G	NA	6	2	C	mfg	N	N		
55. Deleted.												
56. Deleted.												
57. Deleted.												
58. HELB doors	AB	AB	B,S	NA	6	2	C	mfg	N	N		Note am
59. RPF bridge crane	Shared		S	NA	6	2	C	mfg	N	N		
60. RPF building	Shared		S	NA	6	2	C	UBC-76 ACI-318-71	N	N		
CONTAINMENT BUILDING POLAR BRIDGE CRANE												
1. Mechanical components	C	C	B	NA	6	1	6	mfg	N	N		Note q and ab
2. Motors	C	C	B	NA	6	2	E	NEMA MG1	N	N		Note ab
3. Instrumentation and controls			B	NA	6	2	J	mfg	N	N		Note ab
FUEL HANDLING SYSTEM												
1. New and spent fuel storage racks	FB	FB	W,B	NA	0	1	5	mfg	Y	Y	VII	
2. Refueling machine	C	C	W,G	NA	6	2	6	mfg	N	N		Note ab
3. RCC storage station	C	C	W	NA	6	2	6	mfg	N	N		
4. Thimble plug storage rack	C	C	W	NA	6	2	6	mfg	N	N		
5. Integrated head cable assembly	C	C	W	NA	6	2	6	mfg	N	N		
6. Integrated head cable tray	C	C	W	NA	6	2	6	mfg	N	N		
7. Integrated head lifting rig	C	C	W	NA	6	2	6	mfg	N	N		
8. Integrated head lift rods	C	C	W	NA	0	1	5	III-NF	Y	Y		

TABLE 3.2.2-1 (SHEET 91 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
9. Integrated head missile shield	C	C	W	NA	0	1	5	III-NF	Y	Y		
10. Integrated head operator support stand	C	C	W	NA	6	2	6	mfg	N	N		
11. Integrated head package stud support collars	C	C	W	NA	6	2	6	mfg	N	N		
12. Radial arm hoist assembly	C	C	W	NA	6	2	6	mfg	N	N		
13. Radial arm stud tensioner hoist			S	NA	6	2	6	mfg	N	N		
14. Reactor internals lifting rig	C	C	W	NA	6	2	6	mfg	N	N		
15. Fuel handling machine	FB	FB	W	NA	0	1	5	mfg	Y	Y	VII	
16. Spent fuel handling tool	FB	FB	W	NA	0	1	5	mfg	Y	Y	VII	
17. New fuel handling tool	FB	FB	W	NA	6	2	6	mfg	N	N		
18. Fuel transfer tube			W	B	2	1	2	III-MC	Y	Y		
19. Fuel transfer system			W	NA	6	2	6	mfg	N	N		Note ab
20. New fuel elevator	FB	FB	W	NA	6	2	6	mfg	N	N		Note ab
21. Spent fuel cask bridge crane	FB	FB	B	NA	0	1	5	mfg	Y	Y		Note q
RACEWAY, RACEWAY ACCESSORIES, AND FITTINGS FOR												
1. Safety-related power, control, and instrument circuits, in seismic structures			B	NA	6	1	E	mfg	N	N		
2. Nonsafety-related circuits			B	NA	6	2	E	mfg	N	N		

TABLE 3.2.2-1 (SHEET 92 OF 98)

Principal System and Components	(a) Location Unit 1	Unit 2	(b) Source of Supply	(c) Quality Group	(d) VEGP Safety Class	(e) Seismic Category	(f) Codes and Standards Designator	(g) Principal Construction Code	(h) Q-List	(i) Safety Related	(j) Environmental Designator	(k) Comments
3. Safety-related power, control, and instrumentation circuits in nonseismic structures (Reactor trip on turbine trip, steam dump solenoid valves, and turbine impulse chamber pressure transmitters only.)	TB	TB	B	NA	1	1	E/J	mfg	Y	Y		az

TABLE 3.2.2-1 (SHEET 93 OF 98)

FOOTNOTES AND COMMENTS

a. Location

AB	Auxiliary building
AFP	Auxiliary feedwater pumphouse
C	Containment building
CB	Control building
DB	Diesel generator building
EB	Equipment building
FB	Fuel handling building
FPH	Fire protection pumphouses
NSW	Nuclear service cooling water valve house
O	Outdoors onsite
RTB	Radwaste transfer building
TB	Turbine building
TSC	Technical support center
VB	Various building
RPF	Radwaste processing facility

Equipment is located by either room number, elevation, or building. Major buildings and their abbreviations are listed above. These abbreviations are used in the location column; e.g., AB is the auxiliary building, as shown in the following examples: AB-A48 indicates the equipment is located in the auxiliary building, on level A, room number 48; C-171' indicates the equipment is located in the containment building at elevation 171 ft.

For equipment that is common to both units, the location is indicated in the column for Unit 1 with "shared" indicated in the column for Unit 2. Some major valves, such as main steam isolation valves, have been located, but in general, piping, valves, ductwork, instrumentation, etc., have not been located.

b. Source of Supply

G	Georgia Power Company
B	Bechtel
W	Westinghouse
S	Southern Company Services, Inc.
SNC	Southern Nuclear Operating Company, Inc.

The organization which has the principal procurement responsibility is identified as the source of supply. *(Historical) This is historical information to indicate the organization which at the time of construction of Vogtle 1 and 2 had principle procurement responsibility. Updates to this information for future equipment changes will not be maintained.*

TABLE 3.2.2-1 (SHEET 94 OF 98)

c. Quality Group

The quality group classification corresponds to those provided in Regulatory Guide 1.26. NA indicates not applicable and is used for equipment and structures that do not fall under the purview of Regulatory Guide 1.26.

d. VEGP Safety Class

The VEGP nuclear safety class corresponds to the quality group classifications in Regulatory Guide 1.26; i.e., Nuclear Safety Classes 1, 2, 3, and 4 correspond to Quality Groups A, B, C, and D, respectively. Nuclear Safety Class 1 is also assigned to safety-related instruments, controls, and electric components.

e. Seismic Category

Seismic Category 1 is applied to those safety-related structures, systems, and components that must remain functional during and after a safe shutdown earthquake (SSE) according to Regulatory Guide 1.29 and to those nonsafety-related structures, systems, and components that are designed to Seismic Category 1 requirements.

f. Codes and Standards Designator

See paragraph 3.2.2.3.

g. Principal Construction Code

The codes referenced are primary codes only and are defined in table 3.2.2-2. Detailed construction codes are listed in the component specification.

h. Q-List

Y Yes; requires compliance with 10 CFR 50, Appendix B, as implemented in the VEGP and Bechtel/Westinghouse/SCS/SNC quality assurance programs.

N No; not within the scope of 10 CFR 50, Appendix B.

i. Safety Related

Y Yes; safety related.

N No; not safety related.

j. Environmental Designator

The environmental designators are defined in table 3.11.B.1-1. Environmental designators are provided here for only principal safety-related equipment such as pumps, tank, fans, etc.

TABLE 3.2.2-1 (SHEET 95 OF 98)

k. Comments

This column contains a listing of applicable design criteria and other amplifying information.

l. The main control board will be qualified by Westinghouse to Institute of Electrical and Electronics Engineers (IEEE) 344 and 323. Installation of 62J instruments will be equivalent to project Class 61J.

m. The load can be manually applied to the diesel generator by the operator after sequencer loads have been energized. Refer to diesel generator loading table in drawings 1X3D-AA-K02A, 2X3D-AA-K02A, 1X3D-AA-K02B, and 2X3D-AA-K02B.

n. Offsite power equipment is designed and arranged to provide redundancy and separation for increased reliability.

o. Emergency lighting for the control room suspended ceiling, shutdown panels, and diesel and auxiliary feedwater pumphouse panels is non-Class 1E but shall be powered from 1E supply through isolation devices, maintain necessary redundancy, and meet single failure criteria. The emergency lighting distribution panels are non-Class 1E, but are Seismic Category 1. The control room suspended ceiling is designed and constructed to ensure that the ceiling will not fall or compromise the functioning of safety-related equipment during or after an SSE. Emergency lighting for fire protection consists of fixed 8-h-rated sealed beam fixtures with self-contained battery and charger units powered from normal lighting system. Fixtures which only provide for life safety and are not required for safe shutdown or to support station blackout (SBO) are 1 ½-h rated (minimum). Refer to paragraph 8.4.1.1.2.G for SBO emergency lighting requirements and paragraph 9.5.3.2.3.C for fire protection lighting requirements.

p. Post-accident monitoring system (PAMS) instruments are assigned a project classification based on their category as defined in Regulatory Guide 1.97, Revision 2, as indicated below:

Category 1 - All instruments in this category are classified 11J.

Category 2 - Instruments in this category are qualified from the sensor up to and including the channel isolation device as delineated in table 7.5-1.

Category 3 - All instruments in this category are classified 62J.

PAMS recorders are assigned a project classification of 61J regardless of their category, although they are not designed to function during an SSE, only after an SSE.

q. The crane is designed to retain and prevent dropping its design load during and after an SSE.

r. Selected materials, components, parts, appurtenances, and piping subassemblies are procured in accordance with ASME Code, Section III, Class 3; however, the system is

designed and installed in accordance with ANSI B31.1. Conformance with these aspects of the ASME code is required only for initial procurement.

s. Instrumentation and controls which can affect the operation or actuation of a safety-related function must be qualified to requirements of project Class 11J. Other instrumentation and controls, even in safety systems, need not be qualified to project Class 11J requirements if their function is associated with normal plant operations only and serves no safety-related function. Refer to the project instrument index for individual instrument listing. Refer to note p for further clarification.

t. The entire heat exchanger is constructed to ASME Section III requirements to ensure the integrity of the safety-related portion.

u. Portions of the sample system that are part of the pressure boundary of the system being sampled must meet the same quality and code requirements as that sampled system up to and including the first normally shut isolation valve in the sample line.

v. The quality assurance program to be applied to fire protection systems is described in paragraph 9.5.1.1.4.

w. The quality assurance program to be applied to radioactive waste management systems is described in Regulatory Guide 1.143.

x. The Seismic Category 1 fire protection standpipe system serves no safety function but is classified as project class 313 to ensure the implementation of a Seismic Category 1, ASME III-3 design and installation. ASME Code stamping and inclusion in the Final Design Verification Program is not required.

y. Changes to the site grading will be done on an engineered basis so as to assure acceptability of the drainage analysis for the probable maximum precipitation (PMP) event as described in paragraph 2.4.2.3.

z. Selected materials, components, parts, appurtenances, and piping subassemblies are procured in accordance with ASME Code, Section III, Class 3. This conformance with the ASME Code is only required for initial procurement. The system is designed and installed in accordance with ANSI B31.1. Post-installation nondestructive examination of the system in accordance with ASME III, Class 3 is required for the initial installation. Final "N" stamping and associated documentation are not required.

aa. These components are manufactured under the appropriate provisions of WCAP 8370.

ab. The pertinent provisions of the QA program will be applied to these items.

ac. The principal construction code for this tank is ASME Section VIII; however, no code stamp is required since the tank operates at atmospheric pressure.

ad. This portion of the instrument and service air system identifies upgraded piping and valves that function as part of the nonsafety-related post-LOCA containment purge

system. As described in section 6.2.5, the post-LOCA containment purge system is not required following an accident, but provides additional combustible gas control capabilities.

To assure the availability of this function, the associated portions of the instrument and service air system inside the containment have been upgraded to satisfy Seismic Category 1 and ASME III design requirement.

ae. Duct work in this system was upgraded to Seismic 1 requirement to provide additional margin in the structural design. The quality assurance program is not applicable to the duct work.

af. Design and installation to the technical requirement of 1971-S73 ASME Code without "CS" symbol stamp.

ag. These components were originally procured as safety related but do not perform a safety function, are not required to respond to any accident, and are not included in any evaluation of an accident in the UFSAR. A portion of the waste gas processing system, and also the RHT vent eductor, were reclassified to project class 417 as defined on P&IDs 1/2X4DB128, 1/2X4DB129, 1X4DB141, 1X6AK09-176, 1X6AK09-177, 2X6AK09-178, 2X6AK09-179, AX6AK09-180, and AX6AK09-181. The pertinent provisions of the quality assurance program will be applied to the waste gas processing system.

ah. Leak testing of nonfiltration systems and portions of filtration systems (i.e., flexible connections and terminating duct pieces and accessories at inlets, outlets, and exhaust shafts) is performed per SMACNA requirements using ANSI N510 requirements that address methods for performing leak tests.

ai. Duct stiffener angles may be attached to the duct using huck bolts in safety-related nonfiltration systems or any portion of nonsafety-related systems.

aj. Deleted.

ak. The boron injection tank is not provided for Unit 2.

al. The accelerograph is purchased 61J but powered and mounted 62J.

am. The quality assurance program to be applied to HELB doors is the same as that applied to fire protection systems described in paragraph 9.5.1.1.4.

an. The internals have been removed from filter. Filter housing remains in piping system.

ao. The filter element may be removed by cutting element in half and replacing cover assembly. Note that the cover assembly seal includes the upper portion of the filter element basket.

TABLE 3.2.2-1 (SHEET 98 OF 98)

ap. The liquid hydrogen storage tank has been removed and the hydrogen requirements are provided from a gaseous tube trailer through the discharging stanchion.

aq. The spray additive portion of the containment spray system has been eliminated. The associated components are abandoned in place.

ar. The control room HVAC system (essential portion) return air fans are retired in place.

as. The drain isolation valves for ESF equipment rooms and negative pressure boundary areas are procured as project class 313, but are installed as project class 414. These valves are required to be locked closed to ensure validity of the offsite dose exposure analysis as tabulated in table 15.6.5-6.

at. Drain isolation valves for the ESF equipment rooms may be left open during modes 5 and 6. The exceptions for CVCS and RHR equipment rooms are described below.

a. For RHR equipment rooms, the drain isolation valves of one train may remain open during modes 5 and 6 if that train is out of service and the drain valves of the other train remain locked closed. This will ensure a potential flood in one train room would not affect the other train.

b. For CVCS centrifugal charging pump rooms during modes 5 and 6, the equipment drain isolation valves of one train may remain open if that train is out of service and the equipment drain isolation valves of the other train remain locked in the closed position. For these rooms during modes 5 and 6, the floor drain isolation valves of one train may remain open if that train is out of service and the floor drain isolation valves of the other train remain locked closed. This will ensure a potential flood in one train room would not affect the other train.

c. For the CVCS normal charging pump room during modes 5 and 6, the equipment drain isolation valve(s) may remain open if the normal charging pump is out of service and the equipment drain isolation valves of the inservice CVCS centrifugal charging pump train(s) remain locked in the closed position. For these rooms during modes 5 and 6, the floor drain isolation valves may remain open if the normal charging pump is out of service and the floor drain isolation valves of the inservice CVCS centrifugal charging pump train(s) remain locked closed. This will ensure a potential flood in the normal charging pump room would not affect the inservice CVCS centrifugal charging pump train(s).

au. Recycle evaporator package is abandoned in place.

av. Abandoned in place.

aw. Waste evaporator is abandoned in place.

ax. The seismic category of the cabinet is nonseismic 2. However, through analysis the cabinet can withstand a safe shutdown earthquake.

TABLE 3.2.2-2 (SHEET 1 OF 2)

PRINCIPAL CODES AND STANDARDS

I	ASME Boiler and Pressure Vessel Code, Section I.
III-1,2,3, MC, NF,	ASME Boiler and Pressure Vessel Code, CS Section III, Subsections NB, NC, ND, NE, NF, and NG.
VIII	ASME Boiler and Pressure Vessel Code, Section VIII, Division 1.
B31.1	ANSI B31.1.0, Power Piping.
AISC-69	American Institute of Steel Construction, Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, adopted February 12, 1969, with Supplements 1, 2, and 3.
AISI	American Iron and Steel Institute, Specification for the Design of Cold-Formed Steel Structural Members, 1968, Design of Light Gage Cold-Formed Stainless Steel Structural Members, 1968.
AMCA	Air Moving and Conditioning Association.
ACI 318-71	American Concrete Institute, Building Code Requirements for Reinforced Concrete, including 1974 Supplement.
ANSI N509	American National Standard Institute, Nuclear Power Plant Air Cleaning Units and Components.
API-620	American Petroleum Institute, Recommended Rules for Design and Construction of Large, Low Pressure Storage Tanks.
API-650	American Petroleum Institute, Welded Steel Tanks for Oil Storage.
ARI	Air Conditioning and Refrigeration Institute.
ASHRAE	American Society of Heating, Refrigerating and Air Conditioning Engineers.
DEMA	Diesel Engine Manufacturer Association, Standard Practices for Stationary Diesel and Gas Engines, 1971.
mfg	Manufacturer's standard. Design requirements specified by designer with appropriate consideration of the intended service and operating conditions.
NEMA MG1	National Electric Manufacturers Association, 1972, Motors and Generators.
NFPA	National Fire Protection Association.

TABLE 3.2.2-2 (SHEET 2 OF 2)

SMACNA	Sheet Metal and Air Conditioning Contractors National Association, Inc.
TEMA C,R	Tubular Exchanger Manufacturers Association, Class C or R.
UBC	Uniform Building Code.
UL	Underwriters' Laboratories.

TABLE 3.2.2-3

SUMMARY OF CONSTRUCTION[a] CODES AND STANDARDS FOR VEGP COMPONENTS BY NRC QUALITY CLASSIFICATION SYSTEM[b]

Components	Quality Group A	Quality Group B	Quality Group C	Quality Group D
Pressure vessels	ASME Boiler and Pressure Vessel Code, Section III, Division 1, Subsection NB, Class 1, nuclear power plant components	ASME Boiler and Pressure Vessel Code, Section III, Division 1, Subsection NC, Class 2, nuclear power plant components	ASME Boiler and Pressure Vessel Code, Section III, Division 1, Subsection ND, Class 3, nuclear power plant components[c]	ASME Boiler and Pressure Vessel Code, Section VIII, Division 1
Piping	As above	As above	As above	ANSI B31.1 power piping
Pumps	As above	As above	As above	Manufacturer's standards
Valves	As above	As above	As above	ANSI B31.1 power piping
Atmospheric storage tanks	Not applicable	As above	As above	API-650
0- to 15-psig storage tanks	Not applicable	As above	As above	API-620
Supports	As above except Subsection NF	As above except Subsection NF	As above except Subsection NF	Manufacturer's standards
Metal containment components	Not applicable	As above except Subsection NE, Class MC	Not applicable	Not applicable
Core support structures	Not applicable	As above except Subsection NG	Not applicable	Not applicable

a. As defined in Subarticle NCA-1110 of Section III of the ASME Boiler and Pressure Vessel Code, construction is an all-inclusive term comprising materials, design, fabrication, examination, testing, inspection, and certification required in the manufacture and installation of components.

b. As defined in Regulatory Guide 1.26, the NRC quality classification system identifies on a functional basis components of fluid systems by Quality Groups A, B, C, and D.

c. The specific applicability of ASME Code Cases is covered separately in tables 1.9-1 through 1.9-3.

3.3 WIND AND TORNADO LOADINGS

3.3.1 WIND LOADINGS

The wind loadings for Seismic Category 1 structures are in accordance with American National Standards Institute (ANSI) A58.1, Building Code Requirements for Minimum Design Loads in Buildings and Other Structures.(1)

3.3.1.1 Design Wind Velocity

The design wind velocity for all Seismic Category 1 structures is 110 mph at 30 ft above grade for a 100-year mean recurrence interval. The vertical velocity profiles and gust factors are in accordance with subsection 6.3.4 of reference 1.

3.3.1.2 Determination of Applied Forces

The procedures utilized in transforming the wind velocity into an effective pressure to be applied to structures and parts and portions of structures follow the guidelines of reference 1. For a design wind velocity of V_{30} mph specified at a height of 30 ft above grade the velocity pressure, q_{30}, is given by:

$$q_{30} = 0.00256 (V_{30})^2 \text{ lb/ft}^2$$

The design wind pressure and the pressure distribution are obtained using the provisions of reference 1 for Exposure C, which is applicable for flat open terrain and BC-TOP-3A.(2)

3.3.1.3 References

1. "The American National Standard Building Code Requirements for Minimum Design Loads in Buildings and Other Structures," ANSI A58.1-72.

2. Topical Report, "Tornado and Extreme Wind Design Criteria for Nuclear Power Plants," Bechtel Power Corporation, BC-TOP-3A, Rev. 3, San Francisco, California, August 1974.

3.3.2 TORNADO LOADINGS

Seismic Category 1 structures, housing safety-related equipment, systems, and components, are designed to withstand the effects due to the design basis tornado as described in the following paragraphs.

3.3.2.1 Applicable Design Parameters

The design parameters applicable to the design basis tornado are as follows:

- Maximum peripheral tangential velocity - 290 mph.

- Translational velocity - 70 mph maximum/5 mph minimum.
- Maximum wind velocity - 360 mph.
- Radius from the center of the tornado, where the maximum wind velocity occurs - 150 ft.
- Atmospheric pressure drop - 3 psi.
- Rate of pressure drop - 2 psi/s.

These parameters conform to those given in Regulatory Guide 1.76 for Region I. The design basis tornado missiles are discussed in paragraph 3.5.1.4.

3.3.2.2 Determination of Forces on Structures

The procedures specified in BC-TOP-3A[1] are used to transform the tornado wind loading and differential pressure loading into effective loads on structures.

The dynamic wind pressure is applied to the structure in the same manner as the wind loads described in paragraph 3.3.1.2, with the exception that the gust factor and the variation of wind speed with height do not apply. Loading combinations and load factors used are as specified in reference 1 and are as follows:

$W_t = W_w$

$W_t = W_p$

$W_t = W_m$

$W_t = W_w + 0.5\ W_p$

$W_t = W_w + W_m$

$W_t = W_w + 0.5\ W_p + W_m$

where:

W_t = total tornado load.

W_w = total wind load.

W_p = total differential pressure load.

W_m = total missile load.

The maximum pressure drop of 3 psi, applicable to a nonvented structure, is used for W_p, unless a lower value is justified using the provisions of reference 1, for partially vented structures. When the tornado loading includes the missile load, the structure locally may go in the plastic range due to the missile impact. The procedure for analyzing local missile effects is presented in appendix 3C.

The analysis of the nonsafety-related equipment building shows that it will not collapse on adjacent Seismic Category 1 structures, equipment, systems, or components due to tornado loading.

3.3.2.3 Effect of Failure of Structures or Components Not Designed for Tornado Loads

The non-Seismic Category 1 structures, equipment, systems, and components not designed for tornado loadings are investigated to ensure the following:

A. These structures, equipment, systems, and components cannot produce missiles that have more severe effects than the tornado-generated missiles discussed in paragraph 3.5.1.4.

B. Their failure will not affect the integrity of adjacent Seismic Category 1 structures. This design ensures that Seismic Category 1 structures, equipment, systems, and components required for safe shutdown after a tornado will perform their intended functions.

The analyses of the turbine building, radwaste transfer building, radwaste tunnel, alternate radwaste building, and radwaste processing facility (Seismic Category 2 structures) show that they will not jeopardize adjacent Seismic Category 1 structures when subjected to the tornado loads described in paragraph 3.3.2.2.

3.3.2.4 Reference

1. Topical Report, "Tornado and Extreme Wind Design Criteria for Nuclear Power Plants," Bechtel Power Corporation, BC-TOP-3A, Rev. 3, San Francisco, California, August 1974.

3.4 WATER LEVEL (FLOOD) DESIGN

The flooding of a nuclear power plant from natural causes can be attributed to probable maximum flood (PMF), site and adjacent area probable maximum precipitation (PMP) runoff, and ground water. Criteria for the design basis flood conform to the requirements of Regulatory Guide 1.59, Design Basis Floods for Nuclear Power Plants, and Regulatory Guide 1.102, Flood Protection for Nuclear Power Plants.

3.4.1 FLOOD PROTECTION

3.4.1.1 Flood Protection Measures for Seismic Category 1 Structures

The Seismic Category 1 structures, systems, and components identified in table 3.2.2-1 are designed to withstand the effects of natural phenomena, such as flooding and ground water level. A description of the structures is provided in sub-sections 3.8.1, 3.8.3, and 3.8.4.

3.4.1.2 Flood Protection from Natural Causes

No flooding due to the PMF can occur, because the finished grade levels of the VEGP are located above the PMF level of el 165 ft msl. The PMF results from river flooding, upstream dam failure, and other natural causes. (See subsection 2.4.3.) No Seismic Category 1 structures, systems, or components can be affected by this maximum flood condition, because the plant nominal finished grade elevation is 219 ft 6 in.

No flooding can occur from the PMP. Water from roof drains and/or scuppers and runoff from the plant site and adjacent areas is conveyed to catch basins, underground pipes, or directly to open ditches by sloping the tributary surface area. The site is graded to offer protection to the Seismic Category 1 structures by a minimum of 1-percent surface slope.

A high ground water level has been established at el 165 ft msl (paragraph 2.5.1.2). The basement levels of the containment, auxiliary and fuel handling buildings, and electrical and piping tunnels, which are located below the high ground water table, are protected against ground water flooding. Prevention of ground water entry is primarily afforded by the thick (minimum 24 in.) concrete walls and floors. In addition, nonsafety-related waterproofing treatments are provided on exterior walls of safety-related structures up to plant grade elevation. Two types of waterproofing are used:

- Waterproof membrane.
- A chemical waterproofing treatment.

One waterstop is provided at each construction joint below el 170 ft, except in the nuclear service cooling water towers where two waterstops are provided at each construction joint below el 220 ft. Two waterstops are provided at each seismic separation joint below el 170 ft, and one waterstop is provided in between el 170 and 220 ft. A typical waterstop in a seismic separation joint is described in subsection 3.8.5. The seismic joints are covered by a flashing to prevent water from entering the joint from above. There are no penetrations through exterior walls or basemats into the soil below the high ground water level. The only exterior personnel or equipment access to the Seismic Category 1 structures is at the finished grade level or above.

3.4.1.2.1 Flood Protection From Component Failures

Each area of the plant is reviewed to determine the postulated fluid system failure, including non-Seismic Category 1 and nontornado-protected tanks, vessels, and other process equipment, which results in the most adverse flooding conditions. Flooding due to failure of non-Seismic Category 1 tanks and vessels located in outside areas will not impact safety-related equipment, since flood water would be directed away from buildings to catch basins and open ditches. In addition, no safety-related equipment would be affected by flooding due to failure of non-Seismic Category 1 tanks and vessels in the auxiliary building. Waterproof doors, curbs, wall penetrations, seals, or drainage systems are provided for safety-related equipment to mitigate the consequences of such failures. The criteria used in the evaluation of the most adverse failure of a fluid system are discussed in appendix 3F.

Failure of the natural draft cooling tower or circulating water system in the plant yard will not cause adverse effects to any safety-related components required to mitigate the consequences of the event and/or safely shutdown the plant (i.e., essential equipment). Part of the cooling tower basin is below ground level. In case of failure, the water above ground level (~310,000 ft^3) will empty on the surroundings and flow away from the power block, any Category 1 structures, or any essential equipment due to the slope of the yard grade. All the water will eventually flow into the yard drainage system. A postulated worst condition crack on the circulating water system pipeline outside the turbine building would result in a conservatively calculated leakage rate of 10,621 ft^3/min. At 45 psig it is assumed that the water will break the backfill and flow in the direction of least resistance, up toward ground level.

Once on ground level, the water will be channeled away from the power block due to the slope of the yard grade, and all water will flow into the yard drainage system. There are no essential components in that area. At this time appropriate measures would be taken to stop the waterflow from the cooling tower basin to the crack.

In the unlikely event that a crack on the circulating water system outside the turbine building is not detected by plant operations, the integrity of the safety-related seismic Category 1 structures, buried piping, and tunnels will not be impaired through soil erosion, since they are remotely located, and there are other intervening nonsafety-related structures in between.

3.4.1.2.2 Flood Protection Procedures

The VEGP is designed so that the maximum water levels considered due to natural phenomena do not jeopardize the safety of the plant or the ability to conduct a safe shutdown.

3.4.1.3 Permanent Dewatering System

Only the lower levels of the containment, the auxiliary building, and the tunnels beneath the fuel handling building and control building are subject to inseepage due to their location below the high ground water level. Ground water entry is precluded by the thick concrete walls and floors, waterproofing, and waterstops. No permanent dewatering system is required.

3.4.2 ANALYTICAL AND TEST PROCEDURES

The foundation slabs and exterior walls of the structures are designed to resist the upward and the lateral pressures caused by the high ground water level. The vertical hydrostatic pressure acting uniformly at the bottom of the structures is the product of the height to the high ground

water level and the unit weight of water assumed as 62 lb/ft^3. The horizontal hydrostatic pressure acting on the exterior walls varies with height, from the maximum at the bottom of the wall to zero at the maximum ground water level. Minimum factors of safety for overturning, sliding, and flotation are described in subsection 3.8.5. There are no dynamic water forces associated with the high ground water level. Dynamic forces associated with the probable maximum flood or probable maximum precipitation are not factors in the analysis or design of Category 1 structures, since the finished grade is adequately sloped and is located at an elevation above the maximum flood level.

There are no safety-related hydraulic structures at VEGP.

3.5 MISSILE PROTECTION

In accordance with the requirements of 10 CFR 50, Appendix A, GDC-2 and 4, adequate missile protection is provided to ensure that those portions of the safety-related structures, systems, or components whose failure would result in the failure of the integrity of the reactor coolant system pressure boundary, reduce to an unacceptable level the functioning of any plant feature required for a safe shutdown, or lead to unacceptable offsite radiological consequences, are designed and constructed so as not to fail or cause such a failure in the event of a postulated credible missile impact. Conformance to the requirements of Regulatory Guides 1.13, 1.14, 1.27, 1.76, 1.115, and 1.117 is discussed in section 1.9. The following sections provide the bases for the selection of the postulated missiles, protection requirements for external missiles, and details of the missile barrier design. Safety-related systems or components as described above are protected by locating them within missile-proof structures, by providing separation, or by providing missile shields or barriers. Nonsafety-related structures, systems, and components are protected from internally generated missiles if their failure by postulated missile impact could prevent the required safety function of other safety-related structures, systems, and components.

3.5.1 MISSILE SELECTION AND DESCRIPTION

The following sources are considered for the generation of missiles:

- Internally generated missiles:
  - Internally generated missiles outside containment.
  - Internally generated missiles inside containment.
- Turbine missiles.
- Externally generated missiles:
  - Missiles generated by natural phenomena.
  - Missiles generated by events near the site.
  - Aircraft hazards.
  - Gravity-generated missiles.

The systems located both inside and outside of the containment have been examined to identify and classify potential missiles.

The basic approach is to ensure design adequacy against generation of missiles rather than to allow missile formation and design plant features to contain their effects. In those cases where missile formation does occur, plant features are designed to contain their effects.

3.5.1.1 Internally Generated Missiles (Outside Containment)

There are two general sources of postulated missiles outside the containment which are potentially generated as a result of plant operation:

- Rotating component failures.
- Pressurized component failures.

Excluded as sources of postulated missiles are:

- Those components which operate approximately 2 percent or less of the time (a tabulation of these components is provided in table 3.5.1-8), or;
- Those components, provided in the line designation list, located in high energy fluid systems which are at high energy conditions for 2 percent or less of the time the system is in operation, or;
- Those components, provided in the line designation list, located in high energy fluid systems which are at high energy conditions when in operation but operate for less than 1 percent of the plant operating time.

A tabulation of safety-related structures, systems, and components and their locations, seismic categories, and quality group classifications is given in table 3.2.2-1. General arrangement and section detail drawings are located in section 1.2. The results of the analysis of the effects of missiles are given in table 3.5.1-1.

3.5.1.1.1 Rotating Component Failure Missiles

A tabulation of missiles generated by postulated failures of rotating components, their sources and characteristics, location, and provided missile protection is given in table 3.5.1-1.

Missile selection is based on the following conditions:

A. All rotating components that are operated during normal operating plant conditions are considered to be potential missiles if the energy of the missile is sufficient to perforate the housing.

B. The energy in a rotating part associated with component failure is assumed to occur at 120-percent overspeed for turbine-driven components and at maximum operating speed for electrically-driven components.

C. Components within one train of a redundant system are not protected from potential rotating missiles originating from the same train. Components within the other train are protected by complete separation and compartmentalization.

3.5.1.1.2 Pressurized Component Failure Missiles

Based on the design features noted below and review of the plant areas outside the containment containing pressurized components, it is concluded that there are no pressurized components whose failure will result in postulated missiles affecting the safety-related systems, structures, and components required for safe shutdown of the reactor. The design features of the pressurized components and the basis for the missile selection are described below.

A. Pressurized components in systems which qualify as high-energy systems (as defined in section 3.6) are evaluated as to their potential for becoming missiles.

B. Temperature or other detectors installed in high-energy piping are evaluated as potential missiles if failure of a threaded connection would cause their ejection. Thermowells retained by circumferential, pressure-retaining welds are not considered credible missiles because of their conservative design and weld quality inspections.

C. Where auxiliary fittings such as thermocouple wells, pressure gages, vents, drains, and test connections are attached to piping or process equipment by threaded connections only, they are postulated as missiles. When such fittings are attached by welding and the completed joint has a greater design strength than the parent metal, they are not postulated as missiles.

D. Valves of American National Standards Institute (ANSI) rating 900 psig and above, constructed in accordance with Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code, are primarily pressure seal, bonnet-type valves with the exception that certain small bore valves in this ANSI rating range may be the bolted-bonnet style. For pressure seal bonnet valves, valve bonnets are prevented from becoming missiles by the retaining ring, which would have to fail in shear, and by the yoke, which would capture the bonnet or reduce bonnet energy. Because of the highly conservative design of the retaining ring (safety factors in excess of eight may be used), bonnet ejection is highly improbable; hence, bonnets are not considered credible missiles for these valves.

E. Most valves of ANSI rating 600 psig and below are valves with bolted bonnets or pressure seal, bonnet-type valves. Valve bonnets are prevented from becoming missiles by limiting stresses in the bonnet-to-body bolting material by rules set forth in the ASME B&PV Code, Section III, and by designing flanges in accordance with applicable code requirements. Even if bolt failure were to occur, the likelihood of all bolts experiencing a simultaneous complete severance failure is very remote. The widespread use of valves with bolted bonnets and the low historical incidence of complete severance valve bonnet failures confirm that bolted valve bonnets need not be considered as credible missiles.(1)

F. Valve stems are not considered as potential missiles if at least one feature, in addition to the stem threads, is included in their design to prevent ejection. For example, valves with backseats are prevented from becoming missiles by this feature. In addition, air-or motor-operated valve stems are effectively restrained by the valve operators.

G. Nuts, bolts, nut and bolt combinations, and nut and stud combinations have only a small amount of stored energy and thus are not considered potential missiles.

H. Normally closed gate valves are not considered as potential missile sources, since the force of the fluid acts perpendicularly to the disc, stem, and operator.

I. Components within one train of a system containing redundant trains are not protected from potential pressurized missiles originating from the same train due to complete separation and compartmentalization.

The conclusion, based on design features noted above, that valve bonnets are not credible missiles is also supported by industry experience.

It is necessary to consider the possible modes of rupture of valves to estimate the likelihood that, given a rupture, a missile would be ejected. As has been shown by Nuclear Plant Reliability Data System data and general industry experience, rupture is most likely to take the

form of a through-wall crack, which would be detected as a leak long before it could propagate into a serious loss of fluid or missile-generating failure. To be a source of a significant missile, such a crack would have to occur in the bonnet area of a valve and would have to be a circumferential crack. With the above probability for any such rupture, it is not reasonably credible that such a particular crack could occur and remain undetected for a sufficient time to propagate into a missile-generating condition.

Stem ejection is a possible source of missiles, but because the stem is attached firmly to the valve internals as well as the driving and pressure-retaining mechanism in the majority of large valves, it is highly unlikely.

3.5.1.2 Internally Generated Missiles (Inside Containment)

The general sources and exclusions (see table 3.5.1-8) of postulated missiles outside the containment (paragraph 3.5.1.1) also apply to inside the containment. The results of the analysis of the effects of the missiles are given in tables 3.5.1-2 and 3.5.1-3. For the reactor coolant pressure boundary (RCPB), the selection of potential missiles is based on the application of single-failure criteria to the normal retention features of plant equipment for which there is a source of energy capable of creating a missile in the event of the postulated removal of the normal retention features. Where redundancy is provided by the normal retention features, such that sufficient retention capability remains to prevent creation of a missile in the event of a postulated failure of a single retention feature, no potential missile is postulated.

3.5.1.2.1 Control Rod Drive Mechanisms

Gross failure of a control rod drive mechanism (CRDM) housing sufficient to allow a control rod to be rapidly ejected from the core is not considered credible for the following reasons:

A. Control rod drive mechanisms are shop hydrotested at 4100 $\pm$75 psi.

B. Control rod drive mechanism housings are individually hydrotested to 3107 psi after they are installed on the reactor vessel to the head adapters and are checked again during the hydrotest of the completed reactor coolant system.

C. Control rod drive mechanism housings are made of type 304 stainless steel. This material exhibits excellent notch toughness at all temperatures that will be encountered.

D. Stress levels in the mechanisms are not affected by system transients at power or by thermal movement of the coolant loops.

However, it is postulated that the top plug on the CRDM will become loose and will be forced upward by the water jet. The following sequence of events is assumed:

1. The drive shaft and control rod cluster are forced out of the core by the differential pressure of 2500 psi across the drive shaft. The drive shaft and control rod cluster, latched together, are assumed fully inserted when the accident starts.

2. After approximately 12 ft of travel, the rod cluster control spider hits the underside of the upper support plate (part of the integrated head).

3. Upon impact the flexure arms in the coupling joining the drive shaft and control cluster fracture, completely freeing the drive shaft from the control rod cluster. It is assumed that the control cluster would be completely stopped by the upper support plate; however, the drive shaft would continue to be accelerated upward, hitting the missile shield provided.

The CRDM missiles are summarized in table 3.5.1-2. The velocity of the missiles has been calculated by balancing the forces due to the water jet. No spreading of the water jet has been assumed. These missiles are contained by the integrated head missile shield.

3.5.1.2.2 Valves

Valves have been examined to identify potential missiles. As a result of this review, there are no credible failures that could result in missile formation. Therefore, valves are not considered as credible sources of missiles. Motor-operated and air-operated valves contain design features which effectively preclude the ejection of valve stems.

Valves with a nominal diameter larger than 2 in. are designed against bonnet-to-body connection failure and subsequent bonnet ejection by means of the following:

A. Compliance with the ASME Code, Section III.

B. Control of load during tightening of bonnet-to-body bolted connections.

Reactor coolant pressure retaining parts are constructed in accordance with the ASME B&PV Code, Section III, Class 1. The valves are hydrostatically tested in accordance with the ASME Code, Section III.

In the special case of those valves located on the top of the pressurizer, which extends above the operating deck, certain vertical missiles, although not considered credible, are postulated; and protection is provided by the 3-ft-thick concrete roof slab, which prevents potential damage to the containment liner, engineered safeguards pipes, and components located outside the pressurizer compartments.

The missile characteristics of the valves in the region where the pressurizer extends above the operating deck are given in table 3.5.1-3.

3.5.1.2.3 Temperature and Pressure Sensors

The only credible source of jet-propelled missiles from the reactor coolant piping and piping systems connected to the reactor coolant system is that represented by the temperature and pressure sensor assemblies. The resistance temperature sensor assemblies can be of two types, with well and without well. Two rupture locations have been postulated: one around the welding between the boss and the pipe wall; another at the welding (or thread) between the temperature element assembly and the boss for the without-well element and the welding (or thread) between the well and the boss for the with-well element.

A temperature sensor is installed on the reactor coolant pumps close to the radial bearing assembly. A hole is drilled in the gasket and sealed on the internal end of a steel plate. In evaluating missile potential, it is assumed that this plate could break and the pipe plug on the external end of the hole could become a missile.

The missile characteristics of the piping temperature sensor assemblies are given in table 3.5.1-3. A 10° expansion, half- angle water jet has been assumed. The missile characteristics of the

piping pressure element assemblies are less severe than those of table 3.5.1-3 and are not of concern from a penetration standpoint.

3.5.1.2.4 Other Missiles

The missile characteristics of the reactor coolant pump temperature sensor, the instrumentation well of the pressurizer, and the pressurizer heaters are given in table 3.5.1-3.

Pressurizer heaters are potential missiles; but inasmuch as they would be ejected in a downward direction, no damage to safety-related structures, systems, and components inside the containment would occur.

The pressurizer relief tank rupture discs are designed such that their failure will not result in the formation of missiles. With rupture, the disc will split into quadrants that will be retained by the disc circumference. The tank is located low in the containment outside the secondary shield wall, and disc rupture will not cause failure to either the primary or secondary systems.

Based on the design features and the analysis presented in the preceding sections, it is concluded that because of compartmentalization, protective barriers, redundancy, and low kinetic energy associated with missiles, the intended safety function of the essential structures, systems, or components will not be impaired by any type of rotating or pressurized missile source.

3.5.1.3 Turbine Missiles

The turbine-generator stores large amounts of rotational kinetic energy in its rotor. In the unlikely event of a major mechanical failure, this energy may be transformed into both rotational and translational energy of rotor fragments. These fragments may impact the surrounding stationary parts. If the energy-absorbing capability of these stationary turbine- generator parts is insufficient, external missiles will be released. These ejected missiles may impact various plant structures, including those housing safety-related equipment. Paragraphs 10.2.3.6 and 10.2.4.6 describe the inspection requirements and the testing of valves which prevent turbine overspeed that would cause the missile generation.

3.5.1.3.1 Turbine Placement and Orientation

The placement and orientation of the turbine-generators is shown on drawing AX6DD303.

3.5.1.3.2 Target Description

Drawings AX6DD304, AX6DD305, AX6DD306, AX6DD307, and AX6DD308 show the physical location of all targets considered in this analysis. The safety-related components that are protected from turbine missiles include the components that are protected from tornado missiles in accordance with Regulatory Guide 1.117.

3.5.1.3.3 Low-Pressure Turbine Rotor Types

The turbine-generators for VEGP are manufactured by General Electric (GE) and are described in section 10.2. General Electric's experience and calculations show that, in the improbable

event of a rotor fracture, the substantial fragments of the high-pressure turbine and generator rotors will be contained within their respective casings.(2)

The low-pressure turbine rotors for Unit 1 are manufactured from monoblock forgings. The monoblock rotors have bucket attachment areas integral to the shaft rather than keyed wheels.

The Unit 2 low-pressure rotors are built-up rotors with shrunk-on wheels with axial keyways.

3.5.1.3.4 Unit 1 Low-Pressure Turbine Missile Probability Analysis

The methodology for missile generation probability includes consideration of the probability of unit overspeed, wheel materials, in-service inspection capabilities, and the potential for wheel containment by stationary turbine structures. The analysis methodology considers two fundamental failure modes that can lead to missile generation, brittle fracture failures, and ductile tensile failures. These two failure modes are statistically independent.(3)

The brittle-fracture failure mechanism is due to the growth of keyway stress corrosion cracks to critical size. Since the monoblock rotors on Unit 1 contain no wheel keyways, no missile generation will occur due to brittle fracture. The Unit 1 monoblock rotors do not have shrunk-on wheels, therefore, only the ductile failure portion of the methodology is used.

The probability of ductile failure is a function of speed, temperature, and material tensile strength. The turbine control system is designed to limit peak overspeed to 120 percent of rated. Under 120 percent speed, rotor design stresses are below the ultimate material strength, thus the probability of a ductile failure is negligible at speeds under 120 percent. The GE probabilistic analysis of turbine overspeed was documented in the GE 1984 NRC report and referenced in Supplementary Report GET-8039 dated September 1993, and is applicable to units with low-pressure monoblock rotors. The overspeed analysis considers the characteristics of the turbine control system, the unit configuration, and test requirements for the steam valves and other overspeed protection devices. This overspeed analysis showed that the probability of attaining a given overspeed decreases rapidly as the overspeed value increases. As long as the control system is maintained in accordance with GE's recommendations, the annual probability of attaining an overspeed of 120 percent or greater is less than 2×10^{-6}.

Unit 1 main turbine controls was upgraded with a General Electric Mark Vle control system using triple modular redundancy for in creased reliability over the original GE Mark II controls. Overspeed protective trips are generated by two sets of triple redundant speed pickups using a highly reliable and redundant trip manifold assembly (TMA). GE analysis "Control System Upgrade Impact on the Probability of Turbine Missile Generation" (reference 5) concluded that the probability of an overspeed event caused by a control system failure to be less than the original Mark II control system. Since the control system overspeed failure probability is only a small portion of the total overspeed probability, the overall annual probability of attaining an overspeed of 120 percent or greater remains less than 2×10^{-6} (effective after U1 2017 outage).

All of the components of the monoblock rotors have sufficient margin to tensile strength (at design component temperatures) to support operating speeds well in excess of 120 percent of rated. The limiting components, per design, for the low-pressure rotors are the last stage buckets, which have overspeed capability of 168 percent.

Since ductile failure is only possible at speeds significantly greater than 120 percent and since the turbine control system keeps the probability of speeds over 120 percent below 2×10^{-6}, the probability of missile generation is well below 2×10^{-6}.

The turbine missile analysis considers the following probabilities:

P1 = The probability of missile genesis due to turbine failure which causes fragment ejection through turbine casing.

P2 = The probability that a fragment strikes a specified target given its generation and ejection.

P3 = The probability that the fragment strike damages its target in a manner leading to unacceptable consequences.

P4 = The overall probability that a particular target suffers unacceptable consequences because of turbine failure.

The probability that a particular safety-related or important-to-safety target suffers unacceptable consequences because of turbine failure is:

P4 = P1 x P2 x P3

Therefore, for the monoblock rotor design, the value of P1 is:

P1 = 2 x 10-6

Since the probability of missile generation (P_1) is sufficiently low that when combined with P_2 and P_3 would result in a value of P_4 that meets the NRC acceptance criteria of 1×10^{-7} for target damage with unacceptable consequences, GE does not calculate values for P_2 and P_3. In NUREG-1048, Safety Evaluation Report related to the operation of Hope Creek Generating Station, Supplement 6, July 1986, the NRC makes the following statements in Appendix U, Probability of Missile Generation in General Electric Nuclear Turbines:

> "…in the evaluation of P_4 ($P_1 \times P_2 \times P_3$), the probability of unacceptable damage to safety-related systems from potential turbine missiles, the NRC staff is giving credit for the product of the strike and damage probabilities of $10^{-3}yr^{-1}$ for a favorably oriented turbine and $10^{-2}yr^{-1}$ for an unfavorably oriented turbine, and is discouraging the elaborate calculation of these values.
>
> The NRC staff believes that maintaining an initial small value of P_1 through turbine testing and inspection is a reliable means of ensuring that the objectives precluding turbine missiles and unacceptable damage to safety-related structures, systems, and components can be met. It simplifies and improves procedures for evaluating turbine missile risks and ensures that the public health and safety is maintained."

The above reference provides generic NRC acceptance regarding the use of a nominal value in lieu of plant specific values for the strike and damage probability in the calculation of the overall probability of unacceptable damage due to a turbine missile. Therefore, the value of P_4 for the monoblock rotors will be:

$P_4 = P_1 (1 \times 10^{-2})$

$P_4 = 2 \times 10^{-8}$

3.5.1.3.5 Unit 2 Low-Pressure Turbine Missile Probability Analysis

In the low-pressure turbine design, the energy stored in the hypothetical fragments of the wheels is of the same order of magnitude as the energy-absorbing capability of the stationary parts. The probability of missile generation is based on three major components: 1) probability of overspeed, 2) estimation of wheel burst probability as function of speed, and 3) probability of a wheel fragment penetrating the casing.[3,4] Appropriate information regarding missile fragment nomenclature, size, shape, weight, energy, and velocity is presented in table 5-2 of the GE report.[3]

Turbine missiles are ejected fragments of the turbine wheels or surrounding casing which originate because of brittle fracture at normal rated speed (low-speed burst) or due to ductile fracture during turbine runaway (high-speed burst). Missiles may be ejected at any angle of the 360° arc about the turbine axis. The ejection path will not always be perfectly normal to the turbine axis but may vary from -5° to +5° of the normal for the interior turbine wheels, according to GE data. For the two outer turbine wheels, GE postulates a range of missile ejection angles from -25° to +25° of the normal to the turbine axis.

This probability analysis considers both low-trajectory and high-trajectory missiles. The missile ejection angles are illustrated in figure 3.5.1-1. The vertical angle ϕ is measured about the turbine axis from the horizontal plane; ψ is the angle measured from the normal to the turbine axis; and θ is the projection of ψ on the horizontal plane. The horizontal angle θ is related to ϕ and ψ by the formula:

$$\theta = \tan^{-1}\left(\frac{\tan\psi}{\cos\phi}\right) \qquad (1)$$

From equation 1 it is seen that the angular range of missile ejection measured on the horizontal plane increases with increasing ϕ, the vertical angle of ejection. For example, for a missile ejected in the horizontal plane, at ϕ = 0°, the horizontal range of ejection angles varies from -5° to +5° for interior turbine wheels. At ϕ = 45°, the angular range measured at the horizontal plane varies from -7° to +7°; and at ϕ = 90°, it ranges from -90° to +90°. It is theoretically possible for a missile to strike a target located in line with the turbine axis, although the probability of strike is much lower than for targets located on either side of the axis.

3.5.1.3.5.1 Probabilities Considered. The following probabilities are considered in the determination of the likelihood of a turbine missile accident leading to damage of structures, systems, or components required for safe plant shutdown:

P_1	=	The probability of missile genesis due to turbine failure which causes fragment ejection through turbine casing.
P_2	=	The probability that a fragment strikes a specified target given its generation and ejection.
P_3	=	The probability that the fragment strike damages its target in a manner leading to unacceptable consequences.
P_4	=	The overall probability that a particular target suffers unacceptable consequences because of turbine failure.

The probability analysis is performed for the layout of the two generating units as shown on drawing AX6DD303.

3.5.1.3.5.2 Probability of Missile Genesis (P1). The probability of missile genesis has been determined by GE for each low pressure turbine wheel. Turbine overspeed probability, wheel burst probability, and casing penetration probability are the major components of the missile genesis probability.[3,4] Turbine overspeed probability considers abnormal events such as full load rejection and control system failures which could cause an overspeed. Wheel burst probability considers turbine speed, wheel temperature, and wheel keyway stress corrosion cracking.

Probability of casing penetration considers the kinetic energy of the wheel fragment at the instant of burst as well as the energy absorbing capability of the low pressure turbine stationary components.

The NRC missile genesis is $P_1 = 1 \times 10^{-4}$/year and is used in the turbine missile analysis to satisfy the Regulatory Guide 1.115 requirement.

3.5.1.3.5.3 Probability of Missile Strike (P2). Calculation of P_2: Neglecting the effect of air resistance, a missile trajectory is determined by the initial ejection vector from the turbine casing. The direction of the ejection vector is defined by two angles: ϕ, which is measured about the turbine axis, and ψ, which is measured from the plane normal to the turbine axis. The magnitude of the ejection vector is V, the ejection velocity from the casing. Functions must be specified, $P(\phi)$, $P(\psi)$, and P(V), which determine the distribution of the missile ejection probability over the range of three variables.

The ejection probability distribution $P(\phi)$ is assumed to be uniform over the 360° arc about the turbine axis:

$$P(\phi)\, d\phi = \frac{d\phi}{2\pi} \qquad (2)$$

The probability distribution $P(\psi)$ is considered to be uniform within some specified angular limits:

$$P(\psi)\, d\psi = \frac{d\psi}{\psi\,\mathrm{max} - \psi\,\mathrm{min}},\ \psi\ \mathrm{min} < \psi < \psi\ \mathrm{max} \qquad (3)$$

The limits ψ_{min} to ψ_{max} are typically -5° to +5° for missiles ejected from the interior turbine wheel and -25° to +25° for missiles ejected from the end turbine wheels.

The ejection probability distribution P(V) is normally assumed to be uniform over the specified range of ejection velocities V_{min} to V_{max}:

$$P(V)dV = \frac{dV}{V\,\mathrm{max} - V\,\mathrm{min}},\ V_{min} < V < V_{max} \qquad (4)$$

The principle of the P_2 calculation is to determine, using the basic equations of missile ballistics, the ranges of the variables ϕ, ψ, and V which determine trajectories intersecting the specified target structure. The strike probability is determined by integrating the product of the three ejection probability distributions over the ranges of the variables corresponding to target strike:

$$P_2 = \int_{\phi_1}^{\phi_2} \int_{\psi_1^{(\phi)}}^{\psi_2^{(\phi)}} \int_{V_{1(\phi,\psi)}}^{V_{2(\phi,\psi)}} P(\phi)\, P(\psi)\, P\,(V)\, dV\, d\psi \pi d\phi \qquad (5)$$

The integral is evaluated by the Bechtel computer code TURMIS (turbine missile). Discrete ejection directions are evaluated by first specifying a ϕ_i, for which the limits ψ_1 (ϕ_i) and ψ_2 (ϕ_i) may be computed corresponding to target strike. Discrete values ψ_j within the range $\psi_i(\phi_i)$ to $\psi_2(\phi_i)$ are then specified. Given the values ϕ_i and ψ_j, the limits $V_i(\phi_i, \psi_j)$ and $V_2(\phi_i, \psi_j)$ corresponding to target strike are then computed, and the integral over V may be evaluated analytically. The range of velocity V_{min} to V_{max} is illustrated in figure 3.5.1-2.

3.5.1.3.5.4 Probability of Damage to Target Structure (P_3). Missiles striking a target structure may cause damage to safety-related systems contained inside. The damage incurred may be due to missile penetration or spalling of concrete fragments from the interior surface of the structures. The Ballistic Research Laboratory equations with a safety factor of 1.2 have been adopted to determine whether or not spalling or perforation will occur upon missile impact of a specified concrete slab to ensure that the safety-related functions of the systems are not impaired.

3.5.1.3.5.5 Analytical Results. Calculation of P_4: The value of P_4 for a particular target structure and a particular turbine failure mode is taken as $P_1 \times P_2 \times P_3$ for the worst missile.

The P_4 value for the plant is determined by summation of P_4 values corresponding to the critical failure mode for all targets on the plant site. Table 3.5.1-4 lists missile targets, strike probabilities, and turbine missile damage probabilities, in case the turbine in Unit 2 fails. Table 3.5.1-4 shows the P_4 values, which include both the high- and low-trajectory missiles (0.40×10^{-7} versus 1.0×10^{-7} allowable).

3.5.1.3.6 Turbine Overspeed Protection

A description of the turbine overspeed protection system, in terms of redundancy, diversity, component reliability, and testing procedures, is provided in subsection 10.2.2.

3.5.1.3.7 Turbine Valve Testing

A discussion of the turbine valve testing is provided in subsection 10.2.3.

3.5.1.3.8 Turbine Characteristics

Turbine data pertinent to the evaluation of its failure characteristics, including a description of its overall configuration, major components (e.g., steam valves, reheaters, etc.), rotor materials and their properties, steam environment (e.g., pressure, temperature, quality, chemistry), and other appropriate properties, are provided in section 10.2. Turbine operational and transient characteristics, including turbine startup and trip environments as well as their overspeed parameters, also are provided in section 10.2.

3.5.1.4 Missiles Generated by Natural Phenomena

The credible missiles at VEGP created by natural phenomena are those generated by tornadoes.

At the construction permit stage, VEGP was required to postulate the tornado missiles described in table 3.5.1-5. VEGP was not required to design to the missile spectra specified in paragraph 3.5.1.4 of the Standard Review Plan dated November 24, 1975. However, to further demonstrate the adequacy of the VEGP design for tornado missiles, the missile characteristics of the steel rod and the utility pole, as specified in the Standard Review Plan (November 24, 1975) for missiles C and F, respectively, are considered. The tornado missiles considered in the VEGP design are provided in table 3.5.1-6.

The methodology used to design the Category 1 structures to provide adequate protection for the safety-related equipment, system, and components is described in appendix 3C.

Safety-related systems and components are protected by missile barriers. The barriers provided are listed in table 3.5.1-7. Where concrete exterior walls and roofs are used as barriers to offer missile protection, such walls have a 24-in. minimum thickness, while the roofs are at least 21 in. thick. The concrete has a 28-day compressive strength of at least 4000 psi (91-day strength for concrete containing pozzolan). Where the interior walls and slabs having concrete compressive strength of 5000 psi are used as missile barriers, such walls have an 18-in. minimum thickness, while the slabs are at least 14 in. thick.

3.5.1.5 Missiles Generated by Events Near the Site

As described in subsection 2.2.3, there are no credible site proximity missiles created by events near the site.

3.5.1.6 Aircraft Hazards

There are no airports or airport approaches within 10 miles of the site; there are no airways within 2 miles of the site. For the airports greater than 10 miles from the site, none has projected operations per year greater than 1000 d^2 movements, where d is the distance in miles from the site. Available military aerial navigation charts for Fort Gordon (U.S. Army)show no low-level flight or landing patterns near the plant. Thus, there are no credible aircraft hazards to the VEGP site.

3.5.1.7 Gravity-Generated Missiles

The occurrence of falling objects as a result of seismic events is prevented by adequately supporting equipment in areas where the possibility of interaction exists. The occurrence of falling objects as a result of the failure of a crane or hoist is discussed in subsections 9.1.4 and 9.1.5.

3.5.1.8 Standard Review Plan Evaluation

The tornado missile spectrum used for VEGP differs from that of the Standard Review Plan.

At the construction permit stage, VEGP was required to postulate the tornado missiles described in table 3.5.1-5. VEGP was not required to design to the missile spectra specified in paragraph 3.5.1.4 of the Standard Review Plan dated November 24, 1975. However, to further demonstrate the adequacy of the VEGP design for tornado missiles, the missile characteristics of the steel rod and the utility pole, as specified in the Standard Review Plan (November 24,

1975) for missiles C and F, respectively, are considered. The tornado missiles considered in the VEGP design are provided in table 3.5.1-6.

3.5.1.9 References

1. Kilsby, E. R. Jr., "Reactor Primary - Piping-System Rupture Studies," Nuclear Safety, Vol 7, Winter 1965-1966, p 185.
2. Downs, J. E., "Hypothetical Turbine Missiles – Probability of Occurrence," General Electric Company Memo Report, March 14, 1973. Data cited applies to 43-inch last stage blading.
3. General Electric, "Probability of Missile Generation in General Electric Nuclear Turbines," January 1984. Proprietary Document.
4. General Electric, "Probability of Missile Generation in General Electric Nuclear Turbines, Supplementary Report: Steam Valve Surveillance Test Interval Extension," September 1993. Proprietary Document.
5. General Electric, “Control System Upgrade Impact on the Probability of Turbine Missile Generation,” drawing number AX5AA11-00029.

3.5.2 STRUCTURES, SYSTEMS, AND COMPONENTS TO BE PROTECTED FROM EXTERNALLY GENERATED MISSILES

3.5.2.1 General

The sources of missiles which, if generated, could affect the safety of the plant are considered in subsection 3.5.1. Safety-related structures, systems, and components are designed to withstand the impact of postulated missiles, are physically separated from the source of missiles, or are protected by a missile barrier.

3.5.2.2 Missile Barriers Within Containment

The secondary shield walls, the refueling canal walls, the various structural beams, and the operating floor act as missile barriers separating reactor coolant loops from other protected components and missile sources. These barriers also protect the reactor coolant pressure boundary (RCPB) in each loop from those identified missiles generated elsewhere in the containment building while protecting the RCPB in each loop from externally generated missiles. The feedwater system is routed so that it is not affected by potential missiles.

Except for short piping runs in the safety injection system (SIS), which must supply cooling water to the reactor coolant system after a loss of coolant accident, the emergency safety features are located outside the secondary shield. The SIS lines which penetrate the secondary shield do so in the vicinity of the loop segment to which they are attached.

A missile shield structure is provided over the control rod drive mechanisms (CRDMs) to block any identified missiles generated in that location. The design of the missile shield is discussed in subsection 3.5.3. The control rod drives are protected from horizontal missiles by the refueling canal walls that extend vertically above the CRDMs. The head vent and letdown system piping is the only high-energy piping located close to the CRDMs. (No potential missile

sources exist in the system.) A roof slab is provided to protect against identified missiles that originate in the region where the pressurizer extends above the operating floor.

Missile barriers are provided, as required, to prevent missiles generated by the failure of main steam or feedwater components inside the containment from causing loss of integrity to the containment liner, isolation system, or steam system associated with another steam generator, or from causing loss of function to other required systems or components inside the containment in accordance with the missile protection design criteria previously listed in subsection 3.5.1.

3.5.2.3 Barriers for Missiles Generated Outside of Plant Structures

The protective structures, shields, and missile barriers designed to provide protection against identified missiles generated outside these structures, shields, and missile barriers are listed in table 3.5.1-6. The missile barriers listed are designed for the tornado and accident missiles described in subsection 3.5.1, utilizing the procedures stated in subsection 3.5.3.

3.5.2.4 Missile Barriers Within Plant Structures Other Than Containment

Missile barriers are provided within plant structures outside the containment in conformance with the missile protection design criteria discussed in section 3.5. For the pressurized and rotating component failure missiles that originate outside the containment, identified in subsection 3.5.1, the following steps are taken to assure that the missile protection design criteria are met.

A. Missiles are categorized according to the system in which they originate.

B. The components that must be protected from a missile are identified in accordance with the missile protection design criteria given in subsection 3.5.1.

3.5.3 BARRIER DESIGN PROCEDURES

Missile barriers and protective structures are designed to withstand and absorb missile impact loads in order to prevent damage to safety-related components.

With the exception of the nuclear service cooling water (NSCW) tower fan cells, protection of essential safety-related systems or components against tornado missiles that could enter through any openings in the exterior walls or roofs of Category 1 structures is provided as follows.

Barriers are provided for the openings to ensure that safety-related systems and components are protected from postulated credible missile impact. The following design features provide missile protection:

1. Missile proof doors.

2. Steel plate missile shields.

3. Concrete missile shields.

4. For small openings, the space within the opening being sufficiently occupied by piping, pipe support, or other substantial intervening commodities.

Interior walls and slabs are treated as barriers for systems or components located in the interior rooms, if missile protection is not provided by exterior walls and slabs or other barriers. Any openings in the exterior walls or slabs and the interior walls or slabs that may be credible paths for missile entry are investigated to ensure that the appropriate level of localized protection is provided if necessary.

The NSCW towers are inherently protected against direct horizontal missiles by the towers' concrete construction. The minimum height a missile would have to obtain to enter the cooling tower vertically and strike a fan is approximately 45 ft above grade, which eliminates heavier missiles (such as a utility pole or automobile) from consideration. Since each fan has its own opening, a single missile can damage only a single fan. A detailed probabilistic study was performed to determine the risk of the NSCW towers not being available during and following a tornado. This study demonstrated that even with all incorporated conservatisms the frequency of tornado missiles disabling the NSCW system (loss of one tower for maintenance, and missiles disabling two or more fans in the single operating tower) is lower than the acceptance criterion of 10^{-7} per year given in Standard Review Plan Section 2.2.3. Therefore, additional tornado missile protection is not required for the NSCW tower fans.

The procedures by which each structure or barrier is designed to resist the tornado missile hazards described in paragraph 3.5.1.4 are presented in appendix 3C. Appendix 3C is also applicable to the other missile hazards of subsection 3.5.1, provided such missiles display parameters similar to those created by tornadoes. For those possible missiles not similar to tornado-borne missiles, the design is accomplished using similar principles.

In general, Westinghouse-supplied equipment is not designed to withstand the impact of postulated missiles; therefore, Bechtel has considered the effects of postulated missiles and provided the necessary protection to safety-related components as determined by the design bases provided in subsection 3.5.1.

The exception is the control rod drive mechanism (CRDM) missile shield, which is supplied by Westinghouse as part of the integrated head.

A missile shield structure is provided over the CRDMs to block missiles that might be associated with a fracture of the pressure housing of any mechanism. This missile shield is a reinforced steel structure attached to the reactor vessel head and located above the CRDMs. Each CRDM housing is terminated with a small tapered pin that penetrates the missile shield through a slightly larger diameter hole to direct the ejected CRDM missile into the shield. This prevents any missile from missing or ricocheting from the shield to strike the containment liner or other CRDMs.

Missile shield penetrations are given in tables 3.5.1-2 and 3.5.1-3 using the Ballistic Research Laboratories formula for steel. The steel missile shield has an effective thickness of approximately 3 in.

For the case of housing plug and drive shaft impact, which is the design case, it is assumed that the plug partially perforates the missile shield. The drive shaft then hits the plug and further penetrates the steel missile shield. The resultant penetration into the shield is 0.773 in.; therefore, the effective thickness of the steel missile shield is more than three times the combined penetration for the design case.

The CRDM missile shield is also designed to withstand the dynamic impact loads due to the missile and the water jet.

TABLE 3.5.1-1 (SHEET 1 OF 5)

INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT ROTATING COMPONENT FAILURES[c]

Missile Identification	Source of Missile	Location[d]	Missile Characteristics			Calculated Maximum Steel Perforation Depth (in.)	Casing Thickness (in.)	Casing Perforation	Missile Residual Velocity After Casing Perforation	Calculated Thickness of Surrounding Material to Prevent		Missile Protection Provided
			Velocity (ft/s)	Equiv. Dia. (in.)	Mass (lbm)					Concrete Spalling (in.)	Steel Perforation (in.)	
Impeller	CCW drain tank pump	Aux. bldg. level D room 75	107.0	5.65	6.9	0.17	0.25	No	None	None	None	None
Fan blade	Air handling unit cooling	Aux. bldg. level D	91.8	0.43	0.22	0.033	0.049	No	None	None	None	None
Fan blade	Air handling unit cooling	Aux. bldg. level D room 79	106.9	0.62	0.36	0.038	0.111	No	None	None	None	None
Fan blade	Air handling unit cooling	Aux. bldg. level D room 128/ 130	54	0.16	0.42	0.0143	0.041	No	None	None	None	None
Impeller	CVCS cent. charging pump train A	Aux. bldg. level C room 115	144	2.25	32.72	0.32	0.35	No	None	None	None	None
Impeller	CVCS cent. charging pump train B	Aux. bldg. level C room 118	144	2.25	32.72	0.32	0.35	No	None	None	None	None
Impeller	CVCS cent. normal charging pump	Aux. bldg. level C room 111	128	3.66	10.00	0.08	3.18	No	None	None	None	None
Fan blade	SGBD heat exchanger room cooler fan	Aux. bldg. level C room 108/125	98	0.86	0.20	0.017	0.048	No	None	None	None	None

TABLE 3.5.1-1 (SHEET 2 OF 5)

Missile Identification	Source of Missile	Location[d]	Velocity (ft/s)	Missile Characteristics Equiv. Dia. (in.)	Mass (lbm)	Calculated Maximum Steel Perforation Depth (in.)	Casing Thickness (in.)	Casing Perforation	Missile Residual Velocity After Casing Perforation	Calculated Thickness of Surrounding Material to Prevent Concrete Spalling (in.)	Steel Perforation (in.)	Missile Protection Provided
Fan blade	Air handling unit train A	Aux. bldg. level C room UC-C14	105	0.62	0.36	0.038	0.111	No	None	None	None	None
Fan blade	Air handling unit train B	Aux. bldg. level C room UC-C14	105	0.62	0.36	0.038	0.111	No	None	None	None	None
Impeller	SGB drain pump	Aux. bldg. level C room 108	112	1.75	1.6	0.19	0.25	No	None	None	None	None
Impeller	Boron inj. recirc. pump train A	Aux. bldg. level B room R-B05	65.5	0.365	1	0.07	0.375	No	None	None	None	None
Impeller	Boron inj. recirc. pump train B	Aux. bldg. level B room R-B06	65.5	0.365	1	0.07	0.375	No	None	None	None	None
Fan blade	Air handling unit train B	Aux. bldg. level B room R-B17	44.8	0.16	0.042	0.014	0.041	No	None	None	None	None
Fan blade	Air handling unit	Aux. bldg. room R-B13	44.8	0.16	0.042	0.014	0.041	No	None	None	None	None
Impeller	Aux. comp. cooling water pump	Aux. bldg. level B room R-B23	98	1.14	75	0.65	0.75	No	None	None	None	None

TABLE 3.5.1-1 (SHEET 3 OF 5)

Missile Identification	Source of Missile	Location[d]	Velocity (ft/s)	Missile Characteristics Equiv. Dia. (in.)	Mass (lbm)	Calculated Maximum Steel Perforation Depth (in.)	Casing Thickness (in.)	Casing Perforation	Missile Residual Velocity After Casing Perforation	Calculated Thickness of Surrounding Material to Prevent Concrete Spalling (in.)	Steel Perforation (in.)	Missile Protection Provided
Impeller	Aux. comp. cooling water pump	Aux. bldg. level B room R-B24	98	1.14	75	0.65	0.75	No	None	None	None	None
Fan blade	Air handling unit	Aux. bldg. level A room R-A03/ R-A05	106.9	0.62	0.36	0.038	0.111	No	None	None	None	None
Impeller	Refueling water purification pump	Aux. bldg. level A room A40A	85.8	0.888	3.09	0.08	0.375	No	None	None	None	None
Fan blade	Air handling unit train A	Aux. bldg. level A room R-A53	96.7	0.68	0.60	0.046	0.049	No	None	None	None	None
Impeller	Spent fuel pool skimmer pump	Aux. bldg. level A room R-A53	137.31	2.925	7.635	0.087	0.3125	No	None	None	None	None
Impeller	Spent fuel pool pump train A	Aux. bldg. level A room A-53	77.7	0.98	16.5	0.2	0.25	No	None	None	None	None
Fan blade	MCC room cooler train A	Aux. bldg. level 1 room 116, 118	44.2	16	0.42	0.013	0.041	No	None	None	None	None
Fan blade	Rail corr. ac unit[e]	Aux. bldg. level 1	101	1.83	4.5	0.065	0.078	No	None	None	None	None
Fan blade	AB cont. exhaust units	Aux. bldg. level 2 room 212	254	1.58	4.44	0.255	0.1875	Yes	154.2	1.26[a]	0.131[b]	Yes[f]

TABLE 3.5.1-1 (SHEET 4 OF 5)

Missile Identification	Source of Missile	Location[d]	Missile Characteristics: Velocity (ft/s)	Equiv. Dia. (in.)	Mass (lbm)	Calculated Maximum Steel Perforation Depth (in.)	Casing Thickness (in.)	Casing Perforation	Missile Residual Velocity After Casing Perforation	Calculated Thickness of Surrounding Material to Prevent: Concrete Spalling (in.)	Steel Perforation (in.)	Missile Protection Provided
Fan blade	AB cont. exhaust units	Aux. bldg. level 2 room 221 (Unit 2)	263	1.58	4.44	0.267	0.1875	Yes	168.6	1.42[a]	0.147[b]	Yes[f]
Fan blade	Elect. swgr and MCC room cooler	Aux. bldg. level 2 room 212	109.5	0.591	0.242	0.032	0.028	Yes	46	0.081[a]	0.01[b]	None
Impeller	Turbine-driven pump	Aux. feedwater pumphouse room 106	150.0	0.67	21	0.84	2.0	No	None	None	None	None
Impeller	Motor-driven pumps	Aux. feedwater pumphouse room 101/ 102	164.85	0.513	21	1.26	1.44	No	None	None	None	None
Turbine disk	Steam turbine	Aux. feedwater pumphouse room 106	297.8	3.7	30	0.48	0.812	No	None	None	None	None
Impeller	Fan - a/c unit [e]	Control building room 226 relay room	49.7	0.164	0.021	0.008	0.041	No	None	None	None	None
Impeller	Centrifugal fan-filter unit [e]	Control building room 248 (HVAC)	222.916	1.066	1.532	0.157	0.165	No	None	None	None	None
Fan blade	Normal exhaust unit [e]	Fuel handling bldg. room 301	209.6	2.02	6	0.188	0.375	No	None	None	None	None

TABLE 3.5.1-1 (SHEET 5 OF 5)

Missile Identification	Source of Missile	Location[d]	Missile Characteristics: Velocity (ft/s)	Equiv. Dia. (in.)	Mass (lbm)	Calculated Maximum Steel Perforation Depth (in.)	Casing Thickness (in.)	Casing Perforation	Missile Residual Velocity After Casing Perforation	Calculated Thickness of Surrounding Material to Prevent: Concrete Spalling (in.)	Steel Perforation (in.)	Missile Protection Provided
Fan blade	Elevator room chiller unit	Fuel handling building room 127 railroad corridor	66	0.71	1.04	0.04	0.049	No	None	None	None	None

a. Missile protection is provided by the structural concrete.

b. Thickness of the piping prevents failure.

c. This table lists the rotating component failures for which a specific casing perforation calculation was performed.

d. The locations provided are for unit 1. The unit 2 locations are similar.

e. The missile source is common to both units.

f. Missile protection is provided for essential chilled water lines 1/2-1592-109-4, 1-1592-054-4, and 2-1592-110-4.

TABLE 3.5.1-2

INTERNALLY GENERATED MISSILES INSIDE CONTAINMENT
ROTATING COMPONENT FAILURES[b]

Postulated Missile	Weight (lb)	$Thrust_2$ Area (in.)	$Impact_2$ Area (in.)	Impact Velocity (ft/s)	Kinetic Energy (ft-lb)	Protection Provided
CRDM housing plug	50	4.91	0.87	40	1242	Integrated head missile shield
Control rod drive shaft	165	2.40	3.56	100	25,620	
Control rod drive shaft and mechanism	1610	12.57	1.37	12	3600	
Reactor coolant drain tank pump impeller	2.55	-	-	209.6	-	None[a] Casing is not perforated
Containment preaccess filter unit fan blade	6	-	-	118	-	None[a] Casing is not perforated

a. The analysis performed on these rotating components is similar to the analysis presented in table 3.5.1-1.

b. This table lists the rotating component failures for which a specific calculation was performed.

TABLE 3.5.1-3 (SHEET 1 OF 2)

INTERNALLY GENERATED MISSILES INSIDE CONTAINMENT
PRESSURIZED COMPONENT FAILURES

		Valve Missile Characteristics				
Missile Identification	Weight (lb)	Flow Discharge Area ($in.^2$)	Thrust Area ($in.^2$)	Impact Area ($in.^2$)	Velocity (ft/s)	Protection Provided
Safety-relief valve bonnet	350	2.86	80	24	110	3 ft 0 in. concrete pressurizer roof slab
3-in. motor-operated isolation valve bonnet (plus motor and stem)	400	5.5	113	28.3	135	3 ft 0 in. concrete pressurizer roof slab
2-in. air-operated valve bonnet (plus stem)	75	1.8	20.7	20	115	3 ft 0 in. concrete pressurizer roof slab
3-in. air-operated spray valve bonnet (plus stem)	120	5.5	50.3	50	190	3 ft 0 in. concrete pressurizer roof slab
4-in. air-operated spray valve bonnet (plus stem)	200	9.3	50.3	50.0	190	3 ft 0 in. concrete pressurizer roof slab

TABLE 3.5.1-3 (SHEET 2 OF 2)

Piping Temperature Element Assembly Missile Characteristics

Characteristics	Without Well	With Well
For a tear around the weld between the boss and the pipe		
Flow discharge area (in.^{2})	0.11	0.60
Thrust area (in.^{2})	7.1	9.6
Missile weight (lb)	11.0	15.2
Area of impact (in.^{2})	3.14	3.14
(psi)	3.15	4.84
Velocity (ft/s)	20	120
For a tear at the junction between the temperature element assembly and the boss for the without-well element and at the junction between the boss and the well for the with-well element		
Flow discharge area (in.^{2})	0.11	0.60
Thrust area (in.^{2})	3.14	3.14
Missile weight (lb)	4.0	6.1
Area of impact (in.^{2})	3.14	3.14
Weight to impact area ratio (psi)	1.27	1.94
Velocity (ft/s)	75	120

Characteristics of Other Missiles Postulated Within Reactor Containment

	Reactor Coolant Pump Temperature Element	Instrument Well of Pressurizer	Pressurizer Heaters
Weight (lb)	0.25	5.5	15
Discharge area (in.^{2})	0.50	0.442	0.61
Thrust area (in.^{2})	0.50	1.35	2.4
Impact area (in.^{2})	0.50	1.35	2.4
Velocity (ft/s)	260	100	55

TABLE 3.5.1-4 (SHEET 1 OF 2)

TURBINE MISSILE STRIKE AND DAMAGE PROBABILITIES
PER MISSILE FRAGMENT FROM UNIT 2

I. Turbine Missile Genesis Probability (P_1) = 1 x 10^{-4} per year.

II. Missile Damage Probabilities (P_4) per year.

Location Unit	Target Building or Structure	Strike and Damage Probabilities (P_2) x (P_3) x 10^{-3} per year	Missile Damage Probabilities (P_4) x (10^{-7}) per year
2	Containment	0.130	0.130
1 & 2	Control building	0.167	0.167
2	Main steam valve room (north)	0.005	0.005
1 & 2	Fuel handling building	0.080	0.080
1 & 2	Auxiliary building	0.180	0.180
2	Main steam valve room (south)	0.012	0.012
2	Comp. cooling heat exch. A & B	0.037	0.037
2	Condenser storage tank	0	0
2	Condenser storage tank	0	0
2	Auxiliary feedwater line	0.020	0.020
2	Refueling water storage tank	0	0
2	Reactor makeup water storage tank	0	0

TABLE 3.5.1-4 (SHEET 2 OF 2)

Location Unit	Target Building or Structure	Strike and Damage Probabilities $(P_2) \times (P_3) \times 10^{-3}$ per year	Missile Damage Probabilities $(P_4) \times (10^{-7})$ per year
2	Diesel generator building	0.123	0.123
2	Auxiliary feedwater pumphouse	0.035	0.035
1	Containment	0	0
1	Main steam valve room (north)	0.005	0.005
1	Main steam valve room (south)	0.010	0.010
1	Comp. cooling heat exchange.	0.032	0.032
1	Condenser storage tank	0	0
1	Condenser storage tank	0	0
1	Auxiliary feedwater line	0.012	0.012

TABLE 3.5.1-5

TORNADO MISSILES POSTULATED
AT THE CONSTRUCTION PERMIT STAGE

Missile	Weight (lb)	Design Missile Velocity (ft/s)	Height at Which Attained (ft)
Wooden plank, 4 in. x 12 in. x 12 ft	200.0	200	216
Steel pipe, 3-in. diameter, schedule 40, 10 ft long	78.5	200	212
Steel rod, 1-in. diameter, 3 ft long	8.0	160	114
Steel pipe, 6-in. diameter, schedule 40, 15 ft long	285.0	160	101
Steel pipe, 12-in. diameter, schedule 40, 15 ft long	744.0	150	46
Utility pole, 13-1/2 in. diameter, 35 ft long	1490.0	100	Ground level
Automobile, frontal area 20 ft^2	4000.0	75	Ground level

TABLE 3.5.1-6

TORNADO MISSILES CONSIDERED
IN THE VEGP DESIGN

Description of Missile	Weight (lb)	Height Limit (ft)	End-On Horizontal Velocity (ft/s)	End-On Vertical Velocity (ft/s)
Wooden plank, 4 in. x 12 in. x 12 ft	200	216	200	160
Steel pipe, 3-in. diameter, schedule 40, 10 ft long	78.5	212	200	160
Steel rod, 1-in. diameter, 3 ft long	8	Unlimited	317	254
Steel pipe, 6-in. diameter, schedule 40, 15 ft long	285	101	160	128
Steel pipe, 12-in. diameter, schedule 40, 15 ft long	744	46	150	120
Utility pole, 13-1/2-in. diameter, 35 ft long	1490	30[(a)]	211	169
Automobile, frontal area 20 ft^2	4000	Ground Level	75	60

a. To 30 ft above all grade levels within 1/2 mile of facility structures.

TABLE 3.5.1-7 (SHEET 1 OF 3)

PROTECTED SYSTEM AND COMPONENT BARRIERS AGAINST EXTERNALLY GENERATED MISSILES

Protected Systems and Components	Missile Barrier	Minimum Concrete Thickness (in.) Walls	Roof	Floor	Design Concrete Strength (psi)
Reactor equipment, reactor coolant system, containment piping and valves, containment electrical, instrumentation, and control systems and containment engineered safety features actuation systems, transfer tube	Containment shell and dome	45	45	-	6000
	Containment basemat	-	-	126	5000
	Internal structures				
	Primary shield wall	Varies 102 min			5000
	Secondary shield wall	36	-	-	5000
	Floor at el 220 ft	-	-	24	5000
Penetrations in containment shell	Missile shield wall/ building walls	24	21	-	5000
Control room and protected electrical, instrumentation control, and ventilation equipment in control building	Control building	24	21	-	4000
Safety injection, containment spray cooling water, ventilation, electrical, instrumentation and control equipment	Auxiliary building	24	21	-	5000
Reactor auxiliaries, e.g., chemical and volume control system, boric acid storage tanks and transfer pumps, and CCW System	Auxiliary building	24[(a)]	21	-	5000
Spent fuel pool, transfer tube	Fuel handling building	24	21	-	4000
	Fuel pool walls	60	-	-	4000

TABLE 3.5.1-7 (SHEET 2 OF 3)

Protected Systems and Components	Missile Barrier	Minimum Concrete Thickness (in.) Walls	Roof	Floor	Design Concrete Strength (psi)
Diesel generators, diesel generator fuel oil system, combustion air intake	Diesel generator building	24	21	-	4000
Diesel generator combustion air exhaust	Concrete barrier (horizontal missile) Steel Plate (vertical missile)	24		-	6000
Diesel fuel storage tank, diesel fuel transfer pumps and pump motors	Diesel fuel storage tank pumphouse	24	21	-	4000
Main steam line isolation valves	Auxiliary and control building main steam valve rooms	24	21	-	5000-auxiliary building 4000-control building
Category 1 water storage tanks	Cylindrical walls and sloping roof	24	21	-	4000
Category 1 water storage pumps, valves, and piping	Enclosures adjoining tanks	24	21	-	4000
Nuclear service cooling water tower fan motors	Enclosures	24	21	-	4000
Nuclear service cooling water pumps	Nuclear service cooling tower valve houses	24	21	-	4000
Auxiliary feedwater pumps, motors, valves, and piping	Auxiliary feedwater pumphouse	24	21	-	4000
Category 1 piping and electrical cables	Category 1 tunnels, or buried a minimum 6 ft backfill cover or 21 in. concrete cover or a total 6 ft combination of backfill and concrete	24	21	-	4000
Auxiliary building HVAC intakes and exhausts	Auxiliary building	24	21	-	5000

TABLE 3.5.1-7 (SHEET 3 OF 3)

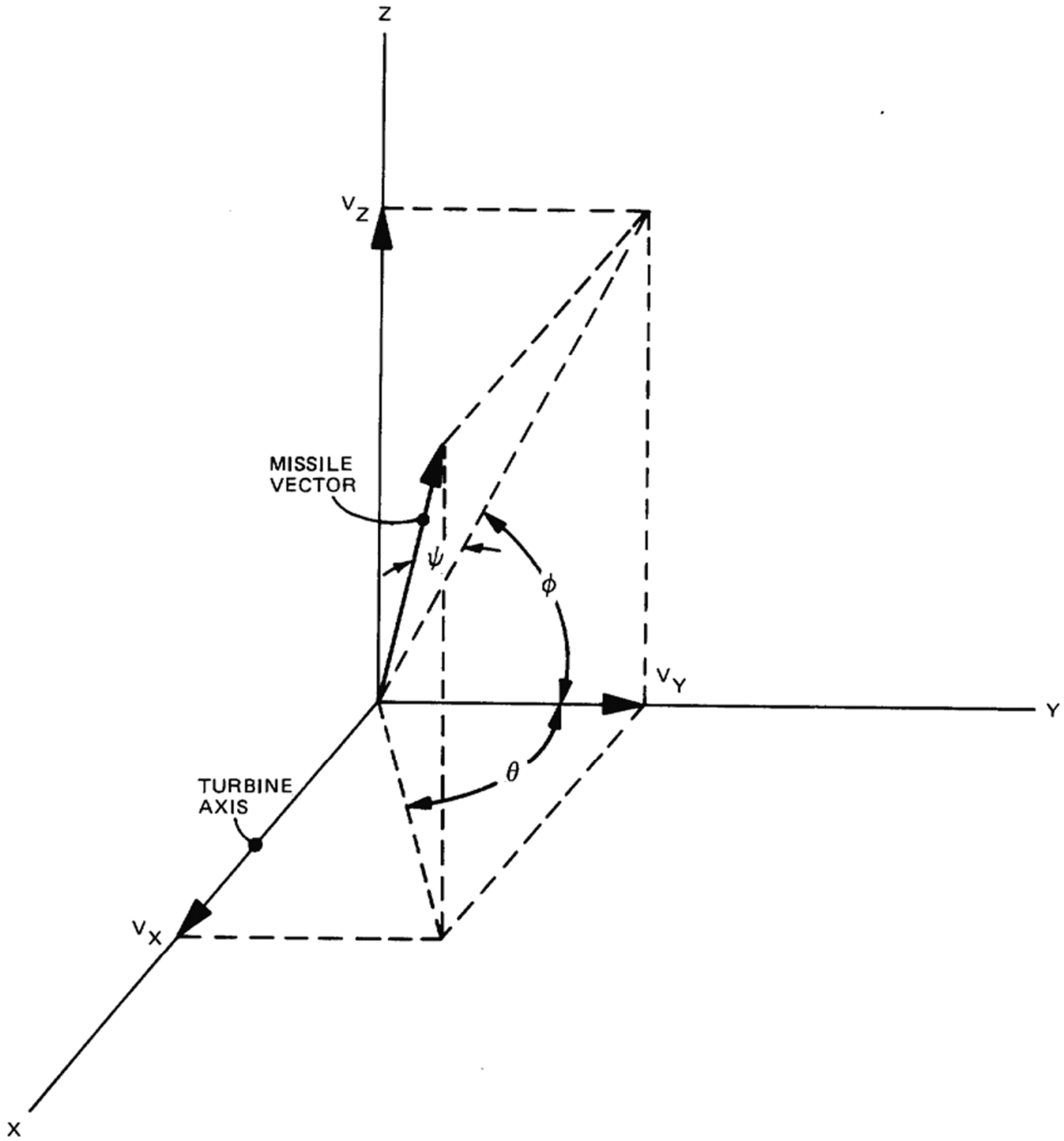
Protected Systems and Components	Missile Barrier	Minimum Concrete Thickness (in.) Walls	Roof	Floor	Design Concrete Strength (psi)
Control building HVAC intakes and exhausts	Control building	24	21	-	4000
Fuel handling building HVAC intakes and exhausts	Fuel handling building	24	21	-	4000
Diesel generator building HVAC intakes and exhausts	Diesel generator building	24	21	-	4000
Auxiliary feedwater pumphouse HVAC intakes and exhausts	Auxiliary feedwater pumphouse	24	21	-	4000
Auxiliary feedwater piping	Control building main steam valve room entrance	24	21	-	4000

a. Interior walls and roof slabs credited with providing missile protection each have a minimum concrete thickness of 18 in. and 14 in. respectively and design strength of 5000 psi.

TABLE 3.5.1-8

INTERNAL ROTATING COMPONENTS EXCLUDED
AS MISSILE GENERATION SOURCES

TAG NUMBER	DESCRIPTION	LOCATION
1301-P4-010 through 013	Wet Layup Recirculation Pump	Containment
1516-B7-001 & -002	Containment Post LOCA Cavity Purge Fans	Containment
2203-P6-001	Fuel Transfer System Hydraulic Pump	Containment
1204-P6-003 & -004	Safety Injection Pumps	Auxiliary Bldg
1206-P6-001 & -002	Containment Spray Pumps	Auxiliary Bldg
1208-P6-006 & -007	Boric Acid Transfer Pumps	Auxiliary Bldg
1555-A7-015 & -016	Safety Injection Pump Room Cooler Fans	Auxiliary Bldg
1561-N7-001 & -002	Piping Penetration Filtration Unit Fans	Auxiliary Bldg
None Provided	Safety Injection Lube Oil Pump	Auxiliary Bldg
2203-P6-002	Fuel Transfer System Hydraulic Pump	Fuel Handling Bldg
2403-G4-001-P27& -002-P27	Jacket Water Chemical Addition Pump	Diesel Generator Bldg



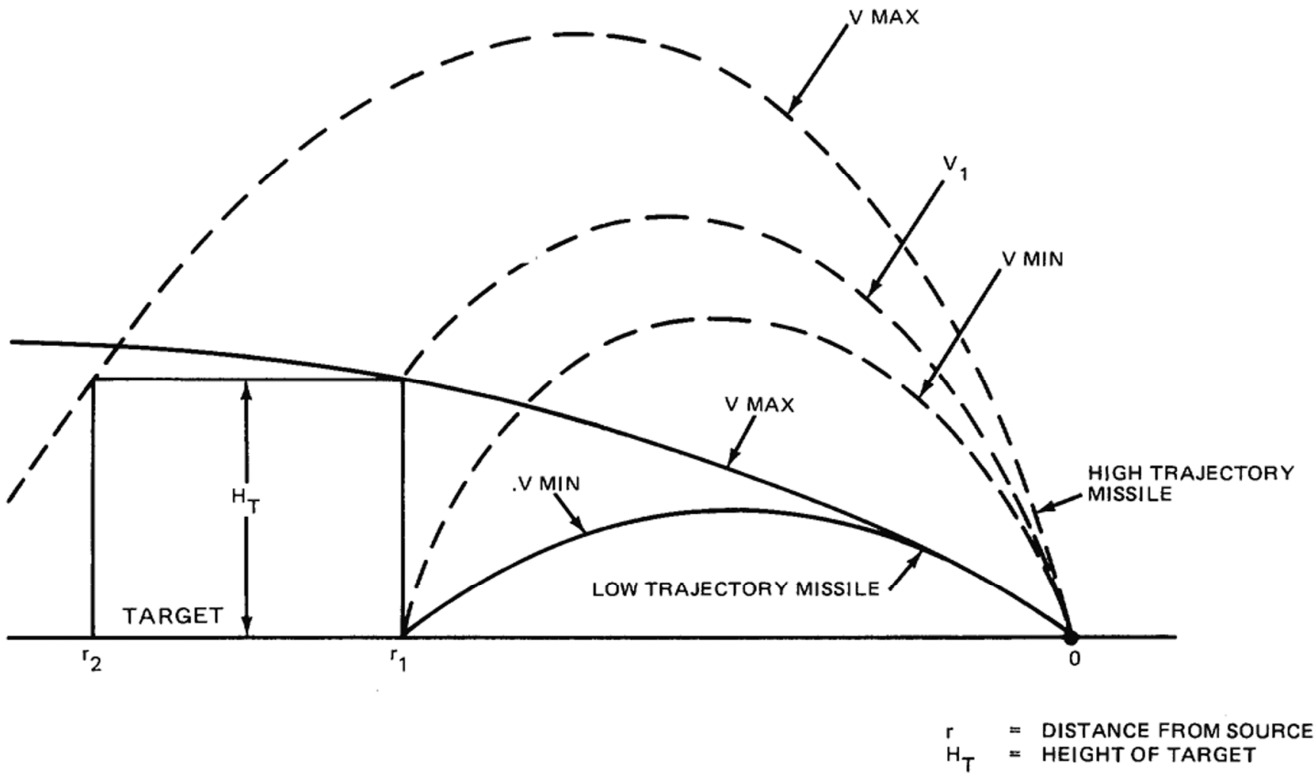
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

MISSILE EJECTION ANGLES

FIGURE 3.5.1–1



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

MISSLE TRAJECTORIES

FIGURE 3.5.1–2

3.6 PROTECTION AGAINST THE DYNAMIC EFFECTS ASSOCIATED WITH THE POSTULATED RUPTURE OF PIPING

Pipe failure protection is provided in accordance with the requirements of 10 CFR 50, Appendix A, General Design Criterion 4.

Postulated breaks in the reactor coolant loop (RCL), except for branch line connections, have been eliminated for both Unit 1 and Unit 2 by reference 3. Subsequent to the General Design Criterion 4 final rule change (52 FR 41288, October 27, 1987), postulated breaks in the RCL branch lines (pressurizer surge line, accumulator line, and residual heat removal (RHR) line for Unit 2 and the pressurizer surge line for Unit 1) were eliminated by application of leak-before-break technology as presented in references 4, 5, 6, 9, and 10. Approval of the elimination of breaks in these Unit 2 branch lines is given in the Vogtle Safety Evaluation Report, Supplement 7, dated November 19, 1987. The necessary information supporting the elimination of breaks in the Unit 1 pressurizer surge line was submitted to the NRC via reference 11.

In the event of a high- or moderate-energy pipe failure within the plant, adequate protection is provided to ensure that those essential structures, systems, or components are not impacted by the effects of postulated piping failure. Essential systems and components are those required to shut down the reactor and mitigate the consequences of the postulated piping failure.

Appendix 3F, Hazards Analysis, provides several examples of the evaluations made of the effects of postulated pipe failures within the plant. The following sections provide the bases for selection of the pipe failures, the determination of the resultant effects, and details of the protection requirements.

3.6.1 POSTULATED PIPING FAILURES IN FLUID SYSTEMS INSIDE AND OUTSIDE CONTAINMENT

Table 3.6.1-1 provides a matrix of plant systems that indicates their classification: high-energy, moderate-energy, essential, or nonessential. Selection of pipe failure locations and evaluation of the consequences on nearby essential systems, components, and structures are presented in subsection 3.6.2 and are in accordance with the requirements of 10 CFR 50, Appendix A, General Design Criterion, and Nuclear Regulatory Commission (NRC) Branch Technical Positions ASB 3-1 and MEB 3-1.

For the reactor coolant loop, reference 1 provides the original criteria for postulating breaks in the reactor coolant loop. Subsequent elimination of postulated pipe breaks in the RCL and Class 1 branch lines is discussed above.

3.6.1.1 Design Bases

The following design bases relate to the evaluation of the effects of the pipe failures determined in subsection 3.6.2:

A. The selection of the failure type is based on whether the system is high or moderate energy during normal operating conditions of the system.

   High-energy piping includes those systems or portions of systems in which the maximum normal operating temperature exceeds 200°F or the maximum normal operating pressure exceeds 275 psig.

Piping systems or portions of systems pressurized above atmospheric pressure during normal plant conditions and not identified as high energy are considered moderate energy.

Piping systems that exceed 200°F or 275 psig for about 2 percent or less of the time the system is in operation or that experience high-energy pressures or temperatures for less than 1 percent of the plant operation time are considered moderate energy.

B. The following assumptions are used to determine the thermodynamic state in the piping system for the calculation of fluid reaction forces:

1. For those portions of piping systems normally pressurized during operation at power, the thermodynamic state in the pipe and associated reservoirs are those of full-power operation.

2. For those portions of piping systems only pressurized during other normal plant conditions (e.g., startup, hot standby, reactor cooldown), the thermodynamic state and associated operating condition is determined as the mode giving the highest enthalpy.

C. Moderate-energy pipe cracks are evaluated for spray wetting, flooding, and other environmental effects.

D. Where postulated, each longitudinal or circumferential break in high-energy fluid system piping or leakage crack in moderate-energy fluid system piping is considered separately as a single initial event occurring during normal plant conditions.

E. Offsite power is assumed to be unavailable if an automatic trip of the turbine-generator system or reactor protection system is a direct consequence of the postulated piping failure.

F. A single active component failure is assumed in systems used to mitigate the consequences of the postulated piping failure or to safely shut down the reactor, except as noted in paragraph G below. The single active component failure is assumed to occur in addition to the postulated piping failure and any direct consequences of the piping failure, such as unit trip and loss of offsite power.

G. When the postulated piping failure occurs in one of two or more redundant trains of a dual-purpose, moderate-energy essential system, single failures of components in the other train or trains (and associated supporting train) are not assumed, because the system is designed to Seismic Category 1 standards; powered from both offsite and onsite sources; and constructed, operated, and inspected to quality assurance, testing, and inservice inspection standards appropriate for nuclear safety systems.

H. All available systems, including those actuated by operator actions, are employed to mitigate the consequences of a postulated piping failure to the extent clarified in the following paragraphs:

1. In determining the availability of the systems, account is taken of the postulated failure and its direct consequences, such as unit trip and loss of offsite power, and of the assumed single active component failure and its direct consequences. The feasibility of carrying out operator actions is determined on the basis of ample time and adequate access to

equipment being available for the proposed actions. Although a postulated high/moderate-energy line failure outside the containment may ultimately require a cold shutdown, operation at hot standby is allowed in order for plant personnel to assess the situation and make repairs.

2. The use of non-Seismic Category 1 piping in mitigating the consequence of postulated piping failure outside the containment is clarified in the following paragraphs:

   a. For non-Seismic Category 1 piping failures, it is assumed that a safe shutdown earthquake could be the cause of the failure. Therefore, only Seismic Category 1 equipment can be used to mitigate the consequences of the failure and bring the plant to a safe shutdown.

   b. Category 1 and seismically supported non-Category 1 piping systems located outside the containment are assumed to fail nonmechanistically (i.e., failure is produced by some mechanism other than an earthquake) for the purpose of pipe break hazard analysis. Therefore, non-Category 1 equipment can be used to bring the plant to a safe shutdown following a postulated pipe break event, subject to the power being available to operate such equipment and provided that the radiological consequences are insignificant in comparison to 10 CFR 100 dose guidelines. For example, non-Category 1 equipment may be used in the mitigation of a charging header piping failure, since this event does not cause a unit trip or significant radiological consequences.

I. A whipping pipe is not considered capable of rupturing impacted pipes of equal or greater nominal pipe diameter and equal or greater wall thickness. This is based on the assumption that only piping is determined to do the impacting. A whipping pipe is considered capable of developing a through-wall leakage crack in a pipe of larger nominal pipe size with thinner wall thickness, assuming that only piping is determined to do the impacting. The above criterion is not utilized where the potential exists for valves or other components in the whipping pipe to impact the targets, since these are treated on a case-by-case basis.

J. Pipe whip is assumed to occur in the plane defined by the piping geometry and to cause movement in the direction of the jet reaction.

If unrestrained, a whipping pipe having a constant energy source sufficient to form a plastic hinge is considered to form a plastic hinge and rotate about the nearest rigid pipe whip restraint, anchor, or wall penetration capable of resisting the pipe whip loads. If the direction of the initial pipe movement caused by the thrust force is such that the whipping pipe impacts a flat surface normal to its direction of travel, it is assumed that the pipe comes to rest against that surface, with no pipe whip in other directions.

In general, whipping ends from a pipe break are restrained so that plastic hinge formation is not allowed to occur. Where plastic hinge could be formed, the effects are evaluated. Pipe whip restraints are provided wherever postulated pipe breaks could impair the ability of any essential system or component to perform its intended safety functions listed in section 3.6.

K. The calculation of thrust and jet impingement forces considers any line restrictions (e.g., flow limiter) between the pressure source and break location and the absence of energy reservoirs, as applicable.

L. Pipe breaks are not postulated to occur in pump and valve bodies since the wall thickness exceeds that of connecting pipe.

M. Components impacted by jets from breaks in piping containing high pressure (870 to 2465 psia) steam or subcooled liquid that would flash at the break, such as piping connected to the steam generators or reactor coolant loops, are evaluated as follows:

1. Impacted components within 10 piping diameters of the broken pipe are assumed to fail. Specific jet loads are calculated and evaluated only when failure of the component, when combined with a single active failure, could adversely affect safe shutdown or accident mitigation capability. These jet loads will be calculated in accordance with FSAR paragraph 3.6.2.3.

3. Components beyond 10 diameters of the broken pipe are considered to be undamaged by the jet and are not analyzed. The basis for these criteria is contained in reference 4.

3.6.1.2 Description

Systems, components, and equipment required to perform the functions discussed in section 3.6 (essential systems) are reviewed to ensure conformance with the design bases and to determine their susceptibility to the failure effects. The break and crack locations are determined in accordance with subsection 3.6.2.

A design comparison to NRC Branch Technical Positions ASB 3-1 and MEB 3-1 is provided in tables 3.6.1-2 and 3.6.1-3.

Pressure response analyses are performed for subcompartments containing high-energy piping. For a detailed discussion of the pipe breaks selected and pressure results, refer to paragraph 6.2.1.2 for selected subcompartments inside the containment and to appendix 3F for selected subcompartments outside the containment. Effects of both internal reactor pressure vessel asymmetric pressurization loads and asymmetric compartment pressurization loads inside containment are addressed in paragraph 6.2.1.2. The analytical methods used for pressure response analysis are in accordance with reference 2.

Appendix 3F provides a typical hazards analysis for the effects of postulated pipe breaks on essential systems, components, and structures.

There are no high-energy lines in the proximity of the control room; therefore, there are no effects upon the habitability of the control room resulting from postulated pipe breaks. Further discussion of the control room habitability systems is provided in section 6.4.

3.6.1.3 Safety Evaluation

3.6.1.3.1 General

An analysis of postulated pipe failures is performed to determine the impact of such piping failures on those safety-related systems or components which provide protective actions and are required to mitigate the consequences of the failure. By means of protective measures, such as separation, barriers, and pipe whip restraints, the effects of breaks and cracks are

prevented from damaging essential items to an extent that would impair their essential function or necessary component operability. Typical measures used for protecting the essential systems, components, and equipment are outlined below and are discussed in detail in subsection 3.6.2. The ability of specific safety-related systems to withstand a single active failure concurrent with the postulated event is discussed, as applicable. When the results of the pipe failure effects analysis show that the effects of a postulated pipe failure are isolated, physically remote, or restrained by protective measures from essential systems or components, no further dynamic hazards analysis is performed.

3.6.1.3.2 Protection Mechanisms

The plant layout arrangement is based on maximizing the physical separation of redundant or diverse safety-related components and systems from each other and from nonsafety-related items. Therefore, in the event a pipe failure occurs, there is a minimal effect on other essential systems or components required for safe shutdown of the plant or to mitigate the consequences of the failure.

The effects associated with a particular pipe failure must be mechanistically consistent with the failure. Thus, pipe dimensions, piping layouts, material properties, and equipment arrangements are considered in defining the specific measures for protection against the consequences of postulated failures.

Protection against the dynamic effects of pipe failures is provided in the form of physical separation of systems and components, barriers, equipment shields, and pipe whip restraints. The precise method chosen depends largely upon considerations such as accessibility and maintenance.

A. Separation

The plant arrangement provides separation, to the extent practicable, between redundant safety systems (including their appurtenances) to prevent loss of safety function as a result of hazards for which the system is required to be functional. Separation between redundant safety systems, with their related appurtenances, therefore, is the basic protective measure incorporated in the design to protect against the dynamic effects of postulated pipe failures.

In general, layout of the facility follows a multi-step process to ensure adequate separation:

1. Safety-related systems are located remotely from high-energy piping, where practicable.
2. Redundant safety systems are located in separate compartments.
3. As necessary, specific components are enclosed to retain the redundancy required for those systems that must function as a consequence of specific piping failure.
4. Drainage systems are reviewed to ensure their adequacy for flooding control.

B. Barriers and Shields

Protection requirements are met through the protection afforded by walls, floors, columns, abutments, and foundations. Where adequate protection does not already exist as a result of separation, additional barriers, deflectors, or shields are provided to meet the functional protection requirements.

Inside the containment, the secondary shield wall serves as a barrier between the reactor coolant loops and the containment liner. In addition, the refueling cavity walls, operating floor, and secondary shield walls minimize the possibility of an accident which may occur in any one reactor coolant loop affecting another loop or the containment liner. Those portions of the steam and feedwater lines located within the containment are routed in such a manner that possible interaction between these lines and the reactor coolant piping is minimized. The barriers described above will withstand loadings caused by jet forces and pipe whip impact forces.

Further discussion of barriers and shields is provided in paragraph 3.6.2.4.

C. Piping Restraint Protection

Measures for protection against pipe whip are provided where the unrestrained pipe movement of either end of the ruptured pipe could cause damage at an unacceptable level to any structure, system, or component required to meet the criteria outlined in section 3.6.

The design criteria for and description of pipe whip restraints are given in paragraph 3.6.2.3.

3.6.1.3.3 Specific Protection Considerations

A. Nonessential systems, structures, and components are not required to meet the criteria outlined in section 3.6. However, while none of the above are needed during or following a pipe break event, pipe whip protection is evaluated where a high-energy nonessential system component or nonessential steel failure could initiate a pipe break event in an essential system or component or in another nonessential system whose failure could affect an essential system.

B. High-energy containment penetrations are subject to special protection mechanisms. As discussed in paragraph 3.6.2.1.1.D, isolation restraints are located as close as practical to the containment isolation valves associated with these penetrations. These restraints are provided to maintain the operability of the isolation valves and the integrity of the penetration due to a break either upstream or downstream of the respective isolation restraints.

C. Instrumentation that is required to function following a pipe rupture is protected.

D. High-energy fluid system pipe whip restraints and protective measures are designed so that a postulated break in one pipe cannot, in turn, lead to a rupture of other nearby pipes or components, if the secondary rupture will result in consequences that would be considered unacceptable for the initial postulated break.

E. For any postulated loss-of-coolant accident, the structural and leaktight integrity of the containment is maintained.

F. The escape of steam, water, combustible or corrosive fluids, gases, and heat in the event of a pipe rupture will not preclude:

1. Subsequent access to any areas, as required, to cope with the postulated pipe rupture.
2. Habitability of the control room.

3. The ability of essential instrumentation, electric power supplies, components, and controls to perform their safety functions to the extent necessary to meet the criteria outlined in section 3.6.

3.6.1.4 References

1. "Pipe Breaks for the LOCA Analysis of the Westinghouse Primary Coolant Loop," WCAP-8082-P-A (proprietary) and WCAP-8172-A (nonproprietary), January 1975.
2. "Subcompartment Pressure Analyses," BN-TOP-4, Revision 1, Bechtel Power Corporation, October 1977.
3. Federal Register, Vol. 50, No. 27, February 8, 1985.
4. NUREG/CR-2913, "Two-Phase Jet Loads," January 1983.
5. Federal Register, Vol. 50, FR 5454, February 8, 1985.
6. "Technical Bases for Eliminating Pressurizer Surge Line Rupture as the Structural Design Basis for Vogtle Unit 2," WCAP-11531 (proprietary) and WCAP-11532 (nonproprietary), July 1987 plus Addenda 1 and 2 dated August and September, 1987.
7. "Technical Basis for Eliminating Accumulator Line Rupture as the Structural Design Basis for Vogtle Unit 2, " WCAP-11583 (proprietary) and WCAP-11584 (nonproprietary), October 1987.
8. "Technical Basis for Eliminating RHR Line Rupture as the Structural Design Basis for Vogtle Unit 2," WCAP-11599 (proprietary) and WCAP-11600 (nonproprietary), September 1987.
9. "Evaluation of Thermal Stratification for the Vogtle Unit 2 Pressurizer Surge Line," WCAP-12218 (proprietary) and WCAP-12219 (nonproprietary), dated March 1989.
10. "Supplementary Analysis to Address Thermal Stratification for Vogtle Unit 1 Pressurizer Surge Line," WCAP-12218 Supplement 1 (proprietary) and WCAP-12219 Supplement 1 (nonproprietary), dated December 1989.
11. GPC letter to NRC transmitting WCAP-12218 Supplement 1 and WCAP-12219 Supplement 1.

3.6.2 DETERMINATION OF BREAK LOCATIONS AND DYNAMIC EFFECTS ASSOCIATED WITH THE POSTULATED RUPTURE OF PIPING

This subsection describes the design bases for locating postulated breaks and cracks in high- and moderate-energy piping systems inside and outside of the containment; the procedures used to define the jet thrust reaction at the break location; the procedures used to define the jet impingement loading on adjacent essential structures, systems, or components; pipe whip restraint design; and the protective assembly design.

3.6.2.1 Criteria Used To Define High/Moderate-Energy Break/Crack Locations and Configurations

Nuclear Regulatory Commission (NRC) Branch Technical Position (BTP) MEB 3-1[(1)] is used as the basis of the criteria for the postulation of high-energy pipe breaks except for the reactor

coolant loop piping of Units 1 and 2 and the Class 1 branch line piping of Units 1 and 2 as discussed in section 3.6. Specific moderate-energy pipe crack locations are not ascertained; and, therefore, they are assumed to occur at any location, as described in paragraph 3.6.2.1.2.4.

A postulated high-energy pipe break is defined as a sudden, gross failure of the pressure boundary of a pipe either in the form of a complete circumferential severance (i.e., a guillotine break) or as a sudden longitudinal, uncontrolled crack. For moderate-energy fluid systems, pipe failures are confined to postulation of controlled cracks in piping. The effects of these cracks on the safety-related equipment are analyzed for flooding and wetting only. These cracks do not result in jet impingement or whipping of the cracked piping.

3.6.2.1.1 High-Energy Break Locations

With the exception of those portions of the piping identified in paragraph 3.6.2.1.1.D, breaks are postulated in high-energy piping at the following locations:

A. American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code, Section III, Division 1 - Class 1 Piping

1. The original design criteria postulates a limited number of pipe break locations in the reactor coolant loop[(2)]. Eight of these break locations were eliminated from the plant design basis because of the consideration of the detailed fracture mechanics evaluation of reference 10. This new design was approved by the NRC in reference 9. For the breaks at the residual heat removal line nozzle and accumulator line nozzle, the breaks are postulated on terminal end criteria for Unit 1 only. The pressurizer surge line nozzle break for Units 1 and 2 and the residual heat removal line nozzle and accumulator line nozzle breaks for Unit 2 are eliminated as discussed in section 3.6.

2. Except for pipe breaks which are not postulated in the Class 1 branch lines as discussed above, pipe breaks are postulated to occur at the following locations in Class 1 piping runs or branch runs outside the primary reactor coolant loops (RCL) as follows:

   a. At terminal ends of the piping, including:

      (1) Piping connected to structures, components, or anchors that act as essentially rigid restraints to piping translation and rotational motion due to static or dynamic loading.

      (2) High/moderate-energy boundary such as piping runs which are maintained pressurized during normal plant conditions for only a portion of the run; i.e., up to the first normally closed valve. The terminal end of such piping is the piping connection to the closed valve.

      (3) Branch intersection points are considered a terminal end for the branch line unless the following are met: the branch and the main piping systems are modeled in the same static, dynamic, and thermal analyses; and the branch and main run are of comparable size and fixity; i.e., the nominal size of the branch is at least one-half of that of the main.

b. At all intermediate locations where the following conditions are satisfied:

(1) Any intermediate locations where the maximum stress range as calculated by equation (10) and either (12) or (13) exceeds 2.4 S_m, (where S_m is the design stress intensity) as described in paragraph NB-3653 of the ASME B&PV Code, Section III.

(2) Any intermediate locations where the cumulative usage factor exceeds 0.1.[3] If two intermediate locations cannot be determined by the above criteria, two highest stress locations based on equation (10) in paragraph NB-3653 of the ASME B&PV Code, Section III[3] are selected. If the piping run has only one change or no change of direction, only one intermediate break location is postulated.

c. For those high energy systems identified in references 11 and 12, postulated intermediate break locations selected in accordance with item b.(2) above may be eliminated from design consideration if the following conditions are satisfied:

- Possibility of stress corrosion cracking has been minimized.
- Thermal and vibration induced piping fatigue has been minimized.
- Steam/waterhammer effects have been minimized.

d. As a result of piping reanalysis, the highest stress locations may be shifted. However, once a high-energy piping system has been analyzed and break locations have been identified and evaluated, the original intermediate break locations (selection based on the methodology specified in item b above) remain the same unless one of the following conditions exists:

(1) Maximum stress ranges or cumulative usage factors exceed the threshold levels specified in item b above.

(2) A change is required in pipe parameters, such as a major difference in pipe size, wall thickness, and routing.

B. ASME B&PV Code, Section III - Class 2 and 3 Piping Systems

1. Pipe breaks are postulated to occur at terminal ends.

2. Pipe breaks are postulated at intermediate locations between terminal ends where the maximum stress value, as calculated by the sum of equations (9) and (10) in Subarticle NC-3652 of the ASME B&PV Code, Section III,[3] considering normal and upset plant conditions (i.e., sustained loads, occasional loads, thermal expansion, and an operating basis earthquake (OBE) event) exceed 0.8 ($1.2 S_h + S_A$).

S_h and S_A are the allowable stress at maximum hot temperature and allowable stress range for thermal expansion, respectively, for Class 2 and 3 piping, as defined in Subarticle NC-3600 of the ASME B&PV Code, Section III.

3. In piping systems where the stresses are lower than the limits in item 2 above, a minimum of two intermediate break locations are postulated on the basis of the highest calculated stress levels. Where the piping consists of one change or a straight run with no fittings, welded attachments, or valves, only one location is chosen, based on the highest stress.

4. For those high energy systems identified in references 11 and 12, postulated intermediate break locations selected in accordance with item 3 above may be eliminated from design consideration if the following conditions are satisfied:

   - Possibility of stress corrosion cracking has been minimized.
   - Thermal and vibration-induced piping fatigue has been minimized.
   - Steam/waterhammer effects have been minimized.
   - Welded attachments (W/A) are not located in the vicinity of intermediate break locations as described below:

     a. No W/A is within 5 nominal piping diameters of the highest stress location(s) on high energy main steam and main feedwater systems.

     b. No W/A is within $\sqrt[3]{Rt}$ of the highest stress location(s) on remaining high energy piping systems other than main steam and main feedwater systems.

        R = The mean pipe radius based on ½ nominal pipe diameter and thickness.

        t = The pipe nominal thickness.

     If condition (a) above is not met for high energy main steam and main feedwater systems, a weak link analysis of the W/A support structure shall be made. Pipe break(s) will not be postulated if this analysis shows that the pipe wall at the W/A to pipe interface is not the weak link.

     For all high energy piping systems, if condition (a) or (b) above is not met, additional intermediate pipe breaks shall be postulated if the combined stress (dissipated local plus general) within $\sqrt[3]{Rt}$ of the high stress location exceeds the threshold of 0.8 of the combined equation 9 and 10 allowables.

4. As a result of piping reanalysis, the highest stress locations may be shifted. However, once a high-energy piping system has been analyzed and break locations have been identified and evaluated, the original intermediate break locations (selection based on the methodology specified in item 2 above) remain the same unless one of the following conditions exists:

a. Maximum stress exceeds the threshold level specified in item 2 above.

b. A change is required in pipe parameters, such as a major difference in pipe size, wall thickness, and routing.

Breaks are postulated as stated above in each piping and branch run adjacent to a protective structure or compartment containing essential systems and components required for safe shutdown. Such piping is considered as located adjacent to a protective structure if the distance between the piping and structure is insufficient to preclude impairment of the structure's integrity from the effects of a postulated piping failure, assuming that the piping is unrestrained.

C. Nonnuclear Piping (i.e., not ASME Section III Class 1, 2, or 3)

Breaks in nonnuclear piping are postulated at the following locations in each run:

1. At the locations specified for ASME, Section III(3) Class 2 and 3 piping (refer to paragraph 3.6.2.1.1.B), if the nonnuclear piping is analyzed and supported to withstand full safe shutdown earthquake loadings.
2. In the absence of stress analysis, breaks in nonnuclear piping are postulated at the following locations in each run or branch run:

   a. Terminal ends.

   b. Each intermediate fitting; e.g., short- and long-radius elbows, tees, and reducers; welded attachments; and valves.

D. High-Energy Piping in Containment Penetration Areas

Breaks are not postulated in the portions of Class 2 piping between the containment penetration flued-head and five-way restraints (i.e., break exclusion zone) provided subject piping meets the following provisions:

1. Stresses do not exceed those specified in paragraph 3.6.2.1.1.B.
2. The maximum stress in this piping as calculated by equation (9), per paragraph NC-3652 of ASME Section III when subjected to the combined loadings of internal pressure, deadweight, and pipe rupture outside the protective restraints, does not exceed 1.8 S_h .
3. The number of circumferential and longitudinal piping welds and branch connections is minimized.

   Areas of system piping where no breaks are postulated are as follows:

   a. The main steam piping, from the containment penetration flued head outboard weld, to the upstream weld of the five-way restraint, which is downstream of the main steam isolation valves, including the main steam safety valves and branch piping to the main steam safety valves. This includes approximately 33 ft of piping for each steam line.

b. The main feedwater piping from the containment penetration to the five-way restraint which is upstream of the isolation valve (approximately 33 ft of piping for each feedwater line).

When required for isolation valve operability, structural integrity, or containment integrity, five-way restraints capable of resisting torsional and bending moments produced by a postulated pipe break, either upstream or downstream of the piping and valves which form the containment isolation boundary, are located reasonably close to the isolation valves or penetration.

The five-way restraints do not prevent the access required to conduct inservice inspection examinations specified in Section XI of the ASME Code. Inservice examinations completed during each inspection interval provide 100-percent volumetric examination of circumferential and longitudinal pipe welds within the boundary of these portions of piping during each inspection interval, as described in section 6.6.

Welded attachments to these portions of piping for pipe supports or other purposes are avoided. Where welded attachments are necessary, detailed stress analyses are performed to demonstrate compliance with the limits of paragraph 3.6.2.1.1.

The five-way restraints outside the containment on the main steam and main feedwater lines are located as close as possible to the containment to accommodate the design for the auxiliary building steam tunnel and still minimize stresses.

3.6.2.1.2 Types of Breaks/Cracks Postulated

3.6.2.1.2.1 ASME Section III, Class 1 RCL Piping - High-Energy. The types of breaks postulated in the ASME Section III,(3) Class 1 primary RCL are discussed in paragraph 3.6.2.1.1.A.1.

3.6.2.1.2.2 Piping Other than RCL Piping - High-Energy. The following types of breaks are postulated to occur at the locations determined in accordance with paragraph 3.6.2.1.1.

A. In piping whose nominal diameter is greater than or equal to 4 in., both circumferential and longitudinal breaks are postulated at each selected break location unless eliminated by comparison of longitudinal and axial stresses with the maximum stress as follows:

1. If the maximum stress range exceeds the limits specified in paragraphs 3.6.2.1.1.A.2.b and 3.6.2.1.1.B.2 but the circumferential stress range is at least 1.5 times the axial stress range, only a longitudinal break is postulated.
2. If the maximum stress range exceeds the limits specified in paragraphs 3.6.2.1.1.A.2.b and 3.6.2.1.1.B.2 but the axial stress is at least 1.5 times

the circumferential stress range, only a circumferential break is postulated.

Longitudinal breaks, however, are not postulated at the following locations:

1. Terminal ends.
2. Intermediate points of Class 1 piping systems where the stress range as calculated by equations (10) and either (12) or (13) does not exceed 2.4 S_m as described in paragraph NB-3653 of the ASME B&PV Code, Section III,(3) and/or if the cumulative usage factor does not exceed 0.1.
3. Intermediate points of Class 2 and 3 piping systems where the maximum stress value, as calculated by the sum of equations (9) and (10) described in paragraph NC-3652 of the ASME B&PV Code, Section III,[(3)] does not exceed 0.8 (1.2 S_h + S_A).

B. In piping whose nominal diameter is greater than 1 in. but less than 4 in., only circumferential breaks are postulated at each selected break location.

C. No breaks are postulated for piping whose nominal diameter is 1 in. or less.

D. In the absence of mechanistic break locations, as described in paragraph 3.6.2.1.1. Breaks or critical cracks are postulated in high energy ASME Code, Section III, Class 1, 2, and 3 and ANSI B31.1 piping at locations that result in the most severe environmental consequences.

3.6.2.1.2.3 Nonnuclear Piping - High-Energy. The types of breaks for nonnuclear piping are postulated as discussed in paragraph 3.6.2.1.2.2; the corresponding break locations are determined in accordance with paragraph 3.6.2.1.1.C.

3.6.2.1.2.4 ASME Section III and Nonnuclear Piping - Moderate-Energy. Through-wall leakage cracks are postulated in moderate-energy piping including branch runs larger than 1-in. nominal diameter as clarified below:

A. Through-wall leakage cracks are not required to be postulated in those portions of piping between containment isolation valves, provided they meet the requirements of ASME Code, Section III, Sub-article NE-1120, and are designed so that the maximum stress range does not exceed 0.4 (1.2 S_h + S_A).

B. Through-wall leakage cracks are not required to be postulated in moderate-energy fluid system piping located in an area where a break in the high-energy fluid system is postulated, provided that such cracks do not result in environmental conditions more limiting than the high-energy pipe break.

C. Subject to paragraph D below, through-wall leakage cracks are required to be postulated in:

(1) ASME, B&PV Code, Section III, Division 1 - Class 1 piping where the maximum stress range in the piping is greater than 1.2 S_m.

(2) ASME, B&PV Code, Section III, Division 1 - Class 2 or 3 piping and seismically supported nonnuclear class piping at locations where the maximum stress range in the piping is greater than 0.4 (1.2 S_h + S_A).

D. Individual cracks are not required to be postulated at specific locations determined by stress analyses when a review of the piping layout and plant arrangement drawings shows that the effects of through-wall leakage cracks at any location in the piping designed to seismic or nonseismic standards are isolated or physically remote from structures, systems, and components required for safe shutdown.

To simplify analysis, cracks may be postulated to occur everywhere in moderate-energy piping regardless of the stress analysis results to determine the maximum damage from fluid spraying and flooding, with the consequent hazards or environmental conditions. Flooding effects are determined on the basis of a 30-min operator time required to effect corrective actions. Further discussion of flooding effects is provided in appendix 3F.

3.6.2.1.3 Break/Crack Configuration

3.6.2.1.3.1 High-Energy Break Configuration. Following a circumferential break, the two ends of the broken pipe are assumed to move clear of each other unless physically limited by piping restraints, structural members, or piping stiffness. The effective cross-sectional (inside diameter) flow area of the pipe is used in the jet discharge evaluation. Movement is assumed to be in the direction of the jet reaction initially, with the total path controlled by the piping geometry.

The orientation of a longitudinal break, except when otherwise justified by a detailed stress analysis, is assumed to be at opposing points on a line perpendicular to the plane of a fitting for a nonaxisymmetric fitting and anywhere around the circumference of the fitting for axisymmetric fittings. The flow area of such a break is equal to the cross-sectional flow area of the pipe. Longitudinal and circumferential breaks are not postulated concurrently.

3.6.2.1.3.2 Moderate-Energy Crack Configuration. Moderate-energy crack openings are assumed to be a circular orifice with cross-sectional flow area equal to that of a rectangle one-half the pipe inside diameter in length and one-half pipe wall thickness in width.

3.6.2.2 Analytical Methods To Define Forcing Functions and Response Models

3.6.2.2.1 Forcing Functions for Jet Thrust

To determine the forcing function, the fluid conditions at the upstream source and at the break exit dictate the analytical approach and approximations that are used. For most applications, one of the following situations exists:

- Superheated or saturated steam.
- Saturated or subcooled water.
- Cold water (nonflashing).

Analytical methods for calculation of jet thrust for the above- described situations are discussed in references 4 and 5. For a discussion of the jet thrust forcing functions from RCL breaks, see paragraph 3.6.2.2.1.1.

3.6.2.2.1.1 Time Functions of Jet Thrust Force on Ruptured and Intact RCL Piping. To determine the thrust and reactive force loads to be applied to the RCL during the postulated loss-of-coolant accident (LOCA), it is necessary to have a detailed description of the hydraulic transient. Hydraulic forcing functions are calculated for the intact RCLs as a result of a postulated LOCA. These forces result from the transient flow and pressure histories in the reactor coolant system (RCS). The calculation is performed in two steps. The first step is to calculate the transient pressure, mass flowrates, and thermodynamic properties as a function of time. The second step uses the results obtained from the hydraulic analysis, along with input of areas and direction coordinates, and calculates the time-history of forces at appropriate locations (e.g., elbows) in the RCLs.

The hydraulic model represents the behavior of the coolant fluid within the entire RCS. Key parameters calculated by the hydraulic model are pressure, mass flowrate, and density. These are supplied to the thrust calculation, together with plant layout information, to determine the time-dependent loads exerted by the fluid on the loops. In evaluating the hydraulic forcing functions during a postulated LOCA, the pressure and momentum flux terms are dominant. The inertia and gravitational terms are taken into account in the evaluation of the local fluid conditions in the hydraulic model.

The blowdown hydraulic analysis is required to provide the basic information concerning the dynamic behavior of the reactor core environment for the loop forces. This requires the ability to predict the flow, quality, and pressure of the fluid throughout the reactor system. The MULTIFLEX code[6] was developed with a capability to provide this information.

The MULTIFLEX computer code calculates the hydraulic transients within the entire primary coolant system. This hydraulic program considers a coupled, fluid-structure interaction by accounting for the deflection of the core support barrel. The depressurization of the system is calculated using the method of characteristics applicable to transient flow of a homogenous fluid in thermal equilibrium.

The ability to treat multiple flow branches and a large number of mesh points gives the MULTIFLEX code the flexibility required to represent the various flow passages within the primary RCS. The system geometry is represented by a network of one-dimensional flow passages.

The THRUST computer program was developed to compute the transient (blowdown) hydraulic loads resulting from a LOCA.

The blowdown hydraulic loads on primary loop components are computed from the equation:

The symbols and units are as follows:

$$F = 144A\left[(P - 14.7) + \left(\frac{\dot{m}^2}{\rho g A_m{}^2 144}\right)\right]$$

F = Force (lb_f).

A = Aperture area (ft^2).

P = System pressure (psia).

m = Mass flowrate (lbm/s).

ρ = Density (lbm/ft^3).

g = Gravitational constant = 32.174 ft-lbm/lb -s^2.

A_m = Mass flow area (ft^2).

In the model to compute forcing functions, the RCL system is represented by a model similar to that employed in the blowdown analysis. The entire loop layout is represented in a global coordinate system. Each node is fully described by:

A. Blowdown hydraulic information.

B. The orientation of the streamlines of the force nodes in the system, which includes flow areas, and projection coefficients along the three axes of the global coordinate system.

Each node is modeled as a separate control volume with one or two flow apertures associated with it. Two apertures are used to simulate a change in flow direction and area. Each force is divided into its x, y, and z components using the projection coefficients. The force components are then summed over the total number of apertures in any one node to give a total x force, a total y force, and a total z force. These thrust forces serve as input to the piping/restraint dynamic analysis.

The THRUST code calculates forces exactly the same way as the STHRUST code, which is described in reference 7.

3.6.2.2.2 Dynamic Analysis of the Reactor Coolant Loop Piping, Equipment Supports

The dynamic analysis of the RCL for LOCA loadings is described in section 3.9.

3.6.2.3 Dynamic Analysis Methods To Verify Integrity and Operability

3.6.2.3.1 Dynamic Analysis Methods To Verify Integrity and Operability for Other than RCL

The analytical methods of references 4 and 5 are used to determine the jet impingement effects and loading effects applicable to components and systems resulting from postulated pipe breaks and cracks.

3.6.2.3.2 Dynamic Analysis Methods To Verify Integrity and Operability for the RCL

3.6.2.3.2.1 General. A LOCA is assumed to occur for a branch line break down to the restraint of the second normally open automatic isolation valve (case II, figure 3.6.2-1) on outgoing lines[(1)] and down to and including the second check valve (case III, figure 3.6.2-1) on incoming lines normally with flow. A pipe break beyond the restraint or second check valve does not result in an uncontrolled loss of reactor coolant if either of the two valves in the line closes.

Accordingly, both of the automatic isolation valves are suitably protected and restrained as close to the valves as possible so that a pipe break beyond the restraint does not jeopardize the integrity and operability of the valves. Further, periodic testing capability of the valves to perform their intended function is essential. This criterion takes credit for only one of the two valves performing its intended function. For normally closed isolation or incoming check valves (cases I and IV, figure 3.6.2-1), a LOCA is assumed to occur for pipe breaks on the reactor side of the valve.

Branch lines connected to the RCL are defined as large strictly for the purpose of pipe break criteria if they have an inside diameter greater than 4 in. up to the largest connecting line. Rupture of these lines results in a rapid blowdown from the RCL, and protection is basically provided by the accumulators and the low-head safety injection pumps (residual heat removal pumps).

Branch lines connected to the RCL are defined as small for the purpose of pipe break analysis if they have an inside diameter equal to or less than 4 in. This size is such that emergency core cooling system analyses, using realistic assumptions, show that no clad damage is expected for a break area of up to 12.5 $in.^2$ corresponding to 4 in. inside diameter piping.

Engineered safety features are provided for core cooling and boration, pressure reduction, and activity confinement in the event of a LOCA or steam or feedwater line break accident to ensure that the public is protected in accordance with 10 CFR 100 guidelines. These safety systems are designed to provide protection for an RCS pipe rupture of a limited flow area severance of an RCL, as identified in paragraph 6.2.1.2.

To assure the continued integrity of the essential components and the engineered safety systems, consideration is given to the consequential effects of the pipe break itself to the extent that:

A. The minimum performance capabilities of the engineered safety systems are not reduced below that required to protect against the postulated break.

B. The containment leaktightness is not decreased below the design value if the break leads to a LOCA.[(2)]

C. Propagation of damage is limited in type and/or degree to the extent that:

---

(1) It is assumed that motion of the unsupported line containing the isolation valves can cause failure of the operators of both valves to function.

(2) The containment is here defined as the containment structure liner and penetrations and the steam generator shell, the steam generator steam side instrumentation connections, the steam, feedwater, blowdown, and steam generator drain pipes within the containment structure.

1. A pipe break which is not a LOCA will not cause a LOCA or steam or feedwater line break. However, pipe breaks on the nonreactor side of an RCS pressure boundary may cause the failure of the reactor side of the same pipe, provided the combined failures are evaluated for impact on system performance.

2. An RCS pipe break will not cause a steam or feedwater system pipe break, and vice versa.

3.6.2.3.2.2 Large Branch Lines. Large branch line piping, as defined in paragraph 3.6.2.3.2.1, is restrained[3] to meet the following criteria in addition to items A through C of paragraph 3.6.2.3.2.1 for a pipe break resulting in a LOCA:

A. Propagation of the break to the unaffected loops (except pressurizer spray on Unit 1) is prevented to ensure the delivery capacity of the accumulators and low head pumps.

B. Propagation of the break in the affected loop is permitted to occur but does not exceed 20 percent of the flow area of the line which initially ruptured (except pressurizer spray, safety, and relief lines on Unit 1). The capacity of the accumulators and low head safety injection system establishes the limit for propagation of large break LOCAs (lines 6 in. and greater) to other branch lines in the loop containing the initially postulated pipe break. The limit selected (total area of additional failed lines to be less than 20 percent of initial break area) ensures that the severity of the initial break is not increased to an unacceptable level. For large branch line breaks, the actual capacity of the accumulators and the low head safety injection system is well in excess of that needed for propagation of this magnitude. For this reason, exceptions to the above criteria are allowed on a case basis, provided that the failure of the broken lines is shown to be enveloped by the design basis LOCAs.

3.6.2.3.2.3 Small Branch Lines. Should one of the small pressurized lines, as defined in paragraph 3.6.2.3.2.1, fail and result in a LOCA, the piping is restrained or arranged to meet the following criteria in addition to items A through C of paragraph 3.6.2.3.2.1:

A. Break propagation is limited to the affected leg; i.e., propagation to the other leg of the affected loop and to the other loops is prevented. Damage to the high-head safety injection lines connected to the other leg of the affected loop or to the other loops is prevented.

[3] The Class 1 branch lines in Unit 2 are not restrained because the dynamic effects of pipe rupture in these lines are not in the plant's design basis as discussed in section 3.6.

B. Propagation of the break in the affected leg is permitted but must be limited to a total break area of 12.5 in.^{2} (4-in. inside diameter). The exception to this case is when the initiating small break is a cold leg high-head safety injection line. Further propagation is not permitted for this case. The capacity of the high head safety injection system establishes the limit for propagation of small break LOCAs (lines 4 in. and smaller) to other branch lines in the leg containing the initially postulated pipe break. The limit selected (total area of additional failed lines to be limited to 12.5 in ^{2}) ensures that the sum of the piping failures will not require safety injection delivery in excess of the capacity of the high head system. Although the severity of such an event is significantly less than that of a large break LOCA, the limit is applied to prevent a small LOCA from becoming a large LOCA. Since an initial break in the cold leg high head safety injection line reduces the capacity of the high head injection system, additional propagation for these breaks is not permitted.

C. Propagation of the break to a high-head safety injection line connected to the affected leg is prevented if the line break results in a loss of core cooling capability due to a spilling injection line.

The above criteria (paragraphs 3.6.2.3.2.2 and 3.6.2.3.2.3) are primarily implemented through the use of physical separation of the branch lines. In many cases, the initiating breaks do not whip or impinge on neighboring branch lines. In those cases where whip or impingement does occur, the impacted line is often designed such that failure will not occur. Whip and impingement by some lines are also prevented by pipe whip restraints. If the impact loads are excessive and it cannot be shown that failure would be acceptable (either by the criteria or engineering analysis), then protective barriers are installed to prevent the interaction.

3.6.2.3.2.4 Design and Verification of Adequacy of RCL Components and Supports. The methods described below are used in the Westinghouse design and verification of the adequacy of primary RCL components and supports. It is emphasized that these methods are used only to determine jet impingement loads on RCL components and supports.[(4)]

The design basis postulated pipe rupture locations are determined using the criteria given in paragraph 3.6.2.1. These design basis ruptures are used here as the rupture locations for consideration of jet impingement effects on primary equipment and supports.

A dynamic analysis is used to determine maximum piping displacements at each design basis rupture location. These maximum piping displacements are used to compute the effective rupture flow area at each location. This area and rupture orientation is then used to determine the jet flow pattern and to identify any primary components which are potential targets for jet impingement.

The jet thrust at the point of rupture is based on the fluid pressure and temperature conditions occurring during normal (100 percent) steady-state operating conditions of the plant. At the point of rupture, the jet force is equal and opposite to the jet thrust. The force of the jet is

[(4)] The jet impingement effects of a break in the RCL piping are not considered in the design verification of the RCL component supports (reactor vessel, pressurizer, reactor coolant pump and steam generator supports) in Units 1 and 2. These effects were eliminated by application of leak-before-break technology and their elimination was factored into the design verification calculations. The jet impingement effects of the most limiting branch line breaks, however, are factored into the Unit 1 and 2 RCL component support design verification.

conservatively assumed to be constant throughout the jet flow distance. The subcooled jet is calculated as discussed in reference 4.

If simplified static analysis is performed instead of a dynamic analysis, the above jet load (F_T) is multiplied by a dynamic load factor. For an equivalent static analysis of the target structure, the jet impingement force is multiplied by a dynamic load factor of 1.2 to 2.0, depending upon the time variance of the jet load. This factor assumes that the target can be represented as essentially a one-degree-of-freedom system, and the impingement force is conservatively applied as a step load.

3.6.2.3.3 Types of Pipe Whip Restraints

3.6.2.3.3.1 Pipe Whip Restraints. To satisfy varying requirements of available space, permissible pipe deflection, and equipment operability, the restraints are designed as a combination of an energy-absorbing element and a restraint structure suitable for the geometry required to pass the restraint load from the whipping pipe to the main building structure.

The restraint structure is typically a structural steel frame or truss and the energy-absorbing element is usually either stainless steel U-bars or energy-absorbing material as described below:

A. Stainless Steel U-Bar

This type consists of one or more U-shaped, upset- threaded rods of stainless steel looped around the pipe but not in contact with the pipe to allow unimpeded pipe motion during seismic and thermal movement of the pipe. At rupture, the pipe moves against the U-bars, which absorb the kinetic energy of pipe motion by yielding plastically. A typical example of a U-bar restraint is shown in figure 3.6.2-2.

B. Energy Absorbing Material

This type of restraint consists of a crushable, stainless steel, internally honeycomb-shaped element designed to yield plastically under impact of the whipping pipe. A design hot position gap is provided between the pipe and the energy-absorbing material to allow unimpeded pipe motion during seismic and thermal pipe movements. A typical example of an energy-absorbing material restraint is shown in figure 3.6.2-3.

3.6.2.3.4 Analytical Methods

3.6.2.3.4.1 Pipe Whip Restraints.

A. Location of Restraints

1. For purposes of determining pipe hinge length and thus locating the pipe whip restraints, the plastic moment of the pipe is determined in the following manner:

$$M_p = 1.1\, z_p S_y$$

where:

z_p = Plastic section modulus of pipe $D_m^2 t$.

D_m = Mean piping diameter.

t = Wall thickness.

S_y = Yield stress at pipe operating temperature.

1.1 = 10-percent factor to account for strain hardening.

Pipe whip restraints are located as close to the axis of the reaction thrust force break as practicable. Pipe whip restraints are generally located so that a plastic hinge does not form in the pipe. If, due to physical limitations, pipe whip restraints are located so that a plastic hinge can form, the consequences of the whipping pipe and the jet impingement effect are further investigated. Lateral guides are provided where necessary to predict and control pipe motion.

2. Generally, restraints are designed and located with sufficient clearances between the pipe and the restraint such that they do not interact and cause additional piping stresses. A design hot position gap is provided that will allow maximum predicted thermal, seismic, and seismic anchor movement displacements to occur without interaction.

   Exception to this general criterion may occur when a pipe support and restraint are incorporated into the same structural steel frame, or when a zero design gap is required. In these cases the restraint is included in the piping analysis, if required.

3. In general, the restraints do not prevent the access required to conduct inservice inspection examination of piping welds. When the location of the restraint makes the piping welds inaccessible for inservice inspection, a portion of the restraint is made removable to provide accessibility.

B. Analysis and Design

Analysis and design of pipe whip restraints for postulated pipe break effects are in accordance with reference 4. Specifically, the following criteria are adopted in analysis and design:

1. Pipe whip restraints are designed based on energy absorption principles by considering the elastic-plastic, strain-hardening behavior of the materials used.

2. A rebound factor of 1.1 is applied to the jet thrust force.

3. Except in cases where calculations are performed to verify that a plastic hinge is formed, the energy absorbed by the ruptured pipe is conservatively assumed to be zero; i.e., the thrust force developed goes directly into moving the broken pipe and is not reduced by the force required to bend the pipe.

4. In elastic-plastic design, limits for strains are as follows:

   ε = Allowable strain used in design.

   a. Stainless Steel U-Bars

      $\varepsilon = 0.5\varepsilon_u$

where:

ε_u = ultimate uniform strain of stainless steel (strain at ultimate stress).

b. Energy-Absorbing Material

$\varepsilon = 0.8\varepsilon_u$

where:

ε_u = maximum crushable height at uniform crushable strength.

5. A dynamic increase factor is used for steel which is designed to remain elastic.

3.6.2.4 Protective Assembly Design Criteria

3.6.2.4.1 Jet Impingement Barriers and Shields

Barriers and shields, which may be of either steel or concrete construction, are provided to protect essential equipment including instrumentation from the effects of jet impingement resulting from postulated pipe breaks. Barriers differ from shields in that they may also accept the impact of whipping pipes. Barriers and shields include walls, floors, and structures specifically designed to provide protection from postulated pipe breaks. Barrier and shield design is based on the methods of reference 4, section 3.0, and the elastic-plastic methods for dynamic analysis included in reference 8. Design criteria and loading combinations are in accordance with subsections 3.8.3 and 3.8.4.

3.6.2.4.2 Auxiliary Guardpipes

The use of guardpipes has been minimized by plant arrangement and routing of high-energy piping. Where they are used, guardpipes are designed to withstand all dynamic and environmental effects of postulated breaks of the enclosed pipe. Auxiliary guardpipes are used only if inservice inspection requirements can be satisfied. Design criteria, loading combinations, and methods of analysis are similar to those for barriers and shields described in paragraph 3.6.2.4.1.

3.6.2.5 Material To Be Submitted for the Operating License Review

3.6.2.5.1 Piping Systems Other than RCL

Pipe break locations are obtained in accordance with the criteria of paragraph 3.6.2.1.

High-energy piping, with break locations identified, is provided in isometric drawings submitted by reference 13. Break types, i.e., circumferential or longitudinal, are also shown. The stress results utilized to determine the break types and locations are given, along with the associated stress nodes. High-energy pipe break effects analysis for a selected portion of the plant

(auxiliary building level C, safety-related pump rooms on levels B and D) is discussed room-by-room in table 3F-1.

Moderate-energy piping crack locations are defined in paragraph 3.6.2.1.2.4. Evaluation of the effects of moderate-energy cracks is discussed in appendix 3F.

The augmented inservice inspection plan is discussed in section 6.6.

Pipe whip restraints are designed in accordance with paragraph 3.6.2.3. Pipe whip restraint location and orientation for each high-energy break are shown in reference 13. Barriers and shields are designed in accordance with the criteria of paragraph 3.6.2.4. Jet thrust and impingement forces were determined in accordance with reference 4. Thrust forces for each pipe whip restraint are presented in reference 13. These values are typically calculated without accounting for the resistance losses. If necessary, the thrust force will be reduced to account for the flow resistance losses.

3.6.2.5.2 Reactor Coolant Loop

A. Drawing AX6DD309 and table 3.6.2-3 identify the design basis break locations remaining and orientations for the RCLs, as explained in paragraph 3.6.2.1.1.A.1.

   The primary and secondary stress intensity ranges and the fatigue cumulative usage factors at the design break locations specified in section 3.6.2.1.1.A.1 are not tabulated since selection of these terminal end locations is independent of detailed stress and fatigue analyses.

B. The results of evaluating jet impingement loads associated with the branch line breaks, identified in 3.6.2.1.1.A.1 are provided by reference 13. As described in item C below, these loads are used to determine the adequacy of the primary equipment and supports.

C. Design loading combinations and applicable criteria for ASME Class 1 components and supports are provided in section 3.9. Pipe rupture loads include not only the jet thrust forces acting on the piping but also jet impingement loads on the primary equipment supports.

3.6.2.6 References

1. "Postulated Break and Leakage Locations in Fluid System Piping Outside Containment," NRC Branch Technical Position MEB 3-1, November 1975.

2. "Pipe Breaks for the LOCA Analysis of the Westinghouse Primary Coolant Loop," WCAP-8082-P-A (proprietary) and WCAP-8172-A (nonproprietary), January 1975.

3. ASME Section III, Subsection 3650, Summer 1979 Addendum to the 1977 Code Edition.

4. "Design for Pipe Break Effects," Bechtel Power Corporation, BN-TOP-2, Revision 2, May 1974.

5. Moody, F. J., Fluid Reaction and Impingement Loads, Paper Presented at the ASCE Specialty Conference, Chicago, December 1973.

6. "MULTIFLEX, A FORTRAN-IV Computer Program for Analyzing Thermal-Hydraulic-Structure System Dynamics," WCAP-8708 (proprietary), February 1976, and WCAP-8709 (nonproprietary), February 1976.

7. "Documentation of Selected Westinghouse Structural Analysis Computer Codes," WCAP-8252, Revision 1, May 1977.

8. Biggs, J. M., Introduction to Structural Dynamics, McGraw-Hill Book Company, New York, 1964.

9. Federal Register, 50 FR 5454, February 8, 1985.

10. "Technical Bases for Eliminating Large Primary Loop Pipe Rupture is the Structural Design Basis for Vogtle, Units 1 and 2," WCAP-10551, (Proprietary Class 2), May 1984.

11. Georgia Power Company letter to the Nuclear Regulatory Commission dated April 26, 1984.

12. Georgia Power Company letter to the Nuclear Regulatory Commission dated April 24, 1985.

13. Georgia Power Company letter to the Nuclear Regulatory Commission, ELV-01-388, March 1990.

TABLE 3.6.1-1 (SHEET 1 OF 2)

ESSENTIAL, HIGH-ENERGY, AND MODERATE-ENERGY SYSTEMS

System	Essential[a] Systems	High[b] Energy	Moderate Energy
Reactor coolant	0	0	
Nuclear service cooling water	0		0
Component cooling water	0		0
Safety injection	0	0	
Residual heat removal	0	0[c]	0
Containment spray	0		0
Chemical volume and control	0	0	0
Nuclear sampling		0	
Spent fuel cooling and purification	0		0
Auxiliary component cooling water			0
Main steam		0	0
Auxiliary feedwater	0	0	0
Condensate and main feedwater	0	0	
Auxiliary steam		0	
Steam generator blowdown	0	0	
Safety-related heating, ventilating, and air conditioning	0		0
Essential chilled water	0		0
Waste processing		0	

TABLE 3.6.1-1 (SHEET 2 OF 2)

System	Essential[a] Systems	High[b] Energy	Moderate Energy
Turbine-Generator		0	0
Auxiliary gas	0	0	
Diesel generator and related systems	0		0
Fire protection			0
Instrument and service air			0

a. Not all essential systems are required for all postulated piping failures; e.g., the containment spray system is essential for loss-of-coolant accident and main steam line break inside containment but is nonessential for piping failure outside containment. Not all portions of essential systems are required for postulated piping failure; e.g., the main steam system is only essential from the steam generator to the main steam isolation valves, including the safety and atmospheric steam relief valves.

b. Not all portions of high-energy systems contain high-energy fluid.

c. During the initial phase of cooldown, the residual heat removal system is a high-energy system. For interaction with the redundant train, the residual heat removal system is considered a dual-purpose, moderate-energy system. (See paragraph 3.6.1.1.G.)

TABLE 3.6.1-2 (SHEET 1 OF 2)

DESIGN COMPARISON TO POSITIONS OF NRC BRANCH TECHNICAL POSITIONS ASB 3-1

	Branch Technical Position ASB 3-1		VEGP Design
B.1	Plant Arrangement	B.1	Conforms. See paragraph 3.6.1.3.
	Protection of essential systems and components against postulated piping failures in high- or moderate-energy fluid systems that operate during normal plant conditions and that are located outside of containment should be provided	B.1.a	Conforms. See paragraph 3.6.1.3.2.1.A.
		B.1.a.(1)	Partial conformance as follows: The essential equipment located in the main bundle being uncovered. Main steamline breaks up to 1.0 ft^2 with steam generator tube bundle being uncovered are considered as discussed in paragraph 3.11.B.1.1. The essential equipment is designed to be protected from the jet impingement and pipe
		B.1.b	Conforms. See paragraphs 3.6.13.2.1B; 3.6.1.3.21C; 3.6.1.3.3; and 3.6.2.3
		B.1.c	
		B.1.c.1(a)	Conforms. As part of the design process, the restraint gap is verified large enough to accommodate thermal, seismic, and seismic anchor movements.
		B.1.c.1(b)	Partial conformance. See paragraph 3.6.2.3.3.1.A Additionally, final pipe whip restraint gap will be verified during hot-functional testing and thus will account for any differential settlement. Pipe relaxation is not specifically considered in the VEGP design.
		B.1.c.1(c)	See response to items (a) and (b) above.
		B.1.c.(2)	Conforms. Restraints which do not have adequate inservice inspection pipe weld space requirements are made removable.

TABLE 3.6.1-2 (SHEET 2 OF 2)

	Branch Technical Position ASB 3-1		VEGP Design
B.2	Design Features		
B.2.a	Essential systems and components should be designed to meet the seismic design requirements of Regulatory Guide 1.29.	B.2.a	Conforms, as described in sections 1.9 and 3.2.
B.2.b	Protective structures or compartments, fluid system piping restraints, and other protective measures.	B.2.b	Conforms. See subsections 3.8.3 and 3.8.4 for loading combinations. See paragraphs 3.6.1.1 and 3.6.1.3 for piping restraints and protection measures.
B.2.c	Fluid system piping in containment penetration areas should be designed to meet the break exclusion provisions contained in item B.1.b of BTP MEB 3-1.	B.2.c	Conforms. High-energy piping is designed as per B.1.b of MEB 3-1. Moderate-energy piping is designed as per B.2.B of MEB 3-1. For further information, refer to B.1.a.(1) above and paragraphs 3.6.2.1.1.D and 3.6.2.1.2.4.
B.2.d	Piping classification as required by NRC Regulatory Guide 1.26 should be maintained.	B.2.d.	Conforms. See paragraph 3.6.2.1.1.D.
B.3	Analyses and Effects of Postulated Piping Failures	B.3.a	Conforms. See paragraphs 3.6.1.1.D; 3.6.1.2; 3.6.1.3; and table 3.6.2-2.
		B.3.b.(1)	Conforms. See paragraph 3.6.1.1.E.
		B.3.b.(2)	Conforms. See paragraph 3.6.1.1.F.
		B.3.b.(3)	Conforms. Paragraph 3.6.1.1.G defines a train to include those systems which support its function.
		B.3.b.(4)	Conforms. See paragraph 3.6.1.1.H.
		B.3.c	Conforms. See paragraph 3.6.1.2 and section 6.4.
		B.3.d	Conforms. See paragraph 3.6.1.1.H and section 3.F.2.

TABLE 3.6.1-3 (SHEET 1 OF 3)

DESIGN COMPARISON TO NRC BRANCH TECHNICAL POSITION MEB 3-1[(a)]

	Branch Technical Position MEB.3-1		VEGP Design
B.1	High-Energy Fluid System Piping		
B.1.a	Fluid systems separated from essential systems and components.	B.1.a	Conforms. See paragraph 3.6.1.3.2.
B.1.b	Fluid system piping in containment penetration areas.	B.1.b	Conforms. See paragraph 3.6.2.1.1.D.
		B.1.b.(1)(a)-(c)	There is no Class 1 piping in containment penetration areas in the VEGP.
		B.1.b.(1)(d)	Conforms. See paragraphs 3.6.2.1.1.D and 3.6.2.1.1.B.
		B.1.b.(1)(e)	Conforms. For further discussion see paragraph 3.6.2.1.1.D.
		B.1.b.(2)	Conforms. See paragraph 3.6.2.1.1.D.
		B.1.b.(3)	Conforms. See paragraph 3.6.2.1.1.D. For guardpipes see paragraph 3.6.2.4.2.
		B.1.b.(4)	See conformance statement to ASB 3-1 position B.2.c.(1) and paragraph 3.6.2.1.1.D.
		B.1.b.(5)	High-energy containment flued head penetrations are integrally forged piped fittings. Pipe whip restraints do not require welding directly to the outer surface of the piping, except where such welds are 100-percent volumetrically examined in service and a review for local stresses is performed. The main steam and main feedwater lines outside the containment have an integrally forged pipe fitting as part of the five-way restraints.
		B.1.b.(6)	Conforms. See paragraph 3.6.2.4.2.
		B.1.b.(7)	Conforms. See paragraph 3.6.2.1.1.D.
B.1.c	Postulation of pipe rupture in areas other than containment penetration.	B.1.c	Conforms. See paragraph 3.6.2.1.1.
		B.1.c.(1)(a)-(d)	Partial conformance. If there are no intermediate locations where maximum stress ranges or cumulative usage factors exceed the threshold levels, no intermediate breaks are postulated provided the conditions of paragraph 3.6.2.1.1.A.2.C are satisfied. See paragraph 3.6.2.1.1.A. There are no postulated pipe breaks

TABLE 3.6.1-3 (SHEET 2 OF 3)

	Branch Technical Position MEB.3-1		VEGP Design
			in the reactor coolant loops of Units 1 and 2, nor are breaks postulated in the Class 1 branch lines of Unit 2 as discussed in Section 3.6.
		B.1.c.(2)(a)-(b)	Partial conformance. If there are no intermediate locations where maximum stress ranges or cumulative usage factors exceed the threshold levels, no intermediate breaks are postulated provided the conditions of paragraph 3.6.2.1.1.B.4 are satisfied. See paragraph 3.6.2.1.1.B.
		B.1.c.(3)	Partial conformance. See paragraph 3.6.2.1.1.C.
		B.1.c.(4)	Conforms. See paragraphs 3.6.2.1.1.B and 3.6.2.1.1.C. Nonnuclear high-energy pipes will either be restrained from impacting or affecting the separating structure or the separating structure will be designed for full effects.
		B.1.d.	Conforms. See paragraph 3.6.2.5.
		B.1.e.	Conforms. See paragraph 3.6.2.1.2.2.D. In the absence of mechanistic (terminal end or high stress) break locations. Breaks or critical cracks are postulated in ASME Section 3, Class 1, 2, and 3 and Nonsafety Class Piping, at locations that result in the most severe environmental consequences.
B.2	Moderate-Energy Fluid System Piping	B.2.a	Conforms. See paragraph 3.6.1.3 and appendix 3F.
		B.2.b	Conforms. See paragraph 3.6.2.1.2.4.
		B.2.c.(1)-(2)	Conforms. See paragraphs 3.6.2.1.2.4.
		B.2.d	Conforms. See paragraph 3.6.2.1.2.4.
		B.2.e	Conforms. See paragraph 3.6.1.1.A.
B.3	Type of Breaks and Leakage Cracks in Fluid System Piping	B.3.a.(1)	Conforms. See paragraph 3.6.2.1.2.2.
		B.3.a.(2)	Conforms. All high-energy Class 1, 2, and 3 piping systems are analyzed using industry approved computer programs in piping stress analysis. In the absence of stress analysis, nonnuclear class high-energy piping breaks are postulated at all welds, fittings, welded attachments, etc. See paragraph 3.6.2.1.1.
		B.3.a.(3)	Conforms. See paragraph 3.6.2.1.3.1.
		B.3.a.(4)	See paragraph 3.6.2.2.1.
		B.3.a.(5)	Conforms. See paragraph 3.6.1.1.J.
		B.3.b.(1)	Conforms. See paragraph 3.6.2.1.2.2.
		B.3.b.(2)	Per paragraph 3.6.2.1.2.2, only circumferential breaks are postulated at terminal ends, even if a longitudinal pipe weld is present at that point.

TABLE 3.6.1-3 (SHEET 3 OF 3)

Branch Technical Position MEB.3-1	VEGP Design
B.3.b.(3)	Conforms. See paragraph 3.6.2.1.3.1.
B.3.b.(4)	See paragraphs 3.6.2.2.1, 3.6.1.1.J, and 3.6.1.1.K.
B.3.b.(5)	Conforms. See paragraphs 3.6.1.1.J and 3.6.2.1.3.1.
B.3.c.(1)	Conforms. See paragraph 3.6.2.1.2.4.
B.3.c.(2)	Conforms. See paragraph 3.6.2.1.3.2.
B.3.c.(3)	Conforms. See paragraph 3.6.2.1.2.4.

a. This table summarizes conformance as to ASB 3-1 and MEB 3-1 which also covers the implementation of WCAP-8082-P-A and WCAP-8172-A.

TABLE 3.6.2-1

HIGH-ENERGY PIPE BREAK STRESS ANALYSIS RESULTS

For the high energy pipe break stress analysis results refer to the individual stress isometric drawings submitted in reference 13.

TABLE 3.6.2-2

HAZARD ANALYSIS RESULTS

Deleted. Refer to table 3F-1 and reference 13.

TABLE 3.6.2-3

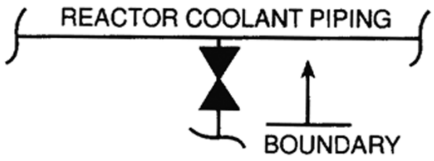
POSTULATED BREAK LOCATIONS FOR THE LOCA ANALYSIS
OF THE PRIMARY COOLANT LOOP(a)

Location of Postulated Rupture	Type	Break Opening Area	Area[(b)]
1. Residual heat removal (RHR) line/primary coolant loop	Guillotine (viewed from the RHR line)	Cross-sectional flow area of the RHR line	86.55 $in.^2$
2. Accumulator (ACC) line/primary coolant loop connection	Guillotine (viewed from the ACC line)	Cross-sectional flow area of the ACC line	60.10 $in.^2$

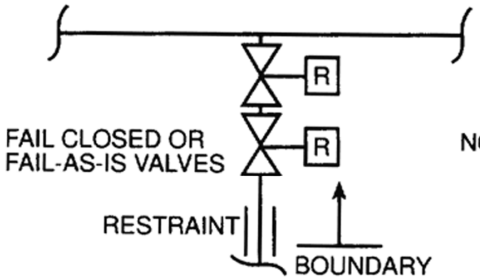
a. Refer to drawing AX6DD309 for location of postulated breaks in reactor coolant lines. This table applies to Unit 1 only. No breaks on Unit 2.

b. Less break opening area is used where justified by analysis, equipment, or consideration of physical restraints such as concrete walls or structural steel.

CASE I OUTGOING LINES WITH NORMALLY CLOSED VALVE

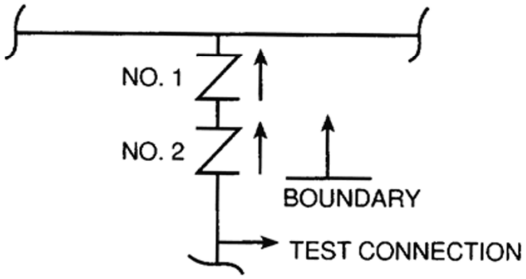


CASE II OUTGOING LINES WITH NORMALLY OPEN VALVES

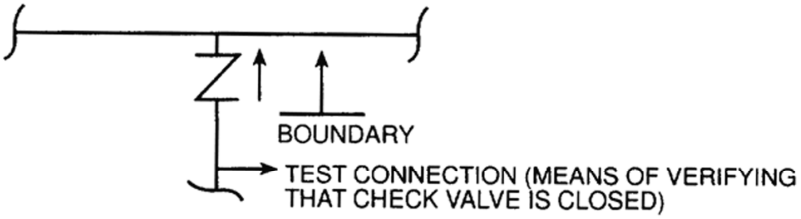


NOTE: THE REACTOR COOLANT PUMP NO. 1 SEAL IS ASSUMED TO BE EQUIVALENT TO FIRST VALVE

CASE III INCOMING LINES NORMALLY WITH FLOW



CASE IV INCOMING LINES NORMALLY WITHOUT FLOW



CASE V ALL INSTRUMENTATION TUBING AND INSTRUMENTS CONNECTED DIRECTLY TO THE REACTOR COOLANT SYSTEM IS CONSIDERED AS A BOUNDARY. HOWEVER, A BREAK WITHIN THIS BOUNDARY RESULTS IN A RELATIVELY SMALL FLOW WHICH CAN NORMALLY BE MADE UP WITH THE CHARGING SYSTEM.

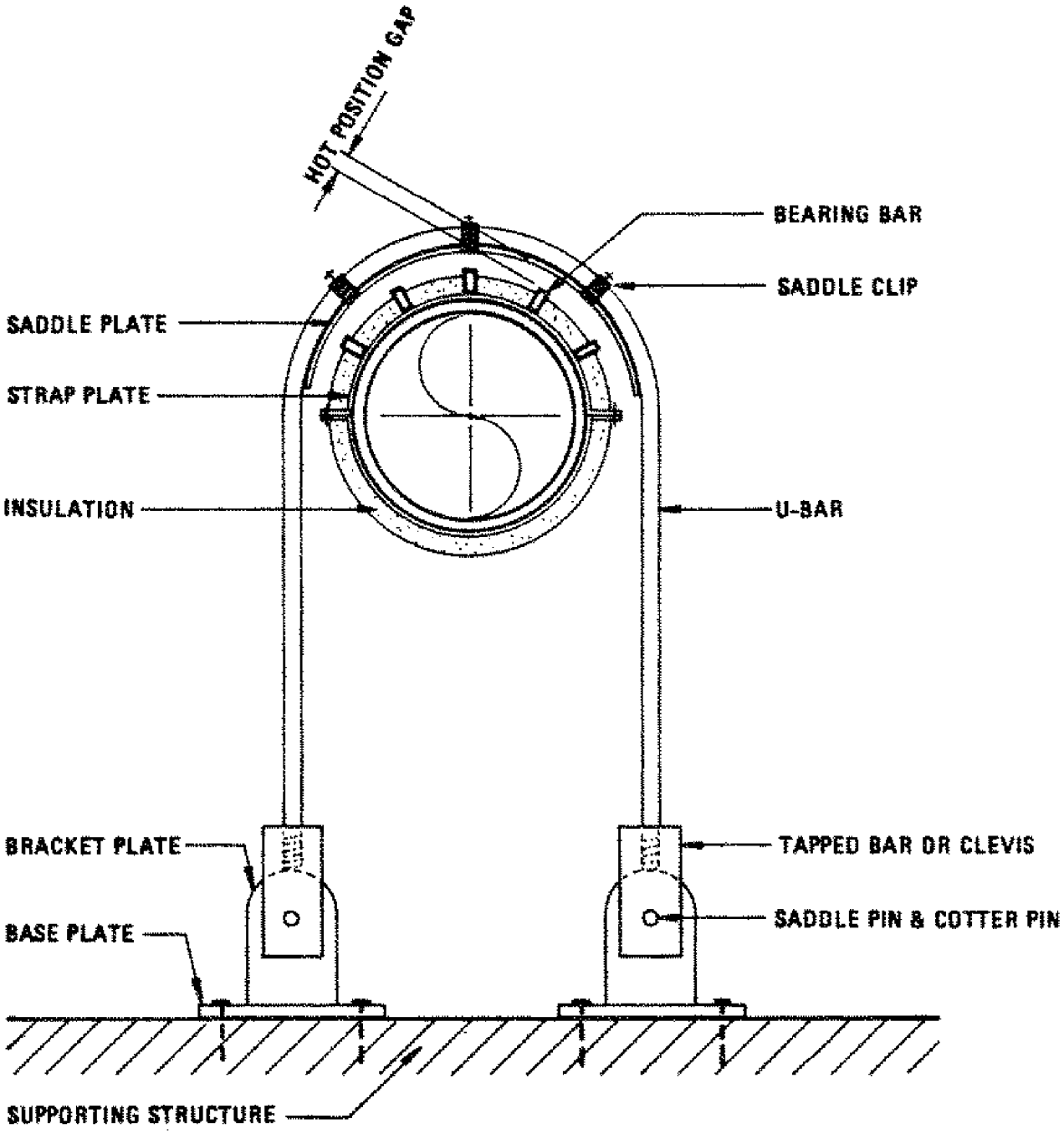
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOSS OF REACTOR COOLANT ACCIDENT
BOUNDARY LIMITS

FIGURE 3.6.2–1



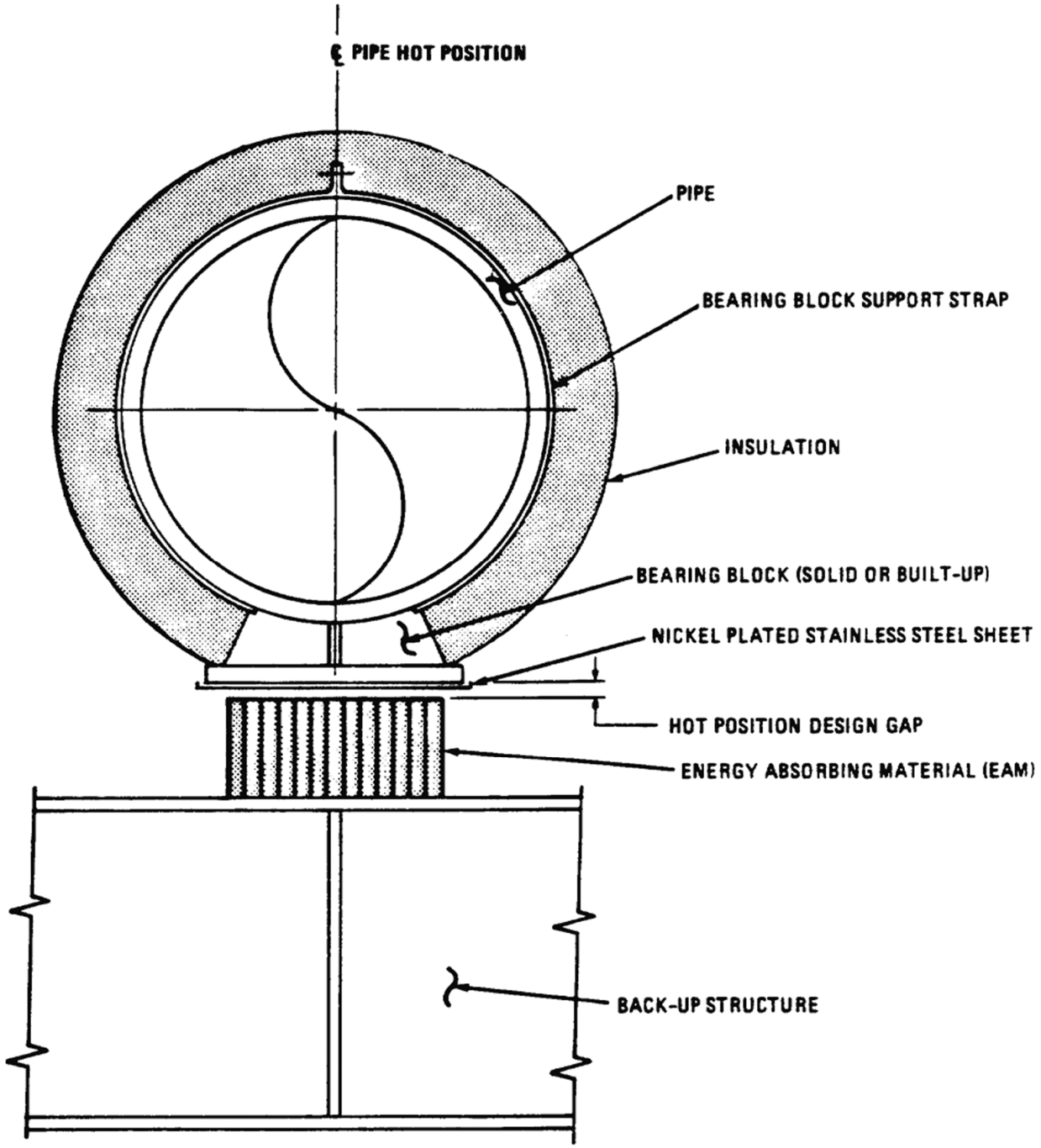
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TYPICAL U–BAR RESTRAINT

FIGURE 3.6.2–2



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TYPICAL EAM RESTRAINT

FIGURE 3.6.2–3

3.7.B SEISMIC DESIGN

All structures, systems, equipment, and components related to plant safety systems are required to have the ability to withstand potential earthquakes. Each structure, system, equipment, and component is placed in the applicable seismic category, depending on its function. A two-level system is used for the seismic classification of structures, systems, and equipment of the facility. A definition of the seismic classifications and a listing of structures, systems, and equipment are included in table 3.2.2-1.

Seismic loadings are characterized by the safe shutdown earthquake (SSE) and the operating basis earthquake (OBE). The SSE is defined as the maximum vibratory ground motion at the plant site that can be reasonably predicted from geologic and seismic evidence. The OBE is that earthquake which, considering the local geology and seismology, can be reasonably expected to occur during the plant life.

All Category 1 structures are designed for SSE and OBE conditions. Category 2 buildings are designed using the Uniform Building Code, 1976 edition. In addition, the radwaste transfer building, the radwaste transfer tunnel, the radwaste solidification building, the basemat of the alternate radwaste building, and the radwaste processing facility are designed to withstand the OBE event. Category 2 structures, systems, equipment, and components, whose failure could result in the loss of required safety functions of adjacent Category 1 structures, systems, equipment, or components shall be either separated by distance or barrier from the affected structure, system, equipment, or component or investigated for SSE loadings to ensure that their failure will not impair the safety-related functions of the adjacent Category 1 structures, systems, equipment, or components.

For Westinghouse-supplied items, refer to section 3.7.N.

3.7.B.1 SEISMIC INPUT

The seismic criteria for VEGP are developed by the Geology Department of the Hydro and Community Facilities Division of the Bechtel Corporation in San Francisco.

The plant site geologic and seismologic investigations are covered in section 2.5. Based on this data, the peak ground accelerations for SSE and OBE are established as 0.20 g and 0.12 g, respectively, as discussed in subsection 2.5.2.

3.7.B.1.1 Design Response Spectra

The VEGP site design response spectra are provided in figures 3.7.B.1-1 and 3.7.B.1-2 for the horizontal and vertical components of the SSE and in figures 3.7.B.1-3 and 3.7.B.1-4 for the horizontal and vertical components of the OBE. The design response spectra are in conformance with Regulatory Guide 1.60, Design Response Spectra for Seismic Design of Nuclear Power Plants.

3.7.B.1.2 Design Time-History

Synthetic earthquake acceleration time-histories are used as the basic input in the dynamic analysis of Category 1 structures under SSE and OBE conditions. The basis for the generation of the synthetic time-histories is discussed in section 2.5 of BC-TOP-4A.[1] Figures 3.7.B.1-5

and 3.7.B.1-6 show the synthetic acceleration time-history motions in the horizontal and vertical directions. Comparison between the free-field time-history response spectra, the design response spectra for both horizontal and vertical motions, and the frequencies at which the spectra values were calculated is provided in section 2.5 of BC-TOP-4A.[1]

In order to satisfy the input requirements of the FLUSH computer program that is used for the soil-structure interaction analyses of the deeply embedded structures, the following operations are performed on the original 4800-point time-histories digitized at 0.005 s to yield 2048-point time-histories digitized at 0.01 s. The time interval of the original 24 s time-histories is increased from 0.005 s to 0.01 s through the use of the computer program SHAKE, and the 20.48-s synthetic time-history motions are obtained by adopting the first 18 s of the time-histories followed by 2.48 s of a quiet zone. The differences between the response spectra derived from these motions and the response spectra obtained from the original time-histories are insignificant.

The synthetic time-history motions are scaled to 0.20 g and 0.12 g to obtain, respectively, the SSE design time-history and the OBE design time-history. The comparison of the response spectra obtained in the free field at the foundation level with the design response spectra is discussed in paragraph 3.7.B.2.4.1.

3.7.B.1.3 Critical Damping Values

Energy dissipation in a structural system is represented by equivalent viscous dampers. Evaluation of the damping coefficients is based on material, loading conditions, and type of connections used in the structural system. The damping values used in the dynamic analysis are those in table 3.7.B.1-1. The values are the same as those provided in Regulatory Guide 1.61, Damping Values for Seismic Design of Nuclear Power Plants, with the exception of damping values for cable trays and supports. The damping values for cable trays and supports are values based on test results[2] and were approved by the Nuclear Regulatory Commission in reference 3. The reports requested in reference 3 were provided to the commission in reference 4.[2] Damping values for heating, ventilation, and air-conditioning ducts and supports are those indicated in table 3.7.B.1-1 for welded or bolted steel structures, as applicable.

For Seismic analysis of ASME Boiler and Pressure Vessel Code, Section III, Division 1, Code Class 1, 2, and 3 piping systems, ASME Code Case N-411 damping values presented in figure 3.7.B.1-11 may be used in lieu of the values given in table 3.7.B.1-1, provided the following commitments are satisfied:

- Increased pipe deflections due to greater piping flexibility shall not violate project separation criteria.
- Criteria outlined in Regulatory Guide 1.61 shall not mix with the criteria of Code Case N-411 for a given piping analysis.
- As part of the integrated piping analysis/as-built reconciliation program, increased piping displacements and clearances shall be reviewed for acceptance.
- The N-411 values for seismic analysis apply to the primary loop piping systems and to other piping systems. These damping values, illustrated in figure 3.7.B.1-11, may be utilized only for piping systems analyzed by the response spectra method. Damping values, illustrated in table 3.7.B.1-1 shall be retained for piping analysis which utilizes the time-history integration method.

- With the exception of those stress calculations described in reference 9, Code Case N-411 damping values are not used in conjunction with multiple response spectrum methodology piping analysis.

The damping values presented in figure 3.7.B.1-11 were approved by the Nuclear Regulatory Commission in reference 8.

Consistent with Regulatory Guide 1.61, damping values higher than those listed in table 3.7.B.1-1 may be used if justified by test results.

For soil damping, strain-dependent damping values are used as given in figures 3.7.B.1-8, 3.7.B.1-9, and 3.7.B.1-10.

The damping values for compacted sand backfill shown in figure 3.7.B.1-8 are average values based on test results.(5) The clay marl bearing stratum is highly overconsolidated and has undrained shear strengths in excess of 10 k/ft^2. Because of this, the damping in this stratum would be somewhat lower than those for soft clays. The damping values shown in figure 3.7.B.1-9 are considered appropriate for the hard clay marl bearing stratum encountered at the site. The damping values for the lower sand stratum shown in figure 3.7.B.1-10 are based on the Seed and Idriss curve.(6)

3.7.B.1.4 Supporting Media for Seismic Category 1 Structures

The depth of bedrock below the plant site is approximately 950 ft, as defined in paragraph 2.5.1.2.1.3. The nominal finished grade level is el 220 ft. The explored depth at the site indicates an overburden which may be divided into the three distinct soil strata listed below:

- Upper sand stratum: sands and clayey sands, varying from loose to dense, to a depth of 75 to 90 ft.
- Clay bearing stratum: very hard, sandy, calcareous clay marl about 65 ft thick.
- Lower sand stratum: clean to silty, medium- to fine-grained dense sands below the marl to undetermined depth.

In the power block area containing the Category 1 structures, the upper sand stratum material and approximately the top 5 ft of clay marl bearing stratum are removed approximately to el 130 ft. Select engineered compacted sand backfill is placed on the top of the clay marl bearing stratum up to the design elevations of Category 1 structure foundations with the exception of the auxiliary building and the nuclear service cooling water (NSCW) towers, which are founded on the clay bearing stratum. Select compacted sand backfill is also placed on the sides of Category 1 structures up to grade elevation.

Values of low strain shear modulus (at shear strain less than or equal to 10^{-4} percent) of the compacted backfill are computed using the expression $G = (1000\, K_2\, (\sigma'_m)^{1/2}\ lb/ft^2\ )$, where G is the shear modulus in lb/ft^2; σ'_m is the mean principal effective stress in lb/ft^2; and K_2, the parameter reflecting primarily the effect of void ratio or relative density and the strain amplitude of the motions, is taken as 80.(5) The low strain shear moduli of the clay bearing stratum and the lower sand stratum are computed using the average measured shear wave velocities of 1700 ft/s and 1800 ft/s, respectively.(7) The unit weights of compacted backfill in moist and saturated conditions are 123 lb/ft^3 and 133 lb/ft^3, respectively. The saturated unit weight of clay bearing stratum and lower sand stratum is 115 lb/ft^3.

Table 3.7.B.1-2 includes foundation embedment depth, width of the structural foundation, and total structure height for Category 1 structures. Subsection 2.5.4 describes the soil properties, shear wave velocity, shear modulus, and density.

3.7.B.1.5 Standard Review Plan Evaluation

For deeply embedded Seismic Category 1 structures, the design ground motion (control motion) is applied at the finished grade level in the free field, instead of applied at the foundation levels of Category 1 structures in the free field.

Refer to paragraph 3.7.B.2.1 for an evaluation of the Standard Review Plan differences.

3.7.B.1.6 References

1. "Topical Report Seismic Analysis of Structures and Equipment for Nuclear Power Plants, Revision 3," BC-TOP-4A, November 1974.
2. "Cable Tray and Conduit Raceway Seismic Test Program, Release 4," Report 1053-21.1-4, ANCO Engineers, Inc., December 15, 1978.
3. NRC letter, Docket Nos. 50-424 and 50-425, dated February 12, 1982.
4. Georgia Power Company letter, Docket Nos. 50-424 and 50-425, to NRC, dated March 5, 1982.
5. Alvin W. Vogtle Nuclear Plant, Report on Dynamic Properties for Compacted Backfill, Bechtel Incorporated, Los Angeles, California, February 1978.
6. Seed, H.B., and Idriss, I.M., "Soil Moduli and Damping Factors for Dynamic Response Analysis," Earthquake Engineering Research Center, EERC 70-10, University of California, Berkeley, California, December 1970.
7. Alvin W. Vogtle Nuclear Plant, Report on Foundation Investigations, Volume 1, Bechtel Incorporated, San Francisco, California, July 1974.
8. NRC letter, Docket Nos. 50-424 and 50-425, to Georgia Power Company, dated March 18, 1985.
9. Georgia Power Company letter (GN-1257) Docket No. 50-424 and 50-425, to NRC, Dated December 22, 1986.

3.7.N SEISMIC DESIGN

This section describes the seismic design methods for nuclear steam supply system (NSSS) equipment.

In addition to the steady-state loads imposed on the system under normal operating conditions, the design of equipment and equipment supports requires consideration of abnormal loading conditions such as earthquakes. Seismic loadings are characterized by the safe shutdown earthquake (SSE) and operating basis earthquake (OBE). The SSE is defined as the maximum vibratory ground motion at the plant site that can reasonably be predicted from geologic and seismic evidence. The OBE is that earthquake which, considering the local geology and seismology, can be reasonably expected to occur during the plant life.

For the OBE loading condition, the NSSS is designed to be capable of continued safe operation. The design for the SSE is intended to ensure the following:

A. That the integrity of the reactor coolant pressure boundary is not compromised.

B. That the capability to shut down the reactor and maintain it in a safe condition is not compromised.

C. That the capability to prevent or mitigate the consequences of accidents that could result in potential offsite exposures comparable to the guideline exposures of 10 CFR 100 is not compromised.

The seismic qualification of safety-related instrumentation and electrical equipment is discussed in sections 3.10.N and 3.10.B. The safety class definitions and classification lists are given in section 3.2.

3.7.N.1 SEISMIC INPUT

3.7.N.1.1 Design Response Spectra

Refer to paragraph 3.7.B.1.1.

3.7.N.1.2 Design Time-History

The seismic analysis of the reactor coolant system utilizes nonlinear, three-dimensional, time-history dynamic analysis methods and a coupled building internals structure/reactor coolant loop model. The coupled model is subjected to three components of earthquake simultaneously at the basemat. The six time-history components, the north-south and east-west horizontal directions and the vertical direction, are statistically independent and are applied simultaneously for 10 s of OBE and SSE, which represents the most severe portion of the earthquake acceleration.

In order to perform the coupled building/loop analysis, it is necessary to construct a synthesized time-history motion. These time-histories are developed to conform to the three translational and three rotational response spectra from the soil-structure interaction analyses of the containment building described in section 3.7.B. The generation of these motions is accomplished by modifying existing earthquake records using spectral raising and suppressing techniques. With data from a real earthquake as input, spectral raising is accomplished by adding to the original time-history a function at the frequency of interest with a phase angle such that the response spectra value will be increased a desired amount. The time when the maximum vibration occurred will be the same. In this way, the characteristics of the required time-history will only be slightly altered. Spectral suppression is carried out by passing the time-history through a linearly damped oscillator connected in series to a second damper. This damping arrangement will reduce the response spectral value, locally, at the natural frequency of the oscillator to the desired amount. The repetitious application of the raising and suppressing techniques is used to arrive at a time-history motion whose response spectrum is sufficiently close to the design spectrum.

The statistical independence of the six synthetic acceleration time-histories is established by comparing statistical properties of the synthetic time-histories with properties derived from recorded earthquake accelerograms. In particular, the values of the normalized correlation coefficient at zero time delay and the average value of the coherence function over the seismic

frequency range are calculated for the synthetic and real time-histories and shown to be comparable.

3.7.N.1.3 Critical Damping Values

The damping values given in table 3.7.N.1-1 are used in the systems analysis of Westinghouse equipment. These are consistent with the damping values recommended in Regulatory Guide 1.61 except in the case of the primary coolant loop system components and large piping (excluding reactor pressure vessel internals), for which the damping values of 2 and 4 percent are used as established in testing programs reported in reference 1.

As an alternative to Regulatory Guide 1.61, the damping values from ASME Code Case N-411 are used for piping system analysis with response spectra analysis techniques per reference 5. The Code Case N-411 frequency dependent damping values are illustrated in figure 3.7.N.1-1. No mixture of Regulatory Guide 1.61 criteria with the N-411 criteria is allowed in a given piping analysis. As part of the integrated piping analysis/as-built reconciliation program, GPC was assured that piping displacements and clearances are acceptable when N-411 criteria is applied. The damping values for control rod drive mechanisms (CRDMs) and the fuel assemblies of the NSSS, when used in seismic analysis, are in conformance with the values for welded and/or bolted steel structures (as appropriate) listed in Regulatory Guide 1.61. Consistent with Regulatory Guide 1.61, damping values higher than those listed in table 3.7.N.1-1 may be used if justified by test results.

Tests on fuel assembly bundles have justified conservative component damping values of 7 percent for OBE and 10 percent for SSE to be used in the fuel assembly component qualification. Documentation of the fuel assembly tests is provided in reference 2.

The damping values used in component analysis of CRDMs and their seismic supports were developed through a testing program performed by Westinghouse. The program consisted of transient vibration tests in which the CRDM was deflected a specified initial amount and suddenly released. A logarithmic decrement analysis of the decaying transient provides the effective damping of the assembly. Documentation of the CRDM tests is provided in references 3 and 4.

The test results indicated that the damping would be greater than 8 percent for both the OBE and the SSE based on a comparison between typical deflections during these seismic events to the initial deflections of the mechanisms in the test. Component damping values of 5 percent are, therefore, conservative for both OBE and SSE.

3.7.N.1.4 Supporting Media for Seismic Category 1 Structures

Refer to paragraph 3.7.B.1.4.

3.7.N.1.5 References

1. "Damping Values of Nuclear Plant Components," WCAP-7921-AR, May 1974.
2. Gesinki, T. L., and Chiang, D., "Safety Analysis of the 17x17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident," WCAP-8236 (Proprietary), December 1973, and WCAP-8288 (Nonproprietary), January 1974.

3. Obermeyer, F. D., "Effective Structural Damping of the KEP L105 CRDM," WCAP-7427 (Proprietary), January 1970.

4. Obermeyer, F. D., WCAP-7427, Addendum 1 (Proprietary) December 1970.

5. "Piping Design Criteria for Vogtle, Units 1 and 2", NRC letter, T. M. Novak to D. O. Foster, March 18, 1985.

3.7.B.2 SEISMIC SYSTEM ANALYSIS

Category 1 structures, systems, and components are classified in accordance with Regulatory Guide 1.29. Seismic systems are defined as Category 1 structures considered in conjunction with foundation media in forming a soil-structure interaction model. All Category 1 structures not designated as seismic systems and all Category 1 systems, such as heating, ventilation, and air-conditioning systems, electrical cable tray, and piping are considered as seismic subsystems, and their analyses are described in subsection 3.7.B.3.

3.7.B.2.1 Seismic Analysis Methods

The acceptance criteria of Sections 3.7.1 and 3.7.2 of the Standard Review Plan (SRP) (7/81) require that the control motion be applied at the foundation levels of Category 1 structures in the free field.

In addition, the acceptance criteria of Section 3.7.2 require that modeling methods for implementing the soil structure interaction analysis include both the half space and finite boundaries approaches and that Category 1 structures, systems, and components be designed to accommodate responses obtained by one of the following:

- Envelope of results of the two methods.
- Results of one method with conservative design considerations of the effects from the use of the other method.
- Combination of the above with the provision of adequate conservatism in the design.

In the VEGP design, for shallowly embedded Category 1 structures, the control motion is applied at foundation levels of the structures in the free field, and soil-structure interaction analyses are performed using the impedance (half space) method. For deeply embedded Category 1 structures, the control motion is applied at the finished grade level in the free field, and soil-structure interaction analyses are performed using the finite element method.

VEGP design methods are based on the SRP (11/24/75) in effect in the years 1977 and 1978, during which period the VEGP design methods evolved from the discussions held during meetings with the Nuclear Regulatory Commission (NRC). The concerns expressed by the NRC staff in these meetings were addressed in Preliminary Safety Analysis Report (PSAR) Supplements 3 and 4 and in the

Georgia Power Company (GPC) letter to the NRC dated February 20, 1978, in which VEGP committed to multiply the envelope in-structure response spectra for the deeply embedded Category 1 structures by a scaling factor of 1.5, consistent with the SRP acceptance criteria stated above. (The scaling factor value of 1.5 was later incorporated through PSAR Supplement No. 5 of November 17, 1978.)

The NRC, in its letter to GPC dated March 27, 1978, accepted the seismic design methods proposed in PSAR Supplements 3 and 4 with the additional information provided in the GPC

letter of February 20, 1978, on the scaling factor, subject to the completion of a confirmatory study and a sensitivity study.

The confirmatory study addressed the NRC staff's concerns on comparing the results of the two methods of soil-structure interaction analyses. The sensitivity study provided the justification for applying the deconvolved control motions at the foundation levels of deeply embedded Category 1 structures.

In the confirmatory study, the response spectra calculated from the finite element method of soil-structure interaction using the VEGP design procedure were compared with those obtained using the impedance (half space) method to confirm that the VEGP seismic design methods are founded on conservative design bases. The objective of the sensitivity study was to demonstrate that the deconvolution procedure in conjunction with the specification of the control motion (based on Regulatory Guide 1.60 response spectra) at the grade level in the free field, is applicable for the VEGP site.

The NRC also requested that the seismic analysis include consideration of torsional moment no less than that required by the Uniform Building Code (to account for the seismic wave propagation effects), in addition to the effects resulting from the eccentricity between the center of mass and center of rigidity at each level.

The reports on the confirmatory study and the sensitivity study together with the description of the methodology to account for torsion caused by the seismic wave propagation effects were submitted to the NRC in the GPC letter dated November 13, 1978. These reports are reproduced in section 3D.2.

For the confirmatory study, the containment building and the control building were analyzed to compute the in-structure response spectra using the VEGP design procedure for the finite element method of soil-structure interaction analysis of deeply embedded structures, applying the control motion at the finished grade level in the free field. This included the multiplication of the envelope in-structure response spectra by a scaling factor of 1.5. The calculated response spectra were compared with those obtained from the impedance (half space) method, wherein the control motion was applied at the foundation levels of the structures in the free field. The comparison showed that, in general, the response spectra from the VEGP design procedure and those from the impedance method exhibited similar characteristics in terms of accelerations and frequencies, despite the fact that there are differences in the methods of modeling and analysis. Additionally, a comparison of the VEGP design in-structure response spectra with the impedance method response spectra presented provided in the confirmatory study is provided in section 3D.3.

For the sensitivity study, a series of ground response studies were performed to determine the variation of ground motion with depth at the VEGP site. Recorded seismic rock motions were used to establish bedrock motions, which were then propagated upward through the site soils from the underlying bedrock. Recorded seismic motions on soil deposits with generally similar characteristics to those at the VEGP site were deconvolved through the soil profile. The computed responses from these two procedures were compared with each other and with the results of the seismic motions used in the VEGP design, namely the Regulatory Guide 1.60 design spectra applied at the grade level in free field and the motions at the foundation levels obtained by the deconvolution procedure.

The sensitivity study in its draft form was presented to the NRC staff on July 12, 1978, and was accepted by the NRC staff with comments. The staff's comments were incorporated into the report. The good agreement between the response spectra obtained from the analyses of the actual recorded earthquake motions and the response spectra obtained by the deconvolution of

the Regulatory Guide 1.60 response spectra supports the use of the later spectra and motions obtained by its deconvolution for the VEGP design.

Therefore, the VEGP seismic analyses are based on the design methods presented in PSAR Supplements 3, 4, and 5 and are supported by the reports submitted to the NRC in the GPC letter dated November 13, 1978. The VEGP seismic design methods are discussed in detail in the following sections.

The seismic systems are analyzed for operating basis earthquake (OBE) and safe shutdown earthquake (SSE) conditions. Depending on whether the structure has shallow embedment or deep embedment, different methods are used for the seismic analysis. As shown in table 3.7.B.2-1, structures are classified as deeply embedded, shallowly embedded, or buried.

For shallowly embedded structures, foundation torsion, rocking, and translation effects are taken into consideration in the computation of equivalent soil spring constants and damping coefficients, as described in BC-TOP-4A.(1) A modal time-history analysis method is used for the analysis of these structures. The number of masses used, the number of modes considered, and the combination of modal responses are in accordance with BC-TOP-4A. For deeply embedded structures, foundation rocking and translation effects are accounted for through modeling the soil around the structures as finite elements. A complex response time-history analysis method is used for the analysis of deeply embedded structures. The number of masses used to represent the structure are as described in BC-TOP-4A. The buried structures essentially move with the ground, and the response of the structure is the same as the ground response. The hydrodynamic effects, if any, are modeled based on TID-7024.(2) Typical models of a shallowly embedded structure and deeply embedded structures are shown in figures 3.7.B.2-1 through 3.7.B.2-4.

3.7.B.2.2 Natural Frequencies and Response Loads

The seismic system analyses of the containment building and other major Category 1 structures are performed using a complex response time-history method. The final response spectra generated from this method is indicative of the frequency content of the soil-structure system. Natural frequencies, mode shapes, or modal responses are not obtained in this method as in the modal response spectra analysis method. Accelerations at the selected levels of major Seismic Category 1 structures are presented in table 3.7.B.2-2. These are indicative of response loads for major Seismic Category 1 structures. Selected response spectra determined by seismic analysis for major Seismic Category 1 structures are given in section 3D.1.

3.7.B.2.3 Procedure Used for Modeling

Shallowly embedded structures and deeply embedded structures, as listed in table 3.7.B.2-1, have different methods of modeling for the dynamic seismic analysis. For structures that have shallow embedment, the lumped parameter method is used to represent the soil-structure interaction. For structures that are deeply embedded, the finite element method is used to represent the soil-structure interaction. In both the methods, the buildings are modeled using beam elements and lumped masses.

3.7.B.2.4 Soil-Structure Interaction

The extent of structural embedment is listed in table 3.7.B.1-2. The depth of soil over rock and the soil stratum layering are described in paragraph 3.7.B.1.4.

3.7.B.2.4.1 Deeply Embedded Structures

The finite element method is used for structures having deep embedment to account for embedment effects and the effects of structure-to-structure interaction. The analytical model is provided with transmitting boundaries both on the left and right sides. The model consists of two types of elements: displacement-compatible isoparametric quadrilateral elements (solid elements) and linear bending elements (beam elements). Usage of transmitting boundaries, elements, and analytical techniques are described by Lysmer, et al.[5] The computer program FLUSH of the same reference is used to perform the analysis.

Soil properties such as shear moduli, Poisson's ratios, and densities for the various soil strata are established from the soils investigation and additional soil testing to establish the dynamic properties of compacted backfill as described in paragraph 3.7.B.1.4. The strain dependency of shear moduli and damping ratios for compacted sand backfill, clay marl bearing stratum, and the lower sand stratum are based on the standard curves proposed by Seed and Idriss[6] with appropriate modifications to account for the in situ soil conditions and backfill characteristics.

In the analysis, the strain-dependent shear moduli as shown in figures 3.7.B.2-5 through 3.7.B.2-7 are used.

In figure 3.7.B.2-5, the curve showing the variation of shear modulus with shear strain is shown for the compacted sand backfill. The curve is based on the average values obtained from test results.[7] The variation of shear modulus with shear strain for the clay marl bearing stratum is shown in figure 3.7.B.2-6. Because the marl is essentially a hard clay, the shear modulus will decrease with increasing shear strain but at a lesser rate than that applicable for soft clays. The shear modulus variation shown is therefore appropriate for the site condition and will be used in the analysis. The variation of shear modulus with shear strain for the lower sand stratum is shown in figure 3.7.B.2-7. This is based on the standard curve proposed by Seed and Idriss.

To account for the variation in soil properties, shear moduli with upper-bound values equal to 1.5 times the mean values and lower-bound values equal to the mean values divided by 1.5 are used in the analysis. The mean values of low strain shear moduli are computed as described in paragraph 3.7.B.1.4.

As discussed in paragraph 3.7.B.1.3, the damping values for the compacted sand backfill, the clay marl bearing stratum, and the lower sand stratum shown in figures 3.7.B.1-8 through 3.7.B.1-10 are used in the analysis.

In general, the soil properties are nonlinear in character. An iterative process is used to obtain equivalent linear properties which are strain dependent. The methods generally used for such an analysis are included in the computer program FLUSH.[5]

In the analyses for the vertical component of the earthquake, the soil properties for the layers below the water table are based on the iterated strain-dependent soil properties or a compression wave velocity of 5000 ft/s, whichever is greater. This is consistent with the assumption that, in saturated soils, the compression wave would travel with the compression wave velocity of the soil medium or the compression wave velocity of water, whichever is greater. The compression wave velocity of water has been assumed to be 5000 ft/s.

The generation of design time-history motions is described in paragraph 3.7.B.1.2. This ground motion is defined for the free field and applied at the finished grade level (el 220 ft 0 in.) of the site.

The time-history at the base of the idealized soil profile is obtained through deconvolution analysis of the design time-history specified at finished grade level, using appropriate soil properties. The time-history thus obtained is applied at the base of the soil-structure interaction system with appropriate soil properties for soil-structure interaction analysis. The resulting time-history responses are used to generate the in-structure response spectra at selected floor elevations. The analysis is performed with consideration given to the variation of soil parameters as indicated above using appropriate cutoff frequencies such that the acceleration profile in the free field is realistic. The envelope in-structure response spectra are developed by enveloping the response spectra obtained by considering the variation of soil properties. The envelope in-structure response spectra curves are multiplied by the scaling factor of 1.5, the basis for which is described in the following paragraph.

Response spectra corresponding to the free field time-history motions calculated at the elevations of Category 1 structural foundations are generated. Considering the variation of soil properties, envelope response spectra for each Category 1 foundation level are developed. The comparison of the envelope response spectra thus obtained in the free field at the foundation levels of deeply embedded Category 1 structures with 60 percent of the design response spectra is provided in figures 3.7.B.2-8 through 3.7.B.2-27. A scaling factor of 1.5 is selected so that when the envelope response spectra curves are multiplied by the scaling factor, the 60-percent design spectra curves are essentially enveloped.

The dynamic analysis performed using the computer program FLUSH is two dimensional, and any three-dimensional analysis is an approximation. The procedure for computing the three-dimensional response of the structures using a two-dimensional soil model is described below. This procedure combines a two-dimensional finite element representation of soil with a three-dimensional representation of structures.

First, a three-dimensional lumped mass model of the structure is created and expressed in the form of stiffness and mass matrices. A two-dimensional model of the soil with the structure removed is prepared and all nodes in contact with the structure (henceforth called common nodes) identified (figure 3.7.B.2-28).

The structure nodes associated with the common nodes have degrees of freedom only in the plane of the soil model in order that the FLUSH program can be executed. The reduction of degrees of freedom of the structure common nodes is accomplished through a mathematical transformation. There is no requirement that the degrees of freedom for the remaining structure nodes be reduced.

After the common degrees of freedom have been made compatible both in the structure and in the soil, then the total soil-structure system is assembled in global matrices and the solution is accomplished by FLUSH, as in a standard finite-element problem.

The response of the structure nodes that are not associated with the common nodes is in three orthogonal directions due to excitation in any one direction, and hence the codirectional responses due to both of the horizontal earthquakes and vertical earthquake can be obtained directly. This approach inherently accounts for the torsional effects in the structure.

In the soil finite element model, the side transmitting boundaries are located three elements away from the structures. This is consistent with the FLUSH program recommendations. The bottom boundary for the FLUSH model is taken so that it is at least at a depth one-half the model dimension of the basemat below that basemat level.

A total of six mathematical models are employed in the analysis of deeply embedded structures as shown in figure 3.7.B.2-2. The first is an east-west model which includes the auxiliary building. The second is also an east-west model which consists of the containment Unit 2, the fuel handling building, and the containment Unit 1. The effect of the diesel generator buildings on the response of the containment is accounted for by modeling their inertial properties with structural layers in the soil finite-element model. The third is an east-west model which includes the control building. The fourth is a north-south model which includes the auxiliary building, containment Unit 1, the control building, and the turbine building. Since the turbine building is a non-Category 1 structure, it is only necessary to consider its effect on adjacent Category 1 structures. Therefore, it is sufficient to model it as a structural layer in the soil finite-element model with proper inertial properties, without modeling the superstructure. The fifth is also a north-south model which includes the auxiliary building, the fuel handling building, the control building, and the turbine building. The sixth is a model which includes a nuclear service cooling tower. Considering the plant layout, it is assumed that there is no significant interaction between each of the nuclear service cooling water (NSCW) towers and the rest of the structures. Two typical FLUSH models are shown in figures 3.7.B.2-3 and 3.7.B.2-4.

The NSCW valve house is a single-story structure with insignificant equipment loads. The structural response of the valve house is dictated by the driving influence of the adjacent massive NSCW tower. Because of the relatively small size of the valve house and its close proximity to the tower, the response spectra of the tower at grade level is used as the response spectra of the valve house.

3.7.B.2.4.2 Shallowly Embedded Structures

For structures founded on the ground surface or having shallow embedment, the lumped parameter approach is used to represent the soil-structure interaction. The details of this method are described in BC-TOP-4A.[1] Strain-dependent soil properties are used in the computation of the impedances. The procedures by which soil layering effects are considered are discussed in appendix 3E. A typical model is shown in figure 3.7.B.2-1.

3.7.B.2.4.3 Buried Structures

Buried structures are surrounded by soil around their perimeters and essentially move with the ground. The response of the structure is the same as that of the ground. As an added conservatism, the seismic response spectra for the Category 1 tunnels are developed from the free field response spectra by multiplying them by a factor of 1.25.

3.7.B.2.4.4 Floor Flexibility

The impedance method (lumped parameter method) of soil-structure interaction analysis is employed to account for the effects of containment internal steel structure flexibility on the response spectra used for equipment qualification and to address, if significant, the coupling effects between the equipment and the steel structure. In addition, the impedance method is used to address the vertical floor flexibility effects in the other structures.

3.7.B.2.5 Development of Floor Response Spectra

Floor response spectra for Category 1 structures are developed using the time-history analysis. Both the horizontal and vertical floor response spectra are computed from the time-history motions at the various floors or other required locations. These motions are obtained from the time-history analysis of the structures due to each of the three orthogonal earthquake components under OBE and SSE events. The floor response spectra are computed at the frequencies given in table 3.7.B.2-3. These frequencies were selected using the suggested frequency intervals in Regulatory Guide 1.122. Since the natural frequencies are not computed in the FLUSH analyses and since the intervals between the selected frequencies are small, the natural frequencies of the system are not specifically included in this table of frequencies.

3.7.B.2.6 Three Components of Earthquake Motion

The three component earthquake effects are combined using the square root of the sum of the squares of the applicable maximum codirectional responses as described in section 4.3 of BC-TOP-4A[(1)] and is in conformance with Regulatory Guide 1.92.

3.7.B.2.7 Combination of Modal Responses

The combination of modal responses is performed as described in section 4.2 of BC-TOP-4A[(1)] and is in conformance with Regulatory Guide 1.92.

3.7.B.2.8 Interaction of Non-Category 1 Structures with Seismic Category 1 Structures

The equipment building, which forms part of the control building and fuel handling building, is designed to withstand the seismic loadings to the same criteria specified for Category 1 structures.

Other than Category 2 tunnels, the turbine building and radwaste transfer building are the only non-Category 1 structures adjacent to Category 1 structures. These are analyzed to demonstrate that under the SSE loadings they will not collapse on any Category 1 structure. In the FLUSH models, the turbine building basemat is modeled as a structural layer with proper inertial properties, and the rest of the turbine structure is modeled by a lumped mass. Shallowly embedded structures are not influenced by adjacent non-Category 1 structures.

3.7.B.2.9 Effects of Parameter Variations on Floor Response Spectra

The in-structure response spectra computed from the time-history acceleration response generally reflect two parameters:

- The amplification of the free field input produced by the soil-structure system.
- The frequency content associated with these amplification regions.

The effects of parameter variations on floor response spectra are accounted for by broadening the peaks associated with each structural frequency by +15 percent.

3.7.B.2.10 Use of Constant Vertical Static Factors

No constant vertical static factors are used for Category 1 structures. The same method of analysis is used for both vertical and horizontal directions.

3.7.B.2.11 Method Used To Account for Torsional Effects

The NRC requested[3] that all safety-related structures, systems, and components be designed to resist a static seismic torsional moment not less than that required by the Uniform Building Code. To accommodate this request, the following methodology is used. The methodology used to account for torsion caused by the seismic wave propagation effects is also described in the GPC letter[4] to the NRC; it is used both for deeply embedded and shallowly embedded structures.

3.7.B.2.11.1 Category 1 Structures

The seismic analyses of the structures are performed on the three-dimensional structure models that account for the eccentricities between the centers of mass and the centers of rigidity of the structures. The accelerations obtained from these analyses at all levels are first calculated. Then, in the design the actual eccentricity is increased by 5 percent of the maximum plan dimension at that level, and the design static seismic torsional moment is computed as the product of the augmented eccentricity and the story shear. This applies to the two orthogonal horizontal directions.

3.7.B.2.11.2 Category 1 Equipment, Systems, and Components

The intent of the additional torsional requirement is to account for the torsional motion imparted to the structure due to the effects of seismic wave propagation. Thus, this would affect only the horizontal in-structure response spectra used for equipment qualification. The procedure used to obtain the effect of this torsional ground motion is described below.

A three-dimensional lumped parameter model of the structure with soil springs is utilized to compute the torsional spectra. The structure model accounts for the eccentricities between the centers of mass and the centers of rigidity of the structure. The translational as well as the rotational stiffness and inertial characteristics are modeled. The foundation impedances consist of three translational (two horizontal and one vertical) and three rotational (two rocking and one torsional) springs and are based on the mean soil properties.

The model is analyzed for the design horizontal ground motion time-history conforming to the Regulatory Guide 1.60 horizontal response spectra applied in the free field at the foundation level of the structure. The base shear computed from this analysis, multiplied by 5 percent of the maximum plan dimension at the foundation level, yields the incremental static torsional moment (T_s) at that level.

Then a torsional ground motion time-history conforming to the Regulatory Guide 1.60 horizontal response spectra is applied again in the free field at the foundation level of the structure. The maximum dynamic torsional moment (T_d) at the base of the structure is computed from this dynamic analysis.

The magnitude of the torsional ground motion is adjusted so that T_d at the base of the structure resulting from the torsional ground motion analysis is equal to the T_s resulting from the 5-

percent eccentricity. The resulting time-history response from the torsional degree of freedom of the base node would then represent the torsional response of the basemat. Multiplying this by the distance along the north-south/east-west direction of the extreme point in the building to the lumped mass node would give the maximum possible additional east-west/north-south horizontal time-history response of the floor. From this, the additional horizontal in-structure response spectra can be computed.

The torsional responses of the nodes at different levels of the building from the torsional ground motion analysis are used with the respective extreme point distances along the north-south/east-west direction to compute the additional horizontal in-structure response spectra at these levels.

The amplification of the torsional response of the structure as a function of height tends to be smaller than the amplification of the horizontal response of the structure. Therefore, as an added conservatism, the torsional input ground motion is increased so that the ratio between the maximum torsional acceleration at a given node (caused by the torsional ground motion) to the maximum horizontal acceleration at the node (caused by the horizontal ground motion) is maintained the same as at the foundation level of the structure.

The computed additional horizontal in-structure response spectra to account for the torsional ground motion effects are added absolutely to the horizontal in-structure response spectra obtained using the methods described in paragraphs 3.7.B.2.4, 3.7.B.2.5, and 3.7.B.2.6, before the broadening of the peaks and smoothing of the curves are done. The peaks of the response spectra resulting from the addition of these two spectra are then broadened and the curves smoothed to arrive at the final design in-structure response spectra for the horizontal direction.

In the computation of the additional horizontal in-structure response spectra to be used for an equipment mounted on a specific location, the actual distance of this location from the lumped mass node may be used instead of the extreme point distance at that level.

The method applied to the NSCW tower, which contains a large water mass, is described below.

Once the magnitude of the torsional ground motion is established so that the Td at the base of the structure resulting from the torsional ground motion analysis is equal to the Ts resulting from the 5-percent eccentricity, the ratio of the horizontal acceleration at the extreme point in the basemat caused by the torsional ground motion to the maximum horizontal acceleration at the basemat caused by the horizontal ground motion is computed. The additional in-structure response spectra used to account for the torsional ground motion effects are computed by multiplying the horizontal in-structure response spectra developed using the methods described in paragraphs 3.7.B.2.4, 3.7.B.2.5, and 3.7.B.2.6 by this ratio.

3.7.B.2.12 Comparison of Responses

This section is not applicable to the VEGP, since only the time-history method is used in the seismic analysis.

3.7.B.2.13 Methods for Seismic Analysis of Dams

Since no dams are utilized directly or indirectly to provide water for the cooling system, this section is not applicable to this power plant.

3.7.B.2.14 Determination of Seismic Category 1 Structure Overturning Moments

The effects of overturning moments are evaluated by the methods described in section 4.4 of BC-TOP-4A.(1) This section includes a description of the methods used to compute foundation reactions and to account for vertical earthquake effects. The three components of the earthquake motion are taken into consideration in the overturning moments by use of the seismic accelerations.

3.7.B.2.15 Analysis Procedure for Damping

The analysis procedure used to account for damping in different elements of the model of a coupled system and the criteria used to account for composite damping are described in sections 3.2 and 3.3 of BC-TOP-4A.(1)

3.7.B.2.16 Standard Review Plan Evaluation

Differences are noted under Standard Review Plan 3.7.1. The Standard Review Plan states that modeling methods for implementing the soil-structure interaction analysis should include both the half-space and finite boundaries approaches. The VEGP soil-structure interaction analysis uses the finite element method for deeply embedded structures and the half-space method for shallowly embedded structures.

Refer to paragraph 3.7.B.2.1 for an evaluation of the Standard Review Plan differences.

3.7.B.2.17 References

1. "Topical Report Seismic Analysis of Structures and Equipment for Nuclear Power Plants," Bechtel Power Corporation, BC-TOP-4A, Revision 3, San Francisco, November 1974.
2. "Nuclear Reactors and Earthquakes," AEC Publication TID-7024, August 1963.
3. Nuclear Regulatory Commission Letter (Docket Nos. 50-424 and 50-425) Dated March 27, 1978, to Georgia Power Company.
4. Georgia Power Company Letter Dated November 13, 1978, to Nuclear Regulatory Commission.
5. Lysmer, J., et al., "Efficient Finite-Element Analysis of Seismic Structure-Soil-Structure Interaction," Earthquake Engineering Research Center, University of California, Report No. EERC 75-34, Berkeley, California, November 1975.
6. Seed, H. B., and Idriss, I. M., "Soil Moduli and Damping Factors for Dynamic Response Analysis," Earthquake Engineering Research Center, University of California, Report No. EERC 70-10, Berkeley, California, December 1970.
7. Alvin W. Vogtle Nuclear Plant, "Report on Dynamic Properties for Compacted Backfill," Bechtel Incorporated, Los Angeles, February 1978.

3.7.N.2 SEISMIC SYSTEM ANALYSIS

This section describes the methods of seismic analysis performed for safety-related components and systems within the Westinghouse scope.

3.7.N.2.1 Seismic Analysis Methods

Those components and systems that must remain functional in the event of a safe shutdown earthquake (SSE) (Seismic Category 1) are identified by applying the criteria of subsection 3.2.1.

In general, the dynamic analyses were performed using response spectrum analysis, integration of the uncoupled modal equations, direct integration of the coupled equations of motion, or nonlinear modal superposition. (See paragraph 3.7.N.2.1.5.)

3.7.N.2.1.1 Dynamic Analysis: Mathematical Model

The first step in any dynamic analysis is to model the structure or component, i.e., convert the real structure or component into a system of masses, springs, and dashpots suitable for mathematical analysis. The essence of this step is to select a model such that the displacements obtained will be a good representation of the motion of the structure or component. Stated differently, the true inertia forces should not be altered so as to appreciably affect the internal stresses in the structure or component. Some typical modeling techniques are presented in reference 1.

Equation of Motion

$$m_r \ddot{y}_r + \sum^{i} c_{ri}\dot{u}_i + \sum^{i} k_{ri}u_i = 0 \qquad (1)$$

Consider the multidegree-of-freedom system shown in figure 3.7.N.2-1. Making a force balance on each mass point r, the equations of motion can be written in the form:

where:

m_r = the value of the mass or mass moment of rotational inertia at mass point r.

$\ddot{y}_r$ = absolute translational or angular acceleration of mass point r.

c_{ri} = damping coefficient: external force or moment required at mass point r to produce a unit translational or angular velocity at mass point i, maintaining zero translational or angular velocity at all other mass points. Force or moment is positive in the direction of positive translational or angular velocity.

$\dot{u}_i$ = translational or angular velocity of mass point i relative to the base.

k_{ri} = stiffness coefficient: the external force (moment) required at mass point r to produce a unit deflection (rotation) at mass point i, maintaining zero displacement (rotation) at all other mass points. Force (moment) is positive in the direction of positive displacement (rotation).

u_i = displacement (rotation) of mass point i relative to the base.

Since:

$$\ddot{y}_r = \ddot{u}_r + \ddot{y}_s \qquad (2)$$

where:

$\ddot{y}_s$ = absolute translational (angular) acceleration of the base.

$\ddot{u}_r$ = translational (angular) acceleration of mass point r relative to the base.

Equation 1 can be written as:

$$m_r\ddot{u}_r + \sum^{i} c_{ri}\dot{u}_i + \sum^{i} K_{ri}u_i = -m_r\ddot{y}_s \quad (3)$$

For a single-degree-of-freedom system with displacement u, mass m, damping c, and stiffness k, the corresponding equation of motion is:

$$m\ddot{u} + c\ddot{u} + ku = -m\ddot{y}_s \quad (4)$$

3.7.N.2.1.2 Modal Analysis

Natural Frequencies and Mode Shapes. The first step in the modal analysis method is to establish the normal modes, which are determined by the eigen solution of equation 3. The right side and the damping term are set equal to zero for this purpose as illustrated in reference 2 (pages 83 through 111). Thus, equation 3 becomes:

$$m_r\ddot{u}_r + \sum^{i} K_{ri}u_i = 0 \quad (5)$$

The equation given for each mass point r in equation 5 can be written as a system of equations in matrix form as:

$$(M)\{\ddot{\Delta}\} + (K)\{\Delta\} = 0 \quad (6)$$

where:

(M) = mass and rotational inertia matrix.

$\{\Delta\}$ = column matrix of the general displacement or rotation at each mass point relative to the base.

(K) = square stiffness matrix.

$\{\ddot{\Delta}\}$ = column matrix of general translational and angular accelerations at each mass point relative to the base, $d^2\{\Delta\}/dt^2$.

Harmonic motion is assumed, and the $\{\Delta\}$ is expressed as:

$$\{\Delta\} = \{\delta\} \sin \omega t \quad (7)$$

where:

$\{\delta\}$ = column matrix of the spatial displacement or rotation at each mass point relative to the base.

ω = natural frequency of harmonic motion in radians per second.

The displacement function and its second derivative are substituted into equation 6 and yield:

$$(K)\{\delta\} = \omega^2(M)\{\delta\} \quad (8)$$

The determinant $|(K) - \omega^2 (M)|$ is set equal to zero and is then solved for the natural frequencies. The associated mode shapes are then obtained from equation 8. This yields n natural frequencies and mode shapes where n equals the number of dynamic degrees of freedom of the system. The mode shapes are all orthogonal to each other and are sometimes referred to as normal mode vibrations. For a single-degree-of-freedom system, the stiffness matrix and mass matrix are single terms and the determinant $|(K) - \delta^2 (M)|$ when set equal to zero yields simply:

$$k - \omega^2 m = 0$$

or:

$$\omega = \sqrt{\frac{k}{m}} \qquad (9)$$

where:

ω = natural angular frequency in radians per second.

The natural frequency in cycles per second is, therefore:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad (10)$$

To find the mode shapes, the natural frequency corresponding to a particular mode, ω_n, can be substituted in equation 8.

Modal Equations. The response of a structure or component is always some combination of its normal modes. Accuracy can usually be obtained by using only the first few modes of vibration. In the normal mode method, the mode shapes are used as principal coordinates to reduce the equations of motion to a set of uncoupled differential equations that describe the motion of each mode n. These equations may be written as (reference 2, pages 116 through 125):

$$\ddot{A}_n + 2\omega_n p_n \dot{A}_n + \omega_n^2 A_n = \Gamma_n \ddot{y}_s \qquad (11)$$

where the modal replacement or rotation, An, is related to the displacement or rotation of mass point r in mode n, urn by the equation:

$$U_{rn} = A_n \phi_{rn} \qquad (12)$$

where:

ω_n = natural frequency of mode n in radians per second.
p_n = critical damping ratio of mode n.
Γ_n = modal participation factor of mode n given by:

$$\Gamma_n = \frac{\sum_n^n m_r \phi r'n}{\sum_n^n m_r \phi_{rn}^2} \qquad (13)$$

where:

ϕ'_{rn} = value of the ϕ_{rn} in the direction of the earthquake.

The essence of the modal analysis lies in the fact that equation 11 is analogous to the equation of motion for a single-degree-of-freedom system that will be developed from equation 4. Dividing equation 4 by m gives:

$$\ddot{u} + \frac{c}{m}\dot{u} + \frac{k}{m}u = -\ddot{y}_s \quad (14)$$

The critical damping ratio of the single-degree-of-freedom system, p, is defined by the equation:

$$p = \frac{c}{c_c} \quad (15)$$

where the critical damping coefficient is given by the expression:

$$c_c = 2\,m\omega \quad (16)$$

Substituting equation 16 into equation 15 and solving for c/m gives:

$$\frac{c}{m} = 2\omega p \quad (17)$$

Substituting this expression and the expression for k/m given by equation 9 into equation 14 gives:

$$\ddot{u} + 2\omega p\dot{u} + \omega^2 u = -\ddot{y}_S \quad (18)$$

Note the similarity of equations 11 and 18. Thus, each mode may be analyzed as though it were a single-degree-of-freedom system and all modes were independent of each other. By this method a fraction of critical damping, i.e., c/c, may be assigned to each mode, and it is not necessary to identify or evaluate individual damping coefficients, i.e., c. However, assigning only a single damping ratio to each mode is not appropriate for a slightly damped structure supported by a massive, moderately damped structure. There are several methods which can be used to incorporate damping in a structural system.

One method is to develop and analyze separate mathematical models for both structures using their respective damping values. The massive, moderately damped support structure is analyzed first. The calculated response at the support points for the slightly damped structures is used as a forcing function for the subsequent detailed analysis. A second method is to inspect the mode shapes to determine which modes correspond to the slightly damped structure and then use the damping associated with the structure having predominant motion. A third method is to use the Rayleigh damping method based on computed modal energy distribution. In yet another method, the damping value for a given mode is derived from the calculation of the composite modal damping which is based on the distribution of energy in the structure for that mode.

3.7.N.2.1.3 Integration of Modal Equations

This method can be separated into the following two basic parts:

A. Integration procedure for the uncoupled modal equation 11 to obtain the modal displacement and accelerations as a function of time.

B. Use of these modal displacements and accelerations to obtain the total displacements, accelerations, forces, and stresses.

Integration Procedure. Integration of these uncoupled modal equations is done by step-by-step numerical integration. The step-by-step numerical integration procedure consists of selecting a suitable time interval, Δt, and calculating modal acceleration, A_n, modal velocity, A_n, and modal displacement, A_n, at discrete time stations Δt apart, starting at t = 0 and continuing through the range of interest for a given time-history of base acceleration.

Total Displacements, Accelerations, Forces, and Stresses. From the modal displacements and accelerations, the total displacements, accelerations, forces, and stresses can be determined as follows:

A. Displacement of mass point r in mode n as a function of time is given by equation 12 as:

$u_{rn} = A_n \phi_{rn}$

with the corresponding acceleration of mass point r in mode n as:

$u_{rn} = \ddot{A}_n \phi_{rn}$

B. The displacement and acceleration values obtained for the various modes are superimposed algebraically to give the total displacement and acceleration at each time interval.

C. The total acceleration at each time interval is multiplied by the mass to give an equivalent static force. Stresses are calculated by applying these forces to the model or from the displacements at each time interval.

3.7.N.2.1.4 Integration of Coupled Equations of Motion

The dynamic transient analysis is a time-history solution of the response of a given structure to known forces and/or displacement forcing functions. The structure may include linear or nonlinear elements, gaps, interfaces, plastic elements, and viscous and Coulomb dampers. Nodal displacements, nodal forces, pressure, and/or temperatures may be considered as forcing functions. Nodal displacements and elemental stresses for the complete structure are calculated as functions of time.

The basic equations for the dynamic analysis are as follows:

$$(M)\ \{\ddot{x}\} + (C)\ \{\dot{x}\} + (K)\ \{x\} = \{F(t)\} \qquad (22)$$

where the terms are as defined earlier; {F(t)} may include the effects of applied displacements, forces, pressures, temperatures, or nonlinear effects such as plasticity and dynamic elements with gaps. Options of translational accelerations input to a structural system and the inclusion of static deformation and/or preload may be considered in the nonlinear dynamic transient analysis. The option of translational input such as uniform base motion to a structural system is considered by introducing an inertia force term of -(M){z} to the right side of the basic equation 22, i.e.,

$$(M)\ \{\ddot{x}\} + (C)\ \{\dot{x}\} + (K)\{x\} = \{F\} - (M)\{\ddot{z}\} \qquad (23)$$

The vector $\{\ddot{z}\}$ is defined by its components $\ddot{z}_i$ where: i refers to each degree of freedom of the system; $\ddot{z}_i$ is equal to a_1, a_2, or a_3 if the ith degree of freedom is aligned with the direction of the system translational acceleration a_1, a_2, or a_3, respectively; and $\ddot{z}_i$ is equal to 0 if the ith degree

of freedom is not aligned with any direction of the system transitional acceleration. Typical application of this option is a structural system subjected to a seismic excitation of a given ground acceleration record. The displacement {x} obtained from the solution of equation 23 is the displacement relative to the ground.

The option of the inclusion of initial static deformation or preload in a nonlinear transient dynamic structural analysis is considered by solving the static problem prior to the dynamic analysis. At each state of integration in transient analysis, the portion of internal forces due to static deformation is always balanced by the portion of the forces which are statically applied. Hence, only the portion of the forces that deviate from the static loads will produce dynamic effects. The output of this analysis is the total result due to static- and dynamic-applied loads.

3.7.N.2.1.5 Nonlinear Modal Superposition

In the nonlinear modal superposition method, the nonlinearities are presented as pseudoforces. The mass and stiffness matrices are calculated only once; the corresponding mode shapes and natural frequencies are associated with the linear system, simulating initial states of the undamped structure with an external force acting on it. This state of the structure is hereafter referred to as the reference state. During the time-history analysis, as the nonlinear behavior comes into action, the true frequencies and mode shapes change. The effect of the variation of the true frequencies and mode shapes from the original ones is represented by pseudoforces on the right side of the equation of motion.

The generalized equation of motion for a nonlinear structure is:

$$[M]\{\ddot{X}\} = (C_{n1})\{\dot{X}\} + (K_{n1})\{X\} = \{F\} \quad (24)$$

where:

$[M]$ = mass matrix.
$[C_{n1}]$ = nonlinear damping matrix, dependent upon velocity and displacement.
$[K_{n1}]$ = nonlinear stiffness matrix, dependent upon displacement.
$\{X\}, \{\dot{X}\}, \{\ddot{X}\}, \{F\}$ = displacement, velocity, acceleration and applied-force vector

Let

$$[C_{n1}] = [C] + [\overline{C}]$$
$$[K_{n1}] = [K] + [\overline{K}] \quad (25)$$

where [C] and [K] are the damping and stiffness matrices representing the reference state of structure; [C] and [K] are the damping and stiffness matrices, dependent on the velocity and displacement. Substitution of equations 25 into equation 24 gives:

$$[M]\{\ddot{X}\} + [C]\{\dot{X}\} + (K)\{X\} = \{F\} - \{F_{nl}\} \quad (26)$$

where the pseudoforce vector is defined by

$$\{F_{nl}\} = [\overline{C}]\ \{\dot{X}\} + [\overline{K}]\ \{X\} \quad (27)$$

The homogeneous, undamped equation of motion representing the references state of the structure is:

$$[M]\{\ddot{X}\} + (K)\{X\} = \{0\} \quad (28)$$

Let (ω) and (ϕ) be the natural frequency and normalized mode shape matrix. The following transformation

$$\{X\} = (\phi)\{q\} \quad (29)$$

is substituted in equation 26, resulting in the following uncoupled nodal equations:

$$\{\ddot{q}\} + 2\zeta_j\omega_j\{\dot{q}\} + (\omega j^2)\{q\} = \{Q\} - \{Q_{nl}\} \quad (30)$$

where:

ζ_j = percentage of the critical damping for the 7th mode.

$\{Q\}$ = $(Q)^T\{F\}$ = generalized applied – force vector.

$\{Q_{nl}\}^I$ = $(Q)^T\{F_{nl}\}^I$ = generalized pseudoforce vector.

Arrays {q}, $\{\dot{q}\}$, and $\{\ddot{q}\}$ are the modal displacement, velocity, and acceleration vectors, respectively. The generalized pseudoforce vector is a function of displacement and velocity. For a given time step, it can be approximated by a Taylor series.

For a given time step, modal equations of motion are integrated analytically. Then the displacement and velocities of the nodes associates with nonlinear elements are calculated. This information is used to calculate the generalized pseudoforce vector and its time derivatives. Then the modal equations are integrated for the next time step.

3.7.N.2.2 Natural Frequencies and Response Loads

Not applicable.

3.7.N.2.3 Procedures Used for Modeling

Procedures used for modeling are discussed in paragraph 3.7.N.2.1.1.

3.7.N.2.4 Soil-Structure Interaction

Refer to paragraph 3.7.B.2.4.

3.7.N.2.5 Development of Floor Response Spectra

Refer to paragraph 3.7.B.2.5.

3.7.N.2.6 Three Components of Earthquake Motion

The seismic design of the piping and equipment includes the effect of the seismic response of the supports, equipment, structures, and components. The system and equipment response is determined using three earthquake components, two horizontal and one vertical. The seismic input described in subsection 3.7.N.1 is the basis for generating these three input components.

In computing the system and equipment response by the response spectrum modal analysis, the methods of 3.7.N.2.7 are used to combine all significant modal responses to obtain the combined unidirectional responses.

The combined total response is then calculated using the square root of the sum of the squares (SRSS) formula applied to the resultant codirectional responses. For instance, for each item of interest, such as displacement, force, stresses, etc., the total response is obtained by applying the SRSS method. The mathematical expression for this method (with R as the item of interest) is:

$$R_C = \left(\sum_{T=1}^{3} R_T^2 \right)^{1/2} \quad (31)$$

where

$$R_T = \left(\sum_{i=1}^{N} R_{Ti}^2 \right)^{1/2} \quad (32)$$

and

R_c = total combined response at a point.

R_T = value of combined response of direction T.

R_{Ti} = value of response for direction T, mode i.

N = total number of modes considered.

Westinghouse employs methods of combining modal responses that have been accepted by the Nuclear Regulatory Commission as an alternative to Regulatory Guide 1.92 (section 1.9).

The subscripts can be reversed without changing the results of the combination.

For systems having modes with closely spaced frequencies, R_T in equation 31 is determined by equation 33 in paragraph 3.7.N.2.7.

The system and equipment response can also be determined using time-history analyses.

When a time-history analysis is performed, the two horizontal and the vertical time-history components are applied simultaneously.

3.7.N.2.7 Combination of Modal Responses

The total codirectional seismic response is obtained by combining the individual modal responses utilizing the SRSS method. For systems having modes with closely spaced frequencies, this method is modified to include the possible effect of these modes. The groups of closely spaced modes are chosen such that the difference between the frequency of the first mode and the last mode in the group does not exceed 10 percent of the lower frequency. Groups are formed starting from the lowest frequency and working toward successively higher frequencies. No one frequency is in more than one group. The combined total response for systems that have such closely spaced modal frequencies is obtained by adding to the SRSS of all modes the product of the responses of the modes in each group of closely spaced modes and a coupling factor ε. This can be represented mathematically as:

$$R_T^2 = \sum_{i=1}^{N} R_i^2 + 2\sum_{j=1}^{S} \sum_{K=M_j}^{N_j-1} \sum_{\ell=K+1}^{N_j} R_K R_\ell \varepsilon K_\ell \quad (33)$$

where:

R_T = total unidirectional response.

R_i = absolute value of response of mode i.

N = total number of modes considered.

S = number of groups of closely spaced modes.

M_j = lowest modal number associated with group j of closely spaced modes.

N = highest modal number associated with group j of closely spaced modes.

εK_ℓ = coupling factor with

$$\varepsilon K_\ell = \left\{ 1 + \frac{(\omega'_K - \omega'_\ell)^2}{(\beta'_K \omega_K + \beta_\ell \omega_\ell)} \right\}^{-1} \quad \text{and} \quad (34)$$

$$\omega'_K = \omega_K (1 - (\beta'_K)^2)^{1/2} \quad (35)$$

$$\beta'_K = \beta_K + \frac{2}{\omega_K t_d} \quad (36)$$

where:

ω_K = frequency of closely spaced mode K.

β_K = fraction of critical damping in closely spaced mode K.

t_d = duration of the earthquake.

3.7.N.2.8 Interaction of Non-Category 1 Structures with Seismic Category 1 Structures

Refer to paragraph 3.7.B.2.8.

3.7.N.2.9 Effects of Parameter Variations on Floor Response Spectra

Refer to paragraph 3.7.B.2.9.

3.7.N.2.10 Use of Constant Vertical Static Factors

Constant vertical static factors are not used as the vertical floor response load for the seismic design of safety classed systems and components within Westinghouse's scope of responsibility. All such systems and components are analyzed in the vertical direction.

3.7.N.2.11 Methods Used to Account for Torsional Effects

In the coupled building internals structure/reactor coolant loop model, the torsional effects are accounted for by including the torsional component due to earthquake motions along with the three translational and two rocking components as described in paragraphs 3.7.N.1.2 and 3.7.B.2.11.

3.7.N.2.12 Comparison of Responses

Not applicable.

3.7.N.2.13 Methods for Seismic Analysis of Dams

Refer to paragraph 3.7.B.2.13.

3.7.N.2.14 Determination of Seismic Category 1 Structure Overturning Moments

Refer to paragraph 3.7.B.2.14.

3.7.N.2.15 Analysis Procedure for Damping

Procedures for damping are discussed in paragraph 3.7.N.1.3.

3.7.N.2.16 Standard Review Plan Evaluation

Closely spaced modes should be combined in accordance with procedures stated in Regulatory Guide 1.92. The VEGP uses the "pi epsilon" method for closely spaced mode combinations.

Westinghouse combines closely spaced modes using the "epsilon" method (section 3.7). The Westinghouse method for combining closely spaced modes represents an alternative to Regulatory Guide 1.92. This method has been accepted by the Nuclear Regulatory Commission's Structural Engineering and Mechanical Engineering Branch on specific plant dockets. Most recently, the Westinghouse position on combining closely spaced modes has been accepted on other dockets.

3.7.N.2.17 References

A. Lin, C. W., How to Lump the Masses - A Guide to the Piping Seismic Analysis, ASME Paper 74-NE-7 Presented at the Pressure Vessels and Piping Conference, Miami, June 1974.

B. Biggs, J. M., Introduction to Structural Dynamics, McGraw- Hill, New York, 1974.

3.7.B.3 SEISMIC SUBSYSTEM ANALYSIS

Seismic subsystems are those systems whose models lack soil/structure interaction relationships. Therefore, all Category 1 structures not designated as seismic systems and all Category 1 systems and equipment are considered seismic subsystems.

3.7.B.3.1 Seismic Analysis Methods

Methods used for seismic analysis of seismic subsystems are described in this paragraph.

3.7.B.3.1.1 Category 1 Structures

3.7.B.3.1.1.1 Containment Building. The shell and basemat internal forces and moments produced by the operating basis earthquake (OBE) and safe shutdown earthquake (SSE) are determined by statically applying, at different elevations, the acceleration values obtained from the dynamic time-history analysis described in subsection 3.7.B.2. The containment building finite element model used for the static analysis is described in subsection 3.8.1.

3.7.B.3.1.1.2 Containment Internal Structure. The concrete internal structure internal forces and moments produced by the OBE and SSE are determined by statically applying, at different elevations, the acceleration values obtained from the dynamic time-history analysis described in subsection 3.7.B.2. The containment internal structure finite element model used for the static analysis is described in subsection 3.8.3.

The structural steel access platform framing system outside the secondary shield wall is modeled as subsystem supported at the containment basemat and the secondary shield walls. The frequencies and mode shapes for this subsystem were used in a response spectrum analysis using an envelope spectra of the concrete internals structure obtained from the time-history analysis described in subsection 3.7.B.2.

3.7.B.3.1.1.3 Auxiliary Building. Forces and moments produced by the OBE and SSE in shear walls, slabs, and basemat are determined by statically applying acceleration values obtained from the dynamic time-history analysis described in subsection 3.7.B.2, simultaneously, at appropriate slab elevations. Further discussion on the methods used in the analysis is provided in subsection 3.8.4.

3.7.B.3.1.1.1 Fuel Handling Building. Forces and moments provided by the OBE and SSE in shear walls, slabs, and basemat are determined by two methods. Elevations 220 ft 0 in. and below are analyzed using a finite element model. Elevations 220 ft 0 in. and above are analyzed using hand calculations to distribute the seismic loads. For both methods of analysis, forces and moments are determined by statically applying acceleration values obtained from the dynamic time-history analysis described in subsection 3.7.B.2, simultaneously, at appropriate slab elevations.

3.7.B.3.1.1.5 Control Building. The forces and moments produced by the OBE and SSE are computed using the procedure described in paragraph 3.7.B.3.1.1.3.

3.7.B.3.1.1.6 Other Category 1 Structures. This paragraph describes the seismic analysis methods used for the following structures:

- Nuclear service cooling water (NSCW) towers and valve houses.
- Refueling, reactor makeup, and condensate water storage tanks.

- Diesel generator buildings.
- Diesel fuel oil storage tank pumphouses.
- Auxiliary feedwater pumphouses.
- All tunnels.

All structures are analyzed using the OBE and SSE acceleration values obtained from the dynamic time-history analysis described in subsection 3.7.B.2. All structures are analyzed using equivalent statically applied accelerations with the exception of the NSCW valve houses. The NSCW valve houses are analyzed using the response spectrum method. Finite element computer models are used for the analysis of the NSCW towers and valve houses. Classical shell solutions are used in the analysis of the water storage tanks. Conventional hand analysis methods are used to distribute the forces in the other structures.

3.7.B.3.1.2 Category 1 Systems

For the analysis of electrical cable trays and tray supports and heating, ventilation, and air conditioning (HVAC) ducts and duct supports, the modal response spectrum analysis method or the equivalent static load method is used. This method is described in paragraph 3.7.B.3.5.

For the analysis of piping systems, the floor response spectra of the specific level of the appropriate building are used as the seismic input. The generation of response spectra and the damping values used are described in subsection 3.7.B.1. The methods used for analyzing piping systems are described in sections 2.0, 4.0, and Appendix D of BP-TOP-1.[(2)]

3.7.B.3.1.3 Category 1 Subsystems and Components

Either modal response spectrum analysis method or equivalent static load method is used for the analyses of Category 1 sub- systems and components. The criteria for the number of masses used, the number of modes considered, and the combination of modal responses are in accordance with BC-TOP-4A.[(1)] The equivalent static load method is described in paragraph 3.7.B.3.5.

3.7.B.3.2 Determination of Number of Earthquake Cycles

The procedures used to determine the number of earthquake cycles for piping during seismic events are discussed in section 6.2 of BP-TOP-1.[(2)] Structures and equipment designed on the basis of analysis are not fatigue controlled since most stress and strain reversals occur only a small number of times. The allowable stresses used in the design of these structures are not affected by 50 maximum stress cycles that could be caused by the OBEs. The number of earthquake cycles simulated in the equipment qualified by testing is discussed in sections 3.9 and 3.10.

3.7.B.3.3 Procedure Used for Modeling

Sections 2.0 and 3.0 of BP-TOP-1 discuss the techniques and procedures used to model Seismic Category 1 piping other than the buried type.[(2)]

3.7.B.3.4 Basis for Selection of Frequencies

Piping system frequencies are calculated and dynamic inter- actions with support structures are accounted for in accordance with section 2 of BP-TOP-1.(2)

3.7.B.3.5 Use of Equivalent Static Load Method of Analysis

The equivalent static load method involves equivalent horizontal and vertical static forces applied at the center of gravity of various masses. The equivalent force at a mass location is computed as the product of the mass and the seismic acceleration value applicable to that mass location.

The magnitude of the seismic acceleration is established on the basis of the dynamic response characteristics of the component. Components which can be adequately characterized as a single-degree-of-freedom system are designed for accelerations associated with their natural frequency. Seismic acceleration values used for design of multidegree-of-freedom systems, which may be in the resonance region of the amplified response spectra curves, are the peak acceleration values multiplied by a factor of 1.5 unless a lower factor is justified. In lieu of using the peak acceleration value, the actual frequency may be calculated and the corresponding acceleration value may be used. In this case, the calculated frequency must be higher than that frequency related to the peak acceleration; otherwise, the peak acceleration value is used in design. For systems and components which have fundamental frequencies of 33 Hz or greater, the zero period acceleration is taken as the seismic acceleration value.

The above equivalent static load method of analysis is used for design of platforms, electrical cable trays and supports, conduits and supports, HVAC ducts and supports, simple piping systems, and other substructures.

The equivalent static load method of analysis can also be used for design of complex piping systems, with significant responses at several vibrational frequencies. In this case, the static load factor of 1.7 shall be applied to the peak accelerations of the applicable floor response spectra. Use of this method for complex piping systems is limited to small-bore (2 in. or less in diameter), non-NSSS piping.

In lieu of the equivalent static load method, the method stated in BP-TOP-1, section 2.3.2 and Appendix D (2) may be used for piping.

3.7.B.3.6 Three Components of Earthquake Motion

Section 5.1 of BP-TOP-1 provides the criteria used to combine the results of horizontal and vertical seismic responses for piping systems.(2)

For the structures, systems, and equipment qualified by analysis, the three component earthquake effects are combined using the square root of the sum of the squares method, as described in paragraph 3.7.B.2.6, or an equivalent method yielding essentially the same results. For equipment qualified by testing, the three component earthquake effects are considered using the guidelines provided in Institute of Electrical and Electronics Engineers Standard 344-1975.(3)

3.7.B.3.7 Combination of Modal Responses

Sections 5.1 and 5.2 of BP-TOP-1 describe the criteria used for Category 1 piping systems.(2)

The combination of modal responses for other structural systems is performed as described in section 4.2 of BC-TOP-4A[1] and is in conformance with Regulatory Guide 1.92.

3.7.B.3.8 Analytical Procedures for Piping Systems

Section 2.0 of BP-TOP-1 describes the design criteria and the analytical techniques applicable to piping systems. Section 3.3 of BP-TOP-1 discusses selection of response spectra, including those situations where the piping system spans a large difference in elevation or the piping system goes between two different structures. In addition to the seismic spectrum selection techniques identified in section 3.3 of BP-TOP-1, multiple response spectrum methodology may also be used. With the exception of those stress calculations described in reference 4, this methodology is not used in conjunction with ASME Code Case N-411 Damping values. Section 4.0 of BP-TOP-1 also discusses relative displacements between piping and support points.[2]

3.7.B.3.9 Multiply-Supported Equipment and Components with Distinct Inputs

Section 4.0 of BP-TOP-1 discusses the methods used for piping systems.[2]

3.7.B.3.10 Use of Constant Vertical Static Factors

A constant seismic vertical load factor is not used for the seismic design of Seismic Category 1 structures, systems, components, and equipment. Use of equivalent static load method of analysis is described in paragraph 3.7.B.3.5.

3.7.B.3.11 Torsional Effects of Eccentric Masses

The significant torsional effects of valves and other eccentric masses are taken into account in the seismic piping analysis by the techniques discussed in section 3.2 of BP-TOP-1.[2]

3.7.B.3.12 Buried Seismic Category 1 Piping Systems and Tunnels

Section 6.0 of BC-TOP-4A discusses the techniques used to calculate the stresses from seismic loadings for buried seismic piping.[1] The buried Seismic Category 1 piping is designed to remain functional when subjected to seismic loads. This is accomplished by limiting the calculated stresses in the pipe material under all loading combinations, including earthquake, as discussed below. The sum of the stresses produced by internal and/or external pressure and those produced by seismic forces shall not exceed 2.4 (S) for the SSE or 1.2 (S) for the OBE. "S " indicates the allowable stresses prescribed as S in tables I-7-1, I-7-2, and I-7-3 of Appendix I of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Section III. Where American National Standards Institute Standard B31.1.0 is used, the allowable stresses are indicated in tables A-1, A-2, and A-3 of Appendix A of that code.

The methods of analysis for Category 1 buried tunnels are discussed in subsection 3.7.B.2.

Buried electrical duct banks containing Class 1E electrical circuits are designed to meet Seismic Category 1 requirements. Engineering analysis is based on the buried structure analysis procedures outlined in Chapter 6 of BC-TOP-4A.[1] Duct banks are provided with a seismic separation at structure, manhole, and intersection interfaces. The duct banks are placed at

depths consistent with protection requirements for missile penetration and large surface surcharge loadings.

3.7.B.3.13 Interaction of Other Piping with Seismic Category 1 Piping

Nonsafety-related piping systems which are classified as Seismic Category 1 are analyzed as described in paragraph 3.7.B.3.8. The results of the analysis are evaluated using the ASME Code, Section III, Class 2 and 3 equations with appropriate load combinations as described in subsection 3.9.B.3.

Section 3.4 of BP-TOP-1 describes the techniques used to consider the interaction of Seismic Category 1 piping with non-Seismic Category 1 piping.(2)

3.7.B.3.14 Seismic Analyses for Reactor Internals

Refer to 3.7.N.3.14.

3.7.B.3.15 Analysis Procedure for Damping

Section 2.4 of BP-TOP-1 describes the procedure used to account for damping of Category 1 piping systems.(2) The analysis procedures used for structures are described in paragraph 3.7.B.2.15.

3.7.B.3.16 References

1. "Seismic Analyses of Structures and Equipment for Nuclear Power Plants," Bechtel Power Corporation, BC-TOP-4A, Revision 3, San Francisco, November 1974.
2. "Seismic Analysis of Piping Systems," Bechtel Power Corporation, BP-TOP-1, Revision 3, San Francisco, January 1976.
3. "Recommended Practices for Seismic Qualification for Class 1E Equipment for Nuclear Power Generating Stations," Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975.
4. Georgia Power Company Letter (GN-1257) Docket No 50.424 and 50-425, To NRC, dated December 22, 1986.

3.7.N.3 SEISMIC SUBSYSTEM ANALYSIS

This section describes the seismic analysis performed on subsystems within the Westinghouse scope of responsibility.

3.7.N.3.1 Seismic Analysis Methods

Seismic analysis methods for subsystems within the Westinghouse scope of responsibility are given in paragraphs 3.7.N.2.1 and 3.7.N.3.5.

3.7.N.3.2 Determination of Number of Earthquake Cycles

The operating basis earthquake (OBE) is conservatively assumed to occur five times over the life of the plant. A time-history study has been conducted to arrive at a realistic number of maximum stress cycles per OBE occurrence for all Westinghouse systems and components.

This evaluation considered both the equipment and its supporting building structure as single-degree-of-freedom systems, which tend to produce a more uniform and unattenuated response than a complex interacting system. The natural frequencies for the building equipment are conservatively chosen to coincide.

As a result of this study, 10 maximum stress cycles for flexible equipment (natural frequencies less than 33 Hz) and 5 maximum stress cycles for rigid equipment (natural frequencies greater than 33 Hz) for each OBE occurrence are used for fatigue evaluation of Westinghouse systems and components.

3.7.N.3.3 Procedure Used for Modeling

Refer to paragraph 3.7.N.2.1 for modeling procedures for subsystems in the Westinghouse scope of responsibility.

3.7.N.3.4 Basis for Selection of Frequencies

In the analysis of Class 1 branch lines attached to the reactor coolant loop (including the surge line), the frequencies of these lines are controlled to avoid the peak building frequencies and the lowest fundamental frequencies of the primary equipment, if necessary, to maintain equipment and support loads within allowable limits.

There are no specific design criteria that attempt to cause the fundamental frequencies of nuclear steam supply system (NSSS) equipment to be different from the forcing frequencies of the supporting structures. The effect of the equipment fundamental frequencies relative to the supporting structure forcing frequencies is, however, considered in the analysis of the NSSS equipment.

3.7.N.3.5 Use of Equivalent Static Load Method of Analysis

The static load equivalent or static analysis method involves the multiplication of the total weight of the equipment or component member by the specified seismic acceleration coefficient. The magnitude of the seismic acceleration coefficient is established on the basis of the expected dynamic response characteristics of the component. Components that can be adequately characterized as single-degree-of-freedom systems are considered to have a modal participation factor of 1. Seismic acceleration coefficients for multidegree-of-freedom systems that may be in the resonance region of the amplified response spectra curves are increased by 50 percent to account conservatively for the increased modal participation.

3.7.N.3.6 Three Components of Earthquake Motion

Methods used to account for three components of earthquake motion for subsystems in the Westinghouse scope of responsibility are given in paragraph 3.7.N.2.6.

3.7.N.3.7 Combination of Modal Responses

Methods used to combine modal responses for subsystems in the Westinghouse scope of responsibility are given in paragraph 3.7.N.2.7.

3.7.N.3.8 Analytical Procedures for Piping

The Class 1 piping systems are analyzed according to the rules of the American Society of Mechanical Engineers (ASME) Code, Section III. When response spectrum methods are used to evaluate piping systems supported at different elevations, the following procedures are used. The effect of differential seismic movement of piping supports is included in the piping analysis according to the rules of the ASME Code, Section III. According to ASME definitions, these displacements cause secondary stresses in the piping system.

In the response spectrum dynamic analysis for evaluation of piping systems supported at different elevations, spectra that envelop the floor response spectra corresponding to the applicable support locations are used. Westinghouse does not have in its scope of analysis any piping systems interconnected between buildings.

3.7.N.3.9 Multiple Supported Equipment Components with Distinct Inputs

When response spectrum methods are used to evaluate reactor coolant system (RCS) primary components interconnected between floors, the procedure outlined in the following paragraphs is used. The primary components of the RCS are supported at no more than two floor elevations.

A. A dynamic response spectrum analysis is first made assuming there is no relative displacement between support points. The response spectra used in this analysis is the most severe floor response spectra.

B. Secondly, the effect of differential seismic movement of components interconnected between floors is considered statically in the detailed component analysis. Per ASME Code rules, the stress caused by differential seismic motion is clearly secondary for piping (NB-3650) and component supports (NF-3231). For components, the differential motion is evaluated as a free end displacement, per ASME III NB-3213.19.

C. The results of these two steps, the dynamic inertia analysis and the static differential motion analysis, are combined absolutely with due consideration for the ASME classification of the stresses.

3.7.N.3.10 Use of Constant Vertical Static Factors

Constant vertical load factors are not used as the vertical floor response load for the seismic design of safety-related components and equipment within the Westinghouse scope of responsibility.

3.7.N.3.11 Torsional Effects of Eccentric Masses

The effect of eccentric masses, such as valves and valve operators, is considered in the seismic piping analyses. These eccentric masses are modeled in the system analysis, and the torsional effects caused by them are evaluated and included in the total system response. The total response must meet the limits of the criteria applicable to the safety class of the piping.

3.7.N.3.12 Buried Seismic Category 1 Piping Systems and Tunnels

Refer to paragraph 3.7.B.3.12.

3.7.N.3.13 Interaction of Other Piping with Seismic Category 1 Piping

Refer to paragraph 3.7.B.3.13.

3.7.N.3.14 Seismic Analyses for Reactor Internals

Fuel assembly component stresses induced by horizontal seismic disturbances are analyzed through the use of finite element computer modeling.

The time-histories were developed from the plant specific response spectra for the OBE and the safe shutdown earthquake at the required elevations. The reactor internals and the fuel assemblies are modeled as spring and lumped-mass systems or beam elements. The seismic response of the fuel assemblies is analyzed to determine design adequacy. A detailed discussion of the analyses performed for typical fuel assemblies is contained in reference 1.

Fuel assembly lateral structural damping obtained experimentally is also presented in reference 1 (figure B-4). The data indicate that the damping values exceeded 10 percent of the critically damped value at fuel assembly displacements greater than 0.11 in. Although the distribution of fuel assembly amplitudes decreases as one approaches the center of the core, the amplitude for the minimum displacement fuel assembly is well above 0.11 in. for the SSE.

Fuel assembly displacement time-history for the SSE seismic input is illustrated in reference 1 (figure 2-3). The fuel assembly amplitude resulting from the time-history response is used to determine the various fuel assembly component stresses.

The control rod drive mechanisms (CRDMs) are seismically analyzed to confirm that system stresses under the combined loading conditions, as described in subsection 3.9.N.1, do not exceed allowable levels as defined by ASME Code, Section III, for upset and faulted conditions. The CRDM is mathematically modeled as a system of lumped and distributed masses. The model is analyzed under appropriate seismic excitation, and the resultant seismic bending moments along the length of the CRDM are calculated. The corresponding stresses are then combined with the stresses from the other loadings required, and the combination is shown to meet the requirements of the ASME Code, Section III.

3.7.N.3.15 Analysis Procedure for Damping

Analysis procedures for damping for subsystems in the Westinghouse scope of responsibility are given in paragraph 3.7.N.1.3.

3.7.N.3.16 References

1. "Safety Analysis of the 8-Grid 17x17 Fuel Assembly for Combined Seismic and Loss of Coolant Accidents," WCAP-8236, Addendum 1 (Proprietary), March 1974, and WCAP-8288, Addendum 1 (Nonproprietary), April 1974.

3.7.N.3.17 Bibliography

Thomas, T. H., et al., "Nuclear Reactors and Earthquakes," U.S. Atomic Energy Commission, TID-7024, Washington, D.C., August 1963.

3.7.4 SEISMIC INSTRUMENTATION

3.7.4.1 Comparison with Regulatory Guide 1.12

The seismic monitoring system for VEGP consists of digital time history recorders that have remote accelerometers connected to them. Each recorder receives the acceleration signal from one triaxial accelerometer model FBA-3. The recorder stores the data on a Personal Computer Memory Card International Association (PCMCIA) card. The data can be retrieved automatically or manually. Each recorder has its own clock and the time is recorded.

The central controller unit, which consists of a computer, keyboard display, and printer, retrieves the data from the recorders, analyses it, and compares the result with the site response spectra for each location of the accelerometers. A printed report is provided showing the time-history, the peak acceleration value recorded at each location of the accelerometers, the response spectra, and the operating basis earthquake (OBE) exceedance. The recorders and the central controller are connected to an alarm panel. The alarm panel provides dry contact closures for different alarms to be sent to the control room. These alarms are:

- Event detected.
- OBE exceedance.
- Loss of dc.
- Loss of ac.
- System Health.

The system is in full compliance with the requirements of the Regulatory Guide 1.12, Rev.2, March 1997. The recording, analysis, and alarm system is housed in a seismically-braced cabinet, furnished for these instruments and devices necessary for system testing, annunciating, calibration, and control. This panel is located in the control room.

Since both units share common buildings and the expected seismic response is the same for both containments, only one complete set of seismic instrumentation is provided for the site in conformance with American National Standards Institute Standard N18.5, Section 4.4. Additional seismic instrumentation is installed to better evaluate the effect of an earthquake on building structures.

3.7.4.2 Location and Description of Instrumentation

The following instrumentation and associated equipment are used to measure plant response to earthquake motion:

- Seven triaxial strong motion accelerometers (SMA) Model FBA-3 connected to seven digital recorders. The new system monitors each accelerometer and when the acceleration level exceeds a preset threshold, it triggers and starts recording the event using a user-set pre-event and post-event memory. The total recording time for each digital recorder is 30 minutes.
- One cabinet to house the alarm panel, the digital recorders, the central controller, and the Uninterruptible Power Supply (UPS). The new central controller will provide the functions of detecting an event, retrieving the data, computing the response spectra for a given damping value, comparing the computed response spectra with the corresponding site response spectra, declaring an OBE exceedance alarm based on the OBE exceedance criteria, detecting a system health problem, and setting an alarm. All alarms are set through the alarm panel.

The UPS provides power for the central controller. Each digital recorder has its own battery and charger. In case of a loss of ac, the digital recorders will continue to operate for another 30 h, and the central controller will operate for 1 h. (See figure 3.7.4-1.) The seismic instrumentation is normally powered from the non-Class 1E, uninterruptible 120-V-ac power supply. However, the power supply for the supplemental SMA is powered by a non-class 1E 120-V-ac power supply. The system also has a self-contained power supply with a 30-min capacity for those cases where normal power is lost. Inservice surveillance for those instruments required to meet the recommendations of Regulatory Guide 1.12 is addressed in the Technical Requirements Manual. The remainder of the instrumentation is addressed in plant procedures.

3.7.4.2.1 Strong Motion Accelerometers

Each accelerometer provides an analog signal (voltage) directly proportional with the acceleration at the measuring points. This signal is recorded in digital Format (18-bit resolution) and analyzed, and the analysis results are printed. Each sensor unit contains three accelerometers mounted in a mutually orthogonal array. Accelerometers have their principal axes oriented identically, with one horizontal axis parallel to the major horizontal axis assumed in the seismic analysis.

One SMA is located in the field at approximately 225 ft from the containment. A second SMA is located in the Unit 1 containment tendon gallery such that it measures the input vibratory motion on the basemat. A third SMA is located on the Unit 1 containment building operating floor. A fourth SMA is located on el 197 ft 6 in. in Unit 1. A fifth SMA is located in the auxiliary building floor on level 1. A sixth SMA is installed on the basemat of the auxiliary building. The seventh SMA is on the slab floor of the diesel generator building.

Each sensor is used by the new system as a potential system trigger.

An eighth SMA is located in the ‘free field’ at approximately 2000 ft from containment. Independent to the primary seismic monitoring system, the supplemental ‘free field’ SMA is equipped with a self-contained seismic trigger and 18-bit digital recorder. The software provided with the central controller may be used to analyze the data provided by this stand-alone recorder and generate an analysis report similar to the one generated by the seismic monitoring system.

3.7.4.2.2 Response Spectrum Analyzer

Following the detection of an event, the central controller will send the event alarm, retrieve the data, analyze the data, and compare the result with the site OBE response spectrum for each accelerometer location. The OBE exceedance alarm will be set based on this comparison. For this configuration, it will take between 10 and 30 minutes to have the data analyzed and the report printed automatically.

3.7.4.2.3 System Control Panel

The cabinet in the control room houses the recorders, the alarm panel, and the central controller. It will provide a digital time-history, record, and complete analysis automatically in conjunction with the accelerometers. All required alarms will be set based on selected criteria.

3.7.4.3 Control Room Operator Notification

Exceedance of the preset values at the location of the accelerometers will provide a dry contact closure through the alarm panel. This can be used to activate corresponding alarms in the control room in accordance with the requirements of the Regulatory Guide 1.12, rev.2. These initial setpoints are based on experience in existing plants and may be changed once significant plant operating data have been obtained, which indicate that a different setpoint would provide a better SMA system operation.

The peak acceleration level at each location of the accelerometers, on each axis is available immediately and automatically after data analysis for each axis (in less than 30 minutes).

Response spectra for each axis of each accelerometer is available automatically after data analysis (in less than 30 minutes). For the free field, the response spectra can be computed and be available in the same format, depending on how fast the data is retrieved from the stand-alone recorder. The comparison with the OBE or safe shutdown earthquake (SSE) site spectra is done automatically, and the OBE exceedance is indicated based on the selected criteria. The system provides acceleration response spectra, velocity response spectra, and cumulative absolute velocity (CAV) computation automatically after the event (maximum 30 minutes). As mentioned before, separate alarms are sent to the control room if the threshold value is exceeded at the location of an accelerometer.

3.7.4.4 Comparison of Measured and Predicted Responses

The plan for utilization of the seismic data includes both the function of the operator and engineering to evaluate the effects of an earthquake on the plant. A detailed description of the data flow is provided in applicable plant procedures.

Initial determination of the earthquake level is performed automatically immediately after the earthquake by comparing the measured response spectra from the containment tendon gallery (and all other accelerometer locations) with the OBE and SSE response spectra for the corresponding location. If the measured spectra exceed the OBE response spectra, the plant will be shut down and a detailed analysis of the earthquake motion will be undertaken.

After an earthquake, the data from the seismic recorders and recording instruments are reviewed. The data from these instruments are analyzed automatically and a printed report generated by the central controller to obtain the seismic accelerations experienced at the

location of major Category 1 structures and equipment. The measured responses spectra from the SMAs, at the location of each Seismic Category 1 structure and system are compared automatically with those used in the design to determine whether the OBE has been exceeded.

TABLE 3.7.B.1-1

DAMPING VALUES FOR FIXED BASE STRUCTURES AND COMPONENTS[a]

	Percent of Critical Damping Per Mode	
Structure of Component	OBE	SSE
Equipment and large-diameter piping systems (pipe diameter in excess of 12 in.)[b]	2	3
Small-diameter piping systems (pipe diameter equal to or less than 12 in.)[b]	1	2
Welded steel structures	2	4
Bolted steel structures	4	7
Prestressed concrete structures	2	5
Reinforced concrete structures	4	7
Electrical cable trays and supports	(See figure 3.7.B.1-7.)	

a. Damping values for foundation material, used in foundation-structure interaction analysis, are not included in this table.

b. In lieu of these values, for ASME Boiler and Pressure Vessel Code, Section III, Division 1, Code Class 1, 2, and 3 piping systems, the damping values provided in figure 3.7.1-11 may be used per Code Case N-411.

TABLE 3.7.B.1-2

EMBEDMENT DEPTHS OF CATEGORY 1 STRUCTURES[c]

Structure	Foundation[a] Embedment Depth (ft)	Least Foundation Width (ft)	Structure[b] Height (ft)
Containment building	61	154	243
Auxiliary building	111	129	179
Control building	47	148[d]	140
Fuel handling building	66	76	134
NSCW towers	89	100	136
Diesel generator building	9	92	71
Condensate storage tanks	4	63	60
Refueling water storage tank	3	62	66
Reactor makeup water tank	2	51	46
Auxiliary feedwater pumphouse	7	40	31
NSCW valve house	20	20	50

a. Distance from bottom of foundation to grade level.

b. Distance from bottom of foundation to highest point of structure.

c. Buried structures are not included in this table.

d. Typical width for most parts of the foundation.

TABLE 3.7.N.1-1

DAMPING VALUES USED FOR SEISMIC SYSTEMS ANALYSIS

Item	Upset Conditions (OBE)	Faulted Condition (SSE, DBA)[a]
Primary coolant loop system components and large piping[b][c]	2	4
Small piping [c]	1	2
Welded steel structures	2	4
Bolted and/or riveted steel structures	4	7
Reinforced concrete structures	4	7

a. Design basis accident.

b. Applicable to 12-in. or larger diameter piping including pressurizer surge line piping.

c. As an alternative, ASME Code Case N-411 is used with response spectra analysis techniques. See figure 3.7.N.1-1.

TABLE 3.7.B.2-1

CLASSIFICATION OF SEISMIC SYSTEMS

Deeply embedded structures

- Containment building and containment internal structures
- Auxiliary building
- Control building[(a)]
- Fuel handling building[(a)]
- NSCW towers
- NSCW tower valve house

Shallowly embedded structures

- Diesel generator building
- Condensate storage tanks
- Refueling water storage tank
- Reactor makeup water tank
- Auxiliary feedwater pumphouse

Buried structures

- Category 1 tunnels
- Diesel fuel oil storage tank pumphouse
- Category 1 buried piping

a. Includes those portions of the equipment building supported by this structure.

TABLE 3.7.B.2-2

ACCELERATIONS OF MAJOR SEISMIC CATEGORY 1 STRUCTURES

Node Elevation	SSE Acceleration (g) Horizontal	SSE Acceleration (g) Vertical	OBE Acceleration (g) Horizontal	OBE Acceleration (g) Vertical
Containment internals				
169 ft 0 in. (basemat)	0.21	0.38	0.14	0.23
261 ft 0 in.	0.50	0.50	0.35	0.35
Containment shell				
169 ft 0 in. (basemat)	0.21	0.38	0.14	0.23
258 ft 0 in.	0.31	0.41	0.22	0.27
361 ft 0 in. (spring line)	0.45	0.43	0.30	0.29
Auxiliary building				
119 ft 3 in. (basemat)	0.20	0.28	0.12	0.18
220 ft 0 in.	0.28	0.30	0.18	0.20
288 ft 3 in. (roof)	0.38	0.35	0.24	0.22
Control building				
180 ft 0 in. (basemat)	0.26	0.40	0.17	0.24
220 ft 0 in.	0.29	0.41	0.19	0.25
280 ft 0 in. (roof)	0.73	0.88	0.53	0.69
Fuel handling building				
160 ft 0 in. (basemat)	0.24	0.38	0.16	0.24
220 ft 0 in.	0.39	0.42	0.25	0.29
288 ft 2 in. (roof)	0.60	0.48	0.41	0.33
Diesel generator building				
219 ft 0 in. (basemat)	0.26	0.30	0.16	0.19
274 ft 0 in. (roof)	0.34	0.32	0.21	0.19

TABLE 3.7.B.2-3

THE FREQUENCIES FOR FLOOR RESPONSE SPECTRA CALCULATIONS (Hz)

0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2
1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3
2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.15	3.3	3.45	3.6
3.8	4.0	4.2	4.4	4.6	4.8	5.0	5.25	5.5	5.75	6.0
6.25	6.5	6.75	7.0	7.25	7.5	7.75	8.0	8.5	9.0	9.5
10	10.5	11	11.5	12	12.5	13	13.5	14	14.5	15
16	17	18	20	22	25	28	31	34		

TABLE 3.7.4-1 (SHEET 1 OF 2)

SEISMIC MONITORING INSTRUMENTATION REQUIREMENTS

	Triaxial Time-History Accelerograph			Triaxial Response Spectrum Recorder			Triaxial Peak Accelerograph			Triaxial Seismic Switch			Seismic Trigger		
Instrumentation Location	RG 1.12 Req.	SRP Req.	VEGP	RG 1.12 Req.	SRP Req.	VEGP	RG 1.12 Req.	SRP Req.	VEGP	RG 1.12 Req.	SRP Req.	VEGP	RG 1.12 Req.	SRP Req.	VEGP
I. Field															
Requirement - Location															
1. Field – 225 ft from structure	1	1[a]	1(1)[g][e][h]												
2. Free field – Approximately2000 ft from structure			1(2)[g][e]												
II. Inside Containment															
Requirement - Location															
1. Basemat - Tendon gallery	1	1[a]	1[g](2)[d]	1[a]	1[a]	0[d]									
2. Structure - Operating floor el 220 ft	1	1	1[g](3)												
3. Structure - Bioshield wall el 202 ft			1[g](4)[d]	1	1	0[d]									
III. Outside Containment															
Requirement - Location															
1. Seismic Cat. 1 piping support or floor - Aux. bldg. floor el 220 ft			1[d](6)[g]	C	C	0[d]									
2. Independent Seismic Cat. 1 foundation struct. - Aux. bldg. slab			1[d](7)[g]	C	C	0[d]									
3. Diesel generator bldg. el 220 ft	0	0	1[g]												

Note: A response spectrum analyzer is located in the control room.

TABLE 3.7.4-1 (SHEET 2 OF 2)

a. Sensor shall have control room readout.

b. Audible and visual alarm in the control room.

c. Denotes one of two locations.

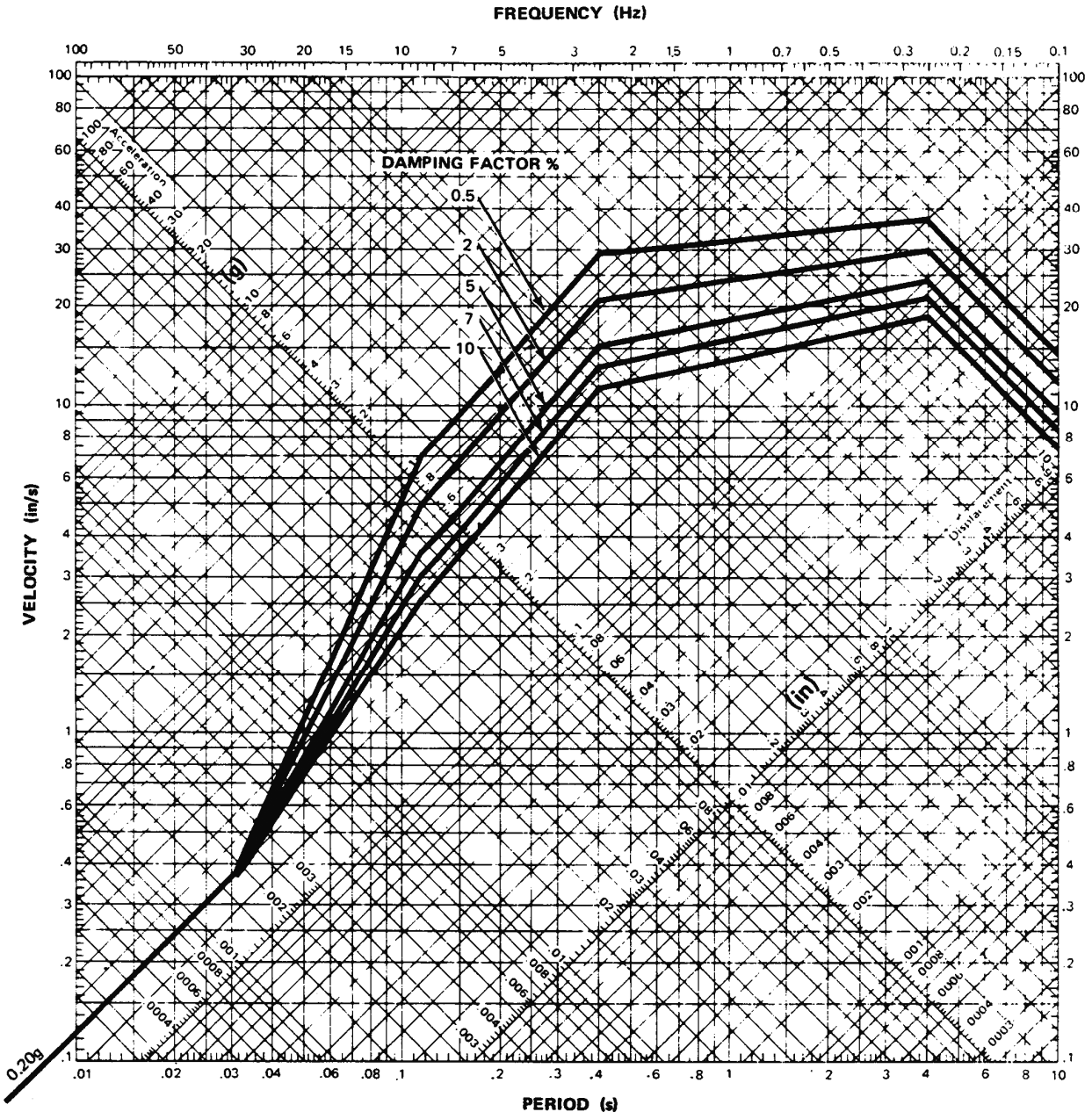
d. Response-spectrum analyzer preferred instead of response-spectrum recorder.

e. Parenthetical numbers show instrument locations in figure 3.7.4-1.

f. Starts time-history accelerograph.

g. Absolute acceleration data from each accelerometer are recorded and manually fed into a playback response spectrum analyzer to provide control room acceleration data and response spectrum for each sensor.

h. When utilizing data from this SMA, potential effects due to soil/structure interaction with adjacent structures will be considered.



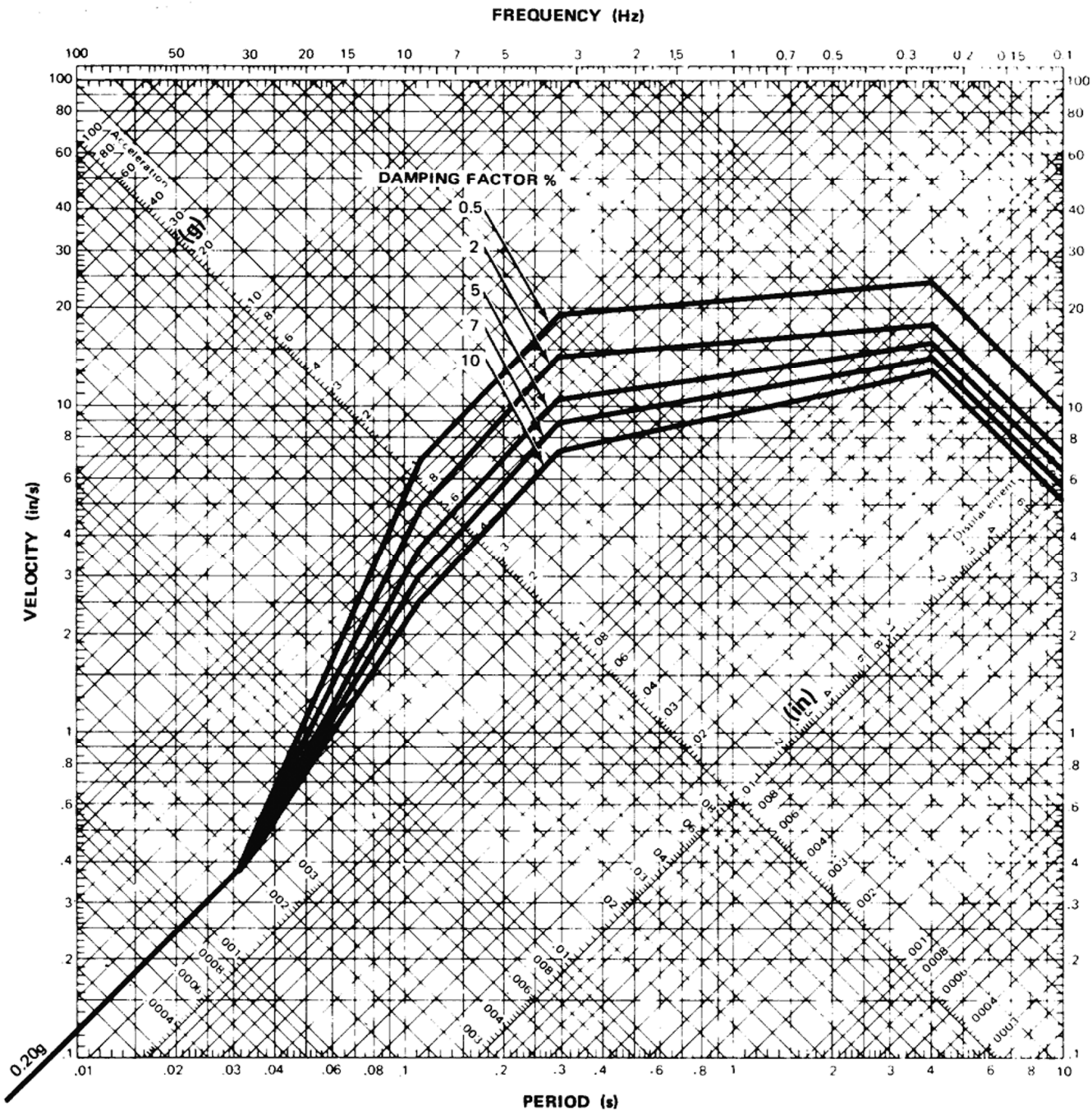
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SAFE SHUTDOWN EARTHQUAKE
HORIZONTAL RESPONSE SPECTRA

FIGURE 3.7.B.1–1



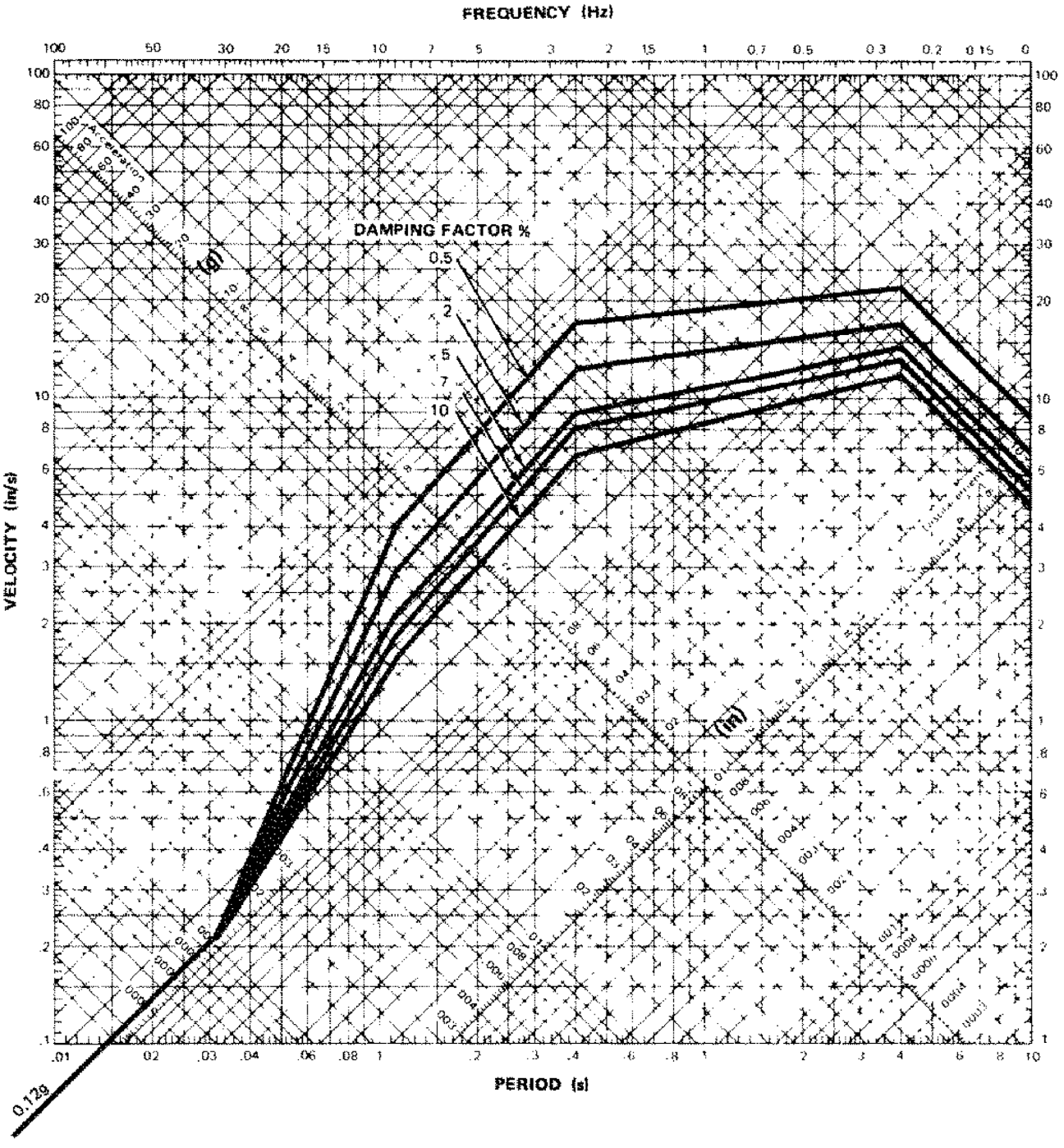
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SAFE SHUTDOWN EARTHQUAKE
VERTICAL RESPONSE SPECTRA

FIGURE 3.7.B.1–2



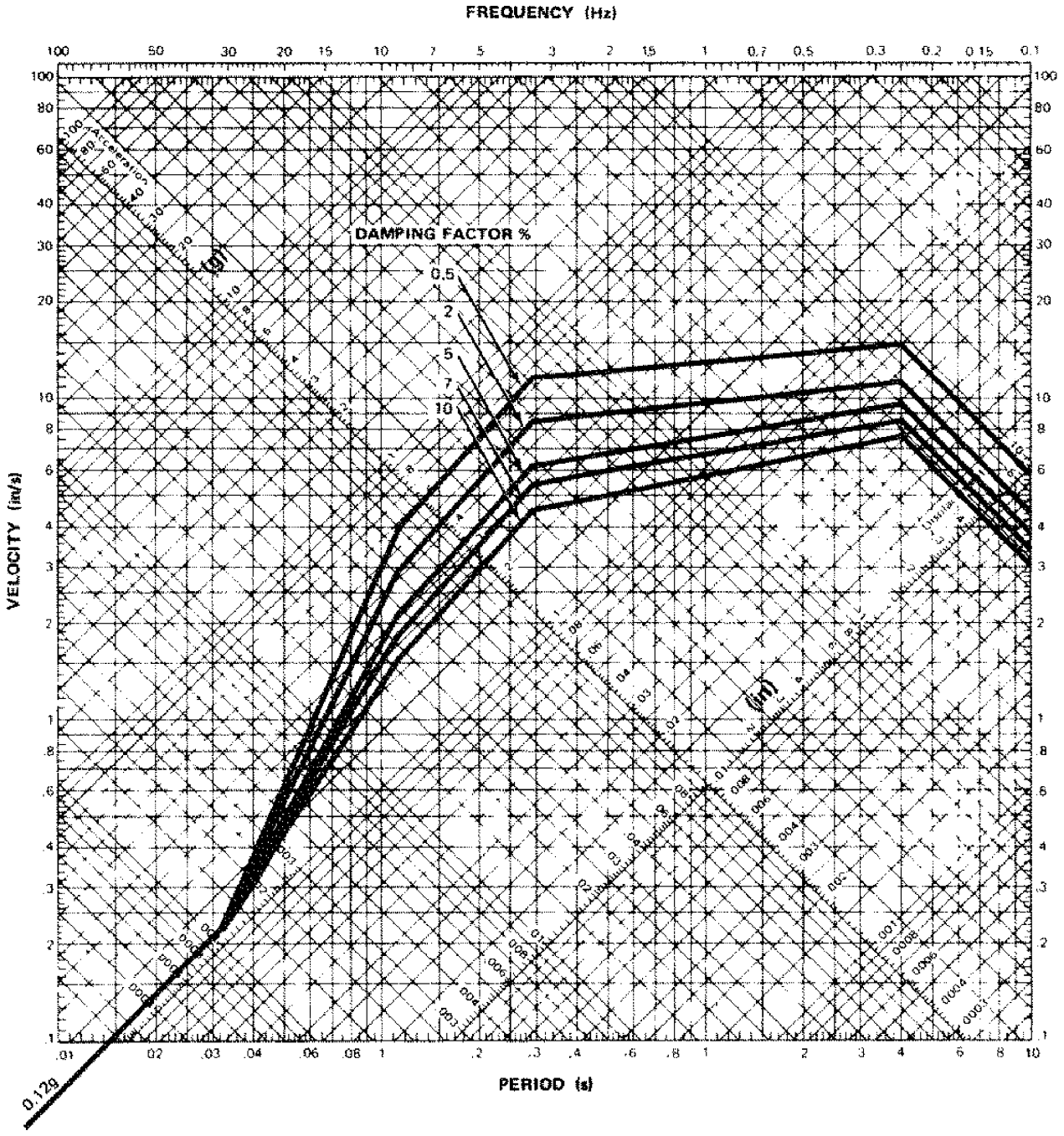
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

OPERATING BASIS EARTHQUAKE
HORIZONTAL RESPONSE SPECTRA

FIGURE 3.7.B.1–3



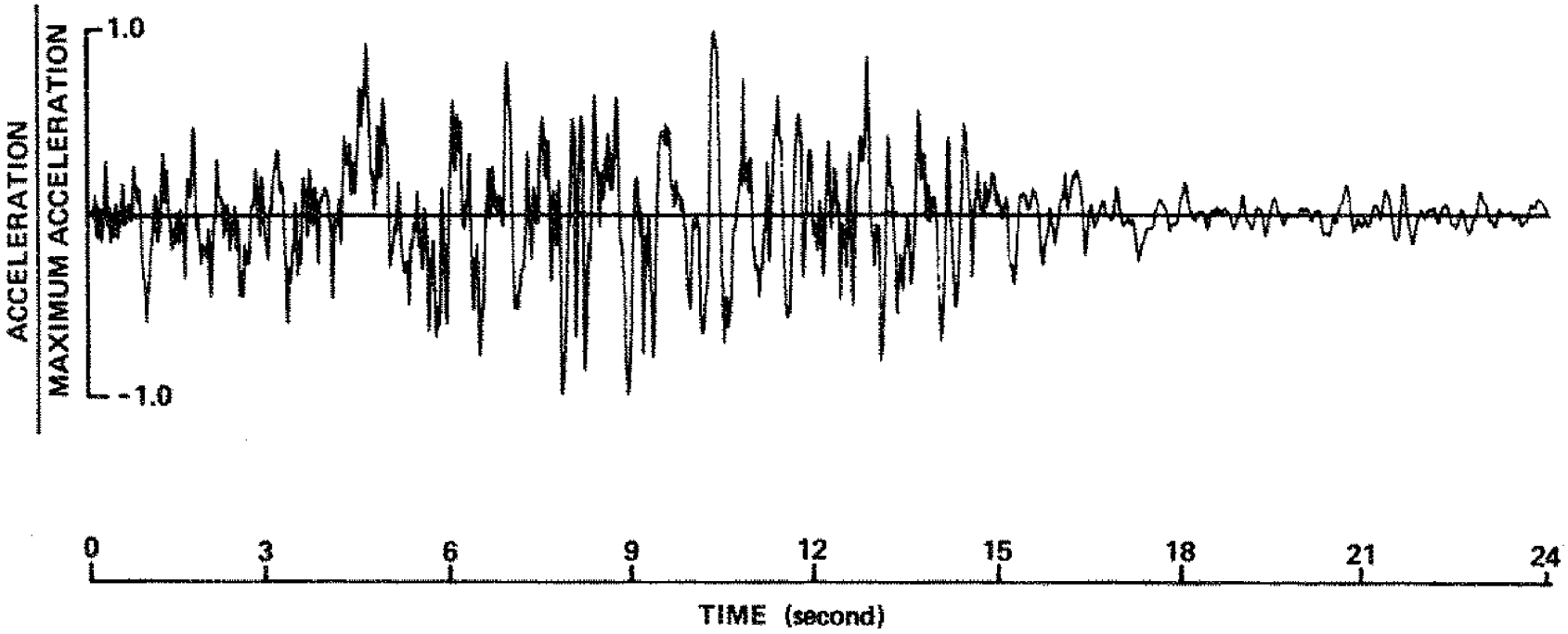
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

OPERATING BASIS EARTHQUAKE
VERTICAL RESPONSE SPECTRA

FIGURE 3.7.B.1–4



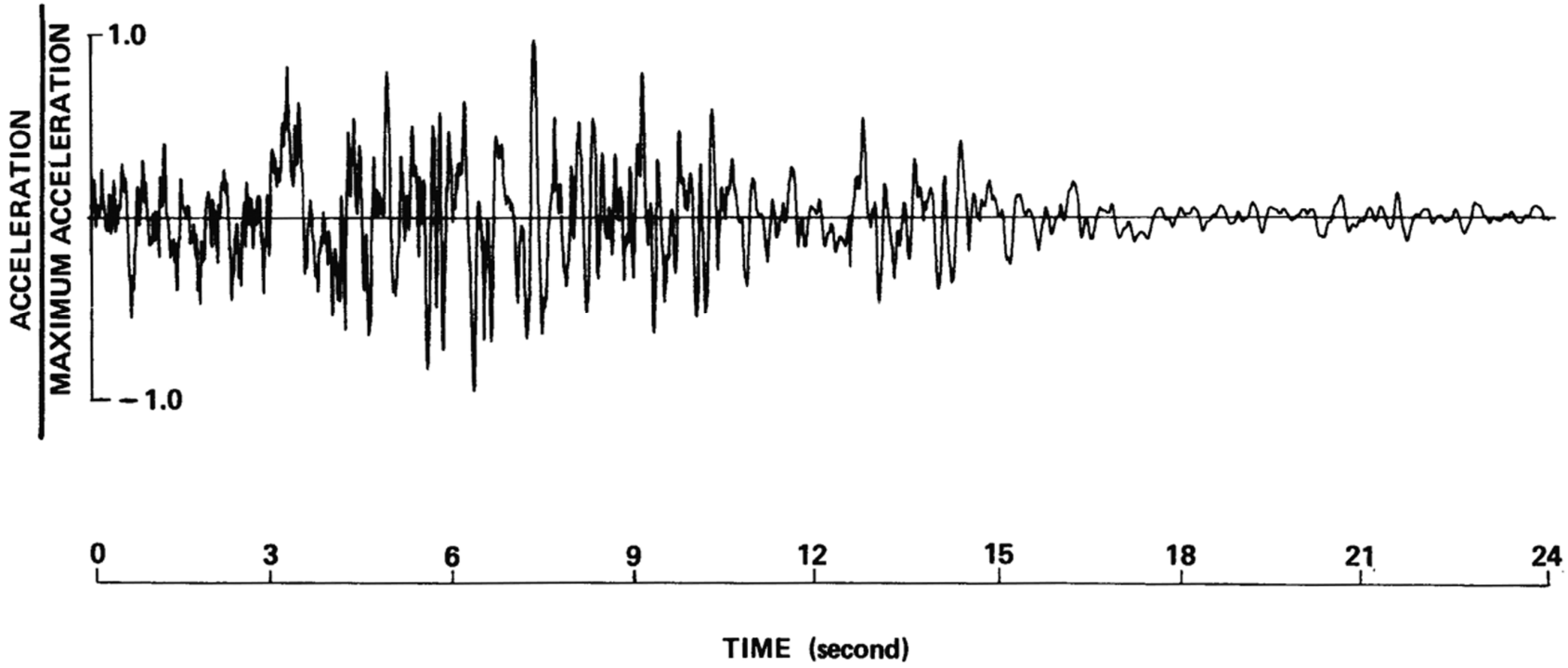
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SYNTHETIC TIME HISTORY
HORIZONTAL DIRECTION

FIGURE 3.7.B.1–5



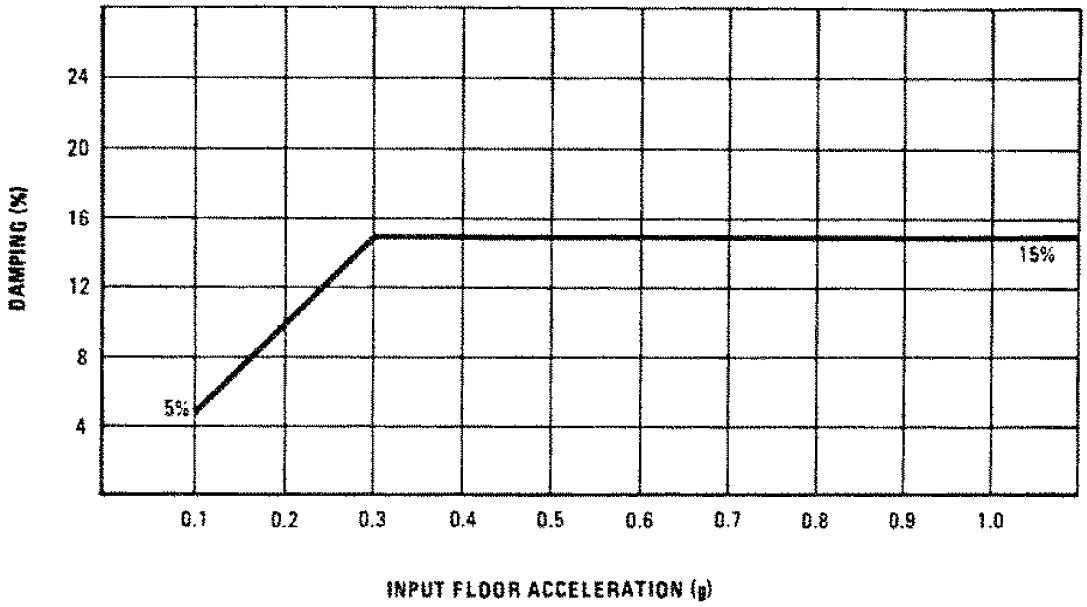
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SYNTHETIC TIME HISTORY
VERTICAL DIRECTION

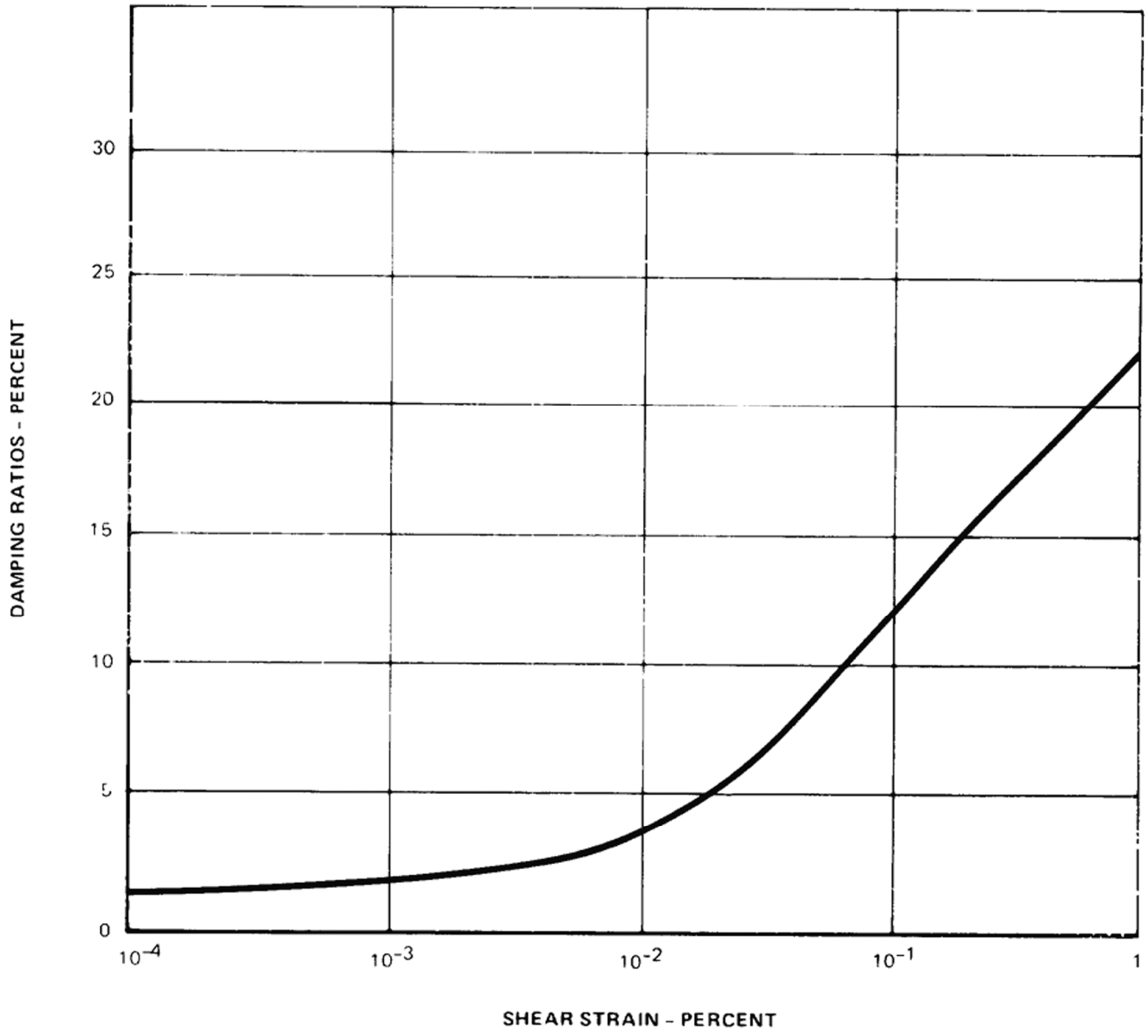
FIGURE 3.7.B.1–6



NOTE:

THE CURVE SHOWN IS APPLICABLE FOR 50% TO FULLY LOADED TRAYS. FOR UNLOADED TRAYS, DAMPING VALUES OF 2% FOR OBE AND 4% FOR SSE ARE APPLICABLE.

FOR TRAYS LOADED LESS THAN 50%, LINEAR INTERPOLATION IS USED TO DETERMINE THE ASSOCIATED DESIGN DAMPING VALUES.



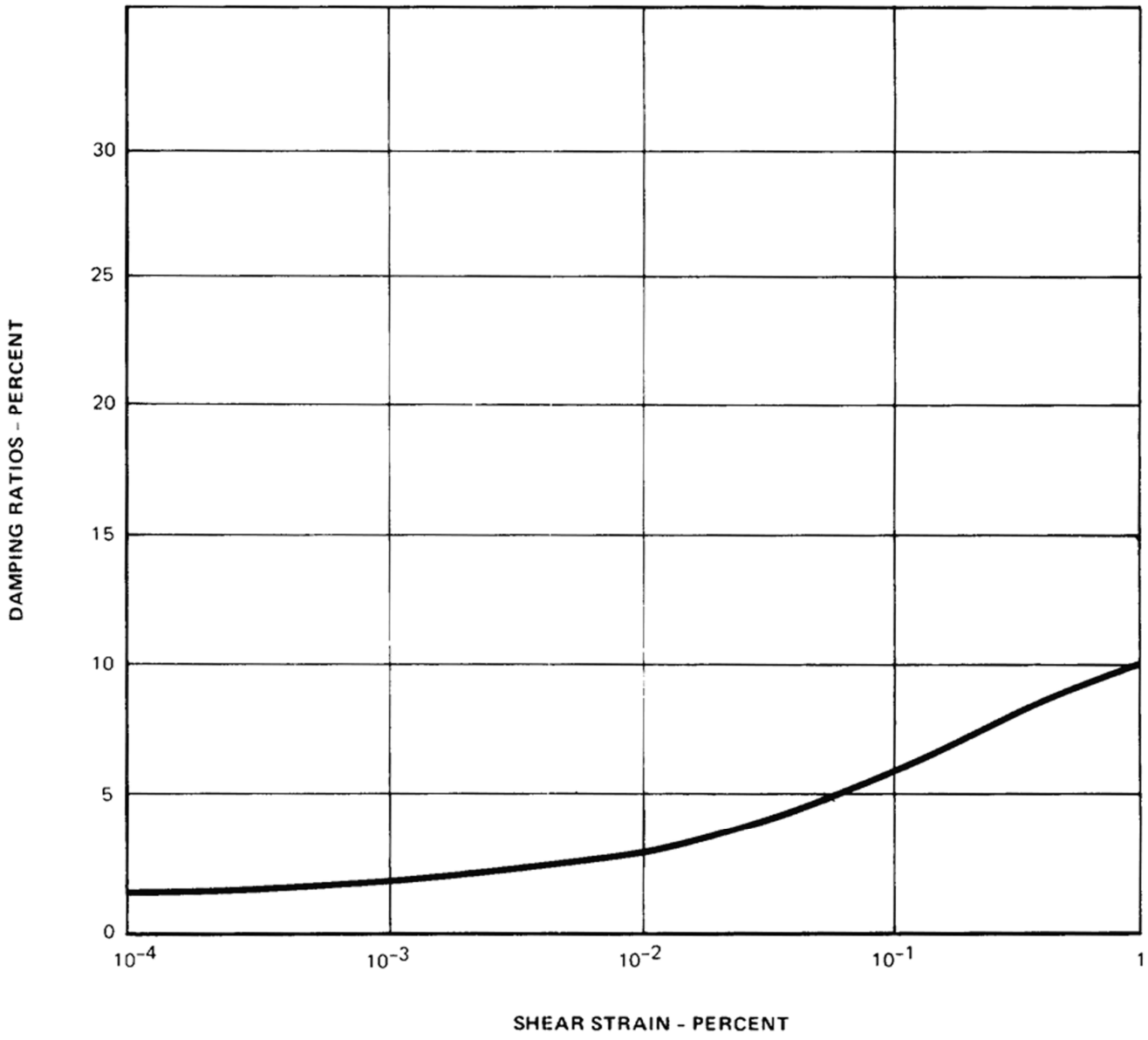
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

DAMPING RATIOS VS SHEAR STRAIN
FOR COMPACTED SAND BACKFILL

FIGURE 3.7.B.1–8



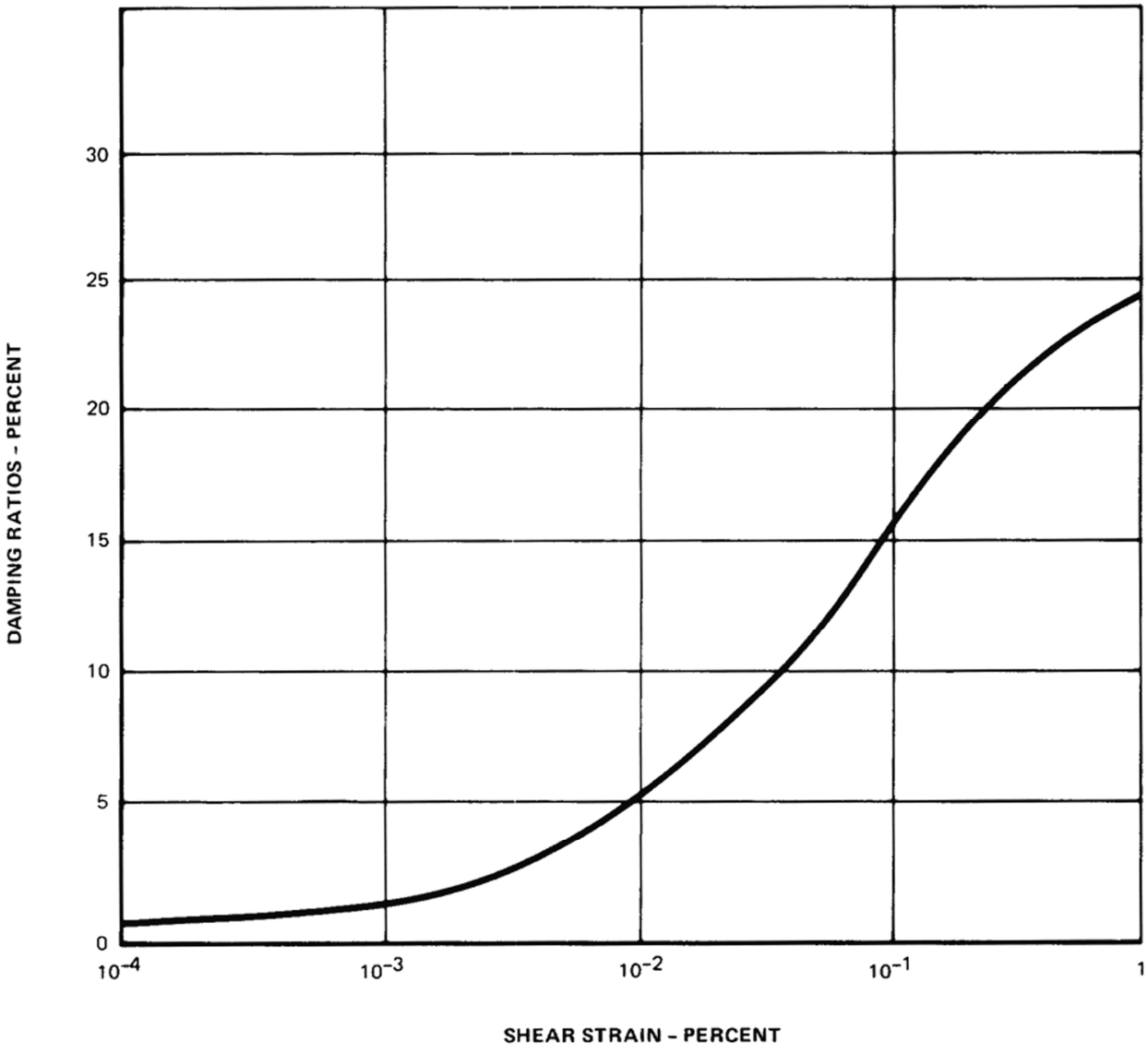
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

DAMPING RATIOS VS SHEAR STRAIN
FOR CLAY MARL BEARING STRATUM

FIGURE 3.7.B.1–9



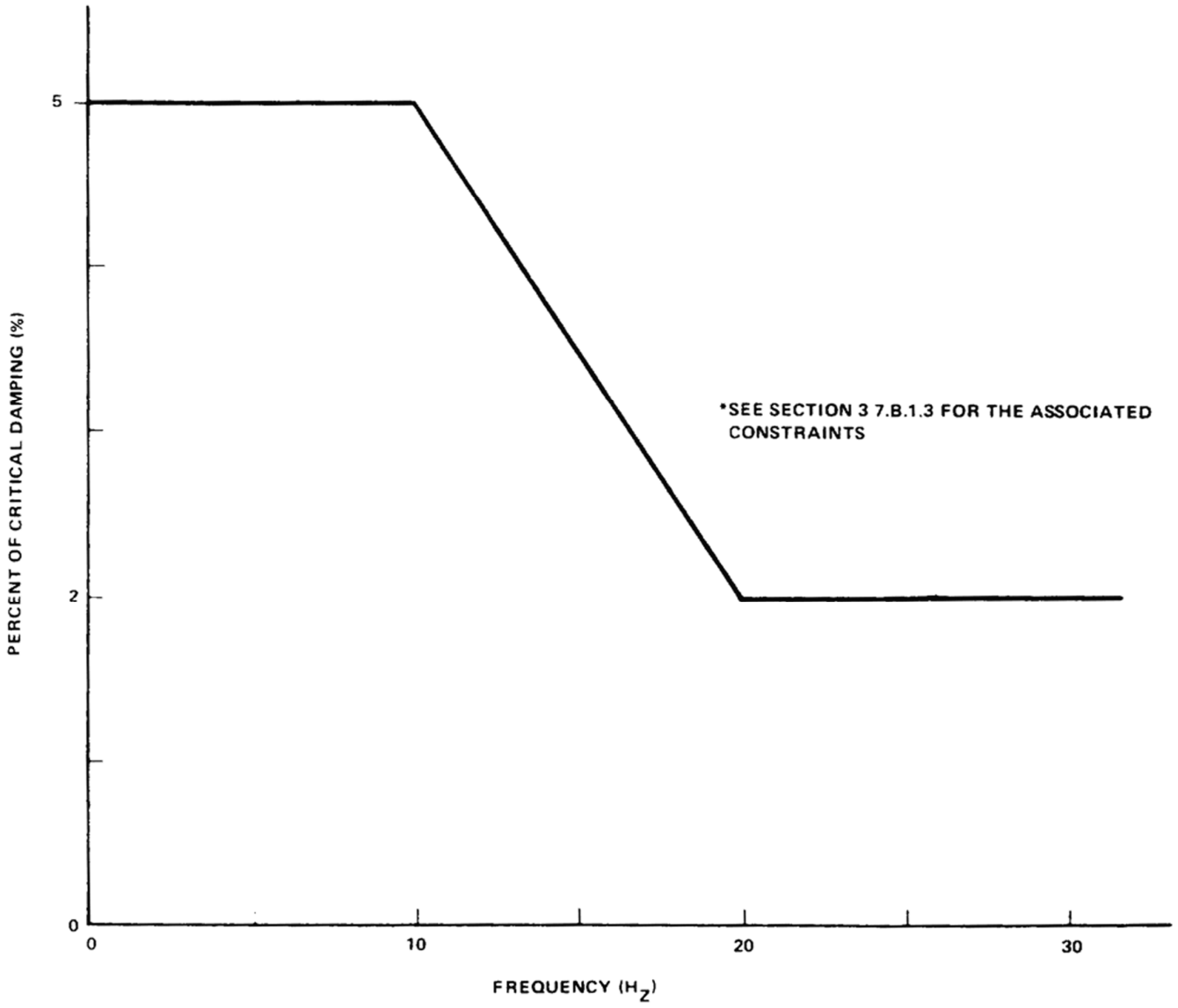
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

DAMPING RATIOS VS SHEAR STRAIN
FOR LOWER SAND STRATUM

FIGURE 3.7.B.1–10



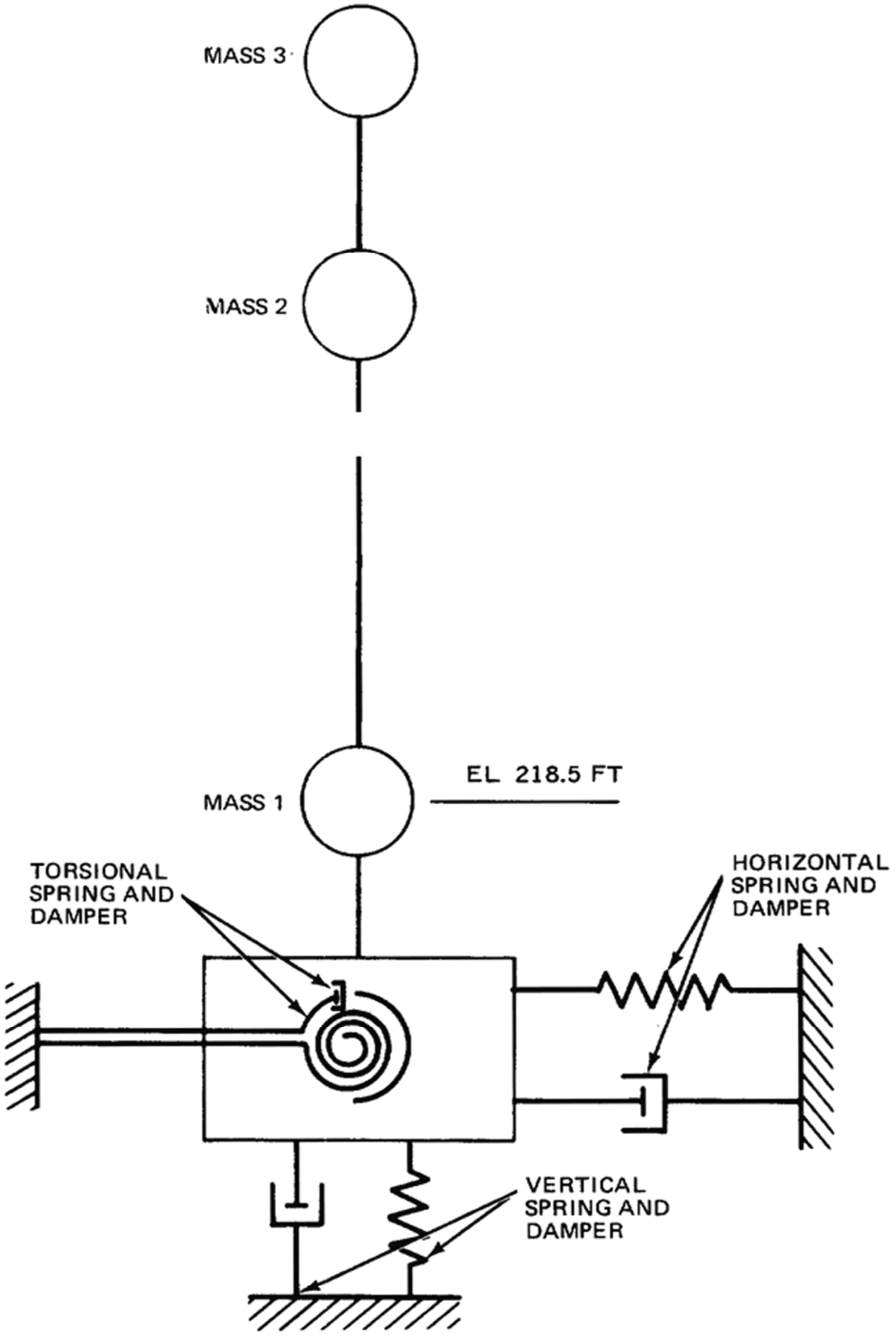
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

ALTERNATE DAMPING VALUES FOR ASME B&PV CODE, SECTION III, DIVISION 1, CODE CLASS 1, 2 AND 3 PIPING SYSTEMS

FIGURE 3.7.B.1–11



THE MODEL SHOWN IS TWO DIMENSIONAL FOR CLARITY. THE ACTUAL MODEL USED IS THREE DIMENSIONAL, WITH 6 (3 TRANSLATIONAL AND 3 ROTATIONAL) DEGREES OF FREEDOM PER NODE.

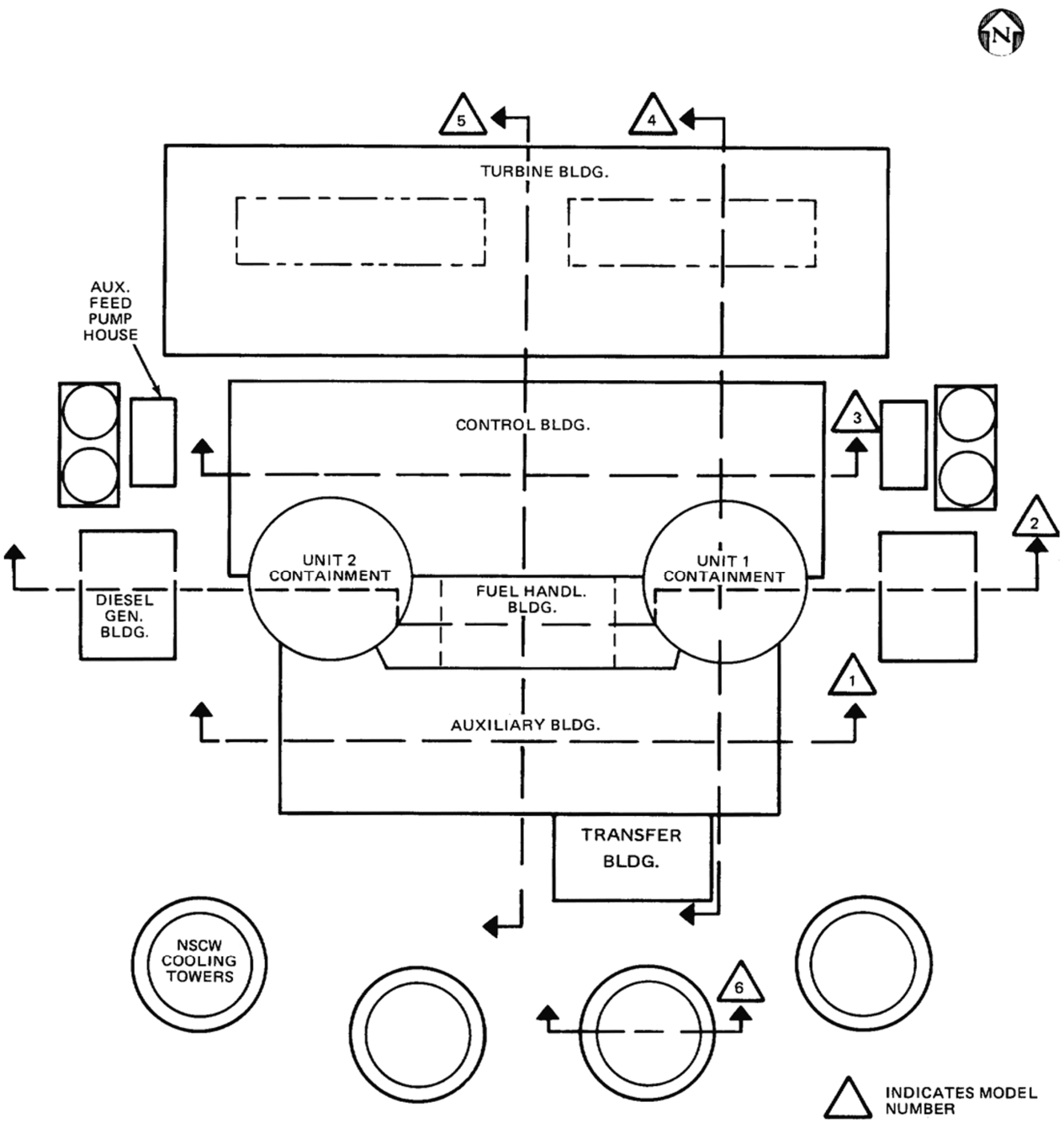
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LUMPED PARAMETER MODEL OF DIESEL GENERATOR BUILDING

FIGURE 3.7.B.2–1



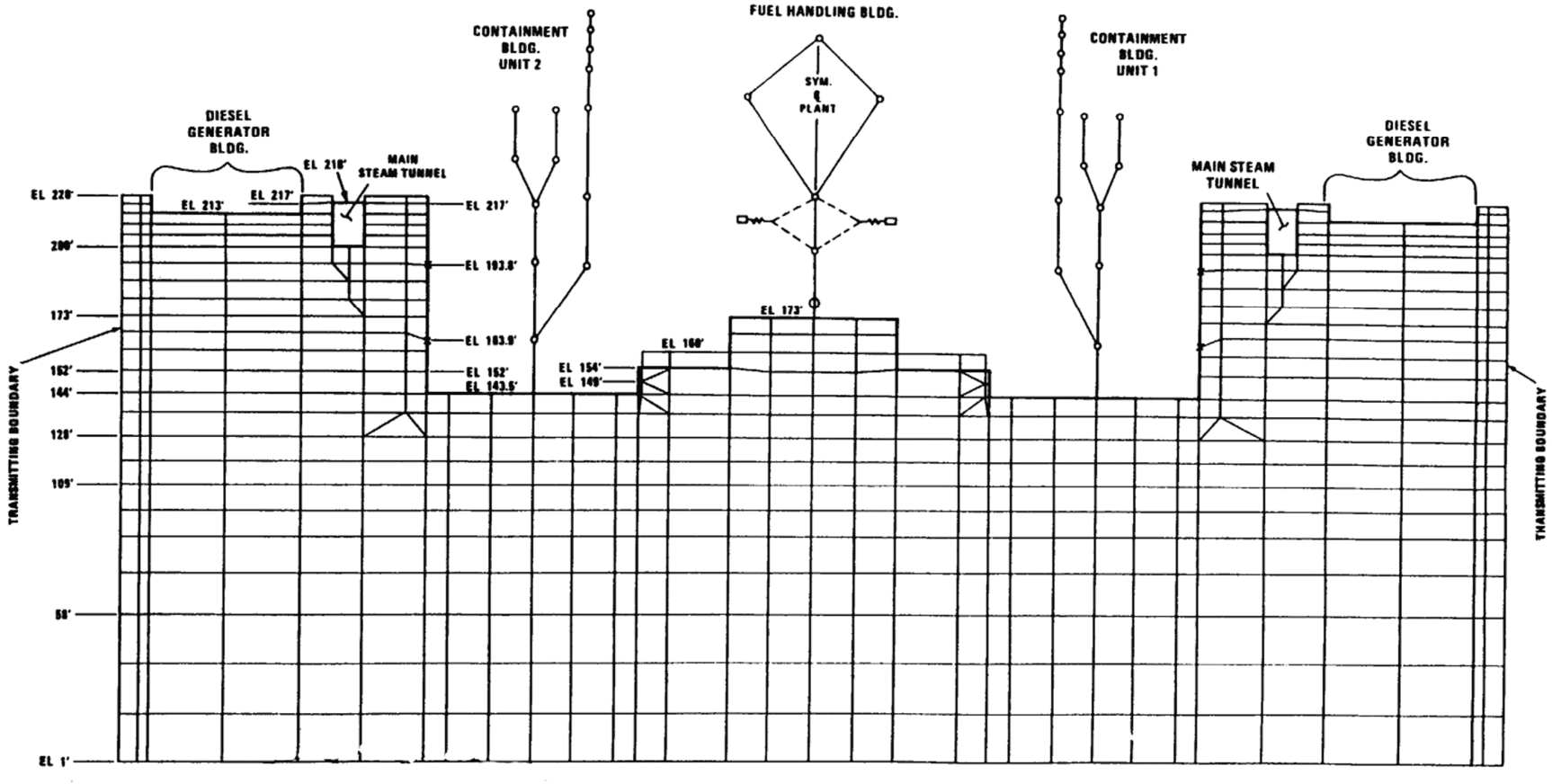
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

POWER BLOCK PLAN VIEW SHOWING SECTIONS FOR FINITE ELEMENT SOIL–STRUCTURE INTERACTION FLUSH MODELS

FIGURE 3.7.B.2–2



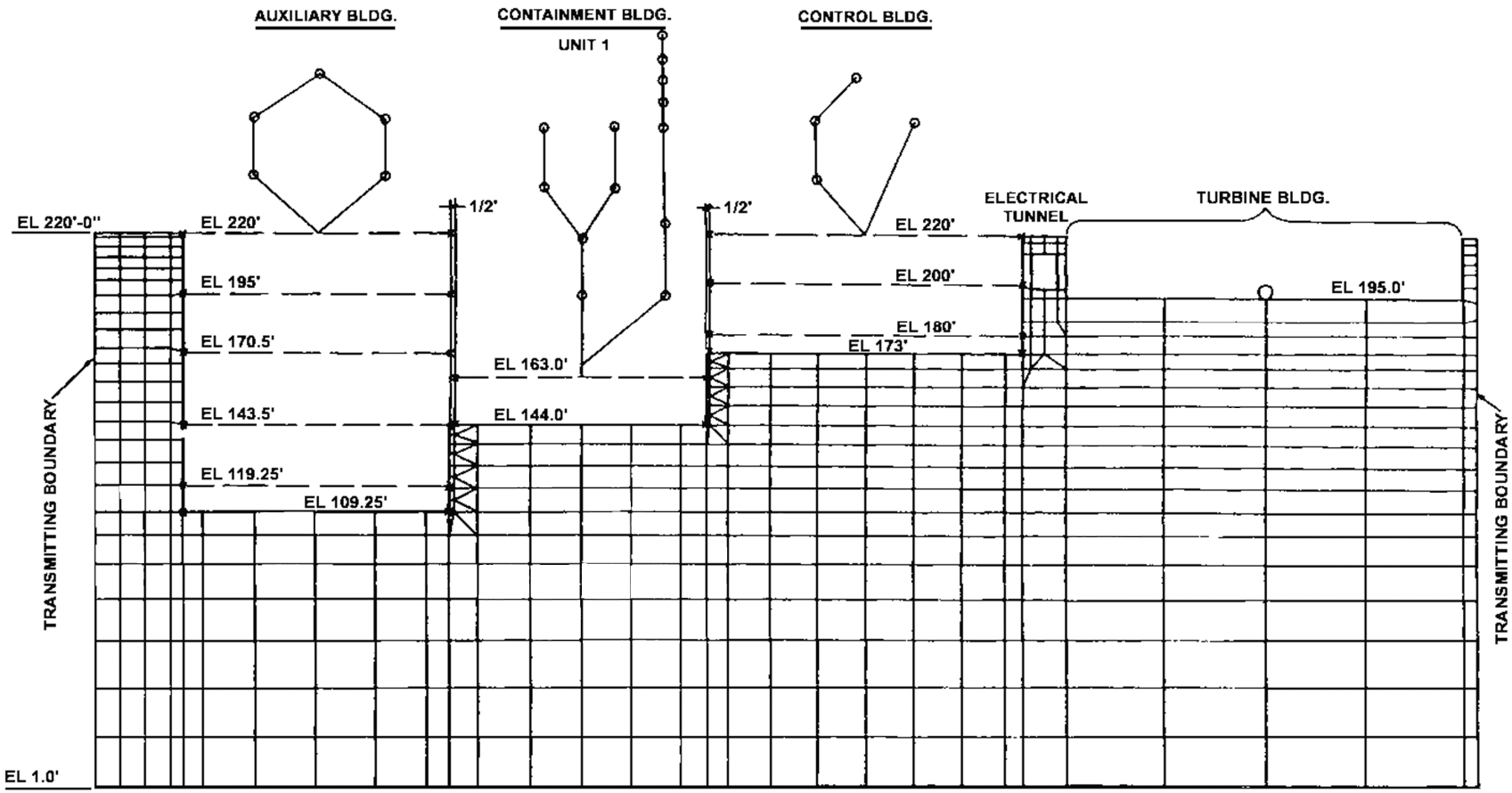
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

FLUSH MODEL SECTION 2
OF FIGURE 3.7.B.2–2

FIGURE 3.7.B.2–3



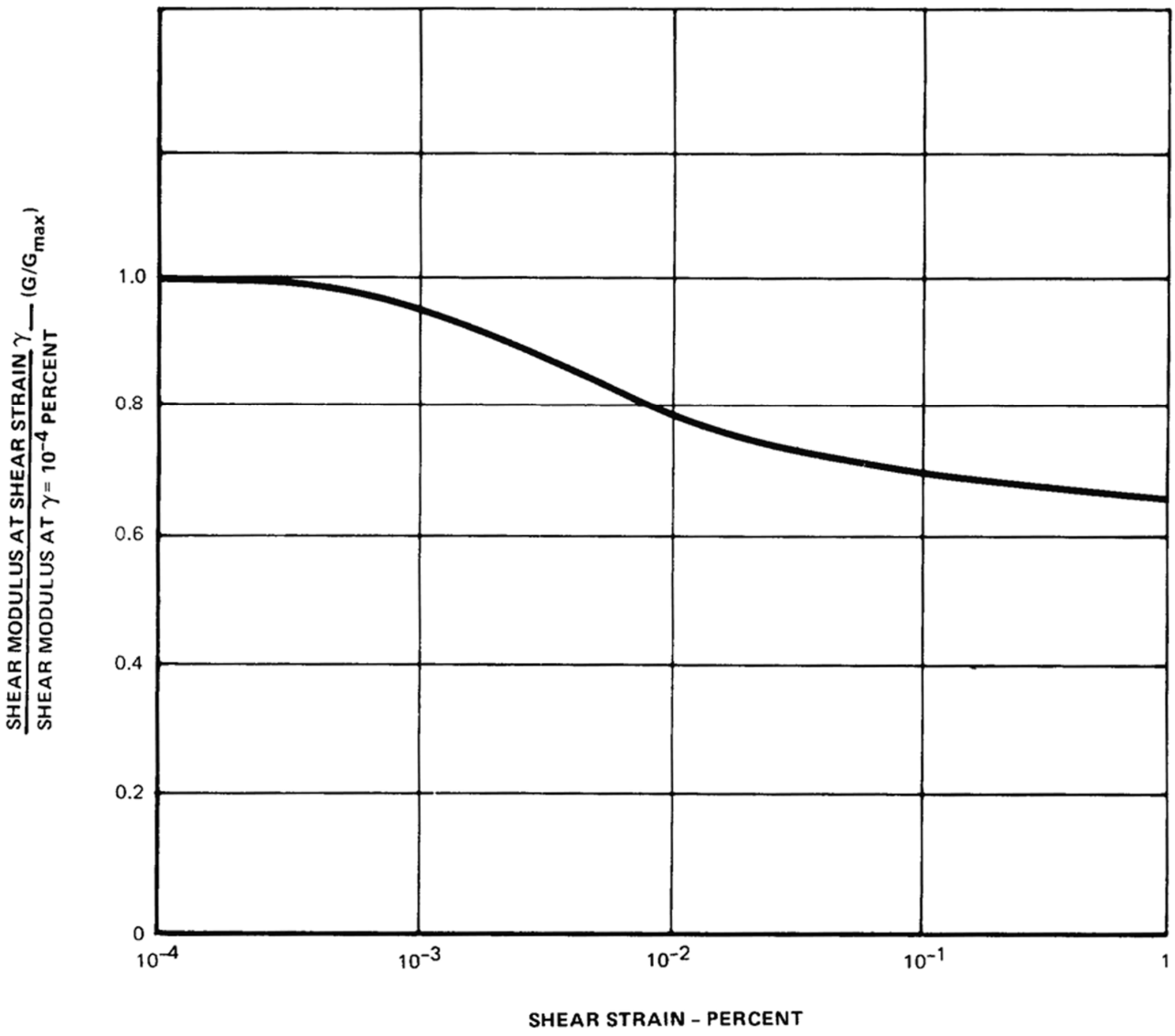
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

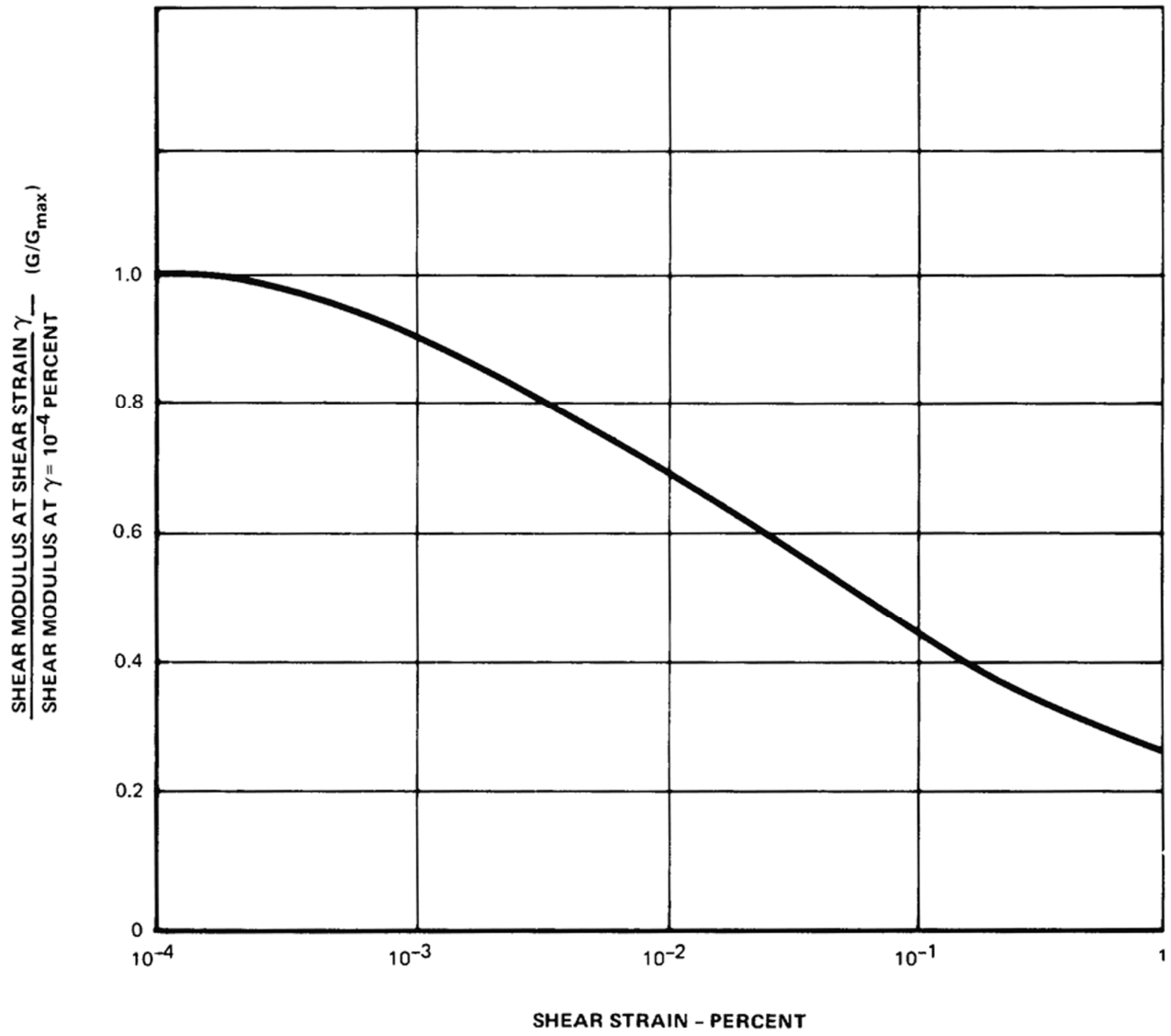
FLUSH MODEL ALONG SECTION 4
OF FIGURE 3.7.B.2–2

FIGURE 3.7.B.2–4



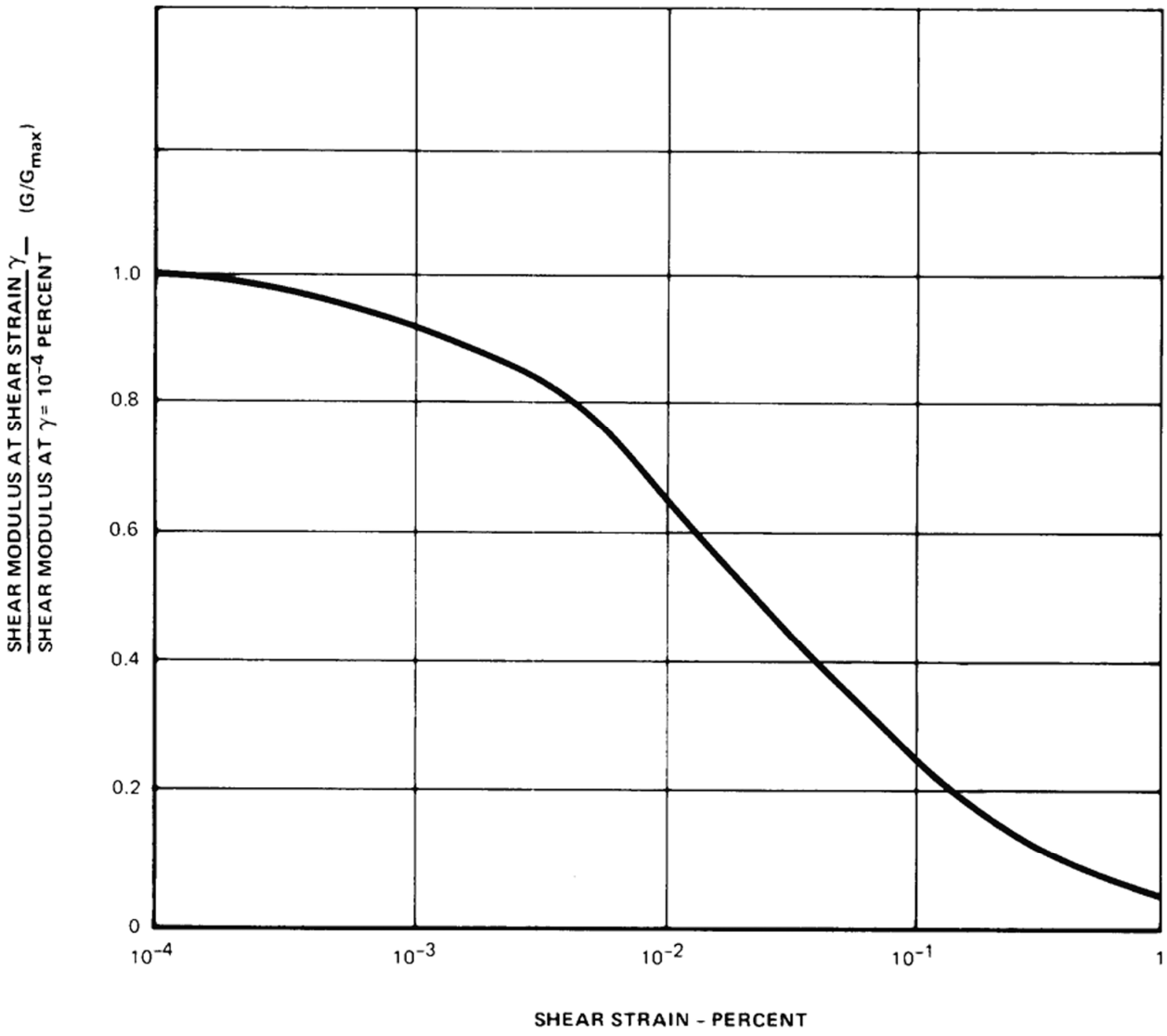
REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	G/G_{MAX} VS SHEAR STRAIN FOR COMPACTED SAND BACKFILL
	FIGURE 3.7.B.2–5



REV 14 10/07

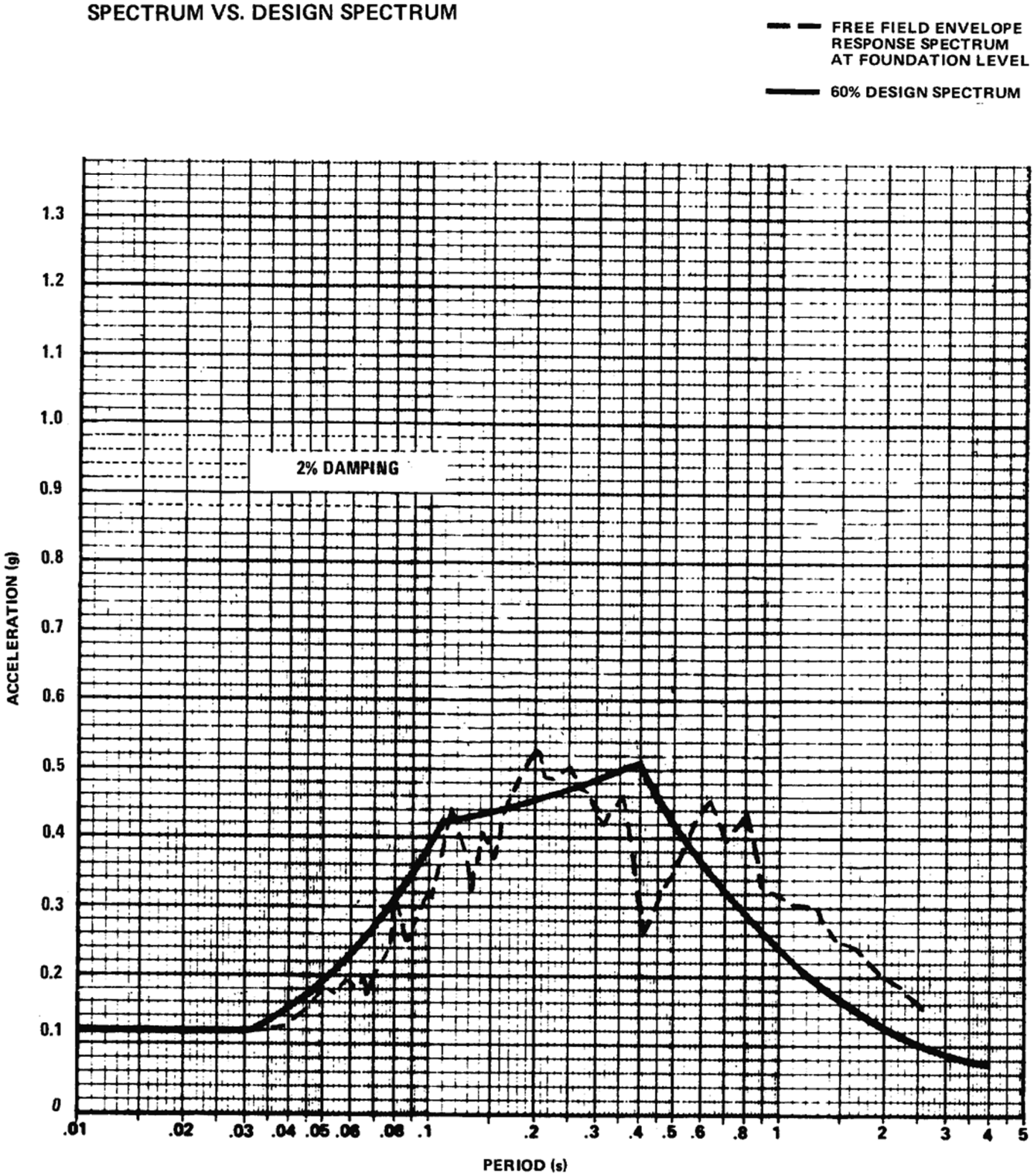
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	G/G_{MAX} VS SHEAR STRAIN FOR SATURATED CLAY MARL–BEARING STRATUM
	FIGURE 3.7.B.2–6



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	G/G_{MAX} VS SHEAR STRAIN FOR LOWER SAND STRATUM
	FIGURE 3.7.B.2–7

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

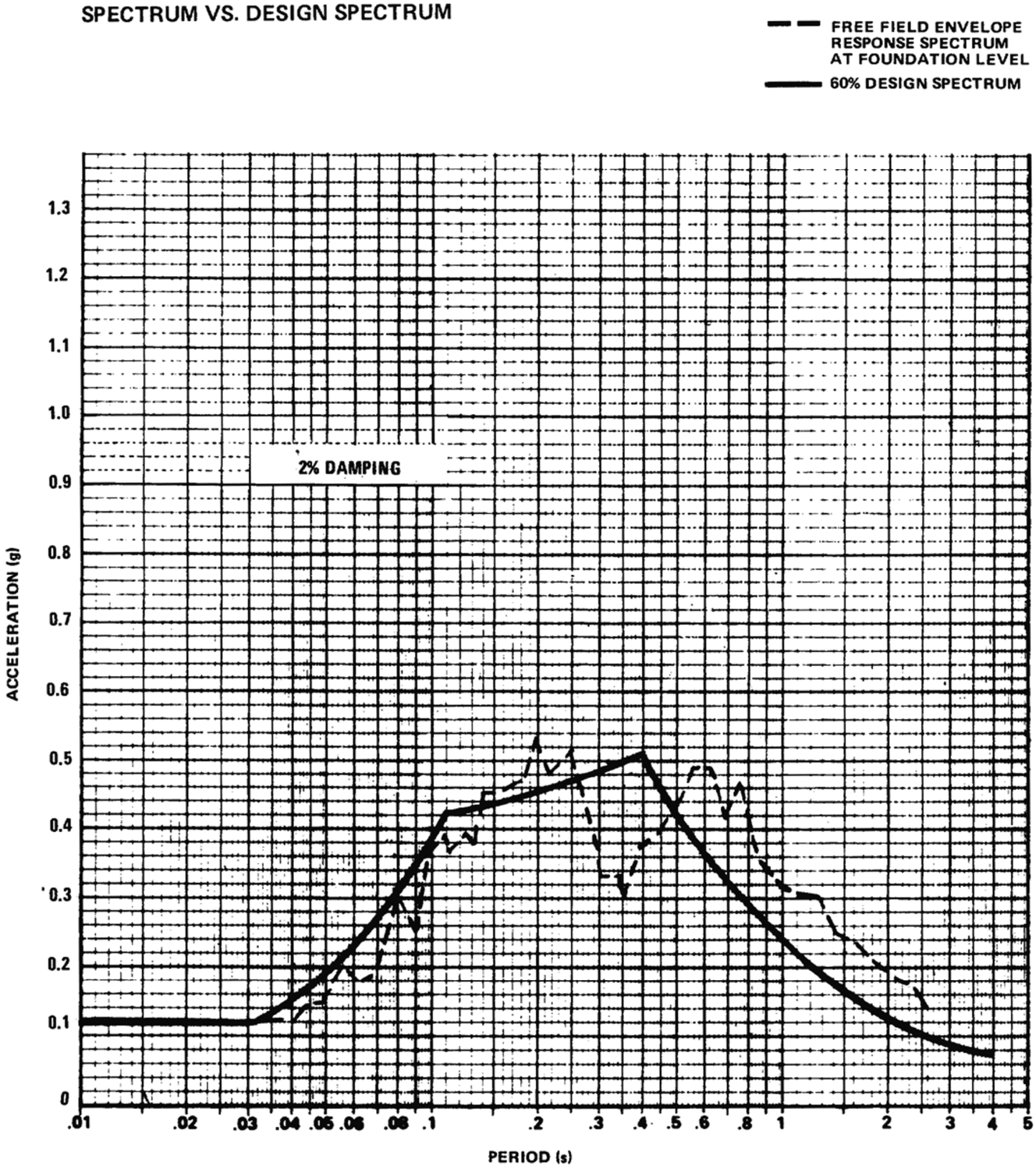
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AUXILIARY BUILDING
el 109 ft 0 in., SSE HORIZONTAL

FIGURE 3.7.B.2–8

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

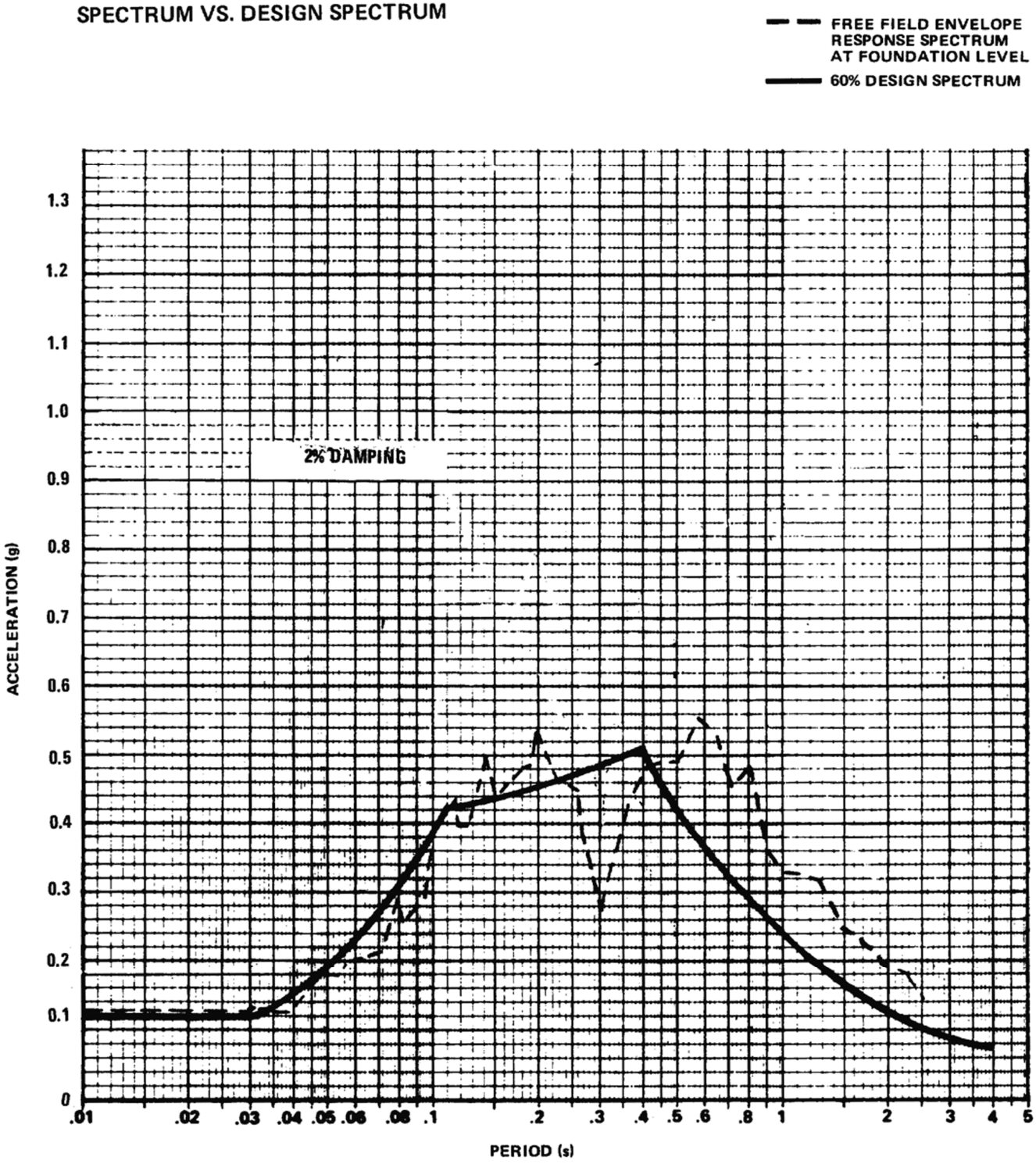
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

NUCLEAR SERVICE COOLING WATER TOWER
el 128 ft 0 in., SSE HORIZONTAL

FIGURE 3.7.B.2–9

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

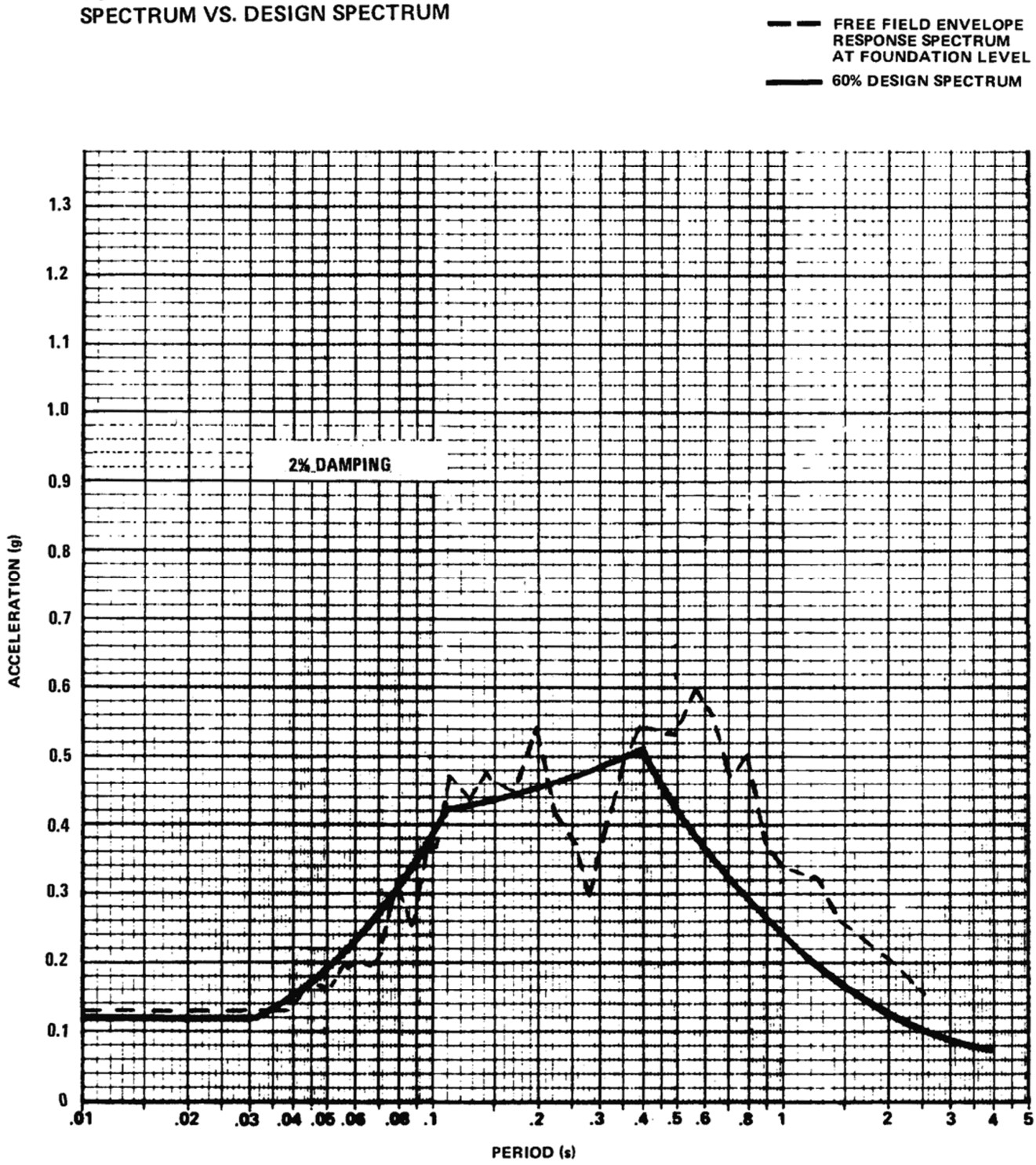
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT BUILDING
el 144 ft 0 in., SSE HORIZONTAL

FIGURE 3.7.B.2–10

COMPARISON – FREE FIELD
SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

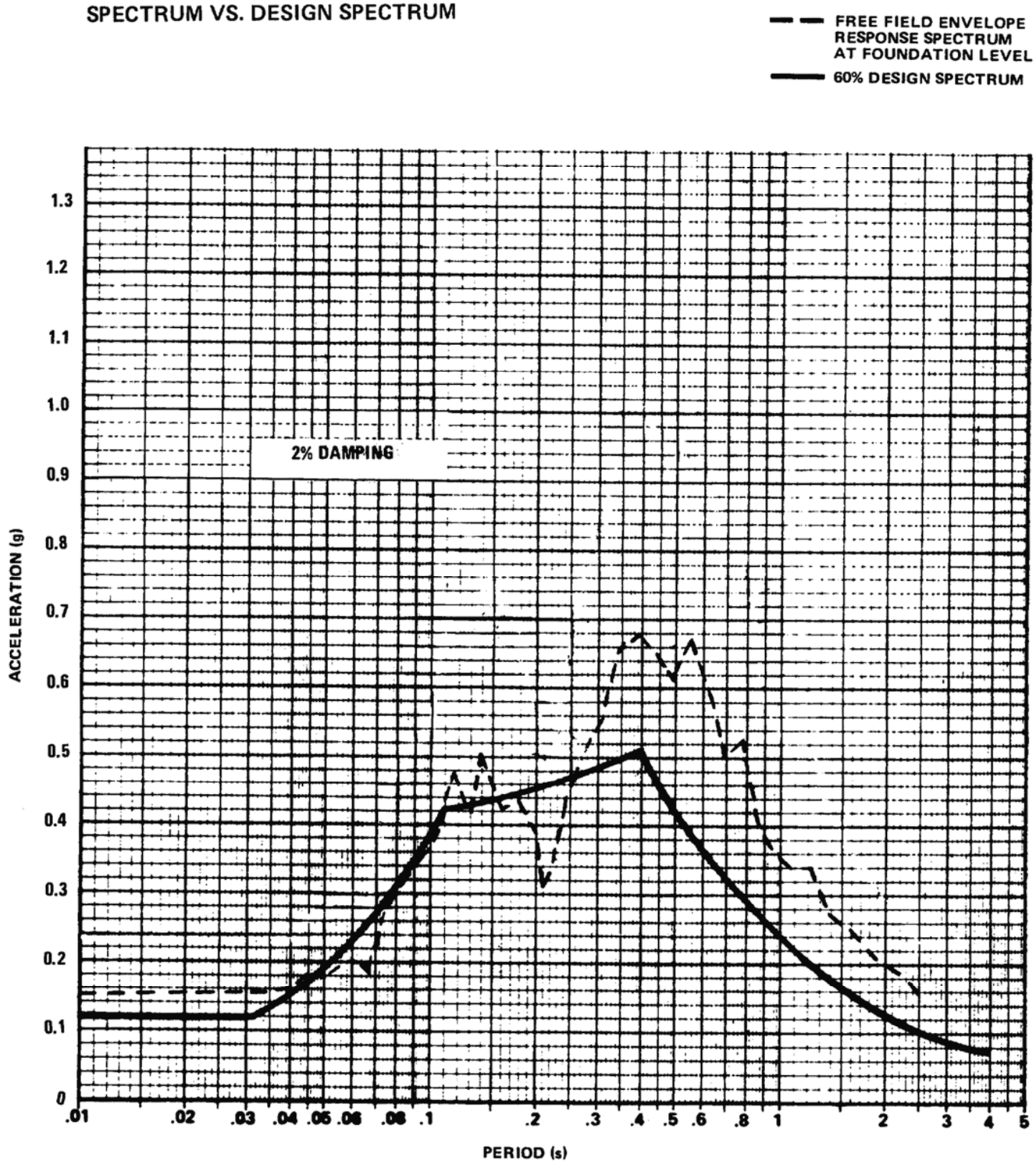
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

FUEL HANDLING BUILDING
el 152 ft 0 in., SSE HORIZONTAL

FIGURE 3.7.B.2–11

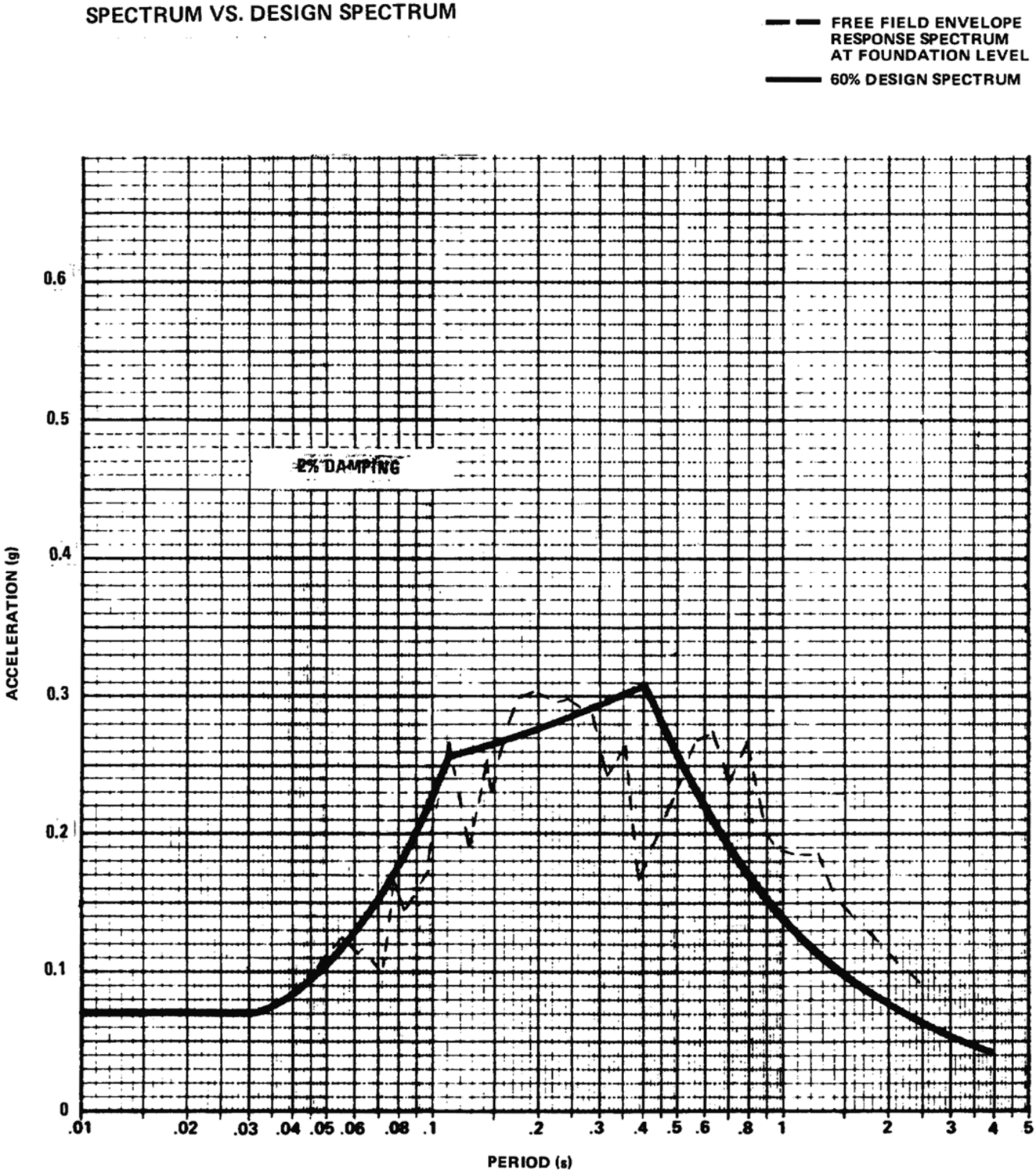
COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	CONTROL BUILDING el 173 ft 0 in., SSE HORIZONTAL
	FIGURE 3.7.B.2–12

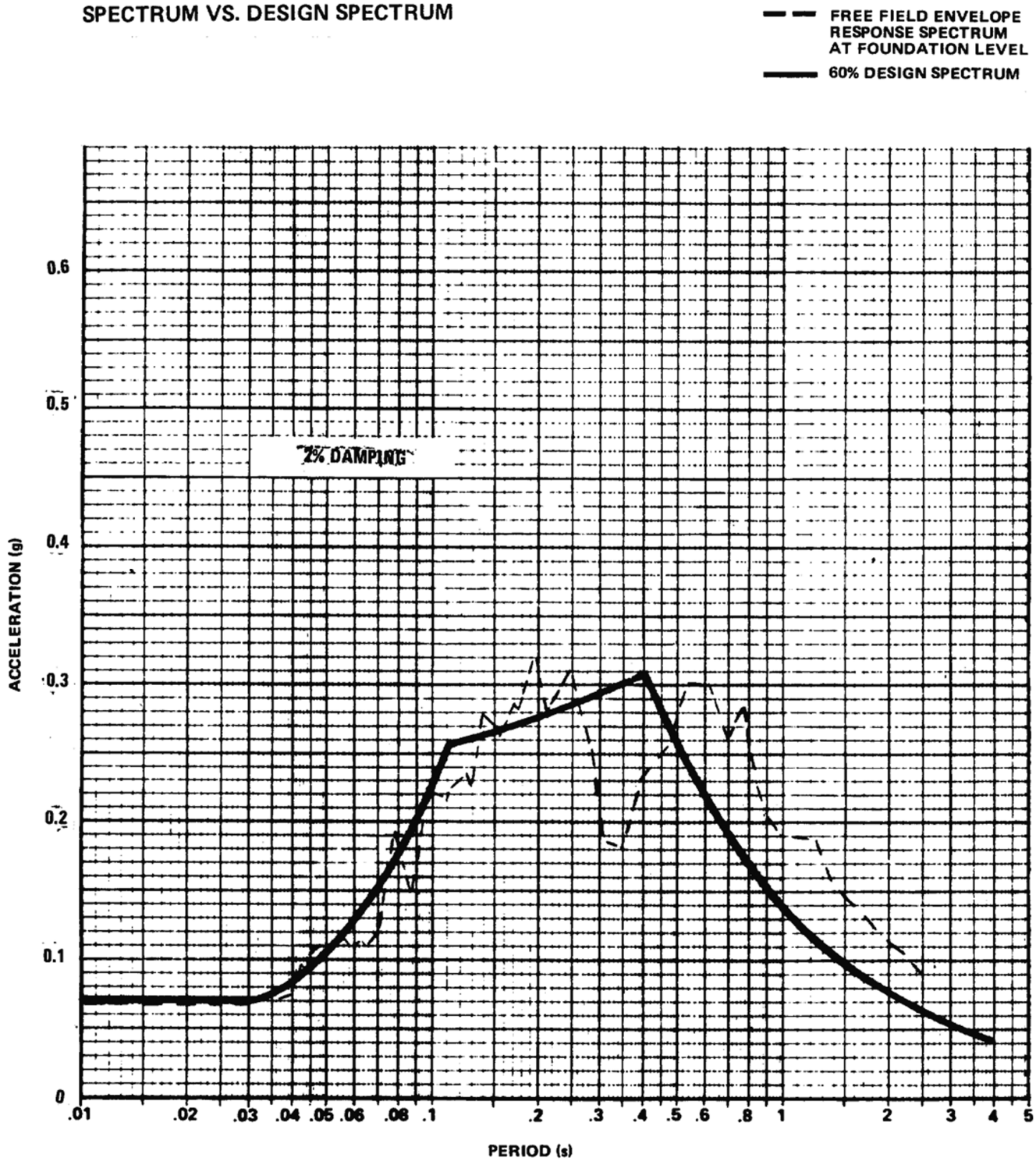
COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	AUXILIARY BUILDING el 109 ft 0 in., OBE HORIZONTAL
	FIGURE 3.7.B.2–13

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

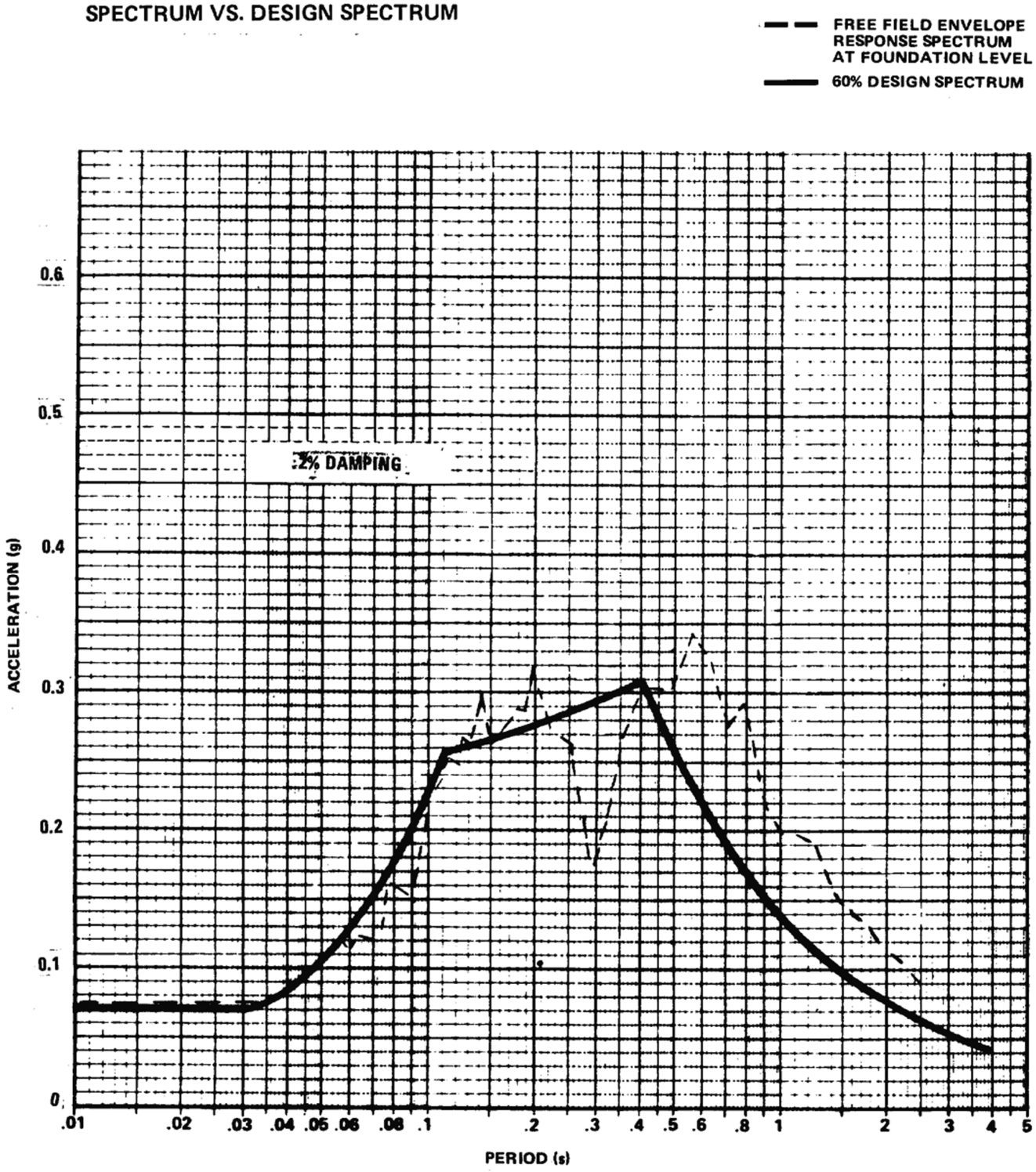
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

NUCLEAR SERVICE COOLING WATER TOWER
el 128 ft 0 in., OBE HORIZONTAL

FIGURE 3.7.B.2–14

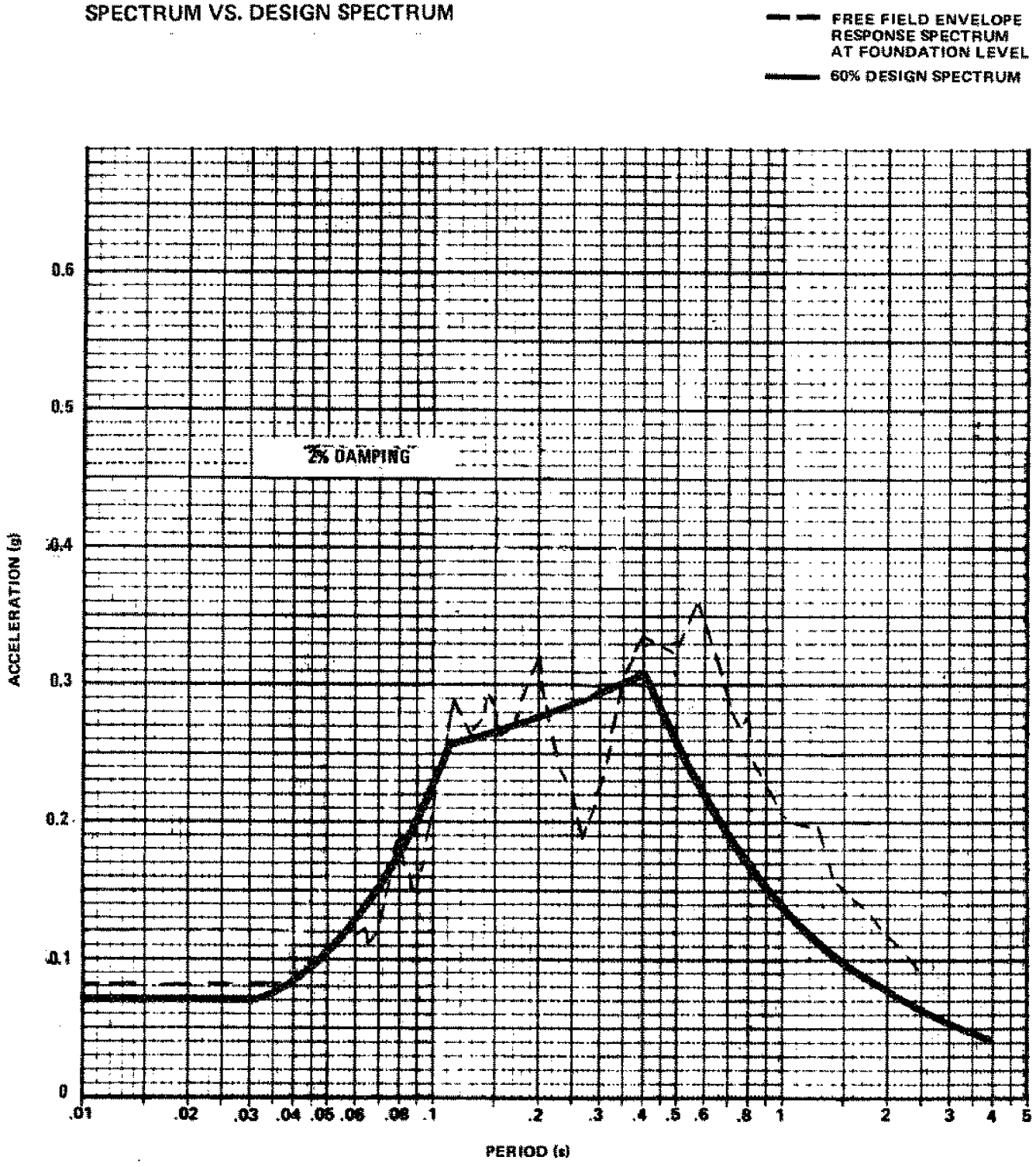
COMPARISON – FREE FIELD
SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT BUILDING el 144 ft 0 in., OBE HORIZONTAL
	FIGURE 3.7.B.2–15

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

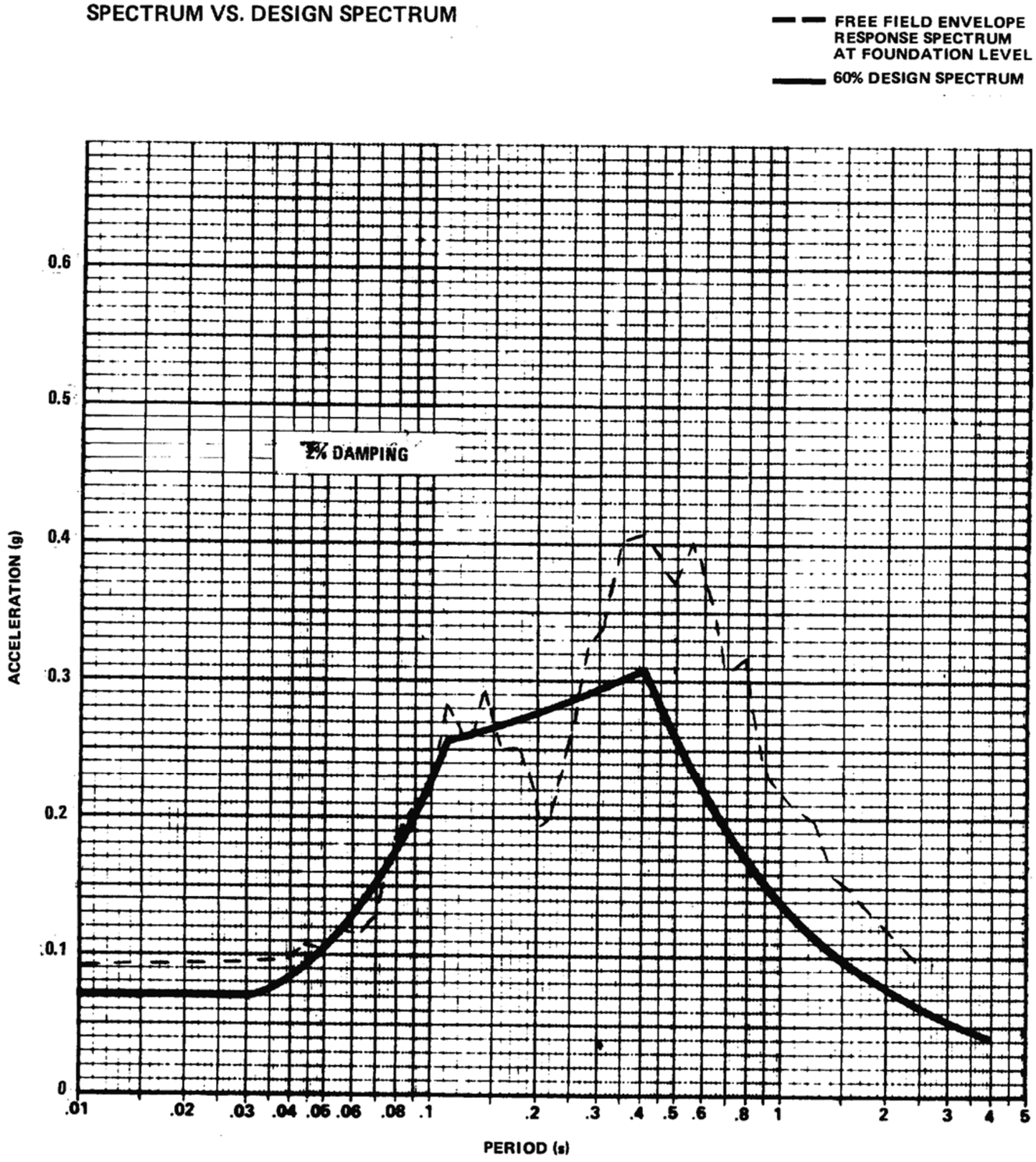
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

FUEL HANDLING BUILDING
el 152 ft 0 in., OBE HORIZONTAL

FIGURE 3.7.B.2–16

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

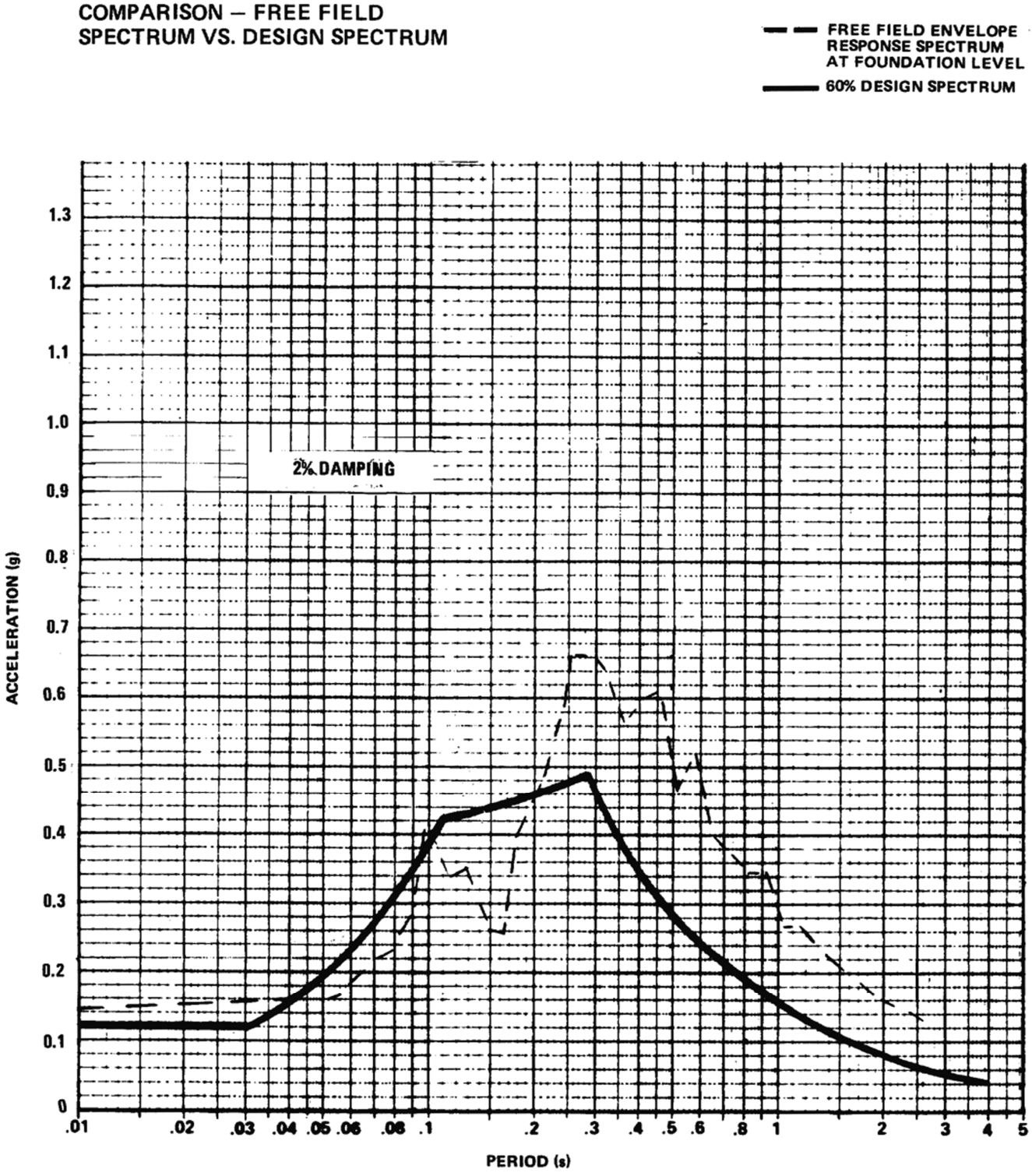
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTROL BUILDING
el 173 ft 0 in., OBE HORIZONTAL

FIGURE 3.7.B.2–17

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

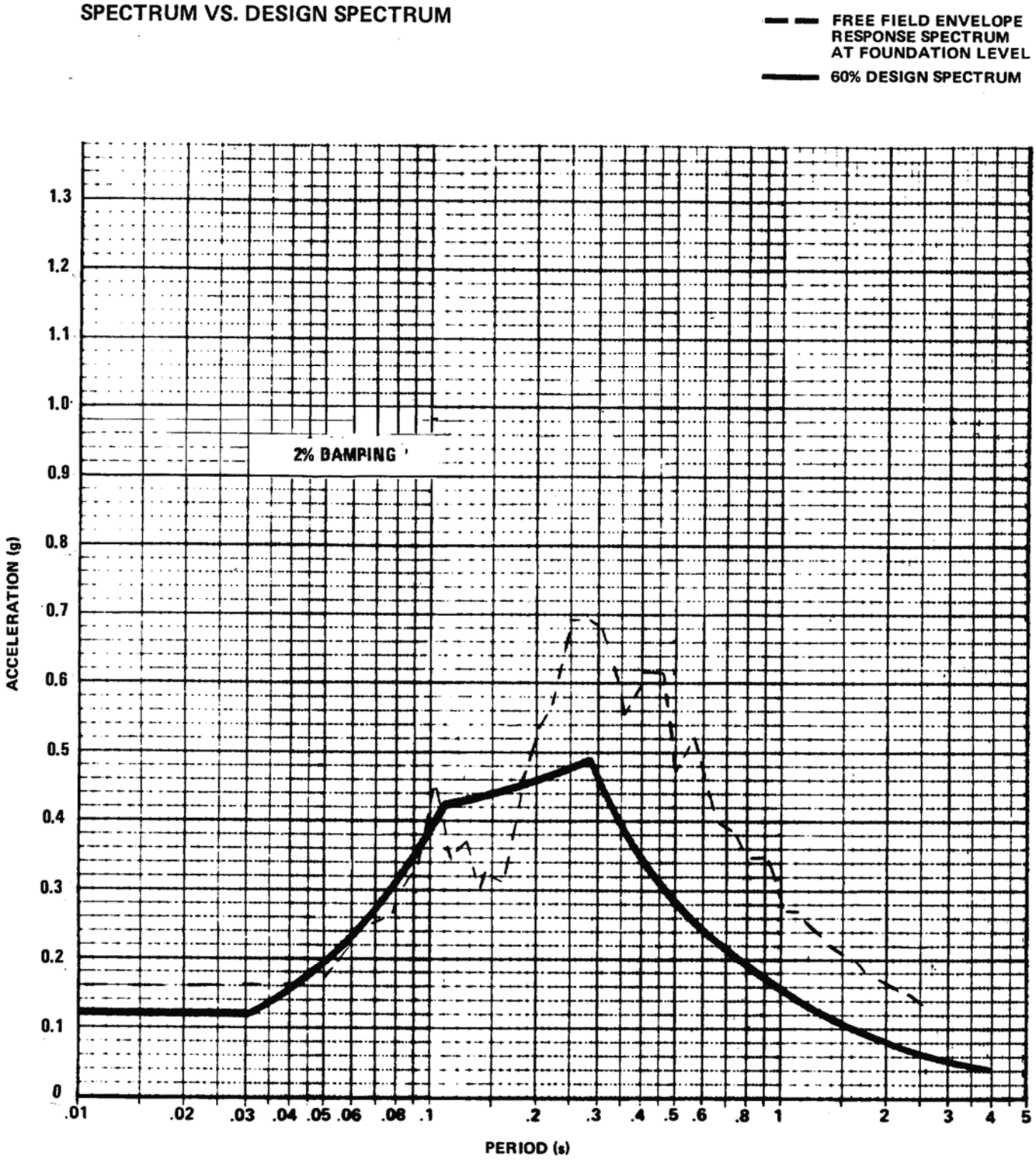
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AUXILIARY BUILDING
el 109 ft 0 in., SSE VERTICAL

FIGURE 3.7.B.2–18

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

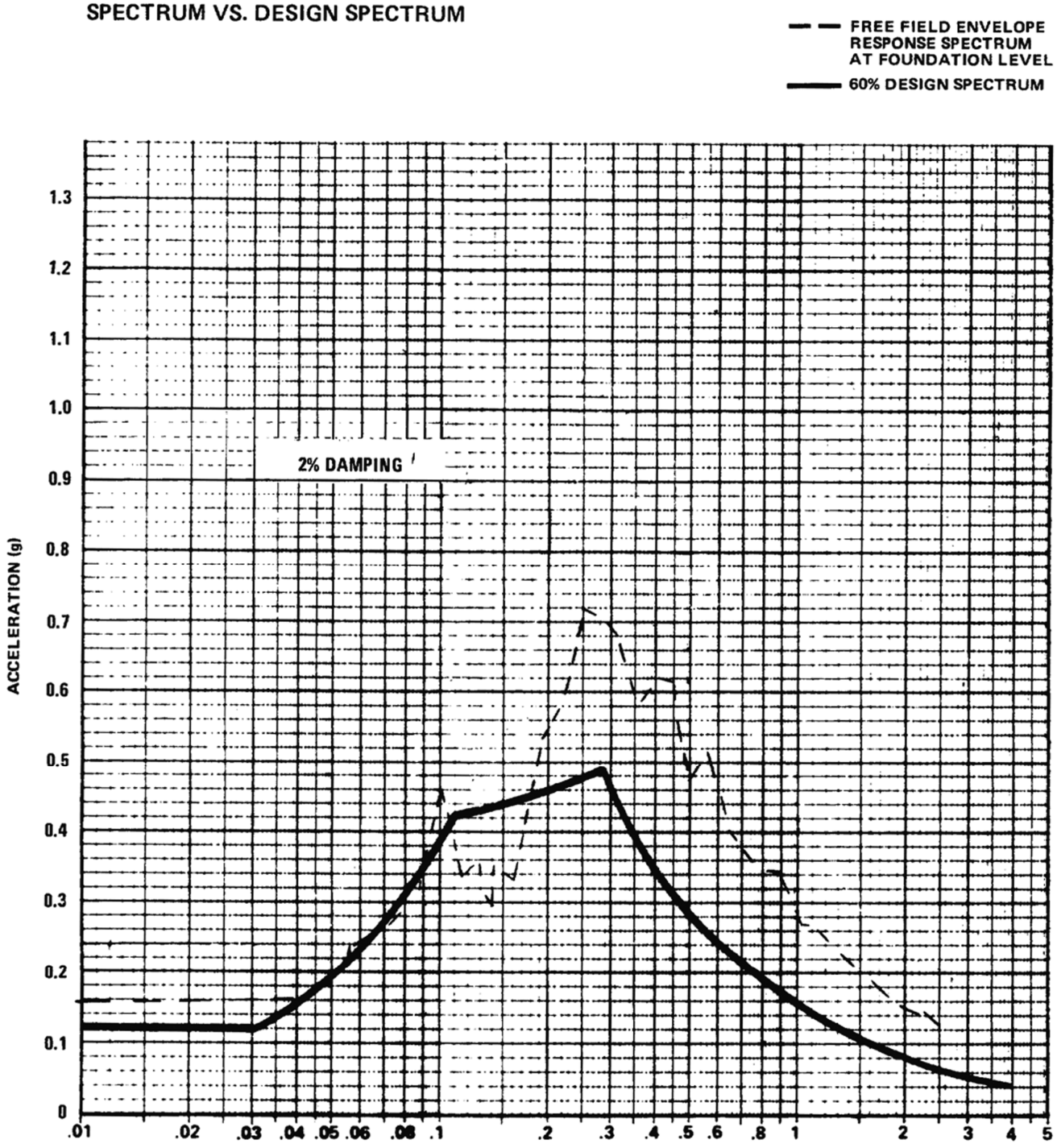
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

NUCLEAR SERVICE COOLING WATER TOWER
el 128 ft 0 in., SSE VERTICAL

FIGURE 3.7.B.2–19

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

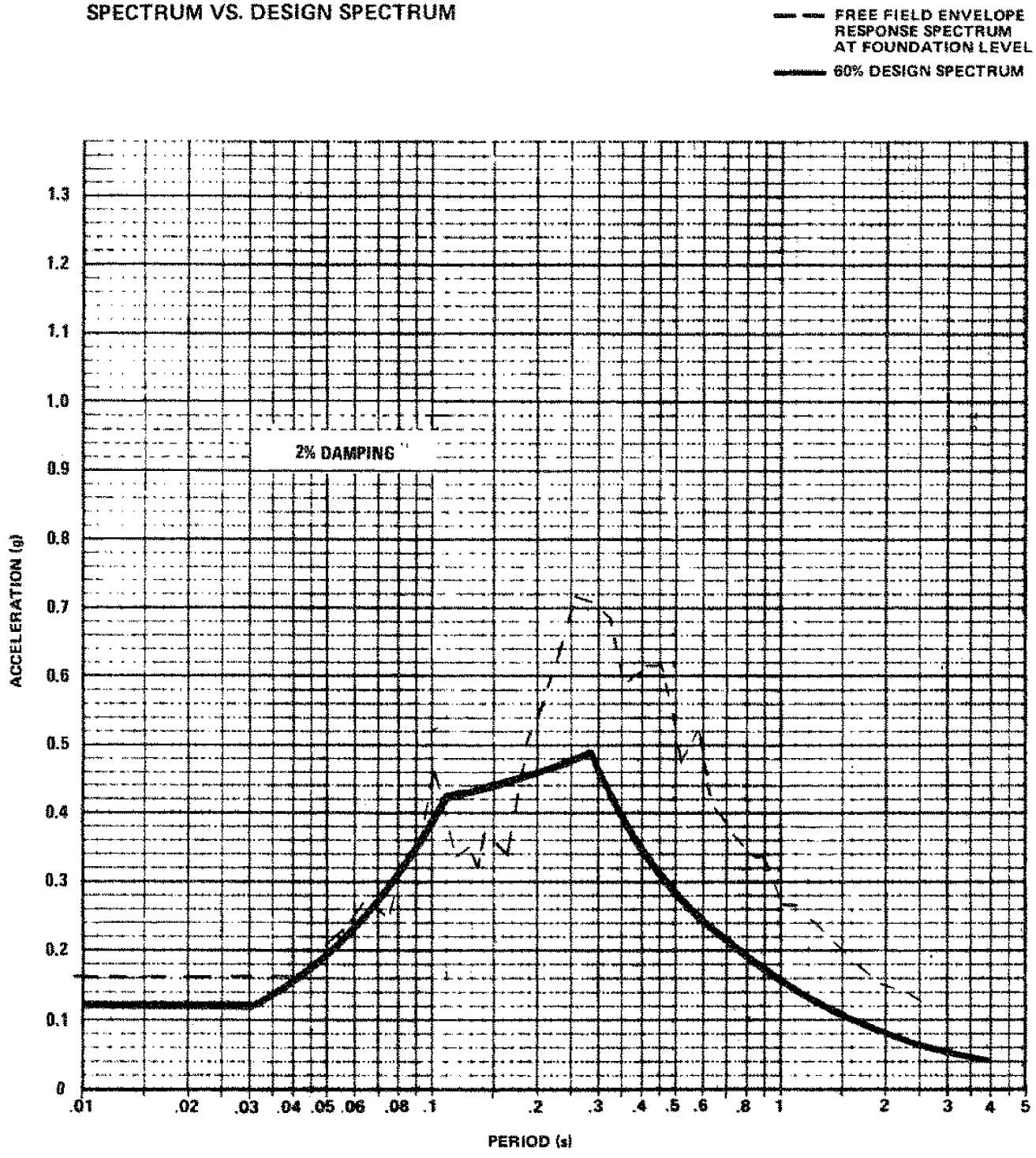
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT BUILDING
el 144 ft 0 in., SSE VERTICAL

FIGURE 3.7.B.2–20

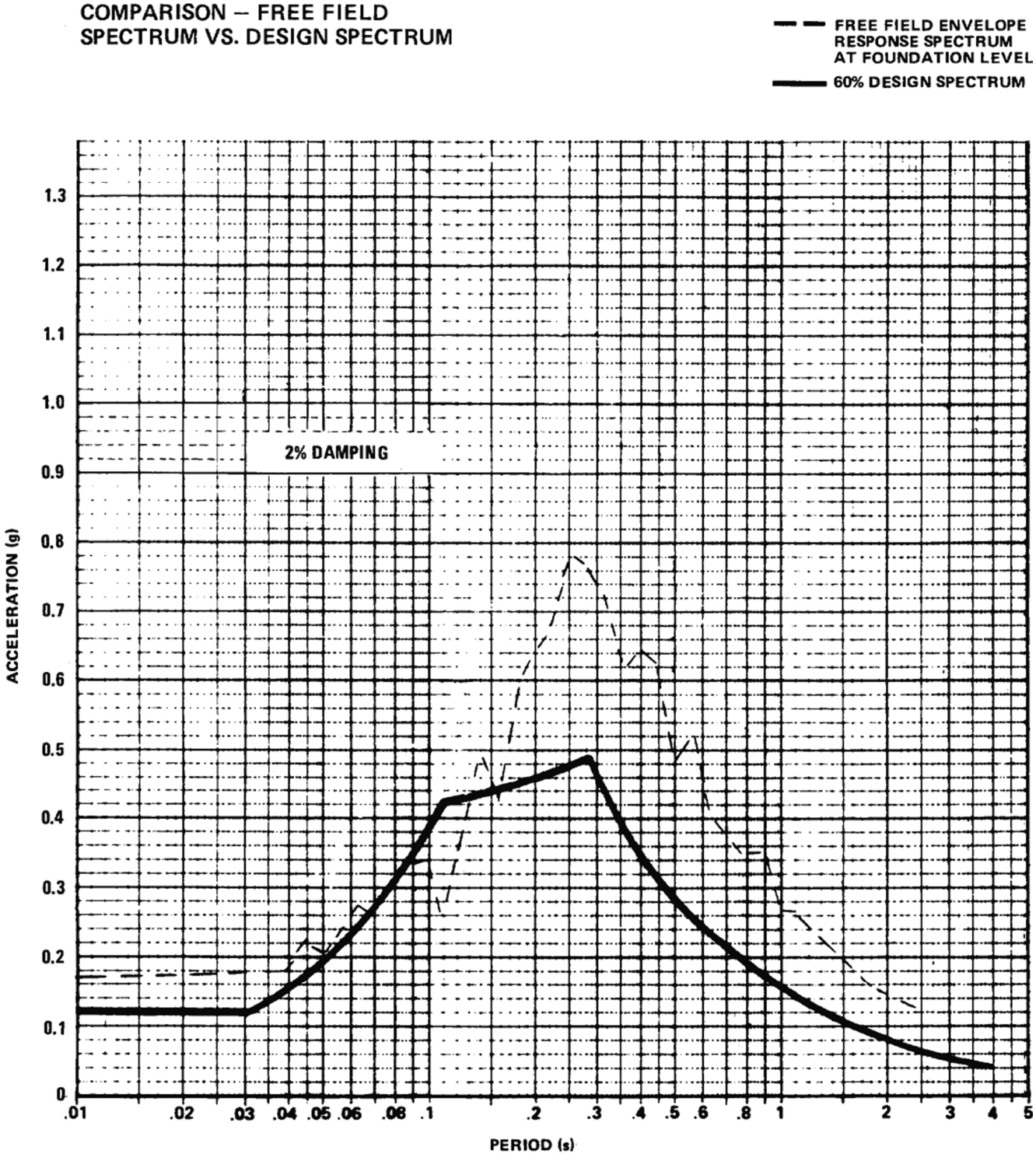
COMPARISON – FREE FIELD
SPECTRUM VS. DESIGN SPECTRUM



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	FUEL HANDLING BUILDING el 152 ft 0 in., SSE VERTICAL
	FIGURE 3.7.B.2–21

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



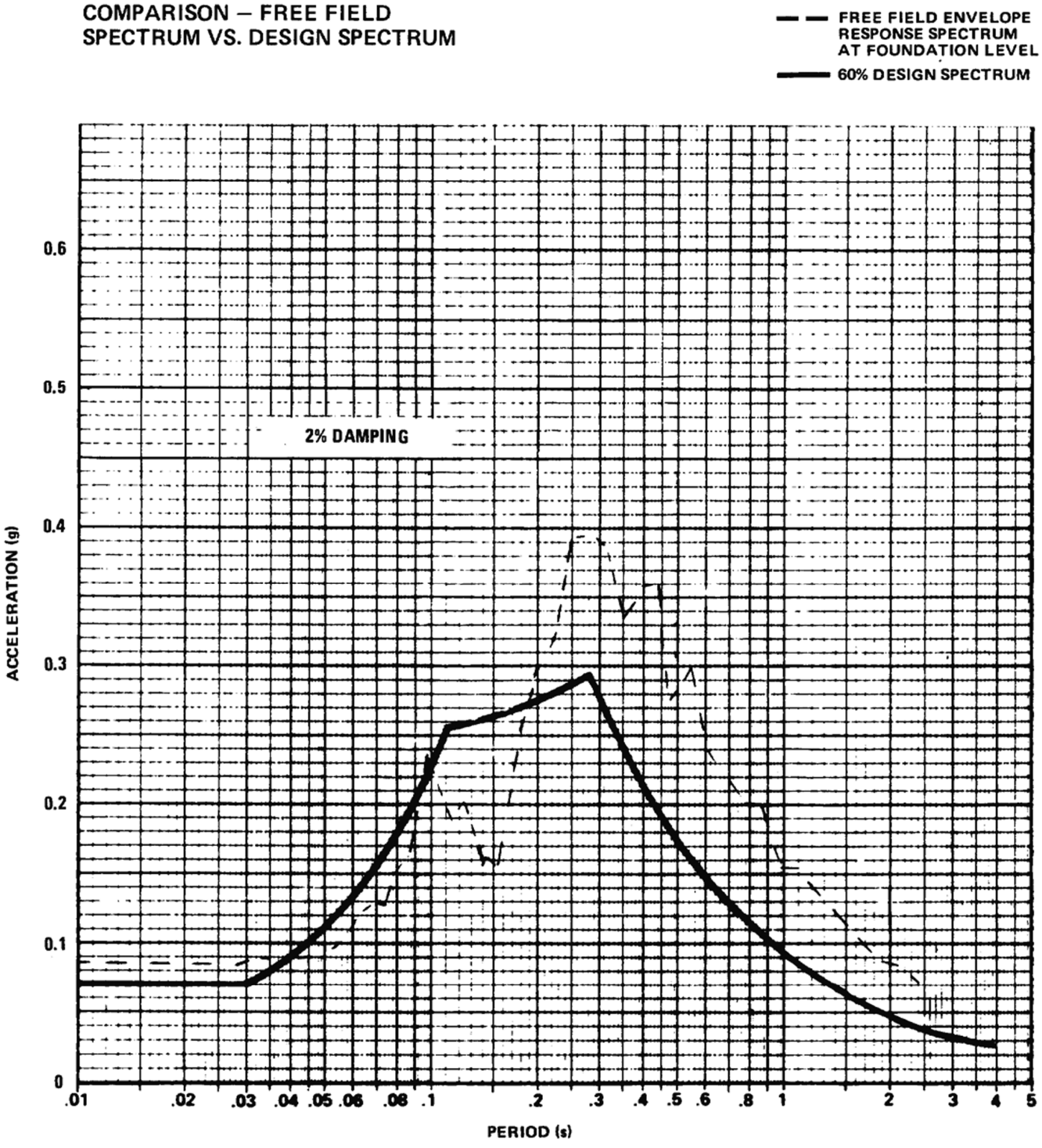
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTROL BUILDING
el 173 ft 0 in., SSE VERTICAL

FIGURE 3.7.B.2–22



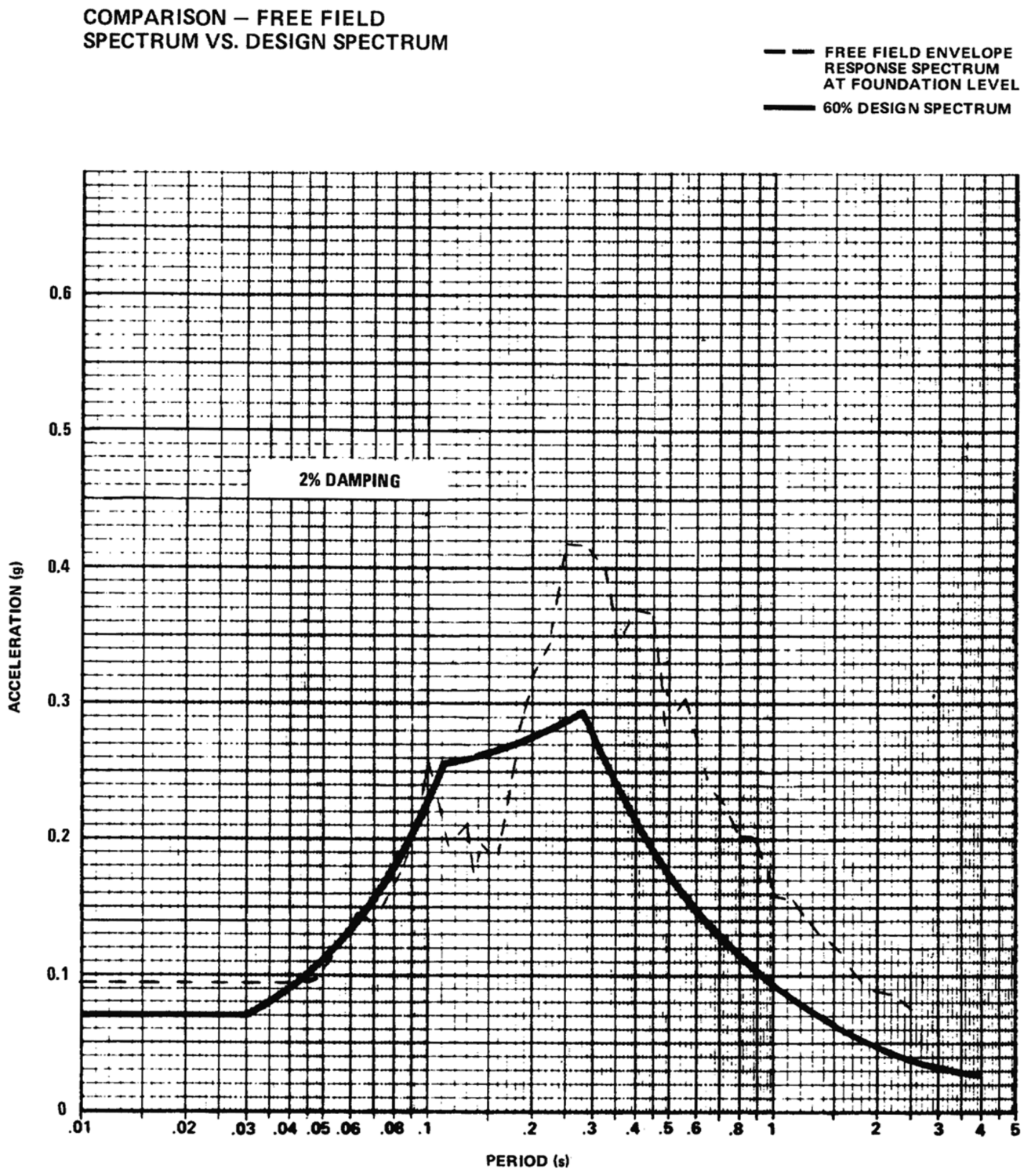
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

AUXILIARY BUILDING
el 109 ft 0 in., OBE VERTICAL

FIGURE 3.7.B.2–23



REV 14 10/07

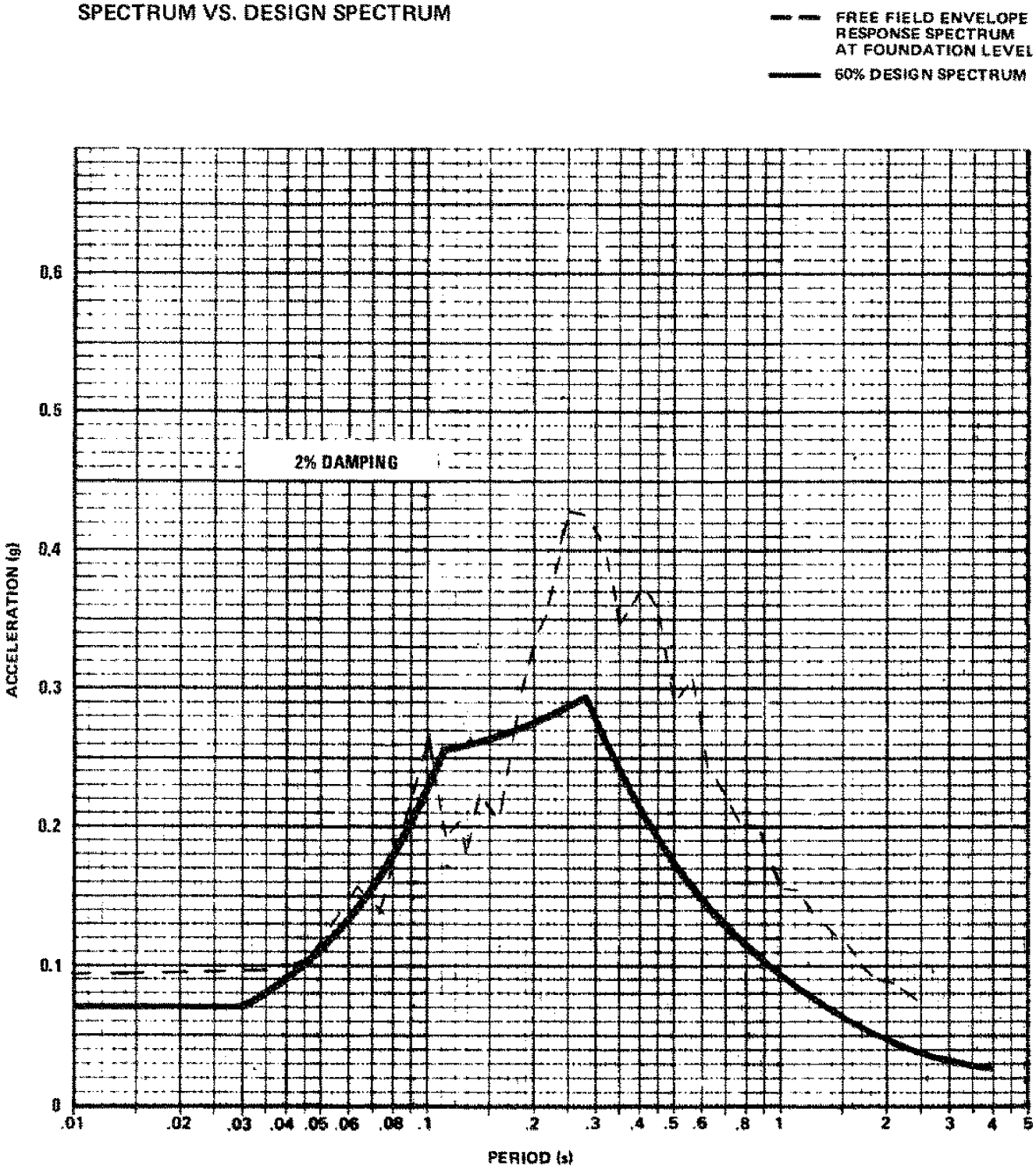
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

NUCLEAR SERVICE COOLING WATER TOWER
el 128 ft 0 in., OBE VERTICAL

FIGURE 3.7.B.2–24

COMPARISON – FREE FIELD
SPECTRUM VS. DESIGN SPECTRUM



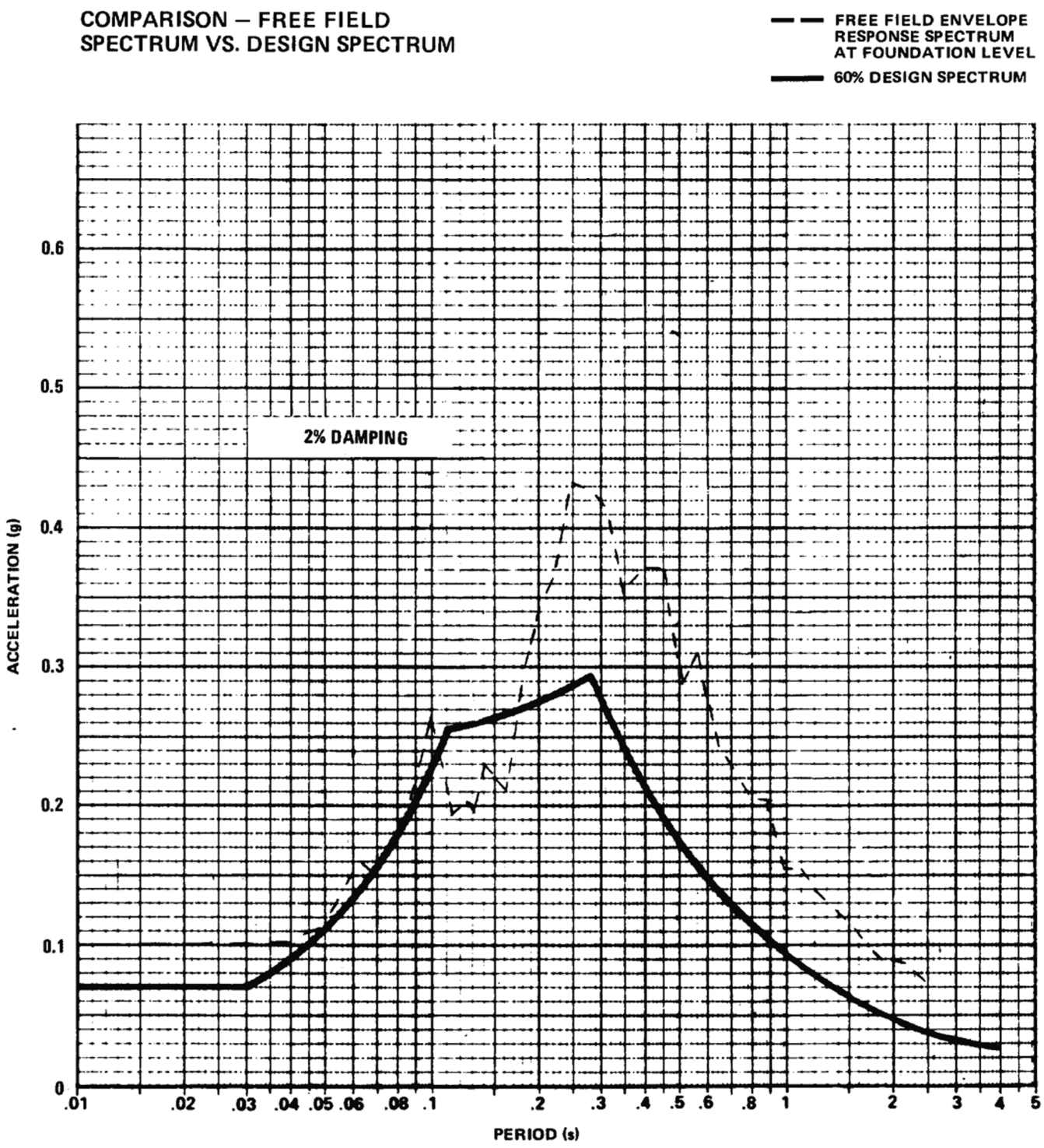
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT BUILDING
el 144 ft 0 in., OBE VERTICAL

FIGURE 3.7.B.2–25



REV 14 10/07

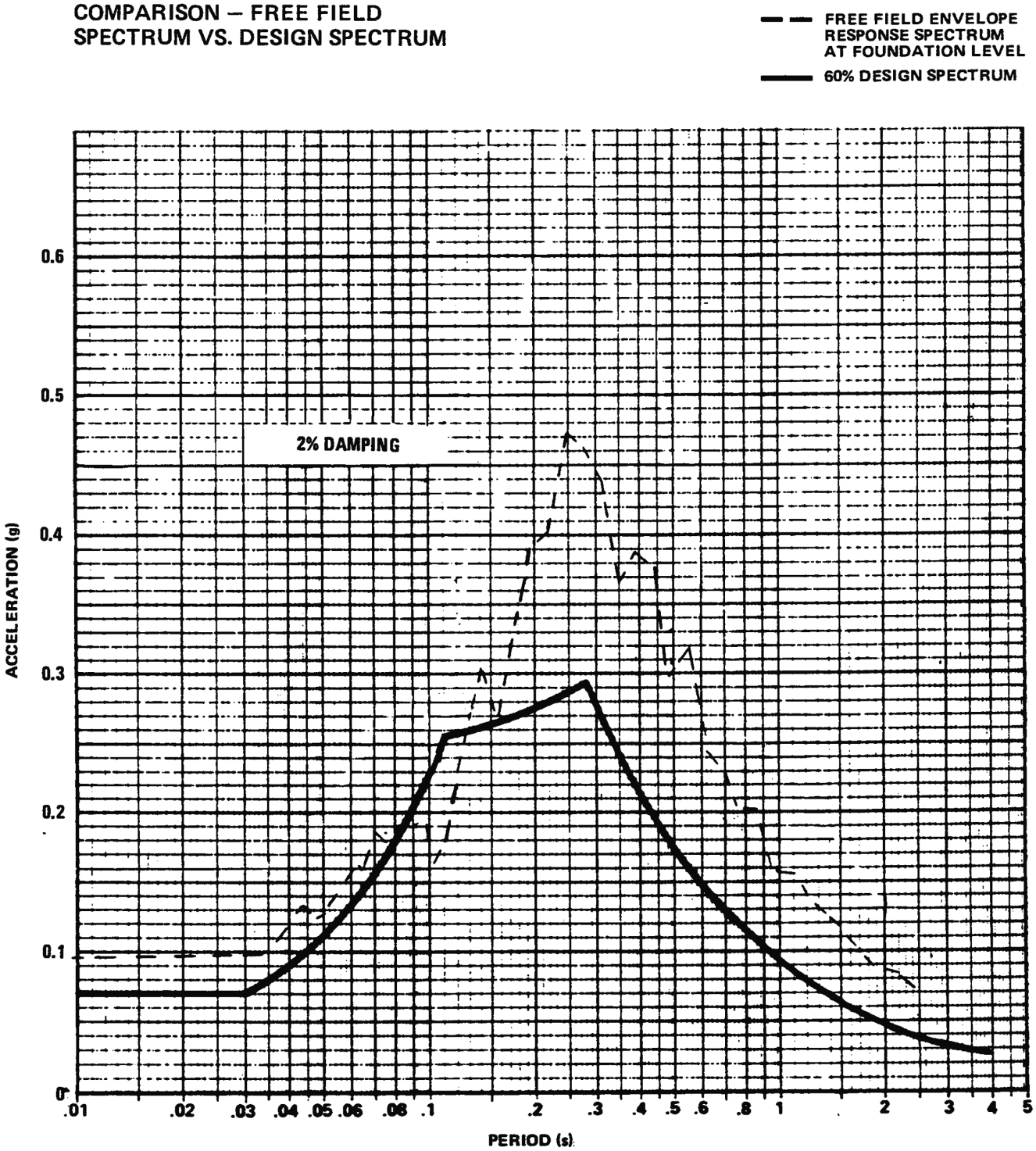
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

FUEL HANDLING BUILDING
el 152 ft 0 in., OBE VERTICAL

FIGURE 3.7.B.2–26

COMPARISON – FREE FIELD SPECTRUM VS. DESIGN SPECTRUM



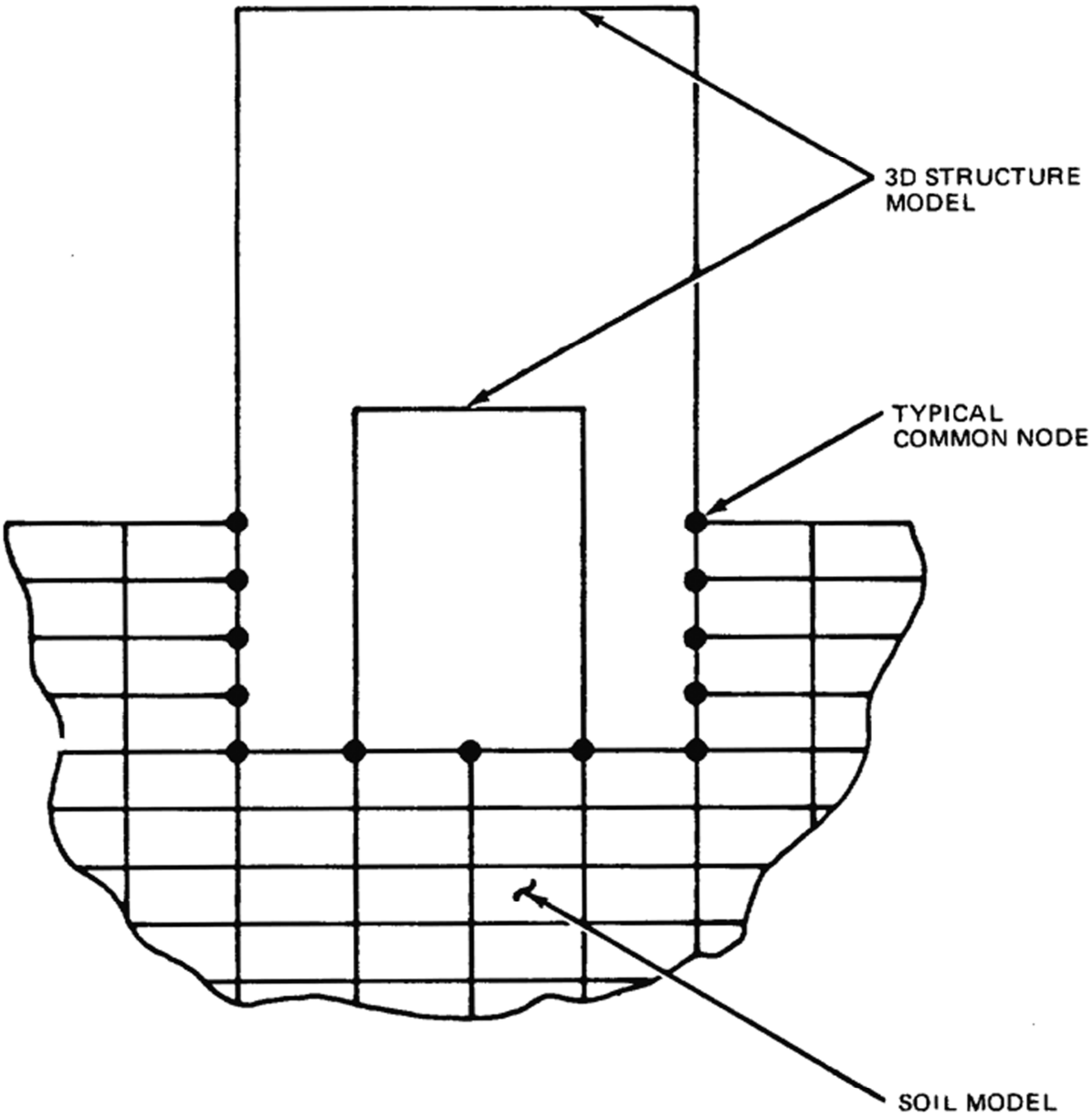
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTROL BUILDING
el 173 ft 0 in., OBE VERTICAL

FIGURE 3.7.B.2–27



ILLUSTRATIVE FIGURE

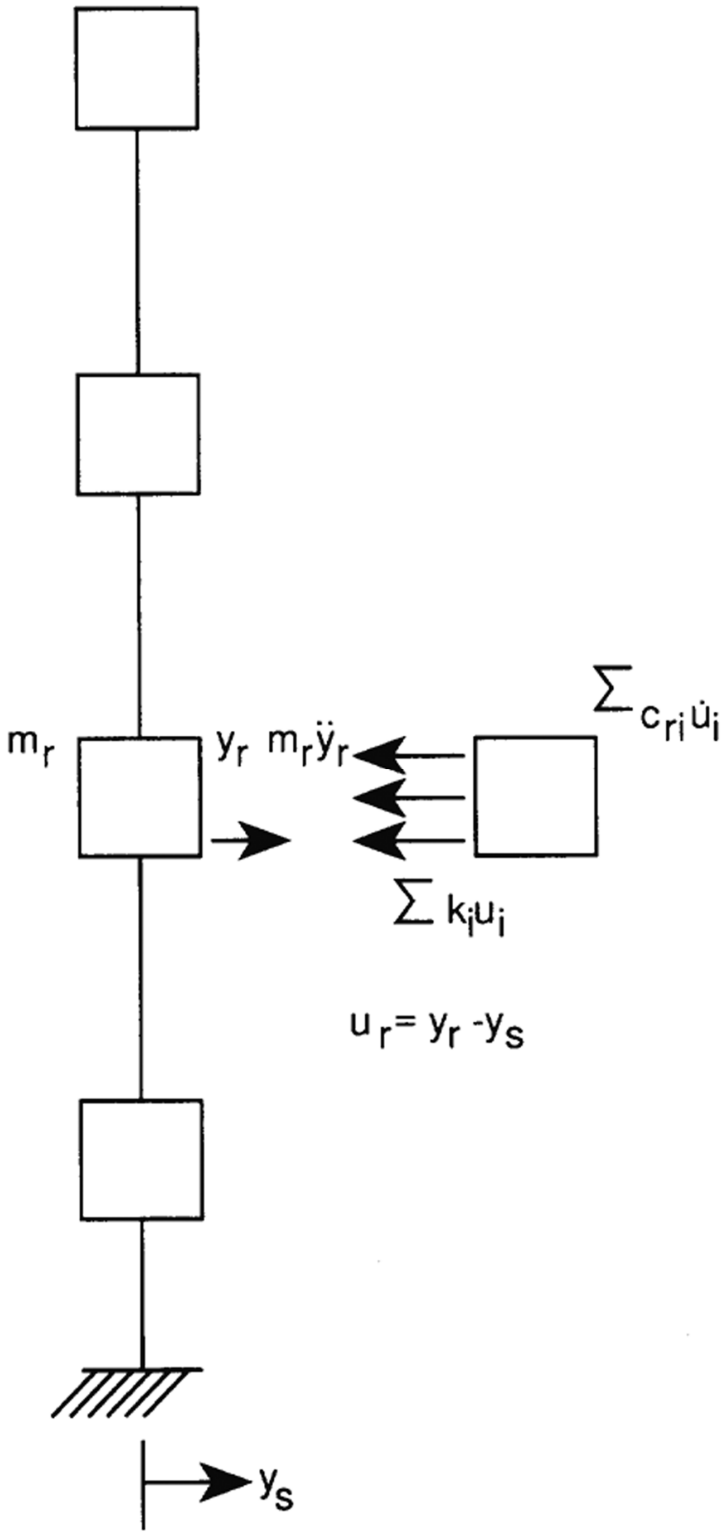
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

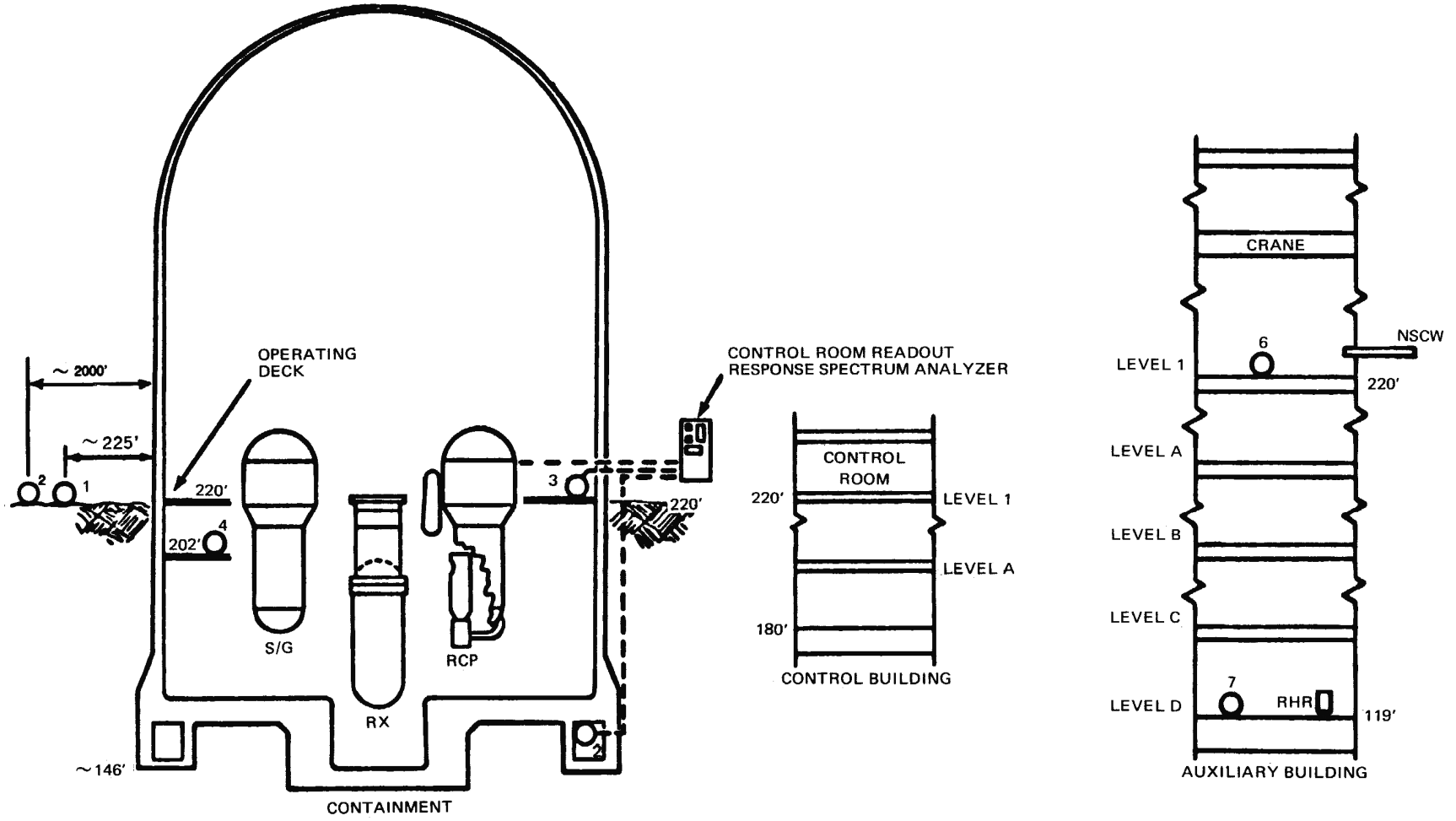
SOIL–STRUCTURE INTERACTION
MODEL COMMON NODES

FIGURE 3.7.B.2–28



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	MILTIDEGREE–OF–FREEDOM SYSTEM FIGURE 3.7.N.2–1



NOTE: NUMBERS DEFINED IN TABLE 3.7.4-1

REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SEISMIC MONITORING
INSTRUMENT LOCATIONS

FIGURE 3.7.4–1

3.8 DESIGN OF CATEGORY 1 STRUCTURES

This section provides information concerning the containments and their internal structures and other Category 1 structures and their foundations and supports.

3.8.1 CONCRETE CONTAINMENT

This section provides the following information on the concrete containment:

A. The physical description.

B. The applicable design codes, standards, and specifications.

C. The loading criteria, including loads and load combinations.

D. The design and analysis procedures.

E. The structural acceptance criteria.

F. The materials, quality control programs, and special construction techniques.

G. The testing and inservice inspection programs.

3.8.1.1 Description of the Containment

The containment (drawing 1X2D01A001) consists of a prestressed reinforced concrete cylinder and hemispherical dome supported on a flat, conventionally reinforced concrete basemat with a central cavity and instrumentation tunnel to house the reactor vessel. The inside face of the containment is lined with steel plates welded together to form a leaktight barrier. The liner is typically 1/4 in. thick and is thickened locally around penetrations, basemat anchorages, and large brackets. The liner plate, including the thickened plate areas, is anchored to the concrete. Leak chase channels are provided at seam welds which are inaccessible after construction. The liner plate system is shown in drawing 1X2D01J015.

3.8.1.1.1 Containment Foundation

The foundation consists of a circular basemat which is 154 ft 6 in. in diameter and 10 ft 6 in. thick, with a minimum thickness of the concrete basemat of 8 ft. 3 in. The minimum thickness of the concrete instrumentation cavity is 8 ft.

Typical basemat reinforcing is shown in drawings 1P01C016, 1P01C018, and 1P01C017.

Attached to the underside of the basemat, at its periphery, is a tendon gallery, 10 ft wide by approximately 9 ft 6 in. high. The gallery provides access to the vertical tendons for installation, tensioning, and inservice inspection.

Vertical shafts surround each of the shell buttresses to provide access for installation, tensioning, and inservice inspection of the hoop tendons. Access shaft No. 1 is attached to the basemat and tendon gallery; it provides access to the tendon gallery. Above the basemat, this access shaft is structurally isolated from the containment shell. Access shafts Nos. 2 and 3 are integral portions of adjoining buildings and are also structurally isolated from the containment.

A steel liner plate with a leak chase system covers the top of the basemat. A structural concrete fill slab, 2 ft 9 in. thick, is placed on top of the liner plate to protect the liner from damage during erection and maintenance.

Anchorage of interior structures (drawings 1X2D48E003, 1X2D48E004, 1X2D48E005, 1X2D48E008, and 1X2D48E007) through the floor liner into the basemat is accomplished by welding cadweld connector sleeves to opposite sides of thickened sections of liner plate. Further discussion is provided in subsection 3.8.3.

3.8.1.1.2 Containment Shell (Cylinder and Dome)

The cylinder and dome, both 3 ft 9 in. thick, are provided with reinforcing steel for load resistance and crack control. Typical shell reinforcing, including reinforcing around piping penetrations, the equipment hatch, and at the buttresses, is shown in drawings 1P01W077, 1P01W081, 1P01W072, 1P01W058, and 1P01W059. A haunch, varying in thickness from 3 ft 9 in. to 4 ft 9 in., is provided at the junction of the cylinder and basemat. (See drawing 1X2D01A001.) At the equipment hatch penetration, the concrete shell is thickened locally on the inside face. (See drawings 1X2D01J017 and 1X2D01J018.) The shell liner plate and its anchorage and stiffening system are shown in drawing 1X2D01J015.

A two-way ungrouted prestressing system is used; it consists of vertical inverted U-shaped tendons and circumferential hoop tendons. The tendon arrangement is shown on drawings 1X2D01K002, 1X2D01K003, 1X2D01K005, 1X2D01K006, 1X2D01K007, and 1X2D01K008, and the anchorage is schematically shown on drawing AX6DD313. Each tendon consists of 55 strands with end anchor components, consisting of anchor heads, wedges, and retainer plates, utilizing the VSL Corporation anchorage and post-tensioning techniques. The tendons are installed in metal sheaths which form ducts through the concrete. The sheaths are filled with grease to protect the tendons from corrosion.

Attachments to the shell, such as brackets for the support of the polar crane, electrical conduit or cable tray, spray piping, and ventilation, are anchored into the shell concrete. For details of the polar crane bracket, see drawing 1X2D01J022.

Steel components of the containment that resist pressure and are not backed by concrete are discussed in subsection 3.8.2.

3.8.1.2 Applicable Codes, Standards, and Specifications

The following documents are applicable to the design, materials, fabrication, construction, inspection, testing, and inservice surveillance of the containment structure.

3.8.1.2.1 Codes and Standards

The following code is applicable to containment design:

- American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code (hereinafter called ASME Code), Section III, Division 2, 1975 Edition through the Winter 1975 Addenda, Article CC-3000 only.

3.8.1.2.2 Regulations

The following regulations apply to containment design:

- 10 CFR 50, Domestic Licensing of Production and Utilization Facilities.

3.8.1.2.3 General Design Criteria (GDC)

The following GDC apply:

- GDC 1, 2, 4, 16, and 50 of Appendix A to 10 CFR 50.

3.8.1.2.4 Regulatory Guides

Applicable dates and revisions are provided in section 1.9.

- 1.10, Mechanical (Cadweld) Splices in Reinforcing Bars of Category 1 Concrete Structures.
- 1.15, Testing of Reinforcing Bars for Category 1 Concrete Structures.
- 1.18, Structural Acceptance Test for Concrete Primary Reactor Containments.
- 1.19, Nondestructive Examination of Primary Containment Liner Welds.
- 1.35, Inservice Inspection of Ungrouted Tendons in Prestressed Concrete Containment Structures.
- 1.55, Concrete Placement in Category 1 Structures.
- 1.94, Quality Assurance Requirements for Installation, Inspection, and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants.
- 1.103, Post-Tensioned Prestressing Systems for Concrete Reactor Vessels and Containments.
- 1.136, Materials, Construction, and Testing of Concrete Containments.

3.8.1.2.5 Industry Standards

Nationally recognized industry standards, such as American Society of Testing Materials (ASTM), American Concrete Institute (ACI), and American Iron and Steel Institute (AISI), are used to specify material properties, testing procedures, fabrication, and construction methods.

3.8.1.3 Loads and Load Combinations

The loads and load combinations used in the analysis and design of the containment are in conformance with the requirements of ASME Code, Section III, Division 2, Subarticle CC-3200, except as discussed below. The load symbol definitions and load combinations are shown in table 3.8.1-1.

A. The word "or" has been inserted between E_o and W in the third abnormal/severe environmental combination to correct an error in the ASME Code. This error has also been recognized in section 3.8.1 of the Standard Review Plan (July 1981).

B. W loads have been included in the construction combination.

C. R_o loads have been included in the extreme environmental load combinations.

D. An additional combination accounting for the effects of the blast load has been included in the extreme environmental factored load combination.

The blast load is conservatively taken as a peak positive incident over pressure, as discussed in subsection 2.2.3. This is considered in the design as an equivalent static pressure of 2 psi.

The determination of wind and tornado loads, flood design loads, and seismic loads is discussed in sections 3.3, 3.4, and 3.7, respectively. Missile effects and pipe rupture effects are discussed in sections 3.5 and 3.6, respectively.

The relationship between the level of prestressing and the containment design pressure is discussed in subsection 6.2.1 of BC-TOP-5-A.

Pressure transients resulting from the design basis accident (DBA) and other lesser accidents serve as the design basis for the containment design pressure of 52 psig.

Temperature gradients through the containment shell for loss-of-coolant accident (LOCA) and main steam line break conditions are shown in figure 3.8.1-1. The variation of temperature with time and the expansion of the liner plate with temperature are considered in determining thermal effects.

Post-LOCA flooding of the containment is postulated to reach el 181 ft 2 in. and is considered as a hydrostatic load.

As described in sections 3.4 and 3.5, there are no credible loads induced by floods, potential aircraft crashes, missiles generated from nearby military installations, or turbine missiles.

3.8.1.4 Design and Analysis Procedures

The design and analysis procedures for the containment generally conform to the requirements of ASME Code, Section III, Division 2, Article CC-3000. Exceptions to this article of the code are as follows:

A. Changes to the load combination table are noted in paragraph 3.8.1.3.

B. In subparagraph CC-3421.4.1(b), the second sentence is taken to read "If M' is negative, use equation (4)." (This conforms to an error correction made by the Winter 1976 Addenda.)

C. In subparagraph CC-3421.4.2, the second sentence is taken to read "For $\rho >$ 0.003, $K = 1.75 - (0.036/\eta\rho) + 4.0\ \eta\rho$, but not less than 0.6 inch." (This conforms to an error correction made by the Summer 1977 Addenda.)

D. The increase in allowable stress permitted for combinations which include W or Eo loads (Table CC-3431-1) is not used.

The guidelines for design presented in BC-TOP-5-A, sections 6.2 and 6.3, are used.

The design of tendon end anchorage zones and reinforcement for the buttresses are as discussed in BC-TOP-7, BC-TOP-8, and BC-TOP-5-A, section 6.6.

3.8.1.4.1 Analysis Procedures

The containment shell and basemat are analyzed employing two three-dimensional finite element computer models. One model is used to obtain results for use in the shell design. The other model is used to obtain results for use in the basemat design. Both models represent the cylinder, dome, basemat, and internal structures. For the model used to obtain results for shell design, the basemat is modeled as a flat disc. The cylinder, dome, and basemat are modeled using shell elements. The mesh near the basemat junction and near the springline is made finer than the mesh for the remainder of the structure to obtain accurate results at these discontinuity locations. The model used to obtain results for the basemat design is discussed in paragraph 3.8.5.4.1. A separate finite element model is used for the thickened shell area around the equipment hatch.

Classical theory and methods are used in the analysis of other local areas such as the personnel and escape locks and small penetrations.

3.8.1.4.2 Boundary Conditions and Load Treatment

Both containment models are supported on translational linear soil springs which simulate the VEGP foundation condition. The model of the local area around the equipment hatch uses results from the analysis of the overall structure as boundary conditions. All loads, axisymmetric and nonaxisymmetric, are applied to the three-dimensional models as nodal loads, element loads, or accelerations.

3.8.1.4.3 Creep, Shrinkage, and Cracking Considerations

In the design of the containment post-tensioning system, conservative values for creep and shrinkage are used, based on past experience. The values used are verified by the evaluation of tests performed on samples of the concrete used in the shell. Concrete crack control is accomplished by following the provisions of paragraph CC-3535.

3.8.1.4.4 Computer Programs

The BSAP, BSAP-POST, and TENDON computer codes are used in the containment design and analysis. Descriptions of these computer codes are provided in appendix 3B. The seismic-related codes employed for dynamic analyses are discussed in subsections 3.7.B.2 and 3.7.N.2.

3.8.1.4.5 Tangential Shear

The ASME Code has no provision for allowable stresses for tangential shear in prestressed concrete. The design procedure use for the VEGP containment shell provides a conservative and viable technique for accounting for the effects of tangential shear. Reinforcement for tangential shear is provided by defining "equivalent" membrane forces:

For seismic loads:

$$N_{ht} = \left(N_h^2 + V_u^2\right)^{1/2}$$

$$N_{vt} = \left(N_v^2 + V_u^2\right)^{1/2}$$

For wind, tornado, or blast loads:

$$N_{ht} = N_h + V_u$$

$$N_{vt} = N_v + V_u$$

where:

N_{ht}, N_{vt} = "equivalent" membrane forces in the horizontal and vertical directions, respectively.

N_h, N_v = membrane forces, due to the same loading causing the tangential shear, in the horizontal and vertical directions, respectively.

Vu = tangential shear force due to seismic, wind, tornado, or blast loads.

3.8.1.4.6 Variation in Physical Material Properties

In the design and analysis of the containment, consideration is given to the effects of possible variations in the physical properties of materials on the analytical results. The variations in physical properties are accounted for by using allowable stress levels, below ultimate strength, for the design of the structure under full service and factored load conditions.

3.8.1.4.7 Liner Plate and Anchorage System

The details which depict the typical liner plate system with anchorages are shown in drawing 1X2D01J015. The design and analysis methods of the liner and its anchorage system are covered in BC-TOP-1. The relative strength of the liner plate against buckling as compared with its anchor and anchor welds is discussed in BC-TOP-1. BC-TOP-1 provides sample calculations that demonstrate that the strength of the anchor and the anchor welds is sufficient to preclude any possibility of overall buckling failure as a result of anchor pullout.

The elastic and plastic solutions are used to analyze the stresses in the anchors and concrete resulting from a buckled panel of the liner. The panel and anchors are modeled as a series of springs. The stresses in the anchors and concrete are obtained by using the strains from all loads being considered, equilibrium of forces, and compatibility of deformations. BC-TOP-1 provides details and further discussions.

3.8.1.4.8 Ultimate Capacity of the Containment

The containment ultimate capacity was calculated and presented in the containment design report submitted under separate cover on October 31, 1984, by GPC Letter GN-433, from Foster to Director NRR.

3.8.1.5 Structural Acceptance Criteria

The fundamental acceptance criterion for the completed containment building is successful completion of the structural integrity test where measured responses are required to be within the limits predicted by analyses. The limits are based on test load combinations and code values for stress, strain, or gross deformation for the range of material properties and construction tolerances specified, as described in paragraph 3.8.1.6.

The limits for allowable stresses and strains are given in ASME Code, Subarticle CC-3400, and are compatible with nationally recognized codes of practice. In this way, the margins of safety associated with the design and construction of the containment building are, as a minimum, the accepted margins associated with nationally recognized codes of practice.

The structural integrity test is planned to yield information on both the overall response of the containment and the response of localized areas. This information, together with the test information documented in BC-TOP-7 and BC-TOP-8, provides direct experimental evidence that the containment structure can withstand the design internal pressure.

The design and analysis methods, as well as the type of construction and construction materials, are chosen to allow assessment of the capability of the structure throughout its service life[a]. Additionally, surveillance testing provides further assurances of the continuing ability of the structure to meet its design functions.

3.8.1.5.1 Factors of Safety and Margin of Safety

Design of the containment is based on ASME Code, Section III, Division 2, Article CC-3000; therefore, adequate margins of safety exist when the containment is subject to the load combinations prescribed by the code.

3.8.1.5.2 Allowable Stresses

The allowable stresses for service load and factored load conditions in concrete, reinforcing steel, and the tendon system are as specified in ASME Code, Section III, Division 2, Subarticle CC-3400. The design for tangential shear in prestressed concrete is discussed in paragraph 3.8.1.4.5.

3.8.1.6 Materials, Quality Control, and Special Construction Techniques

This section contains information relating to the materials, quality control program, and special construction techniques used in the fabrication and construction of the containment.

3.8.1.6.1 Concrete

The compressive strength of concrete used for the containment is as follows:

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. The containment tendon prestress analysis was evaluated as a time-limited aging analysis (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.4.

Basemat, basemat fill slab, tendon gallery, and access shaft No. 1, f'_C = 5000 psi.

Cylinder and dome f'_C = 6000 psi.

The test age of concrete using pozzolan in the concrete mix is designated as 91 days. The test age of concrete without pozzolan in the concrete mix is designated as the normal 28 days. These strength designations are in accordance with ACI 318-71.

Structural concrete is batched and placed in accordance with the Recommended Practice for Measuring, Mixing, Transporting, and Placing Concrete (ACI 304) and Building Code Requirements for Reinforced Concrete (ACI 318-71).

A. Cement

Cement is Type II conforming to the Specification for Portland Cement (ASTM C 150). The cement contains no more than 0.60 percent by weight of alkalis calculated as sodium oxide plus 0.658 potassium oxide. Certified copies of material test reports showing the chemical composition and physical properties are obtained for each load of cement delivered.

B. Aggregates

All aggregates conform to standard specifications for concrete aggregates (ASTM C 33). In addition to the specified gradation, the fine aggregate (sand) has a fineness modulus of not less than 2.3 nor more than 3.1 during normal operations. The average of five successive tests is required to be between 2.5 and 3.0. At least four of five successive test samples should not vary in fineness modulus more than 0.20 from the moving average established by the last five tests. Coarse aggregates may be rejected if the loss, when subjected to the Los Angeles Abrasion Test, using grading A, exceeds 40 percent by weight at 500 revolutions.

Acceptance of source and aggregates is based on the following tests:

ASTM No.	Name
C 131	Los Angeles Abrasion Test
C 142	Clay Lumps and Friable Particles
C 117	Material Finer Than No. 200 Sieve
C 123	Lightweight Pieces
C 40	Organic Impurities
C 235	Scratch Hardness
C 289	Potential Reactivity (Chemical)
D 1411	Water Soluble Chlorides
C 125	Definition of Terms Relating to Concrete and Concrete Aggregates

ASTM No.	Name
C 127	Specific Gravity and Absorption of Coarse Aggregates
C 128	Specific Gravity and Absorption of Fine Aggregates
C 136	Sieve Analysis
C 88	Soundness
C 295	Petrographic Examination

C. Water

Water and ice for mixing shall be clean, with a total solids content of not more than 2000 ppm as measured by ASTM D 1888. The mixing water, including that contained as free water in aggregate, shall contain not more than 250 ppm of chlorides as Cl^- as determined by ASTM D 512. Chloride ions contained in the aggregate shall be included in calculating the total chloride ion content of the mixing water. The chloride content contributed by the aggregate shall be determined in accordance with ASTM D 1411.

D. Admixtures

The concrete may also contain an air-entraining admixture and/or a water-reducing and retarding admixture. The air-entraining admixture is in accordance with the Specification of Air Entraining Admixtures for Concrete (ASTM C 260). It is capable of entraining 3 to 6 percent air, is completely water soluble, and is completely dissolved when it enters the batch. Superplasticizers, entraining from 1.5 to 4.5 percent air, may be used in concrete mixes (f′ = 5000 psi, maximum) for congested areas to improve workability and prevent the formation of voids around reinforcement. The water-reducing admixture conforms to standard specification for Chemical Admixtures for Concrete (ASTM C 494), Types A and D. Type A is used when average ambient temperature for the daylight period is below 70°F. Type D is used when average ambient air temperature for the daylight period is 70°F and above. Pozzolans, if used, conform to Specifications for Fly Ash and Raw or Calcined Natural Pozzolans for Use in Portland Cement Concrete (ASTM C 618) except that the loss on ignition shall be limited to 6 percent. Admixtures containing more than 1 percent by weight chloride ions are not used.

E. Concrete Mix Design and Testing

Concrete mixes are designed in accordance with ACI 211.1, Recommended Practice for Selecting Proportions for Normal and Heavy Weight Concrete, using materials qualified and accepted for this work.

An independent testing laboratory at the site designs and tests the concrete mixes. Only mixes meeting the design requirements specified for concrete are used.

3.8.1.6.2 Reinforcing Steel

Reinforcing bars for concrete are deformed bars meeting requirements of Specification for Deformed and Plain Billet Steel Bars for Concrete Reinforcement (ASTM A 615), Grade 60.

Mill test reports, in accordance with ASTM A 615, are obtained from the reinforcing steel supplier to substantiate specification requirements.

In addition, tonnage of reinforcing steel of each size and grade for user tests on full-diameter specimens is in accordance with Regulatory Guide 1.15.

The test procedures are in accordance with ASTM A 370, and acceptance standards are in accordance with ASTM A 615.

3.8.1.6.3 Prestressing System

A description of the prestressing system is given in paragraph 3.8.1.1.2. Additional material properties for each component of the prestressing system are described as follows:

A. Prestressing Strands

Strands are nominal 1/2-in. diameter, of the seven-wire low-relaxation type, having a guaranteed minimum ultimate tensile strength, f_{pu}, of 270,000 psi or greater based on the nominal steel area of the strand. The minimum yield stress is .90 f_{pu}. All strands conform to ASTM A 416 for weldless grade.

B. Other Prestressing System Components

Materials for the other components required for the prestressing system conform with the following steel specifications:

Bearing plates	ASTM A 537 Class 1 carbon-manganese-silicon steel
Trumpets	ASTM A 569
Trumpet extensions	ASTM A 513
Anchor heads	AISI 1026
Shims	ASTM A 572
Wedges	DIN 17210 and AISI 8620
Tendon sheathing (semirigid corrugated tubing)	ASTM A 527

3.8.1.6.4 Liner Plate

A. The containment structure is lined with 1/4-in.-thick welded steel plate, except in limited areas where thickened plate is utilized, and conforms with the requirements of ASME SA-285, Low and Intermediate Tensile Strength Carbon Steel Plates for Pressure Vessels, Grade A, to ensure a leaktight membrane.

This steel has a minimum yield strength of 24,000 psi and a minimum elongation in an 8-in. specimen of 27 percent.

The materials, fabrication, and erection of the containment reactor cavity liner plate system (el 142 ft 6 in. to 169 ft 0 in.) and thickened basemat liner plates, cadwelds, and embeds at el 169 ft 0 in. are in accordance with 1971 ASME Code, through Summer 1973 Addenda, Section III, Division 1; Section V; Section VIII, Division 1; and Section IX.

The materials, fabrication, and erection of the containment structure 1/4-in. basemat liner plates at the basemat level, el 169 ft 0 in., and all of the cylinder wall and dome liner plate system, support brackets, and penetrations integral with the liner plate are in accordance with 1974 ASME Code, through Summer 1975 Addenda, Section III, Division 1, Subsection NE; Section V; Section VIII, Division 1; and Section IX.

B. Liner plate greater than 1/4 in. thick conforms to the requirements of ASME SA-516, Grade 70, or SA-36 and satisfies the impact requirements specified in ASME Section III, Division 1, Subsubarticle NE-2320, for materials to be impact tested at a maximum temperature of 30°F below the lowest service metal temperature.

C. Materials for the equipment hatch and personnel airlocks, process pipe penetration assemblies, electrical penetration assemblies, and fuel transfer tube housing bellows assemblies are listed in paragraph 3.8.2.6.

3.8.1.6.5 Liner Plate Anchors and Associated Hardware

The following parts wherever used conform to the material requirements of the standards listed below:

A. Anchor bolts and machine bolts - ASTM A 307.

B. High-strength bolts - ASTM A 325.

C. High-strength anchor bolts and studs - ASME SA-540, CL1 and CL4.

D. Nelson concrete anchor studs Type H4 - ASTM A 108.

E. Cadweld connectors - AISI 1018 or AISI 1026.

F. Unistruts - ASTM A 570, Grade C.

G. Heavy hex nuts for high-strength bolts - ASTM A 194, Grade 2H.

H. Hardened washers ASTM A 325, Rockwell Hardness C 38, or as indicated on design drawings.

I. Long hex couplings - ASTM A 307, Grade A, may be substituted for ASTM A 36.

J. Leak chase test couplings - carbon steel ASME SA-234.

3.8.1.6.6 Structural Steel

A. The following parts conform to the requirements of ASTM A 36:

1. Liner plate angle, flat bar, tee, and channel stiffeners.

2. Support brackets not penetrating the liner plate. (ASME SA-516, Grade 70 may be substituted for ASTM A 36).
3. Framing, bracing, and tie rods.
4. Embedded floor beams.
5. Backing strips.
6. Leak chase system channels and angles.

B. Polar crane support bracket plates conform to ASME SA-516, Grade 70. Material for parts of the bracket that penetrate the thickened liner and exceed 5/8 in. in thickness are impact tested at a maximum temperature of -13°F, per ASME Code, Section III, Division 1, Subsection NE, Subparagraph NE-2321.2.

C. Nuclear steam supply system (NSSS) embeds located below top of basemat and penetrating the pressure boundary conform to ASTM A 588, F = 50.0 ksi minimum.

D. Penetration sleeve material conforms to the requirements of the standard specifications listed below. The lowest service metal temperature is 17°F, and the maximum test temperature is -13°F.

1. Penetration sleeves - ASME SA-333, Grade 1 or 6; or ASME SA-516, Grade 70.
2. Penetration sleeve reinforcing (thickened liner plate) - ASME SA-516, Grade 70.
3. Anchor rings, stiffener rings, and gussets - ASME SA-516, Grade 70.

3.8.1.6.7 Tendon Corrosion-Retarding Compound

The tendon corrosion-retarding compound used for sheathing filler material is Visconorust 2090P-4.

3.8.1.6.8 Quality Control

The quality control program for fabrication, construction, testing, and examination is discussed in this paragraph. Conformance to Regulatory Guide 1.94 is as described in paragraph 3.8.3.6.2.C.

Quality control procedures are implemented during construction and inspection. The quality control procedures are specified in the specifications covering the fabrication, furnishing, and installation of each structural component and provide inspection and documentation to assure that the codes and construction practices are met.

The quality control program includes an organizational scheme which functions in accordance with policies delineated in the Bechtel Quality Assurance Manual, Vogtle Field Procedure Manual, and Bechtel Procurement Supplier Quality Manual.

A. Tests for Concrete

Concrete for the containment structure is tested in accordance with ACI 318-71. Concrete placement is accomplished in accordance with Regulatory Guide 1.55 as discussed in section 1.9.

B. Tests for Reinforcing Steel

Reinforcing steel is tested in accordance with Regulatory Guide 1.15.

C. Tests for Mechanical Splices

Control of cadweld mechanical splices for reinforcement utilizing filler metal and enclosing sleeve is in accordance with Regulatory Guide 1.10.

D. Test Procedures for the Liner Plate

The nondestructive examination of the liner plate welds is in accordance with Regulatory Guide 1.19 with exceptions noted in section 1.9.

1. Liner Plate and Thickened Liner Plate

The liner plate is 1/4 in. thick. Thickened liner plates are used as penetration sleeve reinforcing plates and as a part of the bracket and attachment assemblies. Thickened floor liner plate is ultrasonically inspected in accordance with ASME SA-435, except that inspection covers 100 percent of the plate area. Thickened plates over 5/8 in. in thickness require Charpy V-notch impact tests in accordance with ASTM A 593-72. Testing is as specified in paragraph 3.8.1.6.4.B. Structural steel members and electrical ground rods are not classified as thickened liner plate.

2. Penetration Sleeves

Sleeves over 5/8 in. in thickness require the same Charpy V-notch impact tests as for thickened liner plates in paragraph 3.8.1.6.4.B, except for minimum requirements of ASME SA-333, Grades 1 and 6. See paragraph 3.8.3.6.

3. The quality control procedures that ensure the suitability of the steel plate material are discussed as follows:

a. For brackets and attachments that are not continuous through the liner plate:

(1) The liner plate greater than 5/8 in. in thickness is ultrasonically examined for delaminations.

(2) The strength in the through-the-thickness direction is taken as one-half of that in the transverse direction unless tests are performed to justify higher values.

b. Ultrasonic examination is required for the floor liner plate greater than 1/4 in. in thickness.

E. Control Procedures for the Containment Liner Plate Attachments

The quality control procedures associated with penetrations, attachments, and hardware are incorporated in the quality control procedures for the liner plate.

F. Control Procedures for Structural and Miscellaneous Steel

Quality control procedures for structural and miscellaneous steel and the associated structural welding conform to the requirements specified in paragraph 3.8.1.6.6.

G. Control Tests and Inspection of Prestressing System

The following quality control procedures are used:

1. Prestressing Strands

   a. Each tendon is individually identified and traceable to the heat numbers of the wire utilized in its buildup. Physical test reports supporting the integrity of each heat of material are reviewed as a condition of acceptance.

   b. Specimens are cut from each reel of strand and tension tested to rupture, to assure compliance to specifications.

   c. Strands are examined for workmanship and quality prior to fabrication of the tendon.

2. Bearing Plates and Transitions

   a. Verification is made that the bearing plate material complies with that specified on the drawings. Compliance is evidenced by mill test reports traceable to the heat number by a serial number permanently marked on each bearing plate.

   b. Plates are examined for workmanship and quality. Cracks, burrs, corrosion, and other defects are not acceptable.

3. Anchor Head

   Raw material is accompanied by mill certificates and subjected to receiving inspection.

4. System Performance Tests

   a. Static Tensile Test

      Typical anchorage and tendon details are tested to show that the anchorage develops the minimum guaranteed ultimate strength of the tendon.

   b. Dynamic Tensile Test

      High and low cycle dynamic tensile tests are performed on specimens having at least 10 percent of the steel area of the full-size production tendons. The high cycle test has 500,000 cycles of stress variation, from 60 percent to 66 percent of the minimum ultimate tensile strength. The low cycle test has 50 cycles of stress variation, from 40 percent to 80 percent of the minimum ultimate tensile strength. These tests demonstrate that the prestressing system is able to withstand cyclic loadings without failure.

c. Low Temperature Test

Documentary evidence of certified testing at temperatures below the lowest anticipated service temperature substantiates that the anchorage assembly, including the bearing plate, is capable of transmitting the ultimate load of the tendon into the structure without brittle fracture. BC-TOP-8 presents an example of accepted documentation for the cold environment test.

H. Liner Plate Erection Tolerances

Cylinder and dome liner plates are used as forms and erection precedes the concrete placements. Tolerances for erection of the liner plate and the penetration sleeve assemblies are specified as follows:

1. Tolerances for Liner Plate (Including Floor, Cylinder, and Dome)

The following tolerances apply to the fabrication of the liner plate:

a. The radial location of any point on the cylindrical liner shell does not vary from the design radius by more than ±3 in. Measurement is made at 30° spacings for each 10 ft of rise. For the hemispherical dome roof, the actual radius does not exceed +8 in. or -12 in. from the design radius of 70 ft 0 in. measured from the springline el 327 ft 9 in.

b. Misalignment of butt-welded joints is not more than 1/16 in. for 1/4-in. plates. On plates thicker than 1/4 in., the alignment tolerance is as specified in the ASME code.

c. A 15-ft-long template curved to the required radius does not show deviations of erected liner plate from template of more than 1 in. when placed against the completed surface of the shell within a single plate section and not closer than 12 in. to a weld seam. When the template is placed across one or more weld seams, the deviation does not exceed 1/2 in. The effect of change in plate thickness or of weld reinforcement is excluded when determining deviations.

d. A 15-in.-long template curved to the required radius does not show deviations of erected liner plate from template of more than 1/8 in. inward and 3/8 in. outward when placed against the completed surface of the shell within a single plate section and not closer than 12 in. to a weld seam.

e. The deviation in the liner plate between liner plate stiffeners does not exceed ±1/8 in. when referenced to the theoretical surface.

f. Sharp bends are not permitted unless provision has been made for them in the design. A sharp bend is defined as any local bend that deviates from the design radius or a vertical straight edge by an offset of more than 1/2 in. in 1 ft. The template used to measure the local deviations is 1 to 2 ft longer than the area of the deviation itself.

g. The slope of the cylindrical wall liner plate, referred to true vertical, does not exceed 1/180 within any 10-ft ring. The overall vertical out-of-plumb of the cylindrical liner plate at springline el 327 ft 9 in. does not exceed 3 in.

h. An individual wall liner plate section does not deviate from a vertical straight template by more than ±3/4 in. in 10 ft.

i. Dimensions of all liner plates as cut are within ±1/8 in. of the design dimensions.

j. Liner plate sections are fabricated in the lengths necessary for the most efficient methods of handling and erection.

2. Tolerances for Penetration Sleeve Assemblies

The following requirements and those of items 1.b, c, g, and i immediately above apply for penetration sleeve assemblies:

a. A 30-in.-long template curved to the required radius does not show deviations of more than 3/4 in. when placed against the completed surface of the liner plate within a single plate section.

b. Alignment of the axis of cylindrical liner plate penetrations as erected does not vary by more than 1° for pipe diameters greater than 12 in. or by more than 2° for pipe diameters less than or equal to 12 in. Individual penetrations and penetrations other than the main steam and feedwater penetrations in thickened liner plates having multiple penetrations are located within ±1 in. of their design elevations and circumferential locations in the cylindrical liner plate. The location of the main steam and feedwater penetrations in the shell are within ±1/2 in. of the design location.

c. The location of penetrations in a thickened liner plate having multiple penetrations is within ±1/4 in. of the dimensions shown on the design drawings, relative to the thickened liner plate edges.

3. Miscellaneous Tolerances

The following tolerances apply to various items comprising the liner plate system:

a. The location of the crane support brackets with respect to azimuth and elevation are within ±1/2 in. of the design location. All other brackets located in the cylindrical shell are located within ±1 in. of the design location.

b. The liner plate stiffeners are placed so that the relative locations with respect to each other are within ±1/2 in. of the dimensions shown.

c. The radius of curvature of the channels, angles, etc., that maintain the shell curvature are within ±5 percent of the design radius shown.

d. The field placement of the embedded floor beams and thickened floor liner plate is within ±1/2 in. of the theoretical location in plan and are level within ±1/4 in. overall and ±1/8 in. within any 20-ft arc.

e. The locations of cadwelds and threaded sleeves on the thickened floor liner plate are within ±1/8 in. of the theoretical location. The thickened plate dimensions are within ±1/8 in. of dimensions given on the engineering design drawings.

I. Reinforcing Steel Placement Tolerances

The tolerances for placing reinforcing steel conform to the following:

1. In concrete members where the effective depth, d, is less than 24 in., the maximum vertical cover does not exceed the minimum cover by more than 1 in.
2. In concrete members where the effective depth, d, is 24 in. or more, the maximum vertical cover does not exceed the minimum cover by more than 2 in.
3. For bars placed crosswise of members, the total number of bars is maintained and spaced evenly within 2 in.
4. Longitudinal location of bends and ends of bars is ±3 in., provided the minimum cover is maintained, except at discontinuous ends of members, where the tolerance shall be ±1/2 in.

J. Erection Tolerances for the Prestressing System

1. Alignment of tendon sheathing is maintained within ± 3/4-in., except as noted on the design drawings.
2. Bearing plate assemblies are positioned within ± 1/2 in., unless otherwise shown on erection drawings.

3.8.1.6.9 Special Construction Techniques

Because of their more efficient usage on cylindrical surfaces, 10-ft-high steel jump forms are employed on the exterior surfaces of the containment cylinder. This construction technique does not affect the structural integrity of the completed structures.

3.8.1.7 Testing and Inservice Inspection Requirements

3.8.1.7.1 Structural Integrity Pressure Test

Following construction, each containment is proof tested at 60 psi, which is 115 percent of the design pressure. During this test, deflection measurements and concrete crack inspections are

made to determine that the actual structural response is within the limits predicted by the design analyses.

The VEGP containments are considered as nonprototype structures and need not be strain measured. The San Onofre Nuclear Generating Station (SONGS) Unit 2 containment is chosen as the prototype structure, based on the following comparison:

		VEGP Units 1 and 2	SONGS Unit 2
A.	Shape	Vertical cylinder with hemispherical dome	Vertical cylinder with hemispherical dome
B.	Size:		
	Inside diameter	140 ft 0 in.	150 ft 0 in.
	Cylinder height (basemat to springline)	158 ft 9 in.	97 ft 0 in.
C.	Design pressure	52 psi	60 psi
D.	Basemat thickness	10 ft 6 in.	9 ft 0 in.
E.	Number of wall buttresses	3	3
F.	Number of dome buttresses	3	3
G.	Tendon style	55 each 1/2-in. 7-wire strands	55 each 1/2-in. 7-wire strands

The test procedure conforms to the requirements of Regulatory Guide 1.18 except as noted in section 1.9. Section 9 of BC-TOP-5-A also describes test results obtained using a typical procedure as well as those obtained from early tests where a substantial amount of strain information was collected.

3.8.1.7.2 Long-Term Surveillance

The long-term surveillance program consists of evaluating the general condition of the post-tensioning system. Data on wire corrosion level and tendon lift-off forces are obtained and analyzed. The surveillance program conforms to ASME Code, Section XI, Subsection IWL and applicable addenda as required by 10 CFR 50.55a except where an exemption, relief, or an alternative has been authorized by the NRC.

This surveillance program provides assurances of the continuing ability of the structure to meet the design functions as stated in paragraph 3.8.1.5.

Only the Unit 1 containment is subject to the complete surveillance program; every tendon has provision for removal with detensionable anchorage assemblies.

The Unit 2 containment has permanent anchorage assemblies (nondetensionable). Refer to drawing AX6DD313.

The ASME Section XI Inservice Inspector Program's IWE and IWL inspections are credited as license renewal aging management activities (see subsections 19.2.30 and 19.2.31).

3.8.1.8 Standard Review Plan Evaluation

The Standard Review Plan specifies the ASME Section III, Division 2 Code for concrete vessels and containments as the acceptable code. Only Article CC-3000 of this code is considered for the containment design. VEGP takes some alternative positions to Regulatory Guides 1.18, 1.19, 1.55, 1.94, and 1.136.

The ASME Section III, Division 2 Code was not in effect at the time the VEGP construction permit was issued. As directed by the Nuclear Regulatory Commission (Preliminary Safety Analysis Report paragraph 3.8.1.5.2) and consistent with subsection 3.8.1 of the Safety Evaluation Report, the design of the containment is based on Article CC-3000 of the ASME Code. The design procedures and construction practices delineated in the FSAR ensure that the containment structure will withstand the specified design conditions without impairment of structural integrity or the performance of required safety functions.

Refer to section 1.9 for VEGP positions on regulatory guides.

3.8.2 STEEL CONTAINMENT

The VEGP does not utilize a steel containment. However, the concrete containment penetration sleeve assemblies, American Society of Mechanical Engineers (ASME) Code Class MC components and parts, and ASME Code Class 2 piping penetrations are discussed in this subsection.

3.8.2.1 Description of Penetration Sleeve Assemblies, Class MC Components and Parts, and Class 2 Piping Penetrations

A penetration sleeve assembly is a steel weldment composed of a cylindrical steel sleeve (welded or seamless), stiffening ring plate(s) with or without gusset(s), anchors, and a portion of the thickened liner plate.

The sleeve assemblies are field-welded to the steel liner plate and are embedded in the concrete containment wall. Generally, several inches of sleeve extends into the containment, past the surface of the liner plate backed by concrete and acts as a nozzle for later component, part, or piping attachment.

Sleeve assemblies installed as spares for future use are capped with shop-welded closure plates. (See Type V, drawings 1X4DL4A14 and 2X4DL4A14.) The sleeve assemblies (capped or uncapped) are considered to be parts of the containment liner plate system (paragraph 3.8.1.4) and are not subject to ASME Section III, Division 1, Class MC requirements.

Class MC components or parts of the containment structure are those code-stamped items that function as part of the pressure boundary and are not backed by concrete, and which are attached to the penetration sleeve assemblies during construction of the containment vessel.

The components include the personnel and escape lock assemblies. The parts consist of the equipment hatch assembly, fuel transfer tube housing and bellows assemblies, electrical penetration assemblies, and the isolation valve encapsulation vessel assemblies.

Process piping assemblies are field welded to the penetration sleeve assemblies and are not deemed components or parts of the containment. All piping attached to the containment wall penetration sleeves is ASME III Code Class 2.

3.8.2.1.1 Equipment Hatch and Personnel Lock Assemblies

A 20-ft circular equipment hatch, a 9-ft 10-in. circular personnel lock, and a 5-ft 9-in. circular personnel lock penetrate the containment wall and are welded to the penetration sleeves. Hatch and lock doors are provided with concentric double-sealing gaskets, with provision for leak testing as described in paragraph 6.2.6.2. The equipment hatch and personnel lock penetration details are shown on drawings 1X2D01J017 and 1X2D01J018.

The smaller of the personnel locks is for emergency escape purposes. Each personnel lock has a door at each end and is an ASME Code Class MC component. For the two locks, that portion of the penetration sleeve assembly between the liner plate and the ASME III Code Class MC jurisdictional boundary is considered part of the liner plate and is not subject to ASME Class MC requirements.

During plant operation, the two doors of each personnel lock are interlocked to prevent both being opened simultaneously. Quick-acting equalizing valves connect the personnel lock with the interior and exterior of the containment to equalize pressures. Provision is made to bypass the interlock system during plant cold shutdown to facilitate ingress and egress.

3.8.2.1.2 Fuel Transfer Tube and Housing Bellows Assembly

One fuel transfer tube penetration per containment is provided for refueling. An ASME III Code Class MC one-piece inner pipe acts as the transfer tube, through which both new and spent fuel rod assemblies are passed. The scheme is shown in drawing 1X2D48J006. The tube is fitted with a double-gasketed blind flange at the containment refueling canal end and a gate valve at the fuel transfer canal end within the fuel handling building. This arrangement prevents leakage through the tube.

Three separate penetration sleeve assemblies, joined by four bellows assemblies, act as a transfer tube housing. The penetration sleeve assemblies permit the transfer tube to penetrate the refueling canal wall, the containment shell, and the exterior wall of the fuel handling building, while maintaining a pressure-retaining boundary at each wall. The sleeve assemblies are considered part of the liner plate and are not subject to ASME, Section III, Code Class MC requirements.

The transfer tube is supported by the two outboard bellows assemblies. The refueling canal end employs a gimbals yoke (swivel support) mechanism, while the fuel handling building end uses a pin-connected, sliding support bearing.

The ASME III Code Class MC bellows assemblies allow thermal expansion of the transfer tube and housing and permit differential movement between buildings.

3.8.2.1.3 Electrical Penetration Assemblies

Shop-welded weld neck flanges with caged nuts are provided on the outer ends of all penetration sleeves designed to accept the ASME III Code Class MC bolt-on electrical penetration assemblies. In addition, field-welded slip-on ring plates are installed as flanges on

the inner ends of the penetration sleeves to receive bolt-on electrical enclosures. See drawings 1X2D01J019, 1X2D01J020, 1X4DL4A14, 2X2D01J019, and 2X4DL4A14.

Canister-type penetrations are used for medium-voltage power conduction through the containment wall. The stainless steel canisters are factory-sealed units, each consisting of a hollow cylinder with a header plate at each end, through which the conductors are run. One header plate is bolted to the exterior penetration sleeve flange. The header plate mating the flange is provided with two concentric O-ring gaskets, and the annular space between the O-rings is continuously pressurized during plant operation. The canister reposes within the penetration sleeve and is capable of being pressurized through a test connection. All other electrical penetrations are of the modular type, consisting of one header plate, which is bolted to the exterior penetration sleeve flange. Two concentric O-rings gasket the mating pieces, and the annular space is also pressurized during plant operation. Feed-through modules carry the electrical conductors (including fiber optics). The feed-through conductors are mated to the modules by double high-temperature thermoplastic seals with interspace pressure connections. These connections are manifolded so that all feed-through modules for a single penetration may be pressurized at the same time and throughout plant operation. The modules pass through the header plate and are sealed with metal compression fittings.

3.8.2.1.4 Isolation Valve Encapsulation Vessel Assemblies

Two containment spray isolation valve encapsulation vessels and two residual heat removal isolation valve encapsulation vessels are provided for each containment. Although one pair of vessels (one containment spray and one residual heat removal) is present in the auxiliary building and one pair of vessels is present in the fuel handling building, respectively, these vessels are leaktight housings. As such, these vessels and their bellows expansion joints are subject to ASME III Class MC requirements and leaktight tests. However, the encapsulation vessel design is such that they do not communicate directly with the containment atmosphere. The containment spray isolation valve encapsulation vessel is a steel tank, with an inside diameter of 4 ft and a height of approximately 10 ft 8 in. (See drawing 1X4AH04-25.) The residual heat removal isolation valve encapsulation vessel is a similar steel tank, with an inside diameter of 5 ft 6 in. and a height of approximately 11 ft.

3.8.2.1.5 Process Pipe Penetration Assemblies

The closure for process piping to the liner plate is made with a special flued head welded into the piping system and to the penetration sleeve assembly. In certain cases, the piping assemblies include an intermediate sleeve which is inserted between the penetration sleeve assembly and the flued head. In the case of piping carrying hot fluid, the pipe is insulated and the flued head/penetration is designed to prevent excessive concrete temperatures and to prevent excessive heat losses from the fluid. The main high-temperature piping consists of four penetrations for feedwater and four penetrations for main steam, with a maximum operating temperature range between 445°F and 557°F. Thermal insulation is provided on the outside diameter of each line. The penetration assembly is designed to limit the concrete maximum temperature to 200°F.

For typical details of process pipe penetration assemblies used through the containment wall, refer to types I, II, III, IV, and VI on drawings 1X4DL4A14 and 2X4DL4A14. The type IV arrangement allows the piping to be welded directly to the penetration sleeve. Such a juncture is used for the containment purge lines.

3.8.2.2 Applicable Codes, Standards, and Specifications

The following are applicable codes and Regulatory Guides. See section 1.9 for additional discussion of Regulatory Guides.

- ASME Code, Section III, Division 1, Subsections NC, NE, NF. (See paragraph 3.8.2.4 for dates.)
- Regulatory Guide 1.57, Design Limits and Loading Combinations for Metal Primary Reactor Containment System Components.
- 10 CFR 50, Appendix J.

3.8.2.3 Loads and Load Combinations

The loads and load combinations for the Class MC items conform with Article NE-3000 of the ASME Code and Regulatory Guide1.57. The component supports associated with the fuel transfer tube housing bellows assemblies conform with ASME Code Article NF-3000.

ASME Code, Article NC-3000, Section III, Division 1, Subsection NC applies to Class 2 piping.

3.8.2.4 Design and Analysis Procedures

The procedure for each component, part, or piping assembly is given in the following paragraphs. Since the penetration sleeve assemblies are considered portions of the containment liner plate system, their design and analysis are discussed in paragraph 3.8.1.4. Closure plates for spare penetration sleeves are designed and fabricated to the applicable rules of ASME Code, Section III, Division 1, Code Class MC, without stamping.

3.8.2.4.1 Equipment Hatch and Personnel Lock Assemblies

As Class MC items, these assemblies are designed and analyzed in accordance with Article NE-3000 of the 1974 ASME Code, Subsection NE, through Summer 1975 Addenda.

3.8.2.4.2 Fuel Transfer Tube and Housing Bellows Assemblies

As Class MC parts, these assemblies are designed and analyzed in accordance with Article NE-3000 of the 1974 ASME Code, through Winter 1976 Addenda. The tube supports come under the jurisdiction of ASME Code Subsection NF (same addenda) since these are designated as component supports.

3.8.2.4.3 Electrical Penetration Assemblies

As Class MC parts, these assemblies are designed and analyzed in accordance with Article NE-3000 of the 1977 ASME Code, through Summer 1979 Addenda.

3.8.2.4.4 Isolation Valve Encapsulation Vessel Assemblies

As Class MC parts, these assemblies are designed and analyzed in accordance with Article NE-3000 of the 1974 ASME Code, through Summer 1976 Addenda.

3.8.2.4.5 Process Pipe Penetration Assemblies [a]

Flued heads (forgings) and the basic Type IV arrangement used in penetration assemblies as Class 2 items are designed and analyzed in accordance with Article NC-3000 of the 1977 ASME Code, through Summer 1978 Addenda. The entire emergency sump line penetration assemblies are also Class 2.

3.8.2.5 Structural Acceptance Criteria

The fundamental acceptance criterion for the completed containment is successful completion of the structural integrity test.

The structural acceptance criteria for steel items other than the liner plate and sleeve assemblies include allowable stress values, deformation limits, and factors of safety and are established in accordance with ASME Section III, Division 1, Subsections NC (Class 2 piping) and NE (Class MC components or parts) as applicable. No permanent deformations are allowed under any loading condition.

The steel items, which are an integral part of the containment pressure boundary, are designed to meet minimum leakage rate requirements. The leakage rate shall not exceed the acceptable value indicated in the applicable technical specification.

The allowable peak stresses, both primary and secondary, and buckling criteria, are provided in the NE Subsection.

3.8.2.6 Materials, Quality Control, and Special Construction Techniques

A. The materials of the Class MC items listed in paragraph 3.8.2.1 comply with the requirements of ASME Code, Section III, Division 1, Subsection NE, Article NE-2000. The Class 2 piping assemblies comply with Subsection NC, Article NC-2000.

B. The quality control program involving welding procedures, erection tolerances, and nondestructive examination of both shop- and field-fabricated welds is in conformance with Articles NE-4000 and NE-5000 or NC-4000 and NC-5000 of the ASME Code, as applicable.

C. There are no special construction techniques used on the Class MC items or Class 2 piping assemblies.

[a] Fatigue of the pipe penetration assemblies was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.5.

3.8.2.7 Testing and Inservice Inspection Requirements

Testing of the Class MC items and Class 2 piping assemblies forming the pressure boundary within the containment conforms with Articles NE-6000 and NC-6000 of the ASME Code, respectively, with exception. Class 2 pneumatic requirements do not satisfy Article NC-6000, as the containment structural integrity test uses 1.15 times the design pressure, instead of an ASME Code value of 1.25. Regulatory Guide 1.18 is discussed in section 1.9.

Periodic testing of those Class MC items or Class 2 piping assemblies listed in paragraph 3.8.2.1 is planned in accordance with 10 CFR 50, Appendix J.

ASME III, Class MC components will be examined in accordance with the requirements of and frequency specified by the ASME Section XI Code, Subsection IWE except where relief has been authorized by the NRC. ASME III, Class 2 items are examined in accordance with section 6.6.

The 10 CFR 50 Appendix J testing and the ASME Section XI Inservice Inspection Program's IWE inspections are credited as license renewal aging management activities (see subsections 19.2.29 and 19.2.30).

3.8.2.8 Standard Review Plan Evaluation

The loading combinations specified in the design of Class MC components or parts differ from those in the Standard Review Plan.

The specifications for Class MC items require compliance with Article NE-3000 of the ASME Code. Even though the design calculations for some of the Class MC items do not specifically address the Standard Review Plan load combinations, the loads and load combinations conform with Article NE-3000 of the ASME Code and Regulatory Guide 1.57. Refer to section 1.9 for the VEGP position on Regulatory Guide 1.57.

3.8.3 CONCRETE AND STEEL INTERNAL STRUCTURES OF CONCRETE CONTAINMENT

3.8.3.1 Description of the Internal Structures

The internal structures are those concrete or steel structures inside of (not part of) the containment pressure boundary which support the reactor coolant system (RCS) components and related piping systems and equipment. The concrete structures also provide radiation shielding. The internal structures consist of the primary shield wall, secondary shield and pressurizer compartment walls, refueling canal walls, operating floor, intermediate slabs and platforms, and the polar crane runway girders. The polar crane brackets and equipment hatch hoist brackets are considered liner plate components, as described in paragraphs 3.8.1.1.2 and 3.8.1.6.6.B.

Component supports are those steel members designed to transmit loads from the RCS to the load-carrying building structures.

The RCS component supports are on drawing AX6DD312. Component supports are further described in subsection 5.4.14. A description of each major structure is provided below.

3.8.3.1.1 Reactor Pressure Vessel (RPV) Support System

The RPV is supported by four seats under two hot leg and two cold leg nozzles which are spaced approximately 90° apart in the primary shield wall. Reactor pressure vessel support is designed in such a way as to provide for radial thermal growth of the RCS, including the RPV, but so as to prevent the vessel from lateral and torsional movement during a loss-of-coolant accident (LOCA). The vertical loads are carried by the support seats to the embedded steel weldments under each support, while the radial and tangential loads are carried by the embedded steel weldments in the primary shield wall placed radially and tangentially to the wall. Reactor pressure vessel support seats and the associated embedded weldments are shown in drawings 1X2D48A018 and 1X2D48A019. A description of the primary shield wall and reactor cavity is provided in paragraph 3.8.3.1.4.

3.8.3.1.2 Steam Generator Support System

The steam generator support system is shown on drawing AX6DD312. The steam generator is vertically supported by four steel columns, pinned at both ends and bolted to support pads on the vessel and basemat embeds. (See drawings 1X2D01J008, 1X2D48E003, 1X2D48E004, and 1X2D48E005.) A pipe restraint is provided on the hot leg of the steam generator inlet pipe to prevent the formation of a plastic hinge at the primary shield wall and to limit the break area for a steam generator inlet nozzle break. A lower lateral component support is supplied by bearing blocks and a steel beam which spans the inside of the compartment walls. (See drawing AX6DD312.) The upper lateral component support consists of a bearing ring located near the center of gravity of the steam generator. The bearing ring is in turn restrained by a combination of hydraulic snubbers and a hard stop in the direction of thermal growth and by hard stops in the perpendicular direction. The steam generator is supported such that a main steam line or feedwater line break does not result in a break in the RCS or vice versa.

3.8.3.1.3 Reactor Coolant Pump Support System

The reactor coolant pump component supports consist of three steel columns, pinned at both ends and bolted to support pads on the pump and basemat embeds. Drawing AX6DD312 shows the general arrangement and design features. Horizontal steel tie rods, anchored to the primary and secondary shield walls, are provided for lateral support.

3.8.3.1.4 Primary Shield Wall and Reactor Cavity

The primary shield wall, a quasi-cylindrical, reinforced concrete structure extending from el 169 ft 0 in. to el 194 ft 1 3/4 in., provides a support system for the RPV. The primary shield wall and reactor cavity are illustrated on drawings AX6DD312, AX6DD314, 1X2D48A018, and 1X2D48A019. The primary shield wall is anchored into the basemat of the containment with reinforcing steel. Continuity of reinforcement across the pressure boundary is achieved with B-series cadweld splices welded to both sides of the thickened liner plate. (See drawing AX6DD314.)

Only a small vent area around each RPV nozzle has been provided to limit the flow of steam/water into the void space between the outside of the RPV and the primary shield wall, which therefore limits the differential pressure loadings on the RPV. An annular inspection gallery with eight entry ports is provided in the primary shield wall as an access to nozzle welds

for inservice inspection. Coolant pipe penetrations through the primary shield wall are provided with special restraints to limit the postulated break area.

3.8.3.1.5 Secondary Shield Walls

The secondary shield is a series of reinforced concrete walls which, together with the refueling canal walls and primary shield wall, enclose the steam generators. The secondary shield walls are anchored into the basemat of the containment in a manner similar to the primary shield walls to allow for load transfer to the basemat. (See drawing AX6DD314.) Each of the four secondary shield compartments provides support and houses a steam generator, a reactor coolant pump, and nuclear steam supply piping.

3.8.3.1.6 Pressurizer Supports

The pressurizer is supported on a steel ring bearing plate bolted to the flange of the pressurizer support skirt. This ring, in turn, rests on a structural steel frame which is attached to steel embeds in the pressurizer compartment walls. (See drawing AX6DD312.) The pressurizer compartment is a rectangular, reinforced concrete structure built integrally with the secondary shield wall on the outside of the No. 4 loop steam generator compartment. (See drawing AX6DD312.) The pressurizer is also supported laterally at an upper level by four stops projecting from embeds within the pressurizer compartment walls.

3.8.3.1.7 Refueling Canal

The refueling canal is a reinforced concrete structure extending above and to the side of the primary shield wall between el 194 ft 1 3/4 in. (el 179 ft 9 3/4 in. at its extreme low point) and el 220 ft 0 in. It provides for the underwater transfer of fuel assemblies, and for the storage of the reactor internals after removal from the reactor. The entire refueling canal is lined with 1/4-in.-thick stainless steel plate. The refueling canal is shown on drawings AX6DD312, 1X2D48E004, and 1X2D48E005.

3.8.3.1.8 Operating Floor

The operating floor, shown in drawing 1X2D48E005, consists of concrete slabs and steel beams with steel grating. The grating covers the space between the secondary shield wall and the containment shell and covers the reactor coolant pumps hatches. The concrete slabs are supported by the refueling canal walls and the secondary shield walls and by steel columns originating in the fill slab (el 171 ft 9 in.).

3.8.3.1.9 Internal Steel Framing

There are no intermediate floors, but a number of steel access platforms and concrete equipment support areas exist above and below the operating floor. (See drawings 1X2D48E008 and 1X2D48E007.) Steel framing exists at four main lower intermediate elevations to support piping; cable trays; heating, ventilation, and air-conditioning (HVAC) ducts; grating floors; and miscellaneous equipment. Steel framing is also provided at two elevations

above the operating floor to support the containment coolers, containment auxiliary coolers, preaccess filtration units, and hydrogen recombiners.

3.8.3.1.10 Reactor Missile Shield

The missile shield is a circular steel plate, approximately 15 ft in diameter, mounted within the integrated head of the reactor vessel. The missile shield is used to prevent any postulated missiles from the reactor vessel head appendages from penetrating other RCS pressure boundaries or the containment structure. In addition to this function, the missile shield also transfers the reactor vessel head load to the lifting rig. The missile shield also provides seismic support for the control rod drive mechanisms (CRDMs). During plant operation, the missile shield can be easily detached from the lifting ring assembly to provide access to the CRDMs.

3.8.3.1.11 Polar Crane Supporting Elements

The polar crane runway box girders, which support the crane rails, are chords traversing the containment circumference and are supported by the crane brackets. (See drawing 1X2D01J022.)

3.8.3.2 Applicable Codes, Standards, and Specifications

The following documents are applicable to the design, materials, fabrication, construction, inspection, and testing of the containment internal structures.

3.8.3.2.1 Codes and Standards

The following codes and standards apply:

- American Concrete Institute (ACI), Building Code Requirements for Reinforced Concrete, Standard 318-71 including the 1974 supplement.
- American Institute of Steel Construction (AISC), Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, adopted February 12, 1969, and including supplements 1, 2, and 3.
- American Society of Mechanical Engineers (ASME), Boiler and Pressure Vessel Code (hereinafter called the ASME Code), Section III, Division 1, Subsection NF (Component Supports), 1977 ASME Code including the Winter 1977 Addenda.
- American National Standards Institute (ANSI), ANSI N45.2.5-1974, Supplementary Quality Assurance Requirements for Installation, Inspection, and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants.

3.8.3.2.2 Regulations

Regulations applicable to the internal structures are included in those listed in paragraph 3.8.1.2.2.

3.8.3.2.3 General Design Criteria

The design of the containment internal structures complies with General Design Criteria 1, 2, 4, 5, and 50 of Appendix A to 10 CFR 50 and 10 CFR 50.55a.

3.8.3.2.4 Nuclear Regulatory Commission (NRC) Regulatory Guides

The following regulatory guides apply:

- Regulatory Guide 1.10, Mechanical (Cadweld) Splices in Reinforcing Bars of Category 1 Concrete Structures.
- Regulatory Guide 1.15, Testing of Reinforcing Bars for Category 1 Concrete Structures.
- Regulatory Guide 1.55, Concrete Placement in Category 1 Structures.
- Regulatory Guide 1.94, Quality Assurance Requirements for Installation, Inspection and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants.
- Regulatory Guide 1.142, Safety-Related Concrete Structures for Nuclear Power Plants.

The conformance to these regulatory guides is discussed in section 1.9.

3.8.3.2.5 Industry Standards

Nationally recognized industry standards, such as American Society of Testing Materials (ASTM), ACI, American Iron and Steel Institute (AISI), and American Welding Society (AWS), are used to specify material properties, testing procedures, fabrication, and construction methods.

3.8.3.3 Loads and Load Combinations

The loads and load combinations are the same as for other Category 1 structures described in paragraph 3.8.4.3 and the associated tables, except for the following modifications:

> Wind loads (W), tornado loads (W_t), blast loads (B), and precipitation loads (N) are not applicable to the design of the internal structures because of the protection provided by the containment shell; therefore, these loading terms have been excluded in the load combinations for the internal structure.

The RCS component supports are in conformance with the ASME Code, Section III, Division 1, Subsection NF, Component Supports.

3.8.3.4 Design and Analysis Procedures

The internal concrete structure is designed by the strength method as specified in the ACI Building Code Requirements for Reinforced Concrete, Standard 318-71, including the 1974 supplement.

The internal structural steel is designed in accordance with the AISC 1969 Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, including supplements 1, 2,

and 3. Specific components as described in paragraphs 3.8.3.4.1, 3.8.3.4.2, 3.8.3.4.3, and 3.8.3.4.6 are governed by the ASME Code, Subsection NF, instead of the AISC specification.

Two three-dimensional finite element mathematical models are created for computer analysis of the major concrete internal structures. One model comprises only the primary shield wall. The second encompasses the secondary shield walls, pressurizer compartment walls, and refueling canal walls.

The procedures described in section 3.7 are employed to obtain the seismic accelerations at various levels of the internal structures. The generation of the magnitudes of the accident loads due to pipe break is discussed in Containment Functional Design, subsection 6.2.1. The design for the effects of postulated pipe breaks is performed as specified in subsection 3.6.2.

Nuclear steam supply system (NSSS) support loads from the steam generator and reactor coolant pump columns and the crossover leg horizontal run restraint are transmitted into the basemat by embedded steel weldments. Anchor bolt sleeves which cross the pressure boundary form the continuity. (See drawing 1X2D01J008.) Models and methods of analysis for the RCS component supports are described in paragraph 3.9.N.1.4.4.

3.8.3.4.1 Reactor Pressure Vessel Support System [a]

The reactor pressure vessel supports embedded within the primary shield wall are procured in accordance with ASME Code, Section III, Division 1, Subsection NF; however, since they are outside the ASME jurisdictional boundary, their design follows AISC specifications.

3.8.3.4.2 Steam Generator Support System

The embeds for the steam generator supports are designed and procured in the manner described in paragraph 3.8.3.4.1. Lateral support loads on the concrete walls are discussed in paragraph 3.8.3.4.5.

3.8.3.4.3 Reactor Coolant Pump Support System

The embeds for the reactor coolant pumps are designed and procured in the manner described in paragraph 3.8.3.4.1.

3.8.3.4.4 Primary Shield Wall and Reactor Cavity

The primary shield wall is modeled separately using eight-node brick finite elements and is analyzed using the BSAP computer program described in appendix 3B. Seismic loads are derived from the acceleration values obtained using the procedures specified in section 3.7; accident loads are calculated from data described in subsection 6.2.1. The load combinations are discussed in paragraph 3.8.3.3. The loads are applied to a finite element model similar to that described in paragraph 3.8.3.4.5. During normal plant operation, a thermal gradient across the wall is generated by the heat of the reactor and the attenuation of gamma and neutron radiation originating from the reactor core. An insulation and cooling system is provided to reduce the severity of this gradient by limiting the temperature at the inside concrete face of the

[a] Fatigue of the reactor vessel supports was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in paragraph 19.4.2.4.

wall to 150°F. Localized areas, such as around a penetration, are allowed to have increased operating temperatures (not to exceed 200°F).

Should a LOCA occur, the steady-state temperature distribution across the primary shield wall, acting in conjunction with the accident pressure differential, is considered in the primary shield wall design. Design bases for differential pressure analyses within the interior subcompartments are discussed in paragraph 6.2.1.2. The peak pressure differential is of short duration, since equalization immediately begins to take place through its passages into the steam generator compartments and the free volume of the containment. The low thermal conductivity of the primary shield wall prevents occurrence of rapid changes in the temperature profile. Changes in the temperature distribution start from the steady-state temperature profile and proceed towards a more uniform distribution, due to heating on both wall faces. As such, the initial temperature effects on the primary shield wall due to a LOCA are considered negligible and the operating condition temperature differential is used for design.

3.8.3.4.5 Secondary Shield Walls

The secondary shield wall, in conjunction with the pressurizer compartment and the refueling canal, is modeled using both triangular and rectangular plate finite elements and beam, brick, and boundary finite elements and is analyzed using the BSAP computer program. Forces on the finite elements are introduced in the finite element model either as uniformly distributed or nodal loads. Seismic load input is a constant acceleration. Jet impingement and accident pressures occur as uniformly distributed surface loads while pipe reaction and pipe whip accident loads are nodal loads.

The secondary shield walls are mainly subjected to the lateral loadings of the steam generators, reactor coolant pumps, pressurizer, and compartment pressures due to accident conditions.

3.8.3.4.6 Pressurizer Support

The steel frame and its anchorage into the concrete walls are analyzed by conventional methods and designed in accordance with the AISC specification.

The bolts attaching the pressurizer base to the steel frame conform to the ASME Code, Section III, Division 1, Subsection NF, for material, design, and fabrication.

The finite element model for the pressurizer compartment walls is incorporated in the secondary shield wall finite element model.

3.8.3.4.7 Refueling Canal

For the refueling condition, the walls are designed for the hydrostatic head due to the maximum depth of water in the canal during refueling, including the hydrodynamic pressure effects due to the postulated seismic events, as well as a 140°F maximum water temperature. The steam generator compartment pressure loads due to postulated pipe rupture and hydrostatic head are not considered to occur simultaneously.

The NRC publication TID 7024(1) is used for computing hydro-dynamic loads imposed on the refueling canal walls by sloshing water during seismic events.

3.8.3.4.8 Operating Floor

Concrete floor slabs are analyzed by conventional beam and slab methods and designed in accordance with the ACI Code. For analysis and design of the operating floor structural steel, see paragraph 3.8.3.4.9.

3.8.3.4.9 Intermediate Slabs and Platforms

Structural steel in the containment internals is modeled using beam and truss finite elements and is analyzed using the BSAP computer program mentioned in appendix 3B. The steel is designed in accordance with the AISC specification.

Internal slabs are analyzed by conventional methods and designed in accordance with the ACI Code.

3.8.3.4.10 Polar Crane Runway Girders

The runway box girders are designed to AISC specifications. The runway brackets are described in paragraph 3.8.1.6.6.B.

The polar crane seismic loading is based on the seismic response of the containment at the crane support level.

The design of the polar crane is described in subsection 9.1.5.

3.8.3.5 Structural Acceptance Criteria

The structural acceptance criteria for the concrete and steel internal structures are described in paragraph 3.8.4.5.

3.8.3.6 Materials, Quality Control, and Special Construction Techniques

3.8.3.6.1 Materials

The following basic materials are used in the construction of the internal structures:

A. Concrete f'_c = 5000 psi (91-day strength with pozzolans; 28-day strength without pozzolans)

B. Concrete ingredients, reinforcing steel, cadweld splices, structural and miscellaneous steel, bolts, anchors, unistruts, and shear studs all follow those applicable specifications listed in paragraph 3.8.1.6.

   Superplasticizer admixtures are added to the concrete mixes used in congested areas to facilitate placement and prevent the formation of voids around reinforcement. These admixtures do not affect concrete quality and require an air content range of 1.5 to 4.5 percent. Other air-entrained concrete has a 3- to 6-percent air content range.

C. The refueling canal is lined with welded 1/4-in.-thick stainless steel plate. The liner plate, attachments thereto (exposed on the water side), and exposed insert plates are stainless steel material conforming to ASTM A 240, Type 304L. Exposed mechanical fasteners are also fabricated from equivalent corrosion-resistant stainless steel materials.

D. NSSS embedded items above the basemat level conform to SA-537, Class 1.

E. SA-516, Grade 70, unless otherwise noted on the design drawings, is the material used to fabricate the crossover leg horizontal run restraint steel weldments.

F. The material and quality control requirements for concrete comply with ACI 318-71. Structural steel is in conformance with the 1969 AISC specification.

G. The steel linear supports of the RCS are in compliance with Subsection NF of the ASME Code, Section III, Division 1.

H. The quality control for the internal structures is per Regulatory Guide 1.55, Concrete Placement in Category 1 Structures.

I. Structural steel is A36 material. A325 or A490 bolts are used for structural column or beam connections; A36 or A193 material is used for column anchor bolts.

3.8.3.6.2 Quality Control

A. Seam and plug welds of the stainless steel liner plate are tested by the vacuum box method or the helium mass spectrometer method. Where this is not practical, the liquid penetrant test is employed. Other nondestructive examination is in accordance with that described in paragraph 3.8.1.6.8.

B. Construction tolerances for the internal structures and reinforcing steel tolerances are discussed in paragraph 3.8.1.6.8.

Tolerances applicable to the refueling canal are as follows:

1. Location of embedded backing bars: ±1/2 in. of horizontal and vertical design locations.
2. Flatness of concrete wall surfaces: 3/8 in. measured in 5 ft in any direction.
3. Plumbness of concrete walls: 1/4 in. in 10 ft.
4. The extent of compliance with ANSI N45.2.5-1974 is as described in paragraph 1.9.94.2.

3.8.3.6.3 Special Construction Techniques

There are no special construction techniques used in the construction of the internal structures.

3.8.3.7 Testing and Inservice Inspection Requirements

A formal program of testing and inservice surveillance is not required for the internal structures. Tests and inspections for the RCS component supports are discussed in subsection 5.4.14.

For the period of extended operation, periodic inspections of the Category 1 containment internal structures by the Structural Monitoring Program are required license renewal aging management program activities (see subsection 19.2.32).

3.8.3.8 Standard Review Plan Evaluation

The Standard Review Plan specifies ACI 349, augmented by Regulatory Guide 1.142, as the acceptable code for design of concrete structures. The Standard Review Plan also specifies the load combinations that would result from the use of ACI 349, as modified by Regulatory Guide 1.142. VEGP design is based on ACI 318-71 and is in conformance with the load combinations specified in the Standard Review Plan. VEGP takes some alternatives to Regulatory Guides 1.55, 1.94, and 1.142.

Code ACI 349 was not in effect at the time the VEGP construction permit was issued. In accordance with the Preliminary Safety Analysis Report commitment, ACI 318-71 was used. The differences between ACI 349 and ACI 318 are minor except for the load combination equations which, in the case of VEGP, are in conformance with the Standard Review Plan. Thus, the design procedures and construction practices delineated in the FSAR ensure that the structure will withstand the specified design conditions without impairment of structural integrity or the performance of required safety functions.

Refer to section 1.9 for VEGP positions on regulatory guides.

3.8.3.9 Reference

1. "Nuclear Reactors and Earthquakes," NRC Report TID-7024, August 1963.

3.8.4 OTHER SEISMIC CATEGORY 1 STRUCTURES

Category 1 structures, other than the two containments and their internal structures, include the auxiliary building, fuel handling building, control building, two diesel generator buildings, two auxiliary feedwater pumphouses, four nuclear service cooling water (NSCW) towers and valve houses, two refueling water storage tanks, two reactor makeup water storage tanks, four condensate storage tanks, two diesel fuel storage tank pumphouses, and Category 1 tunnels.

3.8.4.1 Description of the Structures

3.8.4.1.1 Equipment Building

The equipment building is composed of portions of the control building and fuel handling building, as shown in drawings 1X4DE313, 1X4DE314, 1X4DE320, and 1X4DE322.

3.8.4.1.2 Auxiliary Building

The auxiliary building is a seven-story reinforced concrete building common to the two-unit plant. It is located south of the fuel handling building. Three stories are above grade; four are subterranean. There are two penetration areas, one on the south side of each containment. This building houses the chemical and volume control system (CVCS); emergency core cooling system (ECCS); residual heat removal (RHR) system; heating, ventilation, and air-conditioning (HVAC) facilities; and other associated equipment. There is a railroad access at grade and crane facilities for shipping and receiving of new and spent fuel. Tendon access shafts No. 2, which provide access to the containment tendon galleries and buttresses for each unit, are formed by portions of the auxiliary building and fuel handling building. Drawings 1X4DE315, 1X4DE318, 1X4DE321, 1X4DE323, 1X4DE324, and 1X4DE325 show the general layout and geometric description of the building. Directly south of each containment is a main steam isolation valve (MSIV) room.

3.8.4.1.3 Fuel Handling Building

The fuel handling building is a five-story, boxlike, reinforced concrete structure common to the two-unit plant. It is completely surrounded by other Category 1 buildings. It is south of the control building, north of the auxiliary building, and between the two containment structures.

The building has a center section which consists of two spent fuel storage pools, a new fuel storage area, cask loading pit, fuel transfer canals, and cask washdown area. The height of this section extends approximately 69 ft above grade and 40 ft below grade. On the east and west sides of the center section are penetration areas which provide access to the containment structures. These areas extend from 60 ft below grade to grade level.

The spent fuel storage pools have concrete walls and floors and are lined on the inside surfaces with 1/4-in.-thick stainless steel plate for leak prevention.

The new fuel storage area is a separate reinforced concrete pit providing temporary dry storage for the new fuel assemblies. An equipment and cask cleaning area is located adjacent to the spent fuel pools and new fuel pit.

The fuel transfer canal system is provided for transport of the new and spent fuel assemblies between the fuel handling building and the two containment buildings. A flexible transfer tube is provided for transfer of the fuel assemblies between the fuel handling building and each of the containment buildings. A mechanical fuel transfer system has been provided which is capable of moving a single fuel assembly in a horizontal orientation along the transfer canal and through the transfer tube.

The fuel handling building contains a bridge crane at el 220 ft which travels the length of the building (east-west), for handling of the spent fuel assemblies. A bridge crane runs north-south in the center of the building for lifting and transporting spent fuel casks. These cranes are discussed at length in section 9.1.

Tendon access shafts No. 2, which provide access to the containment building tendon galleries and buttresses for each unit, are formed by portions of the auxiliary and fuel handling buildings.

Drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE317, 1X4DE320, and 1X4DE322 show the general layout of the fuel handling building.

3.8.4.1.4 Control Building

The control building is a six-story, deeply embedded, reinforced concrete structure common to the two-unit plant. It is situated north of and adjacent to the fuel handling building and the two containment buildings. It is supported on a mat foundation 40 ft below grade. The boxlike center section has three upper levels extending to 60 ft above grade. A partial fourth level extends an additional 20 ft. Penetration areas east and west of the center section provide access to the two containment buildings. These are the primary areas for routing of electrical and control systems cable into the containment. Directly north of each containment building is an MSIV room which extends 40 ft above grade.

The floor at grade is principally occupied by the control room and technical support center. The floors immediately above and below grade house the cable spreading rooms. The lowest level houses switchgear and HVAC equipment. The third and fourth floors mainly contain HVAC equipment, while the fourth floor is primarily occupied by nonsafety-related components.

Access shafts No. 3, providing access to the containment tendon gallery and one buttress for each unit, are formed by portions of the control building.

Drawings 1X4DE312, 1X4DE313, 1X4DE314, 1X4DE317, 1X4DE320, and 1X4DE322 show the general layout of the control building.

3.8.4.1.5 Diesel Generator Buildings

The diesel generator buildings (one for each unit) are two-story rectangular, reinforced concrete structures approximately 92 ft by 114 ft in plan. They are approximately 60 ft in height and are supported on basemat foundations at grade. Each building is divided into two isolated bays. The first floor of each bay houses a diesel generator and a 5-ton bridge crane for equipment removal. The second floor houses air handling, exhaust, and silencing equipment. Drawing 1X4DE327 shows the general layout of the diesel generator buildings.

3.8.4.1.6 Auxiliary Feedwater Pumphouses

The auxiliary feedwater pumphouses (one for each unit) are one- story, rectangular, reinforced concrete structures approximately 40 ft by 74 ft in plan. They extend 22 ft above grade and are supported on basemat foundations 4 ft below grade. Four interior walls provide separation for the steam and electric driven pumps. Roof hatches are present for pump access. Drawing 1X4DE316 shows the general layout of the pumphouses.

3.8.4.1.7 NSCW Towers and Valve Houses

The NSCW cooling towers are mechanical draft cylindrical towers. They function as water storage reservoirs as well as a mechanical heat removal system. There are four NSCW cooling towers (two per unit). They are deeply embedded and are identical. They are separated by a minimum of 77 ft at the closest point. The NSCW cooling towers are reinforced concrete structures. Each consists of a 112-ft-high cylindrical shell, a 9-ft-thick mat foundation, and a 2-ft-thick flat roof deck. The shell wall is 3 ft thick above el 155 ft 5 1/2 in. and 5 ft thick below. The elevation of the top of the basemat is 137 ft 0 in. The top of the roof deck is at el 250 ft 11 in. Grade elevation is approximately 218 ft 0 in. The interior diameter of the cylindrical shell is 88 ft. Each tower has a nominal storage capacity of 3,670,000 gal and is normally filled with water to el 217 ft 9 in.

Large rectangular openings in the cylindrical shell are provided for air intake. The openings are 12 ft high with a bottom el of 218 ft 3 in. The openings are spaced around the entire circumference of each tower. The width varies, with an average of about 8 ft.

Inside the upper part of each tower, two perpendicular cross walls divide the towers into four separate cells. The cross walls are 2 ft 3 in. thick and extend from el 209 ft 9 in. to the roof. The cross walls support the roof deck and are designed as deep beams. At the top of each of the four cells, a large circular opening has been provided in the roof slab for air discharge.

Two levels of concrete beam grids are provided within each of the four cells. They are supported by the cross walls and cylindrical shell and are located at el 235 ft 8 in. and 242 ft 11 in. They support fill, spray distribution piping, eliminators, and other internals necessary to distribute and cool incoming water and to reduce vapor loss from the towers.

Surrounding each of the four roof slab openings on each tower and extending 14 ft above the roof deck are concrete fan stacks. The fan blades are centered in the fan stacks. The diameter of the fan stacks varies in order to enhance fan and airflow performance. The fan motors are mounted on the roof deck outside of the fan stacks and are protected with concrete housing structures.

A concrete splash ring (slab and wall) surrounds each of the towers at grade. The wall is located 12 ft from the exterior of the cylindrical shell and extends to a minimum of 12 in. above the top of the air intake openings.

A concrete buttress is provided on one side of each tower. It extends 10 ft 10 1/2 in. out from the 3-ft cylinder wall and is 38 ft 2 in. wide. It runs from el 218 ft 3 in. down to the basemat. The buttress encloses the four pump wells and supports the pumps.

Drawing 1X2D05E001 shows the general layout of the NSCW towers.

Each NSCW tower has an adjoining one-story, reinforced concrete, boxlike valve house. The valve house extends 30 ft above grade and is founded on a 6-ft-thick basemat foundation 14 ft below grade. The valve house also consists of two reinforced concrete missile shield slabs at el 248 ft 6 in. and el 241 ft 6 in. which protect piping entering the cooling tower from the valve house and the NSCW pumps mounted on the tower buttress. In plan, the wall patterns are irregular and follow the curvature of the cooling tower peripheries. The floor is split-level, allowing for pits to house process piping, valves, and equipment. Drawing 1X2D05E001 shows the general layout of the NSCW towers and the valve houses.

3.8.4.1.8 Category 1 Water Storage Tanks

The storage tanks for refueling water (one for each unit), reactor makeup water (one for each unit), and condensate water (two for each unit) are cylindrical, reinforced concrete shells. They are each covered with concrete roofs. They are supported on basemat foundations at grade. The condensate storage tanks are paired and share a common basemat. The tank capacities and dimensions are as follows:

Tank	Nominal Capacity	Inside Diameter	Height
Refueling water	715,500 gal	48 ft	60 ft
Reactor makeup water	165,000 gal	33 ft	40 ft
Condensate water	480,000 gal	44 ft	54 ft

The water side of the wall and floor of each of the tanks is lined with continuous stainless steel liner plate to ensure leaktight integrity. The refueling water, reactor makeup water and the

condensate storage tanks have perimeter dikes for retention of spilled water. Each pair of condensate water tanks have a common reinforced concrete valve house which provides missile protection for piping and equipment. The reactor makeup water and condensate water storage tanks are fitted with floating diaphragms which minimize oxygen absorption. Drawings 1X2D28A001, 1X2D29A001, and 1X2D30A001 show the general layout of the Category 1 tanks.

3.8.4.1.9 Diesel Fuel Storage Tank Pumphouses

The diesel fuel storage tank pumphouses (two for each unit) shelter pumps and valves for the buried diesel fuel storage tanks. The reinforced concrete pumphouses straddle the tanks and extend 3 ft above grade except for a common entry between each pair of pumphouses, which extends 14 ft above grade. Each pumphouse foundation consists of wall strip footings. The pumphouses are boxlike with work space levels above the top of the tanks. Drawings 1X4DE327 and 1X4DE330 show the general layout.

3.8.4.1.10 Category 1 Tunnels

The Category 1 tunnels are boxlike, reinforced concrete structures, either completely buried or with their roofs exposed at, or near, grade level. The tunnels house piping and electrical trays. Drawings 1X2D44A005 and AX2D11A048 show the partial layout and sections of the tunnels.

3.8.4.1.11 Masonry Walls

There are a few nonseismic masonry walls in Category 1 structures at VEGP. Three-hour-rated fire barriers (nonseismic) are located in seismic category 1 areas. Certain access openings are sealed with concrete units for radiation shielding and maintenance purposes. These concrete units and fire barriers are held captively in place by structural elements such as steel angles, concrete reinforcement bars, or steel beams. Seismic category 1 components are not supported from, on, or by these barriers.

3.8.4.2 Applicable Codes, Standards, and Specifications

The following documents are applicable to the design, materials, fabrication, construction, inspection, and testing.

3.8.4.2.1 Codes and Standards

The following codes and standards apply:

- American Concrete Institute (ACI), Building Code Requirements for Reinforced Concrete, ACI 318-71, including 1974 Supplement.
- American Institute of Steel Construction (AISC), Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings, adopted February 12, 1969, and Supplements Nos. 1, 2, and 3.

3.8.4.2.2 Regulations

The following regulation applies:

- 10 CFR 50.

3.8.4.2.3 General Design Criteria (GDC)

The following criteria apply:

- GDC 1, 2, 4, and 5 of Appendix A, 10 CFR 50.

3.8.4.2.4 Nuclear Regulatory Commission (NRC) Regulatory Guides

Conformance to the following Regulatory Guides is discussed in section 1.9:

- 1.10 Mechanical (Cadweld) Splices in Reinforced Bars of Category 1 Concrete Containment Structures.
- 1.15 Testing of Reinforcing Bars for Category 1 Concrete Structures.
- 1.55 Concrete Placement in Category 1 Structures.
- 1.91 Evaluations of Explosions Postulated To Occur on Transportation Routes Near Nuclear Power Plants.
- 1.194 Quality Assurance Requirements for Installation, Inspection, and Structural Steel During the Construction Phase of Nuclear Power Plants.
- 1.142 Safety-Related Concrete Structures for Nuclear Power Plants Other than Reactor Vessels and Containments.
- 1.143 Design Guidance for Radioactive Waste Management Systems, Structures, and Components Installed in Light-Water-Cooled Nuclear Power Plants.

3.8.4.2.5 Industry Standards

Nationally recognized industry standards, such as American Society for Testing and Materials, American Concrete Institute, and American Iron and Steel Institute, are used to specify material properties, testing procedures, fabrication, and construction methods.

3.8.4.3 Loads and Load Combinations

3.8.4.3.1 Loads

The loads considered are normal loads, severe environmental loads, extreme environmental loads, abnormal loads, and potential site proximity loads.

3.8.4.3.1.1 Normal Loads. Normal loads are those loads to be encountered, as specified, during initial load construction stages, during test conditions, and later, during normal plant operation and shutdown. They include the following:

D Dead loads or their related internal moments and forces, including any permanent loads except prestressing forces, including hydrostatic loads.

L Live loads or their related internal moments and forces, including any movable equipment loads and other loads which vary with intensity and occurrence, such as: floor area occupancy loads, moveable equipment loads, equipment laydown loads, nuclear fuel casks and fuel cask equipment loads, vehicular traffic loads, railroad equipment loads, and lateral soil pressures. Live load intensity varies depending upon the status of plant operation (i.e, prefuel load condition versus post-fuel load condition), the level of functional activity for a specific area, and the type of structural element. The prefuel load maximum design live loads for floor area occupancy include such things as temporary storage during construction of bulk materials and equipment, and maximum occupancy due to workers, tool boxes, scaffolding, and localized cribbing. The post-fuel load maximum anticipated live loads for floor area occupancy, pertaining to loading combinations including earthquake generated loads, are 100 percent of the live load occupancy levels expected to be present during plant operation, and are considered to be not less than 25 percent of the prefuel load maximum design live load for floor area occupancy.

T_o Thermal effects and loads during normal operating or shutdown conditions, based on the most critical transient or steady-state condition.

R_o Pipe reactions during normal operating or shutdown conditions, based on the most critical transient or steady-state conditions.

3.8.4.3.1.2 Severe Environmental Loads. Severe environmental loads are those loads to be infrequently encountered during plant life. Included in this category are:

E Loads generated by the operating basis earthquake (OBE). These include the associated hydrodynamic and dynamic incremental soil pressures.

W Loads generated by the design wind specified for the plant.

3.8.4.3.1.3 Extreme Environmental Loads. Extreme environmental loads are those loads which are credible but are highly improbable. They include:

E' Loads generated by the safe shutdown earthquake (SSE). These include the associated hydrodynamic and dynamic incremental soil pressures.

W_t Loads generated by the design tornado specified for the plant. Tornado load effects consist of tornado wind load (W_w), tornado differential pressure load (W_p), and tornado missile load (W_m). In general, the controlling load combination for tornado load effects in the design of structures or elements is:

$$W_t = W_w + 0.5\ W_p + W_m$$

N Loads generated by the probable maximum precipitation.

B Loads generated by postulated blast along transportation routes.

3.8.4.3.1.4 Abnormal Loads. Abnormal loads are those loads generated by a postulated high-energy pipe break accident within a building and/or compartment thereof. Included in this category are the following:

P_a Pressure load within or across a compartment and/or building, generated by the postulated break.

T_a Thermal loads generated by the postulated break and including T_o.

R_a Pipe and equipment reactions under thermal conditions generated by the postulated break and including R_o.

Y_r Load on a structure generated by the reaction on a ruptured high-energy pipe during the postulated event.

Y_j Load on a structure generated by the jet impingement from a ruptured high-energy pipe during the postulated break.

Y_m Load on a structure or pipe restraint resulting from the impact of a ruptured high-energy pipe during the postulated event.

3.8.4.3.1.5 Site Proximity Loads. Potential loads induced by floods, explosions, and aircraft hazards have been discussed in paragraphs 3.4.1.1, 3.5.1.5, and 3.5.1.6. Loads due to projectiles and missiles from nearby missile bases, in addition to those discussed, are not considered credible in the design of Category 1 structures.

3.8.4.3.1.6 Dynamic Effects of Loads. The dynamic effects from the impulsive and impactive loads caused by P_a, T_a, R_a, Y_r, Y_j, Y_m, B, and tornado missiles are considered by one of the following methods:

- Applying an appropriate dynamic load factor to the peak value of the transient load.
- Using impulse, momentum, and energy balance techniques.
- Performing a time-history dynamic analysis.

Elastoplastic behavior may be assumed with appropriate ductility ratios, provided excessive deflections will not result in loss of function of any safety-related system.

Dynamic increase factors appropriate for the strain rates involved may be applied to static material strengths of steel and concrete for purposes for determining section strength.

3.8.4.3.2 Load Combinations

3.8.4.3.2.1 Steel Structures. The steel structures and components are designed in accordance with elastic working stress design methods of Part 1 of the AISC specification, using the load combinations specifications in table 3.8.4-1.

If plastic design methods are used, the design is performed in accordance with the plastic design methods of Part 2 of the AISC specification, using the load combinations specified in table 3.8.4-2.

3.8.4.3.2.2 Concrete Structures. The concrete structures and components are designed in accordance with the strength design methods of ACI 318, using the load combinations specified in table 3.8.4-3.

3.8.4.4 Design and Analysis Procedures

The procedures used for the analysis and design of Category 1 structures (other than the containment and its internal structures) are described in this paragraph. The bases of design for the tornado, pipe break, and seismic effects are discussed in sections 3.5, 3.6, and 3.7, respectively. Evaluations for the postulated explosions are discussed in subsection 2.2.3. The foundation design is described in subsection 3.8.5. Each of these structures is founded on its own foundation, separated by a seismic gap to prevent interaction between the structures under seismic events. Computer programs used in the analysis and design of structures are described in appendix 3B.

The seismic design of the radwaste transfer building, radwaste transfer tunnel, and radwaste solidification building, for which Regulatory Guide 1.143 is applicable, are described in section 3.7.

3.8.4.4.1 Basic Structural Systems

Category 1 structures can be classified as three basic types of structures:

A. Shear Wall Structures

Included in this group are the auxiliary building, fuel handling building, control building, diesel generator buildings, auxiliary feedwater pumphouses, and NSCW valve houses.

B. Shell Structures

These include the NSCW towers, refueling water storage tanks (RWSTs), reactor makeup water tanks, and condensate storage tanks.

C. Rigid Box Structures

This system is used for buried structures such as the tunnels and diesel fuel oil storage tank pumphouses.

3.8.4.4.1.1 Shear Wall Structures. Concrete shear wall structures consist of vertical shear/bearing walls and horizontal slabs. Lateral loads are distributed to shear walls based on principles of relative rigidity; the more rigid shear walls resist proportionally more lateral load. Shear wall rigidities may be determined by hand calculations using traditional pair analysis assuming fixed-fixed or pinned-fixed boundary conditions, or shear distribution may be determined by a finite element model analysis. Vertical axial loading of the shear walls is determined based on slab tributary area carried by the wall. Additionally, out-of-plane bending and shear loads, such as seismic, lateral earth pressure, hydrostatic, hydrodynamic, and wind pressure, are evaluated and considered in the shear wall design. Slabs are designed for vertical and in-plane horizontal diaphragm loading.

The load distribution analysis and design of shear walls and slabs are in accordance with ACI 318 code.

3.8.4.4.1.2 Shell Structures. Shells are curved plates which, in circular structures, resist out-of-plane loading through axial (circumferential and meridional) forces. Bending stresses are important primarily at boundaries and discontinuities. The Category 1 tanks and NSCW towers fit well into this type of analysis.

The three Category 1 water tanks are structurally similar. Each has a foundation basemat, cylindrical tank walls, and a relatively flat circular roof. Taking advantage of symmetry, the tanks are analyzed, by hand, as simple axisymmetric cylindrical shells. Bending stresses are analyzed at the wall- to-roof and wall-to-basemat boundaries. Concrete design is done in accordance with the ACI 318 Code. Hydrodynamic forces are analyzed as outlined in reference 1.

The primary structural components of the NSCW towers are the basemat, cylindrical shell, fan deck, and cross walls. Axisymmetric shell analysis is not realistic due to the many discontinuities and irregularities in the structure (large openings, intermediate beam-grid systems, cross walls, varying cylinder wall thickness, and irregular fan deck geometry). The NSCW towers are therefore analyzed by three-dimensional finite element modeling. The BSAP computer program is used with beam, brick, and shell elements. A static analysis is performed using seismic loads developed per section 3.7, hydrodynamic loads calculated as outlined in reference 1, lateral earth pressure loads, and other applicable loads. The concrete design is performed using the computer program OPTCON.

A second, more detailed, finite element model is used to analyze the fan deck and fan stacks. The BSAP computer program is used with beam and LCCT9 plate elements. Static and dynamic analyses are performed. The fan deck is treated as a structural subsystem of the NSCW towers. The floor response spectra at the fan deck level are used as the seismic input for the dynamic analysis. All concrete design is done manually, ACI Code 381.

3.8.4.4.1.3 Rigid Box Structures. Buried structures such as the tunnels and diesel fuel storage tank pumphouses are rigid box systems. They are primarily designed to resist bending and shearing stresses resulting from out-of-plane forces. Static and dynamic soil pressures comprise a large portion of the loading. Analysis is done manually, using conventional moment distribution techniques. Concrete design is performed in accordance with ACI 318 Code.

3.8.4.5 Structural Acceptance Criteria

The analysis and design of concrete structures are in accordance with ACI 318-71 (with 1974 Supplement). ACI 349-76 is not employed, except for limited cases after the construction phase, since it was not published at the VEGP construction permit stage (June 1974). The structural steel design is based on the AISC 1969 specification and the three subsequent supplements. Welding of structural steel, miscellaneous steel, raceway supports, and HVAC duct supports is performed in accordance with the Structural Welding Code, AWS D1.1, 1975 or later edition. The AWS D1.1 Code edition invoked is the edition in effect at the purchase order date for material specifications and the initial issue date for construction specifications, unless otherwise stated in the specifications. Changes in the code edition by construction or the supplier require engineering approval. The VEGP acceptance criteria for visual inspection of structural welds is described in section 1.9.94.2. The load combinations and the strength limits are described in paragraph 3.8.4.3 and the associated tables.

3.8.4.6 Materials, Quality Control, and Special Construction Techniques

Materials, quality control, and construction techniques follow those described in paragraph 3.8.3.6.

The minimum design compressive strengths of the Category 1 reinforced concrete structures, other than the containment and its internal structures, are provided below. These are 91-day strengths when pozzolans are used and 28-day strengths without pozzolans.

Structure	f_C'
Auxiliary building	5000 psi
All others	4000 psi

Superplasticizer admixtures are added to the concrete mixes used in congested areas to improve flow and prevent the formation of voids around reinforcement. These admixtures do not affect concrete quality; they require an air content range of 1.5 to 4.5 percent. Other air-entrained concrete has a 3- to 6-percent air content range.

In selected areas (e.g., exhaust silencer pads, exhaust stack missile barriers, and roof slabs in locations of barriers, located in diesel generator building), heat resistance concrete is used. The compressive strength for this concrete is 6000 psi at 24 h.

3.8.4.7 Testing and Inservice Inspection Requirements

There are no testing or inservice inspection requirements for the Category 1 structures except for the containment structure.

For the Category 1 structures required to be maintained as Category 1, periodic inspections performed under the Structural Monitoring Program are required license renewal aging

management program activities for the period of extended operation (see subsection 19.2.32). The noncivil features of the outdoor tanks (e.g., fluid-retaining) are age-managed separately from these structural inspection programs as part of the associated fluid systems.

3.8.4.8 Standard Review Plan Evaluation

The Standard Review Plan specifies ACI 349, augmented by Regulatory Guide 1.142, as the acceptable code for design of concrete structures. The Standard Review Plan also specifies the load combinations that would result from the use of ACI 349, as modified by Regulatory Guide 1.142. VEGP design is based on ACI 318-71 and is in conformance with the load combinations specified in the Standard Review Plan. VEGP also takes some alternative positions to Regulatory Guide 1.143.

Code ACI 349 was not in effect at the time the VEGP construction permit was issued. In accordance with the Preliminary Safety Analysis Report commitment, ACI 318-71 was used. The differences between ACI 349 and ACI 318 are minor except for the load combination equations which, in the case of VEGP, are in conformance with the Standard Review Plan. Thus, the design procedures and construction practices delineated in the FSAR ensure that the structure will withstand the specified design conditions without impairment of structural integrity or the performance of required safety functions.

Refer to section 1.9 for VEGP positions on regulatory guides.

3.8.4.9 Reference

1. "Nuclear Reactors and Earthquakes," NRC Report TID-7024, August 1963.

3.8.5 FOUNDATIONS

3.8.5.1 Description of the Foundations

All Category 1 structures have reinforced concrete foundations. The majority employ thick basemats. Only the diesel fuel storage tank pumphouse and Category 1 tunnels utilize different footing types.

Major adjoining buildings are structurally separated from neighboring buildings by a seismic separation gap which typically is 5.5 in. This separation provides ample space to prevent interaction between the buildings during a seismic event. A typical seismic separation joint incorporating a waterstop is shown in drawing AX2D94V031.

Resistance to sliding of the massive concrete basemat structures is provided by passive soil pressure and soil friction. This is adequate to provide the required factor of safety against lateral movement under the most stringent loading conditions.

For ease of construction, Category 1 structures are built on mud mats. The mud mats are of lean, nonstructural concrete of approximately 6-in. thickness and rest upon the load-bearing soil. Waterproofing treatment is employed as described in paragraph 3.4.1.1.1. The waterproofing material is applied directly to the mud mats prior to placing the foundations. Vertical surfaces are waterproofed on the earth side prior to backfilling.

The computer programs used in foundation analyses, either BSAP or STRUDL-II, are described in appendix 3B.

3.8.5.1.1 Containment

The containment foundation is a mat and is described in subsection 3.8.1.

3.8.5.1.2 Containment Internal Structures

The containment internal structures are supported by the containment basemat foundation, which is described in subsection 3.8.1.

3.8.5.1.3 Auxiliary Building

This structure is founded on a 10-ft-thick mat, continuous over the plan of the building.

3.8.5.1.4 Fuel Handling Building

The fuel handling building foundation consists of three mats, each 6 ft. thick. The two wing area mats are placed at lower elevations than the central mat. The central mat, at its top, is at el 179 ft 0.5 in. Portions of the Category 1 tunnels are attached to the underside of the central mat.

3.8.5.1.5 Control Building

The control building foundation mat is 7 ft thick and is similar to the auxiliary building basemat. Portions of the Category 1 tunnels are attached to the underside of the basemat.

3.8.5.1.6 Diesel Generator Building

This structure is supported by a 9-ft-thick mat. The diesel generators rest on pads which are integral with the basemat.

3.8.5.1.7 Auxiliary Feedwater Pumphouse

This structure is supported on a 3-ft-thick mat foundation with a pit provided for valves and piping.

3.8.5.1.8 Nuclear Service Cooling Water Tower and Valve House

The foundation for the nuclear service cooling water (NSCW) tower consists of a 9-ft-thick circular mat, 100 ft in diameter. The NSCW valve house is supported on a 6-ft-thick mat foundation.

3.8.5.1.9 Category 1 Water Tanks

Each reactor makeup water storage tank and each refueling water storage tank (RWST) is supported by a separate foundation mat.

Each pair of condensate water storage tanks is supported by a combined foundation mat.

The approximate mat dimensions are:

- Refueling water tank, 62 ft square x 4 ft thick.
- Reactor makeup water tank, 51 ft square x 4 ft thick.
- Condensate storage tanks, 63 ft x 115 ft x 4 ft thick.

The dikes for Category 1 storage tanks are constructed of reinforced concrete and are integral portions of the basemats.

3.8.5.1.10 Diesel Fuel Oil Storage Tank Pumphouse

This building is supported on wall strip footings with the depth of footings extending approximately 10 ft into the supporting soil. The interior excavations are occupied by the fuel oil storage tanks and then backfilled. Resistance to horizontal loads is sustained by passive pressure of the soil.

3.8.5.1.11 Category 1 Tunnels

The Category 1 tunnels are box-shaped culverts with moment-resistant joints. Sliding resistance is afforded by passive soil resistance.

3.8.5.2 Applicable Codes, Standards, and Specifications

For the foundations of the containments and internal structures, refer to paragraph 3.8.1.2. For the other Category 1 structures, refer to paragraph 3.8.4.2.

3.8.5.3 Loads and Load Combinations

Foundation loads and load combinations for the containment and its internal structures are described in paragraph 3.8.1.3.

The foundation loads and load combinations for the other Category 1 structures are described in paragraph 3.8.4.3.

3.8.5.4 Design and Analysis Procedures

The basemat is analyzed either using a finite element model or conventional methods, such as treating the mat as a plate on an elastic foundation. In the finite element model, the soil underneath the basemat is represented by discrete soil springs attached to the nodes of the model. The spring constants are functions of the tributary areas of the finite elements and the coefficients of subgrade reaction.

The heavy mat footings preclude significant differential settlements within a particular structure.

The effects of differential settlements on the VEGP structures have been shown to be negligible (reference response to Q241.18). The maximum total predicted settlements of the power block structures are given in figure 2.5.4-1.

In the shear wall structures, diaphragm action of the horizontal slabs carries lateral forces (causing overturning moments) to the shear walls, which in turn distribute loads to the basemats. The following shear wall structures exhibit this action:

- Auxiliary building.
- Fuel handling building.
- Control building.
- Diesel generator buildings.
- Auxiliary feedwater pumphouses.
- NSCW valve houses.

The diesel fuel oil storage tank pumphouse is a box-like structure which transfers the lateral forces through the shear walls to its wall strip footings.

In the tank structures, including the NSCW towers, the concrete shells sustain both the vertical and lateral loads and transfer them to the mat foundations. The containment shells likewise deliver vertical and lateral forces to their basemats.

The rigid frame actions of the Category 1 tunnels transfer lateral forces to the supporting soil.

The three-dimensional nature of earthquakes, as described in paragraph 3.7.B.2.14, is used as seismic input when considering the effects of overturning and sliding.

3.8.5.4.1 Containment

For the basemat analysis, the basemat and reactor cavity are modeled with brick elements. The cylinder and dome are modeled with plate elements.

The tendon gallery is designed as an inverted rigid frame, supported with fixed ends at the underside of the containment basemat. An access shaft is attached to the basemat and tendon gallery and is designed as a retaining well.

3.8.5.4.2 Containment Internal Structures

The containment internal concrete structures are modeled with plate elements.

3.8.5.4.3 Auxiliary Building

The basemat is analyzed using a finite element analysis, with the basemat modeled with plate elements.

3.8.5.4.4 Fuel Handling Building

Plate finite elements are used in the model of the fuel handling building.

3.8.5.4.5 Control Building

The control building foundation is modeled with plate elements.

3.8.5.4.6 Diesel Generator Building

The foundation is designed by hand calculations and a finite element model using beam elements.

3.8.5.4.7 Auxiliary Feedwater Pumphouse

The basemat is designed using hand calculations.

3.8.5.4.8 Nuclear Service Cooling Water Tower and Valve House

The cooling tower and valve house basemats are modeled with plate elements.

3.8.5.4.9 Category 1 Water Tanks

The condensate storage tank basemat is modeled using plate elements. The basemats of refueling and reactor makeup storage tanks are designed by conventional methods using hand calculations.

3.8.5.4.10 Diesel Fuel Storage Tank Pumphouses

The wall strip footings are designed by conventional methods using hand calculations.

3.8.5.4.11 Category 1 Tunnels

The tunnels are designed as either box culverts using hand methods or as inverted rigid frames if attached to the underside of basemats.

3.8.5.5 Structural Acceptance Criteria

The structural acceptance criteria for the containment foundation follow those given for the buildings in paragraph 3.8.1.5. Refer to paragraph 3.8.4.5 for the foundations, also use the same structural acceptance criteria as their superstructures. The limiting conditions for the foundation medium together with a comparison of actual capacity and estimated structure loads are found in subsection 2.5.4. The minimum factors of safety against overturning, sliding, and flotation for Category 1 buildings are given in table 3.8.5-1.

3.8.5.6 Materials, Quality Control, and Special Construction Techniques

The materials and quality control program for the foundations are the same as those listed for the structures. The foundations for the containments and internal structures follow paragraph

3.8.1.6, while the other Category 1 structure foundations conform with those shown in paragraph 3.8.4.6.

There are no special construction techniques employed in construction of the foundations.

3.8.5.7 Testing and Inservice Inspection Requirements

Testing and inservice surveillance other than obtaining and recording foundation settlement data is not required or planned for the structural foundations. The containment basemat, being a portion of the pressure boundary is, however, subject to the requirements specified in paragraph 3.8.1.7.

Settlement monitoring is achieved by use of over 100 markers placed in the basemats and building columns of the Category 1 structures and the turbine building. Two permanent benchmarks, founded in the marl, act as reference points for settlement observations.

For the period of extended operation, periodic inspections of the Category 1 structures required to be maintained as Category 1 (see paragraph 3.8.4.7) are required license renewal aging management program activities to manage aging of foundations (see subsection 19.2.32).

3.8.5.8 Standard Review Plan Evaluation

The Standard Review Plan specifies American Society of Mechanical Engineers (ASME) Section III, Division 2 Code for the containment foundation and American Concrete Institute (ACI) 349 for the other Seismic Category 1 foundations as the applicable code. The design of the VEGP containment foundation is based on Article CC-3000 of the ASME Code. The design of other Seismic Category 1 foundations is based on ACI 318-71. The Standard Review Plan specifies the codes, standards, and guides listed in its sections 3.8.1 and 3.8.3 as acceptable for the containment foundation and other Seismic Category 1 foundations, respectively.

Code ACI 349 was not in effect at the time the VEGP construction permit was issued. In accordance with the Preliminary Safety Analysis Report commitment, ACI 318-71 was used. The differences between ACI 349 and ACI 318 are minor except for the load combination equations which, in the case of VEGP, are in conformance with the Standard Review Plan. Thus, the design procedures and construction practices delineated in the FSAR ensure that the structure will withstand the specified design conditions without impairment of structural integrity or the performance of required safety functions.

Refer to section 1.9 for VEGP positions on regulatory guides.

3.8.5.9 Reference

1. Bechtel Power Corporation, Inc., Report on Foundation Investigations, Vol 1, VEGP, San Francisco, California, July 1974.

TABLE 3.8.1-1 (SHEET 1 OF 2)

CONTAINMENT LOAD COMBINATIONS AND LOAD FACTORS

Category	D[(a)]	L[(b)]	F	Pt	Pa	Tt	To	Ta	Eo	Ess	W	Wt	Ro	Ra	Rr	Pv	B
Service																	
Test	1.0	1.0	1.0	1.0	--	1.0	--	--	--	--	--	--	--	--	--	--	--
Construction	1.0	1.0	1.0	--	--	--	1.0	--	--	--	1.0	--	--	--	--	--	--
Normal	1.0	1.0	1.0	--	--	--	1.0	--	--	--	--	--	1.0	--	--	1.0	--
Severe environmental	1.0	1.0	1.0	--	--	--	1.0	--	1.0	--	--	--	1.0	--	--	1.0	--
	1.0	1.0	1.0	--	--	--	1.0	--	--	--	1.0	--	1.0	--	--	1.0	--
Factored																	
Severe environmental	1.0	1.3	1.0	--	--	--	1.0	--	1.5	--	--	--	1.0	--	--	1.0	--
	1.0	1.3	1.0	--	--	--	1.0	--	--	--	1.5	--	1.0	--	--	1.0	--
Extreme environmental	1.0	1.0	1.0	--	--	--	1.0	--	--	1.0	--	--	1.0	--	--	1.0	--
	1.0	1.0	1.0	--	--	--	1.0	--	--	--	--	1.0	1.0	--	--	1.0	--
	1.0	1.0	1.0	--	--	--	1.0	--	--	--	--	--	1.0	--	--	1.0	1.0
Abnormal	1.0	1.0	1.0	--	1.5	--	--	1.0	--	--	--	--	--	1.0	--	--	--
	1.0	1.0	1.0	--	1.0	--	--	1.0	--	--	--	--	--	1.25	--	--	--
Abnormal/severe environmental	1.0	1.0	1.0	--	1.25	--	--	1.0	1.25	--	--	--	--	1.0	--	--	--
	1.0	1.0	1.0	--	1.25	--	--	1.0	--	--	1.25	--	--	1.0	--	--	--
	1.0	1.0	1.0	--	--	--	1.0	--	1.0	or	1.0	--	--	--	--	--	--
Abnormal/extreme environmental	1.0	1.0	1.0	--	1.0	--	--	1.0	--	1.0	--	--	--	1.0	1.0	--	--

a. Symbols:

- D = dead loads.
- F = prestressing force loads.
- E_o = operating basis earthquake.
- E_{ss} = safe shutdown earthquake.
- L = live load.
- P_v = pressure load due to pressure variation during normal operating conditions.
- P_t = test pressure load.
- P_a = accident/incident maximum pressure.
- R_o = piping loads during operating conditions.

TABLE 3.8.1-1 (SHEET 2 OF 2)

R_a	=	piping loads due to increased temperature resulting from design accident.
T_a	=	accident/incident temperature.
T_o	=	operating temperature.
T_t	=	test temperature.
W	=	wind loads.
W_t	=	tornado loads.
R_r	=	the local effects due to pipe rupture. (See CC-3220, ASME Section III, Division 2.)
B	=	blast load due to postulated site-proximity explosion.

b. Includes all temporary construction loading during and after construction of containment.

TABLE 3.8.4-1[A]

OTHER CATEGORY 1 STRUCTURES
STEEL DESIGN LOAD COMBINATIONS
ELASTIC METHOD

	EQN	D	L	P_a	T_o	T_a	E	E'	W	W_t	R_o	R_a	Y_j	Y_r	Y_m	N	B	Strength Limit(f_s)
Service Load Conditions																		
	1	1.0	1.0															1.0
	2	1.0	1.0				1.0											1.0
	3	1.0	1.0						1.0									1.0
	4	1.0	1.0		1.0						1.0							1.5
	5	1.0	1.0		1.0		1.0				1.0							1.5
	6	1.0	1.0		1.0				1.0		1.0							1.5
Factored Load																		
	7	1.0	1.0		1.0			1.0			1.0							1.6
(See note b.)	8	1.0	1.0		1.0					1.0	1.0							1.6
	9	1.0	1.0	1.0		1.0						1.0						1.6
(See notes c and d.)	10	1.0	1.0	1.0		1.0	1.0					1.0	1.0	1.0	1.0			1.6
(See notes c and d.)	11	1.0	1.0	1.0		1.0		1.0				1.0	1.0	1.0	1.0			1.7
	12	1.0	1.0		1.0						1.0						1.0	1.6
	13	1.0	1.0		1.0						1.0					1.0		1.6

a. See paragraph 3.8.4.3 for definition of load symbols including the definition of L during prefuel load and post-fuel load conditions. f_s is the allowable stress for the elastic design method defined in Part 1 of the AISC, "Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings." The one-third increase in allowable stresses permitted for seismic or wind loadings is not considered.

b. When considering tornado missile load, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without the tornado missile load is also to be considered.

c. When considering Y , Y_r, and Y_m loads, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without Y_j, Y_r, and Y_m is also to be considered.

d. For this load combination, in computing the required section strength, the plastic section modulus of steel shapes, except for those which do not meet the AISC criteria for compact sections, may be used.

TABLE 3.8.4-2[(a)]

OTHER CATEGORY 1 STRUCTURES
STEEL DESIGN LOAD COMBINATIONS
PLASTIC METHOD

	EQN	D	L	P_a	T_o	T_a	E	E'	W	W_t	R_o	R_a	Y_j	Y_r	Y_m	N	B	Strength Limit(Y)
Service Load Conditions																		
	1	1.7	1.7															1.0
	2	1.7	1.7				1.7											1.0
	3	1.7	1.7						1.7									1.0
	4	1.3	1.3		1.3						1.3							1.0
	5	1.3	1.3		1.3		1.3				1.3							1.0
	6	1.3	1.3		1.3				1.3		1.3							1.0
Factored Load Conditions																		
	7	1.0	1.0		1.0			1.0			1.0							1.0
(See note b.)	8	1.0	1.0		1.0					1.0	1.0							1.0
	9	1.0	1.0	1.5		1.0						1.0						1.0
(See note c.)	10	1.0	1.0	1.25		1.0	1.25					1.0	1.0	1.0	1.0			1.0
(See note c.)	11	1.0	1.0	1.0		1.0		1.0				1.0	1.0	1.0	1.0			1.0
	12	1.0	1.0		1.0							1.0					1.0	1.0
	13	1.0	1.0		1.0							1.0				1.0		1.0

a. See paragraph 3.8.4.3 for definition of load symbols including the definition of L during prefuel load and post-fuel load conditions. Y is the required section strength based on plastic design methods described in Part 2 of the AISC "Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings."

b. When considering tornado missile load, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without the tornado missile load is also to be considered.

c. When considering Y_j, Y_r, and Y_m loads, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without Y_j, Y_r, and Y_m is also to be considered.

TABLE 3.8.4-3[(a)]

OTHER CATEGORY 1 STRUCTURES
CONCRETE DESIGN LOAD COMBINATIONS
STRENGTH METHOD

	EQN	D	L	P_a	T_o	T_a	E	E'	W	W_t	R_o	R_a	Y_j	Y_r	Y_m	N	B	Strength Limit
Service Load Conditions																		
	1	1.4	1.7															U
(See note b.)	2	1.4	1.7						1.7									U
(See note c.)	3	1.4	1.7				1.9											U
	4	1.05	1.275		1.275							1.275				U		
	5	1.05	1.275		1.275					1.275		1.275			U			
	6	1.05	1.275		1.275		1.425				1.275				U			
Factored Load Conditions																		
	7	1.0	1.0		1.0			1.0			1.0							U
(See note d.)	8	1.0	1.0		1.0					1.0	1.0							U
	9	1.0	1.0	1.5		1.0						1.0						U
(See note e.)	10	1.0	1.0	1.25		1.0	1.25					1.0	1.0	1.0	1.0			U
(See note f.)	11	1.0	1.0	1.0		1.0		1.0				1.0	1.0	1.0	1.0			U
	12	1.0	1.0		1.0						1.0						1.0	U
	13	1.0	1.0		1.0						1.0					1.0		U

a. See paragraph 3.8.4.3 for definition of load symbols including the definition of L during prefuel load and post-fuel load conditions. U is the required strength based on strength method per ACI 318-71.

b. Unless this equation is more severe, the load combination 1.2D+1.7W is also to be considered.

c. Unless this equation is more severe, the load combination 1.2D+1.9E is also to be considered.

d. When considering tornado missile load, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without the tornado missile load is also to be considered.

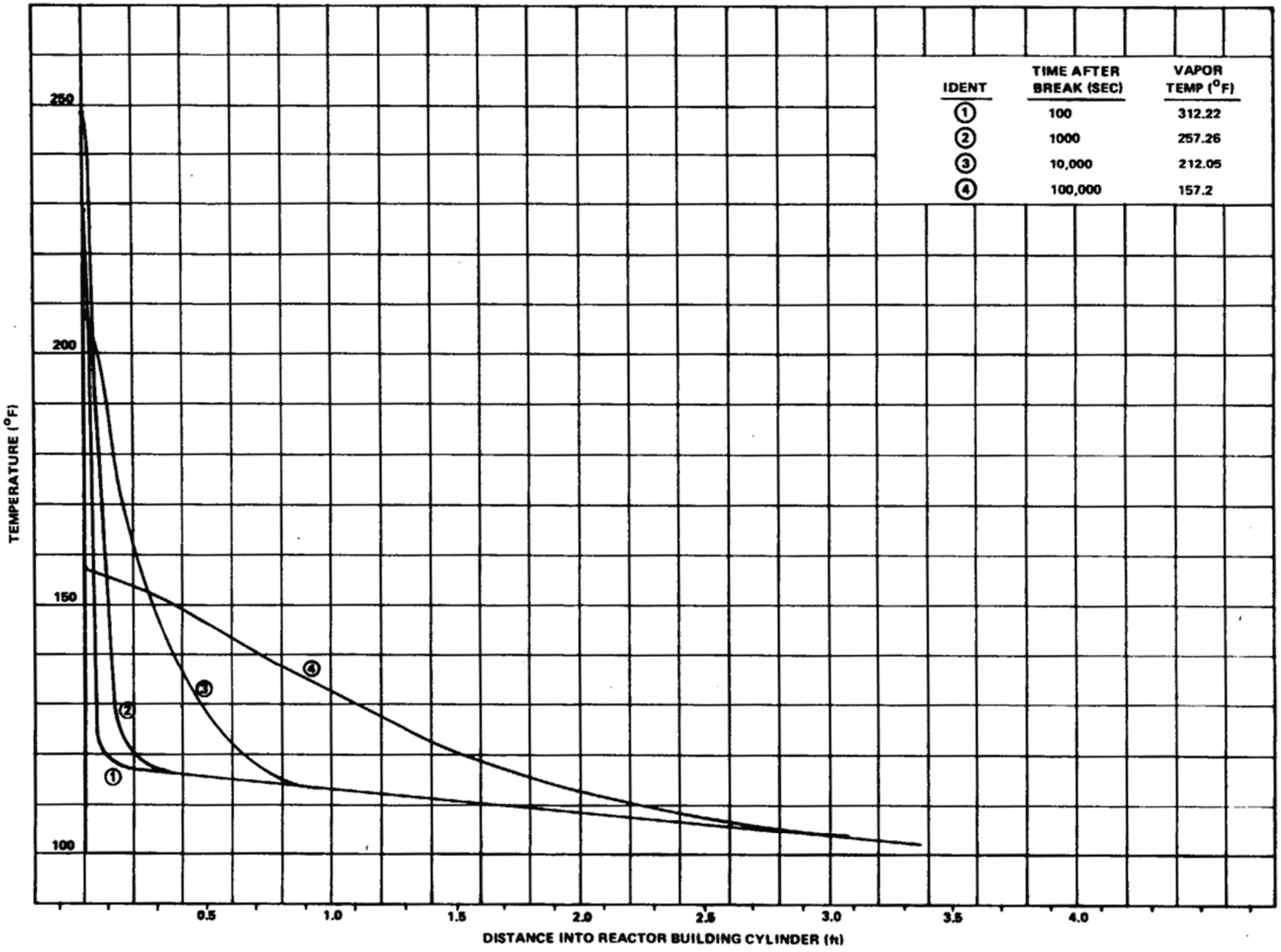
e. When considering Y_j, Y_r, and Y_m loads, local section strength may be exceeded provided there will be no loss of function of any safety-related system. In such cases, this load combination without Y_j, Y_r, and Y_m is also to be considered.

TABLE 3.8.5-1

MINIMUM DESIGN FACTORS OF SAFETY

For Combination[a]	Overturning	Sliding	Flotation
D+H+E	1.5	1.5	-
D+H+W	1.5	1.5	-
D+H+E'	1.1	1.1	-
D+H+W	1.1	1.1	-
D+F'	-		1.5

a. H = Lateral earth pressure.
F' = Flood loads.
D = Dead loads.
E = Loads generated by the operating basis earthquake (OBE).
E' = Loads generated by the safe shutdown earthquake (SSE).
W = Wind loads.
W_t = Tornado loads.



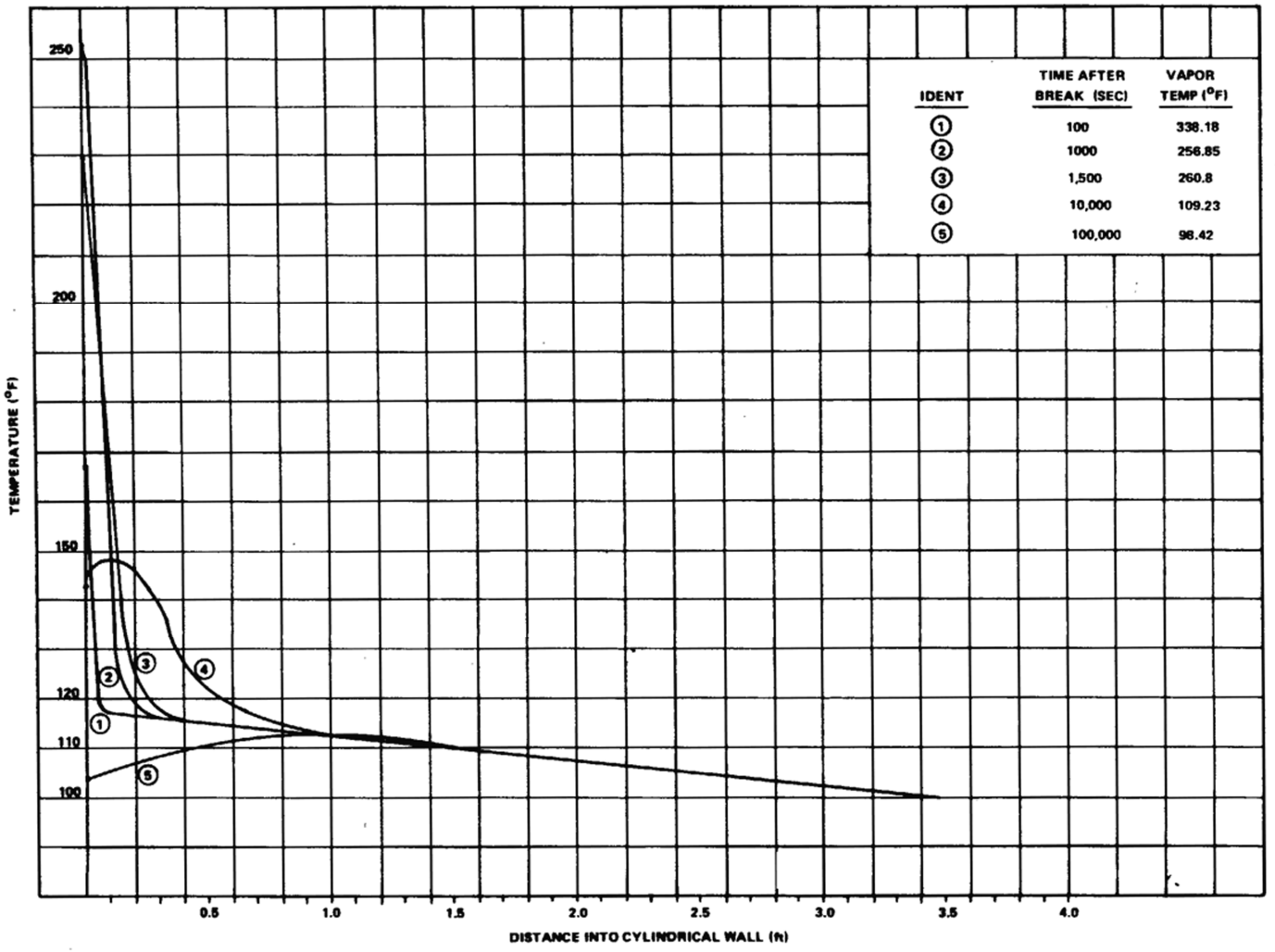
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TEMPERATURE GRADIENTS THROUGH CONTAINMENT WALL FOR LOCA

FIGURE 3.8.1–1 (SHEET 1 OF 2)



REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TEMPERATURE GRADIENTS THROUGH
CONTAINMENT WALL FOR
MAIN STEAM LINE BREAK (MSLB)

FIGURE 3.8.1–1 (SHEET 2 OF 2)

3.9.B MECHANICAL SYSTEMS AND COMPONENTS

3.9.B.1 SPECIAL TOPICS FOR MECHANICAL COMPONENTS

3.9.B.1.1 Design Transients

Refer to paragraph 3.9.N.1.1 for a description of the operating conditions considered in the design of the reactor coolant system (RCS), RCS component supports, and reactor internals. Class 1 piping systems are designed and analyzed using the design transients described in paragraph 3.9.N.1.1.

Class 2 and 3 piping systems and components do not require thermal transient analysis. Class 2 and 3 piping systems and components are designed and analyzed for dynamic transients, as listed in subsection 3.9.B.2.

3.9.B.1.2 Computer Programs Used in Analysis

For nuclear steam supply systems (NSSSs) and Class 1 branch lines, refer to paragraph 3.9.N.1.2.

3.9.B.1.2.1 Seismic Category 1 Items Other Than NSSS Scope

Analysis of piping systems and supports not included in the NSSS scope is performed by use of the following proprietary computer programs:

- ME-101.
- ME-210.
- BSAP (CE800, CE212, and CE217).
- GENERAL FRAME ANALYSIS.
- ANSYS.
- BISEPS (ME-140 and ME-240).
- BASE PLATE PROGRAM (ME-035)
- ME-105
- FAPPS (ME-150)
- PIPESTRESS

The following computer programs have been used in dynamic and static analyses to determine mechanical loads, stresses, and deformations of balance of plant components. These programs are described and verified in reference 2.

- WESTDYN and its associated pre- and post-processors - static and dynamic analyses of redundant piping systems.

- ITCHVALVE - transient hydraulic analyses for a piping system.
- FORFUN - calculation of unbalanced hydraulic forces between two piping elbows as a function of time.
- PIPSAN - finite beam elements to model, analyze, and evaluate three-dimensional linear elastic structures.
- STRUDL - linear elastic two- and three-dimensional analysis of frame or truss structures.
- WECAN - performs finite element structural analyses.
- WESTAT - equivalent static analyses of redundant piping systems.

These proprietary computer programs conform to the requirements of 10 CFR 50, Appendix B, Section III, and are verified as described below.

3.9.B.1.2.1.1 ME-101 Program, Linear Elastic Analysis of Piping System. The ME-101 program is used to determine stresses and loads in the piping systems due to restrained thermal expansion, deadweight, seismic inertia and anchor movements, externally applied loads such as jet-loads, and transient forcing functions such as created by fast relief valve opening and closing, fast check valve closure after pipe breaks in main feedwater line, fast valve closure in main steam line, etc. ME-101 analyzes piping systems in accordance with ANSI and ASME codes.

The ME-101 program is a finite element computer program which performs linear elastic analysis of piping systems using the stiffness method of finite element analysis; the displacements of the joints of a given structure are considered basic unknowns. The dynamic analysis by the modal synthesis method utilizes known maximum accelerations produced in a single degree of freedom model of a certain frequency. The principal program assumptions are as follows:

A. It is a linearly elastic structure.

B. Simultaneous displacement of all supports is described by a single time-dependent function.

C. Lumped mass model satisfactorily replaces the continuous structure.

D. Modal synthesis is applicable.

E. Rotational inertia of the masses has negligible effect.

The results obtained from pipe stress program ME-101 have been compared with the following:

A. ME-632, computer program, seismic analysis of piping systems, VERB MOD8, Bechtel International Corporation, San Francisco, California, 1976.

B. ASME Benchmark problem results, Pressure Vessel and Piping 1972 computer programs verification, American Society of Mechanical Engineers.

C. Longhand calculations--ME-101 is compatible with NRC Regulatory Guide 1.92. A synthesis of closely spaced modes is provided based on equation 4 of Regulatory Guide 1.92.

The verification report is on file at Bechtel.

3.9.B.1.2.1.2 ME-210 Program, Local Stress in Cylindrical Shells Due to External Loading. The ME-210 computes the local stresses in cylindrical shells that results from external loadings. The program is based on Welding Research Council Bulletin 107, August 1965. The program has been verified based upon hand calculations.

3.9.B.1.2.1.3 BSAP. This program is used in calculating moments, forces, displacements, and stiffness of the support components of Seismic Category 1 piping. A description and verification of this program is included in appendix 3B.

3.9.B.1.2.1.4 General Frame Analysis. This program uses a matrix method for solutions. The program solves two-dimensional structural frames used for pipe support and supporting structures.

3.9.B.1.2.1.5 ANSYS. The ANSYS program is a general purpose computer program for solution of several classes of engineering problems. ANSYS includes capabilities for structural analysis, including static elastic, plastic, creep, dynamic and dynamic plastic analyses, large deflection and stability analyses, one-dimensional fluid flow analyses, and heat transfer analysis including conduction and convection. The ANSYS has been developed and verified by Swanson Analysis System, Inc.

3.9.B.1.2.1.6 BISEPS Program, Bechtel Interactive System for Engineering Pipe Support. The BISEPS program is an integrated computer system used in the design of pipe supports. This program is divided into two subprograms:

A. BISEPS STAND (ME-140) performs the sizing of support hardwares, standard steel configurations, and welds in accordance with the standard support design specification. BISEPS STAND also generates a design "hard copy" or sketch showing the bill of materials, location plan and elevation, and other information used in the design, such as loads, movements, etc.

B. BISEPS FRAME (ME-240) performs only the sizing of nonstandard or skew steel configurations or any configuration not scoped within the BISEPS STAND program.

The program verification report is on file with Bechtel data processing.

3.9.B.1.2.1.7 ME-105 Program, OPTIPIPE. The ME-105 is a preprocessing program which provides automated data collection and data entry for the ME-101 program. The verification report is on file at Bechtel.

3.9.B.1.2.1.8 BASE PLATE (ME-035). The BASE PLATE program is a finite element program used to design and analyze pipe support flexible base plates on geometrically nonlinear foundations. The program performs geometry calculations to generate the finite element model and creates data sets for output report tables and the determination of the deformed and/or undeformed geometry configuration. The verification report is on file at Bechtel.

The Base Plate code was verified by comparing test problem results with the results of CDC Base Plate II code runs for the same problems. The code capabilities investigated include:

(1) Calculation of maximum bolt pullout, bolt SRSS shear force, and bolt safety factor

(2) Maximum principal stress of plates

(3) Weld shear stresses

(4) Pressure on concrete

In each case the results of the Base Plate code were consistent with the CDC benchmark.

3.9.B.1.2.1.9 ME-150 Program, FAPPS. The ME-150 optimizes frame member sizes, welds, base plates, and embedments based upon various user-specified design limitation. The verification report is on file at Bechtel.

The FAPPS code was verified by comparing test problem results with the results of STRUDL code runs for the same problems. The code capabilities compared included stiffness matrix formulation and stress calculation techniques common to both codes. In other applications where FAPPS calculations are not within STRUDL capabilities, hand calculations were performed to ensure proper operation of the code. In each case, the FAPPS results were consistent with the STRUDL results or the hand calculations.

3.9.B.1.2.1.10 PIPESTRESS Program, Linear Elastic Analysis of Piping System. The PIPESTRESS is an interrelated finite element computer program which performs linear elastic analysis of piping systems using the stiffness method of finite element analysis; the displacements of the joints of a given structure are considered basic unknowns. The dynamic analysis by the modal synthesis method utilizes known maximum accelerations produced in a single degree of freedom model of a certain frequency. The principal program assumptions are as follows:

It is a linearly elastic structure. Simultaneous displacement of all supports is described by a single time-dependent function. Lumped mass model satisfactorily replaces the continuous structure. Modal synthesis is applicable. Rotational inertia of the masses has negligible effect.

PIPESTRESS has advanced static and dynamic analysis capabilities including detailed uniform and multilevel response spectrum analyses, time history and fatigue calculations, and multiple load cases and load combinations. Stresses due to internal pressure are calculated according to the code. PIPESTRESS solves static problems by constructing a linear finite element model of the piping system using the load-deflection relationships based on the displacement method. Dynamic analysis calculates bound solutions or time history solutions for dynamic loads, which may be described by response spectra or time history data. The dynamic analysis methods used by PIPESTRESS are Modal Extraction, Single or Multilevel Response Analysis, Multimodal/Multilevel Response Analysis, Generalized Response Analysis, Selective Time History Analysis, Left-Out-Force Method, and Primary and Secondary Terms involved in Multilevel Response Analysis. Thermal transient analysis can be performed using a finite difference approximation to find thermal gradients in the pipe walls, thereby determining the maximum value during the transient analysis of the various stress terms. Fluid properties are calculated as functions of instantaneous transient fluid temperatures and pressures.

PIPESTRESS was benchmarked against all seven test problems in NUREG/CR-1677, BNL-NUREG-51267, Vol. I, and against three test problems in NUREG/CR-1677, BNL-NUREG-

51267, Vol. II. These permitted the verification of the analysis methods implemented in PIPESTRESS. The program was verified under a nuclear quality assurance program established in accordance with 10 CFR 50, Appendix B and 10 CFR 21. The verification was directed by personnel competent in the design of ASME Section III, Nuclear Power Plant Components, Class 1, 2, and 3 nuclear power plant piping under ASME nuclear quality assurance procedures. The program performs calculations in accordance with the requirements and intention of Subarticles NC/ND-3600 of ASME, Section III. PIPESTRESS has been developed and verified by DST Computer Services S. A.

3.9.B.1.3 Experimental Stress Analysis

3.9.B.1.3.1 NSSS and Class 1 Branch Lines

Refer to paragraph 3.9.N.1.3.

3.9.B.1.3.2 Seismic Category 1 Items Other Than NSSSs and Class 1 Branch Lines

Experimental stress analysis methods are not used in the design of code or noncode components for the faulted condition. For code components, the stresses do not exceed the limits of Section III of the ASME Boiler and Pressure Vessel Code.(1)

3.9.B.1.4 Considerations for the Evaluation of the Faulted Condition

3.9.B.1.4.1 Seismic Category 1 Items in the NSSS and Class 1 Branch Lines

Refer to paragraph 3.9.N.1.4.

3.9.B.1.4.2 Seismic Category 1 Items Other Than the NSSS

For statically applied loads, the stress allowables of Appendix F of ASME Section III, Division 1, are used for code components. For noncode components, allowables are based on tests or accepted standards consistent with those in the 1974 edition of Appendix F of ASME Section III, Division 1.

Dynamic loads for components loaded in the elastic range are calculated using dynamic load factors, time-history analysis, or any other method that assumes elastic behavior of the component. A component is assumed to be in the elastic range if yielding across a section does not occur. The limits of the elastic range are defined in Paragraph F-1323 of Appendix F of ASME Section III, Division 1, for code components. Local yielding due to stress concentration is assumed not to affect the validity of the assumptions of elastic behavior. The stress allowables of Appendix F of ASME Section III, Division 1, for elastically analyzed components are used for code components. For noncode components, allowables are based on tests or accepted material standards consistent with those in Appendix F for linear elastically analyzed components.

Analysis concerning the rupture of high-energy piping is addressed in section 3.6.

3.9.B.1.5 Weld Inspection Criteria

3.9.B.1.5.1 ASME Boiler and Pressure Vessel Code Section III/ANSI B31.1 Welding

The acceptance criteria for visual inspection of welds meets, or exceeds, the requirements of ASME Code, Section III, 1977 Edition through Winter 1977 Addenda, or ANSI B31.1 Code, 1977 Edition, through Winter 1977 Addenda, as applicable to the weld joint being evaluated.

3.9.B.1.5.2 AWS D1.1 Welding

The acceptance criteria for visual inspection of welds in pipe support supplementary steel structures which are completed in accordance with AISC [(a)], 7th Edition, and AWS D1.1-75 conform to the requirements of these codes except as noted in paragraph 1.9.94.

3.9.B.1.6 References

1. ASME Boiler and Pressure Vessel Code, Section III, Nuclear Power Plant Components, 1974 edition including Addenda through Summer 1975. For small bore piping analysis only, subartical NC-3600 of the Winter 1981 Addenda to the 1980 Edition for Class 2 piping, and subartical ND-3600 of the Summer 1984 Addenda to the 1983 Edition for Class 3 piping.
2. The description and verification of the computer codes are in compliance with quality assurance requirements (WCAP-9550, WCAP-9565, and WCAP-9805) and are maintained in the applicable central file.

3.9.N MECHANICAL SYSTEMS AND COMPONENTS

3.9.N.1 SPECIAL TOPICS FOR MECHANICAL COMPONENTS

3.9.N.1.1 Design Transients

The following five operating conditions are considered in the design of the reactor coolant system (RCS), RCS component supports, and reactor internals.

A. Normal Conditions

Any conditions in the course of startup, operation in the design power range, hot standby, and system shutdown, other than upset, emergency, faulted, or testing conditions.

B. Upset Conditions (Incidents of Moderate Frequency)

Any deviations from normal conditions anticipated to occur often enough so that design should include a capability to withstand the conditions without operational impairment. The upset conditions include those transients that result from any

[(a)] Steel construction manual.

single operator error or control malfunction, transients caused by a fault in a system component requiring its isolation from the system, and transients due to loss of load or power. Upset conditions include any abnormal incidents not resulting in a forced outage and also forced outages for which the corrective action does not include any repair of mechanical damage. The estimated duration of an upset condition is included in the design specifications.

C. Emergency Conditions (Infrequent Incidents)

Those deviations from normal conditions that require shutdown for correction of the conditions or repair of damage in the system. The conditions have a low probability of occurrence but are included to provide assurance that no gross loss of structural integrity will result as a concomitant effect of any damage developed in the system. The total number of postulated occurrences for such events shall not cause more than 25 stress cycles having an S_a value greater than that for 10^6 cycles from the applicable fatigue design curves of the American Society of Mechanical Engineers (ASME) Code Section III.

D. Faulted Conditions (Limiting Faults)

Those combinations of conditions associated with extremely low probability; postulated events whose consequences are such that the integrity and operability of the nuclear energy system may be impaired to the extent that consideration of public health and safety are involved. Such considerations require compliance with safety criteria as may be specified by jurisdictional authorities.

E. Testing Conditions

Testing conditions are those pressure overload tests, including hydrostatic tests, and pneumatic tests. Other types of tests are classified as normal conditions.

To provide the necessary high degree of integrity for the equipment in the RCS, the transient conditions selected for equipment fatigue evaluation are based upon a conservative estimate of the magnitude and frequency of the temperature and pressure transients which may occur during plant operation. To a large extent, the specific transients to be considered for equipment fatigue analyses are based upon engineering judgment and experience. The transients selected are sufficiently severe or frequent enough to be of possible significance to component cyclic behavior. The transients selected may be regarded as a conservative representation of transients which, used as a basis for component fatigue evaluation, provide confidence that the component is appropriate for its application over the design life of the plant. These transients are described by pertinent variations in pressure, fluid temperature, and fluid flow. The Fatigue Monitoring Program, as described in subsection 19.3.2, will be used to monitor plant transients that are significant contributors to the fatigue cumulative usage factor to ensure that the design limit on fatigue usage is not exceeded during the period of extended operation.

The design transients and the number of cycles of each that is normally used for fatigue evaluations are presented in table 3.9.N.1-1. In accordance with ASME III, emergency and faulted conditions are not included in fatigue evaluations. The cyclic or transient limits for the components listed in table 3.9.N.1-2 shall be monitored in accordance with the Component Cyclic or Transient Limit Program.

The transients and components that are monitored in the Component Cyclic or Transient Limit Program (tables 3.9.N.1-2 and 3.9.N.1-3) are based on the following methodology:

- Class 1 components monitored are determined by comparing both design fatigue usage and projected fatigue usage to a screening value of design cumulative usage fatigue

(CUF) ≤ 0.1. Metal fatigue, including the effect of environmentally assisted fatigue, was evaluated as a time-limited aging analysis in accordance with 10 CFR 54.21. The results for the period of extended operation are summarized in subsection 19.4.2.

- Plant cycles monitored are determined by evaluation of the contribution to fatigue usage of lifetime projected plant cycles for any Class 1 component, as well as a screening level of approximately 10% of design allowable cycles applied to the 60-year cycle projections.
- Fatigue monitoring of the limiting component(s) affected by a cycle may be used to show that the ASME Code acceptance criteria of CUF ≤ 1.0 remains valid even if the assumed number of cycles has been exceeded.
- The screening levels were selected to accommodate the maximum anticipated effect of reactor water environmental factors for a projected 60-year operating period.

A feedwater bypass system, minimizing thermal stratification conditions at the main feedwater nozzle, is a feature of the VEGP steam generators. This system reroutes all feedwater through the auxiliary feedwater nozzle when the steam generators are operating below 12% of full feedwater flow, except as otherwise noted.

For any period during a transient in which total feedwater flowrate drops below about 12% of nominal flow, the total feedwater flow enters through the auxiliary feedwater nozzle. For transients which involve total feedwater flowrates greater than about 12% of nominal feedwater flow, the flow through the auxiliary nozzle is approximately 8% of the total flow in effect at the time; i.e., 92% of total flow is through the main feedwater nozzle, and 8% of total flow is through the auxiliary feedwater nozzle.

3.9.N.1.1.1 Normal Conditions

The following primary system transients are considered normal conditions:

- Reactor coolant pump (RCP) startup and shutdown.
- Plant heatup and cooldown.
- Unit loading and unloading between 0 and 15% of full power.
- Unit loading and unloading at 5% of full power/min.
- Reduced temperature return to power.
- Step-load increase and decrease of 10% of full power.
- Large step-load decrease with steam dump.
- Steady-state fluctuations.
- Boron concentration equalization.
- Feedwater cycling.
- Loop out of service.
- Refueling.
- Turbine roll test.
- Primary side leakage test.

- Secondary side leakage test.

3.9.N.1.1.1.1 RCP Startup and Shutdown. The RCPs are started and stopped during routine operations such as RCS venting, plant heatup and cooldown, and in connection with recovery from certain transients such as loop out of service and loss of power. Other (undefined) circumstances may also require pump starting and stopping.

Of the spectrum of RCS pressure and temperature conditions under which these operations may occur, three conditions have been selected for defining transients:

- Cold condition - 70°F and 400 psig 1000 occurrences.
- Pump restart condition - 100°F and 400 psig 500 occurrences.
- Hot condition - 557°F and 2235 psig 2500 occurrences.

For RCP starting and stopping operations, it is assumed that variations in RCS primary side temperature and in pressurizer pressure and temperature are negligible and that the steam generator secondary side is completely unaffected. The only significant variables are the primary system flow and the pressure changes resulting from the pump operations.

- Minimum pressure required for RCP operation may be as low as 300 psig. Four-hundred psig is considered a conservative value for design purposes.
- These conditions are included to take care of situations requiring stopping and restarting the pumps after plant heatup has commenced and venting of the RCS prior to starting a heatup.

The following cases are considered:

Case 1 - First Pump Startup (Last Pump Shutdown)

This case involves variations in reactor coolant loop (RCL) flow which accompany startup of the first pump, both in the loop containing the pump being started and in the other loops (loops in which the pumps remain idle). This case also involves a higher dynamic pressure loss in the loop containing the pump being started, but the magnitude of the flow change is less than in case 2. For the last pump shutdown case, the transient is the reverse of the first pump startup transient.

Case 2 - Last Pump Startup (First Pump Shutdown)

This case conservatively represents the variations in RCL flow accompanying startup of the second, third, and fourth pumps as applicable. Initially, flow exists through these loops in the reverse direction as the result of starting the first pump. The remaining pumps are then started in sequence and a new equilibrium flow is established. The magnitude of flow reversal is the largest in the loop containing the last pump to be started. For the first pump shutdown case, the transient is the reverse of the last pump startup transient.

The 4000 occurrences listed in table 3.9.N.1-1 include RCP startups and shutdowns associated with RCS heatup and cooldown.

3.9.N.1.1.1.2 Heatup and Cooldown. For purposes of designing the major RCS components, the plant heatup and cooldown operations are conservatively represented by

continuous 100°F/h ramp temperature changes, between the shutdown temperature of 120°F(1) and the no-load temperature of 557°F (for the pressurizer vessel, the design cooldown rate is 200°F/h). The number of plant heatup and cooldown operations is defined as 200 each, which corresponds to 5 occurrences per year for the 40-year plant design life.[a] In practice these operations occur more slowly. Some factors which contribute to the lower rates are as follows:

- Material ductility considerations which limit temperature rates of change as functions of RCS pressure and temperature.
- Slower heatup rates when using pump energy only.
- Interruptions due to factors such as pressurizer steam bubble formation, control rod withdrawal, sampling, oxygen scavenging, and other reactor coolant chemistry adjustments.

RCS temperature can be as low as 70°F during the shutdown period. Between 70°F and 120°F the temperature is assumed to change very slowly, without causing any significant thermal transient effects.

3.9.N.1.1.1.3 Unit Loading and Unloading Between 0 and 15% of Full Power. The unit loading and unloading cases between the 0- and 15% power levels are represented by continuous and uniform ramp power changes, requiring 30 min for loading and 5 min for unloading. During loading, reactor coolant temperatures are increased from the no-load value to the normal load program temperatures at the 15% power level. The reverse temperature change occurs during unloading.

Prior to loading, it is assumed that the plant is at hot standby, with 32°F feedwater cycling. Loading commences and the feedwater temperature is assumed to increase from 32°F to the 15% power value. During each loading cycle, the auxiliary feedwater nozzle experiences two cycles of cold (32°F) feedwater addition. In addition, the main feedwater nozzle experiences a step change from 32°F to a final temperature determined for each loading cycle. The duration of the cold feedwater is 30 to 60 s and is picked to maximize the stresses on the steam generator and nozzles. Subsequent to unloading, feedwater heating is terminated, steam dump is reduced to residual heat removal (RHR) requirements, and feedwater temperature decreases from the 15% power value to 32°F. RCS pressure and pressurizer pressure are assumed to remain constant at the normal operating values during these operations.

The number of these loading and unloading transients is assumed to be 500 each during the 40-year plant design life.[b]

3.9.N.1.1.1.4 Unit Loading and Unloading at 5% of Full Power/min. The unit loading and unloading operations are conservatively represented by continuous and uniform ramp power

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a time-limited aging analysis (TLAA) for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

[b] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

changes of 5%/min between the 15- and 100% power levels. This load swing is the maximum possible consistent with operation under automatic reactor control. The reactor temperature varies with load as prescribed by the reactor control system. The unit loading is accomplished by manual rod control because the automatic control rod withdrawal capability has been disabled.

The number of unloading operations is defined as 13,200, based on one swing per day during the 40-year design life of the plant[a] and assuming a 90% availability factor.

It is also possible that as many as 2000 of the loading operations may be conducted in accordance with the "reduced temperature return to power" transient discussed in the following section. Both of these transients must be evaluated to determine which is more severe for a particular component design. If the reduced temperature mode is the more severe, then 2000 occurrences of that transient should replace 2000 occurrences of the 5% of full power/min loading operation, reducing the number of 5% of full power/min loadings to 11,200.

3.9.N.1.1.1.5 Reduced Temperature Return to Power. The reduced temperature return to power operation is designed to improve the spinning reserve capabilities of the plant during load-follow operations. The transient normally begins at the ebb (50%) of a load-follow cycle and proceeds at a rapid positive rate (typically 5%/min) until the abilities of the control rods and the coolant temperature reduction (negative moderator coefficient) to supply reactivity are exhausted. At that point, further power increases are limited to approximately 1%/min, by the ability of the boron system to dilute the reactor coolant. The reduction in primary coolant temperature is limited by the protection system to about 20°F below the programmed value. The plant loading during load follow operations is accomplished by manual rod control because the automatic control rod withdrawal capability has been disabled.

The reduced temperature return to power operation is not intended for daily use. It is designed to supply additional plant capabilities when required because of network fault or upset condition. Hence, this mode of operation is not expected to be used more than once a week in practice (2000 times in 40 years[a]).

3.9.N.1.1.1.6 Step-Load Increase and Decrease of 10% of Full Power. The ±10% step change in load demand is a transient which is assured to be a change in turbine control valve position due to disturbances in the electrical network into which the plant output is tied. The reactor control system is designed to restore plant equilibrium without reactor trip following a 10% step change in turbine load demand initiated from nuclear plant equilibrium conditions in the range between 15 and 100% of full load, the power range for automatic reactor control. In effect, during load change conditions, the RCS attempts to match turbine and reactor outputs such that peak reactor coolant temperature is minimized and reactor coolant temperature is restored to its programmed setpoint at a sufficiently slow rate to prevent excessive pressurizer pressure decrease. Manual rod control may be needed following a load increase transient because the automatic control rod withdrawal capability has been disabled.

Following a step decrease in turbine load, the secondary side steam pressure and temperature initially increase because the decrease in nuclear power lags the decrease in turbine load. During the same increment of time, the RCS average temperature and pressurizer pressure also increase, but this change lags slightly behind the secondary side change. Because of the coolant temperature increase and the power mismatch between turbine and reactor, the control system automatically inserts the control rods to reduce core power. The reactor coolant

temperature then decreases from its peak value to a value below its initial equilibrium value. The change in reactor coolant average temperature setpoint is made as a function of turbine-generator load, as determined by first-stage turbine pressure measurement. Pressurizer pressure also decreases from its peak value and follows the reactor coolant decreasing temperature trend. At some point during the decreasing pressure transient, the saturated water in the pressurizer begins to flash; this reduces the rate of pressure decrease. Subsequently, the pressurizer heaters come on to restore the pressure to its normal value.

Following a step increase in turbine load, the reverse situation occurs; i.e., the secondary side steam pressure and temperature initially decrease and the reactor coolant average temperature and pressure initially decrease. Manual rod control is used to withdraw the control rods to increase core power. The decreasing pressure transient is reversed by actuation of the pressurizer heaters and eventually the system pressure is restored to its normal value. The reactor coolant average temperature is raised to a value above its initial equilibrium value.

The number of each operation is specified at 2000 times, or 50 times per year for the 40-year plant design life[a].

3.9.N.1.1.1.7 Large Step-Load Decrease with Steam Dump. This transient applies to a step decrease in turbine load from full power of such magnitude that the resultant rapid increase in reactor coolant average temperature and secondary side steam pressure and temperature automatically initiates a secondary side steam dump that prevents both reactor trip and lifting of steam generator safety valves. The VEGP Units 1 and 2 are designed to accept a step decrease of 50% from full power.

Subsequent to the large step-load decrease, reactor power is reduced at a controlled rate, resulting in lower flow through the steam dump system. Another consequence of this event is turbine overspeed to as high as 110% of nominal (controlled overspeed just below the turbine overspeed trip setpoint). This results in proportional increases in generator bus frequency, RCP speed, and reactor coolant flowrate.

The number of occurrences of this transient is specified at 200 times (5 per year) for the 40-year plant design life. [b]

3.9.N.1.1.1.8 Steady-State Fluctuations. It is assumed that reactor coolant pressure and temperature can vary around the nominal (steady-state) values during power operation. These variations can occur at many frequencies, but for design purposes two cases are considered:

- Initial Fluctuations

  Initial fluctuations are due to control rod cycling during the first 20 full-power months of reactor operation. Reactor coolant temperatures are assumed to vary by ±3°F and pressure by ±25 psi once during each 2-min period. The total number of occurrences is

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

[b] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

limited to 1.5 x 10^5. These fluctuations are assumed to occur consecutively, but not simultaneously, with random fluctuations.

- Random Fluctuations

  Reactor coolant temperature is assumed to vary by ±0.5°F and pressure by ±6 psi, once during each 6-min period. The total number of occurrences during the plant design life[a] does not exceed 3.0 x 10^6.

3.9.N.1.1.1.9 Boron Concentration Equalization. Following any large change in boron concentration in the RCS, the pressurizer spray is operated to equalize concentration between the loops and the pressurizer. This can be done by manually operating the pressurizer backup heaters, thus causing a pressure increase and initiation of spray at a pressurizer pressure of approximately 2275 psia. The pressure increases to approximately 2281 psia before being returned to 2250 psia by the proportional spray. The pressure is then maintained at 2250 psia by spray operation, matching the heat input from the backup heaters until the concentration is equalized.

For design purposes, it is assumed that this operation is performed once after each load change (increase and decrease) in the design load follow cycle. With two load changes per day and a 90% plant availability factor over the 40-year design life[a], the total number of occurrences is 26,400.

3.9.N.1.1.1.10 Feedwater Cycling. This transient can occur when the plant is being maintained at hot standby or no-load conditions. It is assumed that the low steam generation rate is made up by intermittent (slug) feeding of 32°F feedwater into the steam generator.

For design purposes, 2000 occurrences are assumed over the life of the plant.[a] Feedwater additions required during plant heatup and cooldown operations are also assumed to be covered by the feedwater cycling transient but with no increase in the total number of cycles.

3.9.N.1.1.1.11 Loop out of Service. The plant may be operated at a reduced power level with a single loop out of service for limited periods of time. This is accomplished by reducing reactor power and tripping a single RCP. Flow increases in the loops which remain in service (active loops), and reverse flow is established in the loop with the idle pump (inactive loop). Flow through the reactor is reduced.

For design purposes, loop shutdown is assumed to occur twice per year or 80 times during the life of the plant[a].

Returning an inactive loop to service involves reducing reactor power to approximately 10% and stabilizing conditions.

Then the inactive RCP is started up and conditions are again stabilized at the same power level. Subsequent return to full power is then conducted in accordance with a normal loading operation.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

Seventy occurrences of loop startup are defined for design purposes. This number is based on the assumption that the inactive pump is inadvertently started up at maximum allowable power level 10 times during the life of the plant.[a] (This transient is covered under upset conditions.)

3.9.N.1.1.1.12 Refueling. At the beginning of the refueling operation, the RCS is assumed to have been cooled down to 140°F. The vessel head is removed, and the refueling canal is filled. This is done by transferring water from the refueling water storage tank, which is outdoors and conservatively assumed to be at 32°F, into the loops by means of the RHR pumps. The refueling water flows directly into the reactor vessel via the accumulator connections and cold legs.

This operation is assumed to occur twice per year or 80 times over the life of the plant[a].

3.9.N.1.1.1.13 Turbine Roll Test. This transient is imposed upon the plant during the hot functional test period for turbine cycle checkout. RCP power is used to heat the reactor coolant to operating temperature (no-load conditions), and the steam generated is used to perform a turbine roll test. However, the plant cooldown during this test exceeds the 100°F/h design rate.

Twenty such test cycles are specified to be performed at the beginning of plant operating life prior to reactor operation. This transient occurs before plant startup, and the number of cycles is therefore independent of other operating transients.

3.9.N.1.1.1.14 Primary Side Leakage Test. A leakage test is performed after each opening of the primary system. During this test, the primary system pressure is raised (for design purposes) to 2500 psia, with the system temperature above the minimum temperature imposed by reactor vessel material ductility requirements, while the system is checked for leaks.

In actual practice, the primary system is pressurized, in accordance with ASME Section XI IWA-5211(a) and IWB-5221(a), as measured at the pressurizer, to prevent the pressurizer safety valves from lifting during the leakage test. In addition, the secondary side of the steam generator must be pressurized so that the pressure differential across the tube sheet does not exceed 1600 psi. This is accomplished with the steam, feedwater, and blowdown lines closed off.

For design purposes, it is assumed that 200 cycles of this test occur during the 40-year design life of the plant.[a]

3.9.N.1.1.1.15 Secondary Side Leakage Test. During the life of the plant it may be necessary to check the secondary side of the steam generator, particularly the manway closure, for leakage. For design purposes, it is assumed that the generator secondary side is pressurized to just below its design pressure to prevent the safety valves from lifting. In order not to exceed a secondary side to primary side pressure differential of 670 psi, the primary side must also be pressurized. The primary system must be above the minimum temperature imposed by reactor vessel material ductility requirements, that is, between 120°F and 250°F.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

It is assumed that this test is performed 80 times during the life of the plant.[a]

3.9.N.1.1.2 Upset Conditions

The following primary system transients are considered upset conditions:

- Loss of load without immediate reactor trip.
- Loss of power.
- Partial loss of flow.
- Reactor trip from full power:
  - With no cooldown.
  - With cooldown and no safety injection (SI).
  - With cooldown and SI.
- Inadvertent RCS depressurization.
- Inadvertent startup of an inactive loop.
- Control rod drop.
- Inadvertent SI actuation.
- Excessive feedwater flow.
- Operating basis earthquake (OBE).
- Excessive bypass feedwater flow transient.
- Reactor Coolant System Cold Overpressurization.

3.9.N.1.1.2.1 Loss of Load Without Immediate Reactor Trip. This transient involves a step decrease in turbine load from full power (turbine trip) without immediate automatic reactor trip. These conditions produce the most severe pressure transient on the RCS under upset conditions. The reactor eventually trips as a consequence of a high pressurizer level trip initiated by the reactor protection system. Since redundant means for tripping the reactor are provided by the reactor protection system, a transient of this nature is not expected, but is included to ensure a conservative design.

The number of occurrences of this transient is specified at 80 times, or twice per year for the 40-year plant design life.[a]

3.9.N.1.1.2.2 Loss of Power. This transient applies to a loss of outside electrical power to the plant, which is assumed to be operating initially at 100% power, followed by reactor and turbine trips. The RCPs are deenergized, as are electrical loads connected to the turbine-generator bus, including the main feedwater and condensate pumps. As the RCPs coast down,

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

RCS flow reaches an equilibrium value under natural circulation. This condition permits removal of core residual heat through the steam generators which by this time are receiving feedwater, assumed to be at 32°F, from the auxiliary feedwater system. For equipment design purposes, it is conservatively assumed that the auxiliary feedwater pumps operate within 1 min following the loss of offsite power. Later in the transient, the auxiliary feedwater pumps are operated under manual control to obtain stable plant conditions. Steam is removed for reactor cooldown through atmospheric power-operated steam relief valves provided for this purpose.

The number of occurrences of this transient is specified at 40 times, or once per year for the 40-year plant design life.[a]

3.9.N.1.1.2.3 Partial Loss of Flow. This transient applies to a partial loss of flow from full power in which an RCP is tripped out of service as the result of a loss of power to that pump. The consequences of such an accident are a reactor trip on low reactor coolant flow, followed by turbine trip and automatic opening of the steam dump system. Flow reversal occurs in the affected loop, which causes reactor coolant at cold leg temperature to pass through the steam generator and be cooled still further. This cooled water then flows through the hot leg piping and enters the reactor vessel outlet nozzles. The net result of the flow reversal is a sizable reduction in the hot leg coolant temperature of the affected loop.

The number of occurrences of this transient is specified as 80, or twice per year for the 40-year plant design life.[a]

3.9.N.1.1.2.4 Reactor Trip from Full Power. Reactor trips from full power, which may occur for a variety of reasons, cause temperature and pressure transients in the RCS. Transients also occur in the secondary side of the steam generator due to continued heat transfer from the reactor coolant through the steam generators. These transients continue until the reactor coolant and steam generator secondary side temperatures are in equilibrium at zero power conditions. Continuation of feedwater flow and controlled steam dump remove the core residual heat and prevent the actuation of steam generator safety valves. The reactor coolant temperature and pressure undergo rapid decreases from full-power values as the reactor protection system causes the control rods to move into the core. For design purposes, reactor trip is assumed to occur a total of 400 times, or 10 times per year over the life of the plant.[a]

The severity of the cooldown transient following a reactor trip depends on the extent of steam generator secondary side cooling. Three basic cooldown cases are considered:

Case 1 - Reactor Trip with No Cooldown

Steam and feedwater flow are both controlled to bring the plant back to the no-load conditions and maintain it at no-load. For design purposes, 230 occurrences of this transient are specified.

Case 2 - Reactor Trip with Cooldown and No SI

For this case, it is assumed that normal feedwater flow continues for approximately 1 min after the reactor trip, maintaining a high heat transfer rate through the steam generator which

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

continues to drive the primary side pressure and temperature down. The RCS pressure decreases to just above the SI setpoint. After the feedwater flow is terminated, the plant is brought back to no-load conditions. For design purposes, 160 occurrences of this transient are specified.

Case 3 - Reactor Trip with Cooldown and SI

This transient is similar to case 2, but it is assumed that the protection system setpoints are such that the RCS pressure decreases to just below the SI setpoint. The high-head safety injection system (SIS) is actuated; its operation lowers the RCS temperature and raises the RCS pressure. After approximately 1 min, main feedwater flow is terminated while auxiliary feedwater flow (actuated on the SI signal) is continued. The plant is brought back to the no-load condition after SI is manually terminated. For design purposes, 10 occurrences of this transient are specified.

It is assumed that the turbine control system operates as designed in 95% of the 400 reactor trip cases. For the remaining 5%, or 20 occurrences, it is conservatively assumed that this system fails, resulting in an emergency turbine overspeed. This situation could be initiated with malfunction of the turbine control system following a large step-load decrease with steam dump, resulting in turbine speed increase past the overspeed trip setpoint. It is assumed that the reactor then trips and that the turbine speed increases to 120% of nominal, with accompanying proportional increases in generator bus frequency, RCP speed, and reactor coolant flowrate.

For design purposes, it is assumed that the emergency turbine overspeed constitutes a special case of the reactor trip with no cooldown transient. Thus, for 20 of the 230 occurrences, the effects of the reactor coolant flow variation are to be considered in addition to the basic pressure and temperature variations.

3.9.N.1.1.2.5 Inadvertent RCS Depressurization. Several events can be postulated as occurring during normal plant operation which cause rapid depressurization of the RCS. These include:

- Actuation of a single pressurizer safety valve.
- Inadvertent opening of one pressurizer power-operated relief valve due to equipment malfunction or operator error.
- Malfunction of a single pressurizer pressure controller causing one power-operated relief valve and two pressurizer spray valves to open.
- Inadvertent opening of one pressurizer spray valve due either to equipment malfunction or operator error.
- Inadvertent auxiliary spray.

Of these events, the pressurizer safety valve actuation causes the most severe depressurization transient and is used as a conservative case.

When a pressurizer safety valve opens and remains open, the system rapidly depressurizes, the reactor trips, and the SIS is actuated. The passive accumulators of the SIS are actuated when RCS pressure decreases by approximately 1600 psi, about 5 min after the depressurization begins. The RCS reaches an equilibrium condition where the water release rate through the open pressurizer safety valve is equivalent to the SI flow. The RCS is also cooled down by the flow through the safety valve, the SI flow, and auxiliary feedwater flow. It is assumed that auxiliary feedwater flow is terminated by the operator 10 min after the depressurization begins.

Eventually, the plant must be taken to a cold shutdown condition, as the operator can take no immediate action to stop the transient and bring the plant to hot standby if the safety valve remains open.

For design purposes, 20 occurrences of this transient are specified.

Although inadvertent auxiliary spray actuations are included among the depressurization transient events covered above, the pressurizer safety valve actuation case selected to represent all the depressurization transients does not involve spray operation. Therefore, for the previous case it is assumed that pressurizer spray is not actuated and that no temperature transients due to flow occur at the spray nozzle.

However, should auxiliary spray flow be initiated inadvertently, it could cause severe thermal shock at the pressurizer spray nozzle and on the pressurizer vessel. Therefore, to ensure a conservative design for these components, an "inadvertent auxiliary spray" transient is defined.

The inadvertent auxiliary spray transient occurs when the auxiliary spray valve is opened during normal plant operation due to failure of a control component or operator error. This introduces cold water into the pressurizer resulting in a sharp pressure decrease and eventually in a low pressure reactor trip. The temperature of the auxiliary spray flow is dependent upon the performance of the regenerative heat exchanger. The most conservative case assumes that the letdown stream is shut off and that unheated charging fluid enters the 653°F pressurizer. It is assumed that the temperature of the spray water is 70°F, and the spray flowrate is equal to the normal charging rate. It is also assumed that auxiliary spray flow continued for 5 min before termination.

The total number of occurrences of this transient during the 40-year design life of the plant[a] is specified as 10.

3.9.N.1.1.2.6 Inadvertent Startup of an Inactive Loop. This transient can occur when a loop is out of service. With the plant operating at maximum allowable power level, the RCP in the inactive loop is started as a result of operator error. Reactor trip occurs on high nuclear flux. All feedwater flow for the inactive loop is through the auxiliary feedwater nozzle.

For design purposes, the inadvertent loop startup transient is assumed to occur 10 times during the life of the plant.[a]

3.9.N.1.1.2.7 Control Rod Drop. This transient occurs if a bank of control rods (worth 1-% reactivity) drops into the fully inserted position due to a single component failure. The reactor is tripped on low pressurizer pressure, depending on time in core life and magnitude of the reactivity insertion. It is assumed that this transient occurs 80 times over the life of the plant.[a]

3.9.N.1.1.2.8 Inadvertent SI Actuation. A spurious SI signal results in an immediate reactor trip followed by actuation of the high-head centrifugal charging pumps. These pumps deliver the contents of the boron injection tank to the RCS cold legs. The initial portion of this transient is similar to the reactor trip from full power with no cooldown. Controlled steam dump and

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

auxiliary feedwater flow after trip removes core residual heat. Reactor coolant temperature and pressure decrease as the control rods move into the core.

Later in the transient, the injected water causes the RCS pressure to increase to the pressurizer power-operated relief valve setpoint and the primary and secondary temperatures to decrease gradually. The transient continues until the operator stops the charging pumps. It is assumed that the plant is then returned to no-load conditions, with pressure and temperature changes controlled within normal limits.

For design purposes, this transient is assumed to occur 60 times during the 40-year design life of the plant.[a]

3.9.N.1.1.2.9 Excessive Feedwater Flow. An excessive feedwater flow transient is defined as a conservative case to cover occurrence of several events of the same general nature. The postulated transient results from inadvertent opening of a feedwater regulating valve while the plant is at the hot standby or no-load condition, with the feedwater, condensate, and heater drain systems in operation.

It is assumed that the stem of a feedwater regulating valve fails, and the valve immediately reaches the full-open position. In the steam generator, directly affected by the malfunctioning valve (failed loop), the feedwater flow step increases from essentially zero flow to the value determined by the system resistance and the developed head of all operating feedwater pumps. Steamflow is assumed to remain at zero and the temperature of the feedwater entering the steam generator is conservatively assumed to be 32°F. Feedwater flow is isolated on a reactor coolant low T_{avg} signal; a low pressurizer pressure signal actuates the SIS. Auxiliary feedwater flow, initiated by the SI signal, is assumed to continue with all pumps discharging into the affected steam generator via the auxiliary feedwater nozzle. It is also assumed, for conservatism in the secondary side analysis, that auxiliary feedwater flows to the steam generators not affected by the malfunctioned valve, in the "unfailed loops." Plant conditions stabilize at the values reached in 600 s, at which time auxiliary feedwater flow is terminated. The plant is then either taken to cold shutdown or returned to the no-load condition at a normal heatup rate with the auxiliary feedwater system under manual control.

For design purposes, this transient is assumed to occur 30 times during the life of the plant.[a]

3.9.N.1.1.2.10 Operating Basis Earthquake. The OBE is that earthquake which can reasonably be expected to occur during the plant life.[a] The number of occurrences for fatigue evaluation is assumed to be 5 earthquakes of 10 cycles each (50 cycles total).

3.9.N.1.1.2.11 Excessive Bypass Feedwater Flow Transient. This transient results from a postulated feedwater bypass valve failure and provides 75% of nominal feedwater flow through the auxiliary feedwater nozzle. Thirty occurrences of this transient are assumed.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

3.9.N.1.1.2.12 RCS Cold Overpressurization. RCS cold overpressurization occurs during startup and shutdown conditions at low temperature, with or without existence of a steam bubble in the pressurizer, and is especially severe when the reactor coolant system is in a water-solid configuration. The event is inadvertent, and usually generated by any one of a variety of malfunctions or operator errors. All events which have occurred to date may be categorized as belonging to either events resulting in the addition of mass (mass input transient) or events resulting in the addition of heat (heat input transient). All these possible transients are represented by composite "umbrella" design transients, referred to here as RCS cold overpressurization.

3.9.N.1.1.3 Emergency Conditions

The following primary system transients are considered emergency conditions:

- Small loss-of-coolant accident (LOCA).
- Small steam line break.
- Complete loss of flow.

3.9.N.1.1.3.1 Small LOCA. For design transient purposes the small LOCA is defined as a break equivalent to the severance of a 1-in. inside diameter (ID) branch connection. (Breaks smaller than 0.375 in. ID can be handled by the normal makeup system and produce no significant fluid systems transients). Breaks which are much larger than 1 in. cause accumulator injection soon after the accident and are regarded as faulted conditions. It is assumed that the SIS is actuated immediately after the break occurs and delivers water at a minimum temperature of 32°F to the RCS.

For design purposes, it is assumed that this transient occurs five times during the life of the plant.[a]

3.9.N.1.1.3.2 Small Steam Line Break. For design transient purposes, a small steam line break is defined as a break equivalent in effect to a steam generator safety valve opening and remaining open.

The following conservative assumptions are made:

- The reactor is initially in a hot, zero-power condition.
- The small steam line break results in immediate reactor trip and SI actuation.
- A large shutdown margin, coupled with no feedback or decay heat, prevents heat generation during the transient.

Operation of the high-head SI/charging pumps repressurizes the RCS within a relatively short time to the actuation pressure of the pressurizer power-operated relief valves. These valves

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

open, and an equilibrium condition is established in which the flow through the valves matches the SI flow input to the RCS.

This transient is assumed to occur five times during the life of the plant.[a]

3.9.N.1.1.3.3 Complete Loss of Flow. This accident involves a complete loss of flow from full power resulting from simultaneous loss of power to all RCPs. The consequences are a reactor trip and turbine trip on undervoltage followed by automatic opening of the steam dump system.

For design purposes, this transient is assumed to occur five times during the plant lifetime.[a]

3.9.N.1.1.4 Faulted Conditions

The following primary system transients are considered faulted conditions, which are evaluated for one occurrence:

- Reactor coolant pipe break (large LOCA).
- Large steam line break.
- Feedwater line break.
- RCP locked rotor.
- Control rod ejection.
- Steam generator tube rupture.
- Simultaneous steam line-feedwater line break.
- Safe shutdown earthquake (SSE).

3.9.N.1.1.4.1 Reactor Coolant Pipe Break (Large LOCA). Following rupture of a reactor coolant pipe resulting in a large loss of coolant, the primary system pressure decreases rapidly causing the primary system temperature to decrease. Because of the rapid blowdown of coolant from the system and the comparatively large heat capacity of the metal sections of the components, it is likely that the metal will remain at or near the operating temperature during the blowdown. The SIS is actuated to introduce water, at an assumed minimum temperature of 32°F, into the RCS. The SI signal also results in reactor and turbine trips.

3.9.N.1.1.4.2 Large Steam Line Break. The transient is based on the complete rupture of a main steam line. The following conservative assumptions are made:

- The plant is initially at the hot, no-load condition.
- The steam line break results in immediate reactor trip and actuation of the SIS.
- Offsite power is lost at the time of the break. All RCPs are deenergized and coolant flow coasts down to the natural circulation value.
- The SIS operates at design capacity and repressurizes the RCS within a relatively short time.

In the analysis the effects of core decay heat, thick metal stored energy, and steam generator reverse heat transfer are included in order to maximize RCS post-break pressurization. These represent the most conservative cases for reactor vessel design and evaluation. The results are also characterized by high feedwater flow.

The worst case large steam break, with respect to the steam generator tubes and tube sheet, occurs outside containment with the plant initially at full power. The affected steam generator will rapidly blow down to atmospheric pressure. It is assumed that the auxiliary feedwater system will deliver 1410 gal/min at 32°F to the affected steam generator via the auxiliary feedwater nozzle. The primary side repressurizes to the pressurizer safety valve set pressure, resulting in a large differential pressure across the tubes and tube sheet.

3.9.N.1.1.4.3 Feedwater Line Break. This accident involves the double-ended rupture of a main feedwater line, resulting in rapid blowdown of the affected steam generator and termination of feedwater flow to the others. The plant is assumed to be operating at an initial power level of 102% of engineered safeguards design rating when the break occurs. Turbine trip, with immediate reactor trip, occurs on a low-low level signal from the faulted steam generator.

The auxiliary feedwater system is actuated within 1 min and supplies two intact steam generators with flow equivalent to the capacity of one motor-driven auxiliary feedwater pump via the auxiliary feedwater nozzle. Loss of the plant from the grid is assumed to cause a loss of offsite power; all RCPs are deenergized and coolant flow coasts down to the natural circulation value. The SIS is actuated and is assumed to deliver maximum safeguards flow until manually shut off at approximately 600 s.

3.9.N.1.1.4.4 RCP Locked Rotor. This accident is based on the instantaneous seizure of an RCP with the plant operating at full power. The locked rotor can occur in any loop. Reactor trip occurs almost immediately, as the result of low coolant flow in the affected loop.

3.9.N.1.1.4.5 Control Rod Ejection. This accident is based on the single most reactive control rod being instantaneously removed from the core. This reactivity insertion in a particular region of the core causes a severe pressure increase in the RCS, such that the pressurizer safety valves lift, and also causes a more severe temperature transient in the loop associated with the affected region than in the other loops. For conservatism, the analysis is based on the reactivity insertion and does not include the mitigating effects (on the pressure transient) of coolant blowdown through the hole in the vessel head vacated by the ejected rod.

3.9.N.1.1.4.6 Steam Generator Tube Rupture. This accident is postulated as the double-ended rupture of a single steam generator tube resulting in decreases in pressurizer level and RCS pressure. Eventually the loss of reactor coolant causes a reactor trip (also a turbine trip) on low pressurizer pressure. The ensuing plant cooldown results in SIS actuation due to low pressurizer pressure. The SI signal automatically starts the auxiliary feedwater pumps and isolates the main feedwater lines. The steam line leading from the affected steam generator is isolated manually. When the pressurizer water level is recovered, the operator stops the SI pumps and uses the auxiliary feedwater system to conduct an orderly cooldown to cold shutdown conditions.

Primary side temperatures are assumed to remain constant at their initial values for 10 min following the rupture. After 10 min, the primary side temperatures and the steam temperature are assumed to vary in the same way as for the reactor trip with cooldown and SI transient. Main feedwater flow consistent with the initial power level is assumed for the first 10 min. After 10 min, main feedwater flow is terminated and auxiliary feedwater flow initiated.

3.9.N.1.1.4.7 Simultaneous Steam Line-Feedwater Line Break. This transient is based on the simultaneous, complete severance of both a main steam line and a feedwater line. It is postulated to occur when the pipe whip or missile resulting from the severance of a steam line results in the complete severance of the smaller feedwater line connected to the same steam generator. Since the velocity of the whip or missile can be very high, the two lines are assumed to break simultaneously. To ensure conservatism, this transient is assumed to occur at any power level between hot, zero, and full power.

3.9.N.1.1.4.8 Safe Shutdown Earthquake. The SSE is defined as the maximum vibratory ground motion which can reasonably be predicted from geologic and seismic evidence. The mechanical dynamic or static equivalent loads due to the vibratory motion of the SSE are considered on a component basis.

3.9.N.1.1.5 Test Conditions

The following primary system transients are considered test conditions:

- Primary side hydrostatic test.
- Secondary side hydrostatic test.
- Tube leakage test.

3.9.N.1.1.5.1 Primary Side Hydrostatic Test. The pressure tests covered by this section include both shop and field hydrostatic tests which occur as a result of component or system testing. This hydro test is performed at a water temperature which is compatible with reactor vessel material ductility requirements and a test pressure of 3107 psig (1.25 times design pressure). In this test, the RCS is pressurized to 3107 psig, consistent with steam generator secondary side pressure of 0 psig.

The RCS is designed for 10 cycles of these hydrostatic tests, which are performed prior to plant startup. The number of cycles is independent of other operating transients.

Additional hydrostatic tests may be performed to meet the inservice inspection requirements of ASME Section XI, subarticle IWB-5200. A total of four such tests is expected. The increase in the fatigue usage factor caused by these tests is easily covered by the conservative number (200) of primary side leakage tests that are considered for design.

3.9.N.1.1.5.2 Secondary Side Hydrostatic Test. The secondary side of the steam generator is pressurized to 1.25 design pressure with a minimum water temperature of 120°F, coincident with the primary side at 0 psig.

For design purposes, it is assumed that the steam generator will experience 10 cycles of this test.

These tests may be performed either prior to plant startup, or subsequently following major repairs, or both. The number of cycles is independent of other operating transients.

3.9.N.1.1.5.3 Tube Leakage Test. During the life of the plant it may be necessary to check the steam generator for tube leakage and tube-to-tube sheet leakage. This is done by visual inspection of the underside (channel head side) of the tube sheet for water leakage, with the secondary side pressurized. Tube leakage tests are performed during plant cold shutdown. For these tests, the secondary side of the steam generator is pressurized with water, initially at a relatively low pressure; and the primary system remains depressurized. The underside of the tube sheet is examined visually for leaks. If any are observed, the secondary side is depressurized and repairs made by tube plugging. The secondary side is then repressurized (to a higher pressure), and the underside of the tube sheet is again checked for leaks. This process is repeated until all the leaks are repaired. The maximum (final) secondary side test pressure reached is 840 psig. Both the primary and secondary sides of the steam generators are at ambient temperatures during these tests.

Test Pressure (psig)	Number of Occurrences
200	400
400	200
600	120
840	80

The total number of tube leakage test cycles is defined as 800 during the 40-year life of the plant.[a]

3.9.N.1.2 Computer Programs Used in Analyses

The following computer programs have been used in dynamic and static analyses to determine mechanical loads, stresses, and deformations of Seismic Category 1 components and equipment. These are described and verified in references 1 and 3.

- WESTDYN - static, dynamic, and fatigue analysis of redundant piping systems.
- FIXFM-3[b] - time-history response of three-dimensional structures.
- WESDYN-2[b] - piping system stress analysis from time-history displacement data.
- THRUST - hydraulic loads on loop components from blowdown information.
- WECAN - finite element structural analysis.
- MULTIFLEX - thermal-hydraulic structure systems dynamics.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

[b] These capabilities have been incorporated into WESTDYN.

The WESTEMS™ computer program (reference 4) was used for the fatigue analysis that considered the application of the mechanical stress improvement process to the Unit 1 and Unit 2 Loops 1, 2, 3, and 4 reactor vessel outlet nozzles.

3.9.N.1.3 Experimental Stress Analysis

No experimental stress analysis methods are used for Seismic Category 1 systems or components. However, Westinghouse makes extensive use of measured results from prototype plants and various scale model tests as discussed in subsection 3.9.N.2.

3.9.N.1.4 Considerations for the Evaluation of the Faulted Condition

3.9.N.1.4.1 Loading Conditions

The structural stress analyses performed on the RCS consider the loadings specified in table 3.9.B.3-1. These loads may result from any, or all of the following: thermal expansion, pressure, weight, OBE, SSE, system operating transients, LOCA loop hydraulic forces, subcompartment pressurization forces, and reactor vessel loads.

A description of the loads used in the analysis of the RCS follows.

A. Pressure

Pressure loading is identified as either membrane design pressure or general operating pressure, depending upon its application. The membrane design pressure is used in connection with the longitudinal pressure stress and minimum wall thickness calculations in accordance with the ASME Code.

The term "operating pressure" is used in connection with determination of the system deflections and support forces. The steady-state operating hydraulic forces based on the system initial pressure are applied as general operating pressure loads to the RCL model at changes in direction or flow area.

B. Weight

A weight analysis is performed to meet code requirements by applying a 1.0-g load downward on the complete piping system. The piping is assigned a distributed mass or weight as a function of its properties. This method provides a distributed loading to the piping system as a function of the weight of the pipe and contained fluid during normal operating conditions.

C. Seismic

The input for the RCS seismic analysis is in the form of six statistically independent time-history inputs: three translational accelerations and three rotational accelerations. These six accelerations are applied simultaneously to the containment basemat.

For the OBE and SSE seismic analyses, 2- and 4% critical damping, respectively, is used in the RCL/supports system analysis.

D. Loss-of-Coolant Accident

Blowdown loads are developed in the RCL as a result of transient flow and pressure fluctuations following a postulated pipe break at the applicable branch nozzles. Structural consideration of dynamic effects of postulated pipe breaks requires postulation of a finite number of break locations. Postulated pipe break locations are given in section 3.6.

Time-history dynamic analysis is performed for these postulated break cases. Hydraulic models are used to generate time dependent hydraulic forcing functions used in the analysis of the RCL for each break case. For a further description of the hydraulic forcing functions, refer to section 3.6.

E. Transients

The ASME Code requires satisfaction of certain requirements relative to operating transient conditions. Operating transients are discussed in paragraph 3.9.N.1.1.

To provide the necessary high degree of integrity for the RCS, the transient conditions selected for fatigue evaluation are based on conservative estimates of the magnitude and anticipated frequency of occurrence of the temperature and pressure transients resulting from various plant operating conditions.

3.9.N.1.4.2 RCL Analytical Models and Methods

The analytical methods used in obtaining the solution consist of the transfer matrix method and stiffness matrix formulation for the static structural analysis, the time-history modal superposition method for seismic dynamic analysis, and time- history integration method for the LOCA dynamic analysis.

The integrated RCL/supports system model is the basic system model used to compute loadings on components, component supports, and piping. The system model includes the stiffness and mass characteristics of the RCL piping and components, the stiffness of supports, and the stiffnesses of auxiliary line piping which affect the system. The deflection solution of the entire system is obtained for the various loading cases from which the internal member forces and piping stresses are calculated.

A. Static

The RCL/supports system model, constructed for the WESTDYN computer program, is represented by an ordered set of data which numerically describes the physical system. Figure 3.9.N.1-1 shows an isometric line schematic of this mathematical model.

The spatial geometric description of the RCL model is based upon the RCL piping layout and equipment drawings. The node point coordinates and incremental lengths of the members are determined from these drawings. Geometrical properties of the piping and elbows along with the modulus of elasticity, the coefficient of thermal expansion, the average temperature change from ambient temperature, and the weight per unit length are specified for each element. The primary equipment supports are represented by stiffness matrices which define restraint characteristics of the supports. Due to the symmetry of the static loadings, the reactor pressure vessel (RPV) centerline is represented by a fixed boundary in the system mathematical model. The vertical thermal growth of

the reactor vessel nozzle centerline is considered in the construction of the model.

The model is made up of a number of sections, each having an overall transfer relationship formed from its group of elements. The linear elastic properties of the section are used to define the stiffness matrix for the section. Using the transfer relationship for a section, the loads required to suppress all deflections at the ends of the section arising from the thermal and boundary forces for the section are obtained. These loads are incorporated into the overall load vector.

After all the sections have been defined in this matter, the overall stiffness matrix and associated load vector to suppress the deflection of all the network points is determined. By inverting the stiffness matrix, the flexibility matrix is determined. The flexibility matrix is multiplied by the negative of the load vector to determine the network point deflections due to the thermal and boundary force effects. Using the general transfer relationship, the deflections and internal forces are then determined at all node points in the system.

The static solutions for weight, thermal, and general pressure loading conditions are obtained by using the WESTDYN computer program.

B. Seismic

The model used in the static analysis is modified for the dynamic analysis by including the mass characteristics of the piping and primary equipment. The containment internals structure and all of the piping loops are included in the coupled building/loop system model. The effect of the equipment motion on the RCL/supports system is obtained by modeling the mass and the stiffness characteristics of the equipment in the overall system model.

Component lateral supports are inactive during plant heatup, cooldown, and normal plant operating conditions. However, these restraints become active during the rapid motions of the RCL components that occur from the dynamic loadings and are represented by stiffness matrices and/or individual tension or compression spring members in the dynamic model. The analyses are performed at the full-power condition.

The total response is obtained using the modal superposition method for time integration of the equations of motion. The results of the analysis are time-history forces and displacements. The time-history displacement response is then used in computing support loads and in performing the RCL piping stress evaluation.

C. Loss-of-Coolant Accident

The mathematical model used in the static analysis is modified for the LOCA analyses by including the mass characteristics of the piping and primary equipment. The natural frequencies and eigenvectors are determined from this.

The time-history hydraulic forces at the node points are combined to obtain the forces and moments acting at the corresponding structural lumped-mass node points.

The dynamic structural solution for the full-power LOCA is obtained by using a modified-predictor-corrector-integration technique and normal mode theory.

When elements of the system can only be represented as single acting members (tension or compression members), they are considered as nonlinear elements,

which are represented mathematically by the combination of a gap, a spring, and a viscous damper. The force in this nonlinear element is treated as an externally applied force in the overall normal mode solution. Multiple nonlinear elements can be applied at the same node, if necessary.

The time-history solution is performed in program FIXFM-3. The input to this program consists of the natural frequencies, normal modes, applied forces, and nonlinear elements. The natural frequencies and normal modes for the modified RCL dynamic model are determined with the WESTDYN program. To properly simulate the release of the strain energy in the pipe, the internal forces in the system at the postulated break location due to the initial steady-state hydraulic forces, thermal forces, and weight forces are determined. The release of the strain energy is accounted for by applying the negative of these internal forces as a step function loading. The initial conditions are equal to zero because the solution is only for the transient problem (the dynamic response of the system from the static equilibrium position). The time-history displacement solution of all dynamic degrees of freedom is obtained using program FIXFM-3 and employing 4% critical damping.

The LOCA displacements of the reactor vessel are applied in time-history form as input to the dynamic analysis of the RCL. The LOCA analysis of the reactor vessel includes all the forces acting on the vessel including internals reactions and loop mechanical loads. The reactor vessel analysis is described in paragraph 3.9.N.1.4.6.

The asymmetric external pressure loads on the RCP and steam generator resulting from a postulated pipe rupture and pressure buildup in the loop compartments are applied to the same integrated RCL/supports system model used to compute loadings on the components, component supports, and reactor coolant piping, as discussed above. Jet impingement loads on the loop piping, components, and supports resulting from postulated pipe ruptures are also applied to the RCL supports model. The response of the entire system is obtained for the various external loading cases from which the internal member forces and piping stresses are calculated. The equipment support loads and piping stresses resulting from the external loading are added to the support loads and piping stresses calculated using the loop LOCA hydraulic forces and RPV motion.

The time-history displacement response of the loop is used in computing support loads and in performing stress evaluation of the RCL piping.

The time-history displacements of the FIXFM-3 (or WESTDYN) program are used as input to WESDYN-2 (or WESTDYN) to determine the internal forces, deflections, and stresses at each end of the piping elements. For this calculation, the displacements are treated as imposed deflections on the RCL masses.

By application of leak before break technology, the dynamic effects of the pipe rupture in the RCL of Units 1 and 2 are eliminated, as are those of selected Class 1 branch lines in Units 1 and 2. Consequently, the dynamic effects of a break in the RCL piping are not considered in the design verification of the RCL component supports.

D. Fatigue

Operating transients in a nuclear power plant cause thermal and/or pressure fluctuations in the reactor coolant fluid. The thermal transients cause time varying temperature distributions across the pipe wall. These temperature distributions resulting in pipe wall stresses may be further subdivided in accordance with the code into three parts: a uniform, a linear, and a nonlinear portion. The uniform portion results in general expansion loads. The linear portion causes a bending moment across the wall, and the nonlinear portion causes a skin stress.

The transients, as defined in paragraph 3.9.N.1.1, are used to define the fluctuations in plant parameters. A one-dimensional finite difference heat transfer program is used to solve the thermal transient problem. The pipe is represented by at least 50 elements through the thickness of the pipe. The convective heat transfer coefficient employed in this program represents the time varying heat transfer due to free and forced convection. The outer surface is assumed to be adiabatic while the inner surface boundary experiences the temperature of the coolant fluid. Fluctuations in the temperature of the coolant fluid produce a temperature distribution through the pipe wall thickness which varies with time. This temperature distribution is used to determine the uniform, linear, and nonlinear portions of the pipe stress for each transient.

A load set is defined as a set of pressure loads, moment loads, and through-wall thermal effects at a given location and time in each transient. The method of load set generation is based on reference 2. The through-wall thermal effects are functions of time and can be subdivided into four parts:

- Average temperature (TA), the average temperature through-wall of the pipe which contributes to general expansion loads.
- Radial linear thermal gradient which contributes to the through-wall bending moment (ΔT1).
- Radial nonlinear thermal gradient (ΔT2) which contributes to a peak stress associated with shearing of the surface.
- Discontinuity temperature (TA - TB) represents the difference in average temperature at the cross sections on each side of a discontinuity.

Each transient is described by at least two load sets representing the maximum and minimum stress state during each transient. The construction of the load sets is accomplished by combining the following to yield the maximum (minimum) stress state during each transient.

- ΔT_1.
- ΔT_2.
- $\alpha_A T_A - \alpha_B T_B$.
- Moment loads due to T_A.
- Pressure loads.

This procedure produces at least twice as many load sets as transients for each point.

For all possible load set combinations, the primary- plus-secondary and peak stress intensities, fatigue reduction factors, and cumulative usage factors are calculated. The WESTDYN program is used to perform this analysis in accordance with ASME III, Subsection NB-3650.

Since it is impossible to predict the order of occurrence of the transients over a 40-year life[a], it is assumed that the transients can occur in any sequence. This is a very conservative assumption.

The combination of load sets yielding the highest alternating stress intensity range is determined and the incremental usage factor calculated. Likewise, the next most severe combination is then determined and the incremental usage factor calculated. This procedure is repeated until all combinations having allowable cycles $<10^6$ are formed. The total cumulative usage factor at a point is the summation of the incremental usage factors.

A fatigue analysis was performed that considered the application of the mechanical stress improvement process to the Unit 1 and Unit 2 Loops 1, 2, 3, and 4 reactor vessel outlet nozzles. The WESTEMS™ computer program was used to perform this fatigue analysis in accordance with ASME III, Subsection NB-3650.

3.9.N.1.4.3 Analytical Methods for RCS Class 1 Branch Lines

The analytical methods used to obtain the solution consist of the transfer matrix method and stiffness matrix formulation for the static structural analysis, the response spectrum method for seismic dynamic analysis, and static or dynamic structural analysis for the effect of the applicable branch nozzle breaks per paragraph 3.6.2.1.1.A.1.

The integrated Class 1 piping/supports system model is the basic system model used to compute loadings on components, component and piping supports, and piping. The system models include the stiffness and mass characteristics of the Class 1 piping components, the RCL, and the stiffness of supports which affect the system response. The deflection solution of the entire system is obtained for the various loading cases from which the internal member forces and piping stresses are calculated.

A. Static

The Class 1 piping system models are constructed for the WESTDYN computer program, which numerically describes the physical system. A network model is made up of a number of sections, each having an overall transfer relationship formed from its group of elements. The linear elastic properties of the section are used to define the characteristic stiffness matrix for the section. Using the transfer relationship for a section, the loads required to suppress all deflections at the ends of the section arising from the thermal and boundary forces for the section are obtained.

After all the sections have been defined in this manner, the overall stiffness matrix and associated load vector to suppress the deflection of all the network points is determined. By inverting the stiffness matrix, the flexibility matrix is determined. The flexibility matrix is multiplied by the negative of the load vector to determine the network point deflections due to the thermal and boundary force effects. Using the general transfer relationship, the deflections and internal forces are then determined at all node points in the system. The support loads are also computed by multiplying the stiffness matrix by the displacement vector at the support point.

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. Metal fatigue, including the effect of the extended operating term on the number of transient cycles or occurrences, was evaluated as a TLAA for license renewal in accordance with 10 CFR Part 54. The results of this evaluation are provided in subsection 19.4.2.

B. Seismic

The models used in the static analyses are modified for use in the dynamic analyses by including the mass characteristics of the piping and equipment.

The lumping of the distributed mass of the piping systems is accomplished by locating the total mass at points in the system which appropriately represent the response of the distributed system. Effects of the primary equipment motion, that is, reactor vessel, steam generator, RCP, and pressurizer on the Class 1 piping system are obtained by modeling the mass and the stiffness characteristics of the primary equipment and loop piping in the overall system model.

The supports are represented by stiffness matrices in the system model for the dynamic analysis. Shock suppressors which resist rapid motions are also included in the analysis. The solution for the seismic disturbance employs the response spectra method. This method employs the lumped mass technique, linear elastic properties, and the principle of modal superposition.

The total response obtained from the seismic analysis consists of two parts: the inertia response of the piping system and the response from differential anchor motions. The stresses resulting from the anchor motions are considered to be secondary and, therefore, are included in the fatigue evaluation.

C. Loss-of-Coolant Accident

The mathematical models used in the seismic analyses of the Class 1 lines are also used for three RCL branch nozzle break effect analyses. To obtain the proper dynamic solution for emergency core cooling system (ECCS) lines attached to the unaffected loops, lines 6 in. and larger attached to the affected broken loop, and the surge line, the time-history deflections from the analysis of the RCL are applied at branch nozzle connections. The motion of the RCL is applied statically to non-ECCS lines attached to the unaffected loops and small lines attached to the affected loop which must maintain structural integrity.

D. Fatigue

A thermal transient heat transfer analysis is performed for each different piping component on all the Class 1 branch lines. The normal, upset, and test condition transients identified in paragraph 3.9.N.1.1 are considered in the fatigue evaluation.

The thermal quantities ΔT_1, ΔT_2, and $\alpha_A T_A - \alpha_B T_B$ are calculated on a time-history basis, using a one-dimensional finite difference heat transfer computer program. Stresses due to these quantities are calculated for each time increment using the methods of NB-3650 of ASME III.

For each thermal transient, two load sets are defined, representing the maximum and minimum stress states for that transient.

As a result of the normal mode spectral technique employed in the seismic analysis, the load components cannot be given signed values. Eight load sets are used to represent all possible sign permutations of the seismic moments at each point, thus ensuring the most conservative combinations of seismic loads are used in the stress evaluation.

The WESTDYN computer program is used to calculate the primary-plus-secondary and peak stress intensity ranges, fatigue reduction factors, and cumulative usage factors for all possible load set combinations. It is

conservatively assumed that the transients can occur in any sequence, thus resulting in the most conservative and restrictive combinations of transients.

The combination of load sets yielding the highest alternating stress intensity range is determined and the incremental usage factor calculated. Likewise, the next most severe combination is then determined and the incremental usage factor calculated. This procedure is repeated until all combinations having allowable cycles $<10^6$ are formed. The total cumulative usage factor at a point is the summation of the incremental usage factors.

3.9.N.1.4.4 Primary Component Supports Models and Methods

The static and dynamic structural analyses employ the matrix method and normal mode theory for the solution of lumped- parameter, multimass structural models. The equipment support structure models are dual-purpose since they are required to quantitatively represent the elastic restraints which the supports impose upon the loop, and to evaluate the individual support member stresses due to the forces imposed upon the supports by the loop.

A description of the supports is found in subsection 5.4.14. Detailed models are developed using beam elements and plate elements, where applicable. The reactor vessel supports are modeled using the WECAN computer program. Steam generator and RCP supports are normally modeled as linear or nonlinear springs.

For each operating condition, the loads (obtained from the RCL analysis) acting on the support structures are appropriately combined. The adequacy of each member of the steam generator supports, RCP supports, and pressurizer supports is verified by solving the ASME III Subsection NF stress and interaction equations. The adequacy of the RPV support structure is verified using the WECAN computer program and comparing the resultant stresses to the criteria given in ASME III, Subsection NF.

3.9.N.1.4.5 Analysis of Primary Components

Equipment that serves as part of the pressure boundary in the RCL includes the steam generators, the RCPs, the pressurizer, and the reactor vessel. This equipment is American Nuclear Society (ANS) Safety Class 1, and the pressure boundary meets the requirements of the ASME Code. This equipment is evaluated for the loading combinations outlined in table 3.9.B.3-1. The equipment is analyzed for:

- The normal loads of weight, pressure, and thermal.
- Mechanical transients of OBE, SSE, and pipe ruptures.
- Pressure and temperature transients outlined in paragraph 3.9.N.1.1.

The results of the RCL analysis are used to determine the loads acting on the equipment nozzles and the support/component interface locations. These loads are supplied for all loading conditions on an "umbrella" load basis. That is, on the basis of previous plant analyses, a set of loads are determined which should be larger than those seen in any single plant analysis. The umbrella loads represent a conservative means of allowing detailed component analysis prior to the completion of the system analysis. Upon completion of the system analysis, conformance is demonstrated between the actual plant loads and the loads used in the analyses of the components. Any deviations where the actual load is larger than the umbrella load are handled by individualized analysis.

Seismic analyses are performed individually for the RCP, the pressurizer, and the steam generator. Detailed and complex dynamic models are used for the dynamic analyses. The response spectra corresponding to the building elevation at the highest component/building attachment elevation is used for the component analysis. Seismic analyses for the steam generator, RCP, and pressurizer are performed using 2% damping for the OBE and 4% damping for the SSE. The RPV is qualified by static stress analysis based on loads that have been derived from dynamic analysis.

The pressure boundary portions of Class 1 valves in the RCS are designed and analyzed according to the requirements of NB-3500 of ASME III. These valves are identified in table 3.9.N.3-2.

Valves in sample lines connected to the RCS are not considered to be ANS Safety Class 1 nor ASME Class 1. This is because the nozzles where the lines connect to the primary system piping are orificed to a 3/8-in. hole. This hole restricts the flow such that loss through a severance of one of these lines can be made up by normal charging flow.

3.9.N.1.4.6 Dynamic Analysis of RPV for Postulated LOCA

3.9.N.1.4.6.1 Introduction. This section presents the method of computing the reactor pressure vessel response to a postulated LOCA. Since VEGP has leak-before-break (LBB), the LOCA analyses of the reactor vessel system due to main line breaks to include the dynamic effects are not required in accordance with the rules of GDC-4. Then the most limiting breaks to be considered are the branch line breaks which consist of accumulator line in the cold leg, pressurizer surge line in the hot leg, and residual heat removal (RHR) line in the hot leg. Of these branch line breaks, the most limiting breaks which were considered for the dynamic analysis of the VEGP reactor vessel system are the accumulator line break and the pressurizer surge line break. These breaks were used for RPV dynamic analysis though the pressurizer line break has been eliminated on both units and the accumulator and RHR line breaks have been eliminated on Unit 2 as described in section 3.6.

The structural analysis of the reactor vessel system for a postulated LOCA considers simultaneous application of the time-history loads resulting from the RCL mechanical loads and internal hydraulic pressure transients. The vessel is restrained by reactor vessel supports beneath four of the reactor vessel nozzles and the RCLs with the primary supports of the steam generators and the RCPs.

3.9.N.1.4.6.2 Loading Conditions. Following a postulated pipe rupture at the applicable branch nozzle breaks, the reactor vessel is excited by time-history forces. As previously mentioned, these forces are the combined effect of two phenomena:

- RCL mechanical loads.
- Reactor internal hydraulic forces.

The RCL mechanical forces are derived from the elastic analysis of the loop piping for the postulated break. The reactions on the nozzles of the RCL piping are applied to the vessel in the RPV blowdown analysis.

The reactor internals hydraulic pressure transients were calculated including the assumption that the structural motion is coupled with the pressure transients. This phenomena has been

referred to as hydroelastic coupling or fluid-structure interaction. The hydraulic analysis considers the fluid-structure interaction of the core barrel by accounting for the deflections of constraining boundaries which are represented by masses and springs. The dynamic response of the core barrel in its beam bending mode responding to blowdown forces compensates for internal pressure variation by increasing the volume of the more highly pressurized regions. The analytical methods used to develop the reactor internals hydraulics are described in WCAP-8709[3] and are discussed in more detail in subsection 3.9.N.2.5.

3.9.N.1.4.6.3 Reactor Vessel and Internals Modeling. The mathematical model of the RPV is a three-dimensional nonlinear finite element model which represents the dynamic characteristics of the reactor vessel and its internals in the six geometric degrees of freedom. The model was developed using the WECAN computer code. The model consists of three concentric structural submodels connected by nonlinear impact elements and stiffness matrices. The first submodel, figure 3.9.N.1-2, represents the reactor vessel shell and associated components. The reactor vessel is restrained by the four reactor vessel supports and by the attached primary coolant piping. Each reactor vessel support is modeled by a linear horizontal stiffness and a vertical impact element. The attached piping is represented by a stiffness matrix.

The second submodel, figure 3.9.N.1-3, represents the reactor core barrel, neutron panels, lower support plate, tie plates, and secondary core support components. This submodel is physically located inside the first and is connected to it by a stiffness matrix at the internals support ledge. Core-barrel- to-vessel-shell impact is represented by nonlinear elements at the core barrel flange, core barrel nozzle, and lower radial support locations.

The third and innermost submodel, figure 3.9.N.1-4, represents the lower support plate, guide tubes, support columns, upper and lower core plates, and fuel. The third submodel is connected to the first and second by stiffness matrices and nonlinear elements.

3.9.N.1.4.6.4 Analytical Methods. The time-history effects of the internals loads and loop mechanical loads are combined and applied simultaneously to the appropriate nodes of the mathematical model of the reactor vessel and internals. The analysis is performed by numerically integrating the differential equations of motion to obtain the transient response. The output of the analysis includes the displacements of the reactor vessel and the loads in the reactor vessel supports which are combined with other applicable faulted condition loads and subsequently used to calculate the stresses in the supports. Also, the reactor vessel displacements are applied as a time-history input to the dynamic RCL blowdown analysis. The resulting loads and stresses in the piping components and supports include both loop blowdown loads and reactor vessel displacements. Thus, the effect of vessel displacements upon loop response and the effect of loop blowdown upon vessel displacements are both evaluated.

3.9.N.1.4.6.5 Analytical Results. The results from the nonlinear time history LOCA analysis show that the reactor pressure vessel system component response (displacements and loads) obtained from the auxiliary line breaks (accumulator line and pressurizer surge line) are enveloped by the response obtained from original analysis of the main line breaks.

3.9.N.1.4.7 Evaluation of Control Rod Drive Mechanisms and Supports

The control rod drive mechanisms (CRDMs) and CRDM support structure are evaluated for the loading combinations outlined in table 3.9.B.3-1. The CRDMs and CRDM support structures are described in section 4.6 and subsection 5.4.14, respectively.

A detailed finite element model of the CRDMs and CRDM supports is constructed using the WECAN computer program with beam, pipe, and spring elements. For the LOCA analysis, nonlinearities in the structure are represented. These include RPV plate impact, tie rods, and lifting leg clevis/RPV head interface. The time-history motion of the reactor vessel head, obtained from the RPV analysis described in 3.9.N.1.4.6, is input to the dynamic model. Maximum forces and moments in the CRDMs and support structure are then determined. For the seismic analysis, the structural model is linearized, and the floor response spectra corresponding to the CRDM tie rod elevation is applied to determine the maximum forces and moments in the structure.

The bending moments calculated for the CRDMs for the various loading conditions are compared with maximum allowable moments determined from a detailed finite element stress evaluation of the CRDMs. Adequacy of the CRDM support structure is verified by comparing the calculated stresses to the criteria given in ASME III, Subsection NF.

3.9.N.1.4.8 Stress Criteria for Class 1 Components and Component Supports

The allowable stresses for ASME Class 1 components and supports are given in tables 3.9.B.3-2 through 3.9.B.3-6.

All Class 1 components and supports are designed and analyzed for the design, normal, upset, and emergency conditions and corresponding service level requirements of the ASME Code Section III. The analysis of test methods and associated stress or load allowable limits that are used in evaluation of faulted conditions are those that are defined in Appendix F of the ASME Code with the following supplementary option.

The test load method given in F-1370(d) is an acceptable method of qualifying components in lieu of satisfying the stress/load limits established for the component analysis.

The reactor vessel support pads are qualified using the test option. The RPV support pads are designed to restrain unidirectional horizontal motion in addition to supporting the vessel. The design of the supports allows radial growth of the vessel but restrains the vessel from horizontal displacements since tangential displacement of the vessel is prevented at each vessel nozzle.

To duplicate the loads that act on the pads during faulted conditions, the tests, which utilized a one-eighth linear scale model, were performed by applying an unidirectional static load to the nozzle pad. The load on the nozzle pad was reacted by a support shoe which was mounted to the test fixture.

The above modeling and application of load thus allows the maximum load capacity of the support pads to be accurately established. The test load, L, was then determined by multiplying the maximum collapse load by 64 (ratio of prototype area to model area) and including temperature effects in accordance with the rules of the ASME Code, Section III.

The loads on the reactor vessel support pads, as calculated in the system analysis for faulted conditions, are limited to the value of 0.80 L_T . The tests performed and the limits established for the test load method ensure that the experimentally obtained value for L_T is accurate and that the support pad design is adequate for its intended function.

Loading combinations for ASME Class 1 components are given in table 3.9.B.3-1.

The methods of load combination for each operating condition are as follows:

A. Design

Loads are combined by algebraic sum.

B. Normal, Upset

These loads are used in the fatigue evaluation in accordance with the methods prescribed in the ASME Code. Loadsets are defined for each transient, including the OBE, and are combined such that the maximum stress ranges are obtained without regard to the order in which the transients occur. (This is discussed in more detail in paragraph 3.9.N.1.4.3). The dynamic loads are combined using the square root of the sum of the squares (SRSS) method.

C. Emergency

Normal operating loads are combined algebraically. The dynamic loads are combined using the SRSS method.

D. Faulted

LOCA and SSE loads are combined using the SRSS method on a load component basis; i.e., the LOCA F_x is combined with the SSE F_x by SRSS, the LOCA F_y is combined with the SSE F_y by SRSS, and likewise for F_z, M_x, M_y, and M_z. The sustained loads, such as weight effects, are combined with the SRSS results by algebraic sum. Other dynamic loads are combined using the SRSS method.

3.9.N.1.5 References

1. The description and verification of the computer codes are in compliance with quality assurance requirements (WCAP-9550, WCAP-9565, and WCAP-9805) and are maintained in the applicable central file.

2. "Sample Analysis of a Class 1 Nuclear Piping System," prepared by ASME Working Group on Piping, ASME Publication, 1972.

3. Takeuchi, K., et al., "Multiflex - A Fortran IV Computer Program for Analyzing Thermal-Hydraulic-Structure System Dynamics," WCAP-8708, February 1976 (Proprietary), and WCAP-8709, February 1976 (Nonproprietary).

4. “Topical Report on ASME Section III Piping and Component Fatigue Analysis Utilizing the WESTEMS™ Computer Code,” WCAP-17577-P-A, Revision 2, October 2013 (Proprietary) and WCAP-17577-NP-A, Revision 2, October 2013 (Nonproprietary).

3.9.B.2 DYNAMIC TESTING AND ANALYSIS

3.9.B.2.1 Piping Vibration, Thermal Expansion, and Dynamic Effects

A preoperational test program as described in section 14.2 is implemented as required by NB-3622.3, NC-3622, and ND-3611 of Section III of the ASME Boiler and Pressure Vessel Code(1) to verify that the piping and piping restraints will withstand dynamic effects due to

transients, such as pump trips and valve trips, and that piping vibrations are within acceptable levels.

The preoperational test program for the Class 1, 2, and 3 and high-energy piping systems is to simulate actual operating modes to demonstrate that the components comprising these systems meet functional design requirements and that piping vibrations are within acceptable levels. Piping systems are checked in three sequential steps or series of tests and inspections.

Construction acceptance, the first step, entails inspection of components for correct installation. During this phase, pipe and equipment supports are checked for correct assembly and setting. The cold locations of reactor coolant system (RCS) components, such as steam generators and reactor coolant pumps, are recorded.

During the second step of testing, plant heatup, the plant is heated to normal operating temperatures. During the heatup, all systems are observed periodically to verify proper expansion; expansion data is recorded at the end of heatup.

During the third step of testing, performance testing, systems are operated and performance of critical pumps, valves, controls, and auxiliary equipment is checked. This phase of testing includes transient tests, such as reactor coolant pump trips, reactor trip, and relief valve testing. During this phase of testing, the piping and piping restraints are observed for vibration and expansion response. Automatic safety devices, control devices, and other major equipment are observed for indications of overstress, excess vibration, overheating, and noise. Each system test includes critical valve operation during transient system modes.

The locations in the piping system selected for observation during the testing, and the respective acceptance criteria, are provided in the detailed preoperational vibration, thermal expansion, and dynamic effects test program plan. These are submitted to the Nuclear Regulatory Commission (NRC) at least 60 days prior to the initiation of the test program.

Provisions are made to verify the operability of essential snubbers by recording hot and cold positions. If vibration during testing exceeds the acceptance criteria, corrective measures are taken and the test rerun to demonstrate adequacy.

Should additional restraints be installed, piping rerouted, or other corrective action taken as a result of the preoperational piping test, the NRC is provided with documentation of such action. The analysis verifying that system response is within acceptable limits will be on file.

Further discussion is warranted in the area of piping vibration because of the nature of the resulting loadings and the methodology used during the testing. The loadings can be placed in two categories, transient-induced vibrations and steady-state vibrations. The first is a dynamic system response to a transient, time-dependent forcing function, such as fast valve closure, while the second is a constant vibration, usually flow induced.

A. Transient Response

1. Dynamic events falling in this category are anticipated operational occurrences. The systems and the transients to be included in the preoperational test program are provided in section 14.2.

2. For those types of transients provided in section 14.2, where a time-dependent dynamic analysis is performed on the system, the stresses thus obtained are combined with system stresses resulting from other operating conditions in accordance with the criteria provided in subsections 3.9.1 and 3.9.3.

3. Details of the program and the pipe monitoring displacement transducers or scratch plates and strain gage or load cells locations, including the criteria for evaluation of data gained, are provided in the test procedures.

B. Steady-State Vibration

1. System vibrations resulting from flow disturbances fall into this category. Positive displacement pumps may cause such flow variation and vibrations.

2. Since the exact nature of the flow disturbance is not known prior to pump operation, no analysis is performed. If the system vibration is evidenced during initial operation, the maximum amplitudes are measured and related to alternating stress intensity levels based on the guidance of ANSI/ASME OM3. The VEGP Preoperational Vibration Monitoring program will include appropriate safety-related instrument lines up to the first anchor. The acceptance criterion is that the maximum alternating stress intensity S_{alt}, calculated from the measured amplitudes, shall be limited as defined below:

a. For ASME Class 1 piping systems:

$$S_{alt} = \frac{C_2 K_2}{Z} M \leq \frac{S_{el}}{\alpha}$$

where

C_2 = secondary stress index as defined in the ASME code.

α = allowable stress reduction factor: 1.3 for materials covered by figure I-9.1; or 1.0 for materials covered by figure I-9.2.1 or I-9.2.2 of the ASME code.

K_2 = local stress index as defined in the ASME code.

M = maximum zero to peak dynamic moment loading due to vibration only, or in combination with other loads as required by the system design specification.

S_{el} = 0.8 S_A where S_A is the alternating stress at 10^6 cycles from Figure I-9.1; or S_A at 10^{11} cycles from Figure I-9.2.2 of the ASME Code. The user shall consider the influence of temperature on the modulus of elasticity.

Z = Section modulus of the pipe.

b. For ASME Class 2 and 3 or ANSI B31.1 piping:

$$S_{alt} = \frac{C_2 K_2}{Z} M \leq \frac{S_{el}}{\alpha}$$

where

$C_2K_2 = 2i$

i = stress intensification factor, as defined in subsection NC and ND of the ASME Code or in ANSI B31.1.

If significant vibration levels are detected during the test program which have not been previously considered in the piping system analysis, consideration should

be given to modifying the design specification to reverify applicable code conformance.

3. When required, additional restraints are provided to reduce stresses to below the acceptance criterion levels.

3.9.B.2.2 Seismic Qualification Testing of Safety-Related Mechanical Equipment

The criteria used to decide whether dynamic testing or analysis will be used to qualify Seismic Category 1 mechanical equipment are as follows:

A. Analysis Without Testing

1. Structural analysis without testing will be used if structural integrity alone can ensure the intended design function. Equipment which falls into this category includes:

   - Piping.
   - Ductwork.
   - Tanks and vessels.
   - Heat exchangers.
   - Filters.
   - Passive valves.

   The seismic analysis of piping is described in section 3.7.B.

2. Dynamic analysis without testing shall be used to qualify heavy machinery too large to be tested. It must be verified that deformations due to seismic loadings will not cause binding of the moving parts to the extent that the component cannot perform its required safety function. Components which fall into this category include:

   - Pumps.
   - Turbines.
   - Generators.
   - Fans.
   - Diesel engines.

   The seismic qualification of pumps is discussed more fully in paragraph 3.9.B.3.2. The procedure discussed therein is applied, with some variations, to other items in this category.

B. Dynamic Testing

Dynamic testing is used for components with mechanisms that must change position in order to perform their required safety function. Such components include:

- Electric motor valve operators.
- Valve limit switches.

- Similar appurtenances for other active mechanical equipment.

The seismic qualification of Seismic Category 1 electrical equipment is discussed in section 3.10.B.

C. Combinations of Analysis with Testing

A combination of analysis, static testing, and dynamic testing is used for seismic qualification of complex equipment. Such equipment includes:

1. Standby diesel generators.
2. Turbine-driven auxiliary feedwater pumps.
3. Main steam and main feedwater isolation valves.
4. Other active valves:
   a. The manufacturer determines the first natural frequency of an active valve by analysis or test. If analysis is used the analytical model is verified by test.
   b. If the first natural frequency is 33 Hz or higher, a static stress analysis and a static deflection test are performed to verify that deformation due to seismic loadings will not cause binding of internal valve parts which prevents valve operations within specified time limits.
   c. The operability of valve actuator and other appurtenances is qualified by dynamic testing.
   d. If the first natural frequency is less than 33 Hz, a dynamic analysis of the piping system using a finite element model of the flexible valve is required to determine the effect of the valve to the response of the overall system.

The seismic qualification of active valves is discussed more fully in paragraph 3.9.B.3.2.

The acceptance criteria which are used are as follows:

1. Tests, when used, demonstrate that the component is not prevented from performing its required safety function during and after the test.
2. Analysis, when used, verifies that stresses do not exceed the allowables specified for the appropriate loading combinations listed in subsection 3.9.B.3 and that deformations do not exceed those that will permit the component to perform its required safety function.

The results of tests and analyses of safety-related mechanical equipment are available for inspection.

3.9.B.2.3 Dynamic Response Analysis of Reactor Internals Operational Flow Transients and Steady-State Conditions

Refer to paragraph 3.9.N.2.3.

3.9.B.2.4 Preoperational Flow-Induced Vibration Testing of Reactor Internals

Refer to paragraph 3.9.N.2.4.

3.9.B.2.5 Dynamic System Analysis of the Reactor Internals Under Faulted Condition

Refer to paragraph 3.9.N.2.5.

3.9.B.2.6 Correlations of Reactor Internals Vibration Tests with the Analytical Results

Refer to paragraph 3.9.N.2.6.

3.9.B.2.7 Reference

1. ASME Boiler and Pressure Vessel Code, Section III, "Nuclear Power Plant Components," 1974 edition through Summer 1975 addenda. For small bore piping analysis only, Subartical NC-3600 of the Winter 1981 Addenda to the 1980 Edition for class 2 piping, and Subartical ND-3600 of the Summer, 1984 Addenda to the 1983 Edition for class 3 piping.

3.9.N.2 DYNAMIC TESTING AND ANALYSIS

3.9.N.2.1 Piping Vibration, Thermal Expansion, and Dynamic Effects

A discussion of the preoperational vibration testing program is provided in subsection 3.9.B.2.

3.9.N.2.2 Seismic Qualification Testing of Safety-Related Mechanical Equipment

The operability of Seismic Category 1 mechanical equipment must be demonstrated if the equipment is determined to be active. The operability of active Class 2 and 3 pumps, active Class 1, 2, and 3 valves and their respective drives, operators, and vital auxiliary equipment is shown by satisfying the criteria given in paragraph 3.9.N.3.2.

The testing procedures used in the seismic qualification of instrumentation and electrical equipment are discussed in section 3.10.N.

Inactive Seismic Category 1 equipment is shown to have structural integrity during all plant conditions by analysis satisfying the stress criteria applicable to the particular piece of equipment or by test showing that the equipment retains its structural integrity under the simulated test environment.

A list of Seismic Category 1 equipment is provided in table 3.2.2-1.

3.9.N.2.3 Dynamic Response Analysis of Reactor Internals Under Operational Flow Transients and Steady-State Conditions

The vibration characteristics and behavior due to flow-induced excitation are very complex and not readily ascertained by analytical means alone. Reactor components are excited by the flowing coolant which causes oscillatory pressures on the surfaces. The integration of these pressures over the applied area should provide the forcing functions to be used in the dynamic analysis of the structures. In view of the complexity of the geometries and the random character of the pressure oscillations, a closed-form solution of the vibratory problem by integration of the differential equation of motion is not always practical and realistic. The determination of the forcing functions as a direct correlation of pressure oscillations cannot be practically performed independently of the dynamic characteristics of the structure. The main objective is to establish the characteristics of the forcing functions that essentially determine the response of the structures. By studying the dynamic properties of the structure from previous analytical and experimental work, the characteristics of the forcing function can be deduced. These studies indicate that the most important forcing functions are flow turbulence and pump-related excitation. The relevance of such excitations depends on many factors, such as type and location of component and flow conditions. The effects of these forcing functions have been studied from tests performed on models and prototype plants as well as component tests.[1][2][3]

The Indian Point No. 2 plant (Docket No. 50-247) has been established as the prototype for a four-loop plant internals verification program and was fully instrumented and tested during hot functional testing. In addition, the Trojan plant (Docket No. 50-344) instrumentation program and the Sequoyah No. 1 plant (Docket No. 50-327) instrumentation program will provide prototype data applicable to VEGP.[1][3]

The VEGP reactor internals are similar to Indian Point No. 2; the only significant differences are the modifications resulting from the use of 17 x 17 fuel, replacement of the annular thermal shield with neutron shielding pads, and the change to the upper head injection (UHI) style inverted top hat support structure configuration. These differences are addressed below.

A. 17 x 17 Fuel

The only structural changes in the internals resulting from the design change from the 15 x 15 to the 17 x 17 fuel assembly are the guide tube and control rod driveline. The new 17 x 17 guide tubes are stronger and more rigid; hence, they are less susceptible to flow-induced vibration. The fuel assembly itself is relatively unchanged in mass and spring from the 15 x 15 fuel assembly vibration characteristics.

B. Neutron Shield Panels Lower Internals

The primary cause of core barrel excitation is flow turbulence, generated in the downcomer annulus.[3] The vibration levels due to core barrel excitation for Trojan and VEGP, both having neutron shielding panels, are expected to be similar. Since the VEGP plants have greater velocities than Trojan, vibration levels due to the core barrel excitation are expected to be somewhat greater than that for Trojan (proportional to flow velocity raised to a small power).[2] However, scale model test results[2] and results from Trojan[1] show that core barrel vibration of plants with neutron shield panels is significantly less than that of plants with thermal shields. This information and the fact that low core barrel stresses and large safety margins were measured at Indian Point No. 2 (thermal shield configuration) lead to the conclusion that stresses approximately equal to those of Indian Point No. 2 result on the VEGP internals with the attendant large safety margins.

C. UHI-Style Inverted Top Hat Upper Support Configuration

The components of the upper internals are excited by turbulent forces due to axial and cross-flows in the upper plenum.(3) Sequoyah and VEGP have the same basic upper internals configuration; therefore, the general vibration behavior is not changed.

Results from Sequoyah 1 plant testing(4) show high factors of safety for upper internals components. These results, supported by scale model test results and analytical work are used to demonstrate the adequacy of the VEGP upper internals.

The original test and analysis of the four-loop configuration are augmented by references 1, 2, and 3 to cover the effects of successive hardware modifications.

3.9.N.2.4 Preoperational Flow-Induced Vibration Testing of Reactor Internals

Because the VEGP reactor internals design configuration is well characterized, as is discussed in paragraph 3.9.N.2.3, it is not considered necessary to conduct instrumented tests of the VEGP hardware. The recommendations of Regulatory Guide 1.20 are satisfied by conducting the confirmatory pre- and post-hot functional examination for integrity. Conformance with Regulatory Guide 1.20 is summarized in section 1.9. This examination includes in excess of 30 features illustrated on drawing AX6DD401 with special emphasis on the following areas:

A. All major load-bearing elements of the reactor internals relied upon to retain the core structure in place.

B. The lateral, vertical, and torsional restraints provided within the vessel.

C. The locking and bolting devices whose failure could adversely affect the structural integrity of the internals.

D. The other locations on the reactor internal components which are similar to those that were examined on the prototype designs.

E. The inside of the vessel is inspected before and after the hot functional test, with all the internals removed, to verify that no loose parts or foreign material is in evidence.

A particularly close inspection is made on the following items or areas, using a 5X or 10X magnifying glass or other appropriate inspection.

A. Lower Internals

1. Upper barrel to flange girth weld.
2. Upper barrel to lower barrel girth weld.
3. Upper core plate aligning pin. Examine bearing surface for shadow marks, burnishing, buffing, or scoring. Inspect welds for integrity.
4. Irradiation specimen guide screw locking devices and dowel pins. Check for lockweld integrity.
5. Baffle assembly locking devices. Check for lockweld integrity.
6. Lower barrel to core support girth weld.
7. Neutron shield panel screw locking devices and dowel pin lockwelds. Examine the interface surfaces for evidence of tightness. Check for lockweld integrity.

8. Radial support key welds.
9. Insert screw locking devices. Examine soundness of lockwelds.
10. Core support columns and instrumentation guide tubes. Check the joints for tightness and soundness of the locking devices.
11. Secondary core support assembly screw locking devices for lockweld integrity.
12. Lower radial support keys and inserts. Examine bearing surfaces for shadow marks, burnishing, buffing, or scoring. Check the integrity of the lockwelds. These members supply the radial and torsional constraint of the internals at the bottom relative to the reactor vessel while permitting axial and radial growth between the two. Subsequent to the hot functional testing, the bearing surfaces of the key and keyway show burnishing, buffing, or shadow marks which indicate pressure loading and relative motion between these parts. Minor scoring of engaging surfaces is also possible and acceptable.
13. Gaps at baffle joint. Check gaps between baffle-to-baffle joints.

B. Upper Internals

1. Thermocouple conduits, clamps, and couplings.
2. Guide tube, support column, and thermocouple assembly locking devices.
3. Support column and thermocouple conduit assembly clamp welds.
4. Upper core plate alignment inserts. Examine bearing surface for shadow marks, burnishing, buffing, or scoring. Check the locking devices for integrity of lockwelds.
5. Thermocouple conduit fitting locktab and clamp welds.
6. Guide tube enclosure and card welds.

Acceptance standards are the same as required in the shop by the original design drawings and specifications.

During the hot functional test, the internals are subjected to a total operating time at greater than normal full-flow conditions (four-pump operation) of at least 240 h. This provides a cyclic loading of approximately 107 cycles on the main structural elements of the internals. In addition, there is some operating time with only one, two, and three pumps operating.

Pre- and post-hot functional inspection results serve to confirm that the internals are well behaved. When no signs of abnormal wear or harmful vibrations are detected and no apparent structural changes take place, the four-loop core support structures are considered to be structurally adequate and sound for operation.

3.9.N.2.5 Dynamic System Analysis of the Reactor Internals Under Faulted Conditions

The reactor internals analysis under faulted events considers the following conditions:

- LOCA (RCL auxiliary line breaks as considered in paragraph 3.9.N.1.4.6.1)
- SSE (Safe Shutdown Earthquake)

The criteria for acceptability in regard to mechanical integrity analyses are that adequate core cooling and core shutdown must be assured. This implies that the deformation of the reactor internals must be sufficiently small so that geometry remains substantially intact. Consequently, the limitations established for the internals are concerned with the deflections and stability of the parts in addition to stress criteria to assure integrity of the components.

3.9.N.2.5.1 Reactor Internals Analysis Methods

The evaluation of the reactor internals is composed of two parts. The first part is the three-dimensional response of the reactor internals resulting from the RCL branch pipe break conditions as mentioned in paragraph 3.9.N.1.4.6. The reactor internals response is taken from the WECAN RPV and internals system response as described in paragraph 3.9.N.1.4.6.3. The second part of this evaluation is the component stress evaluations. Maximum stresses and displacements under LOCA and SSE conditions are obtained for the reactor internal components and are combined by the square root sum of the squares. Then these maximum stresses and displacements are compared to the allowable values for the faulted conditions.

Analysis of the reactor internals for blowdown loads resulting from a loss-of-coolant accident is based on the time-history response of the internals to simultaneously applied blowdown forcing functions. The forcing functions are defined at points in the system where changes in cross-section of direction of flow occur such that differential loads are generated during the blowdown transient. The dynamic mechanical analysis can employ the displacement method, lumped parameters, and stiffness matrix formulations. Because of the complexity of the system and the components, it is necessary to use finite element stress analysis codes to provide more detailed information at various points.

Multiflex, a blowdown digital computer program,[5] which was developed for the purpose of calculating local fluid pressure, flow, and density transients that occur in pressurized water reactor coolant systems during a LOCA, is applied to the subcooled, transition, and saturated two-phase blowdown regimes. This is in contrast to programs such as WHAM,[6] which are applicable only to the subcooled region and which, due to their method of solution, could not be extended into the region in which large changes in the sonic velocities and fluid densities take place. Multiflex is based on the method of characteristics wherein the resulting set of ordinary differential equations, obtained from the laws of conservation of mass, momentum, and energy, are solved numerically, using a fixed mesh in both space and time.

Although spatially one-dimensional conservation laws are employed, the code can be applied to describe three-dimensional system geometries by use of the equivalent piping networks. Such piping networks may contain any number of channels of various diameters, dead ends, branches (with up to six pipes connected to each branch), contractions, expansions, orifices, pumps, and free surfaces (such as in the pressurizer). System losses such as friction, contraction, expansion, etc., are considered.

The Multiflex code evaluates the pressure and velocity transients for a maximum of 2400 locations throughout the system. These pressure and velocity transients are stored as a permanent tape file and are made available to the programs LATFORCE and FORCE 2, which utilize a detailed geometric description in evaluating the loadings on the reactor internals.

Each reactor component for which calculations are required is designated as an element and assigned an element number. Forces acting upon each of the elements are calculated summing the effects of:

A. The pressure differential across the element.

B. Flow stagnation on and unrecovered orifice losses across the element.

C. Friction losses along the element.

Input to the code, in addition to the blowdown pressure and velocity transients, includes the effective area of each element on which the force acts due to the pressure differential across the element, a coefficient to account for flow stagnation and unrecovered orifice losses, and the total area of the element along which the shear forces act.

The reactor internals analysis has been performed using the following assumptions:

- The mechanical and hydraulic analyses have considered the effect of hydroelasticity.
- The reactor internals are represented by concentric pipes, 3-D beams, and a multimass system connected with springs and dashpots simulating the elastic response and the viscous damping of the components.
- The model described is considered to have a sufficient number of degrees of freedom to represent the most important modes of vibration in the horizontal and vertical directions.

The pressure waves generated within the reactor are highly dependent on the location and nature of the postulated pipe failure. In general, the more rapid the severance of the pipe, the more severe the imposed loading on the components. A 1-ms severance time is taken as the limiting case.

After a postulated break at the cold leg branch line, a rarefaction wave propagates along the reactor inlet pipe entering through the inlet nozzle into the region between the core barrel and the reactor vessel. This region is called the downcomer annulus. The initial wave propagates up, around, and down the downcomer annulus, then up through the region circumferentially enclosed by the core barrel; that is, the fuel region.

The region of the downcomer annulus close to the break depressurizes rapidly but, because of restricted flow areas and finite wave speed (approximately 3000 feet per second), the opposite side of the core barrel remains at a high pressure. This results in a net horizontal force on the core barrel and the reactor pressure vessel. As the depressurization wave propagates around the downcomer annulus and up through the core, the barrel differential pressure reduces and, similarly, the resulting hydraulic forces drop.

In the case of a postulated break in the hot leg, the wave follows a similar depressurization path, passing though the outlet nozzle and directly into the upper internals region, depressurizing the core and entering the downcomer annulus from the bottom exit of the core barrel. Thus, after the hot leg break, the downcomer annulus would be depressurized with very little difference in pressure across the outside diameter of the core barrel.

A hot leg break produces less horizontal force because the depressurization wave travels directly to the inside of the core barrel (so that the downcomer annulus is not directly involved) and internal differential pressures are not as large as for the cold leg break. Since the differential pressure is less for a hot leg break, the horizontal force applied to the core barrel is less for a hot leg break than for a cold leg break. For breaks in both the hot leg and cold leg, the depressurization waves would continue to propagate by reflection and translation through the reactor vessel and loops.

3.9.N.2.5.2 Reactor Internals Modeling and Analysis Method

The finite element modeling of the reactor pressure vessel and its internals system has been discussed in paragraph 3.9.N.1.4.6.3 for the nonlinear time history LOCA analysis. The output from this analysis consists of time history component loads and displacements.

If a simultaneous seismic event with the intensity of the safe shutdown earthquake is postulated with the LOCA transient, the combined effect is determined by considering the maximum stresses for each condition and combining them by the square root sum of the squares.

A summary of the analysis for major components is presented in the following paragraphs.

3.9.N.2.5.2.1 Core Barrel. For the hydraulic analysis of the pressure transients during hot leg blowdown, the maximum pressure drop across the barrel is a uniform radial compressive impulse.

The barrel is then analyzed for dynamic buckling, using the following conservative assumptions:

- The effect of the fluid environment is neglected.
- The shell is treated as simply supported.

During cold leg blowdown, the barrel is subjected to a nonaxisymmetric expansion radial impulse which changes as the rarefaction wave propagates both around the barrel and down the outer flow annulus between vessel and barrel.

The analysis of the barrel response to cold leg blowdown is performed as follows:

A. The core barrel is analyzed as a shell with two variable sections to model the support flange and core barrel.

B. The barrel, with the core and neutron shield panels, is analyzed as a beam supported at the top and supported at the bottom by the lower radial support; and the dynamic response is obtained.

3.9.N.2.5.2.2 Guide Tubes. The dynamic loads on rod cluster control assembly (RCCA) guides are more severe for a LOCA caused by hot leg branch pipe rupture than for an accident caused by cold leg branch pipe rupture, since the cold leg break leads to much smaller changes in the transverse coolant flow over the RCCA guides. The guide tubes in closest proximity to the outlet nozzle of the ruptured loop are the most severely loaded during a blowdown. The transverse guide tube forces decrease with increasing distance from the ruptured nozzle location. The stresses due to the SSE (vertical and horizontal components) are combined with the blowdown (LOCA) stresses in order to obtain principal stresses and stress intensities to be compared with the allowables.

3.9.N.2.5.2.3 Upper Support Columns. Upper support columns located close to the broken nozzle during a hot leg break are subjected to transverse loads due to cross-flow. The loads applied to the columns are computed with a method similar to the one used for the guide tubes (i.e., by taking into consideration the increase in flow across the column during the accident). The columns are studied as beams with variable sections, and the resulting stresses are obtained using the reduced section modulus and appropriate stress risers for the various sections.

The stresses due to the SSE (vertical and horizontal components) are combined with the blowdown stresses in order to obtain principal stresses and deflection.

3.9.N.2.5.2.4 Results of Reactor Internals Analysis. The system seismic analysis of the reactor vessel and its internals can either be performed by a response spectrum analysis method or by a time history integration method. Both of these analyses techniques are the NRC accepted methodologies. For certain systems or components, when time dependent seismic response is desired, the nonlinear time history analysis is used. The seismic time history analysis technique is essentially the same as that discussed earlier for the LOCA analysis, except that in seismic analysis, time history accelerations are used as the forcing function. The seismic response is then combined with the LOCA response in a conservative manner in order to obtain the maximum stresses and deflections.

All reactor internal components were found to be within acceptable stress and deflection limits for both the cold leg and hot leg LOCA transients when combined with the SSE condition.

3.9.N.2.5.2.5 Control Rod Insertability. During full power plant operation, all RCCAs and the corresponding drive rod assemblies are held at a fully withdrawn position by their respective control rod drive mechanisms (CRDMs). During certain accident conditions, such as small break LOCA (SBLOCA) and/or an SSE condition, all RCCAs are assumed to drop from their fully inserted position. It has been confirmed that all of the guide tubes are designed to maintain the function of the control rods for SBLOCA (break size of 144 in^2. and smaller). No credit for the function of the control rods is assumed for large breaks (break size areas above 144 in^2.). However, the design of the guide tubes permits control operation in all but four control rod positions, which is sufficient to maintain the core in a subcritical configuration, for break sizes up to a double-ended hot leg break. This double-ended hot leg break imposes the limiting lateral guide tube loading.

3.9.N.2.6 Correlation of Reactor Internals Vibration Tests with the Analytical Results

As stated in paragraph 3.9.N.2.3, it is not considered necessary to conduct instrumented tests of the VEGP reactor vessel internals. References 2 and 3 describe predicted vibration behavior based on studies performed prior to the plant tests. These studies, which utilize analytical models, scale model test results, component tests, and results of previous plant tests, are used to characterize the forcing functions and establish component structural characteristics so that the flow-induced vibratory behavior and response levels for VEGP reactor internals are estimated. These estimates are supported by values deduced from plant test data obtained from the Sequoyah and Trojan internals vibration measurement programs. Adequacy of the VEGP internals is verified by the use of Sequoyah and Trojan results supported by scale model tests and analytical work.

3.9.N.2.7 References

1. Bloyd, C. N., Ciaramitaro, W., and Singleton, N. R., "Verification of Neutron Pad and 17 x 17 Guide Tube Designs by Preoperational Tests on the Trojan 1 Power Plant," WCAP-8766 (Proprietary) and WCAP-8780, (Nonproprietary), May 1976.

2. Lee, H., "Prediction of the Flow-Induced Vibration of Reactor Internals by Scale Model Tests," WCAP-8803-P-A (Proprietary) and WCAP-8317-A (Nonproprietary), July 1975.

3. Bloyd, C. N., and Singleton, N. R., "UHI Plant Internals Vibration Measurement Program and Pre- and Post-Hot Functional Examinations," WCAP-8516-P (Proprietary) and WCAP-8517 (Nonproprietary), March 1975.

4. Altman, D. A., and Singleton, N. R., "Preliminary Results of Sequoyah Unit 1 Internals Vibration Measurement Program," Westinghouse Letter, PEE-RI-957, September 10, 1979, (Proprietary Class 2).

5. Takeuchi, K., et al., Multiflex-A Fortran-IV Computer Program for Analyzing Thermal-Hydraulic-Structure System Dynamics," WCAP-8708-P-A, Volumes 1 and 2 (Proprietary) and WCAP-8709-A, Volumes 1 and 2 (Nonproprietary), February 1976.

6. Fabic, S., "Computer Program WHAM for Calculation of Pressure Velocity and Force Transient in Liquid Filled Piping Networks," Kaiser Engineers Report No. 67-49-R, November 1967.

3.9.B.3 AMERICAN SOCIETY OF MECHANICAL ENGINEERS CODE CLASS 1, 2, AND 3 COMPONENTS, COMPONENT SUPPORTS, AND CORE SUPPORT STRUCTURE

The following subsection, 3.9.3, applies to all components and component supports including Westinghouse scope of supply except as noted.

For plant conditions and loading combinations, the requirements of Regulatory Guide 1.48 are met as summarized in section 1.9.

The following ASME Editions and Addenda are applicable to piping stress analysis and pipe support design:

- The design of Code Class 1 piping is in accordance with Subsection NB of the 1977 Edition, including Addenda through Summer 1979, with the following exception:

  Code Class 1 as-built reconciliation analysis of pipe supports on the pressurizer safety and relief valve system is evaluated in accordance with NF-3000 from the 1983 Edition.

- Code Class 2 and 3 piping is designed to Subsections NC and ND of the 1974 Edition, including Addenda through Summer of 1975 with the following exceptions:

  1. Code Class 2 and 3 flanged joints are evaluated in accordance with NC/ND-3658 from the 1977 Edition with Addenda through Summer 1979.

  2. Coefficients of thermal expansion and modulus of elasticity values may be taken from Appendix I, Table I-5.0 and I-6.0 of the 1977 Edition with Addenda through Summer 1979.

  3. Stress indices and stress intensification factors may be taken from Subparagraph NC/ND-3673.2 of the 1980 Edition, Winter 1980 Addendum. For circumferential fillet welded or socket welded joints, the stress intensification factors may be taken from figure NC/ND-3673.2(b)-1 of the 1983 Edition, Summer 1983 Addendum.

  4. Code Class 2 small bore piping stress analysis may be evaluated in accordance with NC-3652, -3653, 3654, and -3655 from the 1980 Edition with Addenda through Winter 1981, and Code Class 3 small bore piping

stress analysis may be evaluated in accordance with ND-3652, -3653, -3654, and -3655 from the 1983 Edition with Addenda through Summer 1984.

5. Code Class 2 and 3 piping design for single nonrepeated anchor movement (e.g., predicted building settlement) are evaluated in accordance with NC/ND-3650 from the 1977 Edition with Addenda through Summer 1979.

6. Later editions of the ASME Code may be used in accordance with 10 CFR 50.55a(b)(1) when properly reconciled to the Code of Record.

- Code Class 1, 2, and 3 component supports are designed to Subsection NF of the 1974 Edition, including Addenda through Summer of 1975, with the following exceptions:

  1. The stress limitation of 0.3 Sy for subparagraphs NF-3226.5, NF-3321.1(c), NF-3392.1, and Mandatory Appendix XVII, Figure XVII-2211(c) of the 1974 Edition with Addenda through Summer 1975 are excluded.

  2. Nuclear service shock arrestors/snubbers are designed to Subsection NF of the 1977 Edition including Addenda through Winter of 1977.

  3. The jurisdictional boundary between the pressure-retaining component and the component support is established in accordance with the applicable subparagraphs of Paragraphs NF-1131 and NF-1132 of the Code.

  4. Supporting structures are extensions of the building and are not considered within the jurisdiction of ASME Section III, Subsection NF. The component supports are attached to these members. The supporting structures are constructed in accordance with the rules of the AISC codes. The pipe support configurations shown in figures 3.9.B.3-1 through 3.9.B.3-3 are samples and are only intended to show weld locations and NF jurisdictional boundaries. Embedded anchor bolts, washers, nuts, and concrete expansion anchors are defined as part of the building structure.

- In general, the basis used for the design and construction of both the component support and the supplementary steel are as follows.

  1. All materials used in the supporting structural steel members, both safety related and nonsafety related, are ASME Code SA-36 or equivalent ASTM type material and are furnished with certificates of compliance and/or certified material test reports as required by the structural steel procurement/construction specification.

  2. The structural tubing used for pipe support is A-500 GR. B material.

  3. Embedded anchor bolts and concrete expansion anchors are not within the scope of ASME Section III. The stress analysis of the bolt (diameter sizing) is in accordance with Subsection NF using the design loads derived from the component ASME III design specification and the allowable stresses specified in Subsection NF for the selected material.

  4. The embedded anchor bolt material is selected from the materials listed in Appendix I of ASME Section III, Division 1 Code or requested in accordance with NA/NCA-1140. Concrete expansion anchors are made

from materials having mechanical properties adequate to support the expected loadings.

5. Supporting structural steel members are welded by ASME code Section IX qualified welders.

6. Welding procedures are qualified in accordance with ASME Section IX and welding materials conform to ASME Section II, Part C requirements.

7. All supporting structural steel members are examined in accordance with the acceptance criteria and methods of Article NF-5000 of the Code.

8. The component support materials are selected from the materials listed in Appendix I of Section III, Division 1 of the ASME Boiler & Pressure Vessel Code.

9. All supports are designed to carry the loads imposed by the piping systems.

10. The amount of clearance shown on the pipe support drawing is as applicable to the erected (cold) system, but calculated based on the pipe at operating temperature and the framing structure at ambient temperature.

11. Friction is considered if known thermal movement is larger than 1/16 of an inch. The load due to friction is applied at the point of contact in the direction of thermal movement.

    Friction reducing devices are used when engineering judgement indicates the friction load may impose excessive stress in the pipe. The equation for determining friction load is:

    $F = \mu N$

    where:

    F = friction load acting in the direction of pipe thermal movement.

    N = normal load due to thermal and dead weight.

    μ = coefficient of friction.

    The following coefficient of friction (μ) values are used in calculating the friction load:

Carbon steel against carbon steel	=	0.3
Teflon-based low friction plates	=	0.1

3.9.B.3.1 Loading Combinations, Design Transients, and Stress Limits

American Society of Mechanical Engineers (ASME) Class 1, 2, and 3 components and component supports are designed to an appropriate combination of plant conditions and design loadings. The plant conditions are design, normal, upset, emergency, faulted, and thermal conditions. The design loadings are pressure, temperature, deadweight, seismic, and dynamic loads.

The plant conditions and design loading combinations for ASME Code Class 1 piping and supports are indicated in table 3.9.B.3-1. Stress limits for Class 1 piping and supports are indicated in tables 3.9.B.3-2 and 3.9.B.3-6, respectively.

The plant conditions and design loading combinations for ASME Code Class 2 and 3 components and supports are indicated in table 3.9.B.3-1. Stress limits for Class 2 and 3 piping vessels, valves, pumps, and supports are indicated in tables 3.9.B.3-2, 3.9.B.3-3, 3.9.B.3-4, 3.9.B.3-5, and 3.9.B.3-7, respectively.

The methodology used for combining responses meets the requirements of NUREG-0484, Revision 1.

Piping components in essential ASME Code Class 1, 2, and 3 piping systems designed to level C or D service limits will be shown to retain functionability for emergency and faulted plant conditions by meeting the screening criteria found in reference 1.

3.9.B.3.1.1 Design Load Combinations and the Associated Operating Plant Condition

3.9.B.3.1.2 Design and Normal Conditions

The design and normal conditions are as defined in NB-3112 and NB-3113 of ASME Boiler and Pressure Vessel Code, Section III.

3.9.B.3.1.3 Upset Condition

Loads that are considered in upset plant operating conditions (defined in ASME Section III as those having a high probability of occurrence) include the following:

- Operating pressure.
- Operating temperature.
- Deadweight.
- Open relief valve thrust.
- Transient pressure effects.
  - Fast valve closure.
  - Closed relief valve discharges.
- Operating basis earthquake (OBE).

Although the occurrence of an earthquake cannot be considered highly probable, the number of cycles associated with a seismic event are considered with the low stress allowables of the upset plant condition.

It should be noted that two different types of relief valves are categorized, open and closed discharge, as described in paragraph 3.9.B.3.3.

The open discharge relief valve has a continuous blowdown thrust that can occur for a period of tens of seconds to minutes. The associated piping must be designed for this thrust. Since the maximum stress due to the relief valve thrust occurs over a significant period of time, coincident earthquake and relief stresses are assumed.

However, the case for a closed discharge system is much different. During the long-term blowdown following the establishment of steady-state flow, the reactions on the discharge piping, relief valve, and inlet piping are balanced; and no stresses are introduced as a result of relief valve blowdown. The time duration for the stresses induced during the transient preceding steady-state flow is approximately 500 ms. After this period of time, the effects from valve blowdowns are damped out.

The results of time-history dynamic analysis, using very conservative damping ratios (0.5%), show that 3 to 10 stress cycles occur before steady-state blowdown.

It may be argued that an earthquake can cause a plant trip and consequential relief valve actuation. However, the probability of the maximum stresses from these transients occurring at the same location and at the same time is extremely low.

Further, the number of cycles (3 to 10) during which both are occurring also is extremely low.

Therefore, it is concluded that the combination of loads resulting from the OBE and the transient induced by relief valve actuation in a closed discharge system is not an upset plant condition. The same argument is made for the fast valve closure event. An example of this is the trip of the main steam turbine stop valves.

3.9.B.3.1.4 Emergency Condition

Load combinations falling into this category have a low probability of occurrence. Therefore, a higher design stress is allowed since the number of cycles is low. The coincident effects of OBE and transient pressures (discussed under upset) are evaluated as an emergency plant condition.

3.9.B.3.1.5 Faulted Condition

For the faulted condition, ASME piping, vessels, pumps, and valves are analyzed to the design loading combinations shown in tables 3.9.B.3-1.

3.9.B.3.1.6 Design Stress Limits

3.9.B.3.1.6.1 ASME Code Class 1 Piping. Stress limits on ASME Code Class 1 piping, vessels, valves, and pumps are given in tables 3.9.B.3-2, 3.9.B.3-3, 3.9.B.3-4, and 3.9.B.3-5, respectively.

3.9.B.3.1.6.2 ASME Code Class 2 and 3 Piping. Stress limits on piping are given in the subsequent listing:

A. Normal Condition

Calculated stresses due to sustained loads and thermal expansion shall conform to the requirements of ASME Code, Section III, NC-3600 or ND-3600. For calculated stresses due to occasional loads, the following shall be used.

B. Upset Condition

The sum of stresses produced by loading combinations shown in table 3.9.B.3-1 for the upset condition shall not exceed 1.2 times the allowable stress values

given in Tables I-7.1, I-7.2, and I-7.3 of Appendix I of the ASME Code Section III or ND-3600. Under upset conditions equation (9) of NC-3650 or ND-3650 shall be met (permissible pressure ≤1.1 P).

C. Emergency Condition

The sum of the stresses produced by the loading combinations shown in table 3.9.B.3-1 for emergency conditions shall not exceed 1.8 times the allowable stress values given in Tables I-7.1, I-7.2, and I-7.3 of Appendix I of the ASME Code, Section III. Under emergency conditions, Equation (9) of NC-3650 or ND-3650 shall be met using a stress limit of 1.8 S. Equations (8), (10), and (11) shall not be considered.

The permissible pressure shall not exceed 1.5 times the design pressure (P) calculated in accordance with equation (4) of NC-3641.1.

D. Faulted Condition (Code Case 1606)

The sum of the stresses produced by the loading combinations shown in table 3.9.B.3-1 for the faulted condition shall not exceed 2.4 times the allowable stress values given in Tables I-7.1, I-7.2, and I-7.3 of the ASME Code, Section III, Appendix I. Under faulted conditions, Equation (9) of NC-3650 shall be met using a stress limit of 2.4 S. Equations (8), (10), and (11) shall not be considered.

The permissible pressure shall not exceed 2.0 times the design pressure (P) calculated in accordance with Equation (4) of NC-3641.1.

3.9.B.3.1.6.3 ASME Code Class 2 and 3 Components. Stress limits on vessels, valves, and pumps are given in tables 3.9.B.3-3, 3.9.B.3-4, and 3.9.B.3-5, respectively.

3.9.B.3.2 Pump and Valve Operability Assurance

3.9.B.3.2.1 Pumps

The balance of plant (BOP) safety-related active pumps are listed in table 3.9.B.3-8. Safety-related active pumps are subjected to inshop tests which include hydrostatic tests of casing to 150% of the design pressure, and performance tests to determine the following:

- Total developed head.
- Minimum and maximum head.
- Net positive suction head (NPSH) requirements except as noted below.
- Other pump/motor characteristics.

Where applicable, bearing temperature and vibration are monitored during the performance tests. (For the diesel fuel oil transfer pumps, the NPSH data is developed by actual developmental testing of pumps of the same size.) After the pump is installed, it undergoes hydrostatic testing, construction acceptance tests, and preoperational tests. Where applicable, periodic inservice inspection and testing will be performed to verify and assure the functional ability and reliability of the pump for the design life of the plant. Except as noted, those pumps listed in table 3.9.B.3-8 will be included in the Inservice Testing Program (ASME OM Code,

~~Section XI~~). For those pumps which are not included in the Inservice Testing Program and are listed in Table 3.9.B.3-8, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, plant procedures and/or Technical Specifications.

In addition to the required testing, the pumps are designed and supplied in accordance with the following specified criteria:

A. In order to ensure that the active pump will not be damaged during the seismic event, the pump manufacturer must demonstrate by test or analysis that the lowest natural frequency of the pump is greater than 33 Hz. The pump, when having a natural frequency above 33 Hz, will be considered essentially rigid. This frequency is considered sufficiently high to avoid problems with amplification between the component and structure for all seismic areas. A static shaft deflection analysis is performed. The natural frequency of the support is determined and used in conjunction with the project seismic response spectra. The deflection determined from the static shaft analysis is compared with the applicable clearances.

   If the natural frequency is found to be below 33 Hz, a dynamic analysis is performed using a finite element model to determine the amplified input accelerations necessary to perform the shaft analysis. The shaft deflection analyses are performed using the adjusted accelerations and the deflections compared with allowable shaft clearances. Assumptions used for generating the analytical model are verified by test.

B. The maximum seismic nozzle loads are also considered in an analysis of the pump supports to ensure that unacceptable system misalignment cannot occur.

C. To complete the pump qualification, the pump motor and all appurtenances vital to the operation of the pump are independently qualified for operation within their specified environment, as well as during the maximum seismic event in accordance with Institute of Electrical and Electronics Engineers (IEEE) Standard 344-1975.

   From this, it is concluded that the safety-related pump/motor assemblies will not be damaged, will continue operating under safe shutdown earthquake (SSE) loadings, and will perform their intended functions. These proposed requirements take into account the complex characteristics of the pump and are sufficient to demonstrate and assure the seismic operability of the active pumps.

3.9.B.3.2.2 Valves

The BOP safety-related active valves are listed in table 3.9.B.3-9. Safety-related active valves are subjected to the following tests:

- Hydrostatic test in accordance with ASME Code, Section III requirements.
- Main seat leakage tests.
- Functional tests to verify that the valve will open and close within the specified time limits when subjected to the design pressure.
- Operability qualification of motor operators for the environmental conditions over the installed life (i.e., aging, radiation, accident, environment simulation, etc.) in accordance

with the general format and qualification procedure of IEEE 382-1972, IEEE 323-1974, and IEEE 344-1975.

In addition, static deflection tests on active valve assemblies are performed to ensure that valve internals will not bind because of loads experienced during an SSE. Normal and abnormal environmental conditions to which the valves are subjected are listed in section 3.11.B.

After installation, the valves undergo hydrostatic tests, construction acceptance tests, and preopertional tests. Where applicable, periodic inservice tests are performed to verify and ensure the functional ability of the valve. These tests demonstrate reliability of the valve for the design life of the plant. Except as noted, those valves listed in table 3.9.B.3-9 will be included in the Inservice Testing Program (ASME OM Code~~, Section XI~~). For those valves which are not included in the Inservice Testing Program and are listed in Table 3.9.B.3-9, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, plant procedures, and/or Technical Specifications.

The valves are designed using either stress analysis (described by ASME Code, Section III) or standard design rules for minimum wall thickness requirements. On rigid (natural frequency above 33 Hz) active valves with extended top works, an analysis is also performed for static equivalent SSE loads applied at the center of gravity of the extended structure. The maximum stress limits allowed in the analyses are those recommended by the ASME for the particular ASME class of valve analyzed.

In addition to these tests and analyses, the operability of the valve during SSE is demonstrated by satisfying the following criteria:

A. Active valves with extended top works are designed to have a first natural frequency greater than 33 Hz. This may be shown by test or by analysis verified by test.

B. The structural integrity of the valve is qualified by requirements that the nozzle loads due to the SSE (emergency or faulted plant condition) are considered as a normal load in the seismic analysis of the active valve, including supports when required to operate after SSE.

C. The motor operators and other electrical appurtenances necessary for operation are qualified for operability during SSE as described above. Seismic qualification of Seismic Category 1 instrumentation and electrical equipment is presented in section 3.10.

D. The complete valve assembly is qualified by test or analysis or both for operability during the SSE. The valve assembly is only qualified by analysis in cases where structural integrity alone is required, as in the case of a nonactive valve.

For the static deflection test, the valve is mounted in a manner that conservatively represents a typical plant installation. The valve includes the actuator and all appurtenances normally attached to the valve when in service.

The extended top works of the valve are subjected to a statically applied equivalent seismic load of 4.5 g horizontally and vertically for line-mounted valves. Vertical load in the static test may be excluded when it is not a contributing factor. Actual g loadings, if known, may also be used for this test with a safety margin.

The load is applied at the center of gravity of the operator, in the direction of the weakest axis of the yoke. The design pressure of the valve is simultaneously applied to the valve during the static load tests.

The valve is then operated while the equivalent seismic static load is applied, i.e., from the normal operating status to the faulted operating status. The valve must perform its safety-related function within the specified operating time limits.

If the frequency of the valve, by test or analysis, is less than 33 Hz, a dynamic analysis of the piping system using a finite element model of the flexible valve is performed to determine the equivalent acceleration, considering the natural frequency of the valve and the frequency content of the applicable plant floor response spectra. The equivalent acceleration is then used in the static analysis and the valve seismic operability test.

The static deflection test applies only to valves with overhanging structures, e.g., the operator. The testing is conducted on a representative number of valves. Valves from each of the primary safety-related design types (e.g., motor-operated valve, solenoid-operated valve) are tested. Valve sizes that cover the range of sizes in service are qualified by the tests, and the results are used to qualify all valves within the intermediate range of sizes. Stress and deformation analyses are used to support the interpolation.

Degraded conditions as discussed in SRP Section 3.10, paragraph II.1.a(2) are minimized during system design to preclude the presence of debris, impurities, and contaminants in the fluid system. For example, containment sump design considers the presence of debris as described in Final Safety Analysis Report (FSAR) subsection 6.2.2. Other degraded conditions, such as motive power fluctuations, air pressure, etc., are addressed by testing or analysis showing that sufficient margin was included in the design of the equipment to perform its function.

Safety-related valves that can be classified as not having an overhanging structure, such as some check valves, are considered below. With the exceptions of counterweighted check valves, which have overhung structures, check valves are characteristically simple in design. The effects from seismic accelerations or the maximum applied nozzle loads on their operation are negligible. The check valve design is compact, and there are no extended structures or masses whose motion can cause distortions that can restrict operation of the valve. The nozzle loads due to maximum seismic excitation will not affect the functional ability of the valve, since the valve disc is designed to be isolated from the casing wall. The clearance supplied around the disc by the design prevents the disc from becoming bound or restricted because of any casing distortions caused by nozzle loads. Therefore, the structural integrity of these valves is assured by standard design or analysis methods, and the ability of the valve to operate is ensured by the design features. In addition to these design considerations, the valve also undergoes the following tests and analyses:

- Stress analysis as a part of the piping system including the SSE loads.
- Inshop hydrostatic test.
- Inshop seat leakage test.
- Periodic in situ valve exercising and inspection to ensure the functional ability of the valve as noted in table 3.9.B.3-9.

The following criteria are used to qualify BOP active check valves for their service conditions. Load conditions are addressed through the dynamic analysis of the piping in which the valve is installed as described in FSAR subsection 3.9.2. Valve suppliers assure by analysis that the valve is stronger than the pipe (i.e., section modulus is at least 110% of the connected pipe and the allowable stress for the valve body material is equal to or greater than the allowable stress of the connected piping material). Reverse flow conditions (transients) are considered for active valves required to close to perform their safety function. For check valves containing critical environmentally degradable organic components, environmental conditions are also addressed

through stringent selection of materials for use under specific environmental conditions. Individual check valve drawings and bills of material were reviewed to identify all age-sensitive parts and their materials. The parts were then evaluated as to their criticality within the check valves. After identifying the critical parts, the check valve location was reviewed for environmental conditions. Qualification testing of check valves is not required for service conditions because of their design configuration.

Turbulence is minimized as much as practicable by the following guidelines. BOP check valves are specified as equal to the line size. This minimizes pressure loss in the system. Furthermore, piping velocity guidelines, used by the project to produce cost-effective system designs, generally envelop the velocities necessary to fully open system check valves.

Check valves are generally located in horizontal pipe runs and the disks are oriented in a vertical position. Also, check valves are located to minimize the effects of flow disruptive devices while considering other important factors such as containment isolation.

Conformance to Regulatory Guide 1.148, concerning active valve assemblies is addressed in table 3.9.B.3-10. In addition, procurement specifications for replacement components (active safety related) will be consistent with the original purchased equipment to the extent of conformance discussed.

3.9.B.3.3 Design and Installation Criteria, Pressure-Relieving Devices

Pressure vessels are protected by pressure-relieving devices to meet applicable code requirements such as ASME Codes, Sections III and VIII, and American National Standards Institute (ANSI) B31.1 with exception to the following relief systems:

The relief systems discharging into the recycle hold-up tank and the thermal relief systems for the regenerative heat exchanger and residual heat removal loop suction valves have manual block valves installed which are physically locked in the open position. These valves are verified to be locked open by plant procedure to ensure overpressure protection is not defeated during normal plant operations. These manual block valves are typically used to facilitate maintenance or hydrostatic testing.

The design of pressure-relieving devices generally can be grouped in two categories, open and closed systems.

3.9.B.3.3.1 Open System

Design requirements for this piping follow the recommendations of Regulatory Guide 1.67 and its referent, Code Case 1569 (N-40).

The relief valve discharge system to atmosphere for the main steam lines is an example of an open system.

Each main steam line is designed to withstand the maximum loads possible from any or all relief valves discharging at full capacity. The design of a safety and relief valve system includes consideration of all components of the system:

- Safety or relief valve.
- Upstream piping or header.
- Downstream or vent piping.

- System support.
- Structures or building to which the supports are attached.

The most severe load combination is considered as follows:

- Internal pressure.
- Deadweight
- Seismic.
- Thermal.
- Reaction forces of blowing valves including entrainment.

The maximum allowable internal pressure in the main steam piping or header at the safety valve inlet nozzle is 110% of the steam generator shell design pressure as specified by the ASME Code.

Relief valve connections will be spaced on the header so that there is no local interaction.

Reaction force and moment effects on the steam header, supports, and connecting nozzles for each valve blowing and for combinations of valves blowing shall be considered. The steady blowdown load is not transmitted to the header but is carried by the structure, using a piston-type design.

The reaction force of the flowing valve shall be obtained from the valve manufacturer; however, the manufacturer's reaction force is verified by the total hydraulic reaction force analysis for a discharging jet of fluid, comprised of a pressure area contribution and fluid momentum contribution, referring to the outlet plane of the flow geometry.

Dynamic amplification of the reaction force is considered using a dynamic load factor of 2.0.

Stress analysis of the safety and relief valve system is conducted including evaluation of the header local stresses due to reaction moment when applicable. The stresses are categorized according to the appropriate code.

Material thicknesses are selected to accommodate expected loads and maintain stresses within allowable limits.

3.9.B.3.3.2 Closed System

A closed discharge system is characterized by piping between the valve and a tank or some other terminal. Under steady-state conditions, there are no net unbalanced forces. The initial transient response and resulting stresses are determined, using either a time-history computer solution or a conservative equivalent static solution. In calculating initial transient forces, pressure and momentum terms are included. If required, water slug effects are also included.

3.9.B.3.4 Component Supports

3.9.B.3.4.1 Supports Furnished with the Nuclear Steam Supply System (NSSS)

Refer to paragraph 3.9.N.3.4.

3.9.B.3.4.2 Supports Not Furnished with the NSSS

The loadings, as specified in the design specifications, are taken into account in designing component supports for ASME Code constructed items. These loadings include but are not limited to the following:

A. Weight of the component and normal contents under operating and test conditions.

B. Weight of the component support.

C. Superimposed loads and reactions induced by the adjacent system components.

D. Dynamic loads, including loads caused by earthquake vibration.

E. Restrained thermal expansion.

F. Anchor and support movement effects.

The combinations of loadings categorized with respect to plant operating conditions identified as normal, upset, emergency, and faulted which are specified for the design of supports for ASME Code constructed items are presented in table 3.9.B.3-1. The stress limits which are provided for each plant operating condition are provided in tables 3.9.B.3-6 and 3.9.B.3-7.

All ASME Section III, Class 1, 2, and 3 supports are designed as welded attachments to embedded or surface-mounted plates. Bolting for plates is designed according to American Institute of Steel Construction allowables. In no case do the tensile stresses in bolts exceed the yield stress of the bolting material at temperature.

3.9.B.3.4.2.1 Snubbers Used as Component Supports. The location and size of the snubbers are determined by stress analysis. The stress analysis uses the computer program mentioned in subsection 3.9.B.1 and the loading combinationgiven in table 3.9.B.3-1. The location and line of action of a snubber are selected based on the necessity of limiting seismic stresses in the piping and nozzle loads on equipment. Snubbers are chosen in lieu of rigid supports where restricting thermal growth would induce excessive thermal stresses in the piping or nozzle loads or equipment. The snubbers are constructed to ASME Boiler and Pressure Vessel Code, Section III, Subsection NF standards.

Pertinent requirements have been included in design specifications to demonstrate that the snubbers will perform their required safety function. These requirements include:

- Seismic requirements.
- Normal environmental parameters.
- Accident/post-accident environmental parameters.
- Full-scale performance test to measure pertinent performance requirements.
- Instructions for periodic maintenance (in technical manuals).

Both hydraulic and mechanical snubbers are approved for use in the non-NSSS piping support design. Details of the contents of mechanical snubber design specification are provided below. The hydraulic snubber design specification has similar requirements.

The design specification for mechanical snubbers requires consideration of the following:

A. The mechanical snubber is considered a linear support. Design is in accordance with Subarticle NF-3200 of Section III.

B. A certified stress report or certified load capacity data sheet is furnished showing the load capabilities of the snubber. Verification of the load carrying capability of the snubber is in accordance with NF-3132 of Section III.

C. The service loading of the snubber is equal to or less than the design strength established under listing B above for the particular loading condition.

D. The frictional resistance due to normal thermal movement does not exceed 1% or 5 lb, whichever is greater, for acceleration limiting snubbers and 2% or 10 lb, whichever is greater, for velocity limiting snubbers.

E. The total movement during cyclic loading including lost motion and structural deflection, does not exceed +0.06 in. at any load up to rated load when subjected to cyclic loading in the frequency range of 3 to 33 Hz.

F. The snubbers are designed for normal operation with a temperature range of 40° to 140°F and are capable of providing normal performance when exposed to an abnormal environmental temperature of 400°F for a period no longer than 15 min.

G. The total travel of mechanical snubbers shall be equal to or greater than the requirements shown on the pipe support drawings.

H. The design, procurement, manufacture, inspection, handling, testing, storage, and shipping of units and their component parts are performed in accordance with ASME Code and the quality assurance program and the vendor's standard quality assurance procedures.

The design specification requires that an installation manual be provided by the manufacturer to ensure correct installation, including dimensional detailed drawings giving materials of construction with installation and adjustment instruction. Visual confirmation and inspection are required in the field. Also, the hot and cold position of the snubbers will be measured during the preoperational testing stage.

There are no formal provisions for accessibility for inspection, testing, and repair or replacement of snubbers. Snubbers are located in order to most efficiently minimize stresses in the components and piping. However, access will be provided for inspection, testing, repair, or replacement by removing obstructions, if necessary.

All non-NSSS snubbers are of the mechanical or hydraulic type. The fabricator of the mechanical non-NSSS snubbers is the Pacific Scientific Company/Anchor Darling Company. The fabricator of the hydraulic non-NSSS snubbers is Lisega. The function of the mechanical and hydraulic snubbers is shock arrest.

Two types of tests are performed on the snubber.

A. Production tests are made on every unit:

1. Check unit to confirm that acceleration or velocity (as applicable) levels are less than specified maximum.

2. Check unit to confirm that it operates freely over the total stroke.

3. Measure and record the force required to initiate motion over the stroke in tension and compression.

4. Measure and record lost motion of the snubber mechanisms.

B. Qualification tests are performed on randomly selected production models. The tests are used to demonstrate the required load performance (load rating).

These tests include dynamic load cycling, low temperature, high temperature, humidity, salt spray, sand, dust, life test, and faulted load test.

In the piping system seismic stress analysis, the snubbers are modeled as stops. Where necessary, the snubber spring rates are incorporated into the analysis. There is no impact on the performance of the mechanical snubber by entrapped air or temperature on fluid properties.

Hydraulic snubbers approved for use have design features to prevent air entrapment. Changes in hydraulic snubber performance due to temperature effects on fluid viscosity have been evaluated and found to be acceptable.

The recommendations of Regulatory Guide 1.124 applicable to the service limits and loading combinations for Class 1 linear supports are met as discussed in table 3.9.B.3-6.

Supports for active pumps and valves are included in the overall design and qualification of the component.

3.9.B.3.5 Standard Review Plan Evaluation

Pipe support deformation is controlled by the following methods to ensure that pipe deflections are negligible and have no effect upon the operability requirements of active pumps and valves:

A. Actual pipe support stiffness values are included in the piping analysis computer model of Class 1 piping and interfacing non-Class 1 branch lines, up to and including the first anchor point. Some Class 2 and 3 lines are also analyzed in this manner. With this method, calculated piping nozzle loads reflect any effects due to support deformation.

B. For the balance of the ASME Class 2 and 3 lines, pipe support stiffness or deflection is controlled by either of the two following methods to ensure that component nozzle loads are not affected by pipe support deformation:

   1. Pipe support miscellaneous steel deflections are limited for deadweight and seismic loading to 0.04 in. in each restrained direction for large bore piping (nominal diameter > 2.0 in.) and to 0.0625 in. for small bore piping (nominal diameter $\leq$ 2.0 in.). These deflections are defined with respect to the structure (e.g., embed, primary structural members) to which the miscellaneous steel is attached. These deflection limits, which ensure adequate stiffness for seismic analysis, are small enough to also ensure that nozzle loads are unaffected by pipe support deformation.

   2. Some small bore ($\leq$ 2.0 in. diameter) piping supports are designed as essentially rigid structures with fundamental frequencies equal to or greater than 25 Hz.

These deformation limits are derived from the guidelines presented in the Bechtel Pipe Support Design Manual (Volume I-1, Section 3.12.2).

As shown in table 3.9.B.3-1, the design basis pipe break is considered a faulted condition. The faulted condition stresses are adequately conservative to ensure safety-related systems perform their function when subjected to faulted condition loads.

3.9.B.3.6 Reference

1. "Functional Capability Criteria for Essential Mark II Piping," GE Topical Report, NEDO-21985, September 1978 (reference NRC Memorandum to R. L. Tedesco, Assistant Director for Licensing, from J. P. Knight, Assistant Director for Components and Structures Engineering, dated July 1980).

3.9.N.3 AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME) CODE CLASS 1, 2, AND 3 COMPONENTS, COMPONENT SUPPORTS, AND CORE SUPPORT STRUCTURES

The ASME Class components are constructed in accordance with the ASME Code, Section III.

A detailed discussion of ASME Code Class 1 components is provided in subsection 3.9.N.1. For core support structures, design loading conditions are given in paragraph 3.9.5.2. Loading conditions are discussed in paragraph 3.9.5.1.

3.9.N.3.1 Loading Combinations, Design Transients, and Stress Limits (for ASME Code Class 2 and 3 Components)

Design pressure, temperature, and other loading conditions that provide the bases for the design of fluid system Code Class 2 and 3 components are presented in the sections which describe the systems.

3.9.N.3.1.1 Design Loading Combinations

The design loading combinations for ASME Code Class 2 and 3 components and supports are given in table 3.9.B.3-1. The design loading combinations are categorized with respect to normal, upset, emergency, and faulted conditions. The responses for all loading combinations defined in table 3.9.B.3-1 for Class 2 and 3 components are combined by the absolute sum method. Stress limits for each of the loading combinations are component oriented and are presented in table 3.9.B.3-3 for vessels, table 3.9.B.3-5 for pumps, table 3.9.B.3-4 for valves, table 3.9.B.3-7 for component supports, and table 3.9.B.3-2 for piping. Active pumps and valves are discussed in paragraph 3.9.B.3.2 and listed in tables 3.9.B.3-8, 3.9.B.3-9, 3.9.N.3-1, and 3.9.N.3-2.[(a)] Design of component supports is discussed in paragraph 3.9.N.3.4.

3.9.N.3.1.2 Design Stress Limits

The design stress limits established for the components are sufficiently low to ensure that violation of the pressure-retaining boundary does not occur. These limits, for each of the loading combinations, are component oriented and are presented in tables 3.9.B.3-2 through 3.9.B.3-6.

[(a)] Active components are those whose operability is relied upon to perform a safety function (as well as a reactor shutdown function) during transients or events considered in the respective operating condition categories.

Inactive components are those whose operability is not relied upon to perform a safety function during the transients or events considered in the respective operating condition category.

3.9.N.3.2 Pump and Valve Operability Assurance

Refer to paragraph 3.10.N.2.2 for discussion of the pump and valve operability program.

3.9.N.3.3 Design and Installation Details in Mounting of Pressure Relief Devices

Safety valves and relief valves are analyzed in accordance with the ASME Section III Code.

The method of analysis for safety valves and relief valves suitably accounts for the time-history of loads acting immediately following a valve opening, i.e., the first few milliseconds. The fluid-induced forcing functions are calculated for each safety valve and relief valve using one-dimensional equations for the conservation of mass, momentum, and energy.

The calculated forcing functions are applied at locations along the associated piping where a change in fluid flow direction occurs. Application of these forcing functions to the associated piping model constitutes the dynamic time-history analysis, referred to as a hydraulic transient analysis, which calculates the dynamic response of the piping system to the forcing functions. Therefore, a dynamic amplification factor is inherently accounted for in the analyses.

Snubbers or strut-type restraints are used, as required, to maintain the stresses resulting from the loads produced by the sudden opening of a relief or safety valve when combined with stress due to other upset loads within allowable limits of the ASME Section III Code for upset conditions. Also, the analyses show that the loads applied to the nozzles of the safety and relief valves do not exceed the maximum loads specified by the manufacturer.

3.9.N.3.3.1 Pressurizer Safety and Relief System – General Description

A. General Description

Special considerations for pressurizer safety and relief valve systems are discussed here.

The pressurizer safety and relief valve discharge piping systems provide overpressure protection for the reactor coolant system. The three springloaded safety valves, located on top of the pressurizer, are designed to prevent system pressure from exceeding design pressure by more than 10%. The two power-operated relief valves, also located on top of the pressurizer, are designed to prevent system pressure from exceeding the normal operating pressure by more than 100 psi. A water seal is maintained upstream of each valve to minimize leakage. Condensate accumulation on the inlet side of each valve prevents any leakage of hydrogen gas or steam through the valves.

The pressurizer safety valves, manufactured by Crosby, are self-actuated, spring-loaded valves with back pressure compensation. The power-operated relief valves, manufactured by Garrett, are solenoid-operated valves, capable of automatic operation via high- pressure signal or remote manual operation. The safety valves and relief valves are located in the pressurizer cubicle and are supported by the attached piping which, in turn, is supported by a system of beams, struts, and snubbers.

If the pressure exceeds the setpoint and the valves open, the water slug from the loop seal discharges. The water slug, driven by high system pressure, generates transient thrust forces. The valve discharge conditions conservatively considered

in the analysis of the pressurizer safety and relief valve piping systems are as follows:

1. The three safety valves are assumed to open simultaneously while the relief valves remain closed.
2. The two relief valves open simultaneously while the safety valves are closed.

In addition to these two cases which consider water seal discharge (water slug) followed by steam, solid water from the pressurizer (cold overpressure) is also analyzed.

B. Plant Hydraulic Model

When the pressurizer pressure reaches the set pressure (2475 psia for a safety valve and 2350 psia for a relief valve) and the valve opens, the high pressure steam in the pressurizer forces the water in the water seal loop through the valve and down the piping system to the pressurizer relief tank. Additionally, the power-operated relief valves are subjected to water discharge transients when used for cold overpressure mitigation. For the pressurizer safety and relief piping system, analytical hydraulic models are developed to represent the conditions described above. A Westinghouse proprietary computer program ITCHVALVE is used to perform the transient hydraulic analysis for the system. This program uses the Method of Characteristics approach to generate fluid parameters as a function of time. One-dimensional fluid calculations applying both the implicit and explicit characteristic methods are performed. Using this approach, the piping network is input as a series of single pipes. The network is generally joined together at one or more places by two- or three-way junctions.

Each of the single pipes has associated with its friction factors, angles or elevation and flow areas.

Conservation equations are converted into characteristic equations. The ITCHVALVE computer program incorporates special provisions to allow analysis of valve opening and closing situations.

Fluid acceleration inside the pipe generates reaction forces on all segments of the line. Reaction forces resulting from fluid pressure and momentum variations are calculated. These forces can be expressed in terms of the fluid properties available from the transient hydraulic analysis. The unbalanced forces are calculated using the momentum balance method.

C. Valve Thrust Analysis

The mathematical model used in the seismic analysis is modified for the valve thrust analysis to represent the safety and relief valve discharge. The time-history hydraulic forces from the aforementioned hydraulic analysis are then applied to the piping system lump mass points. The dynamic solution for the valve thrust is obtained by using a modified-predictor-corrector- integration technique and normal mode theory.

The time-history solution is found using program FIXFM3. The input to this program consists of natural frequencies, normal modes, and applied forces. The natural frequencies and normal modes for the pressurizer safety and relief line dynamic model are determined with the WESTDYN program.

Subsequently the time-history displacements of the FIXFM3 program are used as input to the WESTDYN2 program to determine the time-history internal forces and deflections at each end of the piping elements. For this calculation, the displacements are treated as imposed deflections on the pressurizer safety and relief line masses. The solution is stored on tape for later use in the piping stress evaluation and piping support load evaluation.

The time-history internal forces and displacements of the WESTDYN2 program are used as input to the POSDYN2 program to determine the maximum forces, moments, and displacements that exist at each end of the piping elements and the maximum loads for piping supports. The results from program POSDYN2 are saved for future use in piping stress analysis and support load evaluation.

The major structural analyses programs utilized in this static and dynamic analyses are described in WCAP-8252. This was reviewed and approved by the Nuclear Regulatory Commission (NRC letter, April 7, 1981, R. L. Tedesco to T. M. Anderson).

D. Comparison to EPRI Test Results

Piping load data have been generated from the tests conducted by EPRI at the Combustion Engineering test facility. Pertinent tests simulating dynamic opening of the safety valves for representative commercial upstream environments were carried out. The resulting downstream piping loadings and responses were measured. Upstream environments for particular valve opening cases of importance, which envelop the commercial scenarios, are as follows:

1. Cold water discharge followed by steam - steam between the pressure source and the loop seal - cold loop seal between the steam and the valve.
2. Hot water discharge followed by steam - steam between the pressure source and the loop seal - hot loop seal between the steam and the valve.
3. Steam discharge - steam between the pressure source and the valve.

A discussion of the methodology for generating the thermal hydraulic forcing functions and a comparison of analytically determined hydraulic force results to test data was presented in the following article:

> L. C. Smith and K. S. Howe, "Comparison of EPRI Safety Valve Test Data with Analytically Determined Hydraulic Results," The Inter- national Conference on Structural Mechanics in Reactor Technology, Chicago, Illinois, August 22-28, 1983, Volume F, 2/6, pp. 89-96.

A discussion of the methodology utilized in performing a safety valve discharge structural analysis and a comparison of analytical results to structural test results were presented in the following article.

> L. C. Smith and T. M. Adams, "Comparison of Analytically Determined Structural Solutions with EPRI Safety Valve Test Results," 4th National Congress on Pressure Vessel and Piping Technology, Portland, Oregon, June 19-24, 1983, PVP-Volume 74, pp. 193-199.

E. System Evaluation

In order to evaluate the pressurizer safety and relief valve piping, appropriate load combinations and acceptance criteria were developed. The load combinations and acceptance criteria are identical to those recommended by the piping subcommittee of the PWR PSARV EPRI test program and are outlined in tables 3.9.N.3-3 and 3.9.N.3-4 with a definition of load abbreviations provided in table 3.9.N.3-5.

The structural evaluation of the Class 1 piping is conducted consistent with the rules outlined in NB-3650 of the ASME Boiler and Pressure Vessel Code Section III. The piping configuration and downcomer pipe support layout is illustrated in figures 3.9.N.3-1 through 3.9.N.3-3. The piping between the valves and the pressurizer relief tank is analyzed to satisfy the requirements of the appropriate equations of the ANSI B31-1 Code. The load combinations defined in tables 3.9.N.3-3 and 3.9.N.3-4 are utilized. All static and dynamic cases are considered in the support design and evaluation. All pressurizer nozzles, valve flanges, and weld attachments are evaluated per ASME Code rules.

A comparison of calculated valve end loads to the design umbrella values tabulated in the piping design specification is conducted to verify acceptability. Maximum principal stress maximizes bending stress and maximum torsional stress are compared to umbrella operability values if valve operability is required to be demonstrated. For loading cases where only structural integrity has to be demonstrated, a comparison of maximum principal stresses is conducted.

The accelerations at the center of gravity of the valve are limited to the values provided in the design specification for dynamic loadings.

In summary, the operability and structural integrity of the as-built system is assured for all applicable loadings and load combinations including all pertinent safety and relief valve discharge cases.

3.9.N.3.4 Component Supports

The criteria for Westinghouse-supplied supports for ASME Code Class 1 mechanical equipment are presented in subsection 3.9.N.1 and table 3.9.B.3-6.

The criteria for Westinghouse-supplied supports for ASME Code Class 2 and 3 mechanical equipment are discussed in the following paragraphs. (See also the summary in table 3.9.B.3-7.)

3.9.N.3.4.1 Supports for Vessels Procured after July 1, 1974

Class 2 and 3 vessel supports are designed and analyzed to the rules and requirements of ASME III, Subsection NF.

3.9.N.3.4.2 Supports for Vessels Procured Prior to July 1, 1974

A. Linear

1. Normal - The allowable stresses of American Institute of Steel Construction (AISC)-69 Part 1 are employed for normal condition allowables.
2. Upset - Stress limits for upset conditions are 33% higher than those specified for normal conditions. This is consistent with paragraph 1.5.6 of AISC-69 Part 1, which permits one-third increase in allowable stresses for wind or seismic loads.
3. Emergency - Not applicable.
4. Faulted - Stress limits for faulted conditions are the same as for the upset condition.

B. Plate and Shell

1. Normal - Normal condition limits are those specified in ASME Section VIII, Division 1 of AISC-69 Part 1.
2. Upset - Stress limits for upset conditions are 33% higher than those specified for normal conditions. This is consistent with paragraph 1.5.6 of AISC Part 1, which permits one-third increase in allowable stresses for wind or seismic loads.
3. Emergency - Not applicable.
4. Faulted - Stress limits for faulted conditions are the same as for the upset condition.

3.9.N.3.4.3 Plate and Shell Supports for Pumps

The stress limits used for ASME Code Class 2 and 3 plate and shell component supports are identical to those used for the supported component. These allowable stresses are such that the design requirements for the components and system structural integrity are maintained.

3.9.N.3.4.4 Snubbers Used as Component Supports

The nuclear steam supply system vendor has included pertinent requirements in design specifications to demonstrate that the snubbers will perform their required safety function. These requirements include:

- Seismic requirements.
- Normal environmental parameters.
- Accident/post-accident environmental parameters.
- Full-scale performance test to measure pertinent performance requirements.
- Instructions for periodic maintenance (in technical manuals).

3.9.N.3.5 Standard Review Plan Evaluation

The VEGP FSAR does not fully specify deformation limits of supports where the component supports may affect the operability of the component. The deformation of component supports for active components is not permitted by Westinghouse.

The design basis pipe break is defined by VEGP as a faulted rather than emergency condition. Westinghouse defines design basis pipe break as a faulted (level D) condition consistent with the classification system for plant conditions in American Nuclear Society Standard 18.2. The faulted condition stress limits are sufficiently conservative to ensure the structural integrity and operability of the components and piping to perform their intended safety functions when subjected to faulted condition loads.

3.9.4 CONTROL ROD DRIVE SYSTEM (CRDS)

3.9.4.1 Descriptive Information of CRDS

3.9.4.1.1 Control Rod Drive Mechanism (CRDM)

CRDMs are located on the dome of the reactor vessel. They are coupled to rod control clusters which have absorber material over the entire length of the control rods. The CRDM is shown in figures 3.9.4-1 and 3.9.4-2.

The primary function of the CRDM is to insert or withdraw rod cluster control assemblies (RCCAs) within the core to control average core temperature and during startup and shutdown to control reactivity.

The CRDM is a magnetically operated jack. A magnetic jack is an arrangement of three electromagnets which are energized in a controlled sequence by a power cycler to insert or withdraw RCCAs in the reactor core in discrete steps. Rapid insertion of the RCCAs occurs when electrical power is interrupted.

The CRDM consists of four separate subassemblies. They are the pressure vessel, coil stack assembly, latch assembly, and the drive rod assembly.

A. The pressure vessel includes a latch housing and a rod travel housing which are connected by a threaded, seal-welded, maintenance joint which facilitates replacement of the latch assembly. The closure at the top of the rod travel housing is a threaded plug with a canopy seal weld for pressure integrity. This closure contains a threaded plug used for venting.

   The latch housing is the lower portion of the vessel and contains the latch assembly. The rod travel housing is the upper portion of the vessel and provides space for the drive rod during its upward movement as the control rods are withdrawn from the core.

B. The coil stack assembly includes the coil housings, electrical conduit and connector, and three operating coils: the stationary gripper coil, the movable gripper coil, and the lift coil.

The coil stack assembly is a separate unit which is installed on the drive mechanism by sliding it over the outside of the latch housing. It rests on the base of the latch housing without mechanical attachment.

Energizing the operating coils causes movement of the pole pieces and latches in the latch assembly.

C. The latch assembly includes the guide tube, stationary pole pieces, movable pole pieces, and two sets of latches: the movable gripper latches and the stationary gripper latches.

The latches engage grooves in the drive rod assembly. The movable gripper latches are moved up or down in 5/8-in. steps by the lift pole to raise or lower the drive rod. The stationary gripper latches hold the drive rod assembly while the movable gripper latches are repositioned for the next 5/8-in. step.

D. The drive rod assembly includes a coupling, a drive rod, a disconnect button, a disconnect rod, and a locking button.

The drive rod has 5/8-in. grooves which receive the latches during holding or moving of the drive rod. The coupling is attached to the drive rod and provides the means for coupling to the RCCA.

The disconnect button, disconnect rod, and locking button provide positive locking of the coupling to the RCCA and permit remote disconnection of the drive rod.

The CRDM is designed to release the drive rod and RCCA during any part of the power cycler sequencing if electrical power to the coils is interrupted. When released from the CRDM, the drive rod and RCCA fall by gravity into a shutdown position. The CRDM is threaded and seal welded on an adapter on top of the reactor vessel and is coupled to the RCCA directly below.

The mechanism is capable of raising or lowering a 360-lb load (which includes the drive rod weight) at a rate of 45 in./min. Withdrawal of the drive rod and RCCA is accomplished by magnetic forces, while insertion is by gravity.

The mechanism internals are designed to operate in 650°F reactor coolant. The pressure vessel is designed to contain reactor coolant at 650°F and 2500 psia. The three operating coils are designed to operate at 392°F with forced-air cooling required to maintain the coil internal temperature at or below 392°F.

The CRDM, shown schematically in figure 3.9.4-2, withdraws and inserts an RCCA as shaped electrical pulses are received by the operating coils. An ON or OFF sequence, repeated by silicon-controlled rectifiers in the power programmer, causes either withdrawal or insertion of the control rod. Position of the control rod is measured by 42 discrete coils mounted on the position indicator assembly surrounding the rod travel housing. Each coil magnetically senses the entry and presence of the top of the ferromagnetic drive rod assembly as it moves through the coil center line.

During plant operation the stationary gripper coil of the drive mechanism holds the RCCA in a static position until a stepping sequence is initiated, at which time the movable gripper coil and lift coil are energized sequentially.

3.9.4.1.2 RCCA Withdrawal

The RCCA is withdrawn by repetition of the following sequence of events (figure 3.9.4-2). The sequence, starting with the stationary gripper energized in the hold position, is as follows:

A. Movable Gripper Coil B - ON

The latch-locking plunger raises and swings the movable gripper latches into the drive rod assembly groove. A 1/16-in. axial clearance exists between the latch teeth and the drive rod.

B. Stationary Gripper Coil A - OFF

The force of gravity, acting upon the drive rod assembly and attached control rod, causes the stationary gripper latches and plunger to move downward 1/16 in. until the load of the drive rod assembly and attached control rod is transferred to the movable gripper latches. The plunger continues to move downward and swings the stationary gripper latches out of the drive rod assembly groove.

C. Lift Coil C - ON

The 5/8-in. gap between the movable gripper pole and the lift pole closes, and the drive rod assembly raises one step length (5/8 in.).

D. Stationary Gripper Coil A - ON

The plunger raises and closes the gap below the stationary gripper pole. The three links, pinned to the plunger, swing the stationary gripper latches into a drive rod assembly groove. The latches contact the drive rod assembly and lift it (and the attached control rod) 1/16 in. The 1/16-in. vertical drive rod assembly movement transfers the drive rod assembly load from the movable gripper latches to the stationary gripper latches.

E. Movable Gripper Coil B - OFF

The latch-locking plunger separates from the movable gripper pole under the force of a spring and gravity. Three links, pinned to the plunger, swing the three movable gripper latches out of the drive rod assembly groove.

F. Lift Coil C - OFF

The gap between the movable gripper pole and lift pole opens. The movable gripper latches drop 5/8 in. to a position adjacent to a drive rod assembly groove.

G. Repeat Step

The sequence described above (items A through F) is termed one step or one cycle. The RCCA moves 5/8 in. for each step or cycle. The sequence is repeated at a rate of up to 72 steps/min, and the drive rod assembly (which has a 5/8-in. groove pitch) is raised 72 grooves/min. The RCCA is thus withdrawn at a rate of up to 45 in./min.

3.9.4.1.3 RCCA Insertion

The sequence for RCCA insertion is similar to that for control rod withdrawal, except that the timing of lift coil C ON and OFF is changed to permit the lowering of the control assembly. The sequence, starting with the stationary gripper energized in the hold position, is as follows:

A. Lift Coil C - ON

The 5/8-in. gap between the movable gripper and lift pole closes. The movable gripper latches are raised to a position adjacent to a drive rod assembly groove.

B. Movable Gripper Coil B - ON

The latch-locking plunger raises and swings the movable gripper latches into a drive rod assembly groove. A 1/16-in. axial clearance exists between the latch teeth and the drive rod assembly.

C. Stationary Gripper Coil A - OFF

The force of gravity, acting upon the drive rod assembly and attached RCCA, causes the stationary gripper latches and plunger to move downward 1/16 in. until the load of the drive rod assembly and attached RCCA is transferred to the movable gripper latches. The plunger continues to move downward and swings the stationary gripper latches out of the drive rod assembly groove.

D. Lift Coil C - OFF

The force of gravity and spring force separate the movable gripper pole from the lift pole, and the drive rod assembly and attached RCCA drop down 5/8 in.

E. Stationary Gripper A - ON

The plunger raises and closes the gap below the stationary gripper pole. The three links, pinned to the plunger, swing the three stationary gripper latches into a drive rod assembly groove. The latches contact the drive rod assembly and lift it (and the attached control rod) 1/16 in. The 1/16-in. vertical drive rod assembly movement transfers the drive rod assembly load from the movable gripper latches to the stationary gripper latches.

F. Movable Gripper Coil B - OFF

The latch-locking plunger separates from the movable gripper pole under the force of a spring and gravity. Three links, pinned to the plunger, swing the three movable gripper latches out of the drive rod assembly groove.

G. Repeat Step

The sequence is repeated, as for RCCA withdrawal, up to 72 times/min which gives an insertion rate of 45 in./min.

3.9.4.1.4 Holding and Tripping of the Control Rods

During most of the plant operating time, the CRDMs hold the RCCAs withdrawn from the core in a static position. In the holding mode, only one coil, stationary gripper coil A, is energized on each mechanism. The drive rod assembly and attached RCCAs hang suspended from the three latches.

If power to the stationary gripper coil is cut off, the combined weights of the drive rod assembly and the RCCA (plus the stationary gripper return spring) are sufficient to move the latches out of the drive rod assembly groove. The control rod falls by gravity into the core. The trip occurs as the magnetic field, holding the stationary gripper plunger half against the stationary gripper pole, collapses, and the stationary gripper plunger half is forced down by the weight of the stationary gripper return spring and the weight acting upon the latches. After the RCCA is released by the

mechanism, it falls freely until the control rods enter the dashpot section of the thimble tubes in the fuel assembly.

3.9.4.2 Applicable CRDS Design Specifications

For those components in the CRDS comprising portions of the reactor coolant pressure boundary (RCPB), conformance with the General Design Criteria and 10 CFR 50, section 50.55a is discussed in sections 3.1 and 5.2. Conformance with Regulatory Guides pertaining to materials suitability is discussed in section 4.5 and subsection 5.2.3.

3.9.4.2.1 Design Bases

Bases for temperature, stress on structural members, and material compatibility are imposed on the design of the reactivity control components.

3.9.4.2.2 Design Stresses

The CRDS is designed to withstand stresses originating from various operating conditions as summarized in table 3.9.B.3-1.

3.9.4.2.2.1 Allowable Stresses. For normal operating conditions Section III of the American Society of Mechanical Engineers (ASME) Code is used. All pressure boundary components are analyzed as Class 1 components.

3.9.4.2.2.2 Dynamic Analysis. The cyclic stresses due to dynamic loads and deflections are combined with the stresses imposed by loads from component weights, hydraulic forces, and thermal gradients for the determination of the total stresses of the CRDS.

3.9.4.2.3 Control Rod Drive Mechanisms

The CRDM pressure housings are Class 1 components designed to meet the stress requirements for normal operating conditions of Section III of the ASME Code. Both static and alternating stress intensities are considered. The stresses originating from the required design transient are included in the analysis.

A dynamic seismic analysis is required on the CRDMs when a seismic disturbance has been postulated to confirm the ability of the pressure housing to meet ASME Code, Section III, allowable stresses and to confirm its ability to trip when subjected to the seismic disturbance.

3.9.4.2.4 CRDM Operational Requirements

The basic operational requirements for the CRDMs are:

A. 5/8-in. step.

B. 147-in. travel.

C. 360-lb maximum load.

D. Step in or out at 45 in./min (72 steps/min).

E. Electrical power interruption initiating release of drive rod assembly.

F. Trip delay time of less than or equal to 150 ms. Free fall of drive rod assembly shall begin less than 150 ms after power interruption, no matter what holding or stepping action is being executed, with any load and coolant temperature of 100°F to 550°F.

G. 40-year design life with normal refurbishment.[a]

3.9.4.3 Design Loads, Stress Limits, and Allowable Deformations

3.9.4.3.1 Pressure Vessel

The pressure-retaining components are analyzed for loads corresponding to normal, upset, emergency, and faulted conditions. The analysis performed depends on the mode of operation under consideration.

The scope of the analysis requires many different techniques and methods, both static and dynamic.

Some of the loads that are considered on each component where applicable are as follows:

- Control rod trip (equivalent static load).
- Differential pressure.
- Spring preloads.
- Coolant flow forces (static).
- Temperature gradients.
- Differences in thermal expansion:
  - Due to temperature differences.
  - Due to expansion of different materials.
- Interference between components.
- Vibration (mechanically or hydraulically induced).
- All operational transients listed in table 3.9.N.1-1.
- Pump overspeed.
- Seismic loads (operating basis earthquake (OBE) and safe shutdown earthquake (SSE)).
- Blowdown forces (due to cold and hot leg break).

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

The main objectives of the analysis are to satisfy allowable stress limits, to ensure an adequate design margin, and to establish deformation limits which are concerned primarily with the functioning of the components. The stress limits are established not only to ensure that peak stresses will not reach unacceptable values but also to limit the amplitude of the oscillatory stress component in consideration of fatigue characteristics of the materials. Standard methods of strength of materials are used to establish the stresses and deflections of these components. The dynamic behavior of the reactivity control components has been studied, using experimental test data and experience from operating reactors.

3.9.4.3.2 Drive Rod Assembly

All postulated failures of the drive rod assemblies, either by fracture or uncoupling, lead to a reduction in reactivity. If the drive rod assembly fractures at any elevation, that portion remaining coupled falls with and is guided by the RCCA. This always results in reactivity decrease.

3.9.4.3.3 Latch Assembly and Coil Stack Assembly

3.9.4.3.3.1 Results of Dimensional and Tolerance Analysis.

With respect to the CRDM system as a whole, critical clearances are present in the following areas:

A. Latch assembly - thermal clearances.

B. Latch arm - drive rod clearances.

C. Coil stack assembly - thermal clearances.

D. Coil fit in coil housing.

The following discussion defines clearances that are designed to provide reliable operation in the CRDM in these four critical areas. These clearances have been proven by life tests and actual field performance at operating plants.

A. Latch Assembly - Thermal Clearances

The magnetic jack has several clearances where parts made of type 410 stainless steel fit over parts made of type 304 stainless steel. Differential thermal expansion is therefore important. Minimum clearances of these parts at 68°F is 0.011 in. At the maximum design temperature of 650°F, minimum clearance is 0.0045 in.; at the maximum expected operating temperatures of 550°F, minimum clearance is 0.0057 in.

B. Latch Arm - Drive Rod Clearances

The CRDM incorporates a load transfer action. The movable or stationary gripper latches are not under load during engagement, as previously explained, due to load transfer action.

Figure 3.9.4-3 shows latch clearance variation with the drive rod as a result of minimum and maximum temperatures. Figure 3.9.4-4 shows clearance variations over the design temperature range.

C. Coil Stack Assembly - Thermal Clearances

The assembly clearances of the coil stack assembly over the latch housing were selected so that the assembly could be removed under all anticipated conditions of thermal expansion.

At 70°F the inside diameter of the coil stack is 7.308 to 7.298 in. The outside diameter of the latch housing is 7.260 to 7.270 in.

Thermal expansion of the mechanism due to operating temperature of the CRDM results in minimum inside diameter of the coil stack being 7.310 in. at 222°F and the maximum latch housing diameter being 7.302 in. at 532°F.

Under the extreme tolerance conditions listed above, it is necessary to allow time for a 70°F coil housing to heat during a replacement operation.

Four coil stack assemblies were removed from four hot CRDMs mounted on 11.035-in. centers on a 550°F test loop, allowed to cool, and then replaced without incident as a test to prove the preceding.

D. Coil Fit in Core Housing

CRDM and coil housing clearances are selected so that coil heatup results in a close to tight fit. This is done to facilitate thermal transfer and coil cooling in a hot CRDM.

3.9.4.4 CRDM Performance Assurance Program

The ability of the pressure housing components to perform throughout the design lifetime as defined in the equipment specification is confirmed by the stress analysis report required by the ASME Code, Section III.

Internal components subjected to wear have withstood a minimum of 3,000,000 steps without refurbishment as confirmed by life tests.(1) Latch assembly inspection is recommended after 2.5×10^6 steps have been accumulated on a single CRDM.

To confirm the mechanical adequacy of the fuel assembly, the CRDM, and the RCCA, functional test programs have been conducted on a full scale 12-ft control rod. The 12-ft prototype assembly was tested under simulated conditions of reactor temperature, pressure, and flow for approximately 1000 h. The prototype mechanism accumulated about 3,000,000 steps and 600 trips. At the end of the test, the CRDM was still operating satisfactorily. A correlation was developed to predict the amplitude of flow-excited vibration of individual fuel rods and fuel assemblies. Inspection of the drive line components did not reveal significant fretting.

These tests include verification that the trip time achieved by the CRDMs meets the design requirement of 2.7 s from start of RCCA motion to dashpot entry. This trip time requirement will be confirmed for each CRDM prior to initial reactor operation and at periodic intervals after initial reactor operation, as required by the Technical Specifications.

There are no significant differences between the prototype CRDMs and the production units. Design materials, tolerances, and fabrication techniques are the same (section 4.5).

These tests have been reported in reference 1.

It is expected that all CRDMs will meet specified operating requirements for the duration of plant life with normal refurbishment. However, a technical specification pertaining to an inoperable

RCCA has been set. Latch assembly inspection is recommended after 2.5 x 10^6 steps have been accumulated on a single CRDM.

If an RCCA cannot be moved by its mechanism, adjustments in the boron concentration ensure that adequate shutdown margin would be achieved following a trip. Thus, inability to move one RCCA can be tolerated. More than one inoperable RCCA could be tolerated but would impose additional demands on the plant operator. Therefore, the number of inoperable RCCAs has been limited to one, as discussed in the Technical Specifications.

In order to demonstrate proper operation of the CRDM and to ensure acceptable core power distributions, RCCA partial movement checks are performed (Technical Specifications). In addition, periodic drop tests of the RCCA are performed at each refueling shutdown to demonstrate continued ability to meet trip time requirements, to ensure core subcriticality after reactor trip, and to limit potential reactivity insertions from a hypothetical RCCA ejection. During these tests, the acceptable drop time of each assembly is not greater than 2.7 s, at full flow and operating temperature, from the beginning of motion to dashpot entry.

Actual experience in operating Westinghouse plants indicates excellent performance of CRDMs.

All units are production tested prior to shipment to confirm ability of the CRDM to meet design specification-operation requirements.

Each production CRDM undergoes a production test as listed below:

Test	Acceptance Criteria
Cold (ambient) hydrostatic	ASME Code, Section III
Confirm step length and load transfer (stationary gripper to movable gripper or movable gripper to stationary gripper)	Step length: 0.625+0.015-in. axial movement Load transfer: 0.047-in. nominal axial movement
Cold (ambient) performance test at design load - five full travel excursions	Operating speed: 45 in./min Trip delay: Free fall to drive rod to begin within 150 ms

3.9.4.5 Reference

1. Cooper, F. W., Jr., "17 x 17 Drive Line Components Tests - Phase 1B 11, 111 D-Loop Drop and Deflection," WCAP-8446 (proprietary) and WCAP-8449 (nonproprietary), December 1974.

3.9.5 REACTOR PRESSURE VESSEL INTERNALS[a]

3.9.5.1 Design Arrangements

The VEGP reactor vessel internals are described as follows:

The components of the reactor internals are divided into three parts consisting of the lower core support assembly (including the entire core barrel and neutron shield pad assembly), the upper core support assembly, and the incore instrumentation support structure. The reactor internals support the core, maintain fuel alignment, limit fuel assembly movement, maintain alignment between fuel assemblies and control rod drive mechanisms (CRDMs), direct coolant flow past the fuel elements, direct coolant flow to the pressure vessel head, provide gamma and neutron shielding, and provide guides for the incore instrumentation. The coolant flows from the vessel inlet nozzles down the annulus between the core barrel and the vessel wall and then into a plenum at the bottom of the vessel. It then reverses and flows up through the core support and through the lower core plate. The lower core plate is sized to provide the desired inlet flow distribution to the core. After passing through the core, the coolant enters the region of the upper support structure and then flows radially to the core barrel outlet nozzles and directly through the vessel outlet nozzles. A small portion of the coolant flows between the baffle plates and the core barrel to provide additional cooling of the barrel. Similarly, a small amount of the entering flow is directed into the vessel head plenum and exits through the vessel outlet nozzles.

3.9.5.1.1 Lower Core Support Assembly

The major containment and support member of the reactor internals is the lower core support assembly, shown in figure 3.9.5-1. This assembly consists of the core barrel, the core baffle, the lower core plate and support columns, the neutron shield pads, and the core support, which is welded to the core barrel. All the major material for this structure is type 304 stainless steel. The lower core support assembly is supported at its upper flange from a ledge in the reactor vessel flange, and its lower end is restrained in its transverse movement by a radial support system attached to the vessel wall. Within the core barrel are an axial baffle and a lower core plate, both of which are attached to the core barrel wall and form the enclosure periphery of the assembled core. The lower core support assembly, and principally the core barrel, serve to provide passageways and control for the coolant flow. The lower core plate is positioned at the bottom level of the core below the baffle plates and provides support and orientation for the fuel assemblies.

The lower core plate is a member through which the necessary flow distribution holes for each fuel assembly are located. Fuel assembly locating pins (two for each assembly) are also inserted into this plate. Columns are placed between this plate and the core support of the core barrel to provide stiffness and to transmit the core load to the core support. Adequate coolant distribution is obtained through the use of the lower core plate and core support.

The neutron shield pad assembly consists of four pads that are bolted and pinned to the outside of the core barrel. These pads are constructed of type 304 stainless steel and are approximately 48 in. wide by 148 in. long by 2.8 in. thick. The pads are located azimuthally to provide the required degree of vessel protection. Specimen guides in which material

[a] Fatigue of the reactor pressure vessel internals is evaluated as a TLAA for license renewal in accordance with 10 CFR 54.21 (see paragraph 19.4.2.6).

surveillance samples can be inserted and irradiated during reactor operation are attached to the pads. The samples are held in the guide by a preloaded spring device at the top and bottom to prevent sample movement. Additional details of the neutron shield pads and irradiation specimen holders are given in reference 1.

Vertically downward loads from weight, fuel assembly preload, control rod dynamic loading, hydraulic loads, and earthquake acceleration are carried by the lower core plate partially into the lower core plate support flange on the core barrel shell and partially through the lower support columns to the core support and thence through the core barrel shell to the core barrel flange supported by the vessel flange. Transverse loads from earthquake acceleration, coolant cross-flow, and vibration are carried by the core barrel shell and distributed between the lower radial support to the vessel wall and to the vessel flange. Transverse loads of the fuel assemblies are transmitted to the core barrel shell by direct connection of the lower core plate to the barrel wall and by upper core plate alignment pins, which are welded into the core barrel.

The main radial support system of the lower end of the core barrel is accomplished by key and keyway joints to the reactor vessel wall. At equally spaced points around the circumference, an Inconel clevis block is welded to the vessel inner diameter. Another Inconel insert block is bolted to each of these blocks and has a keyway geometry. Opposite each of these is a key which is attached to the internals. At assembly, as the internals are lowered into the vessel, the keys engage the keyways in the axial direction. Correct positioning of the internals is ensured by the installation equipment (lifting rig) guide tubes and bushings. With this design, the internals are provided with a support at the furthest extremity and may be viewed as a beam supported at the top and bottom.

Radial and axial expansion of the core barrel are accommodated, but transverse movement of the core barrel is restricted by this design. With this system, cyclic stresses in the internal structures are within American Society of Mechanical Engineers (ASME) Code, Section III, limits. In the event of an abnormal downward vertical displacement of the internals following a hypothetical failure, energy-absorbing devices limit the displacement after contacting the vessel bottom head. The load is then transferred through the energy-absorbing devices of the internals to the vessel.

The energy absorber base plate is contoured on its bottom surface to the reactor vessel bottom geometry. Assuming a downward vertical displacement, the potential energy of the system is absorbed mostly by the strain energy of the energy-absorbing devices.

3.9.5.1.2 Upper Core Support Assembly

The VEGP upper core support assembly, shown in figures 3.9.5-2 and 3.9.5-3, consists of the upper support, the upper core plate, the support columns, and the guide tube assemblies. The support columns establish the spacing between the upper support and the upper core plate. They are fastened at top and bottom to these plates. The support columns transmit the mechanical loadings between the two plates and serve the supplementary function of supporting thermocouples. The guide tube assemblies sheath and guide the control rod drive shafts and control rods. They are fastened to the upper support and are restrained by pins in the upper core plate for proper orientation and support. The pins are held in place by a nut locking device assembly. Operation with a missing nut and broken pin has been evaluated and it was concluded that the guide tube assembly would continue to perform its required function.

The upper core support assembly is positioned in its proper orientation with respect to the lower core support assembly by flat-sided pins in the core barrel flange. At an elevation in the core barrel where the upper core plate is positioned, four equally spaced flat-sided pins are located.

Four mating sets of inserts are located in the upper core plate at the same positions. As the upper support assembly is lowered into the lower support assembly, the inserts engage the flat-sided pins in the axial direction. Lateral displacement of the plate and of the upper support assembly is restricted by this design. Fuel assembly locating pins protrude from the bottom of the upper core plate and engage the fuel assemblies as the upper assembly is lowered into place. Proper alignment of the lower core support assembly, the upper core support assembly, the fuel assemblies, and control rods is thereby ensured by this system of locating pins and guidance arrangement. The upper and lower core support assemblies are restrained from any axial movements by a large circumferential spring, which rests between the upper barrel flange and the upper core support assembly and is compressed by installation of the reactor vessel head.

Vertical loads from weight, earthquake acceleration, hydraulic loads, and fuel assembly preload are transmitted through the upper core plate via the support columns to the upper support and then into the reactor vessel head. Transverse loads from coolant cross-flow, earthquake acceleration, and possible vibrations are distributed by the support columns to the upper support and upper core plate. The upper support plate is particularly stiff to minimize deflection.

3.9.5.1.3 Incore Instrumentation Support Structures

The incore instrumentation support structures consist of an upper system to convey and support thermocouples penetrating the vessel through the head and a lower system to convey and support flux thimbles penetrating the vessel through the bottom. (Figure 7.7.1-9 shows the basic flux-mapping system.)

The upper system utilizes the reactor vessel head penetrations. Instrumentation port columns are slip connected to inline columns that are in turn fastened to the upper support. These port columns protrude through the head penetrations. The thermocouples are carried through these port columns and the upper support at positions above their readout locations. The thermocouple conduits are supported from the columns of the upper core support system. The thermocouple conduits are stainless steel tubes.

In addition to the upper incore instrumentation, there are reactor vessel bottom instrumentation columns which carry the retractable, cold-worked stainless steel flux thimbles that are pushed upward into the reactor core. Conduits extend from the bottom of the reactor vessel down through the concrete shield area and up to the thimble seal table. The minimum bend radii are about 144 in., and the trailing ends of the thimbles (at the seal table) are extracted approximately 15 ft during refueling of the reactor. The thimbles are closed at the leading ends and serve as the pressure barrier between the reactor pressurized water and the containment atmosphere.

Mechanical seals between the retractable thimbles and conduits are provided at the seal table. During normal operation, the retractable thimbles are stationary. They are moved only during refueling or for maintenance, at which time a space of approximately 15 ft above the seal table is cleared for the retraction operation.

The incore instrumentation support structure is designed for support of instrumentation during reactor operation and is rugged enough to resist damage under the conditions imposed during the refueling sequence.

3.9.5.2 Design Loading Conditions

3.9.5.2.1 Normal and Upset Conditions

The normal and upset loading conditions that provide the basis for the design of the reactor internals are:

- Fuel and reactor internals weight.
- Fuel and core component spring forces, including spring preloading forces.
- Differential pressure and coolant flow forces.
- Temperature gradients.
- Vibratory loads including operating basis earthquake (OBE) seismic loads.
- Normal and upset operational thermal transients listed in table 3.9.N.1-1.
- Control rod trip (equivalent static load).
- Loads due to loop(s) out of service.
- Loss of load/pump overspeed.

3.9.5.2.2 Emergency Conditions

The emergency loading conditions that provide the basis for the design of the reactor internals are small loss-of-coolant accident (LOCA), small steam break, and complete loss of flow.

3.9.5.2.3 Faulted Conditions

The faulted loading conditions that provide the basis for the design of the reactor internals are large LOCA and safe shutdown earthquake (SSE).

3.9.5.2.4 Design Loading Categories

The combination of design loadings fit into either the normal, upset, emergency, or faulted conditions as defined in the ASME Code, Section III, as indicated by Figures NG-3221.1 and NG-3224.1 and by Appendix F, Rules for Evaluating Faulted Conditions.

The VEGP units are considered noncode plants with respect to their core support structures and as such are not certified to the requirements of subsection NG of the ASME code. However, in design and manufacture it is the Westinghouse policy to meet the intent of article NG-3000. Concerning other internal structures, article NG-3000 does not specifically apply, but as with core support structures Westinghouse does meet the intent of the code.

3.9.5.3 Design Bases

The scope of the stress analysis problem is large, requiring many different techniques and methods, both static and dynamic. The analysis performed depends on the mode of operation under consideration.

3.9.5.3.1 Allowable Deflections

Loads and deflections imposed on components as a result of shock and vibration are determined analytically and experimentally in both scaled models and operating reactors. The cyclic stresses resulting from these dynamic loads and reflections are combined with the stresses imposed by loads from component weights, hydraulic forces, and thermal gradients for the determination of the total stresses of the internals.

The reactor internals are designed to withstand stresses originating from various operating conditions, as summarized in table 3.9.N.1-1.

For normal operating conditions, downward vertical deflection of the lower core support plate is negligible.

For LOCA plus the SSE condition, the deflection criteria of critical internal structures are the limiting values given in table 3.9.5-1. The corresponding no-loss-of-function limits are included in table 3.9.5-1 for comparison purposes with the allowed criteria.

The criteria for the core drop accident are based upon analyses which determine the total downward displacement of the internal structures following a hypothesized core drop resulting from loss of the normal core barrel supports. The initial clearance between the secondary core support structures and the reactor vessel lower head in the hot condition is approximately 1/2 in. An additional displacement of approximately 3/4 in. would occur from the strain of the energy-absorbing devices of the secondary core support; thus, the total drop distance is about 1 1/4 in., which is insufficient to permit the tips of the rod cluster control assembly (RCCA) to come out of the guide thimble in the fuel assemblies.

Specifically, the secondary core support is a device which will never be used, except during a hypothetical accident involving the core support (core barrel, barrel flange, etc.). There are four supports in each reactor. This structure limits the fall of the core and absorbs much of the energy of the fall which otherwise would be imparted to the vessel. The energy of the fall is calculated assuming a complete and instantaneous failure of the primary core support and is absorbed during the plastic deformation of the controlled volume of stainless steel loaded in tension. The maximum deformation of this austenitic stainless piece is limited to approximately 15%, after which a positive stop is provided to ensure support.

3.9.5.3.2 Mechanical Design Bases

The design bases for the mechanical design of the VEGP reactor vessel internals components are as follows:

A. The reactor internals in conjunction with the fuel assemblies direct reactor coolant through the core to achieve acceptable flow distribution and to restrict bypass flow so that the heat transfer performance requirements are met for all modes of operation. In addition, required cooling for the pressure vessel head is provided so that the temperature differences between the vessel flange and head do not result in leakage from the flange during reactor operation.

B. In addition to neutron shielding provided by the reactor coolant, a separate neutron pad assembly is provided to limit the exposure of the pressure vessel in order to maintain the required ductility of the material for all modes of operation.

C. Provisions are made for installing incore instrumentation useful for the plant operation and vessel material test specimens required for a pressure vessel irradiation surveillance program.

D. The core internals are designed to withstand mechanical loads arising from the OBE, SSE, and pipe ruptures and to meet the requirements of item E below.

E. The reactor has mechanical provisions which are sufficient to adequately support the core and internals and to ensure that the core is intact with acceptable heat transfer geometry following transients arising from abnormal operating conditions.

F. Following the design basis accident, the plant is capable of being shut down and cooled in an orderly fashion so that fuel cladding temperature is kept within specified limits. This implies that the deformation of certain critical reactor internals must be kept sufficiently small to allow core cooling.

The functional limitations for the core structures during the design basis accident are shown in table 3.9.5-1. To ensure no column loading of rod cluster control guide tubes, the upper core plate deflection is limited to not exceed the value shown in table 3.9.5-1.

Details of the dynamic analyses, input forcing functions, and response loadings are presented in subsection 3.9.N.2.

3.9.5.4 Reference

1. Kraus, S., "Neutron Shielding Pads," WCAP-7870, June 1972.

3.9.6 INSERVICE TESTING OF PUMPS AND VALVES

Inservice testing of American Society of Mechanical Engineers (ASME) Code Class 1, 2, and 3 pumps and valves will be performed in accordance with ~~Section XI of the ASME Boiler and Pressure Vessel Code~~the ASME Code for Operation and Maintenance of Nuclear Power Plants (OM Code) and applicable addenda, as required by 10 CFR 50.55a(f) (specific edition and addenda of the code delineated in each program), except where specific written relief has been granted by the Nuclear Regulatory Commission (NRC) by 10 CFR 50.55a(f)(6)(i). Additionally, Class 1 component examinations are addressed in subsection 5.2.4, while Class 2 and 3 component examinations are addressed in section 6.6.

The initial inservice test (IST) programs for each unit were submitted to the NRC prior to their respective commercial operation dates. These IST programs were approved and relief granted pursuant to 10 CFR 50.55a(f)(6)(i) where full compliance with ~~the~~ the code then in effect, ASME Code, Section XI, ~~is~~ was not practical. The IST programs include baseline preservice testing and periodic inservice testing to ensure that all applicable pumps and valves are in a state of operational readiness to perform their safety function throughout the life of the plant.

3.9.6.1 Inservice Testing of Pumps

The IST programs include lists of all safety-related Class 1, 2, and 3 pumps that are provided with an emergency power source and are necessary to safely shut down the plant or mitigate the consequences of an accident. The pump testing portion of the IST program conforms to the requirements of ~~Subsection IWP of~~ the ASME OM Code, ~~Section XI,~~ to the extent practical, and complies with all applicable portions of 10 CFR 50.55a(f). In addition, the hydraulic and mechanical test parameters to be measured or observed are included in the IST programs.

3.9.6.2 Inservice Testing of Valves

The IST programs include lists of all safety-related (i.e., those valves necessary to safely shut down the plant or mitigate the consequences of an accident) Class 1, 2, and 3 valves subject to operational readiness and indicate the test parameters to be measured or observed. The valve testing portion of the IST program conforms to the requirements of ~~Subsection IWV of~~ the ASME OM Code, ~~Section XI,~~ to the extent practical, and complies with all applicable portions of 10 CFR 50.55a(f).

3.9.6.3 Relief Request

Relief from the testing requirements of ~~Section XI~~the ASME OM Code will be requested when full compliance with requirements of the code is not practical. In such cases, specific information will be provided which identifies the applicable code requirements, justification for the relief request, and the testing method to be used as an alternative.

TABLE 3.9.N.1-1 (SHEET 1 OF 3)

SUMMARY OF RCS DESIGN TRANSIENTS

Operating Conditions	Occurrences
RCP startup and shutdown	4000
Heatup and cooldown at 100°F/h (pressurizer cooldown at 200°F/h)	200 (each)
Unit loading and unloading between 0 and 15 percent of full power	500 (each)
Unit loading and unloading at 5 percent of full power/min	13,200 unloading 11,200 loading
Reduced temperature return to power	2000
Step-load increase and decrease of 10 percent of full power	2000 (each)
Large step-load decrease with steam dump	200
Steady-state fluctuations	
Initial fluctuations	1.5×10^5
Random fluctuations	3.0×10^6
Boron concentration equalization	26,400
Feedwater cycling	2000
Loop out of service	
Normal loop shutdown	80
Normal loop startup	70
Refueling	80
Turbine roll test	20
Primary side leakage test	200

TABLE 3.9.N.1-1 (SHEET 2 OF 3)

Upset Conditions	Occurrences
Secondary side leakage test	80
Loss of load without immediate reactor trip	80
Loss of power	40
Partial loss of flow	80
Reactor trip from full power	
With no inadvertent cooldown	230
With cooldown and no SI	160
With cooldown and SI	10
Inadvertent RCS depressurization	20
Inadvertent startup of an inactive loop	10
Control rod drop	80
Inadvertent SI actuation	60
Excessive feedwater flow	30
OBE (5 earthquakes of 10 cycles each)	50 cycles
Excessive bypass feedwater flow	30
RCS cold overpressurization	10
Emergency Conditions	
Small LOCA	5
Small steam line break	5
Complete loss of flow	5
Faulted Conditions	
Reactor coolant pipe break (large LOCA)	1

TABLE 3.9.N.1-1 (SHEET 3 OF 3)

Faulted Conditions (continued)	Occurrences
Large steam line break	1
Feedwater line break	1
RCP locked rotor	1
Control rod ejection	1
Steam generator tube rupture	1
Simultaneous steam line - feedwater line break	1
SSE	1
Test Conditions	
Primary side hydrostatic test	10
Secondary side hydrostatic test	10
Tube leakage test	800

TABLE 3.9.N.1-2 (SHEET 1 OF 3)

MONITORED COMPONENTS CYCLIC OR TRANSIENT LIMITS

	TRANSIENT	DESCRIPTION	CYCLIC OR TRANSIENT DESIGN LIMIT
1	RCS heatup	Heatup cycle: T_{avg} from ≤ 200°F to ≥ 500°F	200 heatup cycles at ≤ 100°F/h
2	RCS cooldown	Cooldown cycle: T_{avg} from ≥ 550°F to ≤ 200°F	200 cooldown cycles at ≤ 100°F/h
3	Pressurizer heatup	Pressurizer heatup cycle temperatures from ≤ 200°F to ≥ 650°F	200 pressurizer heatup cycles at ≤ 200°F/h
4	Pressurizer cooldown	Pressurizer cooldown cycle temperatures from ≥ 650°F to ≤ 200°F	200 pressurizer cooldown cycles at ≤ 200°F/h
5	RCS leak tests	Pressurized to ≥ 2485 psig.	200 leak tests
6	RCP-1 startup/shutdown	RCP-1 starts during routine operations plant heatups/cooldowns and due to loop out of service, LOSP, etc.	1000 cycles
7	RCP-2 startup/shutdown	RCP-2 starts during routine operations, plant heatups/cooldowns and due to loop out of service, LOSP, etc.	1000 cycles
8	RCP-3 startup/shutdown	RCP-3 starts during routine operations, plant heatups/cooldowns and due to loop out of service, LOSP, etc.	1000 cycles
9	RCP-4 startup/shutdown	RCP-4 starts during routine operations, plant heatups/cooldowns and due to loop out of service, LOSP, etc.	1000 cycles
10	Turbine trip without reactor trip	Loss of load ≥ 15% of rated thermal power to 0% of rated thermal power	80 loss of load cycles
11	Loss of offsite power	Loss-of-offsite ac electrical ESF system	40 cycles

TABLE 3.9.N.1-2 (SHEET 2 OF 3)

	COMPONENT	DESCRIPTION	CYCLIC OR TRANSIENT DESIGN LIMIT
12	Loss of charging flow in Loop 1	Charging flow terminated to Loop 1	CUF of Normal Charging Nozzle ≤ 1.0
13	Loss of charging flow in Loop 4	Charging flow terminated to Loop 4	CUF of Alternate Charging Nozzle ≤ 1.0
14	Loss of RCS flow in a loop at power	Loss of only one reactor coolant pump	80 cycles
15	Auxiliary spray with $\Delta T > 320°F$	Spray water/ pressurizer temperature differential > 320°F and < 625°F	10 auxiliary spray actuation cycles
16	Plant loading between 0% and 15%	Represented by continuous and uniform ramp power changes	500 cycles
17	Plant unloading loading between 0% and 15%	Represented by continuous and uniform ramp power changes	500 cycles
18	S/G-1 feedwater cycling	Intermittent slug feeding of 32°F auxiliary feedwater at hot standby or no load conditions	CUF of S/G-1 feedwater nozzle and S/G-1 auxiliary feedwater nozzle ≤ 1.0
19	S/G-2 feedwater cycling	Intermittent slug feeding of 32°F auxiliary feedwater at hot standby or no load conditions	CUF of S/G-2 feedwater nozzle and S/G-2 auxiliary feedwater nozzle ≤ 1.0
20	S/G-3 feedwater cycling	Intermittent slug feeding of 32°F auxiliary feedwater at hot standby or no load conditions	CUF of S/G-3 feedwater nozzle and S/G-3 auxiliary feedwater nozzle ≤ 1.0
21	S/G- 4 feedwater cycling	Intermittent slug feeding of 32°F auxiliary feedwater at hot standby or no load conditions	CUF of S/G-4 feedwater nozzle and S/G-4 auxiliary feedwater nozzle ≤ 1.0
22	Refueling	RCS cooled to 140°F, head removed, refueling canal filled	80

TABLE 3.9.N.1-2 (SHEET 3 OF 3)

	COMPONENT	DESCRIPTION	CYCLIC OR TRANSIENT DESIGN LIMIT
23	Secondary side hydrostatic tests S/G-1	Secondary pressurized to ≥ 1481 psig, min water temperature of 120°F	10
24	Secondary side hydrostatic tests S/G-2	Secondary pressurized to ≥ 1481 psig, min water temperature of 120°F	10
25	Secondary side hydrostatic tests S/G-3	Secondary pressurized to ≥ 1481 psig, min water temperature of 120°F	10
26	Secondary side hydrostatic tests S/G-4	Secondary pressurized to ≥ 1481 psig, min water temperature of 120°F	10
27	Reactor trip (with no cooldown)	No immediate cooldown following trip	230
28	Reactor trip (with cooldown and no SI)	Plant cooled down following trip due to continued FWTR flow	160
29	Reactor trip (with cooldown and SI)	Plant cooled down following trip due to continued FWTR flow with safety injection initiation	10
30	Inadvertent safety injection	Spurious safety injection with actuation of high-head centrifugal charging pumps	60
31	Loss of letdown and RTS	Letdown terminated with continued charging	CUF of Normal and Alternate Charging Nozzles ≤ 1.0

TABLE 3.9.N.1-3

MONITORED COMPONENTS FATIGUE LOCATIONS

	Component
1	Alternate charging nozzle
2	Normal charging nozzle
3	Hot leg surge nozzle
4	Pressurizer spray line
5	Pressurizer spray line nozzle
6	Pressurizer surge nozzle
7	Pressurizer surge line
8	S/G 1 auxiliary feedwater nozzle
9	S/G 2 auxiliary feedwater nozzle
10	S/G 3 auxiliary feedwater nozzle
11	S/G 4 auxiliary feedwater nozzle
12	High pressure SI nozzle
13	RCP inlet nozzle
14	RCP outlet nozzle
15	RHR suction nozzle
16	Reactor vessel flange studs
17	Reactor vessel inlet nozzle
18	Reactor vessel outlet nozzle
19	Excess letdown nozzle
20	S/G 1 feedwater nozzle
21	S/G 2 feedwater nozzle
22	S/G 3 feedwater nozzle
23	S/G 4 feedwater nozzle

TABLE 3.9.B.3-1 (SHEET 1 OF 5)

LOAD COMBINATIONS FOR ASME SECTION III, DIVISION 1,
CODE CLASS 1, 2, AND 3 COMPONENT AND COMPONENT SUPPORTS

Plant Condition	Service Level	Components[a]	Component Supports[a][e][q]
Class 1 Components and Component Supports			
Design		PD	DW
Normal	A	PO + DW + TH[d][e]	DW + TH
Upset	B	PO + DW + OBE[e][f]	DW + TH + OBE + Eq
		PO + DW + (OBE + > RVC)[e][f][j]	DW + TH + [(OBE + Eq) + >RVC][j]
		PO + DW + DU[e]	DW + TH + DU
Emergency	C	PO + DW + DE	DW + TH + DE
		PO + DW + SVC[j]	DW + TH + SVC[j]
Faulted	D	PO + DW + SSE	DW + TH + SSE + Eq SSE
		PO + DW + (SSE + >LOCA)	DW + TH + [(SSE + Eq SSE) + >LOCA]
		PO + DW + (SSE + >RVC)[j]	DW + TH + [(SSE + Eq SSE) + >RVC][j]
		PO + DW + (SSE + >SVC)[j]	DW + TH + [(SSE + Eq SSE) + >SVC][j]
		PO + DW + HEB	DW + TH + HEB
		PO + DW + DF	DW + TH + DF

TABLE 3.9.B.3-1 (SHEET 2 OF 5)

Plant Condition	Service Level	Components[a)(b)]	Component Supports[(p)(q)]
Class 2 and 3 Components and Component Supports			
Design		PD[(c)]	PD + DW[(c)(h)]
Normal		PO + DW	DW + TH + BS
Upset		PO + DW + OBE[(g)]	DW + TH + OBE + Eq + BS
		PO + DW + RVC	DW + TH + RVC + Eq + BS
		P0 + DW + (OBE + >RVO)[(g)]	DW + TH + [(OBE + Eq) + >RVO] + BS
		PO + DW + FV	DW + TH + FV + BS
		PO + DW + DU	DW + TH + DU + BS
Emergency		PO + DW + DE	DW + TH + DE + BS
		PO + DW + (OBE + >RVC)[(k)(g)]	DW + TH + [(OBE + Eq) + >RVC] + BS[(k)]
		PO + DW + SVC[(k)]	DW + TH + SVC + BS[(k)]
Faulted		PO + DW + SSE	DW + TH + SSE + Eq SSE + BS
		PO + DW + (SSE + >LOCA)[(n)]	DW + TH + [(SSE + Eq SSE) + >LOCA] + BS[(n)]
		PO + DW + (SSE + >RVO)	DW + TH + [(SSE + Eq SSE) + >RVO] + BS
		PO + DW + FV1	DW + TH + [(SSE + Eq SSE) + >FV1] + BS
		PO + DW + (SSE + >RVC)[(k)]	DW + TH + [(SSE + Eq SSE) + >RVC] + BS[(k)]
		PO + DW + (SSE + >SVC)[(k)]	DW + TH + [(SSE + Eq SSE) + >SVC] + BS[(k)]
		PO + DW + HEB	DW + TH + HEB + BS
		PO + DW + DF	DW + TH + DF + BS
Thermal		Se + Eq	

a. A "+" represents absolute summation of loading effects other than thermal and weight, which are combined algebraically. Thermal loads are considered only if the sign is the same as the direction for which the loads are calculated. Entire downward weight loads are considered for calculation of downward loads. Seventy-five percent of downward weight loads are considered for calculations of upward loads. A "+ >" represents the square root sum of the squares of loading effects.

b. Also applicable to B31.1 piping; where Seismic Category 1 design requirements do not apply, delete OBE/SSE terminology but include seismic effects per Uniform Building Code requirements.

TABLE 3.9.B.3-1 (SHEET 3 OF 5)

c. Loading effects resulting from internal pressure include those caused by unbalanced expansion joints where applicable; this note applies to all pressure terms (PD or PO).

d. TH (for Class 1 components only) includes localized loadings due to thermal transient events.

e. TH should be added for fatigue evaluation. (Note d applies.)

f. OBE includes effects due to inertially induced motions and seismic anchor motions (for Class 1 fatigue evaluation only); for Class 1 primary stress evaluation, only effects due to inertially induced motions are included in OBE.

g. Loading effects of seismic anchor motions are either included with OBE or added to TH (thermal plant condition).

h. For certain components (e.g., valves designed by standard design rules) utilizing pressure/temperature ratings for design conditions, loading effects due to deadweight are not applicable for the design condition loading combination.

i. For Westinghouse equipment, loads caused by the thermal expansion of attached piping are included in all load combinations.

j. Applicable to pressurizer safety and relief valve piping, Class 1 portion only.

k. Applicable to pressurizer safety and relief valve piping, downstream portion only.

l. Also applicable to Class 2, 3 and ANSI B31.1 piping up to the first anchor, which are included in Class 1.

m. Thermal plant condition. This condition is associated with the thermal expansion stresses and stresses associated with seismic anchor point motion (piping only).

n. For 26-in. mainsteam, 16-in. main feedwater, 6-in. auxiliary feedwater and Class 2 and 3 lines included in Class 1 analysis only.

o. Building settlement piping stress is evaluated independent of plant condition.

p. Piping system loading due to thermal expansion, dynamic events not concurrent with seismic events, and seismic excitation of support mass need not be included in load combinations used to satisfy deflection limits for pipe support miscellaneous steel.

q. The analysis of a piping and support system includes several thermal “cases” which represent different operating conditions of the plant and piping systems.

For example, the analysis of the residual heat removal (RHR) system includes four thermal cases which represent the plant at the following conditions: (1) during RHR operation, (2) during plant at 100% power, (3) during post LOCA cooling, and (4) during an RCS transient event. In addition, the analysis of the safety injection (SI) system includes three thermal cases which represent the plant at the following conditions: (1)during plant at 100% power, (2) during post LOCA cooling, and (3) during an RCS transient event.

Following is the example for faulted conditions for Loop 1 & 4 for RHR piping system due to MSIP loads. (See Westinghouse Calculation Note CN-PAFM-13-10, “Vogtle Unit 1 Auxiliary Piping Evaluation with MSIP Applied to Reactor Vessel Outlet Nozzles,” Rev. 1 (SNC Calculation 1X6MSIP1310, Version 1.0), for details of all load combinations applicable to this note):

Detailed Load Combinations:

Faulted Min = DW + MSIP + TH 100% Power - SRSS [(SSE + SAM) + (LOCA + Jet)]
Faulted Max = DW + MSIP + TH 100% Power + SRSS [(SSE + SAM) + (LOCA + Jet)]

Faulted Min (RHR) = DW + MSIP + Min (TH RHR Normal Operation, or TH RHR Post LOCA) - [(SSE + SAM)]
Faulted Max (RHR) = DW + MSIP + Max (TH RHR Normal Operation, or TH RHR Post LOCA) + [(SSE + SAM)]

TABLE 3.9.B.3-1 (SHEET 4 OF 5)

Abbreviations for Loading Terms

PD	-	Loadings associated with the design pressure.
PO	-	Loadings associated with operating pressures including, where applicable, any transient pressures associated with the loading condition event under consideration.
DW	-	Loadings associated with deadweight and liveweight.
OBE	-	Inertial loadings associated with the OBE.
Eq	-	Anchor point displacement loading associated with OBE earthquake.
AN	-	Single nonrepeated anchor movements (building settlement).
SSE	-	Inertial loadings associated with the SSE.
Eq SSE	-	Anchor point displacement loading associated with SSE.
RVC	-	Transient loadings associated with relief valve blowdown in a closed system.
SVC	-	Transient loading associated with safety valve in a closed system.
RVO	-	Sustained loadings associated with relief valve in an open system.
FV	-	Transient loadings associated with fast valve closure.
TH	-	Loadings associated with thermal expansion. (Appropriate to each plant condition).[q]
Se	-	Thermal expansion stress.
LOCA	-	Loadings associated with pressurizer surge line nozzle, residual heat removal line nozzle, and accumulator line nozzle breaks; such loadings result from the absolute summation of the effects from RCL hydraulic loads, jets, and RCL motions which occur during the break. See section 5.4.14.1.
HEB	-	Loadings associated with high-energy line pipe breaks other than LOCA (includes loadings from jet impingement, pipe motion, and pipe impact).
DU	-	Loadings associated with other transient dynamic event classified as an upset condition.
DE	-	Loadings associated with other transient dynamic event classified as an emergency condition.
DF	-	Loadings associated with other transient dynamic event classified as a faulted condition.
FV1	-	Load resulting from main feedwater check valve closure due to pipe break upstream of check valve.

TABLE 3.9.B.3-1 (SHEET 5 OF 5)

BS	-	Loadings associated with building settlement.
MSIP	-	Loadings associated with axial elongation of the RCS hot leg pipe due to the application of the Mechanical Stress Improvement Process.
RHR	-	Residual heat removal (system).

TABLE 3.9.B.3-2

STRESS CRITERIA FOR ASME III CLASS 1, 2, AND 3 PIPING

Stress Limits[a]

Loading Condition	Class 1	Class 2 and 3[b]
Design	NB-3652 (Design)	NC/ND-3611.2(b) (Design)
Normal	NB-3653 (Level A)	NC/ND-3611.2(c) (1) (Level B)
Upset	NB-3654 (Level B)	NC/ND-3611.2(c) (2) (Level B)
Emergency	NB-3655 (Level C)	NC/ND-3611.2(c) (3) (Level C)
Faulted	NB-3656 (Level D)	NC/ND-3611.2(c) (4) (Level D)

a. Limits identified refer to subsections of the ASME Code, Section III.

b. Stress limits of Code Case 1606-1 have been used, the Code Case limits have subsequently been included in ASME Section III (Winter 76 Addenda).

TABLE 3.9.B.3-3

STRESS CRITERIA FOR ASME III
CLASS 1, 2 AND 3 VESSELS

Stress Limits[a]

Loading Condition	Class 1[b]	Class 2[c][d]	Class 2 and 3[d][e]	Class 2 and 3[f]
Design	NC-3221	NB-3217	NC/ND-3321	The vessel shall conform to the requirements of ASME Section VIII, Division 1.
Normal	NB-3222	NC-3217	NC/ND-3321	The vessel shall conform to the requirements of ASME Section VIII, Division 1.
Upset	NB-3223	NC-3217	NC/ND-3321	$\frac{\sigma_m \leq 1.1\ s}{(\sigma_m \text{ or } \sigma_\ell) \leq 1.65\ s} + \sigma_b$
Emergency	NB-3224	NC-3217	NC/ND-3321	$\frac{\sigma_m \leq 1.5\ s}{(\sigma_m \text{ or } \sigma_\ell) \leq 1.8\ s} + \sigma_b$
Faulted	NB-3225	NC-3217	NC/ND-3321	$\frac{\sigma_m \leq 2.0\ s}{(\sigma_m \text{ or } \sigma_\ell) \leq 2.4\ s} + \sigma_b$

a. Limits identified refer to subsections of the ASME Code, Section III.

b. Limits identified also apply to ASME Code Class 2 vessels designed to Division 2 of Section VIII of the ASME Code.

c. Apply for vessels designed in accordance with ASME III, NC-3200.

d. For those vessels procured prior to the incorporation of stress limits into the ASME Code, Section III, the stress limits of Code Case 1607 were used. The code case limits have subsequently been included in ASME III.

e. Apply for vessels designed in accordance with ASME III, NC-3300 or ND-3300.

f. Apply for vessels designed in accordance with the ASME Code, Division 1 of Section III.

TABLE 3.9.B.3-4

STRESS CRITERIA FOR ASME III CLASS 1, 2, AND 3 VALVES

Loading Condition	Stress Limits[a] Class 1	Class 2 and 3[c]
Design	NB-3520 (Design)	NC/ND-3510 (Design)
Normal	NB-3525 (Level A)	NC/ND-3510 (Level A)
Upset	NB-3525 (Level B)	NC/ND-3520 (Level B)
Emergency	NB-3526 (Level C)	NC/ND-3520 (Level C)
Faulted	(b)	NC/ND-3520 (Level D)

a. Limits identified refer to subsections of the ASME Code, Section III.

b. Class 1 valve service level D criteria:

Active	Inactive
Calculate P_m from paragraph NB3545.1 with internal pressure P_s = 1.25 PS $P_m \leq 1.5\ S_m$	Calculate P_m from paragraph NB3545.1 with internal pressure P_s = 1.50 PS $P_m \leq 2.4\ S_m$ or $0.7\ S_u$
Calculate from S_n from paragraph NB3545.2 with C_p = 1.5; P_s = 1.25PS; Qt2 = 0 P_{ed} = 1.3X value of P from equations of 3545.2(b) (1)	Calculate S_n from paragraph NB3545.2 with C_p = 1.5; P_s = 1.50PS; Qt2 = 0; P_{ed} = 1.3X value of P_{ed} from equations of NB3545.2(b) (1)
$S_n \leq 3\ S_m$	$S_n \leq 3\ S_m$

c. For those valves procured prior to the incorporation of stress limits into the ASME Code, Section III, the stress limits of Code Case N69 (1635-1) were used. The code case limits have subsequently been included in ASME III.

TABLE 3.9.B.3-5

STRESS CRITERIA FOR ASME SECTION III, DIVISION 1,
CODE CLASS 1, 2, AND 3 PUMPS

Loading Condition	Class 1[b]	Stress Limits[a][c] Class 2 and 3 (Active)	Class 2 and 3 (Inactive)
Design	NB-3221 (Design)	ASME Section III NC/ND-3400	ASME Section III NC/ND-3400
Normal	NB-3222 (Level A)	ASME Section III NC/ND-3400	ASME Section III NC/ND-3400
Upset	NB-3223 (Level B)	$\sigma_m \leq 1.0$ S (σ_m or σ_L) + $\sigma_b \leq 1.5$ S $P_{max} \leq 1.1$ PD	$\sigma_m \leq 1.1$ S (σ_m or σ_L) + $\sigma_b \leq 1.65$ S $P_{max} \leq 1.1$ PD
Emergency	NB-3224 (Level C)	$\sigma_m \leq 1.1$ S (σ_m or σ_L) + $\sigma_b \leq 1.65$ S[d] $P_{max} \leq 1.2$ PD	$\sigma_m \leq 1.5$ S (σ_m or σ_L) + $\sigma_b \leq 1.8$ S $P_{max} \leq 1.2$ PD
Faulted	NB-3225 (Level D)	$\sigma_m \leq 1.2$ S (σ_m or σ_L) + $\sigma_b \leq 1.8$ S[d] $P_{max} \leq 1.5$ PD	$\sigma_m \leq 22.0$ S (σ_m or σ_L) + $\sigma_b \leq 2.4$ S $P_{max} \leq 1.5$ PD

S = Allowable stress
σ_m = General primary membrane stress
σ_L = Local membrane stress
σ_b = Primary bending stress
PD = Design pressure
P_{max} = Maximum pressure resulting from upset, emergency, or faulted conditions

a. Limits identified refer to subsections of the ASME Code, Section III.
b. There are no active Class 1 pumps.
c. For those pumps procured prior to the incorporation of stress limits into the ASME Code, Section III, the stress limits of Code Case N-70 (1636-1) were used. The Code Case limits have subsequently been included in ASME III.
d. Calculated stresses will be verified to be below yield stress. Otherwise, an analysis must be performed showing that the faulted loading condition will cause only minor rubbing or interference of rotating and stationary parts for the duration of the earthquake only and that the pump will return to design point operation immediately following the earthquake.

TABLE 3.9.B.3-6

STRESS LIMITS FOR CLASS 1 COMPONENT SUPPORTS

	Loading Conditions				
Support Type	Design	Normal	Upset	Emergency	Faulted
Plate and shell design by analysis	NF-3221	NF-3222	NF-3223	NF-3224	NF-3225
Linear[a]	NF-3231.1[a]	NF-3231.1[a]	NF-3231.1[a]	NF-3231.1[b]	NF-3231.1[c]
Component standard supports design by analysis	NF-3240	NF-3240	NF-3240	NF-3240	NF-3240
Component supports design by load rating	NF-3260	NF-3260	NF-3260	NF-3260	NF-3260

a. Paragraph numbers refer to ASME Code, Section III, Subsection NF.

b. In instances where the determination of allowable stress values utilizes S (ultimate tensile stress) at temperatures not included in ASME III, S shall be calculated using one of the methods provided in Regulatory Guide 1.124, Revision 1.

c. For equipment procured prior to July 1, 1974, the procedures of AISC-69 or ASME Section VIII are used.

TABLE 3.9.B.3-7 (SHEET 1 OF 3)

STRESS LIMITS FOR CLASS 2 AND 3 COMPONENT SUPPORTS

	Loading Conditions				
Support Type	Design	Normal	Upset	Emergency[f]	Faulted[f]
Plate and shell design by analysis	NF-3321.1	NF-3321.2[a]	NF-3321.2[b]	NF-3321.2[c]	NF-3321.2[d]
Linear	NF-3231.1[a]	NF-3231.1[a]	NF-3231.1[a]	NF-3231.1[b]	NF-3231.1[c]
Component standard supports design by analysis	NF-3321.1 or NF-3231.1[a]	NF-3321.2[a] or NF-3231.1[a]	NF-3321.2[b] or NF-3231.1[a]	NF-3321.2[c] or NF-3231.1[b]	NF-3321.1[d] or NF-3231.1[c]
Component supports design by load rating	NF-3260	NF-3260	NF-3260	NF-3260	NF-3260
Pipe Support Member	Allowable stresses on support members are based on the AISC (AISI for Tube Track) specification. The yield stress at a selected environmental temperature is based on ASME Section III, Appendix I, Table 13.0. However, the minimum permitted design temperature shall be the applicable environmental temperature for the location of the support member. The design temperature for the support member in close proximity to piping shall consider the piping line temperature when it exceeds the environmental temperature.				
Welds	Weld stresses for pipe supports are limited to the allowable tensile stress for plates and elements of rolled sections in the through thickness direction which is 0.4Sy. Sy is the material yield strength at a design temperature of 400°F. In some cases, however, due to the manner of loading, the throat of the weld may be the more critically stressed area. Therefore, weld allowables associated with the applicable code (i.e., ASME III, Subsection NF, Table 3292.1-1[e]; ANSI B31.1; AISC) are used.				

a. Paragraph numbers refer to ASME Code, Section III.

b. In instances where the determination of allowable stress values utilizes S_u (ultimate tensile stress) at temperatures not included in ASME III, S_u shall be calculated using one of the methods provided in Regulatory Guide 1.124, Revision 1.

c. For equipment procured prior to July 1, 1974, the procedures of AISC-69 or ASME Section VIII are used as defined in paragraph 3.9.N.3.4.

d. Supports for active pumps utilize the same limits as those used for the pumps.

e. Use of Table 3292.1-1 is approved by NRC letter from Youngblood to Richard Conway, dated September 29, 1986.

f. The allowable stress values for safety-related supports may be increased in accordance with Subsection NF for items within the ASME III jurisdictional boundary when evaluating emergency and faulted conditions. For items outside the ASME III jurisdictional boundary allowable stresses may be increased by 33.3 percent above the normal allowable. NRC shall be notified when cases of non-ASME supports exceeding the normal stress allowables by more than 33.3 percent occur.

TABLE 3.9.B.3-7 (SHEET 2 OF 3)

(Allowable Axial Stress for Different $k\ell/r$ Ratio)

$$\left[1 - \frac{k\ell / (r)^2}{2C_C}\right] Fy$$

$$Fa = \frac{5}{3} + \frac{3(k\ell / r)}{8C_C} - \frac{(k\ell / r)^3}{8C_C{}^3} \qquad \left[\begin{array}{c}\text{AISC Equation}\\ (1.5 - 1)\end{array}\right] \qquad \text{when } k\ell/r \le C_C$$

$$\text{where } C_C = \sqrt{\frac{2\pi^2 E_S}{Fy}} = 131.544$$

$$Fa = \frac{12\pi^2 E_S}{23(k\ell / r)^2} \qquad \left[\begin{array}{c}E\ a\\ (1\quad 2)\end{array}\right] \qquad \text{when } k\ell/r > C_C \qquad ES = 27000 \text{ ksi}$$

Fy = 30.8 ksi

kℓ/r	Fa	kℓ/r	Fa	kℓ/r	Fa	kℓ/r	Fa	kℓ/r	Fa
1	18.448	31	17.078	61	15.037	91	12.432	121	9.283
2	18.415	32	17.020	62	14.959	92	12.336	122	9.168
3	18.381	33	16.961	63	14.880	93	12.239	123	9.052
4	18.346	34	16.902	64	14.800	94	12.142	124	8.936
5	18.310	35	16.841	65	14.720	95	12.044	125	8.819
6	18.273	36	16.780	66	14.640	96	11.946	126	8.701
7	18.236	37	16.719	67	14.559	97	11.846	127	8.582
8	18.197	38	16.656	68	14.477	98	11.747	128	8.463
9	18.158	39	16.593	69	14.395	99	11.647	129	8.343
10	18.117	40	16.530	70	14.312	100	11.546	130	8.223
11	18.076	41	16.465	71	14.228	101	11.444	131	8.101
12	18.034	42	16.400	72	14.144	102	11.342	132	7.797
13	17.991	43	16.334	73	14.059	103	11.239	133	7.860
14	17.947	44	16.268	74	13.974	104	11.136	134	7.743
15	17.903	45	16.200	75	13.888	105	11.032	135	7.629
16	17.857	46	16.133	76	13.801	106	10.928	136	7.517
17	17.811	47	16.064	77	13.714	107	10.823	137	7.408
18	17.763	48	15.995	78	13.626	108	10.717	138	7.301
19	17.716	49	15.925	79	13.538	109	10.610	139	7.196
20	17.667	50	15.854	80	13.449	110	10.503	140	7.094
21	17.617	51	15.783	81	13.360	111	10.396	141	6.993
22	17.567	52	15.711	82	13.269	112	10.287	142	6.895
23	17.515	53	15.639	83	13.179	113	10.178	143	6.799
24	17.463	54	15.566	84	13.087	114	10.069	144	6.705
25	17.411	55	15.492	85	12.996	115	9.958	145	6.613
26	17.357	56	15.418	86	12.903	116	9.847	146	6.522
27	17.303	56	15.343	87	12.810	117	9.736	147	6.434
28	17.248	58	15.267	88	12.716	118	9.623	148	6.347
29	17.192	59	15.191	89	12.622	119	9.511	149	6.262
30	17.135	60	15.114	90	12.527	120	9.397	150	6.179

TABLE 3.9.B.3-7 (SHEET 3 OF 3)

Fa	=	allowable axial stress.
k	=	effective length factor.
ℓ	=	length of member.
r	=	minimum radius of gyration.
C_C	=	column slenderness ratio separating elastic and inelastic buckling.
ES	=	modules of elasticity.
Fy	=	yield stress

TABLE 3.9.B.3-8 (SHEET 1 OF 2)

BOP ACTIVE PUMPS

Pump	Equipment Tag Number	Drawing Number
Nuclear service cooling water (NSCW) pump	1-1202-P4-001	1X4DB133-1
NSCW pump	1-1202-P4-002	1X4DB133-2
NSCW pump	1-1202-P4-003	1X4DB133-1
NSCW pump	1-1202-P4-004	1X4DB133-2
NSCW pump	1-1202-P4-005	1X4DB133-1
NSCW pump	1-1202-P4-006	1X4DB133-2
NSCW transfer pump	1-1202-P4-007	1X4DB133-1
NSCW transfer pump	1-1202-P4-008	1X4DB133-2
Component cooling water (CCW) pump	1-1203-P4-001	1X4DB136
CCW pump	1-1203-P4-002	1X4DB136
CCW pump	1-1203-P4-003	1X4DB136
CCW pump	1-1203-P4-004	1X4DB136
CCW pump	1-1203-P4-005	1X4DB136
CCW pump	1-1203-P4-006	1X4DB136
Auxiliary feedwater turbine-driven pump	1-1302-P4-001	1X4DB161-2
Auxiliary feedwater motor-driven pump	1-1302-P4-002	1X4DB161-2
Auxiliary feedwater motor-driven pump	1-1302-P4-003	1X4DB161-2
Diesel fuel oil storage tank pump[(a)]	1-2403-P4-001	1X4DB170-1
Diesel fuel oil storage tank pump[(a)]	1-2403-P4-002	1X4DB170-1

TABLE 3.9.B.3-8 (SHEET 2 OF 2)

Pump	Equipment Tag Number	Drawing Number
Diesel fuel oil storage tank pump[a]	1-2403-P4-003	1X4DB170-2
Diesel fuel oil storage tank pump[a]	1-2403-P4-004	1X4DB170-2
Control building ESF chilled water pump	1-1592-P7-001	1X4DB221
Control building chilled water pump	1-1592-P7-002	1X4DB221

a. For these pumps, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, plant procedures, and/or Technical Specifications in lieu of the Inservice Testing Program.

TABLE 3.9.B.3-9 (SHEET 1 OF 7)

BOP ACTIVE VALVES

Valve No.	System	Active Function[a]	Drawing Number
HV-10950	SIS	(1)	1X4DB120
HV-10951	SIS	(1)	1X4DB120
HV-10952	SIS	(1)	1X4DB120
HV-10953	SIS	(1)	1X4DB120
HV-10957	SIS	(2)	1X4DB121
HV-10958	SIS	(2)	1X4DB121
1204U4262	SIS	(2)	1X4DB121
1204U4263	SIS	(2)	1X4DB121
HV-1668A	NSCW	(2)(3)	1X4DB133-1
HV-1668B	NSCW	(2)(3)	1X4DB133-1
CV-9446	NSCW	(2)(3)	1X4DB133-1
1202U4025	NSCW	(2)(3)	1X4DB133-1
1202U4031	NSCW	(2)(3)	1X4DB133-1
1202U4035	NSCW	(2)(3)	1X4DB133-1
HV-1669B	NSCW	(2)(3)	1X4DB133-2
HV-1669A	NSCW	(2)(3)	1X4DB133-2
CV-9447	NSCW	(2)(3)	1X4DB133-2
1202U4027	NSCW	(2)(3)	1X4DB133-2
1202U4033	NSCW	(2)(3)	1X4DB133-2
1202U4037	NSCW	(2)(3)	1X4DB133-2
1202U4463	NSCW	(2)(3)	1X4DB134
1202U4464	NSCW	(2)(3)	1X4DB133-2
1202U4465	NSCW	(2)(3)	1X4DB133-2
1202U4466	NSCW	(2)(3)(4)	1X4DB135-1
1202U4467	NSCW	(2)(3)(4)	1X4DB135-1
1202U4468	NSCW	(4)	1X4DB135-1
1202U4469	NSCW	(2)(3)	1X4DB135-1
1202U4470	NSCW	(2)(3)	1X4DB135-1
1202U4471	NSCW	(2)(3)(4)	1X4DB135-2
1202U4472	NSCW	(2)(3)(4)	1X4DB135-2
1202U4473	NSCW	(4)	1X4DB135-2
1202U4474	NSCW	(2)(3)	1X4DB135-2
HV-1806[b][c]	NSCW	(2)(3)	1X4DB135-1
HV-1808[b][c]	NSCW	(2)(3)	1X4DB135-1
HV-1822[b][c]	NSCW	(2)(3)	1X4DB135-1
HV-1830[b][c]	NSCW	(2)(3)	1X4DB135-1
HV-2134	NSCW	(2)(3)	1X4DB135-1

TABLE 3.9.B.3-9 (SHEET 2 OF 7)

Valve No.	System	Active Function	Drawing Number
HV-2138	NSCW	(2)(3)	1X4DB135-1
HV-1807[b) (c)]	NSCW	(2)(3)	1X4DB135-2
HV-1809[b) (c)]	NSCW	(2)(3)	1X4DB135-2
HV-1823[b) (c)]	NSCW	(2)(3)	1X4DB135-2
HV-1831[b) (c)]	NSCW	(2)(3)	1X4DB135-2
HV-2135	NSCW	(2)(3)	1X4DB135-2
HV-2139	NSCW	(2)(3)	1X4DB135-2
HV-1975	ACCW	(4)	1X4DB138-1
HV-1979	ACCW	(4)	1X4DB138-1
HV-1974	ACCW	(4)	1X4DB138-2
HV-1978	ACCW	(4)	1X4DB138-2
HV-19051	ACCW	(4)	1X4DB138-2
HV-19053	ACCW	(4)	1X4DB138-2
HV-19055	ACCW	(4)	1X4DB138-2
HV-19057	ACCW	(4)	1X4DB138-2
1217U4113	ACCW	(1)	1X4DB138-2
1217U4084	ACCW	(4)	1X4DB138-2
1217U4085	ACCW	(4)	1X4DB138-2
1217U4086	ACCW	(4)	1X4DB138-2
1217U4087	ACCW	(4)	1X4DB138-2
HV-3502	Normal sampling	(1)	1X4DB140
HV-3507	Normal sampling	(1)	1X4DB140
HV-3508	Normal sampling	(1)	1X4DB140
HV-3513	Normal sampling	(1)	1X4DB140
HV-3514	Normal sampling	(1)	1X4DB140
HV-3548	Normal sampling	(1)	1X4DB140
HV-780	Drains	(1)	1X4DB143
HV-781	Drains	(1)	1X4DB143
HV-8211	PASS	(1)	1X4DB110
HV-8212	PASS	(1)	1X4DB110
HV-9453	Main steam	(4)	1X4DB159-1
HV-9454	Main steam	(4)	1X4DB159-1
HV-15212C	Main steam	(4)	1X4DB159-1
HV-15212D	Main steam	(4)	1X4DB159-1
HV-3006A/B	Main steam	(4)	1X4DB159-2
HV-3016A/B	Main steam	(4)	1X4DB159-2
HV-3026A/B	Main steam	(4)	1X4DB159-2
HV-3036A/B	Main steam	(4)	1X4DB159-2

TABLE 3.9.B.3-9 (SHEET 3 OF 7)

Valve No.	System	Active Function	Drawing Number
HV-3009	Main steam	(2)	1X4DB159-2
HV-3019	Main steam	(2)	1X4DB159-2
PV-3000	Main steam	(3)	1X4DB159-2
PV-3010	Main steam	(3)	1X4DB159-2
PV-3020	Main steam	(3)	1X4DB159-2
PV-3030	Main steam	(3)	1X4DB159-2
1301U4008	Main steam	(2)	1X4DB159-2
1301U4404	Main steam	(2)	1X4DB159-2
PSV-3001	Main steam	(4)	1X4DB159-2
PSV-3002	Main steam	(4)	1X4DB159-2
PSV-3003	Main steam	(4)	1X4DB159-2
PSV-3004	Main steam	(4)	1X4DB159-2
PSV-3005	Main steam	(4)	1X4DB159-2
PSV-3011	Main steam	(4)	1X4DB159-2
PSV-3012	Main steam	(4)	1X4DB159-2
PSV-3013	Main steam	(4)	1X4DB159-2
PSV-3014	Main steam	(4)	1X4DB159-2
PSV-3015	Main steam	(4)	1X4DB159-2
PSV-3021	Main steam	(4)	1X4DB159-2
PSV-3022	Main steam	(4)	1X4DB159-2
PSV-3023	Main steam	(4)	1X4DB159-2
PSV-3024	Main steam	(4)	1X4DB159-2
PSV-3025	Main steam	(4)	1X4DB159-2
PSV-3031	Main steam	(4)	1X4DB159-2
PSV-3032	Main steam	(4)	1X4DB159-2
PSV-3033	Main steam	(4)	1X4DB159-2
PSV-3034	Main steam	(4)	1X4DB159-2
PSV-3035	Main steam	(4)	1X4DB159-2
HV-9451	Main steam	(4)	1X4DB159-3
HV-9452	Main steam	(4)	1X4DB159-3
HV-15212A	Main steam	(4)	1X4DB159-3
HV-15212B	Main steam	(4)	1X4DB159-3
HV-5113	AFW	(2)(3)	1X4DB161-2
HV-5118	AFW	(2)(3)	1X4DB161-2
HV-5119	AFW	(2)(3)	1X4DB161-2
HV-5120	AFW	(2)(3)	1X4DB161-2
HV-5122	AFW	(2)(3)	1X4DB161-2
HV-5125	AFW	(2)(3)	1X4DB161-2
HV-5127	AFW	(2)(3)	1X4DB161-2
HV-5132	AFW	(2)(3)	1X4DB161-2
HV-5134	AFW	(2)(3)	1X4DB161-2
HV-5137	AFW	(2)(3)	1X4DB161-2

TABLE 3.9.B.3-9 (SHEET 4 OF 7)

Valve No.	System	Active Function	Drawing Number
HV-5139	AFW	(2)(3)	1X4DB161-2
FV-5154	AFW	(2)	1X4DB161-2
FV-5155	AFW	(2)	1X4DB161-2
1302U4013	AFW	(2)(3)	1X4DB161-2
1302U4014	AFW	(2)(3)	1X4DB161-2
1302U4001	AFW	(2)(3)	1X4DB161-2
1302U4002	AFW	(2)(3)	1X4DB161-2
1302U4017	AFW	(2)(3)	1X4DB161-2
1302U4020	AFW	(2)(3)	1X4DB161-2
1302U4023	AFW	(2)(3)	1X4DB161-2
1302U4026	AFW	(2)(3)	1X4DB161-2
1302U4033	AFW	(2)(3)	1X4DB161-2
1302U4037	AFW	(2)(3)	1X4DB161-2
1302U4040	AFW	(2)(3)	1X4DB161-2
1302U4043	AFW	(2)(3)	1X4DB161-2
1302U4046	AFW	(2)(3)	1X4DB161-2
1302U4051	AFW	(2)(3)	1X4DB161-2
1302U4052	AFW	(2)(3)	1X4DB161-2
1302U4058	AFW	(2)(3)	1X4DB161-2
1302U4061	AFW	(2)(3)	1X4DB161-2
HV-5106	AFW	(2)(3)	1X4DB161-3
PV-15129[(b) (d)]	AFW	(2)(3)	1X4DB161-3
SV-15133[(b) (d)]	AFW	(2)(3)	1X4DB161-3
HV-5227	Feedwater	(4)	1X4DB168-3
HV-5228	Feedwater	(4)	1X4DB168-3
HV-5229	Feedwater	(4)	1X4DB168-3
HV-5230	Feedwater	(4)	1X4DB168-3
1305U4071[(b)]	Feedwater	(4)	1X4DB168-3
1305U4073[(b)]	Feedwater	(4)	1X4DB168-3
1305U4075[(b)]	Feedwater	(4)	1X4DB168-3
1305U4077[(b)]	Feedwater	(4)	1X4DB168-3
FV-0510	Feedwater	(4)	1X4DB168-3
FV-0520	Feedwater	(4)	1X4DB168-3
FV-0530	Feedwater	(4)	1X4DB168-3
FV-0540	Feedwater	(4)	1X4DB168-3
LV-5242	Feedwater	(4)	1X4DB168-3
LV-5243	Feedwater	(4)	1X4DB168-3
LV-5244	Feedwater	(4)	1X4DB168-3
LV-5245	Feedwater	(4)	1X4DB168-3
1302U4113	AFW	(2)(3)	1X4DB168-3
1302U4114	AFW	(2)(3)	1X4DB168-3
1302U4115	AFW	(2)(3)	1X4DB168-3
1302U4116	AFW	(2)(3)	1X4DB168-3

TABLE 3.9.B.3-9 (SHEET 5 OF 7)

Valve No.	System	Active Function	Drawing Number
1302U4117	AFW	(4)	1X4DB168-3
1302U4118	AFW	(4)	1X4DB168-3
1302U4119	AFW	(4)	1X4DB168-3
1302U4120	AFW	(4)	1X4DB168-3
1302U4125	AFW	(2)(3)	1X4DB168-3
1302U4126	AFW	(2)(3)	1X4DB168-3
1302U4127	AFW	(2)(3)	1X4DB168-3
1302U4128	AFW	(2)(3)	1X4DB168-3
HV-15196	AFW	(4)	1X4DB168-3
HV-15197	AFW	(4)	1X4DB168-3
HV-15198	AFW	(4)	1X4DB168-3
HV-15199	AFW	(4)	1X4DB168-3
HV-27901	Fire protection	(1)	1X4DB174-4
2301U4036	Fire protection	(1)	1X4DB174-4
HV-9378	Instrument air	(1)	1X4DB186-4
2420U4049	Instrument air	(1)	1X4DB186-4
TV-12124[b) (e)]	Essential chilled water	(4)	1X4DB233
TV-12125[b) (e)]	Essential chilled water	(4)	1X4DB234
1203U4030	CCW	(2)(3)	1X4DB136
1203U4032	CCW	(2)(3)	1X4DB136
1203U4034	CCW	(2)(3)	1X4DB136
1203U4055	CCW	(2)(3)	1X4DB136
1203U4057	CCW	(2)(3)	1X4DB136
1203U4059	CCW	(2)(3)	1X4DB136
1592U4188	Essential chilled water	(4)	1X4DB221
1592U4192	Essential chilled water	(4)	1X4DB221
HV-15214[b)]	CVCS	(4)	1X4DB114

TABLE 3.9.B.3-9 (SHEET 6 OF 7)

Valve No.	System	Active Function	Drawing Number
HV-15216 A/B/C/D	Main steam	(4)	1X4DB159-1,3
HV-13005 A/B	Main steam	(4)	1X4DB159-2
HV-13006 A/B	Main steam	(4)	1X4DB159-2
HV-13007 A/B	Main steam	(4)	1X4DB159-2
HV-13008 A/B	Main steam	(4)	1X4DB159-2
HV-2041	ACCW	(2)(3)	1X4DB138-2
HV-11600	NSCW	(2)(3)	1X4DB133-1
HV-11605	NSCW	(2)(3)	1X4DB133-1
HV-11606	NSCW	(2)(3)	1X4DB133-1
HV-11607	NSCW	(2)(3)	1X4DB133-2
HV-11612	NSCW	(2)(3)	1X4DB133-2
HV-11613	NSCW	(2)(3)	1X4DB133-2
TV-11675[(b)]	NSCW	(2)(3)	1X4DB135-2
TV-11740[(b)]	NSCW	(2)(3)	1X4DB134
1202U4A07	NSCW	(4)	1X4DB133-1
1202U4A08	NSCW	(4)	1X4DB133-1
1202U4A09	NSCW	(4)	1X4DB133-1
1202U4A13	NSCW	(4)	1X4DB133-2
1202U4A14	NSCW	(4)	1X4DB133-2
1202U4A15	NSCW	(4)	1X4DB133-2
HV-2626A	CTB Normal Purge	(5)	1X4DB213-1
HV-2627A	CTB Normal Purge	(5)	1X4DB213-1
HV-2628A	CTB Normal Purge	(5)	1X4DB213-1
HV-2629A	CTB Normal Purge	(5)	1X4DB213-1
HV-2626B	CTB Minipurge	(1)	1X4DB213-1
HV-2627B	CTB Minipurge	(1)	1X4DB213-1
HV-2628B	CTB Minipurge	(1)	1X4DB213-1
HV-2629B	CTB Minipurge	(1)	1X4DB213-1
1513U4001	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
1513U4002	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2790A	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2790B	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2791A	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2791B	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2792A	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2792B	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2793A	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-2793B	CTB Hydrogen Recombiner	(1)(4)	1X4DB213-2
HV-12975	Radiation Monitor	(1)(4)	1X4DB213-2
HV-12976	Radiation Monitor	(1)(4)	1X4DB213-2
HV-12977	Radiation Monitor	(1)(4)	1X4DB213-2
HV-12978	Radiation Monitor	(1)(4)	1X4DB213-2
1302U4085	AFW	(2)(3)	1X4DB161-1
1302U4086	AFW	(2)(3)	1X4DB161-1
1302U4087	AFW	(2)(3)	1X4DB161-1

TABLE 3.9.B.3-9 (SHEET 7 OF 7)

a. Active Function
   (1) Containment isolation.
   (2) Emergency cooling operation.
   (3) Safety-grade cold shutdown.
   (4) Miscellaneous safety-related operations.
   (5) Containment isolation, modes 5 and 6 only.

b. For these valves, the capability to perform their safety-related function will be demonstrated through inclusion in plant maintenance programs, plant procedures, and/or Technical Specifications in lieu of the Inservice Testing (IST) Program.

c. Receives SI signal but does not stroke for an active safety function.

d. These valves are not in the scope of ASME Section III but are included for table completeness.

e. These valves provide an ASME Section III pressure boundary. Valve position is not a time critical function. Operators will have ample time to control cooling.

NOTE: The valve table as shown reflects the IST Program as modified by the second 10-year interval update. The updated IST Program will be fully implemented by May 31, 1998.

TABLE 3.9.B.3-10 (SHEET 1 OF 4)

EVALUATION OF NRC REGULATORY GUIDE 1.148 (MARCH 1981)

Regulatory Guide 1.148 Position	Position on BOP and NSSS Valves
The requirements delineated in ANSI N278.1-1975 (1) are generally acceptable to the NRC staff for functional specifications of active valve assemblies whose operability must be ensured and (2) provide an adequate basis for complying with those requirements of Criterion 1 of Appendix A and Section III of Appendix B to 10 CFR 50 relating to the correct translation into specifications, as supplemented or modified by the following:	In general, the requirements delineated by ANSI N278.1-1975 are addressed as part of the design specification for power-actuated valves designed in accordance with ASME section for Class 1, 2, and 3. Clarifications are given below:
1. Applicability and Relationship With Other Standards	
a. The scope of ANSI N278.1-1975, as stated in Section 1 of the standard, should be supplemented to include active manually operated valve assemblies in systems import to safety.	1.a. No manually operated valves are classified as active. Active valves consist of selected relief valves, check valves, and power-actuated valves. Refer to tables 3.9.N.3-2 and 3.9.B.3-9.
b. The phrase design specification relationship, used in Section 2 of the standard to indicate an inter-relationship with other codes and standards, should be broadly interpreted to include a document that contains sufficient detail to serve as a "complete basis for construction" and may, as appropriate, be called an equipment specification, procurement specification, or some similar term that includes the design specification required by the ASME Code. However, when valve operability is a requirement, the functional specification required by N278.1-1975 should be provided either as part of, or concurrent with, the controlling document that serves as the "complete basis for construction." For ASME Code Class 1, 2, and 3 valves, the "valve design specification" identified in Section 2 of N278.1-1975 should be that required by NCA-3250, Provision of Design Specifications, of Subsection NCA, General Requirements, of Section III of the Code.	1.b. The BOP and NSSS active valve specifications are performance specifications, which are considered a "complete basis for construction." ASME Code Class 1, 2, and 3 valve specifications include design information as required by NCA-3250 of the ASME Code. However, these specifications address installation by reference to drawings and do not include maintenance requirements. Maintenance requirements are included in technical manuals.
c. The valve functional specification should meet the following requirements in the ASME Code:	
(1) It should be uniquely identified and referenced in the design specification in accordance with NCA-3252(a) (6) of Section III of the Code.	1.c. For active valves, one design specification is prepared which meets the applicable ASME Code requirements and contains the functional requirements necessary to ensure that the valve will perform its intended safety function.

TABLE 3.9.B.3-10 (2 OF 4)

Regulatory Guide 1.148 Position	Position on BOP and NSSS Valves
(2) It should reference the applicable design specification to permit identification of both documents.	
(3) It should meet those portions of the filing requirements of NCA-3256 that call for a copy to be filed at the location of the installation and made available to the enforcement authorities having jurisdiction over the plant installation.	
d. The functional specification prepared in accordance with N278.1-1975 for valve assemblies classified as Quality Group D in Regulatory Guide 1.26 should be cross-referenced with the document that serves as a "complete basis for construction." (See Regulatory Position 1.b of this guide.)	1.d. See item 1a.
e. When the valve design specification for ASME Code Class 1, 2, and 3 valve assemblies requires use of ASME Code Cases such as N-62-2, the functional specification should consider this aspect, and usage should be consistent with Regulatory Guide 1.84, Design and Fabrication Code Case Acceptability, ASME Section III Division 1, and Regulatory 1.85, Materials Code Case Acceptability, ASME Section III Division 1.	1.e. Conform.
2. Specific Considerations	
a. Section 3.1, Valve Application Characteristics, should be supplemented by the following:	
(1) "Manually operated valves" (see Regulatory Position 1.a) as a separate item to the listing.	2.a.(1) See position 1.a.
(2) The functional specification should identify the relationship or correspondence between the "application characteristics" of the subject standard and the valve categories A, B, C, and D in IWV-2200 of Subsection IWV, Inservice Testing of Valves in Nuclear Power Plants, of Section XI of the Code.	2.a.(2) Valve specifications do not identify the relationship between the "application characteristics" of the Code. Valve categories are identified for valves subject to Section XI of the Code in the VEGP Preservice Test Program (ISI-P-004).
b. Section 3.2, Structural Requirements, should be supplemented by the following:	2.b.(1) and (2) In general, valve specifications do not completely address the interdependence and number of cycles of the parameters cited or the time relationship between seismic loadings and other loadings.

TABLE 3.9.B.3-10 (3 OF 4)

Regulatory Guide 1.148 Position	Position on BOP and NSSS Valves
(1) The interdependence and number of cycles, if applicable, of time, temperatures, pressures, and dynamic loading resulting from plant transients.	
(2) The time relationship between applied seismic loadings and other concurrent loadings.	
(3) The frequency response spectra for the OBE and the SSE as well as other potential forcing functions such as those from attached piping, pumps, restraints, or other equipment as applied to valves.	2.b.(3) These requirements are provided for active valves in the form of accelerations and nozzle loads which must be met in piping analyses.
(4) The maximum static and dynamic differential pressure (considering all plant operating conditions) that exists across the closure device, including potential water hammer, for which valve assembly operation is to be ensured.	2.b.(4) The maximum differential pressure is provided in specifications for power-actuated valves. Check valve specifications identify the operating and design conditions, but not the maximum differential pressure. The design condition for check valves implies a maximum differential pressure that corresponds to the ANSI pressure-temperature rating in the specifications. Water hammer and other plant conditions are analyzed for selective active valves, if required; but this information is not part of the specification.
c. Section 3.3, Operational Requirements, should be supplemented by the following:	
(1) The first paragraph should be supplemented to require that the desired position of the valve assembly in the event of loss of actuator power be specified, e.g., fail open, fail closed, fail as is.	2.c.(1) Conform.
(2) In item e of Section 3.3.1, Operating Conditions, the phrase "normal and abnormal plant operation" should be interpreted to include the events covered by plant "operational modes (condition)," and by the transient and accident classification of Chapter 15, Accident Analysis, of Regulatory Guide 1.70. The functional specification should state whether the specific valve assembly safety function applies to events defined in the plant operational modes (condition) or in the transient and accident classification. The specification should also indicate whether the actual valves assembly operation (open, close, or regulate fluid flow) occurs during or after the specific event.	2.c.(2) Valve specifications generally do not state whether the valve safety function applies to events defined in the plant operational modes or in the transient and accident classification. The appropriate service conditions are provided in the specification along with the actual valve assembly operation (open, close, or regulate fluid flow).

TABLE 3.9.B.3-10 (4 OF 4)

Regulatory Guide 1.148 Position	Position on BOP and NSSS Valves
(3) Instead of item e of 3.3.2, the following should be used: "e. motor power and duty requirements, including stall current."	2.c.(3) Conform.
(4) Instead of Section 3.3.2.1, Electrical Power for Valve Actuators and Control Elements, the following should be used: "Electrical power shall be identified as ac (single phase or three phase) or dc. Applicable voltage and frequency operating ranges shall be specified."	2.c.(4) Conform.
(5) In Section 3.3.3, Environmental Conditions, the terms "normal" and "abnormal" environmental conditions should be interpreted to mean the environmental conditions that will exist as a result of specified plant conditions. The term "environmental conditions" should be interpreted to include both the environment external to the valve assembly and that of the controlled fluid inside the valve assembly.	2.c.(5) Conform.
d. Section 3.4., Seat Leakage Limits, should be supplemented, as applicable, by the following:	
(1) The leakage limits identified in paragraph b of this section should include identification of fluid, temperature, and differential pressure for which the limit applies.	2.d.(1) Conform.
(5) If valve assembly function requires a limit on overall leakage (e.g., leakage in addition to that of the main seat, such as stem packing and flange), such overall leakage limit should be specified in this section.	2.d.(2) Conform.

TABLE 3.9.N.3-1

ACTIVE PUMPS
ANS

Pump	Item Number	System	Safety Class	Normal Mode	Post-Accident Mode	Basis[a]
Centrifugal charging pumps 1 and 2	APCH	CVCS	2	On/Off	On	(1) (2)
Boric acid transfer pumps 1 and 2	APBA	CVCS	3	On/Off	On/Off	(2)
Residual heat removal pumps 1 and 2	APRH	RHRS	2	Off	On	(1) (2)
Safety injection pumps 1 and 2	APSI	SIS	2	Off	On	(1)
Containment spray pumps 1 and 2	APCS	CSS	2	Off	On	(3)
Spent fuel pool[b] pumps 1 and 2	APSF	SFPCS	3	On	On	Spent fuel decay heat removal

a. Basis are defined as follows:

1. Emergency core cooling system safeguards operation.
2. Safety grade cold shutdown operation.
3. Containment spray system safeguards operation.

b. For these pumps, the capability to perform their safety related function will be demonstrated through inclusion in plant maintenance programs, in lieu of the Inservice Testing Program.

TABLE 3.9.N.3-2 (SHEET 1 OF 8)

ACTIVE VALVES

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis [a]
8010A/B/C	RCS	Self-actuated	6	Safety/1	Closed	(1)
PV-455A	RCS	Solenoid	3	Globe/1	Closed	(1) (4)
PV-456A	RCS	Solenoid	3	Globe/1	Closed	(1) (4)
8000A/B	RCS	Motor	3	Gate/1	Open	(1) (4)
8028	RCS	Air	3	Diaphragm/2	Closed	(2)
8033	RCS	Air	1	Diaphragm/2	Closed	(2)
1201U6112	RCS	ΔP	3	Check/2	NA	(2)
8047	RCS	Air	1	Diaphragm/2	NA	(2)
8095A/B	RCS	Solenoid	1	Globe/1	Closed	(1) (4)
8096A/B	RCS	Solenoid	1	Globe/1	Closed	(1) (4)
HV-442A/B	RCS	Solenoid	1	Globe/2	Closed	(4)
8100	CVCS	Motor	2	Globe/2	Open	(2)
8103A/B/C/D	CVCS	Motor	1/2	Globe	Open	(2) (4)
8104	CVCS	Motor	2	Globe/2	Closed	(4)
8105	CVCS	Motor	3	Gate/2	Open	(2) (3) (4)
8106	CVCS	Motor	3	Gate/2	Open	(3) (4)
8110	CVCS	Motor	2	Globe/2	Open	(3) (4)
8111A/B	CVCS	Motor	2	Globe/2	Open	(3) (4)
8112	CVCS	Motor	2	Globe/2	Open	(2)
1208U4021	CVCS	ΔP	3/4	Check/2	NA	(2)
8116	CVCS	Motor	1	Globe/2	Closed	(4)

TABLE 3.9.N.3-2 (SHEET 2 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
8152	CVCS	Air	3	Globe/2	Open	(2)
8160	CVCS	Air	3	Globe/2	Open	(2)
1208U4437[(b)]	CVCS	ΔP	1/2	Check/1	NA	(1) (3) (4)
1208U4438[(b)]	CVCS	ΔP	1/2	Check/1	NA	(1) (3) (4)
1208U4439[(b)]	CVCS	ΔP	1/2	Check/1	NA	(1) (3) (4)
1208U4440[(b)]	CVCS	ΔP	1/2	Check/1	NA	(1) (3) (4)
1208U4006[(b)]	CVCS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1208U4359[(b)]	CVCS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1208U4360[(b)]	CVCS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1208U4361[(b)]	CVCS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1208U4004	CVCS	ΔP	1 1/2	Check/2	NA	(2) (3) (4)
1208U4353	CVCS	ΔP	1 1/2	Check/2	NA	(2) (3) (4)
1208U4354	CVCS	ΔP	1 1/2	Check/2	NA	(2) (3) (4)
1208U4355	CVCS	ΔP	1 1/2	Check/2	NA	(2) (3) (4)
1208U4033[(b)]	CVCS	ΔP	2	Check/1	NA	(1)
1208U6035[(b)]	CVCS	ΔP	3	Check/1	NA	(1) (4)

TABLE 3.9.N.3-2 (SHEET 3 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
1208U6036[b]	CVCS	ΔP	3	Check/1	NA	(1) (4)
1208U6037[b]	CVCS	ΔP	3	Check/1	NA	(1) (4)
1208U6038[b]	CVCS	ΔP	3	Check/1	NA	(1) (4)
1208U6032	CVCS	ΔP	3	Check/2	NA	(2) (4)
8438[b]	CVCS	Motor	4	Gate/2	Open	(3) (4)
1208U6124	CVCS	ΔP	4	Check/2	NA	(4)
1208U4185	CVCS	ΔP	2	Check/2	NA	(4)
1208U4299	CVCS	ΔP	2	Check/3	NA	(4)
1208U4140	CVCS	ΔP	2	Check/2	NA	(4)
1208U4147	CVCS	ΔP	2	Check/2	NA	(4)
1208U6142	CVCS	ΔP	4	Check/2	NA	(3) (4)
1208U6149	CVCS	ΔP	4	Check/2	NA	(3) (4)
8471A/B	CVCS	Motor	6	Gate/2	Open	(4)
8485A/B[w]	CVCS	Motor	4	Gate/2	Open	(3) (4)
1208U4284	CVCS	ΔP	2	Check/3	NA	(4)
1208U6129	CVCS	ΔP	3	Check/2	NA	(3) (4)
8508A/B	CVCS	Motor	2	Globe/2	Closed/Open	(3)
8509A/B	CVCS	Motor	2	Globe/2	Closed/Open	(3)
1208U6189	CVCS	ΔP	8	Check/2	NA	(3) (4)

TABLE 3.9.N.3-2 (SHEET 4 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
HV-190A/B	CVCS	Solenoid	1	Globe/2	Closed	(4)
LV-112B/C	CVCS	Motor	4	Gate/2	Open	(3) (4)
LV-112D/E	CVCS	Motor	8	Gate/2	Closed	(3) (4)
LV-0459[(b),(c)]	CVCS	Air	3	Globe/1	Open	(1)
LV-0460[(b),(c)]	CVCS	Air	3	Globe/1	Open	(1)
8701A	RHRS	Motor	12	Gate/1	Closed	(1) (2) (4)
8701B	RHRS	Motor	12	Gate/1	Closed	(1) (4)
8702A	RHRS	Motor	12	Gate/1	Closed	(1) (2) (4)
8702B	RHRS	Motor	12	Gate/1	Closed	(1) (4)
1201U4251	RHRS	ΔP	3/4	Check/2	NA	(4)[(d)]
1201U4252	RHRS	ΔP	3/4	Check/2	NA	(4)
8716A/B	RHRS	Motor	8	Gate/2	Open	(3) (4)
1205U6009	RHRS	ΔP	8	Check/2	NA	(3) (4)
1205U6010	RHRS	ΔP	8	Check/2	NA	(3) (4)
FV-610	RHRS	Motor	3	Gate/2	Open/Closed	(3) (4)
FV-611	RHRS	Motor	3	Gate/2	Open/Closed	(3) (4)
8801A/B	SIS	Motor	4	Gate/2	Closed	(2) (3) (4)
8802A/B	SIS	Motor	4	Gate/2	Closed	(2) (3)
8804A/B	SIS	Motor	8	Gate/2	Closed	(3)
8806	SIS	Motor	8	Gate/2	Open	(3)
8807A/B	SIS	Motor	6	Gate/2	Closed	(3)

TABLE 3.9.N.3-2 (SHEET 5 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
8809A/B	SIS	Motor	8	Gate/2	Open	(2) (3) (4)
8811A/B	SIS	Motor	14	Gate/2	Closed	(3)
8812A/B	SIS	Motor	12	Gate/2	Open	(3) (4)
8813	SIS	Motor	2	Globe/2	Open	(3)
8814	SIS	Motor	1 1/2	Globe/2	Open	(3)
1204U6013	SIS	ΔP	3	Check/1	NA	(1) (2) (3) (4)
1204U6147	SIS	ΔP	6	Check/1	NA	(1) (2) (3) (4)
1204U6148	SIS	ΔP	6	Check/1	NA	(1) (2) (3) (4)
1204U6149	SIS	ΔP	6	Check/1	NA	(1) (2) (3) (4)
1204U6150	SIS	ΔP	6	Check/1	NA	(1) (2) (3) (4)
1204U4143	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4144	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4145	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4146	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1205U4122	SIS	ΔP	14	Check/2	NA	(3)
1205U4123	SIS	ΔP	14	Check/2	NA	(3)
8821A/B	SIS	Motor	4	Gate/2	Open	(3)
8823	SIS	Air	3/4	Globe/2	Closed	(2)
8824	SIS	Air	3/4	Globe/2	Closed	(2)
8825	SIS	Air	3/4	Globe/2	Closed	(2)
8835	SIS	Motor	4	Gate/2	Open	(2) (3)
8840	SIS	Motor	12	Gate/2	Closed	(2) (3)

TABLE 3.9.N.3-2 (SHEET 6 OF 8)

Valve Number	System	Actuated By	Size (in.)	Type/ANS Safety Class	Normal Position	Basis
1204U6128	SIS	ΔP	8	Check/1	NA	(1) (2) (3)
1204U6129	SIS	ΔP	8	Check/1	NA	(1) (2) (3)
8843	SIS	Air	3/4	Globe/2	Closed	(2)
8871	SIS	Air	3/4	Globe/2	Closed	(2)
8875A/B/C/D/ E/F/G/H	SIS	Solenoid	1	Globe/2	Closed	(4)
8880	SIS	Air	1	Globe/2	Closed	(2)
8881	SIS	Air	3/4	Globe/2	Closed	(2)
8888	SIS	Air	3/4	Globe/2	Closed	(2)
8890A/B	SIS	Air	3/4	Globe/2	Closed	(2)
1204U4026	SIS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1204U4027	SIS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1204U4028	SIS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1204U4029	SIS	ΔP	1 1/2	Check/1	NA	(1) (3) (4)
1204U4120	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4121	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4122	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4123	SIS	ΔP	2	Check/1	NA	(1) (2) (3)
1204U4093	SIS	ΔP	1 1/2	Check/2	NA	(3)
1204U4094	SIS	ΔP	1 1/2	Check/2	NA	(3)
1204U6098	SIS	ΔP	4	Check/2	NA	(3)
1204U6099	SIS	ΔP	4	Check/2	NA	(3)